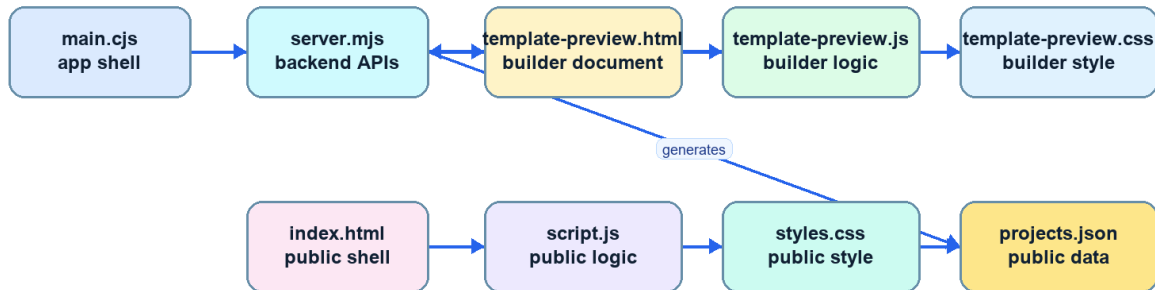


# OMB Portfolio Builder Code Reference

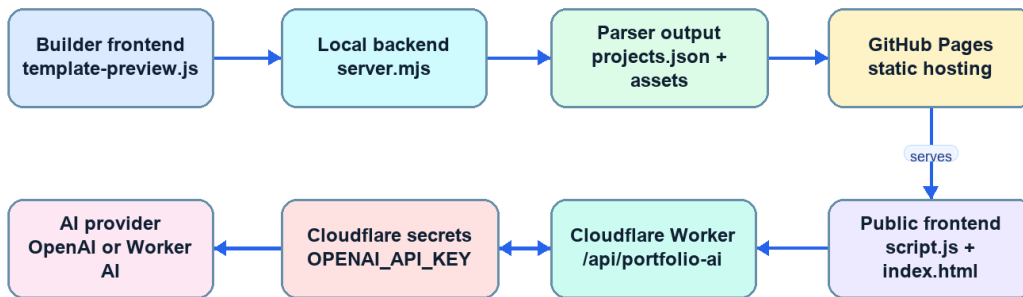
A generated PDF reference that explains the important source files, discovered functions, communication paths, variables, endpoints, installers, AI paths, and compiler paths.

## How Key Files Communicate



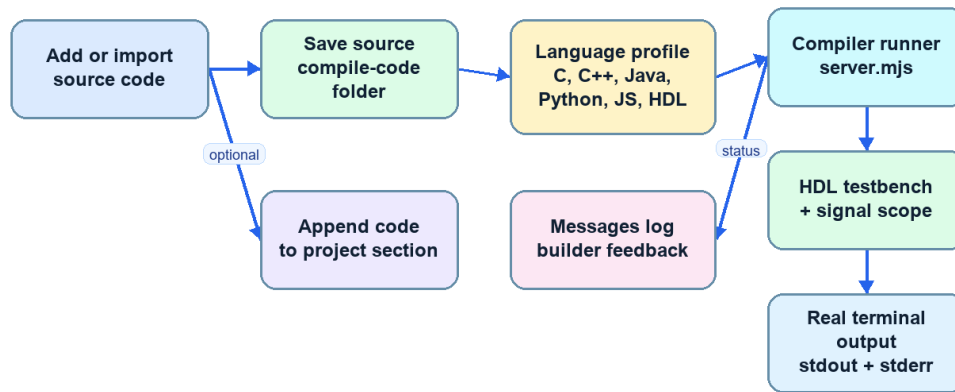
Main source file communication map.

## Frontend, Backend, Cloudflare, And Secrets



Frontend, backend, Cloudflare, and AI boundary.

## Compile Code Workspace



Compile Code workspace and simulator flow.

## Generated file index

| Item                                        | Explanation                                 |
|---------------------------------------------|---------------------------------------------|
| .github/workflows/build-windows-builder.yml | 107 lines, 0 function(s), 13 signal(s).     |
| .github/workflows/main-branch-gate.yml      | 40 lines, 0 function(s), 1 signal(s).       |
| .well-known/security.txt                    | 4 lines, 0 function(s), 2 signal(s).        |
| BRANCHING.md                                | 55 lines, 0 function(s), 6 signal(s).       |
| README.md                                   | 238 lines, 0 function(s), 67 signal(s).     |
| build/installer.nsh                         | 570 lines, 14 function(s), 104 signal(s).   |
| builder-rich-future-sections.js             | 410 lines, 13 function(s), 17 signal(s).    |
| cloudflare/README.md                        | 47 lines, 0 function(s), 16 signal(s).      |
| cloudflare/portfolio-ai-worker.js           | 276 lines, 13 function(s), 45 signal(s).    |
| cloudflare/wrangler.toml                    | 18 lines, 0 function(s), 8 signal(s).       |
| docs/build_complete_guide.py                | 2,412 lines, 75 function(s), 434 signal(s). |
| electronics-search.js                       | 162 lines, 2 function(s), 8 signal(s).      |
| index.html                                  | 268 lines, 0 function(s), 9 signal(s).      |
| installer-notes.txt                         | 23 lines, 0 function(s), 11 signal(s).      |
| main.cjs                                    | 453 lines, 21 function(s), 43 signal(s).    |
| package.json                                | 85 lines, 0 function(s), 16 signal(s).      |
| pnpm-lock.yaml                              | 2,233 lines, 0 function(s), 285 signal(s).  |
| pnpm-workspace.yaml                         | 3 lines, 0 function(s), 1 signal(s).        |
| portfolio-README.md                         | 67 lines, 0 function(s), 10 signal(s).      |
| project-templates.json                      | 184 lines, 0 function(s), 0 signal(s).      |

| Item                  | Explanation                                    |
|-----------------------|------------------------------------------------|
| projects.json         | 46 lines, 0 function(s), 0 signal(s).          |
| robots.txt            | 2 lines, 0 function(s), 0 signal(s).           |
| script.js             | 3,710 lines, 236 function(s), 380 signal(s).   |
| server.mjs            | 3,632 lines, 141 function(s), 566 signal(s).   |
| styles.css            | 2,735 lines, 0 function(s), 260 signal(s).     |
| template-preview.css  | 5,749 lines, 0 function(s), 738 signal(s).     |
| template-preview.html | 1,349 lines, 0 function(s), 124 signal(s).     |
| template-preview.js   | 10,826 lines, 570 function(s), 1973 signal(s). |

# Source File: .github/workflows/build-windows-builder.yml

| Item                | Explanation                                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File                | .github/workflows/build-windows-builder.yml                                                                                                                                                            |
| Purpose             | GitHub Actions workflow that builds the Windows installer and portable application, checks the stable download link, enforces release version bumps, uploads artifacts, and publishes GitHub Releases. |
| Lines               | 107                                                                                                                                                                                                    |
| Size                | 3,867 bytes                                                                                                                                                                                            |
| Talks to            | GitHub/release layer                                                                                                                                                                                   |
| Functions           | 0                                                                                                                                                                                                      |
| Variables/constants | 0                                                                                                                                                                                                      |
| API mentions        | 0                                                                                                                                                                                                      |
| Signals             | 13                                                                                                                                                                                                     |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                             | Explanation                                                                                                                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security or auth at line 11      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: permissions:                                                                                                                    |
| Installer at line 16             | Windows setup, update, shortcut, or installation behavior. Evidence: name: Build Windows installer and portable app                                                                                                    |
| Compiler or simulator at line 23 | Part of the Compile Code workspace or language/tool detection. Evidence: - name: Set up Node.js                                                                                                                        |
| Compiler or simulator at line 24 | Part of the Compile Code workspace or language/tool detection. Evidence: uses: actions/setup-node@v4                                                                                                                   |
| Compiler or simulator at line 26 | Part of the Compile Code workspace or language/tool detection. Evidence: node-version: "24"                                                                                                                            |
| Installer at line 44             | Windows setup, update, shortcut, or installation behavior. Evidence: throw "Windows setup installer was not found."                                                                                                    |
| File read at line 57             | Reads local builder state, project files, website assets, or configuration. Evidence: \$package = Get-Content package.json -Raw   ConvertFrom-Json                                                                     |
| Network request at line 73       | Frontend-backend communication or public web/API calls. Evidence: \$latestInstallerUrl = "https://github.com/otienoma urice/omb-portfolio-builder/releases/latest/download/OMB-Portfolio-Builder-Setup-latest-x64.exe" |
| File read at line 74             | Reads local builder state, project files, website assets, or configuration. Evidence: \$index = Get-Content index.html -Raw                                                                                            |
| Installer at line 75             | Windows setup, update, shortcut, or installation behavior. Evidence: if (\$index -notmatch [regex]::Escape(\$latestInstallerUrl)) {                                                                                    |
| Installer at line 76             | Windows setup, update, shortcut, or installation behavior. Evidence: throw "The portfolio download link must use the stable latest installer URL: \$latestInstallerUrl"                                                |
| Git command at line 83           | Repository state, publishing, branch checks, or release automation. Evidence: git fetch --tags --force                                                                                                                 |
| Git command at line 85           | Repository state, publishing, branch checks, or release automation. Evidence: \$tagExists = [bool](git tag --list \$tag)                                                                                               |

## Representative opening snippet

```
1: name: Build Windows Portfolio Builder
2:
3: on:
4:   workflow_dispatch:
5:   push:
6:     branches:
7:       - main
8:   tags:
9:     - "builder-v*"
```

```

10:
11: permissions:
12:   contents: write
13:
14: jobs:
15:   build-windows:
16:     name: Build Windows installer and portable app
17:     runs-on: windows-latest
18:
19:     steps:
20:       - name: Check out repository
21:         uses: actions/checkout@v4
22:
23:       - name: Set up Node.js
24:         uses: actions/setup-node@v4
25:         with:
26:           node-version: "24"
27:
28:       - name: Set up pnpm
29:         uses: pnpm/action-setup@v4
30:         with:
31:           version: "11.5.3"
32:
33:       - name: Install desktop builder dependencies
34:         run: pnpm install --frozen-lockfile
35:
36:       - name: Build Windows artifacts
37:         run: pnpm run dist
38:
39:       - name: Create stable latest download names
40:         shell: pwsh
41:         run: |
42:           $setup = Get-ChildItem dist -Filter "OMB-Portfolio-Builder-Setup-*-x64.exe" | Sort-Object
LastWriteTime -Descending | Select-Object -First 1
43:           if (-not $setup) {
44:             throw "Windows setup installer was not found."
45:           }
46:           Copy-Item $setup.FullName "dist/OMB-Portfolio-Builder-Setup-latest-x64.exe" -Force
47:
48:           $portable = Get-ChildItem dist -Filter "OMB-Portfolio-Builder-Portable-*-x64.exe" | Sort-Object
LastWriteTime -Descending | Select-Object -First 1
49:           if ($portable) {
50:             Copy-Item $portable.FullName "dist/OMB-Portfolio-Builder-Portable-latest-x64.exe" -Force
51:           }
52:
53:       - name: Resolve release version
54:         id: release_meta
55:         shell: pwsh
56:         run: |
57:           $package = Get-Content package.json -Raw | ConvertFrom-Json
58:           $version = [string]$package.version
59:           if (-not $version) {
60:             throw "package.json version is required before publishing a release."
61:           }
62:           if ($env:GITHUB_REF -like "refs/tags/*") {
63:             $tag = $env:GITHUB_REF_NAME
64:           } else {
65:             $tag = "builder-v$version"
66:           }
67:           "version=$version" >> $env:GITHUB_OUTPUT
68:           "release_tag=$tag" >> $env:GITHUB_OUTPUT
69:
70:       - name: Verify portfolio download link follows latest release
71:         shell: pwsh
72:         run: |

```

# Source File: .github/workflows/main-branch-gate.yml

| Item                | Explanation                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| File                | .github/workflows/main-branch-gate.yml                                                                                               |
| Purpose             | GitHub Actions guard that rejects direct pushes to main and rejects pull requests into main unless the source branch is development. |
| Lines               | 40                                                                                                                                   |
| Size                | 1,259 bytes                                                                                                                          |
| Talks to            | GitHub/release layer                                                                                                                 |
| Functions           | 0                                                                                                                                    |
| Variables/constants | 0                                                                                                                                    |
| API mentions        | 0                                                                                                                                    |
| Signals             | 1                                                                                                                                    |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                        | Explanation                                                                                         |
|-----------------------------|-----------------------------------------------------------------------------------------------------|
| Security or auth at line 11 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: permissions: |

## Representative opening snippet

```
1: name: Main Branch Gate
2:
3: on:
4:   pull_request:
5:     branches:
6:       - main
7:   push:
8:     branches:
9:       - main
10:
11: permissions:
12:   contents: read
13:
14: jobs:
15:   require-development-source:
16:     name: Require development as the main source branch
17:     runs-on: ubuntu-latest
18:
19:     steps:
20:       - name: Verify main only receives development
21:         shell: bash
22:         env:
23:           HEAD_COMMIT_MESSAGE: ${ github.event.head_commit.message }
24:         run: |
25:           if [ "${ github.event_name }" = "push" ]; then
26:             if printf '%s' "$HEAD_COMMIT_MESSAGE" | grep -Eq '^Merge pull request #[0-9]+ from
[^/]+/development$'; then
27:               echo "main was updated by a development release PR."
28:               exit 0
29:             fi
30:
31:             echo "::error::Direct pushes to main are not allowed. Open a pull request from development
into main."
32:             exit 1
33:           fi
34:
35:           if [ "${ github.head_ref }" != "development" ]; then
36:             echo "::error::Pull requests into main must come from development. Merge this branch into
development first, test development, then open the development -> main release pull request."
37:             exit 1
38:           fi
39:
40:           echo "main release source is development."
```

# Source File: .well-known/security.txt

| Item                | Explanation                                                                                                          |
|---------------------|----------------------------------------------------------------------------------------------------------------------|
| File                | .well-known/security.txt                                                                                             |
| Purpose             | Public security contact file used by browsers, researchers, and scanners to find responsible disclosure information. |
| Lines               | 4                                                                                                                    |
| Size                | 170 bytes                                                                                                            |
| Talks to            | Mostly local to its own feature area.                                                                                |
| Functions           | 0                                                                                                                    |
| Variables/constants | 0                                                                                                                    |
| API mentions        | 0                                                                                                                    |
| Signals             | 2                                                                                                                    |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                      | Explanation                                                                                                                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network request at line 3 | Frontend-backend communication or public web/API calls. Evidence: Canonical: <a href="https://mauriceotieno.com/.well-known/security.txt">https://mauriceotieno.com/.well-known/security.txt</a> |
| Network request at line 4 | Frontend-backend communication or public web/API calls. Evidence: Policy: <a href="https://mauriceotieno.com/">https://mauriceotieno.com/</a>                                                    |

## Representative opening snippet

```
1: Contact: mailto:otienomaurice1234@gmail.com
2: Preferred-Languages: en
3: Canonical: https://mauriceotieno.com/.well-known/security.txt
4: Policy: https://mauriceotieno.com/
```

# Source File: BRANCHING.md

| Item                | Explanation                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------|
| File                | BRANCHING.md                                                                                  |
| Purpose             | Human-readable branch model explaining development, main, feature branches, and release flow. |
| Lines               | 55                                                                                            |
| Size                | 3,052 bytes                                                                                   |
| Talks to            | GitHub/release layer                                                                          |
| Functions           | 0                                                                                             |
| Variables/constants | 0                                                                                             |
| API mentions        | 0                                                                                             |
| Signals             | 6                                                                                             |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                   | Explanation                                                                                                        |
|------------------------|--------------------------------------------------------------------------------------------------------------------|
| Git command at line 19 | Repository state, publishing, branch checks, or release automation. Evidence: git checkout development             |
| Git command at line 20 | Repository state, publishing, branch checks, or release automation. Evidence: git pull origin development          |
| Git command at line 21 | Repository state, publishing, branch checks, or release automation. Evidence: git checkout -b feature/my-change    |
| Git command at line 29 | Repository state, publishing, branch checks, or release automation. Evidence: git add .                            |
| Git command at line 30 | Repository state, publishing, branch checks, or release automation. Evidence: git commit -m "Describe the change"  |
| Git command at line 31 | Repository state, publishing, branch checks, or release automation. Evidence: git push -u origin feature/my-change |

## Representative opening snippet

```
1: # Branching Workflow
2:
3: This repository uses a protected release-style workflow.
4:
5: ## Branches
6:
7: - `main` is the consumer branch. It should contain only finished, tested, release-ready code.
8: - `development` is the integration branch. It collects completed work from feature branches before
release.
9: - `feature/<short-name>` branches are used for individual app changes, experiments, fixes, or UI
improvements.
10:
11: `main` must only receive changes from `development`. Feature branches, Codex branches, hotfix branches,
and experiment branches must merge into `development` first. After `development` has collected and verified the
changes, `development` is the only branch that may be merged into `main`.
12:
13: ## Normal Change Flow
14:
15: 1. Start from the latest `development`.
16: 2. Create a focused feature branch:
17:
18: ```powershell
19: git checkout development
20: git pull origin development
21: git checkout -b feature/my-change
22: ```
23:
24: 3. Update the branch register in `README.md` with the new branch name, purpose, source branch, merge
target, and status.
25: 4. Make and test the change.
26: 5. Commit and push the feature branch:
27:
28: ```powershell
29: git add .
```



```
30: git commit -m "Describe the change"
31: git push -u origin feature/my-change
32: ``
33:
34: 6. Merge or pull-request the feature branch into `development`.
35: 7. Update the branch register status when the branch is merged, closed, or released.
36: 8. Test `development`.
37: 9. Before merging `development` into `main`, bump `package.json` to the next release version.
38: 10. Merge `development` into `main` only when the app is ready for consumers. No other source branch
should target `main`.
39: 11. Push `main` to build and publish the `builder-v<version>` release automatically, or create a release
tag from `main`, for example `builder-v<next-version>`.
40:
41: ## Branch Register Rule
42:
43: The README is the human-facing branch register. Every new branch should have a row in `README.md` before
the branch is pushed. This keeps local users, GitHub visitors, and future Codex sessions aligned on what each
branch is for.
44:
45: ## Release Rule
46:
47: Do not make app changes directly on `main`. `main` should move only after `development` has collected and
verified the changes.
48:
49: Pull requests into `main` must use `development` as the source branch. The repository includes
`.github/workflows/main-branch-gate.yml`, which fails pull requests into `main` from any other branch. GitHub
branch protection should also be enabled for `main` so direct pushes are blocked and release changes go through
the development-to-main pull request path.
50:
51: Installed apps detect updates by comparing their current app version with the latest GitHub Release
version. A `main` push must therefore carry a new `package.json` version. The release workflow rejects a `main`
push if the matching `builder-v<version>` tag already exists.
52:
53: ## Current Practice
54:
55: For Codex-assisted work, new app changes should be made on a focused `codex/...` or `feature/...` branch,
merged into `development`, and only then released through `development` into `main`.
```

# Source File: README.md

| Item                | Explanation                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| File                | README.md                                                                                                                                                |
| Purpose             | Main README for the standalone builder application: install, updates, workspaces, text editing, publishing, build commands, branch model, and uninstall. |
| Lines               | 238                                                                                                                                                      |
| Size                | 22,595 bytes                                                                                                                                             |
| Talks to            | Cloudflare/AI layer, GitHub/release layer                                                                                                                |
| Functions           | 0                                                                                                                                                        |
| Variables/constants | 0                                                                                                                                                        |
| API mentions        | 0                                                                                                                                                        |
| Signals             | 67                                                                                                                                                       |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                        | Explanation                                                                                                                                                                                                                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloudflare or AI at line 17 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: It explains the high-level design, generated block diagrams, software-engineering decisions, shell commands, programming syntax, how the important files communicate, what each build tool owns, GitHub workflows, Electron, |
| Installer at line 21        | Windows setup, update, shortcut, or installation behavior. Evidence: Download the latest Windows installer or portable executable from:                                                                                                                                                                             |
| Network request at line 24  | Frontend-backend communication or public web/API calls. Evidence: <a href="https://github.com/otienomaurence/omb-portfolio-builder/releases/latest">https://github.com/otienomaurence/omb-portfolio-builder/releases/latest</a>                                                                                     |
| Installer at line 29        | Windows setup, update, shortcut, or installation behavior. Evidence: - `OMB-Portfolio-Builder-Setup-latest-x64.exe` for the Windows installer.                                                                                                                                                                      |
| Installer at line 32        | Windows setup, update, shortcut, or installation behavior. Evidence: The normal installer default is a per-user AppData location:                                                                                                                                                                                   |
| Installer at line 35        | Windows setup, update, shortcut, or installation behavior. Evidence: C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder                                                                                                                                                                                    |
| Installer at line 41        | Windows setup, update, shortcut, or installation behavior. Evidence: C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder\OMB Portfolio Builder.exe                                                                                                                                                          |
| Installer at line 44        | Windows setup, update, shortcut, or installation behavior. Evidence: The installer uses a Windows wizard with Back, Next, Cancel, progress, and Finish. It can create Desktop and Start Menu shortcuts, and it writes a normal Windows uninstall entry. The default per-user install does not wri                   |
| Git command at line 46      | Repository state, publishing, branch checks, or release automation. Evidence: Installed users do not need Node.js or pnpm. The packaged desktop app includes its runtime. Git for Windows with Git Credential Manager is needed only for publishing to a Git repository. If Git or Git Credential Manager           |
| Installer at line 50        | Windows setup, update, shortcut, or installation behavior. Evidence: The app checks GitHub Releases for newer builder versions. When a newer release exists, the app shows an update dialog. Choosing <b>**Update**</b> downloads the installer, closes the running builder, updates the existing instal            |
| Network request at line 57  | Frontend-backend communication or public web/API calls. Evidence: <a href="https://github.com/otienomaurence/omb-portfolio-builder/releases">https://github.com/otienomaurence/omb-portfolio-builder/releases</a>                                                                                                   |
| Installer at line 62        | Windows setup, update, shortcut, or installation behavior. Evidence: On Windows, the app keeps application files and editable content under AppData:                                                                                                                                                                |
| Installer at line 64        | Windows setup, update, shortcut, or installation behavior. Evidence: - `C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder` stores the installed desktop application.                                                                                                                                      |
| Installer at line 65        | Windows setup, update, shortcut, or installation behavior. Evidence: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\builder` stores the builder workspace, local drafts, uploaded files, templates, previews, and editing tools.                                                                             |

| Item                              | Explanation                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Git command at line 66            | Repository state, publishing, branch checks, or release automation. Evidence: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\portfolio` stores the publish mirror used by Git-based publishing.                                                                                                                                   |
| Installer at line 67              | Windows setup, update, shortcut, or installation behavior. Evidence: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\compile-code` stores project compile-workspace source files and run folders created by the Compile Code tool.                                                                                                 |
| Installer at line 69              | Windows setup, update, shortcut, or installation behavior. Evidence: The installer and updater treat the AppData builder and portfolio folders as managed workspaces, not duplicate app installs.                                                                                                                                        |
| Security or auth at line 71       | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: The builder works offline for editing, project creation, file attachment, local previews, and draft saves. Publishing requires network access and repository write permission.                                                                    |
| Rich text at line 86              | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Project and portfolio-area overview editors support rich text editing with selection-level formatting, links, images, formulas, syntax-highlighted code blocks, hard line breaks, copy, paste, cut, and select all.                                 |
| Rich text at line 88              | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Front-page, profile, contact, fun facts, and project overview fields use the same selection-level text behavior: highlight the exact words to edit, then use the right-click menu or floating selection inspector to change                         |
| Compiler or simulator at line 90  | Part of the Compile Code workspace or language/tool detection. Evidence: Code support is available from the rich editor right-click menu. Use <b>Code block</b> or <b>Paste as code</b> for C, C++, SystemVerilog, LTspice, Java, JavaScript, Python, and HTML. Code blocks store the selected language, pr                              |
| Compiler or simulator at line 92  | Part of the Compile Code workspace or language/tool detection. Evidence: Projects also include a separate <b>Compile Code</b> workspace. This is local builder tooling, not a portfolio text block. Each project behaves like a small IDE workspace: the left explorer lists the source files, the main                                  |
| Compiler or simulator at line 94  | Part of the Compile Code workspace or language/tool detection. Evidence: The Compile Code workspace has separate <b>Compile</b> , <b>Build project</b> , and <b>Run</b> controls. Verilog/SystemVerilog also show <b>Simulate</b> and <b>Show scope</b> . Compile checks or builds the active source. Build project use                  |
| Compiler or simulator at line 98  | Part of the Compile Code workspace or language/tool detection. Evidence: Verilog and SystemVerilog simulation requires a testbench. Mark HDL files as <b>Design</b> or <b>Testbench</b> , or use <b>Add testbench</b> to create a starter bench with <code>\$dumpfile</code> and <code>\$dumpvars</code> . HDL Compile can syntax-check/ |
| Compiler or simulator at line 106 | Part of the Compile Code workspace or language/tool detection. Evidence: For example, a SystemVerilog file attached to a design section might be saved as <code>docs\laser-link\design\filter.sv</code> . The upload status in the builder shows the generated relative path after a file is saved.                                      |
| Rich text at line 112             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Dark mode applies to builder panels, dialogs, guide windows, menus, status chips, and window/icon controls. Editing fields and selected project appearances keep their own colors so typed content, field formatting, and pr                        |
| Git command at line 116           | Repository state, publishing, branch checks, or release automation. Evidence: Publishing is controlled by Git repository write access. The builder does not embed owner passwords or private credentials.                                                                                                                                |
| Git command at line 124           | Repository state, publishing, branch checks, or release automation. Evidence: <a href="https://github.com/YOUR-USERNAME/YOUR-USERNAME.github.io.git">https://github.com/YOUR-USERNAME/YOUR-USERNAME.github.io.git</a>                                                                                                                    |
| Security or auth at line 127      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: 3. Click <b>Authenticate target</b> .                                                                                                                                                                                                             |
| Git command at line 128           | Repository state, publishing, branch checks, or release automation. Evidence: 4. Complete GitHub/Git Credential Manager sign-in if prompted.                                                                                                                                                                                             |
| Security or auth at line 132      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Successful authorization is cached locally per repository and branch for about one day. If the same target receives more than three successful authorizations in one week, the local trust window can extend to 30 days for                       |
| Git command at line 136           | Repository state, publishing, branch checks, or release automation. Evidence: Install Git for Windows, Node.js, and npm. Then run:                                                                                                                                                                                                       |
| Git command at line 139           | Repository state, publishing, branch checks, or release automation. Evidence: git clone <a href="https://github.com/otienomaurice/omb-portfolio-builder.git">https://github.com/otienomaurice/omb-portfolio-builder.git</a>                                                                                                              |
| Installer at line 156             | Windows setup, update, shortcut, or installation behavior. Evidence: npm run installer                                                                                                                                                                                                                                                   |
| Installer at line 168             | Windows setup, update, shortcut, or installation behavior. Evidence: When <code>main</code> receives a release-ready version, GitHub Actions builds the Windows installer and portable executable, creates or updates the <code>builder-v&lt;version&gt;</code> GitHub Release, and uploads stable <code>latest</code> asset names.      |
| Installer at line 186             | Windows setup, update, shortcut, or installation behavior. Evidence:   <code>main</code>   Consumer/release branch for finished installer and public app code.   Initial repository branch   None   Release-ready only                                                                                                                   |
| Rich text at line 192             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence:   <code>codex/manual-update-categories-preview</code>   Adds manual update checks, custom project categories, link-safe rich paste handling, project-only website previews, broad duplicate-install detection, and a main release workf             |

| Item                         | Explanation                                                                                                                                                                                                                                                                                                          |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security or auth at line 193 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence:   `codex/fix-publish-target-auth`   Makes publishing target setup authenticate before saving, resolves the target default branch before access checks, and blocks known local builder workspaces during installation.   `dev` |
| Security or auth at line 194 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence:   `codex/cache-daily-auth-and-restore-profile`   Adds daily and extended publishing authorization caching, clarifies publishing target fields, keeps active-machine profile fields populated, and enforces development-only   |
| Installer at line 195        | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/automatic-installer-update-flow`   Makes the update popup run the installer automatically, strengthens existing-install detection, and publishes stable latest installer asset names for downloads.   `development`                    |
| Installer at line 196        | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/split-builder-portfolio-workspaces`   Separates local builder files from publish-mirror files, syncs publish-safe assets, and bumps the next installer release version.   `development`   `development`   Active                       |
| Installer at line 197        | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/guard-latest-builder-download`   Keeps the stable latest installer asset URL aligned with release assets.   `development`   `development`   Active                                                                                     |
| Security or auth at line 198 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence:   `codex/secure-builder-editor-shell`   Adds native desktop menus, light app chrome, scoped publishing authorization cache, and smoother rich-editor copy, paste, cut, and image placement behavior.   `development`   `dev`  |
| Installer at line 200        | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/local-desktop-shortcut`   Forces installer-created shortcuts onto the physical Windows desktop instead of redirected OneDrive Desktop folders.   `development`   `development`   Active                                                |
| Installer at line 204        | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/fix-update-auto-restart`   Makes the app update flow quit the desktop app cleanly, run the installer, and relaunch the updated executable automatically.   `development`   `development`   Merged into `development`                   |
| Installer at line 205        | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/fix-update-launch-handoff`   Launches the updater through a detached Windows command handoff so the installer/relaunch script survives app shutdown.   `development`   `development`   Merged into `development`                       |
| Rich text at line 208        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence:   `codex/fix-editor-installer-update-flow`   Fixes rich-editor context menus, prevents pasted links from rendering as formulas, adds safe fullscreen exit behavior, and lets installers update in place when old uninstaller    |
| Installer at line 212        | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/skip-old-uninstaller-on-update`   Prevents in-app silent updates from launching stale uninstallers and adds an updater timeout for hung installers.   `development`   `development`   Active                                           |
| Security or auth at line 214 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence:   `codex/security-and-context-menu-reporting`   Improves text-field context menus, adds builder security/download reporting, and prepares publishable website security headers.   `development`   `development`   Active      |
| Rich text at line 215        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence:   `codex/uniform-rich-text-controls`   Makes right-click text formatting consistent for front-page, profile, and contact fields, persists field styles into previews and published portfolios, and expands common color choi    |
| Security or auth at line 216 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence:   `codex/code-block-preferences`   Adds code-block authoring with syntax highlighting, source/basic paste modes, parser rendering, and builder Preferences for light/dark mode.   `codex/rich-profile-frontpage-editors`   `` |
| Rich text at line 217        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence:   `codex/full-builder-dark-mode`   Expands builder dark mode across panels, dialogs, menus, status chips, and icon controls while preserving input-field colors and project appearance templates.   `codex/code-block-prefer    |
| Cloudflare or AI at line 218 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence:   `codex/per-user-update-migration`   Moves the installer/update model to the current user's app location and prefers per-user install records over stale machine-wide records.   `codex/full-builder-dark-mode`   `developm  |
| Installer at line 219        | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/appdata-workspace-root`   Standardizes future installs and managed builder/portfolio workspaces under AppData and migrates legacy OMB app-folder installs forward.   `codex/per-user-update-migration`   `developme                    |
| Installer at line 220        | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/project-code-compile-workspace`   Adds a project-local Compile Code workspace with source import, beautify, compiler detection, save, and terminal-style compile/run output.   `codex/appdata-workspace-root`   `de                    |
| Cloudflare or AI at line 221 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence:   `codex/fix-public-ai-worker-endpoint`   Adds the missing Cloudflare Worker backend files and wires the generated website to `api/portfolio-ai`.   `development`   `development`   Active                                    |
| Cloudflare or AI at line 222 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence:   `codex/fix-ai-worker-cloudflare-fallback`   Lets the website AI answer through Cloudflare Workers AI when no OpenAI Worker secret is configured.   `development`   `development`   Active                                   |

| Item                              | Explanation                                                                                                                                                                                                                                                                                                           |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installer at line 223             | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/bump-builder-version-0.2.26`   Bumps the builder release version so the next approved `development` to `main` release creates a fresh installer tag.   `development`   `development`   Active                                           |
| Compiler or simulator at line 224 | Part of the Compile Code workspace or language/tool detection. Evidence:   `codex/speed-up-compile-runner`   Speeds compiler detection, closes stdin cleanly, reports elapsed compile/run times, shows generated binaries/classes, and wires Icarus Verilog for SystemVerilog simulation.   `development`             |
| Compiler or simulator at line 225 | Part of the Compile Code workspace or language/tool detection. Evidence:   `codex/fix-node-compiler-detection`   Prevents the packaged Electron executable from being mistaken for Node.js in the Compile Code workspace.   `development`   `development`   Active                                                    |
| Compiler or simulator at line 226 | Part of the Compile Code workspace or language/tool detection. Evidence:   `codex/optimize-code-run-cache`   Adds cached compiled artifacts and a separate Rebuild / run command so repeated Java, C, C++, and SystemVerilog runs do not recompile unchanged source.   `development`   `development`                  |
| Compiler or simulator at line 227 | Part of the Compile Code workspace or language/tool detection. Evidence:   `codex/add-verilog-simulator`   Adds Verilog as a first-class compile/simulate language using Icarus Verilog and vvp, with Verilog-specific highlighting, file detection, and cleaner simulator output.   `development`                    |
| Installer at line 231             | Windows setup, update, shortcut, or installation behavior. Evidence:   `codex/expand-complete-guide-detail`   Expands the complete guide with deeper flow walkthroughs, API endpoint details, data contracts, caching detail, installer internals, generated function inventory, and additional b                     |
| Security or auth at line 232      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence:   `codex/hdl-testbench-scope`   Requires HDL simulations to include a testbench, adds HDL design/testbench roles, and renders VCD waveform data in a signal scope.   `development`   `development`   Merged into `development` |
| Security or auth at line 233      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence:   `codex/builder-file-ai-auth-docs-cleanup`   Allows all project evidence file types, cleans publishing target authentication/loading, improves the website AI fallback, adds IDE-style compile workspace controls, and expa   |
| Installer at line 236             | Windows setup, update, shortcut, or installation behavior. Evidence: ## Uninstall                                                                                                                                                                                                                                     |
| Git command at line 238           | Repository state, publishing, branch checks, or release automation. Evidence: Uninstall from Windows **Installed apps** / **Apps & features**. The uninstaller removes OMB Portfolio Builder, its shortcuts, and local app data when Windows requests app-data cleanup. It does not silently remove shared            |

## Representative opening snippet

```

1: # OMB Portfolio Builder
2:
3: OMB Portfolio Builder is a Windows desktop application for creating, editing, previewing, and publishing
a portfolio site from a local workspace. A fresh install starts blank: users add their own profile, sections,
projects, files, images, links, resumes, appearance choices, and publishing target.
4:
5: This README is for the builder application only. Website-specific content, domain notes, public page
layout, and generated portfolio assets belong in the portfolio site's own README.
6:
7: ## Complete Beginner Guide
8:
9: A long beginner-oriented Word guide and a deeper source-code reference are included at:
10:
11: ```text
12: docs/OMB_Portfolio_Builder_Complete_Guide.docx
13: docs/OMB_Portfolio_Builder_Code_Reference.docx
14: docs/OMB_Portfolio_Builder_Code_Reference.pdf
15: ```
16:
17: It explains the high-level design, generated block diagrams, software-engineering decisions, shell
commands, programming syntax, how the important files communicate, what each build tool owns, GitHub workflows,
Electron, the installer, caching, API endpoints, data contracts, frontend/backend/AI/Cloudflare layers, the
website, the parser, publishing, compiler tools, generated function inventories, and file-by-file repository
notes. The generator also writes source-focused Word documents to `docs/code-reference-docx` and PDF documents
to `docs/code-reference-pdf`.
18:
19: ## Install
20:
21: Download the latest Windows installer or portable executable from:
22:
23: ```text
24: https://github.com/otienomaurice/omb-portfolio-builder/releases/latest
25: ```
26:
27: Use:
28:
29: - `OMB-Portfolio-Builder-Setup-latest-x64.exe` for the Windows installer.
30: - `OMB-Portfolio-Builder-Portable-latest-x64.exe` when you want to run without installing.
31:
32: The normal installer default is a per-user AppData location:
33:
34: ```text
35: C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder

```

36: ````  
37:  
38: The installed application executable is:  
39:  
40: ````text  
41: C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder\OMB Portfolio Builder.exe  
42: ````  
43:  
44: The installer uses a Windows wizard with Back, Next, Cancel, progress, and Finish. It can create Desktop and Start Menu shortcuts, and it writes a normal Windows uninstall entry. The default per-user install does not write into `C:\Program Files`, so normal app updates can run without a Windows administrator prompt.  
45:  
46: Installed users do not need Node.js or pnpm. The packaged desktop app includes its runtime. Git for Windows with Git Credential Manager is needed only for publishing to a Git repository. If Git or Git Credential Manager is missing, setup can install Git for Windows; compatible existing tools are skipped rather than removed.  
47:  
48: ## Updates  
49:  
50: The app checks GitHub Releases for newer builder versions. When a newer release exists, the app shows an update dialog. Choosing **Update** downloads the installer, closes the running builder, updates the existing installation in place, and reopens the updated app.  
51:  
52: Setup treats an existing install as an update target, not a second installation. It checks the current user's install first, then legacy machine-wide records, uses the detected install location for updates, avoids creating duplicate builder copies, and stops when the installed version is already current or newer.  
53:  
54: Older releases remain available at:  
55:  
56: ````text  
57: <https://github.com/otienomaurice/omb-portfolio-builder/releases>  
58: ````  
59:  
60: ## Local Workspaces  
61:  
62: On Windows, the app keeps application files and editable content under AppData:  
63:  
64: - `C:\Users\<you>\AppData\Local\Programs\OMB Portfolio Builder` stores the installed desktop application.  
65: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\builder` stores the builder workspace, local drafts, uploaded files, templates, previews, and editing tools.  
66: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\portfolio` stores the publish mirror used by Git-based publishing.  
67: - `C:\Users\<you>\AppData\Local\OMB Portfolio Builder\compile-code` stores project compile-workspace source files and run folders created by the Compile Code tool.  
68:  
69: The installer and updater treat the AppData builder and portfolio folders as managed workspaces, not duplicate app installs.  
70:  
71: The builder works offline for editing, project creation, file attachment, local previews, and draft saves. Publishing requires network access and repository write permission.  
72:

# Source File: build/installer.nsh

| Item                | Explanation                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------|
| File                | build/installer.nsh                                                                                                    |
| Purpose             | NSIS installer customization script for update behavior, shortcut placement, old install detection, and setup details. |
| Lines               | 570                                                                                                                    |
| Size                | 29,416 bytes                                                                                                           |
| Talks to            | local backend, GitHub/release layer                                                                                    |
| Functions           | 14                                                                                                                     |
| Variables/constants | 0                                                                                                                      |
| API mentions        | 0                                                                                                                      |
| Signals             | 104                                                                                                                    |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                   | Explanation                                                                                                                                                                                                                                                                                                              |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installer at line 6    | Windows setup, update, shortcut, or installation behavior. Evidence: <code>!ifndef BUILD_UNINSTALLER</code>                                                                                                                                                                                                              |
| Installer at line 8    | Windows setup, update, shortcut, or installation behavior. Evidence: <code>Var OMB_ShortcutCheckbox</code>                                                                                                                                                                                                               |
| Installer at line 10   | Windows setup, update, shortcut, or installation behavior. Evidence: <code>Var OMB_CreateDesktopShortcutState</code>                                                                                                                                                                                                     |
| Installer at line 24   | Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${NSD_CreateLabel} 0 0 100% 12u "Tools and shortcuts"</code>                                                                                                                                                                                 |
| Git command at line 27 | Repository state, publishing, branch checks, or release automation. Evidence: <code>\${NSD_CreateLabel} 0 18u 100% 36u "The desktop app includes its own runtime. Users do not need to install Node.js or npm.\$\r\nGit for Windows with Git Credential Manager is only needed when publishing to GitHub."</code>        |
| Installer at line 30   | Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${NSD_CreateCheckbox} 0 62u 100% 12u "Create a desktop shortcut"</code>                                                                                                                                                                      |
| Installer at line 31   | Windows setup, update, shortcut, or installation behavior. Evidence: <code>Pop \$OMB_ShortcutCheckbox</code>                                                                                                                                                                                                             |
| Installer at line 32   | Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${If} \$OMB_CreateDesktopShortcutState == ""</code>                                                                                                                                                                                          |
| Installer at line 33   | Windows setup, update, shortcut, or installation behavior. Evidence: <code>StrCpy \$OMB_CreateDesktopShortcutState \${BST_CHECKED}</code>                                                                                                                                                                                |
| Installer at line 35   | Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${If} \$OMB_CreateDesktopShortcutState == \${BST_CHECKED}</code>                                                                                                                                                                             |
| Installer at line 36   | Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${NSD_Check} \$OMB_ShortcutCheckbox</code>                                                                                                                                                                                                   |
| Git command at line 39 | Repository state, publishing, branch checks, or release automation. Evidence: <code>\${NSD_CreateCheckbox} 0 82u 100% 24u "Check publishing tools and install Git for Windows if Git/Git Credential Manager is missing"</code>                                                                                           |
| Git command at line 48 | Repository state, publishing, branch checks, or release automation. Evidence: <code>\${NSD_CreateLabel} 0 114u 100% 38u "Existing shared tools are not uninstalled silently. If a compatible tool is already available, setup skips it. If Git is missing or unusable, setup will try to install Git for Windows"</code> |
| Installer at line 55   | Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${NSD_GetState} \$OMB_ShortcutCheckbox \$OMB_CreateDesktopShortcutState</code>                                                                                                                                                               |
| Git command at line 66 | Repository state, publishing, branch checks, or release automation. Evidence: <code>IfExists "\$PROGRAMFILES64\Git\cmd\git.exe" 0 +2</code>                                                                                                                                                                              |
| Git command at line 68 | Repository state, publishing, branch checks, or release automation. Evidence: <code>IfExists "\$PROGRAMFILES\Git\cmd\git.exe" 0 +2</code>                                                                                                                                                                                |
| Git command at line 70 | Repository state, publishing, branch checks, or release automation. Evidence: <code>IfExists "\$PROGRAMFILES32\Git\cmd\git.exe" 0 +2</code>                                                                                                                                                                              |

| Item                         | Explanation                                                                                                                                                                                                                                                                                               |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Git command at line 72       | Repository state, publishing, branch checks, or release automation. Evidence: IfFileExists "\$LOCALAPPDATA\Programs\Git\cmd\git.exe" 0 +2                                                                                                                                                                 |
| Git command at line 76       | Repository state, publishing, branch checks, or release automation. Evidence: nsExec::ExecToStack 'cmd /c git --version'                                                                                                                                                                                  |
| Security or auth at line 86  | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Function OMBCheckGitCredentialManagerAvailable                                                                                                                                                                     |
| Git command at line 87       | Repository state, publishing, branch checks, or release automation. Evidence: nsExec::ExecToStack 'cmd /c git credential-manager --version'                                                                                                                                                               |
| Security or auth at line 100 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Call OMBCheckGitCredentialManagerAvailable                                                                                                                                                                         |
| Git command at line 105      | Repository state, publishing, branch checks, or release automation. Evidence: DetailPrint "Git for Windows and Git Credential Manager were found. Skipping publishing tool installation."                                                                                                                 |
| Git command at line 114      | Repository state, publishing, branch checks, or release automation. Evidence: MessageBox MB_OK MB_ICONINFORMATION "OMB Portfolio Builder was installed, but Git for Windows was not found and Windows Package Manager is unavailable. Open Publishing target > Install Git inside the app, or install Git |
| Git command at line 118      | Repository state, publishing, branch checks, or release automation. Evidence: DetailPrint "Installing or repairing Git for Windows. This can take several minutes."                                                                                                                                       |
| Git command at line 119      | Repository state, publishing, branch checks, or release automation. Evidence: ExecWait 'winget install --id Git.Git -e --source winget --accept-source-agreements --accept-package-agreements' \$0                                                                                                        |
| Git command at line 121      | Repository state, publishing, branch checks, or release automation. Evidence: MessageBox MB_OK MB_ICONINFORMATION "OMB Portfolio Builder was installed, but Git for Windows setup did not complete. Open Publishing target > Install Git inside the app when you are ready to publish."                   |
| Git command at line 123      | Repository state, publishing, branch checks, or release automation. Evidence: DetailPrint "Git for Windows setup completed."                                                                                                                                                                              |
| Installer at line 149        | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONINFORMATION "OMB Portfolio Builder \$OMB_ExistingInstallVersion is already installed.\$r\$n\$r\$nThis installer is also version \${VERSION}, so setup will stop without changing the app."                   |
| Installer at line 157        | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONINFORMATION "OMB Portfolio Builder \$OMB_ExistingInstallVersion is already installed.\$r\$n\$r\$nThis installer is older (\${VERSION}), so setup will stop without changing the app."                        |
| Installer at line 161        | Windows setup, update, shortcut, or installation behavior. Evidence: DetailPrint "OMB Portfolio Builder \$OMB_ExistingInstallVersion is newer than installer \${VERSION}. Setup stopped."                                                                                                                 |
| Installer at line 168        | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder workspace:\$r\$n\$r\$n\$PROFILE\OMB\builder\desktop-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing this releas        |
| Installer at line 172        | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder workspace:\$r\$n\$r\$n\$DOCUMENTS\OMB\desktop-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing this release. Thi        |
| Installer at line 176        | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder workspace:\$r\$n\$r\$n\$PROFILE\OneDrive\Documents\OMB\desktop-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing         |
| Installer at line 180        | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder workspace:\$r\$n\$r\$n\$PROFILE\Documents\OMB\desktop-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing this rele        |
| Installer at line 184        | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing OMB Portfolio Builder development copy:\$r\$n\$r\$n\$DOCUMENTS\omb-portfolio-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing this release."        |
| Installer at line 188        | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing OMB Portfolio Builder development copy:\$r\$n\$r\$n\$PROFILE\OneDrive\Documents\omb-portfolio-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installi        |
| File read at line 208        | Reads local builder state, project files, website assets, or configuration. Evidence: FileWrite \$0 " try { \$\$packageText = Get-Content -LiteralPath (Join-Path \$\$Directory 'package.json') -Raw -ErrorAction Stop } catch { return \$\$false }\$r\$n"                                                |



| Item                    | Explanation                                                                                                                                                                                                                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| File read at line 215   | Reads local builder state, project files, website assets, or configuration. Evidence: FileWrite \$0 " try { \$\$packageText = Get-Content -LiteralPath (Join-Path \$\$Directory 'package.json') -Raw -ErrorAction Stop } catch { return \$\$false }\$r\$n"                                                    |
| Installer at line 220   | Windows setup, update, shortcut, or installation behavior. Evidence: FileWrite \$0 " \$\$managedRoot = Join-Path \$Env:LOCALAPPDATA 'OMB Portfolio Builder'\$r\$n"                                                                                                                                            |
| Git command at line 227 | Repository state, publishing, branch checks, or release automation. Evidence: FileWrite \$0 " if (\$\$name -in @('node_modules','.git','dist','.vs',\$\$Recycle.Bin','System Volume Information',WinSxS)) { return \$\$true }\$r\$n"                                                                          |
| Installer at line 251   | Windows setup, update, shortcut, or installation behavior. Evidence: FileWrite \$0 "\$\$registryKeys = @(('HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall\*', 'HKCU:\Software\Microsoft\Windows\CurrentVersion\Uninstall\*'))\$r\$n"                                                               |
| Installer at line 261   | Windows setup, update, shortcut, or installation behavior. Evidence: FileWrite \$0 "foreach (\$\$common in @(\$\$env:ProgramFiles, \$\$programFilesX86, (Join-Path \$Env:LOCALAPPDATA 'Programs'), \$Env:USERPROFILE, (Join-Path \$Env:PUBLIC 'Desktop')) { if (\$\$common -and -not \$\$roots.Contains(\$\$c |
| Installer at line 282   | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found another OMB Portfolio Builder copy on this machine.\$r\$n\$r\$n\$R3\$r\$n\$r\$nTo prevent duplicate installations and stale builder features, remove, uninstall, or rename the exis            |
| Installer at line 285   | Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup could not complete the duplicate-installation scan. To avoid duplicate builder copies, setup will stop. Run the installer again as Administrator, or remove old OMB Portfolio Builder co             |
| Installer at line 298   | Windows setup, update, shortcut, or installation behavior. Evidence: Function OMBDisableBuiltinOldUninstallerForInPlaceUpdate                                                                                                                                                                                 |
| Installer at line 299   | Windows setup, update, shortcut, or installation behavior. Evidence: DetailPrint "Preparing in-place update. The old uninstaller handoff will be skipped."                                                                                                                                                    |
| Installer at line 302   | Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "UninstallString"                                                                                                                        |
| Installer at line 303   | Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "QuietUninstallString"                                                                                                                   |
| Installer at line 304   | Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "UninstallString"                                                                                                                        |
| Installer at line 305   | Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "QuietUninstallString"                                                                                                                   |
| Installer at line 308   | Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "UninstallString"                                                                                                                        |
| Installer at line 309   | Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "QuietUninstallString"                                                                                                                   |
| Installer at line 310   | Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "UninstallString"                                                                                                                        |
| Installer at line 311   | Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "QuietUninstallString"                                                                                                                   |
| Installer at line 315   | Windows setup, update, shortcut, or installation behavior. Evidence: Function OMBUninstallExistingInstallIfPresent                                                                                                                                                                                            |
| Installer at line 326   | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R6 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" DisplayVersion                                                                                                                          |
| Installer at line 327   | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" QuietUninstallString                                                                                                                    |
| Installer at line 329   | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" UninstallString                                                                                                                         |
| Installer at line 332   | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R7 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" InstallLocation                                                                                                                         |
| Installer at line 337   | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R6 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" DisplayVersion                                                                                                                          |
| Installer at line 338   | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" QuietUninstallString                                                                                                                    |
| Installer at line 340   | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" UninstallString                                                                                                                         |
| Installer at line 343   | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R7 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" InstallLocation                                                                                                                         |

| Item                  | Explanation                                                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installer at line 350 | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R6 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" DisplayVersion       |
| Installer at line 351 | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" QuietUninstallString |
| Installer at line 353 | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" UninstallString      |
| Installer at line 356 | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R7 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" InstallLocation      |
| Installer at line 360 | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R6 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" DisplayVersion       |
| Installer at line 361 | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" QuietUninstallString |
| Installer at line 363 | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" UninstallString      |
| Installer at line 366 | Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R7 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" InstallLocation      |
| Installer at line 373 | Windows setup, update, shortcut, or installation behavior. Evidence: IfFileExists "\$LOCALAPPDATA\Programs\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3                 |
| Installer at line 374 | Windows setup, update, shortcut, or installation behavior. Evidence: StrCpy \$R7 "\$LOCALAPPDATA\Programs\OMB Portfolio Builder"                                                           |
| Installer at line 375 | Windows setup, update, shortcut, or installation behavior. Evidence: StrCpy \$R5 "\$LOCALAPPDATA\Programs\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" /S'                   |
| Installer at line 378 | Windows setup, update, shortcut, or installation behavior. Evidence: IfFileExists "\$PROGRAMFILES64\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3                        |
| Installer at line 380 | Windows setup, update, shortcut, or installation behavior. Evidence: StrCpy \$R5 "\$PROGRAMFILES64\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" /S'                          |
| Installer at line 383 | Windows setup, update, shortcut, or installation behavior. Evidence: IfFileExists "\$PROGRAMFILES\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3                          |
| Installer at line 385 | Windows setup, update, shortcut, or installation behavior. Evidence: StrCpy \$R5 "\$PROGRAMFILES\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" /S'                            |
| Installer at line 440 | Windows setup, update, shortcut, or installation behavior. Evidence: DetailPrint "Migrating legacy OMB application install to the AppData per-user install location."                      |

## Function-by-function detail

### *OMBToolsPageCreate - lines 17-52*

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 17-52 (36 source lines).                                                                                                                                                                        |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBToolsPageCreate                                                                                                                                                                           |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

17: Function OMBToolsPageCreate
18:   nsDialogs::Create 1018
19:   Pop $OMB_ToolsDialog

```

```

20:     ${If} $OMB_ToolsDialog == error
21:         Abort
22:     ${EndIf}
23:
24:     ${NSD_CreateLabel} 0 0 100% 12u "Tools and shortcuts"
25:     Pop $0
26:
27:     ${NSD_CreateLabel} 0 18u 100% 36u "The desktop app includes its own runtime. Users do not need to
install Node.js or npm.$\r$\nGit for Windows with Git Credential Manager is only needed when publishing to
GitHub."
28:     Pop $0
29:
30:     ${NSD_CreateCheckbox} 0 62u 100% 12u "Create a desktop shortcut"
31:     Pop $OMB_ShortcutCheckbox
32:     ${If} $OMB_CreateDesktopShortcutState == ""
33:         StrCpy $OMB_CreateDesktopShortcutState ${BST_CHECKED}
34:     ${EndIf}
35:     ${If} $OMB_CreateDesktopShortcutState == ${BST_CHECKED}
36:         ${NSD_Check} $OMB_ShortcutCheckbox
37:     ${EndIf}
38:
39:     ${NSD_CreateCheckbox} 0 82u 100% 24u "Check publishing tools and install Git for Windows if Git/Git
Credential Manager is missing"
40:     Pop $OMB_PublishingToolsCheckbox
41:     ${If} $OMB_InstallPublishingToolsState == ""
42:         StrCpy $OMB_InstallPublishingToolsState ${BST_CHECKED}
43:     ${EndIf}
44:     ${If} $OMB_InstallPublishingToolsState == ${BST_CHECKED}
45:         ${NSD_Check} $OMB_PublishingToolsCheckbox
46:     ${EndIf}
47:
48:     ${NSD_CreateLabel} 0 114u 100% 38u "Existing shared tools are not uninstalled silently. If a compatible
tool is already available, setup skips it. If Git is missing or unusable, setup will try to install Git for
Windows side-by-side using Windows Package Manager."
49:     Pop $0
50:
51:     nsDialogs::Show
52: FunctionEnd

```

## OMBToolsPageLeave - lines 54-62

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 54-62 (9 source lines).                                                                                                                                                                         |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBToolsPageLeave                                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

54: Function OMBToolsPageLeave
55:     ${NSD_GetState} $OMB_ShortcutCheckbox $OMB_CreateDesktopShortcutState
56:     ${NSD_GetState} $OMB_PublishingToolsCheckbox $OMB_InstallPublishingToolsState
57:     ${If} $OMB_IsUpdateInstall == "1"
58:     ${AndIf} $INSTDIR != $OMB_ExistingInstallLocation
59:         MessageBox MB_OK|MB_ICONSTOP "OMB Portfolio Builder is already installed on this
machine.$\r$\n$\r$\nUpdates must use the current installed
location:$\r$\n$OMB_ExistingInstallLocation$\r$\n$\r$\nSetup will not create another installed copy in a
different directory."
60:         Abort
61:     ${EndIf}
62: FunctionEnd

```

## OMBCheckGitAvailable - lines 64-84

| Item            | Explanation                                                            |
|-----------------|------------------------------------------------------------------------|
| Source location | Lines 64-84 (21 source lines).                                         |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | Function OMBCheckGitAvailable                                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

64: Function OMBCheckGitAvailable
65:   StrCpy $R9 "0"
66:   IfFileExists "$PROGRAMFILES64\Git\cmd\git.exe" 0 +2
67:   StrCpy $R9 "1"
68:   IfFileExists "$PROGRAMFILES\Git\cmd\git.exe" 0 +2
69:   StrCpy $R9 "1"
70:   IfFileExists "$PROGRAMFILES32\Git\cmd\git.exe" 0 +2
71:   StrCpy $R9 "1"
72:   IfFileExists "$LOCALAPPDATA\Programs\Git\cmd\git.exe" 0 +2
73:   StrCpy $R9 "1"
74:
75:   ${If} $R9 == "0"
76:     nsExec::ExecToStack 'cmd /c git --version'
77:     Pop $0
78:     Pop $1
79:     ${If} $0 == 0
80:       StrCpy $R9 "1"
81:     ${EndIf}
82:   ${EndIf}
83:   Push $R9
84: FunctionEnd

```

## ***OMBCheckGitCredentialManagerAvailable - lines 86-95***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 86-95 (10 source lines).                                                                                                                                                                        |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | Function OMBCheckGitCredentialManagerAvailable                                                                                                                                                        |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

86: Function OMBCheckGitCredentialManagerAvailable
87:   nsExec::ExecToStack 'cmd /c git credential-manager --version'
88:   Pop $0
89:   Pop $1
90:   ${If} $0 == 0
91:     Push "1"
92:   ${Else}
93:     Push "0"
94:   ${EndIf}
95: FunctionEnd

```

## ***OMBInstallGitIfNeeded - lines 97-125***

| Item            | Explanation                     |
|-----------------|---------------------------------|
| Source location | Lines 97-125 (29 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | Function OMBInstallGitIfNeeded                                                                                                                                                                        |
| Touches            | filesystem, network/API requests, authentication/security data                                                                                                                                        |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

97: Function OMBInstallGitIfNeeded
98:   Call OMBCheckGitAvailable
99:   Pop $R9
100:  Call OMBCheckGitCredentialManagerAvailable
101:  Pop $R8
102:
103:   ${If} $R9 == "1"
104:   ${AndIf} $R8 == "1"
105:   DetailPrint "Git for Windows and Git Credential Manager were found. Skipping publishing tool
installation."
106:   Return
107:   ${EndIf}
108:
109:   DetailPrint "Publishing tools are missing or incomplete. Checking Windows Package Manager."
110:   nsExec::ExecToStack 'where.exe winget'
111:   Pop $0
112:   Pop $1
113:   ${If} $0 != 0
114:   MessageBox MB_OK|MB_ICONINFORMATION "OMB Portfolio Builder was installed, but Git for Windows was not
found and Windows Package Manager is unavailable. Open Publishing target > Install Git inside the app, or
install Git for Windows with Git Credential Manager from https://git-scm.com/download/win."
115:   Return
116:   ${EndIf}
117:
118:   DetailPrint "Installing or repairing Git for Windows. This can take several minutes."
119:   ExecWait 'winget install --id Git.Git -e --source winget --accept-source-agreements --accept-package-
agreements' $0
120:   ${If} $0 != 0
121:   MessageBox MB_OK|MB_ICONINFORMATION "OMB Portfolio Builder was installed, but Git for Windows setup
did not complete. Open Publishing target > Install Git inside the app when you are ready to publish."
122:   ${Else}
123:   DetailPrint "Git for Windows setup completed."
124:   ${EndIf}
125: FunctionEnd

```

## OMBReadExistingAppVersion - lines 127-139

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 127-139 (13 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBReadExistingAppVersion                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Write                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

127: Function OMBReadExistingAppVersion
128:   StrCpy $OMB_ExistingInstallVersion ""
129:   IfFileExists "$R7\OMB Portfolio Builder.exe" omb_read_existing_app_version 0
130:   Return

```

```

131:
132:     omb_read_existing_app_version:
133:     nsExec::ExecToStack 'powershell.exe -NoProfile -ExecutionPolicy Bypass -Command "$$p = '$R7\OMB
Portfolio Builder.exe'; try { $$v = (Get-Item -LiteralPath $$p).VersionInfo.FileVersion; if (-not $$v) { $$v =
(Get-Item -LiteralPath $$p).VersionInfo.ProductVersion }; [Console]::Write($$v) } catch { exit 1 }"'
134:     Pop $R4
135:     Pop $R3
136:     ${If} $R4 == 0
137:         StrCpy $OMB_ExistingInstallVersion $R3
138:     ${EndIf}
139: FunctionEnd

```

## OMBStopIfExistingInstallIsCurrent - lines 141-164

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 141-164 (24 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBStopIfExistingInstallIsCurrent                                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | older                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

141: Function OMBStopIfExistingInstallIsCurrent
142:     ${If} $OMB_ExistingInstallVersion == ""
143:         Return
144:     ${EndIf}
145:
146:     ${VersionCompare} "$OMB_ExistingInstallVersion" "${VERSION}" $R4
147:     ${If} $R4 == 0
148:         IfSilent omb_existing_install_current_silent
149:         MessageBox MB_OK|MB_ICONINFORMATION "OMB Portfolio Builder $OMB_ExistingInstallVersion is already
installed.$\r$\n$\r$\nThis installer is also version ${VERSION}, so setup will stop without changing the app."
150:         Quit
151:
152:         omb_existing_install_current_silent:
153:         DetailPrint "OMB Portfolio Builder $OMB_ExistingInstallVersion is already installed. Setup stopped
because the installed version is current."
154:         Quit
155:     ${ElseIf} $R4 == 1
156:         IfSilent omb_existing_install_newer_silent
157:         MessageBox MB_OK|MB_ICONINFORMATION "OMB Portfolio Builder $OMB_ExistingInstallVersion is already
installed.$\r$\n$\r$\nThis installer is older (${VERSION}), so setup will stop without changing the app."
158:         Quit
159:
160:         omb_existing_install_newer_silent:
161:         DetailPrint "OMB Portfolio Builder $OMB_ExistingInstallVersion is newer than installer ${VERSION}.
Setup stopped."
162:         Quit
163:     ${EndIf}
164: FunctionEnd

```

## OMBCheckKnownBuilderWorkspace - lines 166-190

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 166-190 (25 source lines).                                                                       |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | Function OMBCheckKnownBuilderWorkspace                                                                 |
| Touches         | Mostly local calculations or local UI state.                                                           |
| Calls detected  | No obvious named calls detected by the generator.                                                      |
| API endpoints   | None detected inside this function body.                                                               |
| Storage keys    | No local/session storage keys detected.                                                                |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

166: Function OMBCheckKnownBuilderWorkspace
167:   IfFileExists "$PROFILE\OMB\builder\desktop-builder\package.json" 0 +3
168:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder
workspace:$r$\n$r$\n$PROFILE\OMB\builder\desktop-builder$r$\n$r$\nRemove, uninstall, or rename that copy
before installing this release. This prevents an older working builder from staying on the machine beside the
installed app."
169:   Quit
170:
171:   IfFileExists "$DOCUMENTS\OMB\desktop-builder\package.json" 0 +3
172:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder
workspace:$r$\n$r$\n$DOCUMENTS\OMB\desktop-builder$r$\n$r$\nRemove, uninstall, or rename that copy before
installing this release. This prevents an older working builder from staying on the machine beside the installed
app."
173:   Quit
174:
175:   IfFileExists "$PROFILE\OneDrive\Documents\OMB\desktop-builder\package.json" 0 +3
176:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder
workspace:$r$\n$r$\n$PROFILE\OneDrive\Documents\OMB\desktop-builder$r$\n$r$\nRemove, uninstall, or rename
that copy before installing this release. This prevents an older working builder from staying on the machine
beside the installed app."
177:   Quit
178:
179:   IfFileExists "$PROFILE\Documents\OMB\desktop-builder\package.json" 0 +3
180:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder
workspace:$r$\n$r$\n$PROFILE\Documents\OMB\desktop-builder$r$\n$r$\nRemove, uninstall, or rename that copy
before installing this release. This prevents an older working builder from staying on the machine beside the
installed app."
181:   Quit
182:
183:   IfFileExists "$DOCUMENTS\omb-portfolio-builder\package.json" 0 +3
184:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing OMB Portfolio Builder development
copy:$r$\n$r$\n$DOCUMENTS\omb-portfolio-builder$r$\n$r$\nRemove, uninstall, or rename that copy before
installing this release."
185:   Quit
186:
187:   IfFileExists "$PROFILE\OneDrive\Documents\omb-portfolio-builder\package.json" 0 +3
188:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing OMB Portfolio Builder development
copy:$r$\n$r$\n$PROFILE\OneDrive\Documents\omb-portfolio-builder$r$\n$r$\nRemove, uninstall, or rename that
copy before installing this release."
189:   Quit
190: FunctionEnd

```

## OMBWriteDuplicateScanScript - lines 192-272

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 192-272 (81 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBWriteDuplicateScanScript                                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | Hit, IsNullOrEmpty, Contains, Add, DesktopBuilder, not, LiteralPath, BuilderWorkspace, SkipDirectory, Windows, Tree, Push                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 17 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

192: Function OMBWriteDuplicateScanScript
193:   InitPluginsDir
194:   FileOpen $0 "$PLUGINSDIR\omb-duplicate-scan.ps1" w
195:   FileWrite $0 "$$ErrorActionPreference = 'SilentlyContinue'$r$\n"
196:   FileWrite $0 "$$hits = New-Object System.Collections.Generic.List[string]$r$\n"
197:   FileWrite $0 "function Add-Hit([string]$$Kind, [string]$$Candidate) {$r$\n"
198:   FileWrite $0 "  if ([string]::IsNullOrEmpty($$Candidate)) { return }$r$\n"
199:   FileWrite $0 "  try { $$resolved = (Resolve-Path -LiteralPath $$Candidate -ErrorAction Stop).Path }
catch { $$resolved = $$Candidate }$r$\n"

```

```

200: FileWrite $0 " if ($$resolved -match '\\\\OMB-Portfolio-Builder-Setup-([\\]\\]+\\.exe$') { return
}$r$`n"
201: FileWrite $0 " $$line = $$Kind + ' - ' + $$resolved$r$`n"
202: FileWrite $0 " if (-not $$hits.Contains($$line)) { [void]$$hits.Add($$line) }$r$`n"
203: FileWrite $0 "}$r$`n"
204: FileWrite $0 "function Test-DesktopBuilder([string]$$Directory) {$r$`n"
205: FileWrite $0 " if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'package.json'))) { return
$$false }$r$`n"
206: FileWrite $0 " if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'main.cjs'))) { return $$false
}$r$`n"
207: FileWrite $0 " if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'builder-site'))) { return
$$false }$r$`n"
208: FileWrite $0 " try { $$packageText = Get-Content -LiteralPath (Join-Path $$Directory 'package.json')
-Raw -ErrorAction Stop } catch { return $$false }$r$`n"
209: FileWrite $0 " return $$packageText -match 'OMB Portfolio Builder|omb-portfolio-
builder|portfoliobuilder'$r$`n"
210: FileWrite $0 "}$r$`n"
211: FileWrite $0 "function Test-BuilderWorkspace([string]$$Directory) {$r$`n"
212: FileWrite $0 " if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'package.json'))) { return
$$false }$r$`n"
213: FileWrite $0 " if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'main.cjs'))) { return $$false
}$r$`n"
214: FileWrite $0 " if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'server.mjs'))) { return
$$false }$r$`n"
215: FileWrite $0 " try { $$packageText = Get-Content -LiteralPath (Join-Path $$Directory 'package.json')
-Raw -ErrorAction Stop } catch { return $$false }$r$`n"
216: FileWrite $0 " return $$packageText -match 'OMB Portfolio Builder|omb-portfolio-
builder|portfoliobuilder'$r$`n"
217: FileWrite $0 "}$r$`n"
218: FileWrite $0 "function Should-SkipDirectory([string]$$Directory) {$r$`n"
219: FileWrite $0 " $$name = Split-Path -Leaf $$Directory$r$`n"
220: FileWrite $0 " $$managedRoot = Join-Path $Env:LOCALAPPDATA 'OMB Portfolio Builder'$r$`n"
221: FileWrite $0 " $$legacyRoot = Join-Path $Env:USERPROFILE 'OMB'$r$`n"
222: FileWrite $0 " $$managedBuilder = Join-Path $$managedRoot 'builder'$r$`n"
223: FileWrite $0 " $$managedPortfolio = Join-Path $$managedRoot 'portfolio'$r$`n"
224: FileWrite $0 " $$legacyBuilder = Join-Path $$legacyRoot 'builder'$r$`n"
225: FileWrite $0 " $$legacyPortfolio = Join-Path $$legacyRoot 'portfolio'$r$`n"
226: FileWrite $0 " if ($$Directory -ieq $$managedBuilder -or $$Directory -ieq $$managedPortfolio -or
$$Directory -ieq $$legacyBuilder -or $$Directory -ieq $$legacyPortfolio) { return $$true }$r$`n"
227: FileWrite $0 " if ($$name -in @('node_modules', '.git', 'dist', '.vs', '$$Recycle.Bin', 'System Volume
Information', 'WinSxS')) { return $$true }$r$`n"
228: FileWrite $0 " if ($$Directory -match '\\\\Windows(\\\\|$$)') { return $$true }$r$`n"
229: FileWrite $0 " return $$false$r$`n"
230: FileWrite $0 "}$r$`n"
231: FileWrite $0 "function Scan-Tree([string]$$Root) {$r$`n"
232: FileWrite $0 " if (-not (Test-Path -LiteralPath $$Root)) { return }$r$`n"
233: FileWrite $0 " $$stack = New-Object System.Collections.Generic.Stack[string>$r$`n"
234: FileWrite $0 " $$stack.Push($$Root)$r$`n"
235: FileWrite $0 " while ($$stack.Count -gt 0) {$r$`n"
236: FileWrite $0 "     $$dir = $$stack.Pop()$r$`n"
237: FileWrite $0 "     if (Should-SkipDirectory $$dir) { continue }$r$`n"
238: FileWrite $0 "     if ((Split-Path -Leaf $$dir) -ieq 'desktop-builder' -and (Test-DesktopBuilder $$dir))
{ Add-Hit 'Local desktop-builder workspace' $$dir }$r$`n"
239: FileWrite $0 "     if ((Split-Path -Leaf $$dir) -ieq 'builder' -and (Test-BuilderWorkspace $$dir)) {
Add-Hit 'Local builder workspace' $$dir }$r$`n"
240: FileWrite $0 "     try { $$children = Get-ChildItem -LiteralPath $$dir -Force -ErrorAction Stop } catch
{ continue }$r$`n"
241: FileWrite $0 "     foreach ($$child in $$children) {$r$`n"
242: FileWrite $0 "         if ($$child.PSIsContainer) {$r$`n"
243: FileWrite $0 "             if (-not (Should-SkipDirectory $$child.FullName)) {
$$stack.Push($$child.FullName) }$r$`n"
244: FileWrite $0 "             } else {$r$`n"
245: FileWrite $0 "                 if ($$child.Name -ieq 'OMB Portfolio Builder.exe') { Add-Hit 'Installed app
executable' $$child.FullName }$r$`n"
246: FileWrite $0 "                 if ($$child.Name -like 'OMB-Portfolio-Builder-Portable-*.exe') { Add-Hit
'Portable app executable' $$child.FullName }$r$`n"
247: FileWrite $0 "             }$r$`n"
248: FileWrite $0 "         }$r$`n"
249: FileWrite $0 "     }$r$`n"
250: FileWrite $0 " }$r$`n"
251: FileWrite $0 " $$registryKeys = @('HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall\*', 'HKCU:\S
oftware\Microsoft\Windows\CurrentVersion\Uninstall\*)$r$`n"
252: FileWrite $0 " foreach ($$key in $$registryKeys) {$r$`n"
253: FileWrite $0 "     Get-ItemProperty $$key -ErrorAction SilentlyContinue | Where-Object { $_.DisplayName
-like 'OMB Portfolio Builder*' } | ForEach-Object {$r$`n"
254: FileWrite $0 "         if ($$.InstallLocation) { Add-Hit 'Registered Windows installation'
$$.InstallLocation }$r$`n"
255: FileWrite $0 "         elseif ($$.DisplayIcon) { Add-Hit 'Registered Windows installation' $$.DisplayIcon
}$r$`n"
256: FileWrite $0 "     }$r$`n"
257: FileWrite $0 " }$r$`n"
258: FileWrite $0 " $$roots = New-Object System.Collections.Generic.List[string]$r$`n"
259: FileWrite $0 " Get-CimInstance Win32_LogicalDisk -Filter 'DriveType=3' -ErrorAction SilentlyContinue |
ForEach-Object { [void]$$roots.Add(($$.DeviceID + '\')) }$r$`n"
260: FileWrite $0 " $$programFilesX86 = [Environment]::GetEnvironmentVariable('ProgramFiles(x86)')$r$`n"
261: FileWrite $0 " foreach ($$common in @($$Env:ProgramFiles, $$programFilesX86, (Join-Path
$Env:LOCALAPPDATA 'Programs'), $Env:USERPROFILE, (Join-Path $Env:PUBLIC 'Desktop'))) { if ($$common -and -not
$$roots.Contains($$common)) { [void]$$roots.Add($$common) } }$r$`n"
262: FileWrite $0 " foreach ($$root in $$roots) { Scan-Tree $$root }$r$`n"
263: FileWrite $0 " $$unique = $$hits | Sort-Object -Unique$r$`n"
264: FileWrite $0 " if ($$unique.Count -gt 0) {$r$`n"
265: FileWrite $0 "     $$preview = $$unique | Select-Object -First 10$r$`n"

```



```

266:   FileWrite $0 "   if ($$unique.Count -gt 10) { $$preview += ('... and ' + ($$unique.Count - 10) + '
more') }$r$n"
267:   FileWrite $0 "   Write-Output ($$preview -join [Environment]::NewLine)$r$n"
268:   FileWrite $0 "   exit 42$r$n"
269:   FileWrite $0 "}$r$n"
270:   FileWrite $0 "exit 0$r$n"
271:   FileClose $0
272: FunctionEnd

```

## OMBScanForDuplicateBuilderCopies - lines 274-288

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 274-288 (15 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBScanForDuplicateBuilderCopies                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

274: Function OMBScanForDuplicateBuilderCopies
275:   DetailPrint "Scanning this machine for existing OMB Portfolio Builder copies."
276:   Call OMBWriteDuplicateScanScript
277:   nsExec::ExecToStack 'powershell.exe -NoProfile -ExecutionPolicy Bypass -File "$PLUGINDIR\omb-
duplicate-scan.ps1"'
278:   Pop $R4
279:   Pop $R3
280:
281:   ${If} $R4 == 42
282:     MessageBox MB_OK|MB_ICONSTOP "Setup found another OMB Portfolio Builder copy on this
machine.$r$n$R3$r$nTo prevent duplicate installations and stale builder features, remove,
uninstall, or rename the existing copy first, then run this installer again."
283:     Quit
284:   ${ElseIf} $R4 != 0
285:     MessageBox MB_OK|MB_ICONSTOP "Setup could not complete the duplicate-installation scan. To avoid
duplicate builder copies, setup will stop. Run the installer again as Administrator, or remove old OMB Portfolio
Builder copies manually before installing."
286:     Quit
287:   ${EndIf}
288: FunctionEnd

```

## OMBStopBuilderProcessesForUpdate - lines 290-296

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 290-296 (7 source lines).                                                                                                                                                                       |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBStopBuilderProcessesForUpdate                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

290: Function OMBStopBuilderProcessesForUpdate
291:   DetailPrint "Stopping running OMB Portfolio Builder processes before update."
292:   nsExec::ExecToStack 'cmd /c taskkill /IM "OMB Portfolio Builder.exe" /F /T'
293:   Pop $R4

```

```
294:   Pop $R3
295:   Sleep 1500
296: FunctionEnd
```

## ***OMBDisableBuiltInOldUninstallerForInPlaceUpdate - lines 298-313***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 298-313 (16 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBDisableBuiltInOldUninstallerForInPlaceUpdate                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
298: Function OMBDisableBuiltInOldUninstallerForInPlaceUpdate
299:   DetailPrint "Preparing in-place update. The old uninstaller handoff will be skipped."
300:
301:   SetRegView 64
302:   DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"UninstallString"
303:   DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"QuietUninstallString"
304:   DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"UninstallString"
305:   DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"QuietUninstallString"
306:
307:   SetRegView 32
308:   DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"UninstallString"
309:   DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"QuietUninstallString"
310:   DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"UninstallString"
311:   DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"QuietUninstallString"
312:   SetRegView lastused
313: FunctionEnd
```

## ***OMBUninstallExistingInstallIfPresent - lines 315-497***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 315-497 (183 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBUninstallExistingInstallIfPresent                                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
315: Function OMBUninstallExistingInstallIfPresent
316:   StrCpy $OMB_ExistingInstallLocation ""
317:   StrCpy $OMB_ExistingInstallVersion ""
318:   StrCpy $OMB_IsUpdateInstall "0"
319:   StrCpy $OMB_ManualApplicationInstall "0"
```

```

320: StrCpy $R7 ""
321: StrCpy $R6 ""
322: StrCpy $R5 ""
323:
324: SetRegView 64
325: ReadRegStr $R7 HKCU "Software\${APP_GUID}" InstallLocation
326: ReadRegStr $R6 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
DisplayVersion
327: ReadRegStr $R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
QuietUninstallString
328: ${If} $R5 == ""
329: ReadRegStr $R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
UninstallString
330: ${EndIf}
331: ${If} $R7 == ""
332: ReadRegStr $R7 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
InstallLocation
333: ${EndIf}
334:
335: ${If} $R7 == ""
336: ReadRegStr $R7 HKLM "Software\${APP_GUID}" InstallLocation
337: ReadRegStr $R6 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
DisplayVersion
338: ReadRegStr $R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
QuietUninstallString
339: ${If} $R5 == ""
340: ReadRegStr $R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
UninstallString
341: ${EndIf}
342: ${If} $R7 == ""
343: ReadRegStr $R7 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
InstallLocation
344: ${EndIf}
345: ${EndIf}
346:
347: ${If} $R7 == ""
348: SetRegView 32
349: ReadRegStr $R7 HKCU "Software\${APP_GUID}" InstallLocation
350: ReadRegStr $R6 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
DisplayVersion
351: ReadRegStr $R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
QuietUninstallString
352: ${If} $R5 == ""
353: ReadRegStr $R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
UninstallString
354: ${EndIf}
355: ${If} $R7 == ""
356: ReadRegStr $R7 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
InstallLocation
357: ${EndIf}
358: ${If} $R7 == ""
359: ReadRegStr $R7 HKLM "Software\${APP_GUID}" InstallLocation
360: ReadRegStr $R6 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
DisplayVersion
361: ReadRegStr $R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
QuietUninstallString
362: ${If} $R5 == ""
363: ReadRegStr $R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
UninstallString
364: ${EndIf}
365: ${If} $R7 == ""
366: ReadRegStr $R7 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
InstallLocation
367: ${EndIf}
368: ${EndIf}
369: SetRegView lastused
370: ${EndIf}
371:
372: ${If} $R7 == ""
373: IfFileExists "$LOCALAPPDATA\Programs\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3
374: StrCpy $R7 "$LOCALAPPDATA\Programs\OMB Portfolio Builder"
375: StrCpy $R5 "$LOCALAPPDATA\Programs\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" /S'
376: ${EndIf}
377: ${If} $R7 == ""
378: IfFileExists "$PROGRAMFILES64\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3
379: StrCpy $R7 "$PROGRAMFILES64\OMB Portfolio Builder"
380: StrCpy $R5 "$PROGRAMFILES64\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" /S'
381: ${EndIf}
382: ${If} $R7 == ""
383: IfFileExists "$PROGRAMFILES\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3
384: StrCpy $R7 "$PROGRAMFILES\OMB Portfolio Builder"
385: StrCpy $R5 "$PROGRAMFILES\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" /S'
386: ${EndIf}
387: ${If} $R7 == ""
388: IfFileExists "$PROFILE\OMB\application\OMB Portfolio Builder\OMB Portfolio Builder.exe" 0 +4
389: StrCpy $R7 "$PROFILE\OMB\application\OMB Portfolio Builder"
390: StrCpy $R6 "a legacy app-folder copy"
391: StrCpy $OMB_ManualApplicationInstall "1"
392: ${EndIf}
393: ${If} $R7 == ""
394: IfFileExists "$PROFILE\OMB\application\OMB Portfolio Builder.exe" 0 +4
395: StrCpy $R7 "$PROFILE\OMB\application"

```

```

396:         StrCpy $R6 "a standalone app-folder copy"
397:         StrCpy $OMB_ManualApplicationInstall "1"
398:     ${EndIf}
399:     ${If} $R7 == ""
400:         IfFileExists "$PROFILE\OneDrive\Documents\OMB\application\OMB Portfolio Builder\OMB Portfolio
Builder.exe" 0 +4
401:         StrCpy $R7 "$PROFILE\OneDrive\Documents\OMB\application\OMB Portfolio Builder"
402:         StrCpy $R6 "a legacy app-folder copy"
403:         StrCpy $OMB_ManualApplicationInstall "1"
404:     ${EndIf}
405:     ${If} $R7 == ""
406:         IfFileExists "$PROFILE\OneDrive\Documents\OMB\application\OMB Portfolio Builder.exe" 0 +4
407:         StrCpy $R7 "$PROFILE\OneDrive\Documents\OMB\application"
408:         StrCpy $R6 "a standalone app-folder copy"
409:         StrCpy $OMB_ManualApplicationInstall "1"
410:     ${EndIf}
411:     ${If} $R7 == ""
412:         IfFileExists "$PROFILE\OneDrive\Desktop\OMB\application\OMB Portfolio Builder.exe" 0 +4
413:         StrCpy $R7 "$PROFILE\OneDrive\Desktop\OMB\application"
414:         StrCpy $R6 "a standalone app-folder copy"
415:         StrCpy $OMB_ManualApplicationInstall "1"
416:     ${EndIf}
417:     ${If} $R7 == ""
418:         IfFileExists "$DOCUMENTS\OMB\application\OMB Portfolio Builder.exe" 0 +4
419:         StrCpy $R7 "$DOCUMENTS\OMB\application"
420:         StrCpy $R6 "a standalone app-folder copy"
421:         StrCpy $OMB_ManualApplicationInstall "1"
422:     ${EndIf}
423:
424:     ${If} $OMB_ManualApplicationInstall == "1"
425:         StrCpy $R5 ""
426:     ${EndIf}
427:
428:     ${If} $R7 != ""
429:         ${If} $OMB_ManualApplicationInstall == "1"
430:             Call OMBReadExistingAppVersion
431:         ${Else}
432:             StrCpy $OMB_ExistingInstallVersion $R6
433:         ${EndIf}
434:     ${If} $R7 == "$PROFILE\OMB\application\OMB Portfolio Builder"
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

## OMBUninstallPageCreate - lines 546-560

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 546-560 (15 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | Function OMBUninstallPageCreate                                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

546: Function OMBUninstallPageCreate
547:     nsDialogs::Create 1018
548:     Pop $OMB_UninstallDialog
549:     ${If} $OMB_UninstallDialog == error
550:         Abort
551:     ${EndIf}
552:
553:     ${NSD_CreateLabel} 0 0 100% 12u "Uninstall OMB Portfolio Builder"
554:     Pop $0
555:
556:     ${NSD_CreateLabel} 0 20u 100% 90u "Uninstall will remove:$\r$\n\r$\n- OMB Portfolio Builder
application files\r$\n- OMB Portfolio Builder shortcuts\r$\n- Local app data created by this application when
Windows requests app-data cleanup\r$\n\r$\nUninstall will not silently remove shared tools such as Git for
Windows, Node.js, pnpm, browsers, or your Git repositories."
557:     Pop $0
558:
559:     nsDialogs::Show
560: FunctionEnd

```

## Representative opening snippet

```
1: !include LogicLib.nsh
2: !include nsDialogs.nsh
3: !include WinMessages.nsh
4: !include WordFunc.nsh
5:
6: !ifndef BUILD_UNINSTALLER
7: Var OMB_ToolsDialog
8: Var OMB_ShortcutCheckbox
9: Var OMB_PublishingToolsCheckbox
10: Var OMB_CreateDesktopShortcutState
11: Var OMB_InstallPublishingToolsState
12: Var OMB_ExistingInstallLocation
13: Var OMB_ExistingInstallVersion
14: Var OMB_IsUpdateInstall
15: Var OMB_ManualApplicationInstall
16:
17: Function OMBToolsPageCreate
18:   nsDialogs::Create 1018
19:   Pop $OMB_ToolsDialog
20:   ${If} $OMB_ToolsDialog == error
21:     Abort
22:   ${EndIf}
23:
24:   ${NSD_CreateLabel} 0 0 100% 12u "Tools and shortcuts"
25:   Pop $0
26:
27:   ${NSD_CreateLabel} 0 18u 100% 36u "The desktop app includes its own runtime. Users do not need to
install Node.js or npm.$\r$\nGit for Windows with Git Credential Manager is only needed when publishing to
GitHub."
28:   Pop $0
29:
30:   ${NSD_CreateCheckbox} 0 62u 100% 12u "Create a desktop shortcut"
31:   Pop $OMB_ShortcutCheckbox
32:   ${If} $OMB_CreateDesktopShortcutState == ""
33:     StrCpy $OMB_CreateDesktopShortcutState ${BST_CHECKED}
34:   ${EndIf}
35:   ${If} $OMB_CreateDesktopShortcutState == ${BST_CHECKED}
36:     ${NSD_Check} $OMB_ShortcutCheckbox
37:   ${EndIf}
38:
39:   ${NSD_CreateCheckbox} 0 82u 100% 24u "Check publishing tools and install Git for Windows if Git/Git
Credential Manager is missing"
40:   Pop $OMB_PublishingToolsCheckbox
41:   ${If} $OMB_InstallPublishingToolsState == ""
42:     StrCpy $OMB_InstallPublishingToolsState ${BST_CHECKED}
43:   ${EndIf}
44:   ${If} $OMB_InstallPublishingToolsState == ${BST_CHECKED}
45:     ${NSD_Check} $OMB_PublishingToolsCheckbox
46:   ${EndIf}
47:
48:   ${NSD_CreateLabel} 0 114u 100% 38u "Existing shared tools are not uninstalled silently. If a compatible
tool is already available, setup skips it. If Git is missing or unusable, setup will try to install Git for
Windows side-by-side using Windows Package Manager."
49:   Pop $0
50:
51:   nsDialogs::Show
52: FunctionEnd
53:
54: Function OMBToolsPageLeave
55:   ${NSD_GetState} $OMB_ShortcutCheckbox $OMB_CreateDesktopShortcutState
56:   ${NSD_GetState} $OMB_PublishingToolsCheckbox $OMB_InstallPublishingToolsState
57:   ${If} $OMB_IsUpdateInstall == "1"
58:     ${AndIf} $INSTDIR != $OMB_ExistingInstallLocation
59:     MessageBox MB_OK|MB_ICONSTOP "OMB Portfolio Builder is already installed on this
machine.$\r$\n$\r$\nUpdates must use the current installed
location:$\r$\n$OMB_ExistingInstallLocation$\r$\n$\r$\nSetup will not create another installed copy in a
different directory."
60:     Abort
61:   ${EndIf}
62: FunctionEnd
63:
64: Function OMBCheckGitAvailable
65:   StrCpy $R9 "0"
66:   IfFileExists "$PROGRAMFILES64\Git\cmd\git.exe" 0 +2
67:   StrCpy $R9 "1"
68:   IfFileExists "$PROGRAMFILES\Git\cmd\git.exe" 0 +2
69:   StrCpy $R9 "1"
70:   IfFileExists "$PROGRAMFILES32\Git\cmd\git.exe" 0 +2
71:   StrCpy $R9 "1"
72:   IfFileExists "$LOCALAPPDATA\Programs\Git\cmd\git.exe" 0 +2
```

# Source File: builder-rich-future-sections.js

| Item                | Explanation                                                                         |
|---------------------|-------------------------------------------------------------------------------------|
| File                | builder-rich-future-sections.js                                                     |
| Purpose             | Builder-side helper code for richer future portfolio sections and section defaults. |
| Lines               | 410                                                                                 |
| Size                | 21,340 bytes                                                                        |
| Talks to            | Mostly local to its own feature area.                                               |
| Functions           | 13                                                                                  |
| Variables/constants | 42                                                                                  |
| API mentions        | 0                                                                                   |
| Signals             | 17                                                                                  |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                         | Explanation                                                                                                                                                                            |
|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rich text at line 82         | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <button type="button" data-rich-action="font-small">Small</button>                                |
| Rich text at line 83         | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <button type="button" data-rich-action="font-normal">Normal</button>                              |
| Rich text at line 84         | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <button type="button" data-rich-action="font-large">Large</button>                                |
| Rich text at line 97         | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: data-placeholder="Write the overview here. You can paste images directly into this area."         |
| Browser DOM at line 125      | Builder or website interface behavior in the browser/Electron renderer. Evidence: if (editor.dataset.richEditor === "pending-description") {                                           |
| Browser DOM at line 139      | Builder or website interface behavior in the browser/Electron renderer. Evidence: if (editor.dataset.richEditor === "pending-description") {                                           |
| Browser DOM at line 193      | Builder or website interface behavior in the browser/Electron renderer. Evidence: document.querySelector("#pending-editor")?.scrollIntoView({ behavior: "smooth", block: "nearest" }); |
| Browser DOM at line 204      | Builder or website interface behavior in the browser/Electron renderer. Evidence: const type = button.dataset.openEditor;                                                              |
| Browser DOM at line 205      | Builder or website interface behavior in the browser/Electron renderer. Evidence: const mode = button.dataset.mode    "add";                                                           |
| Browser DOM at line 206      | Builder or website interface behavior in the browser/Electron renderer. Evidence: const index = button.dataset.index;                                                                  |
| Browser DOM at line 208      | Builder or website interface behavior in the browser/Electron renderer. Evidence: const section = (project.sections    []).find((item) => item.id === button.dataset.sectionId);       |
| Browser DOM at line 213      | Builder or website interface behavior in the browser/Electron renderer. Evidence: sectionId: button.dataset.sectionId,                                                                 |
| Browser DOM at line 222      | Builder or website interface behavior in the browser/Electron renderer. Evidence: const section = (project.sections    []).find((item) => item.id === button.dataset.sectionId);       |
| Browser DOM at line 226      | Builder or website interface behavior in the browser/Electron renderer. Evidence: sectionId: button.dataset.sectionId,                                                                 |
| Cloudflare or AI at line 238 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: // their richDescription field back to the project model. Fallback to                           |
| Browser DOM at line 245      | Builder or website interface behavior in the browser/Electron renderer. Evidence: const richEditor = form.querySelector("[data-rich-editor='pending-description']");                   |

| Item                    | Explanation                                                                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser DOM at line 393 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.addEventListener("contextmenu", (event) =&gt; {</code> |

## Important constants and variables

| Item                                       | Explanation                                                                                                                                                                     |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| text                                       | Line 34: <code>const text = plainTextFromRich(rich).trim();</code>                                                                                                              |
| label                                      | Line 73: <code>const label = editor?.type === "custom-section" ? "Section overview" : "Subsection overview";</code>                                                             |
| folder                                     | Line 74: <code>const folder = pendingEditorFolder(editor);</code>                                                                                                               |
| originalPopulateRichEditor                 | Line 108: <code>const originalPopulateRichEditor = typeof populateRichEditor === "function" ? populateRichEditor : null;</code>                                                 |
| originalSaveRichEditorToProject            | Line 109: <code>const originalSaveRichEditorToProject = typeof saveRichEditorToProject === "function" ? saveRichEditorToProject : null;</code>                                  |
| originalOpenPendingEditor                  | Line 110: <code>const originalOpenPendingEditor = typeof openPendingEditor === "function" ? openPendingEditor : null;</code>                                                    |
| originalOpenEditorFromButton               | Line 111: <code>const originalOpenEditorFromButton = typeof openEditorFromButton === "function" ? openEditorFromButton : null;</code>                                           |
| originalRenderPendingEditor                | Line 112: <code>const originalRenderPendingEditor = typeof renderPendingEditor === "function" ? renderPendingEditor : null;</code>                                              |
| originalRenderSectionContent               | Line 113: <code>const originalRenderSectionContent = typeof renderSectionContent === "function" ? renderSectionContent : null;</code>                                           |
| originalSavePendingEditor                  | Line 114: <code>const originalSavePendingEditor = typeof savePendingEditor === "function" ? savePendingEditor : null;</code>                                                    |
| originalRenderCustomSection                | Line 115: <code>const originalRenderCustomSection = typeof renderCustomSection === "function" ? renderCustomSection : null;</code>                                              |
| originalParseProjectForPortfolio           | Line 116: <code>const originalParseProjectForPortfolio = typeof parseProjectForPortfolio === "function" ? parseProjectForPortfolio : null;</code>                               |
| originalParsedSectionHasContent            | Line 117: <code>const originalParsedSectionHasContent = typeof parsedSectionHasContent === "function" ? parsedSectionHasContent : null;</code>                                  |
| originalPreviewSectionHasRenderableContent | Line 118: <code>const originalPreviewSectionHasRenderableContent = typeof previewSectionHasRenderableContent === "function" ? previewSectionHasRenderableContent : null;</code> |
| rich                                       | Line 140: <code>const rich = extractRichSummary(editor);</code>                                                                                                                 |
| title                                      | Line 154: <code>const title = pendingEditor.title    "";</code>                                                                                                                 |
| description                                | Line 155: <code>const description = pendingEditor.description    "";</code>                                                                                                     |
| isTool                                     | Line 156: <code>const isTool = pendingEditor.type === "tool";</code>                                                                                                            |
| isSimpleText                               | Line 157: <code>const isSimpleText = pendingEditor.type === "text-array";</code>                                                                                                |
| isSectionDetails                           | Line 158: <code>const isSectionDetails = pendingEditor.type === "custom-section";</code>                                                                                        |
| usesRich                                   | Line 159: <code>const usesRich = pendingEditorUsesRichDescription();</code>                                                                                                     |
| project                                    | Line 202: <code>const project = selectedProject();</code>                                                                                                                       |
| type                                       | Line 204: <code>const type = button.dataset.openEditor;</code>                                                                                                                  |
| mode                                       | Line 205: <code>const mode = button.dataset.mode    "add";</code>                                                                                                               |
| index                                      | Line 206: <code>const index = button.dataset.index;</code>                                                                                                                      |
| section                                    | Line 208: <code>const section = (project.sections    []).find((item) =&gt; item.id === button.dataset.sectionId);</code>                                                        |
| item                                       | Line 209: <code>const item = mode === "edit" ? section?.items?.[Number(index)] : {};</code>                                                                                     |
| section                                    | Line 222: <code>const section = (project.sections    []).find((item) =&gt; item.id === button.dataset.sectionId);</code>                                                        |
| project                                    | Line 242: <code>const project = selectedProject();</code>                                                                                                                       |
| title                                      | Line 244: <code>const title = form.elements.title.value.trim();</code>                                                                                                          |
| richEditor                                 | Line 245: <code>const richEditor = form.querySelector("[data-rich-editor='pending-description']");</code>                                                                       |
| richDescription                            | Line 246: <code>const richDescription = richEditor ? extractRichSummary(richEditor) : null;</code>                                                                              |

| Item          | Explanation                                                                                                                        |
|---------------|------------------------------------------------------------------------------------------------------------------------------------|
| description   | Line 247: const description = richEditor ? richPlainTextOrSpacer(richDescription) : form.elements.description?.value.trim()    ""; |
| section       | Line 253: const section = (project.sections    []).find((item) => item.id === pendingEditor.sectionId);                            |
| nextItem      | Line 256: const nextItem = {                                                                                                       |
| section       | Line 278: const section = (project.sections    []).find((item) => item.id === pendingEditor.sectionId);                            |
| section       | Line 297: const section = (project.sections    []).find((item) => item.id === sectionId);                                          |
| parsed        | Line 352: const parsed = originalParseProjectForPortfolio(project);                                                                |
| sectionId     | Line 355: const sectionId = parsedSection.id.replace("custom:", "");                                                               |
| sourceSection | Line 356: const sourceSection = (project.sections    []).find((section) => section.id === sectionId);                              |
| editor        | Line 394: const editor = event.target.closest("[data-rich-editor]");                                                               |
| block         | Line 395: const block = event.target.closest(".rich-block");                                                                       |

## Event handlers and user interactions

| Item     | Explanation                                           |
|----------|-------------------------------------------------------|
| Line 393 | document.addEventListener("contextmenu", (event) => { |

## Function-by-function detail

### *hasRichContent - lines 23-31*

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 23-31 (9 source lines).                                                                                                                                                                         |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function hasRichContent(rich) {                                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | some                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

23:     function hasRichContent(rich) {
24:         return Boolean(rich?.blocks?.some((block) =>
25:             (block.type === "image" && block.url) ||
26:             (block.type === "formula" && block.formula) ||
27:             block.text ||
28:             block.title ||
29:             block.caption
30:         ));
31:     }

```

### *richPlainTextOrSpacer - lines 33-36*

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 33-36 (4 source lines).                                                                          |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | function richPlainTextOrSpacer(rich) {                                                                 |
| Touches         | Mostly local calculations or local UI state.                                                           |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | plainTextFromRich, trim, hasRichContent                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

33:     function richPlainTextOrSpacer(rich) {
34:         const text = plainTextFromRich(rich).trim();
35:         return text || (hasRichContent(rich) ? "\u200B" : "");
36:     }

```

### ***pendingEditorUsesRichDescription - lines 41-43***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 41-43 (3 source lines).                                                                                                                                                                         |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function pendingEditorUsesRichDescription(editor = pendingEditor) {                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | includes                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

41:     function pendingEditorUsesRichDescription(editor = pendingEditor) {
42:         return Boolean(editor && ["custom", "custom-section"].includes(editor.type));
43:     }

```

### ***pendingEditorFolder - lines 48-57***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 48-57 (10 source lines).                                                                                                                                                                        |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function pendingEditorFolder(editor = pendingEditor) {                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | slugify                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

48:     function pendingEditorFolder(editor = pendingEditor) {
49:         if (!editor) return "custom-overview";
50:         if (editor.type === "custom-section") {
51:             return `custom-${slugify(editor.sectionId || "section")}-overview`;
52:         }
53:         if (editor.type === "custom") {
54:             return `custom-${slugify(editor.sectionId || "section")}-subsection`;
55:         }

```

```

56:         return "custom-overview";
57:     }

```

## pendingEditorRichBlocks - lines 62-65

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 62-65 (4 source lines).                                                                                                                                                                         |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function pendingEditorRichBlocks(editor = pendingEditor) {                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | textBlocksFromPlainText, richBlocksFromValue                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

62:     function pendingEditorRichBlocks(editor = pendingEditor) {
63:         if (!editor) return textBlocksFromPlainText("");
64:         return richBlocksFromValue(editor.richDescription, editor.description || "");
65:     }

```

## renderPendingRichDescriptionEditor - lines 72-103

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 72-103 (32 source lines).                                                                                                                                                                       |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPendingRichDescriptionEditor(editor = pendingEditor) {                                                                                                                                 |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | pendingEditorFolder, escapeHtml                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

72:     function renderPendingRichDescriptionEditor(editor = pendingEditor) {
73:         const label = editor?.type === "custom-section" ? "Section overview" : "Subsection overview";
74:         const folder = pendingEditorFolder(editor);
75:         return `
76:         <div class="pending-rich-description rich-design-summary-field">
77:             <div class="summary-builder-heading">
78:                 <span>${label}</span>
79:             </div>
80:             <div class="rich-summary-panel">
81:                 <div class="rich-summary-toolbar" aria-label="${escapeHtml(label)} tools">
82:                     <button type="button" data-rich-action="font-small">Small</button>
83:                     <button type="button" data-rich-action="font-normal">Normal</button>
84:                     <button type="button" data-rich-action="font-large">Large</button>
85:                     <button type="button" data-rich-action="align-left">Left</button>
86:                     <button type="button" data-rich-action="align-center">Center</button>
87:                     <button type="button" data-rich-action="align-right">Right</button>
88:                     <button type="button" data-rich-action="add-image">Add image</button>
89:                     <button type="button" data-rich-action="add-formula">Add formula</button>
90:                     <button type="button" data-rich-action="edit-block">Edit selected</button>
91:                     <button class="danger-icon" type="button" data-rich-action="delete-block">Delete
selected</button>
92:                 </div>
93:                 <div
94:                     class="rich-summary-editor"
95:                     data-rich-editor="pending-description"
96:                     data-rich-folder="${escapeHtml(folder)}">

```

```

97:         data-placeholder="Write the overview here. You can paste images directly into this area."
98:         aria-label="${escapeHtml(label)} rich editor"
99:     ></div>
100: </div>
101: </div>
102: `;
103: }

```

## section - lines 208-209

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 208-209 (2 source lines).                                                                                                                                                                       |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | <code>const section = (project.sections    []).find((item) =&gt; item.id === button.dataset.sectionId);</code>                                                                                        |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | find                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

208:         const section = (project.sections || []).find((item) => item.id ===
button.dataset.sectionId);
209:         const item = mode === "edit" ? section?.items?.[Number(index)] : {};

```

## section - lines 222-230

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 222-230 (9 source lines).                                                                                                                                                                       |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | <code>const section = (project.sections    []).find((item) =&gt; item.id === button.dataset.sectionId);</code>                                                                                        |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | find, openPendingEditor                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

222:         const section = (project.sections || []).find((item) => item.id ===
button.dataset.sectionId);
223:         openPendingEditor({
224:             type,
225:             mode: "edit",
226:             sectionId: button.dataset.sectionId,
227:             title: section?.title || "",
228:             description: section?.description || "",
229:             richDescription: section?.richDescription || null
230:         });

```

## section - lines 253-262

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 253-262 (10 source lines).                                                                       |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | const section = (project.sections    []).find((item) => item.id === pendingEditor.sectionId);                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

253:         const section = (project.sections || []).find((item) => item.id ===
pendingEditor.sectionId);
254:         if (!section) return;
255:         section.items = section.items || [];
256:         const nextItem = {
257:             title,
258:             description,
259:             richDescription,
260:             type: "Text",
261:             status: "draft"
262:         };

```

## section - lines 278-291

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 278-291 (14 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | const section = (project.sections    []).find((item) => item.id === pendingEditor.sectionId);                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | find, setStatus, scheduleAutosave, renderAll, originalSavePendingEditor                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

278:         const section = (project.sections || []).find((item) => item.id ===
pendingEditor.sectionId);
279:         if (!section) return;
280:         section.title = title;
281:         section.description = description;
282:         section.richDescription = richDescription;
283:         pendingEditor = null;
284:         setStatus("Saved in the project editor. Click Save project to include this version in the
portfolio preview.");
285:         scheduleAutosave();
286:         renderAll();
287:         return;
288:     }
289:     return originalSavePendingEditor(form);
290: };
291: }

```

## section - lines 297-301

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 297-301 (5 source lines).                                                                        |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | const section = (project.sections    []).find((item) => item.id === sectionId);                        |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, escapeHtml                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

297:         const section = (project.sections || []).find((item) => item.id === sectionId);
298:         if (!section) return `<p class="evidence-empty">Section not found.</p>`;
299:         return `
300:         <div class="section-window-heading">
301:         <h3>${escapeHtml(section.title || "Custom section")}</h3>

```

## sourceSection - lines 356-360

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 356-360 (5 source lines).                                                                                                                                                                       |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | const sourceSection = (project.sections    []).find((section) => section.id === sectionId);                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, hasRichContent, clone, richPlainTextOrSpacer                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

356:         const sourceSection = (project.sections || []).find((section) => section.id ===
sectionId);
357:         if (hasRichContent(sourceSection?.richDescription)) {
358:             parsedSection.rich = clone(sourceSection.richDescription);
359:             parsedSection.description = sourceSection.description ||
richPlainTextOrSpacer(sourceSection.richDescription);
360:         }

```

## improveRichContextMenu - lines 392-405

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 392-405 (14 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function improveRichContextMenu() {                                                                                                                                                                   |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | addEventListener, closest, preventDefault, selectRichBlock, currentRichBlock, syncRichContextMenuControls                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

392:     function improveRichContextMenu() {

```

```

393:     document.addEventListener("contextmenu", (event) => {
394:         const editor = event.target.closest("[data-rich-editor]");
395:         const block = event.target.closest(".rich-block");
396:         if (!editor && !block) return;
397:         event.preventDefault();
398:         activeSummaryEditor = editor || block.closest("[data-rich-editor]");
399:         selectRichBlock(block || currentRichBlock(activeSummaryEditor));
400:         syncRichContextMenuControls(activeSummaryBlock);
401:         summaryContextMenu.style.left = `${event.clientX}px`;
402:         summaryContextMenu.style.top = `${event.clientY}px`;
403:         summaryContextMenu.hidden = false;
404:     }, true);
405: }

```

## Representative opening snippet

```

1: // JavaScript source code
2: (() => {
3:     "use strict";
4:
5:     /**
6:      * Extend the portfolio builder with rich description editors for custom
7:      * sections and subsections. This patch adds a new rich editor to the
8:      * pending editor for custom items so that images, formulas and other
9:      * formatted content can be used in overviews of future sections. It
10:     * hooks into existing helper functions such as `populateRichEditor`
11:     * and `saveRichEditorToProject` by overriding them at runtime. To
12:     * avoid interfering with built-in behaviour, everything is wrapped
13:     * inside an IIFE and existing functions are stored before being
14:     * redefined. If a function is not present in the current context
15:     * (because the builder script changed), the patch simply does
16:     * nothing.
17:     */
18:
19:     // Guard against applying the patch multiple times.
20:     if (window.builderRichFutureSectionsPatchLoaded) return;
21:
22:     // Helpers to detect whether an object has meaningful rich content.
23:     function hasRichContent(rich) {
24:         return Boolean(rich?.blocks?.some((block) =>
25:             (block.type === "image" && block.url) ||
26:             (block.type === "formula" && block.formula) ||
27:             block.text ||
28:             block.title ||
29:             block.caption
30:         ));
31:     }
32:
33:     function richPlainTextOrSpacer(rich) {
34:         const text = plainTextFromRich(rich).trim();
35:         return text || (hasRichContent(rich) ? "\u200B" : "");
36:     }
37:
38:     // Determine whether the pending editor should use a rich editor. All
39:     // custom sections and subsections use rich editors. Tools, design
40:     // objects and other built-in forms continue to use plain text areas.
41:     function pendingEditorUsesRichDescription(editor = pendingEditor) {
42:         return Boolean(editor && ["custom", "custom-section"].includes(editor.type));
43:     }
44:
45:     // Build a folder name for images uploaded via the pending editor. For
46:     // custom sections and subsections, use a prefix based on the section
47:     // id so that uploaded images are organised under docs/<project>/<folder>/.
48:     function pendingEditorFolder(editor = pendingEditor) {
49:         if (!editor) return "custom-overview";
50:         if (editor.type === "custom-section") {
51:             return `custom-${slugify(editor.sectionId || "section")}-overview`;
52:         }
53:         if (editor.type === "custom") {
54:             return `custom-${slugify(editor.sectionId || "section")}-subsection`;
55:         }
56:         return "custom-overview";
57:     }
58:
59:     // Derive blocks from a rich value or fallback plain text. When the
60:     // editor is empty, provide a single text block so that the content
61:     // area is editable.
62:     function pendingEditorRichBlocks(editor = pendingEditor) {
63:         if (!editor) return textBlocksFromPlainText("");
64:         return richBlocksFromValue(editor.richDescription, editor.description || "");
65:     }
66:
67:     // Render the rich description field for the pending editor. This
68:     // function builds a UI similar to other rich editors in the builder,
69:     // including a toolbar and editable area with appropriate data
70:     // attributes. The label varies depending on whether we're editing
71:     // a section or a subsection.
72:     function renderPendingRichDescriptionEditor(editor = pendingEditor) {

```

# Source File: cloudflare/README.md

| Item                | Explanation                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------|
| File                | cloudflare/README.md                                                                                   |
| Purpose             | Cloudflare setup notes for deploying the AI Worker endpoint and wiring the public site to the backend. |
| Lines               | 47                                                                                                     |
| Size                | 1,562 bytes                                                                                            |
| Talks to            | Cloudflare/AI layer, GitHub/release layer                                                              |
| Functions           | 0                                                                                                      |
| Variables/constants | 0                                                                                                      |
| API mentions        | 1                                                                                                      |
| Signals             | 16                                                                                                     |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## API endpoints mentioned or called

| Item              | Explanation                              |
|-------------------|------------------------------------------|
| /api/portfolio-ai | Line 32. Handles AI assistant questions. |

## Important implementation signals

| Item                        | Explanation                                                                                                                                                                                                                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloudflare or AI at line 1  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: # Portfolio AI Cloudflare Worker                                                                                                                                                                                             |
| Cloudflare or AI at line 6  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <a href="https://mauriceotieno.com/api/portfolio-ai">https://mauriceotieno.com/api/portfolio-ai</a>                                                                                                                          |
| Cloudflare or AI at line 9  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: The static GitHub Pages site cannot safely hold API keys. The browser sends portfolio context to this Worker. The Worker uses OpenAI when `OPENAI_API_KEY` is configured, and otherwise falls back to Cloudflare Workers AI. |
| Cloudflare or AI at line 13 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: From this `cloudflare` folder:                                                                                                                                                                                               |
| Cloudflare or AI at line 16 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: npx wrangler login                                                                                                                                                                                                           |
| Cloudflare or AI at line 17 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: npx wrangler secret put OPENAI_API_KEY # optional, Workers AI works without it                                                                                                                                               |
| Cloudflare or AI at line 18 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: npx wrangler deploy                                                                                                                                                                                                          |
| Cloudflare or AI at line 21 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: `wrangler.toml` routes `/api/portfolio-ai` on `mauriceotieno.com` and `www.mauriceotieno.com` to this Worker.                                                                                                                |
| Cloudflare or AI at line 23 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ## Cloudflare Requirements                                                                                                                                                                                                   |
| Cloudflare or AI at line 25 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: 1. The domain must be active in Cloudflare.                                                                                                                                                                                  |
| Cloudflare or AI at line 26 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: 2. DNS for `mauriceotieno.com` / `www` must be proxied through Cloudflare, not DNS-only, for Worker routes to run.                                                                                                           |
| Cloudflare or AI at line 27 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: 3. The Worker must have the `AI` binding enabled. This repo's `wrangler.toml` includes it.                                                                                                                                   |
| Cloudflare or AI at line 28 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: 4. Optional: add the `OPENAI_API_KEY` secret if you want the Worker to use OpenAI instead of Workers AI.                                                                                                                     |
| Cloudflare or AI at line 32 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <meta name="portfolio-ai-endpoint" content="/api/portfolio-ai" />                                                                                                                                                            |

| Item                        | Explanation                                                                                                                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloudflare or AI at line 41 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: -Uri "https://mauriceotieno.com/api/portfolio-ai" `                                                              |
| Cloudflare or AI at line 47 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: Expected result: JSON with an `answer` field and a `provider` field such as `cloudflare-workers-ai` or `openai`. |

## Representative opening snippet

```

1: # Portfolio AI Cloudflare Worker
2:
3: This Worker powers the public website endpoint:
4:
5: ```text
6: https://mauriceotieno.com/api/portfolio-ai
7: ```
8:
9: The static GitHub Pages site cannot safely hold API keys. The browser sends portfolio context to this
Worker. The Worker uses OpenAI when `OPENAI_API_KEY` is configured, and otherwise falls back to Cloudflare
Workers AI.
10:
11: ## Deploy
12:
13: From this `cloudflare` folder:
14:
15: ```powershell
16: npx wrangler login
17: npx wrangler secret put OPENAI_API_KEY # optional, Workers AI works without it
18: npx wrangler deploy
19: ```
20:
21: `wrangler.toml` routes `/api/portfolio-ai` on `mauriceotieno.com` and `www.mauriceotieno.com` to this
Worker.
22:
23: ## Cloudflare Requirements
24:
25: 1. The domain must be active in Cloudflare.
26: 2. DNS for `mauriceotieno.com` / `www` must be proxied through Cloudflare, not DNS-only, for Worker
routes to run.
27: 3. The Worker must have the `AI` binding enabled. This repo's `wrangler.toml` includes it.
28: 4. Optional: add the `OPENAI_API_KEY` secret if you want the Worker to use OpenAI instead of Workers AI.
29: 5. Keep the website meta tag:
30:
31: ```html
32: <meta name="portfolio-ai-endpoint" content="/api/portfolio-ai" />
33: ```
34:
35: ## Test
36:
37: After deployment:
38:
39: ```powershell
40: Invoke-RestMethod `
41:   -Uri "https://mauriceotieno.com/api/portfolio-ai" `
42:   -Method Post `
43:   -ContentType "application/json" `
44:   -Body '{"question":"Hi","context":{"profile":{"displayName":"Maurice Otieno}}}'
45: ```
46:
47: Expected result: JSON with an `answer` field and a `provider` field such as `cloudflare-workers-ai` or
`openai`.

```



# Source File: cloudflare/portfolio-ai-worker.js

| Item                | Explanation                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------|
| File                | cloudflare/portfolio-ai-worker.js                                                               |
| Purpose             | Cloudflare Worker backend for the public portfolio AI assistant and fallback intelligence path. |
| Lines               | 276                                                                                             |
| Size                | 10,435 bytes                                                                                    |
| Talks to            | Cloudflare/AI layer, GitHub/release layer                                                       |
| Functions           | 13                                                                                              |
| Variables/constants | 22                                                                                              |
| API mentions        | 1                                                                                               |
| Signals             | 45                                                                                              |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## API endpoints mentioned or called

| Item              | Explanation                               |
|-------------------|-------------------------------------------|
| /api/portfolio-ai | Line 228. Handles AI assistant questions. |

## Important implementation signals

| Item                        | Explanation                                                                                                                                                                                |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network request at line 2   | Frontend-backend communication or public web/API calls. Evidence: "https://mauriceotieno.com",                                                                                             |
| Network request at line 3   | Frontend-backend communication or public web/API calls. Evidence: "https://www.mauriceotieno.com",                                                                                         |
| Network request at line 4   | Frontend-backend communication or public web/API calls. Evidence: "https://otienomaurice.github.io",                                                                                       |
| Network request at line 5   | Frontend-backend communication or public web/API calls. Evidence: "http://localhost:8080",                                                                                                 |
| Network request at line 6   | Frontend-backend communication or public web/API calls. Evidence: "http://localhost:8081",                                                                                                 |
| Network request at line 7   | Frontend-backend communication or public web/API calls. Evidence: "http://127.0.0.1:8080",                                                                                                 |
| Network request at line 8   | Frontend-backend communication or public web/API calls. Evidence: "http://127.0.0.1:8081"                                                                                                  |
| Security or auth at line 11 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: function json(data, status = 200, headers = {}) {                                                   |
| Security or auth at line 14 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: headers: {                                                                                          |
| Local storage at line 16    | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: "Cache-Control": "no-store",                                                                 |
| Security or auth at line 17 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ...headers                                                                                          |
| Security or auth at line 30 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: function corsHeaders(request, env) {                                                                |
| Security or auth at line 31 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const origin = request.headers.get("Origin")    "";                                                 |
| Security or auth at line 36 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Access-Control-Allow-Headers": "Content-Type",                                                     |
| Cloudflare or AI at line 54 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: function extractOpenAiText(data = {}) {                                                             |
| Security or auth at line 89 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Never invent portfolio projects, credentials, files, test results, or private repository access.", |

| Item                         | Explanation                                                                                                                                                                                                                   |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloudflare or AI at line 121 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>async function callOpenAi(body, env) {</code>                                                                                    |
| Cloudflare or AI at line 122 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>const model = env.OPENAI_MODEL    "gpt-4.1-mini";</code>                                                                         |
| Cloudflare or AI at line 124 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>model,</code>                                                                                                                    |
| Cloudflare or AI at line 135 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>max_output_tokens: Number(env.OPENAI_MAX_OUTPUT_TOKENS    1000)</code>                                                           |
| Cloudflare or AI at line 138 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>const response = await fetch("https://api.openai.com/v1/responses", {</code>                                                     |
| Security or auth at line 140 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>headers: {</code>                                                                                                                |
| Cloudflare or AI at line 141 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>"Authorization": `Bearer \${env.OPENAI_API_KEY}`,</code>                                                                         |
| Cloudflare or AI at line 151 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>error: data?.error?.message    "OpenAI request failed."</code>                                                                   |
| Cloudflare or AI at line 156 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>answer: extractOpenAiText(data),</code>                                                                                          |
| Cloudflare or AI at line 157 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>model</code>                                                                                                                     |
| Cloudflare or AI at line 166 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>error: "Cloudflare Workers AI binding is not available for this Worker."</code>                                                  |
| Cloudflare or AI at line 170 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>const model = env.WORKERS_AI_MODEL    "@cf/meta/llama-3.2-3b-instruct";</code>                                                   |
| Cloudflare or AI at line 171 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>const data = await env.AI.run(model, {</code>                                                                                    |
| Cloudflare or AI at line 173 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>max_tokens: Number(env.OPENAI_MAX_OUTPUT_TOKENS    1000)</code>                                                                  |
| Cloudflare or AI at line 178 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>model</code>                                                                                                                     |
| Cloudflare or AI at line 210 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>"The full AI model is not reachable from this Worker right now, so this is a fallback answer from the available context."</code> |
| Cloudflare or AI at line 214 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>"I can answer that in a general way, but the full AI model is not reachable from this Worker right now.",</code>                 |
| Network request at line 223  | Frontend-backend communication or public web/API calls. Evidence: <code>async fetch(request, env) {</code>                                                                                                                    |
| Security or auth at line 224 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>const cors = corsHeaders(request, env);</code>                                                                                   |
| Security or auth at line 225 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <code>if (request.method === "OPTIONS") return new Response(null, { status: 204, headers: cors });</code>                              |
| Cloudflare or AI at line 228 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>if (request.method !== "POST"    !url.pathname.endsWith("/api/portfolio-ai")) {</code>                                           |
| Cloudflare or AI at line 243 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>let provider = env.OPENAI_API_KEY ? "openai" : "cloudflare-workers-ai";</code>                                                   |
| Cloudflare or AI at line 244 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>let result = env.OPENAI_API_KEY</code>                                                                                           |
| Cloudflare or AI at line 245 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>? await callOpenAi({ ...body, question }, env)</code>                                                                            |
| Cloudflare or AI at line 247 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>if (!result.ok &amp;&amp; env.OPENAI_API_KEY &amp;&amp; env.AI) {</code>                                                         |
| Cloudflare or AI at line 248 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>provider = "cloudflare-workers-ai";</code>                                                                                       |
| Cloudflare or AI at line 254 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>model: "worker-fallback",</code>                                                                                                 |
| Cloudflare or AI at line 262 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>model: result.model,</code>                                                                                                      |
| Cloudflare or AI at line 269 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>model: "worker-fallback",</code>                                                                                                 |

## Important constants and variables

| Item                  | Explanation                                                                                            |
|-----------------------|--------------------------------------------------------------------------------------------------------|
| defaultAllowedOrigins | Line 1: const defaultAllowedOrigins = [                                                                |
| origin                | Line 31: const origin = request.headers.get("Origin")    "";                                           |
| allowOrigin           | Line 32: const allowOrigin = allowedOrigins(env).includes(origin) ? origin : defaultAllowedOrigins[0]; |
| parts                 | Line 56: const parts = [];                                                                             |
| messages              | Line 96: const messages = [{ role: "system", content: buildInstructions() }];                          |
| model                 | Line 122: const model = env.OPENAI_MODEL    "gpt-4.1-mini";                                            |
| payload               | Line 123: const payload = {                                                                            |
| response              | Line 138: const response = await fetch("https://api.openai.com/v1/responses", {                        |
| data                  | Line 146: const data = await response.json().catch(() => ({}));                                        |
| model                 | Line 170: const model = env.WORKERS_AI_MODEL    "@cf/meta/llama-3.2-3b-instruct";                      |
| data                  | Line 171: const data = await env.AI.run(model, {                                                       |
| question              | Line 183: const question = String(body.question    "").trim();                                         |
| clean                 | Line 184: const clean = question.toLowerCase();                                                        |
| context               | Line 185: const context = body.context    {};                                                          |
| projects              | Line 186: const projects = Array.isArray(context.projects) ? context.projects : [];                    |
| owner                 | Line 187: const owner = context.owner    context.profile?.displayName    "Maurice Otieno";             |
| cors                  | Line 224: const cors = corsHeaders(request, env);                                                      |
| url                   | Line 227: const url = new URL(request.url);                                                            |
| body                  | Line 232: let body = {};                                                                               |
| question              | Line 239: const question = clamp(body.question, 1200).trim();                                          |
| provider              | Line 243: let provider = env.OPENAI_API_KEY ? "openai" : "cloudflare-workers-ai";                      |
| result                | Line 244: let result = env.OPENAI_API_KEY                                                              |

## JSON/YAML-style keys detected

| Item                         | Explanation                                                 |
|------------------------------|-------------------------------------------------------------|
| Content-Type                 | Line 15: "Content-Type": "application/json; charset=utf-8", |
| Cache-Control                | Line 16: "Cache-Control": "no-store",                       |
| Access-Control-Allow-Origin  | Line 34: "Access-Control-Allow-Origin": allowOrigin,        |
| Access-Control-Allow-Methods | Line 35: "Access-Control-Allow-Methods": "POST, OPTIONS",   |
| Access-Control-Allow-Headers | Line 36: "Access-Control-Allow-Headers": "Content-Type",    |
| Access-Control-Max-Age       | Line 37: "Access-Control-Max-Age": "86400",                 |
| Vary                         | Line 38: "Vary": "Origin"                                   |
| Authorization                | Line 141: "Authorization": `Bearer \${env.OPENAI_API_KEY}`, |
| Content-Type                 | Line 142: "Content-Type": "application/json"                |

## Function-by-function detail

### *json - lines 11-20*

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 11-20 (10 source lines).                                                                                                                                                                        |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function json(data, status = 200, headers = {}) {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Response, stringify                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

11: function json(data, status = 200, headers = {}) {
12:   return new Response(JSON.stringify(data, null, 2), {
13:     status,
14:     headers: {
15:       "Content-Type": "application/json; charset=utf-8",
16:       "Cache-Control": "no-store",
17:       ...headers
18:     }
19:   });
20: }

```

## ***allowedOrigins - lines 22-28***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 22-28 (7 source lines).                                                                                                                                                                         |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function allowedOrigins(env) {                                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, map, trim, filter, concat                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

22: function allowedOrigins(env) {
23:   return String(env.ALLOWED_ORIGINS || "")
24:     .split(",")
25:     .map((origin) => origin.trim())
26:     .filter(Boolean)
27:     .concat(defaultAllowedOrigins);
28: }

```

## ***corsHeaders - lines 30-40***

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 30-40 (11 source lines).                                                                         |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | function corsHeaders(request, env) {                                                                   |
| Touches         | Mostly local calculations or local UI state.                                                           |
| Calls detected  | get, allowedOrigins, includes                                                                          |
| API endpoints   | None detected inside this function body.                                                               |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

30: function corsHeaders(request, env) {
31:   const origin = request.headers.get("Origin") || "";
32:   const allowOrigin = allowedOrigins(env).includes(origin) ? origin : defaultAllowedOrigins[0];
33:   return {
34:     "Access-Control-Allow-Origin": allowOrigin,
35:     "Access-Control-Allow-Methods": "POST, OPTIONS",
36:     "Access-Control-Allow-Headers": "Content-Type",
37:     "Access-Control-Max-Age": "86400",
38:     "Vary": "Origin"
39:   };
40: }

```

## ***clamp - lines 42-44***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 42-44 (3 source lines).                                                                                                                                                                         |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function clamp(value, maxLength) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | slice                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

42: function clamp(value, maxLength) {
43:   return String(value || "").slice(0, maxLength);
44: }

```

## ***cleanConversation - lines 46-52***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 46-52 (7 source lines).                                                                                                                                                                         |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function cleanConversation(conversation = []) {                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isArray, slice, map, clamp, filter                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

46: function cleanConversation(conversation = []) {
47:   if (!Array.isArray(conversation)) return [];
48:   return conversation.slice(-8).map((item) => ({
49:     role: item?.role === "assistant" ? "assistant" : "user",
50:     content: clamp(item?.content || item?.text || "", 1400)

```

```

51:   })).filter((item) => item.content);
52: }

```

## ***extractOpenAiText - lines 54-64***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 54-64 (11 source lines).                                                                                                                                                                        |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function extractOpenAiText(data = {}) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, push, join                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

54: function extractOpenAiText(data = {}) {
55:   if (typeof data.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
56:   const parts = [];
57:   for (const item of data.output || []) {
58:     for (const content of item.content || []) {
59:       if (typeof content.text === "string") parts.push(content.text);
60:       if (typeof content.output_text === "string") parts.push(content.output_text);
61:     }
62:   }
63:   return parts.join("\n").trim();
64: }

```

## ***extractWorkersAiText - lines 66-79***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 66-79 (14 source lines).                                                                                                                                                                        |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function extractWorkersAiText(data = {}) {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, isArray, map, filter, join                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 9 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

66: function extractWorkersAiText(data = {}) {
67:   if (typeof data.response === "string" && data.response.trim()) return data.response.trim();
68:   if (typeof data.result?.response === "string" && data.result.response.trim()) return
data.result.response.trim();
69:   if (typeof data.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
70:   if (Array.isArray(data.output)) {
71:     return data.output.map((item) => {
72:       if (typeof item === "string") return item;
73:       if (typeof item?.text === "string") return item.text;
74:       if (typeof item?.content === "string") return item.content;
75:       return "";
76:     }).filter(Boolean).join("\n").trim();
77:   }
78:   return "";
79: }

```

## ***buildInstructions - lines 81-93***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 81-93 (13 source lines).                                                                                                                                                                        |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function buildInstructions() {                                                                                                                                                                        |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | join                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

81: function buildInstructions() {
82:   return [
83:     "You are the AI assistant for a public engineering portfolio website.",
84:     "Answer naturally like a helpful technical mentor and recruiter-facing portfolio guide.",
85:     "If the question is general, answer the concept directly before mentioning portfolio context.",
86:     "If the question is about the portfolio owner, projects, files, resume, tools, links, or sections,
answer from the supplied portfolio context.",
87:     "Use recent conversation to keep a conversational flow. Greetings should receive short greetings.",
88:     "Only include portfolio links when they are directly relevant to the visitor's question.",
89:     "Never invent portfolio projects, credentials, files, test results, or private repository access.",
90:     "If a requested detail is not in the context, say what is missing and give a useful next step.",
91:     "Keep answers concise, readable, and specific. Use bullets when they make the answer easier to scan."
92:   ].join("\n");
93: }
```

## ***buildMessages - lines 95-105***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 95-105 (11 source lines).                                                                                                                                                                       |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function buildMessages(body = {}) {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | buildInstructions, cleanConversation, push, buildUserPrompt                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

95: function buildMessages(body = {}) {
96:   const messages = [{ role: "system", content: buildInstructions() }];
97:   for (const item of cleanConversation(body.conversation)) {
98:     messages.push({
99:       role: item.role === "assistant" ? "assistant" : "user",
100:      content: item.content
101:     });
102:   }
103:   messages.push({ role: "user", content: buildUserPrompt(body) });
104:   return messages;
105: }
```

## ***buildUserPrompt - lines 107-119***

| Item            | Explanation                                                               |
|-----------------|---------------------------------------------------------------------------|
| Source location | Lines 107-119 (13 source lines).                                          |
| Purpose         | Creates an object, UI structure, process, payload, or generated artifact. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function buildUserPrompt(body = {}) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | clamp, stringify, cleanConversation, join                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

107: function buildUserPrompt(body = {}) {
108:   return [
109:     `Visitor question: ${clamp(body.question, 1200)}\`,
110:     `Question intent: ${clamp(body.intent || "portfolio_specific", 80)}\`,
111:     `Web/public lookup requested by browser: ${body.allowWebSearch === true ? "yes" : "no"}\`,
112:     "",
113:     "Recent conversation JSON:",
114:     clamp(JSON.stringify(cleanConversation(body.conversation), null, 2), 6000),
115:     "",
116:     "Portfolio context JSON:",
117:     clamp(JSON.stringify(body.context || {}, null, 2), 20000)
118:   ].join("\n");
119: }

```

## ***callOpenAi - lines 121-145***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 121-145 (25 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | async function callOpenAi(body, env) {                                                                                                                                                                |
| Touches            | network/API requests, authentication/security data                                                                                                                                                    |
| Calls detected     | buildInstructions, buildUserPrompt, fetch, stringify                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

121: async function callOpenAi(body, env) {
122:   const model = env.OPENAI_MODEL || "gpt-4.1-mini";
123:   const payload = {
124:     model,
125:     input: [
126:       {
127:         role: "developer",
128:         content: [{ type: "input_text", text: buildInstructions() }]
129:       },
130:       {
131:         role: "user",
132:         content: [{ type: "input_text", text: buildUserPrompt(body) }]
133:       }
134:     ],
135:     max_output_tokens: Number(env.OPENAI_MAX_OUTPUT_TOKENS || 1000)
136:   };
137:
138:   const response = await fetch("https://api.openai.com/v1/responses", {
139:     method: "POST",
140:     headers: {
141:       "Authorization": `Bearer ${env.OPENAI_API_KEY}`,
142:       "Content-Type": "application/json"
143:     },
144:     body: JSON.stringify(payload)
145:   });

```



## ***callWorkersAi - lines 161-180***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 161-180 (20 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | async function callWorkersAi(body, env) {                                                                                                                                                             |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | run, buildMessages, extractWorkersAiText                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
161: async function callWorkersAi(body, env) {
162:   if (!env.AI || typeof env.AI.run !== "function") {
163:     return {
164:       ok: false,
165:       status: 503,
166:       error: "Cloudflare Workers AI binding is not available for this worker."
167:     };
168:   }
169:
170:   const model = env.WORKERS_AI_MODEL || "@cf/meta/llama-3.2-3b-instruct";
171:   const data = await env.AI.run(model, {
172:     messages: buildMessages(body),
173:     max_tokens: Number(env.OPENAI_MAX_OUTPUT_TOKENS || 1000)
174:   });
175:   return {
176:     ok: true,
177:     answer: extractWorkersAiText(data),
178:     model
179:   };
180: }
```

## ***fallbackWorkerAnswer - lines 182-220***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 182-220 (39 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | function fallbackWorkerAnswer(body = {}) {                                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim, toLowerCase, isArray, slice, map, filter, join, b                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
182: function fallbackWorkerAnswer(body = {}) {
183:   const question = String(body.question || "").trim();
184:   const clean = question.toLowerCase();
185:   const context = body.context || {};
186:   const projects = Array.isArray(context.projects) ? context.projects : [];
187:   const owner = context.owner || context.profile?.displayName || "Maurice Otieno";
188:
189:   if (/^(hi|hello|hey|yo)\b/.test(clean)) {
190:     return `Hi, what can I do for you? I can help with ${owner}'s projects, files, resume links, or
related electronics concepts.`;
191:   }
192:   if (/^bembedded systems?\b/.test(clean)) {
193:     return [
```

```

194:         "Embedded systems is a field of engineering where software is built into dedicated hardware to
control a device, measure signals, communicate with peripherals, or respond to real-time events.",
195:         "",
196:         "Typical embedded work combines microcontrollers or processors, firmware, timing, interrupts,
communication buses, sensors, actuators, and hardware validation.",
197:         "",
198:         projects.length
199:         ? `In this portfolio, I would connect that explanation back to saved projects such as
${projects.slice(0, 3).map((project) => project.title || project.name).filter(Boolean).join(", ")} when those
projects are relevant.`
200:         : "When project context is available, I connect the explanation to the saved project evidence."
201:     ].join("\n");
202: }
203: if (projects.length &&
/b(project|portfolio|maurice|file|github|resume|tool|design|test)\b/.test(clean)) {
204:     return [
205:         `I can use the portfolio context for ${owner}.`,
206:         "",
207:         "Relevant saved project areas include:",
208:         ...projects.slice(0, 6).map((project) => `- ${project.title || project.name || "Untitled
project"}`),
209:         "",
210:         "The full AI model is not reachable from this Worker right now, so this is a fallback answer from
the available context."
211:     ].join("\n");
212: }
213: return [
214:     "I can answer that in a general way, but the full AI model is not reachable from this Worker right
now.",
215:     "",
216:     "A useful answer should define the concept, explain the main parts, give a practical example, and
connect it to a project artifact when portfolio context is relevant.",
217:     "",
218:     `Question received: ${question}`
219: ].join("\n");
220: }

```

## Representative opening snippet

```

1: const defaultAllowedOrigins = [
2:   "https://mauriceotieno.com",
3:   "https://www.mauriceotieno.com",
4:   "https://otienomaurice.github.io",
5:   "http://localhost:8080",
6:   "http://localhost:8081",
7:   "http://127.0.0.1:8080",
8:   "http://127.0.0.1:8081"
9: ];
10:
11: function json(data, status = 200, headers = {}) {
12:   return new Response(JSON.stringify(data, null, 2), {
13:     status,
14:     headers: {
15:       "Content-Type": "application/json; charset=utf-8",
16:       "Cache-Control": "no-store",
17:       ...headers
18:     }
19:   });
20: }
21:
22: function allowedOrigins(env) {
23:   return String(env.ALLOWED_ORIGINS || "").
24:     .split(",")
25:     .map((origin) => origin.trim())
26:     .filter(Boolean)
27:     .concat(defaultAllowedOrigins);
28: }
29:
30: function corsHeaders(request, env) {
31:   const origin = request.headers.get("Origin") || "";
32:   const allowOrigin = allowedOrigins(env).includes(origin) ? origin : defaultAllowedOrigins[0];
33:   return {
34:     "Access-Control-Allow-Origin": allowOrigin,
35:     "Access-Control-Allow-Methods": "POST, OPTIONS",
36:     "Access-Control-Allow-Headers": "Content-Type",
37:     "Access-Control-Max-Age": "86400",
38:     "Vary": "Origin"
39:   };
40: }
41:
42: function clamp(value, maxLength) {
43:   return String(value || "").slice(0, maxLength);
44: }
45:
46: function cleanConversation(conversation = []) {
47:   if (!Array.isArray(conversation)) return [];
48:   return conversation.slice(-8).map((item) => ({
49:     role: item?.role === "assistant" ? "assistant" : "user",
50:     content: clamp(item?.content || item?.text || "", 1400)
51:   })).filter((item) => item.content);

```

```

52: }
53:
54: function extractOpenAiText(data = {}) {
55:   if (typeof data.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
56:   const parts = [];
57:   for (const item of data.output || []) {
58:     for (const content of item.content || []) {
59:       if (typeof content.text === "string") parts.push(content.text);
60:       if (typeof content.output_text === "string") parts.push(content.output_text);
61:     }
62:   }
63:   return parts.join("\n").trim();
64: }
65:
66: function extractWorkersAiText(data = {}) {
67:   if (typeof data.response === "string" && data.response.trim()) return data.response.trim();
68:   if (typeof data.result?.response === "string" && data.result.response.trim()) return
data.result.response.trim();
69:   if (typeof data.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
70:   if (Array.isArray(data.output)) {
71:     return data.output.map((item) => {
72:       if (typeof item === "string") return item;

```

# Source File: cloudflare/wrangler.toml

| Item                | Explanation                                                  |
|---------------------|--------------------------------------------------------------|
| File                | cloudflare/wrangler.toml                                     |
| Purpose             | Wrangler configuration used to deploy the Cloudflare Worker. |
| Lines               | 18                                                           |
| Size                | 573 bytes                                                    |
| Talks to            | Cloudflare/AI layer, GitHub/release layer                    |
| Functions           | 0                                                            |
| Variables/constants | 0                                                            |
| API mentions        | 0                                                            |
| Signals             | 8                                                            |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                        | Explanation                                                                                                                                                                    |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloudflare or AI at line 1  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: name = "omb-portfolio-ai"                                                               |
| Cloudflare or AI at line 2  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: main = "portfolio-ai-worker.js"                                                         |
| Cloudflare or AI at line 7  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: { pattern = "mauriceotieno.com/api/portfolio-ai", zone_name = "mauriceotieno.com" },    |
| Cloudflare or AI at line 8  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: { pattern = "www.mauriceotieno.com/api/portfolio-ai", zone_name = "mauriceotieno.com" } |
| Cloudflare or AI at line 12 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: OPENAI_MODEL = "gpt-4.1-mini"                                                           |
| Cloudflare or AI at line 13 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: OPENAI_MAX_OUTPUT_TOKENS = "1000"                                                       |
| Cloudflare or AI at line 14 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: WORKERS_AI_MODEL = "@cf/meta/llama-3.2-3b-instruct"                                     |
| Network request at line 15  | Frontend-backend communication or public web/API calls. Evidence: ALLOWED_ORIGINS = "https://mauriceotieno.com,https://www.mauriceotieno.com,https://otienomaurice.github.io"  |

## Representative opening snippet

```
1: name = "omb-portfolio-ai"
2: main = "portfolio-ai-worker.js"
3: compatibility_date = "2026-07-01"
4: workers_dev = true
5:
6: routes = [
7:   { pattern = "mauriceotieno.com/api/portfolio-ai", zone_name = "mauriceotieno.com" },
8:   { pattern = "www.mauriceotieno.com/api/portfolio-ai", zone_name = "mauriceotieno.com" }
9: ]
10:
11: [vars]
12: OPENAI_MODEL = "gpt-4.1-mini"
13: OPENAI_MAX_OUTPUT_TOKENS = "1000"
14: WORKERS_AI_MODEL = "@cf/meta/llama-3.2-3b-instruct"
15: ALLOWED_ORIGINS =
"https://mauriceotieno.com,https://www.mauriceotieno.com,https://otienomaurice.github.io"
16:
17: [ai]
18: binding = "AI"
```

# Source File: docs/build\_complete\_guide.py

| Item                | Explanation                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------|
| File                | docs/build_complete_guide.py                                                                                          |
| Purpose             | Tracked source/configuration file in the OMB Portfolio Builder repository.                                            |
| Lines               | 2,412                                                                                                                 |
| Size                | 163,977 bytes                                                                                                         |
| Talks to            | builder frontend, local backend, public website runtime, portfolio catalog, Cloudflare/AI layer, GitHub/release layer |
| Functions           | 75                                                                                                                    |
| Variables/constants | 0                                                                                                                     |
| API mentions        | 6                                                                                                                     |
| Signals             | 434                                                                                                                   |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Imports, requires, and external dependencies

| Item   | Explanation       |
|--------|-------------------|
| Line 3 | import html       |
| Line 4 | import json       |
| Line 5 | import math       |
| Line 6 | import re         |
| Line 7 | import shutil     |
| Line 8 | import subprocess |
| Line 9 | import textwrap   |

## API endpoints mentioned or called

| Item              | Explanation                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------|
| /api/...          | Line 93. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior. |
| /api/code/save    | Line 449. Saves source code into the local compile workspace.                                             |
| /api/code/compile | Line 642. Compiles or runs a source file and returns terminal output.                                     |
| /api/code/compile | Line 654. Compiles or runs a source file and returns terminal output.                                     |
| /api/portfolio-ai | Line 666. Handles AI assistant questions.                                                                 |
| /api/portfolio-ai | Line 697. Handles AI assistant questions.                                                                 |

## Important implementation signals

| Item                 | Explanation                                                                                                                       |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Rich text at line 12 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: from PIL import Image, ImageDraw, ImageFont  |
| Rich text at line 18 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: from docx.shared import Inches, Pt, RGBColor |
| Rich text at line 21 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: from reportlab.lib import colors             |

| Item                             | Explanation                                                                                                                                                                                                                                                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installer at line 44             | Windows setup, update, shortcut, or installation behavior. Evidence: ".github/workflows/build-windows-builder.yml": "GitHub Actions workflow that builds the Windows installer and portable application, checks the stable download link, enforces release version bumps, uploads artifacts, and                    |
| Git command at line 46           | Repository state, publishing, branch checks, or release automation. Evidence: ".gitignore": "Git ignore rules for build outputs, temporary files, and local-only data that should not be committed.",                                                                                                               |
| Cloudflare or AI at line 49      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "BRANCHING.md": "Human-readable branch model explaining development, main, feature branches, and release flow.",                                                                                                             |
| Git command at line 50           | Repository state, publishing, branch checks, or release automation. Evidence: "Backgrounds/.gitkeep": "Keeps the Backgrounds folder in Git even when no background image files are present.",                                                                                                                       |
| Cloudflare or AI at line 51      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "README.md": "Main README for the standalone builder application: install, updates, workspaces, text editing, publishing, build commands, branch model, and uninstall.",                                                     |
| Cloudflare or AI at line 52      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "_headers": "Cloudflare Pages style HTTP headers for security and caching behavior on the public website.",                                                                                                                  |
| Installer at line 61             | Windows setup, update, shortcut, or installation behavior. Evidence: "assets/omb-app-icon.ico": "Windows ICO application icon used by Electron Builder and shortcuts.",                                                                                                                                             |
| Installer at line 63             | Windows setup, update, shortcut, or installation behavior. Evidence: "build/installer.nsh": "NSIS installer customization script for update behavior, shortcut placement, old install detection, and setup details.",                                                                                               |
| Cloudflare or AI at line 65      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "cloudflare/README.md": "Cloudflare setup notes for deploying the AI Worker endpoint and wiring the public site to the backend.",                                                                                            |
| Cloudflare or AI at line 66      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "cloudflare/portfolio-ai-worker.js": "Cloudflare Worker backend for the public portfolio AI assistant and fallback intelligence path.",                                                                                      |
| Cloudflare or AI at line 67      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "cloudflare/wrangler.toml": "Wrangler configuration used to deploy the Cloudflare Worker.",                                                                                                                                  |
| Git command at line 68           | Repository state, publishing, branch checks, or release automation. Evidence: "docs/.gitkeep": "Keeps the docs folder in Git even before generated documents are added.",                                                                                                                                           |
| Installer at line 71             | Windows setup, update, shortcut, or installation behavior. Evidence: "installer-notes.txt": "Installer license/notes text shown during Windows setup.",                                                                                                                                                             |
| Compiler or simulator at line 73 | Part of the Compile Code workspace or language/tool detection. Evidence: "package.json": "Node/Electron package manifest with scripts, dependencies, version, installer settings, and extra resources.",                                                                                                            |
| Security or auth at line 81      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "server.mjs": "Local backend server used by the builder for drafts, parsing, publishing, compiler execution, authentication checks, and file operations.",                                                                   |
| Git command at line 90           | Repository state, publishing, branch checks, or release automation. Evidence: ("High-Level System Map", "The system has two personalities: a private builder app and a public website. The builder is where content is created and verified. The website is what recruiters see after publishing.", "Owner          |
| Security or auth at line 92      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ("Private Builder Versus Public Website", "The builder includes editing controls, local draft storage, authentication checks, compiler tools, and publishing actions. The public website removes those controls and only dis |
| Git command at line 93           | Repository state, publishing, branch checks, or release automation. Evidence: ("Local Backend Role", "The frontend cannot safely do everything by itself. The local backend handles file writes, repository operations, compiler execution, publishing checks, and generated site output.", "template-prev          |
| Git command at line 95           | Repository state, publishing, branch checks, or release automation. Evidence: ("GitHub Pages Publishing", "The target repository receives static files. GitHub Pages then serves those files over HTTPS. The builder does not run on the public site.", "Local publish mirror -> git commit -> git push ->          |
| Cloudflare or AI at line 96      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("Cloudflare Role", "Cloudflare can sit in front of the domain and can also host Worker endpoints such as the portfolio AI backend. DNS points the custom domain to the website host.", "Visitor DNS lookup -> Cloudflare DN |
| Cloudflare or AI at line 97      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("AI Backend Role", "The website AI should answer from portfolio context when the question is about Maurice's work and use general knowledge when the question is general. The Worker endpoint keeps API keys off the public |
| Installer at line 99             | Windows setup, update, shortcut, or installation behavior. Evidence: ("Installer And Update Role", "The installer places the app in the per-user AppData Programs folder. Updates should replace that install instead of creating duplicate copies.", "GitHub Release -> installer download -> up                   |
| Cloudflare or AI at line 100     | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("Branch Model", "Development collects work. Main is the release branch. Feature branches should merge into development first. Main should receive only development.", "feature branch -> development -> main -> release wor |

| Item                              | Explanation                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security or auth at line 101      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ("Security Boundary", "Editing is local and authenticated. Public visitors should not get editing tools. Publishing requires GitHub write access to the target repository.", "Visitor can read public site\nOwner can edit b |
| Cloudflare or AI at line 109      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "Cloudflare Role": "cloudflare-ai-flow",                                                                                                                                                                                     |
| Cloudflare or AI at line 110      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "AI Backend Role": "cloudflare-ai-flow",                                                                                                                                                                                     |
| Installer at line 112             | Windows setup, update, shortcut, or installation behavior. Evidence: "Installer And Update Role": "release-pipeline",                                                                                                                                                                                               |
| Cloudflare or AI at line 113      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "Branch Model": "release-pipeline",                                                                                                                                                                                          |
| Git command at line 118           | Repository state, publishing, branch checks, or release automation. Evidence: ("git status", "Shows what branch you are on and whether files are changed, staged, or clean.", "Run it before edits, before commits, and before merges."),                                                                           |
| Git command at line 119           | Repository state, publishing, branch checks, or release automation. Evidence: ("git switch development", "Moves your working copy to the development branch.", "Use development as the integration branch before anything goes to main."),                                                                          |
| Git command at line 120           | Repository state, publishing, branch checks, or release automation. Evidence: ("git switch -c codex/example-change", "Creates a new feature branch from the branch you are currently on.", "Use this before changing the builder so development stays stable."),                                                    |
| Git command at line 121           | Repository state, publishing, branch checks, or release automation. Evidence: ("git add <file>", "Stages a changed file for the next commit.", "Only stage files that belong to the current change."),                                                                                                              |
| Git command at line 122           | Repository state, publishing, branch checks, or release automation. Evidence: ("git commit -m \"message\"", "Records staged changes as a named checkpoint.", "Write messages that explain why the change exists."),                                                                                                 |
| Git command at line 123           | Repository state, publishing, branch checks, or release automation. Evidence: ("git push origin <branch>", "Uploads your branch to GitHub.", "This lets GitHub run workflows and enables pull requests."),                                                                                                          |
| Git command at line 124           | Repository state, publishing, branch checks, or release automation. Evidence: ("git pull --ff-only", "Updates the current branch only if Git can fast-forward safely.", "Avoids surprise merge commits."),                                                                                                          |
| Git command at line 125           | Repository state, publishing, branch checks, or release automation. Evidence: ("git fetch --tags --force", "Downloads branch and tag metadata from GitHub.", "Used before checking whether a release tag already exists."),                                                                                         |
| Compiler or simulator at line 126 | Part of the Compile Code workspace or language/tool detection. Evidence: ("pnpm install --frozen-lockfile", "Installs Node dependencies exactly as pinned in pnpm-lock.yaml.", "Makes release builds repeatable."),                                                                                                 |
| Installer at line 129             | Windows setup, update, shortcut, or installation behavior. Evidence: ("pnpm run installer", "Builds the NSIS Windows installer.", "Use this when testing installation and update behavior."),                                                                                                                       |
| Cloudflare or AI at line 132      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("wrangler deploy", "Deploys the Cloudflare Worker backend.", "Use it when the AI endpoint code changes."),                                                                                                                  |
| Compiler or simulator at line 133 | Part of the Compile Code workspace or language/tool detection. Evidence: ("gcc file.c -o app.exe", "Compiles C source into a Windows executable.", "The builder uses similar logic when C tools are available."),                                                                                                   |
| Compiler or simulator at line 135 | Part of the Compile Code workspace or language/tool detection. Evidence: ("javac Main.java", "Compiles Java source into .class bytecode.", "Java has a compile step before running."),                                                                                                                              |
| Compiler or simulator at line 136 | Part of the Compile Code workspace or language/tool detection. Evidence: ("java Main", "Runs compiled Java bytecode.", "The builder shows this output in the terminal panel."),                                                                                                                                     |
| Compiler or simulator at line 137 | Part of the Compile Code workspace or language/tool detection. Evidence: ("python script.py", "Runs Python source directly.", "Python is interpreted by default."),                                                                                                                                                 |
| Compiler or simulator at line 138 | Part of the Compile Code workspace or language/tool detection. Evidence: ("node script.js", "Runs JavaScript through Node.js.", "Useful for builder scripts and JavaScript examples."),                                                                                                                             |
| Compiler or simulator at line 139 | Part of the Compile Code workspace or language/tool detection. Evidence: ("iverilog -g2012 design.sv -o sim.vvp", "Compiles Verilog/SystemVerilog into an Icarus simulation file.", "Prepares HDL for simulation."),                                                                                                |
| Compiler or simulator at line 140 | Part of the Compile Code workspace or language/tool detection. Evidence: ("vvp sim.vvp", "Runs Icarus Verilog simulation output.", "This is how HDL terminal output appears."),                                                                                                                                     |
| Git command at line 141           | Repository state, publishing, branch checks, or release automation. Evidence: ("git push --dry-run origin main", "Tests whether GitHub would accept a push without changing the remote.", "Used during authorization checks."),                                                                                     |
| Git command at line 142           | Repository state, publishing, branch checks, or release automation. Evidence: ("git remote -v", "Shows which GitHub repository the local folder talks to.", "Important for publishing targets."),                                                                                                                   |
| Git command at line 143           | Repository state, publishing, branch checks, or release automation. Evidence: ("git branch --show-current", "Prints the active branch name.", "The builder must know whether it is pushing the right branch."),                                                                                                     |

| Item                              | Explanation                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Git command at line 144           | Repository state, publishing, branch checks, or release automation. Evidence: ("git tag --list builder-v*", "Lists builder release tags."), "Release workflows use this to prevent duplicate releases."),                                                                                                           |
| Installer at line 145             | Windows setup, update, shortcut, or installation behavior. Evidence: ("Get-ChildItem", "PowerShell command that lists files and folders."), "Used to inspect workspaces and installer output."),                                                                                                                    |
| Process execution at line 147     | Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: ("Start-Process", "PowerShell command that starts another program."), "The updater uses detached process launching."),                                                                                                   |
| Cloudflare or AI at line 152      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "Profile identity and contact details", "Front-page hero text", "Fun facts block", "Portfolio areas", "Project categories", "Project creation flow", "Project overview", "Design sections", "Simulation sections", "Files an |
| Installer at line 157             | Windows setup, update, shortcut, or installation behavior. Evidence: ("The app does not open", "Check whether the installed executable exists under AppData Local Programs. If it does, start it directly. If it fails, reinstall from the latest GitHub Release."),                                                |
| Installer at line 158             | Windows setup, update, shortcut, or installation behavior. Evidence: ("Update does not restart", "The update handoff must close Electron, run the installer, and then launch the new executable. Check the updater log and whether another builder process is still running."),                                     |
| Security or auth at line 159      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ("GitHub authentication succeeds but builder cannot push", "Authentication and repository freshness are different. Pull or load from target first if the remote branch is ahead."),                                          |
| Local storage at line 161         | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: ("Website changes show on desktop but not phone", "Check cache, confirm the published files changed, and verify the mobile renderer is using the same projects.json and CSS."),                                       |
| Cloudflare or AI at line 162      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("AI endpoint unavailable", "Verify the Worker route, environment secrets, Cloudflare deployment, and browser network errors."),                                                                                             |
| Local storage at line 163         | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: ("Compiler takes too long", "Check whether the first run is installing or detecting tools. Cached runs should be faster after the first successful compile."),                                                        |
| Compiler or simulator at line 164 | Part of the Compile Code workspace or language/tool detection. Evidence: ("SystemVerilog output missing", "Check that Icarus Verilog and vvp are available and that the file is a simulation testbench or has a runnable top module."),                                                                             |
| Rich text at line 165             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: ("Right-click formatting does not apply", "Confirm the active editor saved its selection before the menu changed font, size, or color."),                                                                                      |
| Rich text at line 166             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: ("Links are treated as formulas", "URL detection should win before formula detection. Pasted http, https, and www links must become normal links."),                                                                           |
| Installer at line 168             | Windows setup, update, shortcut, or installation behavior. Evidence: ("Installer reports an old installation", "The installer searches per-user and legacy locations. Remove stale install records only after confirming the active app path."),                                                                    |
| Git command at line 176           | Repository state, publishing, branch checks, or release automation. Evidence: "What HTML does", "What CSS does", "What JavaScript does", "What JSON does", "What Markdown does", "What YAML does", "What TOML does", "What an executable is", "What an installer is", "What a portable app is", "What a lo          |
| Git command at line 182           | Repository state, publishing, branch checks, or release automation. Evidence: ("Electron vs pure web builder", "Electron gives desktop file access and packaging. A pure web builder would be easier to access anywhere, but browser security would block many local file, compiler, and Git operations.")          |
| Installer at line 184             | Windows setup, update, shortcut, or installation behavior. Evidence: ("Per-user install vs Program Files", "Per-user installs update without admin prompts. Program Files feels traditional but causes UAC friction and update failures."),                                                                         |
| Git command at line 187           | Repository state, publishing, branch checks, or release automation. Evidence: ("Git authentication cache vs always prompt", "Caching reduces annoyance. Always prompting is stricter but makes normal publishing painful."),                                                                                        |
| Cloudflare or AI at line 188      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("Cloudflare Worker AI vs browser-only AI", "Worker AI hides secrets and can access server-side providers. Browser-only AI would expose keys or be limited to local browser models."),                                       |
| Git command at line 203           | Repository state, publishing, branch checks, or release automation. Evidence: ("git", "Git target repo\ncommit + push", 1040, 140, 250, 135, "#FEF3C7"),                                                                                                                                                            |
| Cloudflare or AI at line 206      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("ai", "Cloudflare Worker\nportfolio AI endpoint", 1040, 565, 250, 130, "#CCFBF1"),                                                                                                                                          |
| Git command at line 212           | Repository state, publishing, branch checks, or release automation. Evidence: ("backend", "git", "publishes"),                                                                                                                                                                                                      |
| Git command at line 213           | Repository state, publishing, branch checks, or release automation. Evidence: ("git", "site", "hosts files"),                                                                                                                                                                                                       |
| Cloudflare or AI at line 215      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("site", "ai", "chat API"),                                                                                                                                                                                                  |
| Git command at line 227           | Repository state, publishing, branch checks, or release automation. Evidence: ("system", "Filesystem, Git,\ncompilers, updater", 1045, 250, 300, 140, "#DCFCE7"),                                                                                                                                                   |



| Item                              | Explanation                                                                                                                                                           |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installer at line 269             | Windows setup, update, shortcut, or installation behavior. Evidence: ("release", "GitHub Release\ninstaller + portable", 1015, 370, 300, 130, "#EDE9FE"),             |
| Compiler or simulator at line 287 | Part of the Compile Code workspace or language/tool detection. Evidence: ("lang", "Language profile\nC, C++, Java,\nPython, JS, HDL", 705, 150, 260, 145, "#FEF3C7"), |
| Security or auth at line 289      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ("scope", "HDL testbench\n+ signal scope", 1030, 315, 250, 125, "#DCFCE7"),    |

## JSON/YAML-style keys detected

| Item                                        | Explanation                                                                                                                                                                                       |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| .github/workflows/build-windows-builder.yml | Line 44: ".github/workflows/build-windows-builder.yml": "GitHub Actions workflow that builds the Windows installer and portable application, checks the stable download link, enforces releases", |
| .github/workflows/main-branch-gate.yml      | Line 45: ".github/workflows/main-branch-gate.yml": "GitHub Actions guard that rejects direct pushes to main and rejects pull requests into main unless the source branch is development.",        |
| .gitignore                                  | Line 46: ".gitignore": "Git ignore rules for build outputs, temporary files, and local-only data that should not be committed.",                                                                  |
| .nojekyll                                   | Line 47: ".nojekyll": "Tells GitHub Pages not to process the site with Jekyll, so folders and static files publish exactly as generated.",                                                        |
| .well-known/security.txt                    | Line 48: ".well-known/security.txt": "Public security contact file used by browsers, researchers, and scanners to find responsible disclosure information.",                                      |
| BRANCHING.md                                | Line 49: "BRANCHING.md": "Human-readable branch model explaining development, main, feature branches, and release flow.",                                                                         |
| Backgrounds/.gitkeep                        | Line 50: "Backgrounds/.gitkeep": "Keeps the Backgrounds folder in Git even when no background image files are present.",                                                                          |
| README.md                                   | Line 51: "README.md": "Main README for the standalone builder application: install, updates, workspaces, text editing, publishing, build commands, branch model, and uninstall.",                 |
| _headers                                    | Line 52: "_headers": "Cloudflare Pages style HTTP headers for security and caching behavior on the public website.",                                                                              |
| assets/apple-touch-icon.png                 | Line 53: "assets/apple-touch-icon.png": "Apple touch icon shown when the public site is saved on iOS or compatible launch surfaces.",                                                             |
| assets/favicon-16.png                       | Line 54: "assets/favicon-16.png": "Small browser favicon used by tabs and browser UI.",                                                                                                           |
| assets/favicon-192.png                      | Line 55: "assets/favicon-192.png": "Large web app icon for Android and installable browser surfaces.",                                                                                            |
| assets/favicon-32.png                       | Line 56: "assets/favicon-32.png": "Standard browser favicon size.",                                                                                                                               |
| assets/favicon-48.png                       | Line 57: "assets/favicon-48.png": "Windows/browser icon size used by some desktop surfaces.",                                                                                                     |
| assets/favicon.ico                          | Line 58: "assets/favicon.ico": "ICO favicon bundle used by browsers that request favicon.ico.",                                                                                                   |
| assets/omb-app-icon-256.png                 | Line 59: "assets/omb-app-icon-256.png": "Builder application icon source at 256 pixels.",                                                                                                         |
| assets/omb-app-icon-512.png                 | Line 60: "assets/omb-app-icon-512.png": "Builder application icon source at 512 pixels.",                                                                                                         |
| assets/omb-app-icon.ico                     | Line 61: "assets/omb-app-icon.ico": "Windows ICO application icon used by Electron Builder and shortcuts.",                                                                                       |
| assets/omb-mark.png                         | Line 62: "assets/omb-mark.png": "OMB snake/mark image used for branding in the site and builder.",                                                                                                |
| build/installer.nsh                         | Line 63: "build/installer.nsh": "NSIS installer customization script for update behavior, shortcut placement, old install detection, and setup details.",                                         |
| builder-rich-future-sections.js             | Line 64: "builder-rich-future-sections.js": "Builder-side helper code for richer future portfolio sections and section defaults.",                                                                |
| cloudflare/README.md                        | Line 65: "cloudflare/README.md": "Cloudflare setup notes for deploying the AI Worker endpoint and wiring the public site to the backend.",                                                        |
| cloudflare/portfolio-ai-worker.js           | Line 66: "cloudflare/portfolio-ai-worker.js": "Cloudflare Worker backend for the public portfolio AI assistant and fallback intelligence path.",                                                  |
| cloudflare/wrangler.toml                    | Line 67: "cloudflare/wrangler.toml": "Wrangler configuration used to deploy the Cloudflare Worker.",                                                                                              |
| docs/.gitkeep                               | Line 68: "docs/.gitkeep": "Keeps the docs folder in Git even before generated documents are added.",                                                                                              |
| electronics-search.js                       | Line 69: "electronics-search.js": "Electronics vocabulary and search support used by the website search behavior.",                                                                               |
| index.html                                  | Line 70: "index.html": "Public website HTML shell that recruiters and visitors load in the browser.",                                                                                             |

| Item                      | Explanation                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| installer-notes.txt       | Line 71: "installer-notes.txt": "Installer license/notes text shown during Windows setup.",                                                                                               |
| main.cjs                  | Line 72: "main.cjs": "Electron main process: creates the desktop window, starts the local backend, handles menus, update handoff, and application lifecycle.",                            |
| package.json              | Line 73: "package.json": "Node/Electron package manifest with scripts, dependencies, version, installer settings, and extra resources.",                                                  |
| pnpm-lock.yaml            | Line 74: "pnpm-lock.yaml": "Pinned dependency lockfile so installs and GitHub Actions use the same dependency versions.",                                                                 |
| pnpm-workspace.yaml       | Line 75: "pnpm-workspace.yaml": "pnpm workspace declaration for the repository.",                                                                                                         |
| portfolio-README.md       | Line 76: "portfolio-README.md": "README for the generated public portfolio website rather than the builder application.",                                                                 |
| project-templates.json    | Line 77: "project-templates.json": "Appearance template definitions used by projects to control visual feel rather than content.",                                                        |
| projects.json             | Line 78: "projects.json": "Public project catalog consumed by the website after parsing builder data.",                                                                                   |
| robots.txt                | Line 79: "robots.txt": "Crawler instruction file for search engines.",                                                                                                                    |
| script.js                 | Line 80: "script.js": "Public website JavaScript for navigation, portfolio rendering, search, AI UI, and interactive project windows.",                                                   |
| server.mjs                | Line 81: "server.mjs": "Local backend server used by the builder for drafts, parsing, publishing, compiler execution, authentication checks, and file operations.",                       |
| styles.css                | Line 82: "styles.css": "Public website CSS for layout, contact area, projects, responsive behavior, AI panel, and mobile/desktop rendering.",                                             |
| template-preview.css      | Line 83: "template-preview.css": "Builder application CSS for dialogs, windows, rich editors, compile code, dark mode, guide windows, and local previews.",                               |
| template-preview.html     | Line 84: "template-preview.html": "Builder HTML shell loaded inside the Electron app and local preview runtime.",                                                                         |
| template-preview.js       | Line 85: "template-preview.js": "Builder frontend brain: state handling, rich editors, project builder, previews, parser triggers, publishing UI, compile workspace, and guide windows.", |
| High-Level System Map     | Line 105: "High-Level System Map": "system-overview",                                                                                                                                     |
| Why Electron Was Chosen   | Line 106: "Why Electron Was Chosen": "electron-runtime",                                                                                                                                  |
| Portfolio Parser Role     | Line 107: "Portfolio Parser Role": "builder-to-site-files",                                                                                                                               |
| GitHub Pages Publishing   | Line 108: "GitHub Pages Publishing": "builder-to-site-files",                                                                                                                             |
| Cloudflare Role           | Line 109: "Cloudflare Role": "cloudflare-ai-flow",                                                                                                                                        |
| AI Backend Role           | Line 110: "AI Backend Role": "cloudflare-ai-flow",                                                                                                                                        |
| Compiler Workspace Role   | Line 111: "Compiler Workspace Role": "compile-code-flow",                                                                                                                                 |
| Installer And Update Role | Line 112: "Installer And Update Role": "release-pipeline",                                                                                                                                |
| Branch Model              | Line 113: "Branch Model": "release-pipeline",                                                                                                                                             |
| name                      | Line 196: "name": "system-overview",                                                                                                                                                      |
| title                     | Line 197: "title": "Private Builder To Public Portfolio",                                                                                                                                 |
| nodes                     | Line 198: "nodes": [                                                                                                                                                                      |
| edges                     | Line 208: "edges": [                                                                                                                                                                      |
| name                      | Line 219: "name": "electron-runtime",                                                                                                                                                     |
| title                     | Line 220: "title": "Electron Runtime Inside The Desktop App",                                                                                                                             |
| nodes                     | Line 221: "nodes": [                                                                                                                                                                      |
| edges                     | Line 229: "edges": [                                                                                                                                                                      |
| name                      | Line 239: "name": "builder-to-site-files",                                                                                                                                                |
| title                     | Line 240: "title": "How Builder Edits Become Website Files",                                                                                                                              |
| nodes                     | Line 241: "nodes": [                                                                                                                                                                      |
| edges                     | Line 250: "edges": [                                                                                                                                                                      |

| Item  | Explanation                                     |
|-------|-------------------------------------------------|
| name  | Line 261: "name": "release-pipeline",           |
| title | Line 262: "title": "Builder Release Pipeline",  |
| nodes | Line 263: "nodes": [                            |
| edges | Line 272: "edges": [                            |
| name  | Line 282: "name": "compile-code-flow",          |
| title | Line 283: "title": "Compile Code Workspace",    |
| nodes | Line 284: "nodes": [                            |
| edges | Line 294: "edges": [                            |
| name  | Line 305: "name": "tool-build-map",             |
| title | Line 306: "title": "What Tool Builds What",     |
| nodes | Line 307: "nodes": [                            |
| edges | Line 316: "edges": [                            |
| name  | Line 327: "name": "file-communication-map",     |
| title | Line 328: "title": "How Key Files Communicate", |
| nodes | Line 329: "nodes": [                            |
| edges | Line 340: "edges": [                            |
| name  | Line 353: "name": "cloudflare-ai-flow",         |

Event handlers and user interactions

| Item      | Explanation                                                                                                                                                                        |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Line 454  | "sectionContent.addEventListener('click', async (event) => {\n const button = event.target.closest('[data-compile-run]');\n if (button) await compileActiveFile(project, file);\n} |
| Line 1531 | if ".addEventListener(" in line:                                                                                                                                                   |

Function-by-function detail

tracked\_files - lines 891-895

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 891-895 (5 source lines).                                                                                                                                                                       |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def tracked_files() -> list[str]:                                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | run, strip, splitlines                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

891: def tracked_files() -> list[str]:
892:     result = subprocess.run(["git", "ls-files"], cwd=REPO, text=True, capture_output=True, check=True)
893:     return [line.strip() for line in result.stdout.splitlines() if line.strip()]
894:
895:
```

## ***load\_package - lines 896-902***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 896-902 (7 source lines).                                                                                                                                                                       |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | def load_package() -> dict:                                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | loads, read_text                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
896: def load_package() -> dict:
897:     try:
898:         return json.loads((REPO / "package.json").read_text(encoding="utf-8"))
899:     except Exception:
900:         return {}
901:
902:
```

## ***load\_font - lines 903-910***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 903-910 (8 source lines).                                                                                                                                                                       |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | def load_font(size: int, bold: bool = False) -> ImageFont.ImageFont:                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Path, exists, truetype, str, load_default                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
903: def load_font(size: int, bold: bool = False) -> ImageFont.ImageFont:
904:     font_name = "arialbd.ttf" if bold else "arial.ttf"
905:     font_path = Path(r"C:\Windows\Fonts") / font_name
906:     if font_path.exists():
907:         return ImageFont.truetype(str(font_path), size=size)
908:     return ImageFont.load_default()
909:
910:
```

## ***wrap\_for\_box - lines 911-930***

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 911-930 (20 source lines).                                                                       |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | def wrap_for_box(label: str, font: ImageFont.ImageFont, max_width: int) -> list[str]:                  |
| Touches         | Mostly local calculations or local UI state.                                                           |
| Calls detected  | str, splitlines, split, append, getbbox                                                                |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

911: def wrap_for_box(label: str, font: ImageFont.ImageFont, max_width: int) -> list[str]:
912:     lines: list[str] = []
913:     for part in str(label).splitlines():
914:         words = part.split()
915:         if not words:
916:             lines.append("")
917:             continue
918:         current = words[0]
919:         for word in words[1:]:
920:             candidate = f"{current} {word}"
921:             bbox = font.getbbox(candidate)
922:             if bbox[2] - bbox[0] <= max_width:
923:                 current = candidate
924:             else:
925:                 lines.append(current)
926:                 current = word
927:         lines.append(current)
928:     return lines
929:
930:

```

### ***draw\_arrow - lines 931-943***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 931-943 (13 source lines).                                                                                                                                                                      |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def draw_arrow(draw: ImageDraw.ImageDraw, start: tuple[int, int], end: tuple[int, int], color: str, width: int = 4) -> None:                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | line, atan2, int, cos, sin, polygon                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

931: def draw_arrow(draw: ImageDraw.ImageDraw, start: tuple[int, int], end: tuple[int, int], color: str,
width: int = 4) -> None:
932:     draw.line([start, end], fill=color, width=width)
933:     angle = math.atan2(end[1] - start[1], end[0] - start[0])
934:     length = 18
935:     spread = 0.55
936:     points = [
937:         end,
938:         (int(end[0] - length * math.cos(angle - spread)), int(end[1] - length * math.sin(angle -
spread))),
939:         (int(end[0] - length * math.cos(angle + spread)), int(end[1] - length * math.sin(angle +
spread))),
940:     ]
941:     draw.polygon(points, fill=color)
942:
943:

```

### ***edge\_points - lines 944-959***

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 944-959 (16 source lines).                                                                       |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | def edge_points(source: tuple[int, int, int, int], target: tuple[int, int, int, int]) -> tuple[tuple[int, int], tuple[int, int]]:                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | abs                                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

944: def edge_points(source: tuple[int, int, int, int], target: tuple[int, int, int, int]) -> tuple[tuple[int,
int], tuple[int, int]]:
945:     sx1, sy1, sx2, sy2 = source
946:     tx1, ty1, tx2, ty2 = target
947:     sc = ((sx1 + sx2) // 2, (sy1 + sy2) // 2)
948:     tc = ((tx1 + tx2) // 2, (ty1 + ty2) // 2)
949:     dx = tc[0] - sc[0]
950:     dy = tc[1] - sc[1]
951:     if abs(dx) >= abs(dy):
952:         start = (sx2 if dx >= 0 else sx1, sc[1])
953:         end = (tx1 if dx >= 0 else tx2, tc[1])
954:     else:
955:         start = (sc[0], sy2 if dy >= 0 else sy1)
956:         end = (tc[0], ty1 if dy >= 0 else ty2)
957:     return start, end
958:
959:

```

## ***make\_diagrams - lines 960-1009***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 960-1009 (50 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def make_diagrams() -> dict[str, Path]:                                                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | mkdir, load_font, Draw, rounded_rectangle, text, line, edge_points, draw_arrow, getbbox, hypot, max, wrap_for_box                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

960: def make_diagrams() -> dict[str, Path]:
961:     DIAGRAM_DIR.mkdir(parents=True, exist_ok=True)
962:     title_font = load_font(38, bold=True)
963:     box_font = load_font(24, bold=True)
964:     label_font = load_font(18, bold=False)
965:     diagram_paths: dict[str, Path] = {}
966:
967:     for spec in DIAGRAM_SPECS:
968:         image = Image.new("RGB", (1600, 900), "#F8FBFD")
969:         draw = ImageDraw.Draw(image)
970:         draw.rounded_rectangle((24, 24, 1576, 876), radius=26, outline="#B7D7EA", width=3,
fill="#FFFFFF")
971:         draw.text((60, 48), spec["title"], fill="#0B2545", font=title_font)
972:         draw.line((60, 105, 1540, 105), fill="#B7D7EA", width=3)
973:
974:         boxes: dict[str, tuple[int, int, int, int]] = {}
975:         for node_id, label, x, y, w, h, fill in spec["nodes"]:
976:             boxes[node_id] = (x, y, x + w, y + h)
977:
978:         for source_id, target_id, *maybe_label in spec["edges"]:
979:             source = boxes[source_id]
980:             target = boxes[target_id]
981:             start, end = edge_points(source, target)

```

```

982:         draw_arrow(draw, start, end, "#2563EB", width=4)
983:     if maybe_label:
984:         label = maybe_label[0]
985:         mid = ((start[0] + end[0]) // 2, (start[1] + end[1]) // 2)
986:         text_box = label_font.getbbox(label)
987:         tw = text_box[2] - text_box[0] + 16
988:         th = text_box[3] - text_box[1] + 10
989:         if math.hypot(end[0] - start[0], end[1] - start[1]) > max(tw + 28, 95):
990:             draw.rounded_rectangle((mid[0] - tw // 2, mid[1] - th // 2, mid[0] + tw // 2, mid[1]
+ th // 2), radius=9, fill="#FFF6FF", outline="#BFDDBFE", width=1)
991:             draw.text((mid[0] - tw // 2 + 8, mid[1] - th // 2 + 4), label, fill="#1E3A8A",
font=label_font)
992:
993:     for node_id, label, x, y, w, h, fill in spec["nodes"]:
994:         draw.rounded_rectangle((x, y, x + w, y + h), radius=20, fill=fill, outline="#668EA8",
width=3)
995:         lines = wrap_for_box(label, box_font, w - 28)
996:         line_height = 30
997:         total_h = len(lines) * line_height
998:         start_y = y + (h - total_h) // 2
999:         for index, line in enumerate(lines):
1000:             bbox = box_font.getbbox(line)
1001:             tw = bbox[2] - bbox[0]
1002:             draw.text((x + (w - tw) // 2, start_y + index * line_height), line, fill="#102033",
font=box_font)
1003:
1004:     path = DIAGRAM_DIR / f"{spec['name']}.png"
1005:     image.save(path)
1006:     diagram_paths[spec["name"]] = path
1007:     return diagram_paths
1008:
1009:

```

## add\_diagram - lines 1010-1023

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1010-1023 (14 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_diagram(doc: Document, path: Path, caption: str) -> None:                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | exists, add_paragraph, add_run, add_picture, str, Inches, Pt, RGBColor                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1010: def add_diagram(doc: Document, path: Path, caption: str) -> None:
1011:     if not path.exists():
1012:         return
1013:     paragraph = doc.add_paragraph()
1014:     paragraph.alignment = WD_ALIGN_PARAGRAPH.CENTER
1015:     run = paragraph.add_run()
1016:     run.add_picture(str(path), width=Inches(6.35))
1017:     cap = doc.add_paragraph(caption)
1018:     cap.alignment = WD_ALIGN_PARAGRAPH.CENTER
1019:     for run in cap.runs:
1020:         run.font.size = Pt(9)
1021:         run.font.color.rgb = RGBColor(85, 85, 85)
1022:
1023:

```

## shade - lines 1024-1030

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 1024-1030 (7 source lines).                                                                      |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | def shade(cell, fill: str) -> None:                                                                    |
| Touches         | Mostly local calculations or local UI state.                                                           |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | get_or_add_tcPr, OxmlElement, set, qn, append                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1024: def shade(cell, fill: str) -> None:
1025:     tc_pr = cell._tc.get_or_add_tcPr()
1026:     shd = OxmlElement("w:shd")
1027:     shd.set(qn("w:fill"), fill)
1028:     tc_pr.append(shd)
1029:
1030:

```

## ***cell\_text - lines 1031-1039***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1031-1039 (9 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def cell_text(cell, text: str, bold: bool = False) -> None:                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_run, str, Pt                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1031: def cell_text(cell, text: str, bold: bool = False) -> None:
1032:     cell.text = ""
1033:     run = cell.paragraphs[0].add_run(str(text))
1034:     run.bold = bold
1035:     run.font.name = "Calibri"
1036:     run.font.size = Pt(9)
1037:     cell.vertical_alignment = WD_CELL_VERTICAL_ALIGNMENT.CENTER
1038:
1039:

```

## ***add\_table - lines 1040-1056***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1040-1056 (17 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_table(doc: Document, rows: list[tuple[str, str]]) -> None:                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Inches, cell_text, shade, add_row                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |



Source excerpt:

```
1040: def add_table(doc: Document, rows: list[tuple[str, str]]) -> None:
1041:     table = doc.add_table(rows=1, cols=2)
1042:     table.alignment = WD_TABLE_ALIGNMENT.CENTER
1043:     table.autofit = False
1044:     table.columns[0].width = Inches(1.55)
1045:     table.columns[1].width = Inches(4.95)
1046:     head = table.rows[0].cells
1047:     cell_text(head[0], "Item", True)
1048:     cell_text(head[1], "Explanation", True)
1049:     shade(head[0], "E8EEF5")
1050:     shade(head[1], "E8EEF5")
1051:     for label, value in rows:
1052:         cells = table.add_row().cells
1053:         cell_text(cells[0], label, True)
1054:         cell_text(cells[1], value, False)
1055:
1056:
```

***add\_code - lines 1057-1064***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1057-1064 (8 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_code(doc: Document, text: str) -> None:                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_paragraph, add_run, rstrip, set, qn                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1057: def add_code(doc: Document, text: str) -> None:
1058:     p = doc.add_paragraph()
1059:     p.style = doc.styles["CodeBlock"]
1060:     run = p.add_run((text or "").rstrip())
1061:     run.font.name = "Consolas"
1062:     run._element.rPr.rFonts.set(qn("w: eastAsia"), "Consolas")
1063:
1064:
```

***add\_note - lines 1065-1079***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1065-1079 (15 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_note(doc: Document, title: str, body: str) -> None:                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_table, Inches, cell, shade, add_run, RGBColor, Pt, add_paragraph                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1065: def add_note(doc: Document, title: str, body: str) -> None:
1066:     table = doc.add_table(rows=1, cols=1)
1067:     table.alignment = WD_TABLE_ALIGNMENT.CENTER
1068:     table.autofit = False
1069:     table.columns[0].width = Inches(6.35)
```

```

1070:     cell = table.cell(0, 0)
1071:     shade(cell, "F4F8FC")
1072:     title_run = cell.paragraphs[0].add_run(title)
1073:     title_run.bold = True
1074:     title_run.font.color.rgb = RGBColor(31, 77, 120)
1075:     title_run.font.size = Pt(10)
1076:     paragraph = cell.add_paragraph(body)
1077:     paragraph.style = "Body Text"
1078:
1079:

```

## ***bullets - lines 1080-1084***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1080-1084 (5 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def bullets(doc: Document, items: list[str]) -> None:                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_paragraph                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1080: def bullets(doc: Document, items: list[str]) -> None:
1081:     for item in items:
1082:         doc.add_paragraph(item, style="List Bullet")
1083:
1084:

```

## ***numbers - lines 1085-1089***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1085-1089 (5 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def numbers(doc: Document, items: list[str]) -> None:                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_paragraph                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1085: def numbers(doc: Document, items: list[str]) -> None:
1086:     for item in items:
1087:         doc.add_paragraph(item, style="List Number")
1088:
1089:

```

## ***setup\_document - lines 1090-1139***

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 1090-1139 (50 source lines).                                                                     |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | def setup_document() -> Document:                                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | Document, Inches, set, qn, Pt, from_string, add_style, RGBColor                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1090: def setup_document() -> Document:
1091:     doc = Document()
1092:     section = doc.sections[0]
1093:     section.page_height = Inches(11)
1094:     section.page_width = Inches(8.5)
1095:     for section in doc.sections:
1096:         section.top_margin = Inches(0.65)
1097:         section.bottom_margin = Inches(0.6)
1098:         section.left_margin = Inches(0.65)
1099:         section.right_margin = Inches(0.65)
1100:         section.header_distance = Inches(0.28)
1101:         section.footer_distance = Inches(0.28)
1102:
1103:     normal = doc.styles["Normal"]
1104:     normal.font.name = "Calibri"
1105:     normal._element.rPr.rFonts.set(qn("w:eastAsia"), "Calibri")
1106:     normal.font.size = Pt(10)
1107:     normal.paragraph_format.space_after = Pt(3)
1108:     normal.paragraph_format.line_spacing = 1.08
1109:
1110:     for name, size, color, before, after in [
1111:         ("Heading 1", 14, "2E74B5", 9, 4),
1112:         ("Heading 2", 12, "2E74B5", 7, 3),
1113:         ("Heading 3", 10.5, "1F4D78", 5, 2),
1114:     ]:
1115:         style = doc.styles[name]
1116:         style.font.name = "Calibri"
1117:         style._element.rPr.rFonts.set(qn("w:eastAsia"), "Calibri")
1118:         style.font.size = Pt(size)
1119:         style.font.color.rgb = RGBColor.from_string(color)
1120:         style.paragraph_format.space_before = Pt(before)
1121:         style.paragraph_format.space_after = Pt(after)
1122:
1123:     code = doc.styles.add_style("CodeBlock", 1)
1124:     code.font.name = "Consolas"
1125:     code._element.rPr.rFonts.set(qn("w:eastAsia"), "Consolas")
1126:     code.font.size = Pt(7.2)
1127:     code.paragraph_format.space_before = Pt(2)
1128:     code.paragraph_format.space_after = Pt(3)
1129:     code.paragraph_format.line_spacing = 1.0
1130:
1131:     header = doc.sections[0].header.paragraphs[0]
1132:     header.text = "OMB Portfolio Builder and Website Complete Guide"
1133:     header.alignment = WD_ALIGN_PARAGRAPH.RIGHT
1134:     for run in header.runs:
1135:         run.font.size = Pt(8)
1136:         run.font.color.rgb = RGBColor(85, 85, 85)
1137:     return doc
1138:
1139:

```

## ***add\_cover - lines 1140-1157***

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 1140-1157 (18 source lines).                                                                     |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | def add_cover(doc: Document, package: dict) -> None:                                                   |
| Touches         | filesystem                                                                                             |
| Calls detected  | add_paragraph, add_run, Pt, RGBColor, get, add_note, add_page_break                                    |
| API endpoints   | None detected inside this function body.                                                               |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1140: def add_cover(doc: Document, package: dict) -> None:
1141:     title = doc.add_paragraph()
1142:     title.alignment = WD_ALIGN_PARAGRAPH.CENTER
1143:     run = title.add_run("OMB Portfolio Builder and Website\nComplete Creation Guide")
1144:     run.bold = True
1145:     run.font.size = Pt(26)
1146:     run.font.color.rgb = RGBColor(11, 37, 69)
1147:     subtitle = doc.add_paragraph()
1148:     subtitle.alignment = WD_ALIGN_PARAGRAPH.CENTER
1149:     r = subtitle.add_run("A beginner-friendly manual explaining the desktop app, public website, GitHub
workflows, installer, parser, compiler tools, AI backend, and publishing system.")
1150:     r.font.size = Pt(12)
1151:     meta = doc.add_paragraph()
1152:     meta.alignment = WD_ALIGN_PARAGRAPH.CENTER
1153:     meta.add_run(f"Repository: otienomaurice/omb-portfolio-builder\nVersion documented:
{package.get('version', 'unknown')}\nGenerated: July 4, 2026")
1154:     add_note(doc, "How this guide is written", "This guide starts from the highest-level idea and then
drills down. It intentionally explains terms that engineers usually skip: what Electron is, what a workflow is,
what a local backend is, why Git branches exist, and why the builder and website are separated.")
1155:     doc.add_page_break()
1156:
1157:

```

## ***add\_reading\_map - lines 1158-1171***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1158-1171 (14 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_reading_map(doc: Document) -> None:                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_heading, add_paragraph, add_table, add_page_break                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1158: def add_reading_map(doc: Document) -> None:
1159:     doc.add_heading("Reading Map", level=1)
1160:     doc.add_paragraph("Use this manual in layers. First read the high-level design pages. Then read the
app flow pages. After that, use the file-by-file and command pages as a reference when you need to understand a
specific file or command.")
1161:     add_table(doc, [
1162:         ("Part 1", "High-level architecture and big-picture mental model."),
1163:         ("Part 2", "Builder app internals: Electron, local backend, parser, rich editors, previews,
compile code, and publishing."),
1164:         ("Part 3", "Public website internals: HTML, CSS, JavaScript, projects.json, assets, search, AI,
and mobile behavior."),
1165:         ("Part 4", "GitHub, branches, workflows, releases, installers, and updates."),
1166:         ("Part 5", "Every tracked file explained in plain English."),
1167:         ("Part 6", "Commands, tradeoffs, troubleshooting, and safe operating procedures."),
1168:     ])
1169:     doc.add_page_break()
1170:
1171:

```

## ***add\_entry\_points - lines 1172-1188***

| Item            | Explanation                        |
|-----------------|------------------------------------|
| Source location | Lines 1172-1188 (17 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_entry_points(doc: Document) -> None:                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | add_heading, add_paragraph, add_table, numbers, add_page_break                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1172: def add_entry_points(doc: Document) -> None:
1173:     doc.add_heading("Where Someone Starts: Entry Points", level=1)
1174:     doc.add_paragraph("An entry point is the first file, executable, or function that starts a flow. This
project has several entry points because normal users, developers, GitHub Actions, Cloudflare, and installers
all start in different places.")
1175:     add_table(doc, [(name, f"{entry}. {meaning}")] for name, entry, meaning in ENTRY_POINTS])
1176:     doc.add_heading("Fastest way to orient yourself", level=2)
1177:     numbers(doc, [
1178:         "If you are using the installed app, start with OMB Portfolio Builder.exe.",
1179:         "If you are debugging desktop startup, start with package.json and main.cjs.",
1180:         "If you are debugging local APIs, start with server.mjs.",
1181:         "If you are debugging builder UI, start with template-preview.html, template-preview.js, and
template-preview.css.",
1182:         "If you are debugging the public website, start with index.html, script.js, styles.css, and
projects.json.",
1183:         "If you are debugging installer behavior, start with package.json build.nsis and
build/installer.nsh.",
1184:         "If you are debugging releases, start with .github/workflows/build-windows-builder.yml.",
1185:     ])
1186:     doc.add_page_break()
1187:
1188:

```

## ***add\_flow\_walkthroughs - lines 1189-1199***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1189-1199 (11 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_flow_walkthroughs(doc: Document, diagrams: dict[str, Path]) -> None:                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_heading, add_diagram, add_paragraph, add_table, add_page_break                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1189: def add_flow_walkthroughs(doc: Document, diagrams: dict[str, Path]) -> None:
1190:     for title, diagram_name, steps, debug_note in FLOW_WALKTHROUGHS:
1191:         doc.add_heading(f"Detailed Walkthrough: {title}", level=1)
1192:         add_diagram(doc, diagrams[diagram_name], f"Context diagram for {title}.")
1193:         doc.add_paragraph("This walkthrough follows the real direction of work through the system. Read
it slowly: each line is one handoff from one layer to another.")
1194:         add_table(doc, steps)
1195:         doc.add_heading("Debug note", level=2)
1196:         doc.add_paragraph(debug_note)
1197:         doc.add_page_break()
1198:
1199:

```

## ***add\_data\_contracts - lines 1200-1212***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1200-1212 (13 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_data_contracts(doc: Document) -> None:                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_heading, add_paragraph, add_code, add_page_break                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1200: def add_data_contracts(doc: Document) -> None:
1201:     for title, purpose, example, note in DATA_CONTRACTS:
1202:         doc.add_heading(f"Data Contract: {title}", level=1)
1203:         doc.add_paragraph(purpose)
1204:         doc.add_heading("Example shape", level=2)
1205:         add_code(doc, example)
1206:         doc.add_heading("How to read this", level=2)
1207:         doc.add_paragraph(note)
1208:         doc.add_heading("Why contracts matter", level=2)
1209:         doc.add_paragraph("A data contract is an agreement between two pieces of software. If the
frontend sends one shape and the backend expects another shape, the feature breaks even if both files are
individually valid code.")
1210:         doc.add_page_break()
1211:
1212:

```

## ***endpoint\_purpose - lines 1213-1242***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1213-1242 (30 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def endpoint_purpose(endpoint: str) -> str:                                                                                                                                                           |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 14 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1213: def endpoint_purpose(endpoint: str) -> str:
1214:     if "code/compile" in endpoint:
1215:         return "Compiles or runs a source file and returns terminal output."
1216:     if "code/save" in endpoint:
1217:         return "Saves source code into the local compile workspace."
1218:     if "code/beautify" in endpoint:
1219:         return "Formats source code for cleaner display and editing."
1220:     if "code/tools" in endpoint:
1221:         return "Checks compiler/runtime availability."
1222:     if "code/install" in endpoint:
1223:         return "Attempts to install missing compiler tools."
1224:     if "publish-target" in endpoint:
1225:         return "Reads or writes the Git publishing target and authorization state."
1226:     if "publish" in endpoint:
1227:         return "Publishes generated website files to the target repository."
1228:     if "catalog" in endpoint:
1229:         return "Loads project/catalog data for the builder."
1230:     if "templates" in endpoint:
1231:         return "Loads appearance template definitions."
1232:     if "app-update" in endpoint:
1233:         return "Checks or starts builder app update behavior."

```

```

1234:     if "security-report" in endpoint:
1235:         return "Returns security/download/auth reporting for the builder."
1236:     if "portfolio-ai" in endpoint:
1237:         return "Handles AI assistant questions."
1238:     if "system-check" in endpoint:
1239:         return "Checks local tool/system readiness."
1240:     return "Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact
behavior."
1241:
1242:

```

## ***extract\_api\_endpoints - lines 1243-1255***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1243-1255 (13 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def extract_api_endpoints() -> list[tuple[str, str, str]]:                                                                                                                                            |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | exists, enumerate, read_text, splitlines, findall, rstrip, min, get, str, sorted, items                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1243: def extract_api_endpoints() -> list[tuple[str, str, str]]:
1244:     endpoints: dict[tuple[str, str], int] = {}
1245:     for file_name in ["template-preview.js", "script.js", "server.mjs", "index.html"]:
1246:         path = REPO / file_name
1247:         if not path.exists():
1248:             continue
1249:         for line_no, line in enumerate(path.read_text(encoding="utf-8", errors="replace").splitlines(),
start=1):
1250:             for match in re.findall(r"['\"](/api/[A-Za-z0-9_./-]+)", line):
1251:                 clean = match.rstrip("/")
1252:                 endpoints[(clean, file_name)] = min(line_no, endpoints.get((clean, file_name), line_no))
1253:     return [(endpoint, file_name, str(line_no)) for (endpoint, file_name), line_no in
sorted(endpoints.items())]
1254:
1255:

```

## ***add\_api\_endpoint\_catalog - lines 1256-1278***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1256-1278 (23 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_api_endpoint_catalog(doc: Document) -> None:                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | extract_api_endpoints, add_heading, add_paragraph, add_page_break, endpoint_purpose, add_table                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1256: def add_api_endpoint_catalog(doc: Document) -> None:
1257:     endpoints = extract_api_endpoints()
1258:     doc.add_heading("API Endpoint Catalog", level=1)
1259:     doc.add_paragraph("This catalog is generated from strings found in the source files. It tells you
which /api paths are used and where to start reading when a request fails.")
1260:     if not endpoints:
1261:         doc.add_paragraph("No /api endpoints were found by the scanner.")

```

```

1262:         doc.add_page_break()
1263:         return
1264:     rows = [(endpoint, f"Seen in {file_name} near line {line_no}. Purpose: {endpoint_purpose(endpoint)}")
for endpoint, file_name, line_no in endpoints]
1265:     add_table(doc, rows)
1266:     doc.add_page_break()
1267:     for endpoint, file_name, line_no in endpoints:
1268:         doc.add_heading(f"API Detail: {endpoint}", level=1)
1269:         add_table(doc, [
1270:             ("Where seen", f"{file_name} near line {line_no}"),
1271:             ("Purpose", endpoint_purpose(endpoint)),
1272:             ("Frontend question", "Which button, form, or page action sends this request?"),
1273:             ("Backend question", "Which handler in server.mjs receives this path and what files/tools
does it touch?"),
1274:             ("Debugging", "Check browser network status, response JSON, server logs, and whether stale
cache was avoided with no-store or a timestamp."),
1275:         ])
1276:         doc.add_page_break()
1277:
1278:

```

## function\_purpose - lines 1279-1317

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1279-1317 (39 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def function_purpose(file_name: str, function_name: str) -> str:                                                                                                                                      |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | lower, startswith                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 18 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

### Source excerpt:

```

1279: def function_purpose(file_name: str, function_name: str) -> str:
1280:     name = function_name.lower()
1281:     if name.startswith("render"):
1282:         return "Renders UI, markup, or display output."
1283:     if name.startswith("handle"):
1284:         return "Handles an event, request, command, or user action."
1285:     if name.startswith("update"):
1286:         return "Updates UI, state, cache, or stored data."
1287:     if name.startswith("save") or name.startswith("write") or name.startswith("store"):
1288:         return "Persists data to local state, disk, credentials, or browser storage."
1289:     if name.startswith("read") or name.startswith("load") or name.startswith("fetch"):
1290:         return "Loads data from disk, network, browser storage, or a service."
1291:     if "compile" in name or "compiler" in name:
1292:         return "Part of the Compile Code language/tool/build/run flow."
1293:     if "publish" in name or "git" in name or "remote" in name:
1294:         return "Part of publishing, Git, target repository, or authorization behavior."
1295:     if "auth" in name or "credential" in name or "access" in name:
1296:         return "Part of authentication, authorization, or credential checking."
1297:     if "cache" in name:
1298:         return "Reads, writes, or validates cached state."
1299:     if name.startswith("validate") or name.startswith("normalize") or name.startswith("safe") or
name.startswith("sanitize"):
1300:         return "Validates or normalizes input so later code receives safe predictable values."
1301:     if name.startswith("create") or name.startswith("new") or name.startswith("build"):
1302:         return "Creates an object, UI structure, process, payload, or generated artifact."
1303:     if name.startswith("open") or name.startswith("close") or name.startswith("show") or
name.startswith("hide"):
1304:         return "Controls a window, dialog, panel, menu, or visible state."
1305:     if name.startswith("sync"):
1306:         return "Keeps two places aligned, such as local data and target files."
1307:     if file_name == "main.cjs":
1308:         return "Part of Electron desktop startup, window control, workspace setup, menu behavior, or
update handoff."
1309:     if file_name == "server.mjs":
1310:         return "Part of the local backend API, filesystem, compiler, Git, AI, update, or publish
workflow."
1311:     if file_name == "template-preview.js":
1312:         return "Part of the private builder frontend and editing experience."
1313:     if file_name == "script.js":
1314:         return "Part of the public website runtime."
1315:     return "Named function in the repository. Inspect the surrounding lines to learn its exact inputs and
outputs."

```



1316:  
1317:

## ***line\_excerpt - lines 1352-1361***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1352-1361 (10 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def line_excerpt(lines: list[str], start: int, end: int, max_lines: int = 120) -> str:                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | max, min, len, enumerate, append, join                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1352: def line_excerpt(lines: list[str], start: int, end: int, max_lines: int = 120) -> str:
1353:     selected = lines[max(start - 1, 0): min(end, len(lines))]
1354:     truncated = len(selected) > max_lines
1355:     selected = selected[:max_lines]
1356:     rendered = [f"{start + index:>5}: {line}" for index, line in enumerate(selected)]
1357:     if truncated:
1358:         rendered.append(f"        ... excerpt truncated after {max_lines} lines. Open the source file for
the rest of this function.")
1359:     return "\n".join(rendered)
1360:
1361:
```

## ***find\_block\_end - lines 1362-1377***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1362-1377 (16 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def find_block_end(lines: list[str], start_index: int) -> int:                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | range, min, len, sub, count, startswith                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1362: def find_block_end(lines: list[str], start_index: int) -> int:
1363:     brace_balance = 0
1364:     saw_brace = False
1365:     for index in range(start_index, min(start_index + 500, len(lines))):
1366:         line = re.sub(r"//.*$", "", lines[index])
1367:         brace_balance += line.count("{")
1368:         brace_balance -= line.count("}")
1369:         if "{" in line:
1370:             saw_brace = True
1371:         if saw_brace and brace_balance <= 0 and index > start_index:
1372:             return index + 1
1373:         if not saw_brace and index > start_index and not lines[index].startswith((" ", "\t")):
1374:             return index
1375:     return min(start_index + 80, len(lines))
1376:
1377:
```

## extract\_function\_blocks - lines 1378-1423

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1378-1423 (46 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def extract_function_blocks(repo_file: str, lines: list[str]) -> list[dict]:                                                                                                                          |
| Touches            | browser/Electron DOM, filesystem, network/API requests, authentication/security data                                                                                                                  |
| Calls detected     | enumerate, strip, search, group, lstrip, startswith, next, range, len, min, find_block_end, join                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1378: def extract_function_blocks(repo_file: str, lines: list[str]) -> list[dict]:
1379:     functions = []
1380:     for index, line in enumerate(lines):
1381:         stripped = line.strip()
1382:         for pattern in FUNCTION_PATTERNS:
1383:             match = pattern.search(line)
1384:             if not match:
1385:                 continue
1386:             name = match.group(1)
1387:             if line.lstrip().startswith("Function "):
1388:                 end = next((i + 1 for i in range(index + 1, len(lines)) if
lines[i].lstrip().startswith("FunctionEnd")), min(index + 80, len(lines)))
1389:             elif line.lstrip().startswith("def "):
1390:                 end = next((i for i in range(index + 1, len(lines)) if lines[i] and not
lines[i].startswith((" ", "\t", "@"))), min(index + 120, len(lines)))
1391:             else:
1392:                 end = find_block_end(lines, index)
1393:             body = "\n".join(lines[index:end])
1394:             call_candidates = re.findall(r"\b([A-Za-z_$][\w$]*)s*(", body)
1395:             calls = []
1396:             for candidate in call_candidates:
1397:                 if candidate not in COMMON_CALL_WORDS and candidate != name and candidate not in calls:
1398:                     calls.append(candidate)
1399:             endpoint_hits = sorted(set(match.rstrip("/") for match in
re.findall(r"['\"](/api/[A-Za-z0-9_-./-]+)", body)))
1400:             local_storage_keys = sorted(set(match for match in
re.findall(r"(?:localStorage|sessionStorage)\.(?:getItem|setItem|removeItem)\(['\"](^[^\"']+)", body)))
1401:             functions.append({
1402:                 "line": index + 1,
1403:                 "end_line": end,
1404:                 "name": name,
1405:                 "purpose": function_purpose(Path(repo_file).name, name),
1406:                 "signature": stripped[:240],
1407:                 "excerpt": line_excerpt(lines, index + 1, end, max_lines=120),
1408:                 "line_count": max(end - index, 1),
1409:                 "calls": calls[:30],
1410:                 "endpoints": endpoint_hits[:30],
1411:                 "storage_keys": local_storage_keys[:30],
1412:                 "returns": len(re.findall(r"\breturn\b", body)),
1413:                 "awaits": len(re.findall(r"\bawait\b", body)),
1414:                 "touches_dom":
bool(re.search(r"document\.|querySelector|getElementById|classList|dataset", body)),
1415:                 "touches_files":
bool(re.search(r"readFile|writeFile|appendFile|copyFile|mkdir|rm|unlink|fs\.", body, re.IGNORECASE)),
1416:                 "runs_process": bool(re.search(r"\bspawn\b|\bexecFile\b|\bexec\b", body, re.IGNORECASE)),
1417:                 "uses_network": bool(re.search(r"\bfetch\s*([https?://|api/]", body, re.IGNORECASE)),
1418:                 "uses_security":
bool(re.search(r"auth|token|secret|credential|permission|authorize|scope", body, re.IGNORECASE)),
1419:             })
1420:             break
1421:     return functions
1422:
1423:
```

## collect\_signal\_hits - lines 1424-1441

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 1424-1441 (18 source lines).                                                                     |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | def collect_signal_hits(lines: list[str]) -> list[dict]:                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | enumerate, strip, search, append                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1424: def collect_signal_hits(lines: list[str]) -> list[dict]:
1425:     hits = []
1426:     for line_no, line in enumerate(lines, start=1):
1427:         stripped = line.strip()
1428:         if not stripped:
1429:             continue
1430:         for name, pattern, explanation in SOURCE_SIGNAL_RULES:
1431:             if pattern.search(line):
1432:                 hits.append({
1433:                     "line": line_no,
1434:                     "type": name,
1435:                     "evidence": stripped[:220],
1436:                     "meaning": explanation,
1437:                 })
1438:             break
1439:     return hits
1440:
1441:

```

## ***function\_detail\_rows - lines 1442-1466***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1442-1466 (25 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def function_detail_rows(item: dict) -> list[tuple[str, str]]:                                                                                                                                        |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | append, join, expression, statement                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1442: def function_detail_rows(item: dict) -> list[tuple[str, str]]:
1443:     touches = []
1444:     if item["touches_dom"]:
1445:         touches.append("browser/Electron DOM")
1446:     if item["touches_files"]:
1447:         touches.append("filesystem")
1448:     if item["runs_process"]:
1449:         touches.append("external processes")
1450:     if item["uses_network"]:
1451:         touches.append("network/API requests")
1452:     if item["uses_security"]:
1453:         touches.append("authentication/security data")
1454:     return [
1455:         ("Source location", f"Lines {item['line']}-{item['end_line']} ({item['line_count']} source
lines)."),
1456:         ("Purpose", item["purpose"]),
1457:         ("Declaration", item["signature"]),
1458:         ("Touches", ", ".join(touches) if touches else "Mostly local calculations or local UI state."),
1459:         ("Calls detected", ", ".join(item["calls"][:12]) if item["calls"] else "No obvious named calls
detected by the generator."),
1460:         ("API endpoints", ", ".join(item["endpoints"]) if item["endpoints"] else "None detected inside
this function body."),

```

```

1461:         ("Storage keys", "", ".join(item["storage_keys"]) if item["storage_keys"] else "No local/session
storage keys detected."),
1462:         ("Async/return clues", f"{item['awaits']} await expression(s), {item['returns']} return
statement(s)."),
1463:         ("Debug approach", "Trigger the user action that reaches this function, inspect inputs/state
before the function, then inspect the returned value, DOM update, file write, process output, or API response
after it runs."),
1464:     ]
1465:
1466:

```

## ***extract\_functions - lines 1467-1482***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1467-1482 (16 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def extract_functions() -> list[tuple[str, int, str, str]]:                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | exists, enumerate, read_text, splitlines, search, group, append, function_purpose                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1467: def extract_functions() -> list[tuple[str, int, str, str]]:
1468:     function_rows: list[tuple[str, int, str, str]] = []
1469:     for file_name in ["main.cjs", "server.mjs", "template-preview.js", "script.js",
"cloudflare/portfolio-ai-worker.js", "docs/build_complete_guide.py", "build/installer.nsh"]:
1470:         path = REPO / file_name
1471:         if not path.exists():
1472:             continue
1473:         for line_no, line in enumerate(path.read_text(encoding="utf-8", errors="replace").splitlines(),
start=1):
1474:             for pattern in FUNCTION_PATTERNS:
1475:                 match = pattern.search(line)
1476:                 if match:
1477:                     name = match.group(1)
1478:                     function_rows.append((file_name, line_no, name, function_purpose(file_name, name)))
1479:                     break
1480:     return function_rows
1481:
1482:

```

## ***is\_text\_code\_file - lines 1488-1496***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1488-1496 (9 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def is_text_code_file(repo_file: str) -> bool:                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | startswith, endswith, is_file, lower                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1488: def is_text_code_file(repo_file: str) -> bool:
1489:     path = REPO / repo_file
1490:     if repo_file.startswith(("docs/code-reference/", "docs/code-reference-docx/", "docs/code-reference-
pdf/", "docs/guide-render-check/")):

```

```

1491:         return False
1492:     if repo_file.endswith(("docx", ".pdf", ".png", ".ico")):
1493:         return False
1494:     return path.isfile() and path.suffix.lower() in TEXT_CODE_EXTENSIONS
1495:
1496:

```

## source\_lines - lines 1497-1503

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1497-1503 (7 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def source_lines(repo_file: str) -> list[str]:                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | exists, is_file, read_text, splitlines                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1497: def source_lines(repo_file: str) -> list[str]:
1498:     path = REPO / repo_file
1499:     if not path.exists() or not path.isfile():
1500:         return []
1501:     return path.read_text(encoding="utf-8", errors="replace").splitlines()
1502:
1503:

```

## extract\_source\_facts - lines 1504-1563

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1504-1563 (60 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def extract_source_facts(repo_file: str) -> dict:                                                                                                                                                     |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | source_lines, join, extract_function_blocks, enumerate, strip, match, len, append, group, findall, rstrip, search                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1504: def extract_source_facts(repo_file: str) -> dict:
1505:     lines = source_lines(repo_file)
1506:     text = "\n".join(lines)
1507:     functions = extract_function_blocks(repo_file, lines)
1508:     variables = []
1509:     imports = []
1510:     endpoints = []
1511:     selectors = []
1512:     events = []
1513:     routes = []
1514:     json_keys = []
1515:     for line_no, line in enumerate(lines, start=1):
1516:         stripped = line.strip()
1517:         var_match = re.match(r"^\s*(?:const|let|var)\s+([A-Za-z_$][\w$]*)\s*=", line)
1518:         if var_match and len(variables) < 140:
1519:             variables.append({
1520:                 "line": line_no,
1521:                 "name": var_match.group(1),
1522:                 "statement": stripped[:180],

```

```

1523:         })
1524:         if re.match(r"^s*import\s+", line) or re.match(r"^s*(?:const|let|var)\s+\S+\s*=\s*require\(",
line):
1525:             imports.append({"line": line_no, "statement": stripped[:220]})
1526:         for match in re.findall(r"['\"](\/api\/[A-Za-z0-9_\/-]+)", line):
1527:             endpoints.append({"line": line_no, "endpoint": match.rstrip("/")})
1528:         route_match = re.search(r"\bapp\.(get|post|put|delete|patch)\(['\"]([^\"]+)\)", line)
1529:         if route_match:
1530:             routes.append({"line": line_no, "method": route_match.group(1).upper(), "path":
route_match.group(2)})
1531:         if ".addEventListener(" in line:
1532:             events.append({"line": line_no, "statement": stripped[:180]})
1533:         json_match = re.match(r"^s*"([^\"]+)\s*:', line)
1534:         if json_match and len(json_keys) < 120:
1535:             json_keys.append({"line": line_no, "key": json_match.group(1), "statement": stripped[:180]})
1536:         if repo_file.endswith(".css"):
1537:             selector_match = re.match(r"^s*"([.#A-Za-z0-9_\-:\[\]\(\)\s>+.=\\'"]+)\s*\{", line)
1538:             if selector_match and len(selectors) < 140:
1539:                 selectors.append({"line": line_no, "selector": selector_match.group(1).strip()[:160]})
1540:         return {
1541:             "repo_file": repo_file,
1542:             "line_count": len(lines),
1543:             "size_bytes": (REPO / repo_file).stat().st_size if (REPO / repo_file).exists() else 0,
1544:             "description": FILE_DESCRIPTIONS.get(repo_file, "Tracked source/configuration file in the OMB
Portfolio Builder repository."),
1545:             "functions": functions,
1546:             "variables": variables,
1547:             "imports": imports,
1548:             "endpoints": endpoints,
1549:             "routes": routes,
1550:             "selectors": selectors,
1551:             "events": events,
1552:             "json_keys": json_keys,
1553:             "signals": collect_signal_hits(lines),
1554:             "mentions_server": "server.mjs" in text,
1555:             "mentions_template_preview": "template-preview" in text,
1556:             "mentions_script": "script.js" in text,
1557:             "mentions_projects_json": "projects.json" in text,
1558:             "mentions_github": "github" in text.lower(),
1559:             "mentions_cloudflare": "cloudflare" in text.lower() or "worker" in text.lower(),
1560:             "snippet": line_excerpt(lines, 1, min(72, len(lines)), max_lines=72) if lines else "",
1561:         }
1562:
1563:

```

## fact\_relationship\_summary - lines 1564-1580

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1564-1580 (17 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def fact_relationship_summary(facts: dict) -> str:                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | get, append, join                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1564: def fact_relationship_summary(facts: dict) -> str:
1565:     related = []
1566:     if facts.get("mentions_template_preview"):
1567:         related.append("builder frontend")
1568:     if facts.get("mentions_server"):
1569:         related.append("local backend")
1570:     if facts.get("mentions_script"):
1571:         related.append("public website runtime")
1572:     if facts.get("mentions_projects_json"):
1573:         related.append("portfolio catalog")
1574:     if facts.get("mentions_cloudflare"):
1575:         related.append("Cloudflare/AI layer")
1576:     if facts.get("mentions_github"):
1577:         related.append("GitHub/release layer")
1578:     return ", ".join(related) if related else "Mostly local to its own feature area."
1579:
1580:

```

## ***code\_reference\_name - lines 1581-1584***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1581-1584 (4 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def code_reference_name(repo_file: str, suffix: str) -> str:                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | sub, strip                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1581: def code_reference_name(repo_file: str, suffix: str) -> str:
1582:     return re.sub(r"^[A-Za-z0-9_-]+" , "_" , repo_file).strip("_") + suffix
1583:
1584:
```

## ***remove\_directory - lines 1585-1598***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1585-1598 (14 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def remove_directory(path: Path) -> None:                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | exists, remove_readonly, Path, chmod, rmtree                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1585: def remove_directory(path: Path) -> None:
1586:     if not path.exists():
1587:         return
1588:
1589:     def remove_readonly(function, item_path, excinfo):
1590:         try:
1591:             Path(item_path).chmod(0o700)
1592:             function(item_path)
1593:         except Exception:
1594:             raise excinfo
1595:
1596:     shutil.rmtree(path, onexc=remove_readonly)
1597:
1598:
```

## ***remove\_readonly - lines 1589-1598***

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 1589-1598 (10 source lines).                                                                     |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | def remove_readonly(function, item_path, excinfo):                                                     |
| Touches         | filesystem                                                                                             |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | Path, chmod, rmtree                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1589:     def remove_readonly(function, item_path, excinfo):
1590:         try:
1591:             Path(item_path).chmod(0o700)
1592:             function(item_path)
1593:         except Exception:
1594:             raise excinfo
1595:
1596:     shutil.rmtree(path, onexc=remove_readonly)
1597:
1598:

```

## ***add\_source\_fact\_summary - lines 1599-1612***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1599-1612 (14 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_source_fact_summary(doc: Document, facts: dict) -> None:                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_table, fact_relationship_summary, str, len                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1599: def add_source_fact_summary(doc: Document, facts: dict) -> None:
1600:     add_table(doc, [
1601:         ("File", facts["repo_file"]),
1602:         ("Purpose", facts["description"]),
1603:         ("Lines", f"{facts['line_count']:,}"),
1604:         ("Size", f"{facts['size_bytes']:,} bytes"),
1605:         ("Talks to", fact_relationship_summary(facts)),
1606:         ("Functions", str(len(facts["functions"]))),
1607:         ("Variables/constants", str(len(facts["variables"]))),
1608:         ("API mentions", str(len(facts["endpoints"]))),
1609:         ("Signals", str(len(facts["signals"]))),
1610:     ])
1611:
1612:

```

## ***add\_source\_fact\_sections - lines 1613-1659***

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 1613-1659 (47 source lines).                                                                     |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | def add_source_fact_sections(doc: Document, facts: dict, include_excerpts: bool = True) -> None:       |
| Touches         | filesystem                                                                                             |
| Calls detected  | add_heading, add_paragraph, add_table, endpoint_purpose, function_detail_rows, add_code, numbers       |
| API endpoints   | None detected inside this function body.                                                               |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1613: def add_source_fact_sections(doc: Document, facts: dict, include_excerpts: bool = True) -> None:
1614:     doc.add_heading("How this file participates in the system", level=2)
1615:     doc.add_paragraph("This generated section reads the actual file and points out how it communicates
with other pieces of the app. It is intentionally concrete: line numbers, declarations, endpoints, events,
source excerpts, and debugging cues.")
1616:     if facts["imports"]:
1617:         doc.add_heading("Imports, requires, and external dependencies", level=2)
1618:         add_table(doc, [(f"Line {item['line']}", item["statement"]) for item in facts["imports"][:60]])
1619:     if facts["routes"]:
1620:         doc.add_heading("Backend routes implemented here", level=2)
1621:         add_table(doc, [(f"{item['method']} {item['path']}", f"Line {item['line']}").
{endpoint_purpose(item['path'])}"] for item in facts["routes"][:80]])
1622:     if facts["endpoints"]:
1623:         doc.add_heading("API endpoints mentioned or called", level=2)
1624:         add_table(doc, [(item["endpoint"], f"Line {item['line']}. {endpoint_purpose(item['endpoint'])}"))
for item in facts["endpoints"][:80]])
1625:     if facts["signals"]:
1626:         doc.add_heading("Important implementation signals", level=2)
1627:         add_table(doc, [(f"{item['type']} at line {item['line']}", f"{item['meaning']} Evidence:
{item['evidence']}") for item in facts["signals"][:80]])
1628:     if facts["variables"]:
1629:         doc.add_heading("Important constants and variables", level=2)
1630:         add_table(doc, [(item["name"], f"Line {item['line']}: {item['statement']}") for item in
facts["variables"][:80]])
1631:     if facts["json_keys"]:
1632:         doc.add_heading("JSON/YAML-style keys detected", level=2)
1633:         add_table(doc, [(item["key"], f"Line {item['line']}: {item['statement']}") for item in
facts["json_keys"][:80]])
1634:     if facts["events"]:
1635:         doc.add_heading("Event handlers and user interactions", level=2)
1636:         add_table(doc, [(f"Line {item['line']}", item["statement"]) for item in facts["events"][:80]])
1637:     if facts["selectors"]:
1638:         doc.add_heading("CSS selectors and visual hooks", level=2)
1639:         add_table(doc, [(f"Line {item['line']}", item["selector"]) for item in facts["selectors"][:120]])
1640:     if facts["functions"]:
1641:         doc.add_heading("Function-by-function detail", level=2)
1642:         for item in facts["functions"]:
1643:             doc.add_heading(f"{item['name']} - lines {item['line']}-{item['end_line']}", level=3)
1644:             add_table(doc, function_detail_rows(item))
1645:             if include_excerpts:
1646:                 doc.add_paragraph("Source excerpt:")
1647:                 add_code(doc, item["excerpt"])
1648:         doc.add_heading("Representative opening snippet", level=2)
1649:         add_code(doc, facts["snippet"] or "(empty file)")
1650:         doc.add_heading("Beginner debugging checklist for this file", level=2)
1651:         numbers(doc, [
1652:             "Name the user action, command, or workflow that reaches this file.",
1653:             "Find the exact function or route that owns the behavior.",
1654:             "Check the inputs: DOM state, request body, file path, local storage value, compiler source, or
Git branch.",
1655:             "Check the outputs: returned JSON, updated DOM, written file, terminal output, generated project
catalog, or published website asset.",
1656:             "If the issue crosses a boundary, test the boundary directly: browser console for frontend,
endpoint call for backend, command line for Git/compiler, Cloudflare logs for Worker behavior.",
1657:         ])
1658:     ]
1659:

```

## setup\_code\_reference\_document - lines 1660-1673

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 1660-1673 (14 source lines).                                                                     |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | def setup_code_reference_document(title: str, subtitle: str) -> Document:                              |
| Touches         | Mostly local calculations or local UI state.                                                           |
| Calls detected  | setup_document, add_paragraph, add_run, Pt, RGBColor                                                   |
| API endpoints   | None detected inside this function body.                                                               |
| Storage keys    | No local/session storage keys detected.                                                                |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1660: def setup_code_reference_document(title: str, subtitle: str) -> Document:
1661:     doc = setup_document()
1662:     title_p = doc.add_paragraph()
1663:     title_p.alignment = WD_ALIGN_PARAGRAPH.LEFT
1664:     run = title_p.add_run(title)
1665:     run.bold = True
1666:     run.font.size = Pt(22)
1667:     run.font.color.rgb = RGBColor(11, 37, 69)
1668:     doc.add_paragraph(subtitle)
1669:     doc.add_paragraph(f"Generated from repository state at {REPO}.")
1670:     doc.add_paragraph("")
1671:     return doc
1672:
1673:

```

### ***compact\_generated\_docx - lines 1674-1694***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1674-1694 (21 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def compact_generated_docx(doc: Document, keep_first_page_break: bool = True) -> int:                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | list, any, qn, get, iter, getparent, remove                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1674: def compact_generated_docx(doc: Document, keep_first_page_break: bool = True) -> int:
1675:     """Remove generator-inserted page breaks so pages fill naturally."""
1676:     kept = 0
1677:     removed = 0
1678:     for paragraph in list(doc.paragraphs):
1679:         has_page_break = any(
1680:             element.tag == qn("w:br") and element.get(qn("w:type")) == "page"
1681:             for element in paragraph._element.iter()
1682:         )
1683:         if not has_page_break:
1684:             continue
1685:         if keep_first_page_break and kept == 0:
1686:             kept += 1
1687:             continue
1688:         parent = paragraph._element.getparent()
1689:         if parent is not None:
1690:             parent.remove(paragraph._element)
1691:             removed += 1
1692:     return removed
1693:
1694:

```

### ***write\_single\_code\_reference\_docx - lines 1695-1705***

| Item            | Explanation                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Source location | Lines 1695-1705 (11 source lines).                                            |
| Purpose         | Persists data to local state, disk, credentials, or browser storage.          |
| Declaration     | def write_single_code_reference_docx(facts: dict, output_path: Path) -> None: |
| Touches         | Mostly local calculations or local UI state.                                  |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | setup_code_reference_document, add_source_fact_summary, add_source_fact_sections, compact_generated_docx, save                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1695: def write_single_code_reference_docx(facts: dict, output_path: Path) -> None:
1696:     doc = setup_code_reference_document(
1697:         f"Code Reference: {facts['repo_file']}",
1698:         "A source-level explanation with function details, file communication, variables, endpoints, and
debugging notes.",
1699:     )
1700:     add_source_fact_summary(doc, facts)
1701:     add_source_fact_sections(doc, facts, include_excerpts=True)
1702:     compact_generated_docx(doc, keep_first_page_break=False)
1703:     doc.save(output_path)
1704:
1705:

```

## pdf\_styles - lines 1706-1735

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1706-1735 (30 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def pdf_styles():                                                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getSampleStyleSheet, add, ParagraphStyle, HexColor                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1706: def pdf_styles():
1707:     styles = getSampleStyleSheet()
1708:     styles.add(ParagraphStyle(
1709:         name="CodeTiny",
1710:         parent=styles["Code"],
1711:         fontName="Courier",
1712:         fontSize=6.4,
1713:         leading=7.6,
1714:         backColor=colors.HexColor("#F5F8FB"),
1715:         borderColor=colors.HexColor("#D8E6F0"),
1716:         borderWidth=0.25,
1717:         borderPadding=4,
1718:         spaceAfter=7,
1719:     ))
1720:     styles.add(ParagraphStyle(
1721:         name="SmallBody",
1722:         parent=styles["BodyText"],
1723:         fontSize=8.5,
1724:         leading=11.0,
1725:         spaceAfter=5,
1726:     ))
1727:     styles.add(ParagraphStyle(
1728:         name="TinyTable",
1729:         parent=styles["BodyText"],
1730:         fontSize=7.0,
1731:         leading=8.5,
1732:     ))
1733:     return styles
1734:
1735:

```

## pdf\_paragraph - lines 1736-1739

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1736-1739 (4 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def pdf_paragraph(text: str, style) -> Paragraph:                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Paragraph, escape, str, replace                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1736: def pdf_paragraph(text: str, style) -> Paragraph:
1737:     return Paragraph(html.escape(str(text)).replace("\n", "<br/>"), style)
1738:
1739:
```

## pdf\_code - lines 1740-1748

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1740-1748 (9 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def pdf_code(text: str, styles) -> Preformatted:                                                                                                                                                      |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | in, splitlines, replace, wrap, extend, Preformatted, join                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1740: def pdf_code(text: str, styles) -> Preformatted:
1741:     wrapped_lines = []
1742:     for raw_line in (text or "").splitlines():
1743:         expanded = raw_line.replace("\t", " ")
1744:         chunks = textwrap.wrap(expanded, width=112, replace_whitespace=False, drop_whitespace=False) or
[""]
1745:         wrapped_lines.extend(chunks)
1746:     return Preformatted("\n".join(wrapped_lines), styles["CodeTiny"])
1747:
1748:
```

## pdf\_table - lines 1749-1764

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 1749-1764 (16 source lines).                                                                     |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | def pdf_table(rows: list[tuple[str, str]], styles):                                                    |
| Touches         | Mostly local calculations or local UI state.                                                           |
| Calls detected  | pdf_paragraph, extend, Table, setStyle, TableStyle, HexColor                                           |
| API endpoints   | None detected inside this function body.                                                               |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1749: def pdf_table(rows: list[tuple[str, str]], styles):
1750:     data = [[pdf_paragraph("Item", styles["TinyTable"]), pdf_paragraph("Explanation",
styles["TinyTable"])]
1751:     data.extend([[pdf_paragraph(label, styles["TinyTable"]), pdf_paragraph(value, styles["TinyTable"])]
for label, value in rows])
1752:     table = Table(data, colwidths=[1.45 * inch, 5.25 * inch], repeatRows=1)
1753:     table.setStyle(TableStyle([
1754:         ("BACKGROUND", (0, 0), (-1, 0), colors.HexColor("#E8EEF5")),
1755:         ("GRID", (0, 0), (-1, -1), 0.25, colors.HexColor("#B7CFE0")),
1756:         ("VALIGN", (0, 0), (-1, -1), "TOP"),
1757:         ("LEFTPADDING", (0, 0), (-1, -1), 5),
1758:         ("RIGHTPADDING", (0, 0), (-1, -1), 5),
1759:         ("TOPPADDING", (0, 0), (-1, -1), 4),
1760:         ("BOTTPADDING", (0, 0), (-1, -1), 4),
1761:     ]))
1762:     return table
1763:
1764:

```

## ***add\_pdf\_source\_fact\_sections - lines 1765-1803***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1765-1803 (39 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_pdf_source_fact_sections(story: list, facts: dict, styles, include_excerpts: bool = True) -> None:                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | append, Paragraph, pdf_paragraph, pdf_table, endpoint_purpose, function_detail_rows, pdf_code                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1765: def add_pdf_source_fact_sections(story: list, facts: dict, styles, include_excerpts: bool = True) ->
None:
1766:     story.append(Paragraph("How this file participates in the system", styles["Heading2"]))
1767:     story.append(pdf_paragraph("This generated section reads the actual file and points out line numbers,
declarations, endpoints, events, source excerpts, and debugging cues.", styles["SmallBody"]))
1768:     if facts["imports"]:
1769:         story.append(Paragraph("Imports, requires, and external dependencies", styles["Heading2"]))
1770:         story.append(pdf_table([(f"Line {item['line']}", item["statement"]) for item in
facts["imports"][:60]], styles))
1771:         if facts["routes"]:
1772:             story.append(Paragraph("Backend routes implemented here", styles["Heading2"]))
1773:             story.append(pdf_table([(f"{item['method']} {item['path']}", f"Line {item['line']}.
{endpoint_purpose(item['path'])}") for item in facts["routes"][:80]], styles))
1774:             if facts["endpoints"]:
1775:                 story.append(Paragraph("API endpoints mentioned or called", styles["Heading2"]))
1776:                 story.append(pdf_table([(item["endpoint"], f"Line {item['line']}.
{endpoint_purpose(item['endpoint'])}") for item in facts["endpoints"][:80]], styles))
1777:                 if facts["signals"]:
1778:                     story.append(Paragraph("Important implementation signals", styles["Heading2"]))
1779:                     story.append(pdf_table([(f"{item['type']} at line {item['line']}", f"{item['meaning']} Evidence:
{item['evidence']}") for item in facts["signals"][:80]], styles))
1780:                     if facts["variables"]:
1781:                         story.append(Paragraph("Important constants and variables", styles["Heading2"]))
1782:                         story.append(pdf_table([(item["name"], f"Line {item['line']}: {item['statement']}") for item in
facts["variables"][:80]], styles))
1783:                         if facts["json_keys"]:
1784:                             story.append(Paragraph("JSON/YAML-style keys detected", styles["Heading2"]))
1785:                             story.append(pdf_table([(item["key"], f"Line {item['line']}: {item['statement']}") for item in
facts["json_keys"][:80]], styles))
1786:                             if facts["events"]:
1787:                                 story.append(Paragraph("Event handlers and user interactions", styles["Heading2"]))

```

```

1788:         story.append(pdf_table([(f"Line {item['line']}", item["statement"]) for item in
facts["events"][:80]], styles))
1789:     if facts["selectors"]:
1790:         story.append(Paragraph("CSS selectors and visual hooks", styles["Heading2"]))
1791:         story.append(pdf_table([(f"Line {item['line']}", item["selector"]) for item in
facts["selectors"][:120]], styles))
1792:     if facts["functions"]:
1793:         story.append(Paragraph("Function-by-function detail", styles["Heading2"]))
1794:         for item in facts["functions"]:
1795:             story.append(Paragraph(f"{item['name']} - lines {item['line']}-{item['end_line']}",
styles["Heading3"]))
1796:         story.append(pdf_table(function_detail_rows(item), styles))
1797:         if include_excerpts:
1798:             story.append(pdf_paragraph("Source excerpt:", styles["SmallBody"]))
1799:             story.append(pdf_code(item["excerpt"], styles))
1800:         story.append(Paragraph("Representative opening snippet", styles["Heading2"]))
1801:         story.append(pdf_code(facts["snippet"] or "(empty file)", styles))
1802:
1803:

```

## write\_single\_code\_reference\_pdf - lines 1804-1827

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1804-1827 (24 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | def write_single_code_reference_pdf(facts: dict, output_path: Path) -> None:                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | pdf_styles, Paragraph, pdf_paragraph, pdf_table, fact_relationship_summary, str, len, Spacer, add_pdf_source_fact_sections, SimpleDocTemplate, build                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1804: def write_single_code_reference_pdf(facts: dict, output_path: Path) -> None:
1805:     if not REPORTLAB_AVAILABLE:
1806:         return
1807:     styles = pdf_styles()
1808:     story = [
1809:         Paragraph(f"Code Reference: {facts['repo_file']}", styles["Title"]),
1810:         pdf_paragraph("A source-level explanation with function details, file communication, variables,
endpoints, and debugging notes.", styles["SmallBody"]),
1811:         pdf_table([
1812:             ("File", facts["repo_file"]),
1813:             ("Purpose", facts["description"]),
1814:             ("Lines", f"{facts['line_count']:,}"),
1815:             ("Size", f"{facts['size_bytes']:,} bytes"),
1816:             ("Talks to", fact_relationship_summary(facts)),
1817:             ("Functions", str(len(facts["functions"]))),
1818:             ("Variables/constants", str(len(facts["variables"]))),
1819:             ("API mentions", str(len(facts["endpoints"]))),
1820:             ("Signals", str(len(facts["signals"]))),
1821:         ], styles),
1822:         Spacer(1, 0.12 * inch),
1823:     ]
1824:     add_pdf_source_fact_sections(story, facts, styles, include_excerpts=True)
1825:     SimpleDocTemplate(str(output_path), pagesize=LETTER, rightMargin=0.55 * inch, leftMargin=0.55 * inch,
topMargin=0.55 * inch, bottomMargin=0.55 * inch).build(story)
1826:
1827:

```

## write\_master\_code\_reference\_docx - lines 1828-1856

| Item            | Explanation                                                                                                                     |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 1828-1856 (29 source lines).                                                                                              |
| Purpose         | Persists data to local state, disk, credentials, or browser storage.                                                            |
| Declaration     | def write_master_code_reference_docx(code_files: list[str], facts_by_file: dict[str, dict], diagrams: dict[str, Path]) -> None: |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | network/API requests, authentication/security data                                                                                                                                                    |
| Calls detected     | setup_code_reference_document, add_diagram, add_heading, numbers, add_table, len, signal, add_page_break, add_source_fact_summary, add_source_fact_sections, compact_generated_docx, save             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1828: def write_master_code_reference_docx(code_files: list[str], facts_by_file: dict[str, dict], diagrams:
dict[str, Path]) -> None:
1829:     doc = setup_code_reference_document(
1830:         "OMB Portfolio Builder Code Reference",
1831:         "A generated Word reference that explains the important source files, all discovered functions,
communication paths, variables, endpoints, installers, AI paths, and compiler paths.",
1832:     )
1833:     add_diagram(doc, diagrams["file-communication-map"], "Main source file communication map.")
1834:     add_diagram(doc, diagrams["frontend-backend-cloudflare"], "Frontend, backend, Cloudflare, and AI
boundary.")
1835:     add_diagram(doc, diagrams["compile-code-flow"], "Compile Code workspace and simulator flow.")
1836:     doc.add_heading("How to use this document", level=1)
1837:     numbers(doc, [
1838:         "Start with the file you want to understand.",
1839:         "Read the summary table to understand its role.",
1840:         "Read implementation signals to see whether the file touches Git, files, compilers, Cloudflare,
authentication, DOM, storage, or network calls.",
1841:         "Read function details to understand line ranges, side effects, calls, endpoints, and source
excerpts.",
1842:         "Use the debugging checklist to trace a behavior from click to function to file/API/output.",
1843:     ])
1844:     doc.add_heading("Generated file index", level=1)
1845:     add_table(doc, [(repo_file, f"{facts_by_file[repo_file]['line_count']:,} lines,
{len(facts_by_file[repo_file]['functions'])} function(s), {len(facts_by_file[repo_file]['signals'])}
signal(s).") for repo_file in code_files])
1846:     doc.add_page_break()
1847:     for repo_file in code_files:
1848:         facts = facts_by_file[repo_file]
1849:         doc.add_heading(f"Source File: {repo_file}", level=1)
1850:         add_source_fact_summary(doc, facts)
1851:         add_source_fact_sections(doc, facts, include_excerpts=True)
1852:         doc.add_page_break()
1853:     compact_generated_docx(doc, keep_first_page_break=True)
1854:     doc.save(CODE_REFERENCE_MASTER_DOCX)
1855:
1856:

```

## ***write\_master\_code\_reference\_pdf - lines 1857-1895***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1857-1895 (39 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | def write_master_code_reference_pdf(code_files: list[str], facts_by_file: dict[str, dict], diagrams: dict[str, Path]) -> None:                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | pdf_styles, Paragraph, pdf_paragraph, get, exists, append, PdfImage, str, pdf_table, len, signal, PageBreak                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1857: def write_master_code_reference_pdf(code_files: list[str], facts_by_file: dict[str, dict], diagrams:
dict[str, Path]) -> None:
1858:     if not REPORTLAB_AVAILABLE:
1859:         return
1860:     styles = pdf_styles()

```

```

1861:     story = [
1862:         Paragraph("OMB Portfolio Builder Code Reference", styles["Title"]),
1863:         pdf_paragraph("A generated PDF reference that explains the important source files, discovered
functions, communication paths, variables, endpoints, installers, AI paths, and compiler paths.",
styles["SmallBody"]),
1864:     ]
1865:     for diagram_name, caption in [
1866:         ("file-communication-map", "Main source file communication map."),
1867:         ("frontend-backend-cloudflare", "Frontend, backend, Cloudflare, and AI boundary."),
1868:         ("compile-code-flow", "Compile Code workspace and simulator flow."),
1869:     ]:
1870:         path = diagrams.get(diagram_name)
1871:         if path and path.exists():
1872:             story.append(PdfImage(str(path), width=6.6 * inch, height=3.72 * inch))
1873:             story.append(pdf_paragraph(caption, styles["SmallBody"]))
1874:             story.append(Paragraph("Generated file index", styles["Heading1"]))
1875:             story.append(pdf_table([(repo_file, f"{facts_by_file[repo_file]['line_count']:,}" lines,
{len(facts_by_file[repo_file]['functions'])} function(s), {len(facts_by_file[repo_file]['signals'])}
signal(s).") for repo_file in code_files], styles))
1876:             story.append(PageBreak())
1877:             for repo_file in code_files:
1878:                 facts = facts_by_file[repo_file]
1879:                 story.append(Paragraph(f"Source File: {repo_file}", styles["Heading1"]))
1880:                 story.append(pdf_table([
1881:                     ("File", facts["repo_file"]),
1882:                     ("Purpose", facts["description"]),
1883:                     ("Lines", f"{facts['line_count']:,}" lines),
1884:                     ("Size", f"{facts['size_bytes']:,}" bytes),
1885:                     ("Talks to", fact_relationship_summary(facts)),
1886:                     ("Functions", str(len(facts["functions"]))),
1887:                     ("Variables/constants", str(len(facts["variables"]))),
1888:                     ("API mentions", str(len(facts["endpoints"]))),
1889:                     ("Signals", str(len(facts["signals"]))),
1890:                 ], styles))
1891:             add_pdf_source_fact_sections(story, facts, styles, include_excerpts=True)
1892:             story.append(PageBreak())
1893:             SimpleDocTemplate(str(CODE_REFERENCE_MASTER_PDF), pagesize=LETTER, rightMargin=0.55 * inch,
leftMargin=0.55 * inch, topMargin=0.55 * inch, bottomMargin=0.55 * inch).build(story)
1894:
1895:

```

## write\_code\_reference\_docs - lines 1896-1921

| Item               | Explanation                                                                                                                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1896-1921 (26 source lines).                                                                                                                                                                                                          |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                                                        |
| Declaration        | def write_code_reference_docs(files: list[str], diagrams: dict[str, Path]) -> list[Path]:                                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                                                                  |
| Calls detected     | remove_directory, mkdir, is_text_code_file, extract_source_facts, code_reference_name, write_single_code_reference_docx, append, write_single_code_reference_pdf, exists, write_master_code_reference_docx, write_master_code_reference_pdf |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                    |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                     |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                       |

Source excerpt:

```

1896: def write_code_reference_docs(files: list[str], diagrams: dict[str, Path]) -> list[Path]:
1897:     remove_directory(CODE_REFERENCE_DIR)
1898:     remove_directory(CODE_REFERENCE_DOCX_DIR)
1899:     remove_directory(CODE_REFERENCE_PDF_DIR)
1900:     CODE_REFERENCE_DOCX_DIR.mkdir(parents=True, exist_ok=True)
1901:     CODE_REFERENCE_PDF_DIR.mkdir(parents=True, exist_ok=True)
1902:     code_files = [file for file in files if is_text_code_file(file)]
1903:     facts_by_file = {repo_file: extract_source_facts(repo_file) for repo_file in code_files}
1904:     written: list[Path] = []
1905:     for repo_file in code_files:
1906:         facts = facts_by_file[repo_file]
1907:         docx_path = CODE_REFERENCE_DOCX_DIR / code_reference_name(repo_file, ".docx")
1908:         write_single_code_reference_docx(facts, docx_path)
1909:         written.append(docx_path)
1910:         pdf_path = CODE_REFERENCE_PDF_DIR / code_reference_name(repo_file, ".pdf")
1911:         write_single_code_reference_pdf(facts, pdf_path)
1912:         if pdf_path.exists():
1913:             written.append(pdf_path)
1914:     write_master_code_reference_docx(code_files, facts_by_file, diagrams)
1915:     written.append(CODE_REFERENCE_MASTER_DOCX)

```



```

1916: write_master_code_reference_pdf(code_files, facts_by_file, diagrams)
1917: if CODE_REFERENCE_MASTER_PDF.exists():
1918:     written.append(CODE_REFERENCE_MASTER_PDF)
1919: return written
1920:
1921:

```

## add\_generated\_code\_reference - lines 1922-1971

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1922-1971 (50 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_generated_code_reference(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:                                                                                                 |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | add_heading, write_code_reference_docs, add_paragraph, add_table, str, relative_to, exists, len, add_page_break, is_text_code_file, extract_source_facts, fact_relationship_summary                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1922: def add_generated_code_reference(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:
1923:     doc.add_heading("Generated Word And PDF Code References", level=1)
1924:     written = write_code_reference_docs(files, diagrams)
1925:     doc.add_paragraph("The repository now includes generated Word and PDF code-reference documents rather
than only Markdown notes. These documents are meant for source-level reading beside the actual files. They
explain functions, variables, API endpoints, imports, event handlers, file relationships,
security/authentication cues, compiler calls, installer behavior, and debugging paths.")
1926:     add_table(doc, [
1927:         ("Word folder", str(CODE_REFERENCE_DOCX_DIR.relative_to(REPO))),
1928:         ("PDF folder", str(CODE_REFERENCE_PDF_DIR.relative_to(REPO))),
1929:         ("Master Word reference", str(CODE_REFERENCE_MASTER_DOCX.relative_to(REPO))),
1930:         ("Master PDF reference", str(CODE_REFERENCE_MASTER_PDF.relative_to(REPO)) if
CODE_REFERENCE_MASTER_PDF.exists() else "PDF generation skipped because ReportLab is unavailable."),
1931:         ("Artifacts generated", str(len(written))),
1932:         ("Regeneration command", "python docs/build_complete_guide.py"),
1933:     ])
1934:     doc.add_page_break()
1935:
1936:     for repo_file in [file for file in files if is_text_code_file(file)]:
1937:         facts = extract_source_facts(repo_file)
1938:         doc.add_heading(f"Code Reference: {repo_file}", level=1)
1939:         doc.add_paragraph(facts["description"])
1940:         add_table(doc, [
1941:             ("Lines", f"{facts['line_count']:,}"),
1942:             ("Size", f"{facts['size_bytes']:,} bytes"),
1943:             ("Talks to", fact_relationship_summary(facts)),
1944:             ("Functions discovered", str(len(facts["functions"]))),
1945:             ("Variables discovered", str(len(facts["variables"]))),
1946:             ("API endpoints mentioned", str(len(facts["endpoints"]))),
1947:             ("Implementation signals", str(len(facts["signals"]))),
1948:         ])
1949:         if facts["signals"]:
1950:             doc.add_heading("Signals detected", level=2)
1951:             add_table(doc, [(f"{item['type']} at line {item['line']}", f"{item['meaning']} Evidence:
{item['evidence']}") for item in facts["signals"][:20]])
1952:             if facts["endpoints"]:
1953:                 doc.add_heading("Endpoints in this file", level=2)
1954:                 add_table(doc, [(item["endpoint"], f"Line {item['line']}
{endpoint_purpose(item['endpoint'])}") for item in facts["endpoints"][:20]])
1955:                 if facts["functions"]:
1956:                     doc.add_heading("Function details", level=2)
1957:                     for item in facts["functions"][:60]:
1958:                         doc.add_heading(f"{item['name']} (lines {item['line']}-{item['end_line']})", level=3)
1959:                         add_table(doc, function_detail_rows(item))
1960:                         add_code(doc, item["excerpt"])
1961:                 if facts["variables"]:
1962:                     doc.add_heading("Variables and constants", level=2)
1963:                     add_table(doc, [(item["name"], f"Line {item['line']}: {item['statement']}") for item in
facts["variables"][:20]])
1964:                     if facts["imports"]:
1965:                         doc.add_heading("Imports and dependencies", level=2)
1966:                         add_table(doc, [(f"Line {item['line']}", item["statement"]) for item in
facts["imports"][:20]])
1967:                         doc.add_heading("Opening snippet", level=2)
1968:                         add_code(doc, facts["snippet"] or "(empty file)")

```

```

1969:         doc.add_page_break()
1970:
1971:

```

## ***add\_complete\_function\_inventory - lines 1972-1997***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1972-1997 (26 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_complete_function_inventory(doc: Document) -> None:                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | extract_functions, add_heading, add_paragraph, add_table, str, len, add_page_break, setdefault, append, items, range                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1972: def add_complete_function_inventory(doc: Document) -> None:
1973:     rows = extract_functions()
1974:     doc.add_heading("Generated Function Inventory", level=1)
1975:     doc.add_paragraph("This section is generated from the real source files. It does not replace reading
the code, but it gives you a map of where functions live and what their names suggest they own.")
1976:     add_table(doc, [
1977:         ("Files scanned", "main.cjs, server.mjs, template-preview.js, script.js"),
1978:         ("Functions found", str(len(rows))),
1979:         ("How descriptions are made", "Descriptions are heuristic based on file name and function naming
patterns. For exact behavior, inspect the source around the listed line."),
1980:     ])
1981:     doc.add_page_break()
1982:
1983:     by_file: dict[str, list[tuple[str, int, str, str]]] = {}
1984:     for row in rows:
1985:         by_file.setdefault(row[0], []).append(row)
1986:     for file_name, file_rows in by_file.items():
1987:         doc.add_heading(f"Function Inventory: {file_name}", level=1)
1988:         doc.add_paragraph(f"{file_name} contains {len(file_rows)} named functions discovered by the guide
generator.")
1989:         chunk_size = 45
1990:         for index in range(0, len(file_rows), chunk_size):
1991:             chunk = file_rows[index:index + chunk_size]
1992:             add_table(doc, [(f"{name} (line {line_no})", purpose) for _, line_no, name, purpose in
chunk])
1993:             if index + chunk_size < len(file_rows):
1994:                 doc.add_paragraph("Continued on the next table.")
1995:             doc.add_page_break()
1996:
1997:

```

## ***add\_deep\_detail\_topics - lines 1998-2011***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1998-2011 (14 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_deep_detail_topics(doc: Document) -> None:                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_heading, add_paragraph, bullets, add_page_break                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1998: def add_deep_detail_topics(doc: Document) -> None:
1999:     for title, body in DEEP_DETAIL_TOPICS:
2000:         doc.add_heading(f"Deep Detail: {title}", level=1)
2001:         doc.add_paragraph(body)
2002:         doc.add_heading("How to use this detail", level=2)
2003:         bullets(doc, [
2004:             "Start with the user-visible behavior.",
2005:             "Find the file that owns that behavior.",
2006:             "Trace the data handoff to the next file, endpoint, cache, or generated artifact.",
2007:             "Change the smallest responsible piece and test the flow end to end.",
2008:         ])
2009:         doc.add_page_break()
2010:
2011:

```

## add\_programming\_syntax - lines 2012-2021

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2012-2021 (10 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_programming_syntax(doc: Document) -> None:                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_heading, add_code, add_paragraph, add_page_break                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2012: def add_programming_syntax(doc: Document) -> None:
2013:     for title, snippet, explanation in PROGRAMMING_SYNTAX_LESSONS:
2014:         doc.add_heading(f"Programming Syntax: {title}", level=1)
2015:         add_code(doc, snippet)
2016:         doc.add_paragraph(explanation)
2017:         doc.add_heading("How this appears in the project", level=2)
2018:         doc.add_paragraph("You will see this pattern in main.cjs, server.mjs, template-preview.js,
script.js, Cloudflare Worker code, HTML files, CSS files, JSON catalogs, and GitHub workflow YAML.")
2019:         doc.add_page_break()
2020:
2021:

```

## add\_shell\_syntax - lines 2022-2031

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2022-2031 (10 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_shell_syntax(doc: Document) -> None:                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_heading, add_code, add_paragraph, add_page_break                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2022: def add_shell_syntax(doc: Document) -> None:
2023:     for title, snippet, explanation in SHELL_SYNTAX_LESSONS:
2024:         doc.add_heading(f"Shell Or Script Syntax: {title}", level=1)
2025:         add_code(doc, snippet)
2026:         doc.add_paragraph(explanation)
2027:         doc.add_heading("why this matters", level=2)
2028:         doc.add_paragraph("Shell scripts glue tools together. They are not the app UI, but they install

```

```

tools, build releases, deploy workers, scan directories, and enforce rules.")
2029:     doc.add_page_break()
2030:
2031:

```

## add\_caching\_methods - lines 2032-2056

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2032-2056 (25 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_caching_methods(doc: Document, diagrams: dict[str, Path]) -> None:                                                                                                                            |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | add_heading, add_diagram, add_paragraph, add_table, add_page_break                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2032: def add_caching_methods(doc: Document, diagrams: dict[str, Path]) -> None:
2033:     doc.add_heading("Caching Methods Used In The Builder And Website", level=1)
2034:     add_diagram(doc, diagrams["compile-code-flow"], "Compile artifact caching happens inside the Compile
Code workspace.")
2035:     doc.add_paragraph("Caching means remembering a result so the app does not repeat slow or annoying
work. This project uses several different caches, each with a different safety boundary.")
2036:     add_table(doc, [(name, f"{where} Purpose: {purpose}") for name, where, purpose in CACHING_METHODS])
2037:     doc.add_heading("Important safety rule", level=2)
2038:     doc.add_paragraph("A cache must be scoped. Publishing authorization is scoped to the same repository,
branch, user/machine scope, and expiration time. Compiler caches are scoped to source code, language, compiler
path, runtime path, and file name. UI caches like update snooze are local convenience data and do not grant
publishing access.")
2039:     doc.add_page_break()
2040:     for name, where, purpose in CACHING_METHODS:
2041:         doc.add_heading(f"Cache Detail: {name}", level=1)
2042:         if "Compiler" in name or "Compile" in name or "Terminal" in name or "source" in name:
2043:             add_diagram(doc, diagrams["compile-code-flow"], f"Context diagram for {name}.")
2044:             elif "Publishing" in name or "Extended trust" in name:
2045:                 add_diagram(doc, diagrams["release-pipeline"], f"Publishing authorization cache supports this
release/publish workflow.")
2046:             elif "HTTP" in name:
2047:                 add_diagram(doc, diagrams["frontend-backend-cloudflare"], f"HTTP no-store protects request
freshness across frontend/backend paths.")
2048:                 add_table(doc, [
2049:                     ("Where implemented", where),
2050:                     ("Why it exists", purpose),
2051:                     ("What can go stale", "Cached data can become outdated when files, tools, versions,
repositories, or user choices change."),
2052:                     ("How to refresh", "Use the explicit refresh/rebuild/authenticate action for that feature,
clear the relevant local data, or restart the builder when appropriate."),
2053:                 ])
2054:                 doc.add_page_break()
2055:
2056:

```

## add\_installer\_details - lines 2057-2092

| Item               | Explanation                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2057-2092 (36 source lines).                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration        | def add_installer_details(doc: Document, diagrams: dict[str, Path]) -> None:                           |
| Touches            | filesystem                                                                                             |
| Calls detected     | add_heading, add_diagram, add_paragraph, add_table, numbers, add_page_break                            |
| API endpoints      | None detected inside this function body.                                                               |
| Storage keys       | No local/session storage keys detected.                                                                |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                          |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2057: def add_installer_details(doc: Document, diagrams: dict[str, Path]) -> None:
2058:     doc.add_heading("How The Installer Works", level=1)
2059:     add_diagram(doc, diagrams["release-pipeline"], "Installer and update behavior sits at the end of the
release pipeline.")
2060:     doc.add_paragraph("The installer is produced by electron-builder and customized by NSIS script code
in build/installer.nsh. electron-builder gathers app files and creates the installer shell. NSIS controls
Windows setup pages, duplicate install checks, shortcut behavior, Git installation checks, update behavior, and
uninstall pages.")
2061:     add_table(doc, [
2062:         ("electron-builder", "Reads package.json build settings and packages the Electron app."),
2063:         ("NSIS", "Runs installer wizard logic and custom script functions."),
2064:         ("installer.nsh", "Adds OMB-specific setup pages, duplicate detection, update handling,
shortcuts, and Git checks."),
2065:         ("AppData install", "Default per-user install location avoids Program Files admin prompts."),
2066:         ("Update install", "Existing installation is updated in place instead of creating a duplicate
copy."),
2067:     ])
2068:     doc.add_heading("Installer flow", level=2)
2069:     numbers(doc, [
2070:         "customInit runs first.",
2071:         "The installer checks for an existing registered install.",
2072:         "If a current or newer install exists, setup stops.",
2073:         "If an older install exists, setup treats the run as an update and uses the existing location.",
2074:         "If no install exists, setup scans for duplicate builder copies.",
2075:         "The custom tools page asks about desktop shortcut and publishing tools.",
2076:         "Files are installed into the chosen per-user location.",
2077:         "customInstall creates/removes shortcuts and optionally installs Git for Windows.",
2078:         "The uninstaller has its own page explaining what will and will not be removed.",
2079:     ])
2080:     doc.add_page_break()
2081:     for name, description in INSTALLER_FUNCTIONS:
2082:         doc.add_heading(f"Installer Function: {name}", level=1)
2083:         doc.add_paragraph(description)
2084:         add_table(doc, [
2085:             ("File", "build/installer.nsh"),
2086:             ("Syntax family", "NSIS installer scripting"),
2087:             ("Function job", description),
2088:             ("Beginner note", "Installer functions run during setup/uninstall, not while the normal
builder UI is being used."),
2089:         ])
2090:         doc.add_page_break()
2091:
2092:

```

## ***add\_function\_maps - lines 2093-2120***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2093-2120 (28 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_function_maps(doc: Document, diagrams: dict[str, Path]) -> None:                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | add_heading, add_diagram, add_paragraph, add_table, add_page_break                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2093: def add_function_maps(doc: Document, diagrams: dict[str, Path]) -> None:
2094:     for file_name, group_purpose, functions in FUNCTION_GROUPS:
2095:         doc.add_heading(f"Function Map: {file_name}", level=1)
2096:         if file_name == "main.cjs":
2097:             add_diagram(doc, diagrams["electron-runtime"], "Electron startup functions sit inside this
runtime flow.")
2098:             elif file_name == "server.mjs":
2099:                 add_diagram(doc, diagrams["frontend-backend-cloudflare"], "server.mjs is the local backend
layer in this map.")
2100:             elif file_name == "template-preview.js":

```

```

2101:         add_diagram(doc, diagrams["builder-to-site-files"], "template-preview.js is the builder
frontend that sends edits toward parser output.")
2102:     elif file_name == "script.js":
2103:         add_diagram(doc, diagrams["cloudflare-ai-flow"], "script.js owns the public website behavior,
including the AI chat request path.")
2104:         doc.add_paragraph(group_purpose)
2105:         add_table(doc, [(name, description) for name, description in functions])
2106:         doc.add_heading("How to use this map", level=2)
2107:         doc.add_paragraph("When debugging, find the user action first, then find the function group that
owns it. For example, a compile terminal issue starts in template-preview.js, then calls server.mjs, then
returns output back to template-preview.js.")
2108:         doc.add_page_break()
2109:         for name, description in functions:
2110:             doc.add_heading(f"Function Detail: {file_name} -> {name}", level=1)
2111:             doc.add_paragraph(description)
2112:             add_table(doc, [
2113:                 ("File", file_name),
2114:                 ("Function", name),
2115:                 ("Responsibility", description),
2116:                 ("Debug question", "What input does this function receive, what output or state change
does it produce, and what function calls it next?"),
2117:             ])
2118:             doc.add_page_break()
2119:
2120:

```

## add\_architecture - lines 2121-2140

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2121-2140 (20 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_architecture(doc: Document, diagrams: dict[str, Path]) -> None:                                                                                                                               |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | add_heading, add_paragraph, get, add_diagram, add_code, bullets, add_page_break                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2121: def add_architecture(doc: Document, diagrams: dict[str, Path]) -> None:
2122:     for title, body, diagram in ARCHITECTURE_PAGES:
2123:         doc.add_heading(title, level=1)
2124:         doc.add_paragraph(body)
2125:         diagram_name = ARCHITECTURE_DIAGRAM_BY_TITLE.get(title)
2126:         if diagram_name:
2127:             add_diagram(doc, diagrams[diagram_name], f"Generated block diagram: {title}.")
2128:             doc.add_heading("Text version of the block diagram", level=2)
2129:             add_code(doc, diagram)
2130:             doc.add_heading("Beginner explanation", level=2)
2131:             doc.add_paragraph("Read each arrow as 'this thing talks to the next thing.' The important rule is
that editing happens locally in the builder, while viewing happens publicly on the website after generated
static files are published.")
2132:             doc.add_heading("What can break", level=2)
2133:             bullets(doc, [
2134:                 "If local builder state is not saved, the parser may not see the latest edits.",
2135:                 "If Git authentication or branch state is wrong, Apply to site cannot safely push.",
2136:                 "If public hosting cache is stale, a phone may show an older copy until refresh or cache
expiry.",
2137:             ])
2138:             doc.add_page_break()
2139:
2140:

```

## add\_file\_communication - lines 2141-2163

| Item            | Explanation                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------|
| Source location | Lines 2141-2163 (23 source lines).                                                                     |
| Purpose         | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration     | def add_file_communication(doc: Document, diagrams: dict[str, Path]) -> None:                          |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_heading, add_diagram, add_paragraph, add_table, bullets, add_page_break                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2141: def add_file_communication(doc: Document, diagrams: dict[str, Path]) -> None:
2142:     doc.add_heading("How The Important Files Communicate", level=1)
2143:     add_diagram(doc, diagrams["file-communication-map"], "Generated map of the main builder files and
public website files.")
2144:     doc.add_paragraph("This is the file-level mental model. The desktop app starts in main.cjs, the
backend lives in server.mjs, the builder screen is template-preview.html with template-preview.css and template-
preview.js, and the public website is index.html with styles.css, script.js, projects.json, and assets.")
2145:     add_table(doc, [
2146:         ("main.cjs -> server.mjs", "Electron starts the backend so the builder UI can call local APIs."),
2147:         ("main.cjs -> template-preview.html", "Electron opens the builder window and loads the builder
HTML file."),
2148:         ("template-preview.html -> template-preview.js", "The builder HTML loads the JavaScript that
handles editing, saving, previewing, compile code, and publishing UI."),
2149:         ("template-preview.js -> server.mjs", "The builder uses fetch requests to ask the backend to save
files, parse projects, run Git, run compilers, or publish."),
2150:         ("server.mjs -> projects.json", "The parser writes public project data used by the website."),
2151:         ("index.html -> script.js/styles.css", "The public page loads its behavior and visual design."),
2152:         ("script.js -> projects.json", "The public website reads the generated catalog to render project
windows."),
2153:     ])
2154:     doc.add_heading("Why this communication shape is useful", level=2)
2155:     bullets(doc, [
2156:         "The public website can stay static and fast.",
2157:         "The builder can safely use local-only features without exposing them online.",
2158:         "The parser creates a clean boundary between editable state and visitor-facing output.",
2159:         "The same project data can be previewed locally before it is published.",
2160:     ])
2161:     doc.add_page_break()
2162:
2163:

```

## add\_tool\_build\_map - lines 2164-2173

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2164-2173 (10 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_tool_build_map(doc: Document, diagrams: dict[str, Path]) -> None:                                                                                                                             |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | add_heading, add_diagram, add_paragraph, add_table, add_page_break                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2164: def add_tool_build_map(doc: Document, diagrams: dict[str, Path]) -> None:
2165:     doc.add_heading("What Tool Builds What", level=1)
2166:     add_diagram(doc, diagrams["tool-build-map"], "Generated map showing which tool owns each part of the
build and release process.")
2167:     doc.add_paragraph("A common source of confusion is thinking one tool builds everything. In this
project, each tool has a narrow job. pnpm installs dependencies, Electron runs the app, electron-builder
packages the app, NSIS makes the installer, Git moves files, GitHub Actions automates release builds, Wrangler
deploys Cloudflare Worker code, and compilers run project source code.")
2168:     add_table(doc, [(tool, f"{builds} Used for: {where}") for tool, builds, where in TOOL_BUILD_ROWS])
2169:     doc.add_heading("Beginner shortcut", level=2)
2170:     doc.add_paragraph("When something fails, ask which tool owns that step. Installer failure points
toward NSIS or electron-builder. AI backend failure points toward Cloudflare Worker or Wrangler. Website publish
failure points toward Git, GitHub, branch state, or authentication.")

```

```

2171:         doc.add_page_break()
2172:
2173:

```

## add\_software\_decisions - lines 2174-2192

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2174-2192 (19 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_software_decisions(doc: Document) -> None:                                                                                                                                                    |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | add_heading, add_paragraph, add_table, bullets, add_page_break                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2174: def add_software_decisions(doc: Document) -> None:
2175:     for title, body in SOFTWARE_DECISIONS:
2176:         doc.add_heading(f"Software Engineering Decision: {title}", level=1)
2177:         doc.add_paragraph(body)
2178:         add_table(doc, [
2179:             ("Decision", title),
2180:             ("Reason", body),
2181:             ("What it protects", "Predictable publishing, safer public/private separation, easier
updates, clearer debugging, or better user experience."),
2182:             ("Tradeoff", "The design may add more files or moving parts, but each moving part has a
defined job."),
2183:         ])
2184:         doc.add_heading("How to evaluate this later", level=2)
2185:         bullets(doc, [
2186:             "If the feature becomes hard to maintain, check whether responsibilities are mixed across too
many files.",
2187:             "If users are confused, improve the builder UI or guide rather than exposing implementation
details.",
2188:             "If publishing becomes risky, tighten the parser and authentication boundary first.",
2189:         ])
2190:         doc.add_page_break()
2191:
2192:

```

## add\_system\_layers - lines 2193-2208

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2193-2208 (16 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_system_layers(doc: Document, diagrams: dict[str, Path]) -> None:                                                                                                                              |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | add_heading, add_diagram, add_paragraph, add_page_break, add_table, add_code                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2193: def add_system_layers(doc: Document, diagrams: dict[str, Path]) -> None:
2194:     doc.add_heading("AI, Frontend, Backend, Cloudflare, And Secrets", level=1)
2195:     add_diagram(doc, diagrams["frontend-backend-cloudflare"], "Generated map of the frontend, backend,
Cloudflare Worker, AI provider, and secret boundary.")
2196:     doc.add_paragraph("This section separates the major layers so you know where to look when something
breaks. The frontend draws and captures input. The backend performs protected work. Cloudflare hosts the public

```



```

AI endpoint. Secrets stay server-side.")
2197:     doc.add_page_break()
2198:     for title, body, rows, snippet in SYSTEM_LAYER_SECTIONS:
2199:         doc.add_heading(title, level=1)
2200:         doc.add_paragraph(body)
2201:         add_table(doc, rows)
2202:         doc.add_heading("Representative code or command", level=2)
2203:         add_code(doc, snippet)
2204:         doc.add_heading("Debug question", level=2)
2205:         doc.add_paragraph("Ask which layer owns the problem: did the UI send the request, did the backend
receive it, did the service have the needed secret or tool, and did the response come back in the format the UI
expects?")
2206:     doc.add_page_break()
2207:
2208:

```

## add\_workflows - lines 2209-2242

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2209-2242 (34 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_workflows(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:                                                                                                                |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | add_heading, add_diagram, numbers, add_paragraph, add_page_break, startswith, read_text, get, add_code, join, splittlines                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2209: def add_workflows(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:
2210:     workflows = [
2211:         ("Content Creation Workflow", ["Open the builder.", "Edit profile, front-page text, portfolio
areas, or projects.", "Save project or Save all sections.", "Open preview and inspect the exact public layout.",
"Save draft.", "Apply to site after the target is authenticated."]),
2212:         ("Project Creation Workflow", ["Click Add project.", "Choose category.", "Choose appearance.",
"Enter title.", "Edit overview and sections.", "Add files, images, links, code blocks, and compile-code
evidence.", "Save project so the parser can build the project preview."]),
2213:         ("Publishing Workflow", ["Enter repository target.", "Authenticate GitHub write access.", "Load
from target if the target already has content.", "Save draft.", "Apply to site.", "Git commits and pushes
generated files.", "GitHub Pages or Cloudflare serves the updated website."]),
2214:         ("Update Workflow", ["The installed app checks GitHub Releases.", "If a newer version exists, the
app shows an update dialog.", "The updater downloads the installer.", "The app closes itself.", "The installer
updates the existing AppData install.", "The app relaunches after the installer finishes."]),
2215:         ("Compile Code Workflow", ["Create or import a source file.", "Pick the language.", "For
Verilog/SystemVerilog, mark design files as Design and create or import a Testbench file.", "Keep $dumpfile and
$dumpvars in the HDL testbench so the signal scope can draw waveforms.", "Save source.", "Run beautifier if
desired.", "Compile or run.", "Read terminal output, messages log, and HDL scope when available.", "Append the
source code to a project section when ready."]),
2216:     ]
2217:     for title, steps in workflows:
2218:         doc.add_heading(title, level=1)
2219:         if title in {"Content Creation Workflow", "Project Creation Workflow", "Publishing Workflow"}:
2220:             add_diagram(doc, diagrams["builder-to-site-files"], f"Generated flow diagram supporting
{title}.")
2221:             if title == "Update Workflow":
2222:                 add_diagram(doc, diagrams["release-pipeline"], "Generated release/update pipeline diagram.")
2223:             if title == "Compile Code Workflow":
2224:                 add_diagram(doc, diagrams["compile-code-flow"], "Generated Compile Code workflow diagram.")
2225:             numbers(doc, steps)
2226:             doc.add_heading("Plain-English mental model", level=2)
2227:             doc.add_paragraph("A workflow is just a repeatable recipe. The builder turns complicated actions
into visible buttons so you do not need to remember shell commands every time.")
2228:             doc.add_page_break()
2229:
2230:     for workflow in [f for f in files if f.startswith(".github/workflows/")]:
2231:         text = (REPO / workflow).read_text(encoding="utf-8", errors="replace")
2232:         doc.add_heading(f"GitHub Workflow File: {workflow}", level=1)
2233:         doc.add_paragraph(FILE_DESCRIPTIONS.get(workflow, "GitHub Actions workflow file."))
2234:         doc.add_heading("What a GitHub workflow is", level=2)
2235:         doc.add_paragraph("A workflow is an automated script GitHub runs in the cloud. Instead of you
manually building installers or enforcing branch rules, GitHub reads this YAML file and executes each job on a
temporary machine.")
2236:         doc.add_heading("Important YAML excerpt", level=2)
2237:         add_code(doc, "\n".join(text.splittlines()[0:80]))
2238:         doc.add_heading("Tradeoff", level=2)

```

```

2239:         doc.add_paragraph("Automated workflows reduce mistakes, but a bad workflow can block good merges.
That is why the main branch gate should be strict but understandable.")
2240:         doc.add_page_break()
2241:
2242:

```

## file\_type - lines 2243-2250

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2243-2250 (8 source lines).                                                                                                                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def file_type(path: Path, repo_path: str) -> str:                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | endswith, lower, lstrip                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2243: def file_type(path: Path, repo_path: str) -> str:
2244:     if repo_path.endswith(".gitkeep"):
2245:         return "folder marker"
2246:     if path.suffix:
2247:         return path.suffix.lower().lstrip(".")
2248:     return "no extension"
2249:
2250:

```

## add\_files - lines 2251-2280

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2251-2280 (30 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_files(doc: Document, files: list[str]) -> None:                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_heading, add_paragraph, get, add_table, file_type, stat, exists, is_file, lower, read_text, splitlines, add_code                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2251: def add_files(doc: Document, files: list[str]) -> None:
2252:     text_exts = {".js", ".mjs", ".cjs", ".html", ".css", ".json", ".md", ".txt", ".yaml", ".yml",
".toml", ".nsh", ""}
2253:     for repo_file in files:
2254:         path = REPO / repo_file
2255:         doc.add_heading(f"File Explained: {repo_file}", level=1)
2256:         doc.add_paragraph(FILE_DESCRIPTIONS.get(repo_file, "Tracked repository file used by the builder
or public website.))
2257:         add_table(doc, [
2258:             ("Type", file_type(path, repo_file)),
2259:             ("Size", f"{path.stat().st_size:,} bytes" if path.exists() else "missing locally"),
2260:             ("Used by", "Builder app, public site, installer, Cloudflare, or GitHub workflow depending on
the path."),
2261:             ("Beginner idea", "A repository is a folder Git tracks. This file is one object Git can
version, review, and publish."),
2262:         ])
2263:         if path.exists() and path.is_file() and path.suffix.lower() in text_exts:
2264:             lines = path.read_text(encoding="utf-8", errors="replace").splitlines()
2265:             doc.add_heading("Representative snippet", level=2)

```

```

2266:         add_code(doc, "\n".join(lines[:24]) if lines else "(empty file)")
2267:         if len(lines) > 24:
2268:             doc.add_paragraph(f"Only the first 24 lines are shown here. The full file has
{len(lines):,} lines and should be opened in the repository when editing.")
2269:         else:
2270:             doc.add_heading("Binary or asset note", level=2)
2271:             doc.add_paragraph("This is treated as an asset. The guide explains its purpose, but the raw
binary content is not printed because raw image/icon bytes would not be useful to read in Word.")
2272:             doc.add_heading("Safe editing rule", level=2)
2273:             bullets(doc, [
2274:                 "Make changes on a feature branch from development.",
2275:                 "Preview or test the behavior before merging.",
2276:                 "Do not edit generated/public files by hand if the builder parser is supposed to generate
them.",
2277:             ])
2278:             doc.add_page_break()
2279:
2280:

```

## add\_commands - lines 2281-2295

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2281-2295 (15 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_commands(doc: Document) -> None:                                                                                                                                                              |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | add_heading, add_code, add_table, add_paragraph, add_page_break                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2281: def add_commands(doc: Document) -> None:
2282:     for command, explanation, why in COMMAND_LESSONS:
2283:         doc.add_heading(f"Command Explained: {command}", level=1)
2284:         add_code(doc, command)
2285:         add_table(doc, [
2286:             ("What it does", explanation),
2287:             ("Why it matters", why),
2288:             ("Where used", "Local development, GitHub Actions, builder publishing, installer testing, or
compiler execution."),
2289:             ("Beginner warning", "Run commands from the correct folder. In this project, most development
commands belong in the builder repository root."),
2290:         ])
2291:         doc.add_heading("How to read command output", level=2)
2292:         doc.add_paragraph("Command output is the program talking back to you. Success output usually
names what happened. Error output usually names a missing file, missing tool, denied permission, wrong branch,
or failed compile step.")
2293:         doc.add_page_break()
2294:
2295:

```

## add\_features - lines 2296-2318

| Item               | Explanation                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2296-2318 (23 source lines).                                                                     |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration        | def add_features(doc: Document, diagrams: dict[str, Path]) -> None:                                    |
| Touches            | filesystem, authentication/security data                                                               |
| Calls detected     | add_heading, add_diagram, add_paragraph, lower, add_table, add_page_break                              |
| API endpoints      | None detected inside this function body.                                                               |
| Storage keys       | No local/session storage keys detected.                                                                |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                          |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2296: def add_features(doc: Document, diagrams: dict[str, Path]) -> None:
2297:     for topic in FEATURE_TOPICS:
2298:         doc.add_heading(f"Builder Feature: {topic}", level=1)
2299:         if topic in {"Compile Code workspace", "HDL testbench requirement", "HDL signal scope", "Terminal
output", "Messages log", "Append code to project"}:
2300:             add_diagram(doc, diagrams["compile-code-flow"], f"Context diagram for {topic}.")
2301:         elif topic in {"AI chat", "Cloudflare AI worker"}:
2302:             add_diagram(doc, diagrams["cloudflare-ai-flow"], f"Context diagram for {topic}.")
2303:         elif topic in {"Installer workflow", "Updater workflow", "Release tags", "Branch protection"}:
2304:             add_diagram(doc, diagrams["release-pipeline"], f"Context diagram for {topic}.")
2305:         elif topic in {"Project preview", "Full portfolio preview", "Project parser", "Asset
synchronization"}:
2306:             add_diagram(doc, diagrams["builder-to-site-files"], f"Context diagram for {topic}.")
2307:             doc.add_paragraph(f"This page explains the {topic.lower()} area of the OMB builder and website
system.")
2308:             add_table(doc, [
2309:                 ("Owner view", "The builder exposes controls for creating, editing, saving, previewing, or
publishing this area."),
2310:                 ("Visitor view", "The public website shows only parsed, approved output. Editing controls
stay private."),
2311:                 ("Parser role", "The parser converts local builder state into website-safe data and
markup."),
2312:                 ("Risk", "If this area is not saved before preview or publishing, the public site may show an
older version."),
2313:             ])
2314:             doc.add_heading("Plain-English example", level=2)
2315:             doc.add_paragraph("Think of the builder as the kitchen and the public website as the plate served
to visitors. The parser is the person plating the content neatly. Visitors should never see the prep
controls.")
2316:             doc.add_page_break()
2317:
2318:

```

## ***add\_tradeoffs - lines 2319-2330***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2319-2330 (12 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_tradeoffs(doc: Document) -> None:                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | add_heading, add_paragraph, add_table, add_page_break                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2319: def add_tradeoffs(doc: Document) -> None:
2320:     for title, body in TRADEOFFS:
2321:         doc.add_heading(f"Tradeoff: {title}", level=1)
2322:         doc.add_paragraph(body)
2323:         add_table(doc, [
2324:             ("Benefit", "The chosen direction fits the current builder goal: owner-controlled, local-
first, preview-before-publish portfolio creation."),
2325:             ("Cost", "Every architecture choice gives up something. Usually the cost is complexity, setup
work, or less flexibility in another direction."),
2326:             ("Decision rule", "Prefer the option that keeps publishing safe, previews accurate, and
editing understandable."),
2327:         ])
2328:         doc.add_page_break()
2329:
2330:

```

## ***add\_troubleshooting - lines 2331-2343***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2331-2343 (13 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_troubleshooting(doc: Document) -> None:                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | add_heading, add_paragraph, numbers, add_page_break                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2331: def add_troubleshooting(doc: Document) -> None:
2332:     for title, body in TROUBLESHOOTING_TOPICS:
2333:         doc.add_heading(f"Troubleshooting: {title}", level=1)
2334:         doc.add_paragraph(body)
2335:         numbers(doc, [
2336:             "Read the exact error message before changing anything.",
2337:             "Confirm whether the issue is local builder state, Git state, installer state, public website
cache, or backend service availability.",
2338:             "Fix the smallest proven cause first.",
2339:             "Save a clean checkpoint after the fix works.",
2340:         ])
2341:         doc.add_page_break()
2342:
2343:

```

## ***add\_concepts - lines 2344-2357***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2344-2357 (14 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_concepts(doc: Document) -> None:                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | add_heading, add_paragraph, add_table, add_page_break                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2344: def add_concepts(doc: Document) -> None:
2345:     for concept in CONCEPTS:
2346:         doc.add_heading(f"Beginner Concept: {concept}", level=1)
2347:         doc.add_paragraph(f"{concept} is one of the building blocks behind the OMB builder and portfolio
website. You do not need to memorize every term at once. Learn the role it plays, then come back when you see it
in an error message or file name.")
2348:         add_table(doc, [
2349:             ("Simple meaning", "This concept helps explain how the app, website, GitHub workflow,
installer, or backend behaves."),
2350:             ("Where it appears", "Look for it in README instructions, source files, GitHub Actions logs,
installer behavior, browser behavior, or terminal output."),
2351:             ("How to debug it", "Ask: what is supposed to happen, what actually happened, what file or
service owns that behavior, and what changed most recently?"),
2352:         ])
2353:         doc.add_heading("Tiny mental model", level=2)
2354:         doc.add_paragraph("When confused, draw boxes. Put the user on the left, the thing they click in
the middle, and the file or service that responds on the right. Most software problems become easier once the
boxes are visible.")
2355:         doc.add_page_break()
2356:
2357:

```

## ***add\_closing - lines 2358-2372***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2358-2372 (15 source lines).                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                |
| Declaration        | def add_closing(doc: Document) -> None:                                                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | add_heading, numbers, add_note                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2358: def add_closing(doc: Document) -> None:
2359:     doc.add_heading("Final Operating Checklist", level=1)
2360:     numbers(doc, [
2361:         "Make feature changes on a feature or codex branch from development.",
2362:         "Update README and the in-app guide when behavior changes.",
2363:         "Run local syntax checks and useful smoke tests.",
2364:         "Merge feature work into development.",
2365:         "Only merge development into main for release-ready builds.",
2366:         "Bump package.json before a new main release.",
2367:         "Confirm GitHub Actions publish a fresh installer and stable latest asset.",
2368:         "Open the builder, check update behavior, and verify the public website link downloads the latest
installer."],
2369:     ])
2370:     add_note(doc, "The core rule", "The builder is the source of truth for creating content. The preview
is the truth test before publishing. GitHub is the delivery path. The public website is the final read-only
result.")
2371:
2372:
```

## ***main - lines 2373-2410***

| Item               | Explanation                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2373-2410 (38 source lines).                                                                                                                                                                                    |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.                                                                                                                |
| Declaration        | def main() -> None:                                                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                                          |
| Calls detected     | tracked_files, load_package, make_diagrams, setup_document, add_cover, add_reading_map, add_entry_points, add_architecture, add_flow_walkthroughs, add_file_communication, add_tool_build_map, add_software_decisions |
| API endpoints      | None detected inside this function body.                                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                 |

Source excerpt:

```
2373: def main() -> None:
2374:     files = tracked_files()
2375:     package = load_package()
2376:     diagrams = make_diagrams()
2377:     doc = setup_document()
2378:     add_cover(doc, package)
2379:     add_reading_map(doc)
2380:     add_entry_points(doc)
2381:     add_architecture(doc, diagrams)
2382:     add_flow_walkthroughs(doc, diagrams)
2383:     add_file_communication(doc, diagrams)
2384:     add_tool_build_map(doc, diagrams)
2385:     add_software_decisions(doc)
2386:     add_system_layers(doc, diagrams)
```

```

2387:     add_data_contracts(doc)
2388:     add_api_endpoint_catalog(doc)
2389:     add_programming_syntax(doc)
2390:     add_shell_syntax(doc)
2391:     add_caching_methods(doc, diagrams)
2392:     add_installer_details(doc, diagrams)
2393:     add_function_maps(doc, diagrams)
2394:     add_complete_function_inventory(doc)
2395:     add_generated_code_reference(doc, files, diagrams)
2396:     add_deep_detail_topics(doc)
2397:     add_workflows(doc, files, diagrams)
2398:     add_files(doc, files)
2399:     add_commands(doc)
2400:     add_features(doc, diagrams)
2401:     add_tradeoffs(doc)
2402:     add_troubleshooting(doc)
2403:     add_concepts(doc)
2404:     add_closing(doc)
2405:     removed_breaks = compact_generated_docx(doc, keep_first_page_break=True)
2406:     print(f"Compacted generated DOCX layout by removing {removed_breaks} page breaks.")
2407:     doc.save(OUTPUT)
2408:     print(OUTPUT)
2409:
2410:

```

## Representative opening snippet

```

1: from __future__ import annotations
2:
3: import html
4: import json
5: import math
6: import re
7: import shutil
8: import subprocess
9: import textwrap
10: from pathlib import Path
11:
12: from PIL import Image, ImageDraw, ImageFont
13: from docx import Document
14: from docx.enum.table import WD_CELL_VERTICAL_ALIGNMENT, WD_TABLE_ALIGNMENT
15: from docx.enum.text import WD_ALIGN_PARAGRAPH
16: from docx.oxml import OxmlElement
17: from docx.oxml.ns import qn
18: from docx.shared import Inches, Pt, RGBColor
19:
20: try:
21:     from reportlab.lib import colors
22:     from reportlab.lib.pagesizes import LETTER
23:     from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
24:     from reportlab.lib.units import inch
25:     from reportlab.platypus import Image as PdfImage
26:     from reportlab.platypus import PageBreak, Paragraph, Preformatted, SimpleDocTemplate, Spacer, Table,
TableStyle
27:
28:     REPORTLAB_AVAILABLE = True
29: except ImportError:
30:     REPORTLAB_AVAILABLE = False
31:
32:
33: REPO = Path(__file__).resolve().parents[1]
34: OUTPUT = REPO / "docs" / "OMB_Portfolio_Builder_Complete_Guide.docx"
35: DIAGRAM_DIR = REPO / "docs" / "guide-diagrams"
36: CODE_REFERENCE_DIR = REPO / "docs" / "code-reference"
37: CODE_REFERENCE_DOCX_DIR = REPO / "docs" / "code-reference-docx"
38: CODE_REFERENCE_PDF_DIR = REPO / "docs" / "code-reference-pdf"
39: CODE_REFERENCE_MASTER_DOCX = REPO / "docs" / "OMB_Portfolio_Builder_Code_Reference.docx"
40: CODE_REFERENCE_MASTER_PDF = REPO / "docs" / "OMB_Portfolio_Builder_Code_Reference.pdf"
41:
42:
43: FILE_DESCRIPTIONS = {
44:     ".github/workflows/build-windows-builder.yml": "GitHub Actions workflow that builds the Windows
installer and portable application, checks the stable download link, enforces release version bumps, uploads
artifacts, and publishes GitHub Releases.",
45:     ".github/workflows/main-branch-gate.yml": "GitHub Actions guard that rejects direct pushes to main
and rejects pull requests into main unless the source branch is development.",
46:     ".gitignore": "Git ignore rules for build outputs, temporary files, and local-only data that should
not be committed.",
47:     ".nojekyll": "Tells GitHub Pages not to process the site with Jekyll, so folders and static files
publish exactly as generated.",
48:     ".well-known/security.txt": "Public security contact file used by browsers, researchers, and scanners
to find responsible disclosure information.",
49:     "BRANCHING.md": "Human-readable branch model explaining development, main, feature branches, and
release flow.",
50:     "Backgrounds/.gitkeep": "Keeps the Backgrounds folder in Git even when no background image files are
present.",
51:     "README.md": "Main README for the standalone builder application: install, updates, workspaces, text
editing, publishing, build commands, branch model, and uninstall.",
52:     "_headers": "Cloudflare Pages style HTTP headers for security and caching behavior on the public
website.",

```

```
53: "assets/apple-touch-icon.png": "Apple touch icon shown when the public site is saved on iOS or
compatible launch surfaces.",
54: "assets/favicon-16.png": "Small browser favicon used by tabs and browser UI.",
55: "assets/favicon-192.png": "Large web app icon for Android and installable browser surfaces.",
56: "assets/favicon-32.png": "Standard browser favicon size.",
57: "assets/favicon-48.png": "Windows/browser icon size used by some desktop surfaces.",
58: "assets/favicon.ico": "ICO favicon bundle used by browsers that request favicon.ico.",
59: "assets/omb-app-icon-256.png": "Builder application icon source at 256 pixels.",
60: "assets/omb-app-icon-512.png": "Builder application icon source at 512 pixels.",
61: "assets/omb-app-icon.ico": "Windows ICO application icon used by Electron Builder and shortcuts.",
62: "assets/omb-mark.png": "OMB snake/mark image used for branding in the site and builder.",
63: "build/installer.nsh": "NSIS installer customization script for update behavior, shortcut placement,
old install detection, and setup details.",
64: "builder-rich-future-sections.js": "Builder-side helper code for richer future portfolio sections and
section defaults.",
65: "cloudflare/README.md": "Cloudflare setup notes for deploying the AI Worker endpoint and wiring the
public site to the backend.",
66: "cloudflare/portfolio-ai-worker.js": "Cloudflare Worker backend for the public portfolio AI assistant
and fallback intelligence path.",
67: "cloudflare/wrangler.toml": "Wrangler configuration used to deploy the Cloudflare Worker.",
68: "docs/.gitkeep": "Keeps the docs folder in Git even before generated documents are added.",
69: "electronics-search.js": "Electronics vocabulary and search support used by the website search
behavior.",
70: "index.html": "Public website HTML shell that recruiters and visitors load in the browser.",
71: "installer-notes.txt": "Installer license/notes text shown during Windows setup.",
72: "main.cjs": "Electron main process: creates the desktop window, starts the local backend, handles
menus, update handoff, and application lifecycle.",
```



# Source File: electronics-search.js

| Item                | Explanation                                                                    |
|---------------------|--------------------------------------------------------------------------------|
| File                | electronics-search.js                                                          |
| Purpose             | Electronics vocabulary and search support used by the website search behavior. |
| Lines               | 162                                                                            |
| Size                | 9,856 bytes                                                                    |
| Talks to            | GitHub/release layer                                                           |
| Functions           | 2                                                                              |
| Variables/constants | 16                                                                             |
| API mentions        | 0                                                                              |
| Signals             | 8                                                                              |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                              | Explanation                                                                                                                                                                                              |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rich text at line 8               | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: "phase margin", "slew rate", "common mode", "common mode range", "cmrr", "psrr", "loop gain",                       |
| Security or auth at line 45       | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "oscilloscope", "scope capture", "logic analyzer", "spectrum analyzer", "network analyzer", "function generator", |
| Compiler or simulator at line 59  | Part of the Compile Code workspace or language/tool detection. Evidence: "static timing analysis", "sta", "fpga", "asic", "rtl", "verilog", "systemverilog", "vhdl",                                     |
| Compiler or simulator at line 61  | Part of the Compile Code workspace or language/tool detection. Evidence: "modelsim", "questa", "iverilog", "yosys", "formal verification", "assertion", "coverage", "axi",                               |
| Installer at line 80              | Windows setup, update, shortcut, or installation behavior. Evidence: "potentiometer", "trimmer", "thermistor", "varistor", "photodiode", "phototransistor", "hall effect sensor",                        |
| Cloudflare or AI at line 87       | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "layout", "route", "solder", "assemble", "test", "verify", "model", "analyze", "optimize",                        |
| Compiler or simulator at line 100 | Part of the Compile Code workspace or language/tool detection. Evidence: "ltspice", "spice", "pspice", "ngspice", "matlab", "simulink", "python", "numpy", "scipy",                                      |
| Git command at line 101           | Repository state, publishing, branch checks, or release automation. Evidence: "excel", "kiCad", "kicad", "vivado", "stm32cubeide", "git", "github", "markdown", "c",                                     |

## Important constants and variables

| Item          | Explanation                      |
|---------------|----------------------------------|
| analogTerms   | Line 2: const analogTerms = [    |
| powerTerms    | Line 20: const powerTerms = [    |
| pcbTerms      | Line 33: const pcbTerms = [      |
| testTerms     | Line 44: const testTerms = [     |
| digitalTerms  | Line 55: const digitalTerms = [  |
| embeddedTerms | Line 66: const embeddedTerms = [ |
| components    | Line 76: const components = [    |
| actions       | Line 85: const actions = [       |

| Item             | Explanation                                                                   |
|------------------|-------------------------------------------------------------------------------|
| metrics          | Line 92: const metrics = [                                                    |
| toolTerms        | Line 99: const toolTerms = [                                                  |
| phraseTerms      | Line 105: const phraseTerms = [];                                             |
| categoryKeywords | Line 114: const categoryKeywords = {                                          |
| allKeywords      | Line 120: const allKeywords = [...new Set([                                   |
| haystack         | Line 139: const haystack = [                                                  |
| buckets          | Line 149: const buckets = new Set(categoryKeywords[project?.category]    []); |
| compactTerm      | Line 154: const compactTerm = term.replace(/([a-z0-9+#+]+/g, " ").trim();     |

## JSON/YAML-style keys detected

| Item                | Explanation                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| analog-mixed-signal | Line 115: "analog-mixed-signal": [...analogTerms, ...powerTerms, ...pcbTerms, ...testTerms, ...components, ...metrics], |

## Function-by-function detail

### *normalize - lines 134-136*

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 134-136 (3 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalize(value) {                                                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | toLowerCase                                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

134:   function normalize(value) {
135:     return String(value || "").toLowerCase();
136:   }

```

### *projectSpecificKeywords - lines 138-158*

| Item               | Explanation                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Source location    | Lines 138-158 (21 source lines).                                                                       |
| Purpose            | Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs. |
| Declaration        | function projectSpecificKeywords(project, categoryLabel = "") {                                        |
| Touches            | filesystem                                                                                             |
| Calls detected     | map, join, Set, forEach, includes, toLowerCase, add, replace, trim                                     |
| API endpoints      | None detected inside this function body.                                                               |
| Storage keys       | No local/session storage keys detected.                                                                |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                          |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

138: function projectSpecificKeywords(project, categoryLabel = "") {
139:   const haystack = [
140:     project?.title,
141:     project?.summary,
142:     project?.category,
143:     categoryLabel,
144:     ...(project?.focus || []),
145:     ...(project?.tools || []).map((item) => typeof item === "string" ? item : [item.name, item.title,
item.label, item.description].join(" ")),
146:     ...(project?.languages || [])
147:   ].map(normalize).join(" ");
148:
149:   const buckets = new Set(categoryKeywords[project?.category] || []);
150:   toolTerms.forEach((term) => {
151:     if (haystack.includes(term.toLowerCase())) buckets.add(term);
152:   });
153:   allKeywords.forEach((term) => {
154:     const compactTerm = term.replace(/^[a-z0-9-]+#/g, " ").trim();
155:     if (compactTerm && haystack.includes(compactTerm)) buckets.add(term);
156:   });
157:   return [...buckets];
158: }

```

## Representative opening snippet

```

1: (function () {
2:   const analogTerms = [
3:     "op amp", "operational amplifier", "fully differential op amp", "differential amplifier",
"instrumentation amplifier",
4:     "transimpedance amplifier", "transconductance amplifier", "current mirror", "cascode", "folded
cascode",
5:     "common source", "common gate", "common drain", "emitter follower", "source follower", "differential
pair",
6:     "active load", "gain stage", "output stage", "input stage", "bias circuit", "bandgap reference",
7:     "voltage reference", "current reference", "low noise", "offset voltage", "input bias current", "gain
bandwidth",
8:     "phase margin", "slew rate", "common mode", "common mode range", "cmrr", "psrr", "loop gain",
9:     "stability", "compensation", "miller compensation", "frequency response", "bode plot", "pole zero",
10:    "small signal", "large signal", "noise analysis", "thermal noise", "flicker noise", "shot noise",
11:    "analog filter", "low pass filter", "high pass filter", "band pass filter", "notch filter", "active
filter",
12:    "rc filter", "rlc circuit", "resonance", "q factor", "cutoff frequency", "impedance", "admittance",
13:    "matching network", "attenuator", "buffer", "comparator", "window comparator", "schmitt trigger",
"hysteresis",
14:    "oscillator", "vco", "voltage controlled oscillator", "relaxation oscillator", "ring oscillator",
"crystal oscillator",
15:    "pll", "phase locked loop", "charge pump", "sample and hold", "track and hold", "analog switch",
16:    "multiplexer", "adc", "dac", "data converter", "sigma delta", "successive approximation", "sar adc",
17:    "pipeline adc", "integrator", "differentiator", "rectifier", "precision rectifier", "peak detector"
18:  ];
19:
20:   const powerTerms = [
21:     "ac to dc", "charger", "battery charger", "power supply", "bench supply", "linear regulator", "ldo",
22:     "switching regulator", "buck converter", "boost converter", "buck boost", "flyback", "forward
converter",
23:     "sepic", "inverter", "rectifier bridge", "diode bridge", "full wave rectifier", "half wave
rectifier",
24:     "ripple", "load regulation", "line regulation", "transient response", "power stage", "mosfet driver",
25:     "gate driver", "current limit", "overvoltage", "undervoltage", "reverse polarity", "thermal
shutdown",
26:     "power path", "battery management", "bms", "cc cv", "constant current", "constant voltage", "trickle
charge",
27:     "precharge", "protection circuit", "fuse", "polyfuse", "tvs diode", "esd diode", "snubber", "clamp",
28:     "soft start", "inrush current", "efficiency", "loss budget", "heat sink", "thermal resistance",
"power dissipation",
29:     "load switch", "ideal diode", "current sense", "shunt resistor", "hall sensor", "isolation",
"transformer",
30:     "optoisolator", "dc dc", "ac dc", "emi filter", "common mode choke", "ferrite bead", "ground loop"
31:   ];
32:
33:   const pcbTerms = [
34:     "pcb", "printed circuit board", "schematic", "layout", "footprint", "symbol", "netlist", "bom",
35:     "gerber", "drill file", "pick and place", "assembly", "bring up", "board bring up", "prototype",
36:     "revision", "rev a", "dfm", "dft", "design rule", "clearance", "creepage", "trace width", "via",
37:     "microvia", "blind via", "buried via", "ground plane", "power plane", "stackup", "controlled
impedance",
38:     "differential pair routing", "return path", "star ground", "analog ground", "digital ground", "guard
ring",
39:     "decoupling capacitor", "bypass capacitor", "bulk capacitor", "test point", "connector", "header",
40:     "silkscreen", "solder mask", "copper pour", "thermal relief", "stitching via", "keepout", "fiducial",
41:     "kicad", "altium", "eagle", "orcad", "cadence allegro", "pcbway", "jlcpcb", "osh park"
42:   ];

```

```

43:
44:   const testTerms = [
45:     "oscilloscope", "scope capture", "logic analyzer", "spectrum analyzer", "network analyzer", "function
generator",
46:     "signal generator", "multimeter", "dmm", "electronic load", "power analyzer", "curve tracer",
"probe",
47:     "differential probe", "current probe", "bench test", "validation", "verification",
"characterization",
48:     "measurement", "calibration", "test plan", "test report", "waveform", "transient", "dc sweep", "ac
sweep",
49:     "monte carlo", "corner analysis", "temperature sweep", "load test", "line test", "frequency sweep",
50:     "gain measurement", "phase measurement", "jitter", "rise time", "fall time", "duty cycle", "pwm",
51:     "latency", "throughput", "noise floor", "snr", "thd", "thd+n", "eye diagram", "ber", "debug",
52:     "troubleshooting", "root cause", "failure analysis", "regression test", "unit test", "integration
test"
53:   ];
54:
55:   const digitalTerms = [
56:     "digital design", "logic design", "combinational logic", "sequential logic", "finite state machine",
"fsm",
57:     "flip flop", "latch", "counter", "register", "shift register", "clock divider", "clock domain
crossing",
58:     "cdc", "metastability", "synchronizer", "debounce", "timing closure", "setup time", "hold time",
59:     "static timing analysis", "sta", "fpga", "asic", "rtl", "verilog", "systemverilog", "vhdl",
60:     "testbench", "uvm", "simulation", "synthesis", "place and route", "bitstream", "vivado", "quartus",
61:     "modelsim", "questa", "iverilog", "yosys", "formal verification", "assertion", "coverage", "axi",
62:     "apb", "ahb", "spi controller", "i2c controller", "uart", "fifo", "ram", "rom", "bram",
63:     "pll clock", "mmcm", "serdes", "lvds", "gpio", "interrupt", "dma", "bus protocol"
64:   ];
65:
66:   const embeddedTerms = [
67:     "embedded", "firmware", "microcontroller", "mcu", "stm32", "stm32cubeide", "arm cortex", "cortex m",
68:     "bare metal", "rtos", "freertos", "scheduler", "interrupt service routine", "isr", "timer",
69:     "pwm output", "adc input", "dac output", "gpio", "spi", "i2c", "uart", "usb", "can bus",
70:     "ethernet", "bootloader", "flash memory", "eeprom", "dma", "low power", "sleep mode", "watchdog",
71:     "sensor interface", "motor control", "control loop", "pid", "encoder", "hall sensor", "imu",
72:     "temperature sensor", "pressure sensor", "display", "lcd", "oled", "ble", "wifi", "esp32",

```

# Source File: index.html

| Item                | Explanation                                                                 |
|---------------------|-----------------------------------------------------------------------------|
| File                | index.html                                                                  |
| Purpose             | Public website HTML shell that recruiters and visitors load in the browser. |
| Lines               | 268                                                                         |
| Size                | 10,799 bytes                                                                |
| Talks to            | public website runtime, Cloudflare/AI layer, GitHub/release layer           |
| Functions           | 0                                                                           |
| Variables/constants | 0                                                                           |
| API mentions        | 0                                                                           |
| Signals             | 9                                                                           |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                              | Explanation                                                                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security or auth at line 11       | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <meta name="author" content="Portfolio Owner" />                                                                            |
| Cloudflare or AI at line 12       | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <meta name="portfolio-ai-endpoint" content="https://omb-portfolio-ai.maurice-baraza-otieno.workers.dev/api/portfolio-ai" /> |
| Installer at line 17              | Windows setup, update, shortcut, or installation behavior. Evidence: <link rel="shortcut icon" href="assets/favicon.ico" />                                                                                        |
| Network request at line 38        | Frontend-backend communication or public web/API calls. Evidence: "@context": "https://schema.org",                                                                                                                |
| Compiler or simulator at line 55  | Part of the Compile Code workspace or language/tool detection. Evidence: "Python",                                                                                                                                 |
| Network request at line 126       | Frontend-backend communication or public web/API calls. Evidence: href="https://github.com/otienomaurice/omb-portfolio-builder/releases/latest/download/OMB-Portfolio-Builder-Setup-latest-x64.exe"                |
| Compiler or simulator at line 160 | Part of the Compile Code workspace or language/tool detection. Evidence: <input id="project-search" type="search" placeholder="LTspice, PCB, frequency response, Verilog..." />                                    |
| Cloudflare or AI at line 242      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <div class="ai-assistant-log" id="ai-assistant-log" aria-live="polite" aria-label="AI chat messages"></div>                 |
| Cloudflare or AI at line 253      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <button class="ai-clear-chat" id="ai-clear-chat" type="button">Clear chat</button>                                          |

## JSON/YAML-style keys detected

| Item         | Explanation                                |
|--------------|--------------------------------------------|
| @context     | Line 38: "@context": "https://schema.org", |
| @type        | Line 39: "@type": "ProfilePage",           |
| dateModified | Line 40: "dateModified": "2026-06-04",     |
| mainEntity   | Line 41: "mainEntity": {                   |
| @type        | Line 42: "@type": "Person",                |
| name         | Line 43: "name": "Portfolio Owner",        |
| url          | Line 44: "url": "",                        |

| Item        | Explanation                                         |
|-------------|-----------------------------------------------------|
| image       | Line 45: "image": "",                               |
| jobTitle    | Line 46: "jobTitle": "",                            |
| description | Line 47: "description": "Portfolio owner profile.", |
| sameAs      | Line 48: "sameAs": [],                              |
| knowsAbout  | Line 49: "knowsAbout": [                            |

## Representative opening snippet

```

1: <!DOCTYPE html>
2: <html lang="en">
3:   <head>
4:     <meta charset="UTF-8" />
5:     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6:     <meta
7:       name="description"
8:       content="A portfolio website built with OMB Portfolio Builder for projects, documents, diagrams,
source code, tests, links, and profile information."
9:     />
10:    <meta name="robots" content="index, follow" />
11:    <meta name="author" content="Portfolio Owner" />
12:    <meta name="portfolio-ai-endpoint" content="https://omb-portfolio-ai.maurice-baraza-
otieno.workers.dev/api/portfolio-ai" />
13:    <link rel="canonical" href="" />
14:    <link rel="icon" type="image/png" sizes="16x16" href="assets/favicon-16.png" />
15:    <link rel="icon" type="image/png" sizes="32x32" href="assets/favicon-32.png" />
16:    <link rel="icon" type="image/png" sizes="192x192" href="assets/favicon-192.png" />
17:    <link rel="shortcut icon" href="assets/favicon.ico" />
18:    <link rel="apple-touch-icon" sizes="180x180" href="assets/apple-touch-icon.png" />
19:    <meta property="og:type" content="profile" />
20:    <meta property="og:title" content="Portfolio" />
21:    <meta
22:      property="og:description"
23:      content="Explore projects, documents, tests, PCBs, tools, links, and source code."
24:    />
25:    <meta property="og:url" content="" />
26:    <meta property="og:image" content="" />
27:    <meta name="twitter:card" content="summary_large_image" />
28:    <meta name="twitter:title" content="Portfolio" />
29:    <meta
30:      name="twitter:description"
31:      content="A project portfolio built with OMB Portfolio Builder."
32:    />
33:    <meta name="twitter:image" content="" />
34:    <title>Portfolio</title>
35:    <link rel="stylesheet" href="styles.css?v=2026062203" />
36:    <script type="application/ld+json">
37:      {
38:        "@context": "https://schema.org",
39:        "@type": "ProfilePage",
40:        "dateModified": "2026-06-04",
41:        "mainEntity": {
42:          "@type": "Person",
43:          "name": "Portfolio Owner",
44:          "url": "",
45:          "image": "",
46:          "jobTitle": "",
47:          "description": "Portfolio owner profile.",
48:          "sameAs": [],
49:          "knowsAbout": [
50:            "Embedded systems",
51:            "ASIC design",
52:            "Digital hardware",
53:            "Software engineering",
54:            "Microcontrollers",
55:            "Python",
56:            "C++"
57:          ]
58:        }
59:      }
60:    </script>
61:  </head>
62:  <body>
63:    <header class="site-header" id="top">
64:      <a class="brand" href="#top" aria-label="Go to top">
65:        <span class="brand-lockup" aria-hidden="true">
66:          
67:          <span class="omb-engraving">OMB</span>
68:        </span>
69:        <span>
70:          <strong>Portfolio</strong>
71:          <small>Portfolio</small>
72:        </span>

```

# Source File: installer-notes.txt

| Item                | Explanation                                              |
|---------------------|----------------------------------------------------------|
| File                | installer-notes.txt                                      |
| Purpose             | Installer license/notes text shown during Windows setup. |
| Lines               | 23                                                       |
| Size                | 3,068 bytes                                              |
| Talks to            | GitHub/release layer                                     |
| Functions           | 0                                                        |
| Variables/constants | 0                                                        |
| API mentions        | 0                                                        |
| Signals             | 11                                                       |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                            | Explanation                                                                                                                                                                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installer at line 3             | Windows setup, update, shortcut, or installation behavior. Evidence: This installer includes the desktop application, the local portfolio builder, the local preview server, the website template files, and the Electron runtime needed to run the app.                                                   |
| Compiler or simulator at line 5 | Part of the Compile Code workspace or language/tool detection. Evidence: The app does not require the user to install Node.js or npm. Those are development/build tools only.                                                                                                                              |
| Installer at line 7             | Windows setup, update, shortcut, or installation behavior. Evidence: The installer creates a desktop shortcut by default. Untick the shortcut option during setup if you do not want one.                                                                                                                  |
| Installer at line 9             | Windows setup, update, shortcut, or installation behavior. Evidence: By default, setup installs for the current Windows user at %LOCALAPPDATA%\Programs\OMB Portfolio Builder. The builder workspace and publish mirror live under %LOCALAPPDATA%\OMB Portfolio Builder, so editable builder file          |
| Installer at line 11            | Windows setup, update, shortcut, or installation behavior. Evidence: If OMB Portfolio Builder is already installed through Windows, setup tells the user which version it found, checks the current user's installation before legacy machine-wide records, pins setup to that current installed           |
| Installer at line 13            | Windows setup, update, shortcut, or installation behavior. Evidence: Setup also scans fixed drives for other OMB Portfolio Builder executables, portable copies, and local desktop-builder workspaces. It treats %LOCALAPPDATA%\OMB Portfolio Builder\builder and %LOCALAPPDATA%\OMB Portfolio Bu          |
| Git command at line 15          | Repository state, publishing, branch checks, or release automation. Evidence: Publishing to GitHub Pages requires Git for Windows with Git Credential Manager. During setup, the installer can check publishing tools and install Git for Windows if Git or Git Credential Manager is missing. If compatib |
| Git command at line 17          | Repository state, publishing, branch checks, or release automation. Evidence: For GitHub authentication, open Publishing target inside the app, enter the target repository, then click Save target and authenticate. A GitHub/Git Credential Manager sign-in flow may open in the browser. The builder ke |
| Security or auth at line 19     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Successful publishing authorization is cached locally for about one day for the selected repository and branch.                                                                                                     |
| Installer at line 21            | Windows setup, update, shortcut, or installation behavior. Evidence: Older and newer installers remain available from the GitHub Releases page.                                                                                                                                                            |
| Security or auth at line 23     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Fresh installs start blank. No personal projects, resumes, profile photos, custom domains, or portfolio sections are included until the user adds them or authenticates to load a compatible publishing target.     |

## Representative opening snippet

```
1: OMB Portfolio Builder installation notes
2:
3: This installer includes the desktop application, the local portfolio builder, the local preview server,
```

the website template files, and the Electron runtime needed to run the app.

4:

5: The app does not require the user to install Node.js or npm. Those are development/build tools only.

6:

7: The installer creates a desktop shortcut by default. Untick the shortcut option during setup if you do not want one.

8:

9: By default, setup installs for the current Windows user at %LOCALAPPDATA%\Programs\OMB Portfolio Builder. The builder workspace and publish mirror live under %LOCALAPPDATA%\OMB Portfolio Builder, so editable builder files and generated portfolio output stay together in AppData. This avoids protected Program Files updates and lets the app update itself without needing an administrator prompt for normal releases.

10:

11: If OMB Portfolio Builder is already installed through Windows, setup tells the user which version it found, checks the current user's installation before legacy machine-wide records, pins setup to that current installed location, asks for approval during normal interactive installs, and updates that copy. Setup will not create a second installed copy in a different folder. During an in-app Update action, setup runs silently after the user clicks Update and updates the existing installation in place without running the old uninstaller first or requiring the user to manually uninstall first.

12:

13: Setup also scans fixed drives for other OMB Portfolio Builder executables, portable copies, and local desktop-builder workspaces. It treats %LOCALAPPDATA%\OMB Portfolio Builder\builder and %LOCALAPPDATA%\OMB Portfolio Builder\portfolio as managed app workspaces, not duplicate installs. If it finds another unregistered copy, setup stops and asks the user to remove, uninstall, or rename that copy first. This prevents duplicate builders from leaving one machine with stale editing features.

14:

15: Publishing to GitHub Pages requires Git for Windows with Git Credential Manager. During setup, the installer can check publishing tools and install Git for Windows if Git or Git Credential Manager is missing. If compatible tools already exist, setup skips them. Existing shared tools are not silently removed.

16:

17: For GitHub authentication, open Publishing target inside the app, enter the target repository, then click Save target and authenticate. A GitHub/Git Credential Manager sign-in flow may open in the browser. The builder keeps the previous target until GitHub verifies write access to the new target. After authentication succeeds, click Load from target to refresh from that repository.

18:

19: Successful publishing authorization is cached locally for about one day for the selected repository and branch.

20:

21: Older and newer installers remain available from the GitHub Releases page.

22:

23: Fresh installs start blank. No personal projects, resumes, profile photos, custom domains, or portfolio sections are included until the user adds them or authenticates to load a compatible publishing target.



# Source File: main.cjs

| Item                | Explanation                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| File                | main.cjs                                                                                                                               |
| Purpose             | Electron main process: creates the desktop window, starts the local backend, handles menus, update handoff, and application lifecycle. |
| Lines               | 453                                                                                                                                    |
| Size                | 15,134 bytes                                                                                                                           |
| Talks to            | builder frontend, local backend, public website runtime, portfolio catalog, Cloudflare/AI layer, GitHub/release layer                  |
| Functions           | 21                                                                                                                                     |
| Variables/constants | 42                                                                                                                                     |
| API mentions        | 0                                                                                                                                      |
| Signals             | 43                                                                                                                                     |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Imports, requires, and external dependencies

| Item   | Explanation                              |
|--------|------------------------------------------|
| Line 3 | const fs = require("node:fs");           |
| Line 4 | const fsp = require("node:fs/promises"); |
| Line 5 | const net = require("node:net");         |
| Line 6 | const path = require("node:path");       |

## Important implementation signals

| Item                            | Explanation                                                                                                                                    |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Process execution at line 2     | Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: const { execFile } = require("node:child_process"); |
| Compiler or simulator at line 3 | Part of the Compile Code workspace or language/tool detection. Evidence: const fs = require("node:fs");                                        |
| Compiler or simulator at line 4 | Part of the Compile Code workspace or language/tool detection. Evidence: const fsp = require("node:fs/promises");                              |
| Compiler or simulator at line 5 | Part of the Compile Code workspace or language/tool detection. Evidence: const net = require("node:net");                                      |
| Compiler or simulator at line 6 | Part of the Compile Code workspace or language/tool detection. Evidence: const path = require("node:path");                                    |
| Compiler or simulator at line 7 | Part of the Compile Code workspace or language/tool detection. Evidence: const { pathToFileURL } = require("node:url");                        |
| Compiler or simulator at line 8 | Part of the Compile Code workspace or language/tool detection. Evidence: const { promisify } = require("node:util");                           |
| Process execution at line 13    | Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: const execFileAsync = promisify(execFile);          |
| Security or auth at line 27     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "_headers"                                              |
| Cloudflare or AI at line 38     | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: const directoryRefreshes = ["cloudflare"];              |
| Security or auth at line 48     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "_headers",                                             |
| Git command at line 53          | Repository state, publishing, branch checks, or release automation. Evidence: process.env.GIT_EXE,                                             |
| Git command at line 54          | Repository state, publishing, branch checks, or release automation. Evidence: "git",                                                           |
| Git command at line 55          | Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files\\Git\\cmd\\git.exe",                          |

| Item                        | Explanation                                                                                                                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Git command at line 56      | Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files\\Git\\bin\\git.exe",                                                                                      |
| Git command at line 57      | Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files (x86)\\Git\\cmd\\git.exe",                                                                                |
| Git command at line 58      | Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files (x86)\\Git\\bin\\git.exe"                                                                                 |
| Installer at line 61        | Windows setup, update, shortcut, or installation behavior. Evidence: function dispatchBuilderMenuAction(action) {                                                                                          |
| Installer at line 110       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "New Project", accelerator: "CmdOrCtrl+N", click: () => dispatchBuilderMenuAction({ type: "new-project" }) },                |
| Installer at line 111       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Publishing Target", click: () => dispatchBuilderMenuAction({ type: "publishing-target" }) },                                |
| Installer at line 113       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Save Draft", accelerator: "CmdOrCtrl+S", click: () => dispatchBuilderMenuAction({ type: "save-draft" }) },                  |
| Installer at line 114       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Apply to Site", accelerator: "CmdOrCtrl+Shift+S", click: () => dispatchBuilderMenuAction({ type: "apply-site" }) },         |
| Rich text at line 127       | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: { role: "paste" },                                                                                                    |
| Installer at line 141       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Portfolio Preview", click: () => dispatchBuilderMenuAction({ type: "portfolio-preview" }) },                                |
| Installer at line 142       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Check Updates", click: () => dispatchBuilderMenuAction({ type: "check-updates" }) },                                        |
| Installer at line 154       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Open Preferences", accelerator: "CmdOrCtrl+," , click: () => dispatchBuilderMenuAction({ type: "preferences" }) },          |
| Installer at line 156       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Light Mode", click: () => dispatchBuilderMenuAction({ type: "set-theme", theme: "light" }) },                               |
| Installer at line 157       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Dark Mode", click: () => dispatchBuilderMenuAction({ type: "set-theme", theme: "dark" }) }                                  |
| Installer at line 163       | Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Builder Guide", accelerator: "F1", click: () => dispatchBuilderMenuAction({ type: "builder-guide" }) },                     |
| Network request at line 164 | Frontend-backend communication or public web/API calls. Evidence: { label: "GitHub Releases", click: () => shell.openExternal("https://github.com/otienomaurice/omb-portfolio-builder/releases/latest") }, |
| File write at line 190      | Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(path.dirname(target), { recursive: true });                                    |
| File write at line 191      | Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.copyFile(source, target);                                                            |
| File write at line 196      | Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(target, { recursive: true });                                                  |
| File write at line 212      | Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(path.dirname(target), { recursive: true });                                    |
| Git command at line 241     | Repository state, publishing, branch checks, or release automation. Evidence: throw lastError    new Error("Git executable was not found.");                                                               |
| Git command at line 250     | Repository state, publishing, branch checks, or release automation. Evidence: return fileExists(path.join(workspaceRoot, ".git"));                                                                         |
| File write at line 267      | Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(workspaceRoot, { recursive: true });                                           |
| Installer at line 279       | Windows setup, update, shortcut, or installation behavior. Evidence: const defaultWorkspaceHome = process.platform === "win32" && process.env.LOCALAPPDATA                                                 |
| Installer at line 280       | Windows setup, update, shortcut, or installation behavior. Evidence: ? path.join(process.env.LOCALAPPDATA, defaultWorkspaceHomeName)                                                                       |
| File write at line 286      | Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(workspaceRoot, { recursive: true });                                           |
| File write at line 287      | Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(portfolioRoot, { recursive: true });                                           |
| Network request at line 363 | Frontend-backend communication or public web/API calls. Evidence: return `http://127.0.0.1:\${port}`;                                                                                                      |
| Rich text at line 377       | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: backgroundColor: "#eef8fd",                                                                                           |

## Important constants and variables

| Item                       | Explanation                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------|
| fs                         | Line 3: const fs = require("node:fs");                                                                        |
| fsp                        | Line 4: const fsp = require("node:fs/promises");                                                              |
| net                        | Line 5: const net = require("node:net");                                                                      |
| path                       | Line 6: const path = require("node:path");                                                                    |
| mainWindow                 | Line 10: let mainWindow = null;                                                                               |
| builderOrigin              | Line 11: let builderOrigin = "";                                                                              |
| updateShutdownStarted      | Line 12: let updateShutdownStarted = false;                                                                   |
| execFileAsync              | Line 13: const execFileAsync = promisify(execFile);                                                           |
| defaultRepositoryUrl       | Line 14: const defaultRepositoryUrl = process.env.OMB_BUILDER_REPOSITORY    "";                               |
| defaultWorkspaceHomeName   | Line 15: const defaultWorkspaceHomeName = "OMB Portfolio Builder";                                            |
| rootFilesToRefresh         | Line 17: const rootFilesToRefresh = [                                                                         |
| rootFilesToSeed            | Line 30: const rootFilesToSeed = [                                                                            |
| directoryRefreshes         | Line 38: const directoryRefreshes = ["cloudflare"];                                                           |
| directorySeeds             | Line 39: const directorySeeds = ["assets", "Backgrounds", "docs", ".well-known"];                             |
| portfolioFilesToSeed       | Line 40: const portfolioFilesToSeed = [                                                                       |
| portfolioDirectoriesToSeed | Line 51: const portfolioDirectoriesToSeed = ["assets", "Backgrounds", "docs", ".well-known"];                 |
| gitCandidates              | Line 52: const gitCandidates = [                                                                              |
| payload                    | Line 63: const payload = JSON.stringify(action);                                                              |
| shouldFullscreen           | Line 98: const shouldFullscreen = typeof nextState === "boolean" ? nextState : !mainWindow.isFullScreen();    |
| entries                    | Line 197: const entries = await fsp.readdir(source, { withFileTypes: true });                                 |
| sourcePath                 | Line 199: const sourcePath = path.join(source, entry.name);                                                   |
| targetPath                 | Line 200: const targetPath = path.join(target, entry.name);                                                   |
| entries                    | Line 218: const entries = await fsp.readdir(directory);                                                       |
| lastError                  | Line 226: let lastError = null;                                                                               |
| current                    | Line 254: let current = path.dirname(process.execPath);                                                       |
| next                       | Line 257: const next = path.dirname(current);                                                                 |
| bundledSiteRoot            | Line 278: const bundledSiteRoot = path.join(process.resourcesPath, "site");                                   |
| defaultWorkspaceHome       | Line 279: const defaultWorkspaceHome = process.platform === "win32" && process.env.LOCALAPPDATA               |
| workspaceHome              | Line 282: const workspaceHome = process.env.OMB_WORKSPACE_HOME    defaultWorkspaceHome;                       |
| workspaceRoot              | Line 283: const workspaceRoot = path.join(workspaceHome, "builder");                                          |
| portfolioRoot              | Line 284: const portfolioRoot = process.env.OMB_PORTFOLIO_WORKSPACE    path.join(workspaceHome, "portfolio"); |
| envWorkspace               | Line 315: const envWorkspace = process.env.OMB_BUILDER_WORKSPACE;                                             |
| workspaceRoot              | Line 323: const workspaceRoot = path.resolve(__dirname);                                                      |
| nearbyRepository           | Line 329: const nearbyRepository = findRepositoryNearExecutable();                                            |
| tester                     | Line 341: const tester = net.createServer();                                                                  |
| address                    | Line 344: const address = tester.address();                                                                   |
| port                       | Line 345: const port = typeof address === "object" && address ? address.port : 0;                             |
| port                       | Line 352: const port = await findFreePort();                                                                  |
| serverPath                 | Line 361: const serverPath = path.join(workspaceRoot, "server.mjs");                                          |

| Item                  | Explanation                                                                            |
|-----------------------|----------------------------------------------------------------------------------------|
| iconPath              | Line 367: const iconPath = path.join(workspaceRoot, "assets", "omb-app-icon-256.png"); |
| refreshFullscreenMenu | Line 388: const refreshFullscreenMenu = () => {                                        |
| gotLock               | Line 426: const gotLock = app.requestSingleInstanceLock();                             |

## Function-by-function detail

### *dispatchBuilderMenuAction - lines 61-68*

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 61-68 (8 source lines).                                                                                                                                                                         |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | function dispatchBuilderMenuAction(action) {                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isDestroyed, stringify, executeJavaScript, dispatchEvent, CustomEvent                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

61: function dispatchBuilderMenuAction(action) {
62:   if (!mainWindow || mainWindow.isDestroyed()) return;
63:   const payload = JSON.stringify(action);
64:   mainWindow.webContents.executeJavaScript(
65:     `window.dispatchEvent(new CustomEvent("builder-menu-action", { detail: ${payload} }));`,
66:     true
67:   ).catch(() => {});
68: }

```

### *quitForBuilderUpdate - lines 70-92*

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 70-92 (23 source lines).                                                                                                                                                                        |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | function quitForBuilderUpdate() {                                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isDestroyed, destroy, quit, exit                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

70: function quitForBuilderUpdate() {
71:   if (updateShutdownStarted) return;
72:   updateShutdownStarted = true;
73:   try {
74:     if (mainWindow && !mainWindow.isDestroyed()) {
75:       mainWindow.destroy();
76:     }
77:   } catch {
78:     // The process exit fallback below still guarantees the updater can continue.
79:   }
80:   try {

```

```

81:     app.quit();
82:   } catch {
83:     // Fall through to app.exit.
84:   }
85:   setTimeout(() => {
86:     try {
87:       app.exit(0);
88:     } catch {
89:       process.exit(0);
90:     }
91:   }, 1500);
92: }

```

## setBuilderFullScreen - lines 96-103

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 96-103 (8 source lines).                                                                                                                                                                        |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | function setBuilderFullScreen(nextState = null) {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isDestroyed, isFullScreen, setFullScreen, setMenuBarVisibility, setAutoHideMenuBar, setMenu, createAppMenu                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

96: function setBuilderFullScreen(nextState = null) {
97:   if (!mainWindow || mainWindow.isDestroyed()) return;
98:   const shouldFullscreen = typeof nextState === "boolean" ? nextState : !mainWindow.isFullScreen();
99:   mainWindow.setFullScreen(shouldFullscreen);
100:  mainWindow.setMenuBarVisibility(true);
101:  mainWindow.setAutoHideMenuBar(false);
102:  mainWindow.setMenu(createAppMenu(builderOrigin));
103: }

```

## createAppMenu - lines 105-184

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 105-184 (80 source lines).                                                                                                                                                                      |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function createAppMenu(origin) {                                                                                                                                                                      |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | buildFromTemplate, dispatchBuilderMenuAction, setBuilderFullScreen, openExternal, isDestroyed, isMinimized, restore, focus, fileExists, accessSync                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

105: function createAppMenu(origin) {
106:   return Menu.buildFromTemplate([
107:     {
108:       label: "File",
109:       submenu: [
110:         { label: "New Project", accelerator: "CmdOrCtrl+N", click: () => dispatchBuilderMenuAction({
type: "new-project" }) },
111:         { label: "Publishing Target", click: () => dispatchBuilderMenuAction({ type: "publishing-target"
}) },
112:         { type: "separator" },
113:         { label: "Save Draft", accelerator: "CmdOrCtrl+S", click: () => dispatchBuilderMenuAction({ type:

```

```

"save-draft" }) },
114:     { label: "Apply to Site", accelerator: "CmdOrCtrl+Shift+S", click: () =>
dispatchBuilderMenuAction({ type: "apply-site" }) },
115:     { type: "separator" },
116:     { role: "quit", label: "Exit" }
117:   ]
118: },
119: {
120:   label: "Edit",
121:   submenu: [
122:     { role: "undo" },
123:     { role: "redo" },
124:     { type: "separator" },
125:     { role: "cut" },
126:     { role: "copy" },
127:     { role: "paste" },
128:     { role: "selectAll" }
129:   ]
130: },
131: {
132:   label: "View",
133:   submenu: [
134:     { role: "reload" },
135:     { role: "forceReload" },
136:     { type: "separator" },
137:     { role: "resetZoom" },
138:     { role: "zoomIn" },
139:     { role: "zoomOut" },
140:     { type: "separator" },
141:     { label: "Portfolio Preview", click: () => dispatchBuilderMenuAction({ type: "portfolio-preview"
}) },
142:     { label: "Check Updates", click: () => dispatchBuilderMenuAction({ type: "check-updates" }) },
143:     { type: "separator" },
144:     {
145:       label: mainWindow?.isFullScreen?.() ? "Exit Full Screen" : "Toggle Full Screen",
146:       accelerator: "F11",
147:       click: () => setBuilderFullScreen()
148:     }
149:   ]
150: },
151: {
152:   label: "Preferences",
153:   submenu: [
154:     { label: "Open Preferences", accelerator: "CmdOrCtrl+," , click: () => dispatchBuilderMenuAction({
type: "preferences" }) },
155:     { type: "separator" },
156:     { label: "Light Mode", click: () => dispatchBuilderMenuAction({ type: "set-theme", theme: "light"
}) },
157:     { label: "Dark Mode", click: () => dispatchBuilderMenuAction({ type: "set-theme", theme: "dark"
}) }
158:   ]
159: },
160: {
161:   label: "Help",
162:   submenu: [
163:     { label: "Builder Guide", accelerator: "F1", click: () => dispatchBuilderMenuAction({ type:
"builder-guide" }) },
164:     { label: "GitHub Releases", click: () =>
shell.openExternal("https://github.com/otienomaurice/omb-portfolio-builder/releases/latest") },
165:     {
166:       label: "Focus Builder Window",
167:       click: () => {
168:         if (!mainWindow || mainWindow.isDestroyed()) return;
169:         if (mainWindow.isMinimized()) mainWindow.restore();
170:         mainWindow.focus();
171:       }
172:     }
173:   ]
174: }
175: ]]);
176: }
177:
178: function fileExists(filePath) {
179:   try {
180:     fs.accessSync(filePath);
181:     return true;
182:   } catch {
183:     return false;
184:   }
}

```

## fileExists - lines 178-185

| Item            | Explanation                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------|
| Source location | Lines 178-185 (8 source lines).                                                                      |
| Purpose         | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff. |
| Declaration     | function fileExists(filePath) {                                                                      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | accessSync                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

178: function fileExists(filePath) {
179:   try {
180:     fs.accessSync(filePath);
181:     return true;
182:   } catch {
183:     return false;
184:   }
185: }

```

## ***copyFileIfAvailable - lines 187-192***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 187-192 (6 source lines).                                                                                                                                                                       |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | async function copyFileIfAvailable(source, target, overwrite = false) {                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | fileExists, mkdir, dirname, copyFile                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

187: async function copyFileIfAvailable(source, target, overwrite = false) {
188:   if (!fileExists(source)) return;
189:   if (!overwrite && fileExists(target)) return;
190:   await fsp.mkdir(path.dirname(target), { recursive: true });
191:   await fsp.copyFile(source, target);
192: }

```

## ***copyDirectoryMissingFiles - lines 194-207***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 194-207 (14 source lines).                                                                                                                                                                      |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | async function copyDirectoryMissingFiles(source, target) {                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | fileExists, mkdir, readdir, join, isDirectory, isFile, copyFileIfAvailable                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 4 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
194: async function copyDirectoryMissingFiles(source, target) {
195:   if (!fileExists(source)) return;
196:   await fsp.mkdir(target, { recursive: true });
197:   const entries = await fsp.readdir(source, { withFileTypes: true });
198:   for (const entry of entries) {
199:     const sourcePath = path.join(source, entry.name);
200:     const targetPath = path.join(target, entry.name);
201:     if (entry.isDirectory()) {
202:       await copyDirectoryMissingFiles(sourcePath, targetPath);
203:     } else if (entry.isFile()) {
204:       await copyFileIfAvailable(sourcePath, targetPath, false);
205:     }
206:   }
207: }
```

## ***copyDirectoryRefresh - lines 209-214***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 209-214 (6 source lines).                                                                                                                                                                       |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | async function copyDirectoryRefresh(source, target) {                                                                                                                                                 |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | fileExists, rm, mkdir, dirname, cp                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
209: async function copyDirectoryRefresh(source, target) {
210:   if (!fileExists(source)) return;
211:   await fsp.rm(target, { recursive: true, force: true });
212:   await fsp.mkdir(path.dirname(target), { recursive: true });
213:   await fsp.cp(source, target, { recursive: true, force: true });
214: }
```

## ***directoryIsEmpty - lines 216-223***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 216-223 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | async function directoryIsEmpty(directory) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | readdir                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
216: async function directoryIsEmpty(directory) {
217:   try {
218:     const entries = await fsp.readdir(directory);
219:     return entries.length === 0;
220:   } catch {
221:     return true;
222:   }
223: }
```



## ***runGit - lines 225-242***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 225-242 (18 source lines).                                                                                                                                                                      |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function runGit(args, cwd) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | execFileAsync, Error                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
225: async function runGit(args, cwd) {
226:   let lastError = null;
227:   for (const candidate of gitCandidates) {
228:     try {
229:       await execFileAsync(candidate, args, {
230:         cwd,
231:         maxBuffer: 10 * 1024 * 1024,
232:         windowsHide: true
233:       });
234:       return true;
235:     } catch (error) {
236:       lastError = error;
237:       if (error.code === "ENOENT") continue;
238:       throw error;
239:     }
240:   }
241:   throw lastError || new Error("Git executable was not found.");
242: }
```

## ***workspaceLooksUsable - lines 244-247***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 244-247 (4 source lines).                                                                                                                                                                       |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | function workspaceLooksUsable(workspaceRoot) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | fileExists, join                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
244: function workspaceLooksUsable(workspaceRoot) {
245:   return fileExists(path.join(workspaceRoot, "server.mjs"))
246:     && fileExists(path.join(workspaceRoot, "index.html"));
247: }
```

## ***workspaceIsGitBacked - lines 249-251***

| Item            | Explanation                                                            |
|-----------------|------------------------------------------------------------------------|
| Source location | Lines 249-251 (3 source lines).                                        |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function workspaceIsGitBacked(workspaceRoot) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | fileExists, join                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

249: function workspaceIsGitBacked(workspaceRoot) {
250:   return fileExists(path.join(workspaceRoot, ".git"));
251: }

```

## ***findRepositoryNearExecutable - lines 253-262***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 253-262 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | function findRepositoryNearExecutable() {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | dirname, workspaceLooksUsable, workspaceIsGitBacked                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

253: function findRepositoryNearExecutable() {
254:   let current = path.dirname(process.execPath);
255:   for (let index = 0; index < 8; index += 1) {
256:     if (workspaceLooksUsable(current) && workspaceIsGitBacked(current)) return current;
257:     const next = path.dirname(current);
258:     if (next === current) break;
259:     current = next;
260:   }
261:   return "";
262: }

```

## ***cloneWorkspaceIfPossible - lines 264-275***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 264-275 (12 source lines).                                                                                                                                                                      |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | async function cloneWorkspaceIfPossible(workspaceRoot) {                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | workspaceIsGitBacked, mkdir, directoryIsEmpty, runGit, dirname                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
264: async function cloneWorkspaceIfPossible(workspaceRoot) {
265:   if (!defaultRepositoryUrl) return false;
266:   if (workspaceIsGitBacked(workspaceRoot)) return true;
267:   await fsp.mkdir(workspaceRoot, { recursive: true });
268:   if (!(await directoryIsEmpty(workspaceRoot))) return false;
269:   try {
270:     await runGit(["clone", "--depth", "1", defaultRepositoryUrl, workspaceRoot],
path.dirname(workspaceRoot));
271:     return workspaceIsGitBacked(workspaceRoot);
272:   } catch {
273:     return false;
274:   }
275: }
```

## ***preparePackagedWorkspace - lines 277-312***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 277-312 (36 source lines).                                                                                                                                                                      |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | async function preparePackagedWorkspace() {                                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | join, getPath, mkdir, cloneWorkspaceIfPossible, copyFileIfAvailable, copyDirectoryRefresh, copyDirectoryMissingFiles                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 10 await expression(s), 1 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
277: async function preparePackagedWorkspace() {
278:   const bundledSiteRoot = path.join(process.resourcesPath, "site");
279:   const defaultWorkspaceHome = process.platform === "win32" && process.env.LOCALAPPDATA
280:     ? path.join(process.env.LOCALAPPDATA, defaultWorkspaceHomeName)
281:     : path.join(app.getPath("userData"), "workspace");
282:   const workspaceHome = process.env.OMB_WORKSPACE_HOME || defaultWorkspaceHome;
283:   const workspaceRoot = path.join(workspaceHome, "builder");
284:   const portfolioRoot = process.env.OMB_PORTFOLIO_WORKSPACE || path.join(workspaceHome, "portfolio");
285:
286:   await fsp.mkdir(workspaceRoot, { recursive: true });
287:   await fsp.mkdir(portfolioRoot, { recursive: true });
288:   await cloneWorkspaceIfPossible(portfolioRoot);
289:
290:   for (const fileName of rootFilesToRefresh) {
291:     await copyFileIfAvailable(path.join(bundledSiteRoot, fileName), path.join(workspaceRoot, fileName),
true);
292:   }
293:   for (const fileName of rootFilesToSeed) {
294:     await copyFileIfAvailable(path.join(bundledSiteRoot, fileName), path.join(workspaceRoot, fileName),
false);
295:   }
296:   for (const directoryName of directoryRefreshes) {
297:     await copyDirectoryRefresh(path.join(bundledSiteRoot, directoryName), path.join(workspaceRoot,
directoryName));
298:   }
299:   for (const directoryName of directorySeeds) {
300:     await copyDirectoryMissingFiles(path.join(bundledSiteRoot, directoryName), path.join(workspaceRoot,
directoryName));
301:   }
302:
303:   for (const fileName of portfolioFilesToSeed) {
304:     await copyFileIfAvailable(path.join(bundledSiteRoot, fileName), path.join(portfolioRoot, fileName),
false);
305:   }
306:   await copyFileIfAvailable(path.join(bundledSiteRoot, "portfolio-README.md"), path.join(portfolioRoot,
"README.md"), false);
307:   for (const directoryName of portfolioDirectoriesToSeed) {
308:     await copyDirectoryMissingFiles(path.join(bundledSiteRoot, directoryName), path.join(portfolioRoot,
directoryName));
309:   }
310:
311:   return { workspaceRoot, portfolioRoot };
312: }
```

## ***resolveWorkspaceRoots - lines 314-337***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 314-337 (24 source lines).                                                                                                                                                                      |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | async function resolveWorkspaceRoots() {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | workspaceLooksUsable, join, dirname, resolve, findRepositoryNearExecutable, preparePackagedWorkspace                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

314: async function resolveWorkspaceRoots() {
315:   const envWorkspace = process.env.OMB_BUILDER_WORKSPACE;
316:   if (envWorkspace && workspaceLooksUsable(envWorkspace)) {
317:     return {
318:       workspaceRoot: envWorkspace,
319:       portfolioRoot: process.env.OMB_PORTFOLIO_WORKSPACE || path.join(path.dirname(envWorkspace),
"portfolio")
320:     };
321:   }
322:   if (!app.isPackaged) {
323:     const workspaceRoot = path.resolve(__dirname);
324:     return {
325:       workspaceRoot,
326:       portfolioRoot: process.env.OMB_PORTFOLIO_WORKSPACE || workspaceRoot
327:     };
328:   }
329:   const nearbyRepository = findRepositoryNearExecutable();
330:   if (nearbyRepository) {
331:     return {
332:       workspaceRoot: nearbyRepository,
333:       portfolioRoot: process.env.OMB_PORTFOLIO_WORKSPACE || nearbyRepository
334:     };
335:   }
336:   return preparePackagedWorkspace();
337: }

```

## ***findFreePort - lines 339-349***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 339-349 (11 source lines).                                                                                                                                                                      |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | function findFreePort() {                                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | createServer, once, listen, address, close, resolve                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

339: function findFreePort() {
340:   return new Promise((resolve, reject) => {
341:     const tester = net.createServer();
342:     tester.once("error", reject);
343:     tester.listen(0, "127.0.0.1", () => {
344:       const address = tester.address();
345:       const port = typeof address === "object" && address ? address.port : 0;
346:       tester.close(() => resolve(port));
347:     });
348:   });
349: }

```

## ***startBuilderServer - lines 351-364***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 351-364 (14 source lines).                                                                                                                                                                      |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | async function startBuilderServer(workspaceRoot, portfolioRoot) {                                                                                                                                     |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | findFreePort, getVersion, join, pathToFileURL, now                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
351: async function startBuilderServer(workspaceRoot, portfolioRoot) {
352:   const port = await findFreePort();
353:   process.env.PORT = String(port);
354:   process.env.HOST = "127.0.0.1";
355:   process.env.OMB_DESKTOP_APP = "1";
356:   process.env.OMB_BUILDER_WORKSPACE = workspaceRoot;
357:   process.env.OMB_PORTFOLIO_WORKSPACE = portfolioRoot;
358:   process.env.OMB_BUILDER_REPOSITORY = defaultRepositoryUrl;
359:   process.env.OMB_APP_VERSION = app.getVersion();
360:
361:   const serverPath = path.join(workspaceRoot, "server.mjs");
362:   await import(`${pathToFileURL(serverPath).href}?desktop=${Date.now()}`);
363:   return `http://127.0.0.1:${port}`;
364: }
```

## ***createWindow - lines 366-423***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 366-423 (58 source lines).                                                                                                                                                                      |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function createWindow(workspaceRoot, origin) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | join, BrowserWindow, fileExists, setMenu, createAppMenu, setMenuBarVisibility, setAutoHideMenuBar, isDestroyed, on, preventDefault, setBuilderFullScreen, isFullScreen                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
366: function createWindow(workspaceRoot, origin) {
367:   const iconPath = path.join(workspaceRoot, "assets", "omb-app-icon-256.png");
368:   nativeTheme.themeSource = "light";
369:   mainWindow = new BrowserWindow({
370:     width: 1440,
371:     height: 960,
372:     minWidth: 980,
373:     minHeight: 680,
374:     title: "OMB Portfolio Builder",
375:     autoHideMenuBar: false,
376:     icon: fileExists(iconPath) ? iconPath : undefined,
377:     backgroundColor: "#eef8fd",
378:     webPreferences: {
379:       contextIsolation: true,
380:       nodeIntegration: false,
381:       sandbox: true
382:     }
383:   });
```

```

384: mainWindow.setMenu(createAppMenu(origin));
385: mainWindow.setMenuBarVisibility(true);
386: mainWindow.setAutoHideMenuBar(false);
387:
388: const refreshFullscreenMenu = () => {
389:   if (mainWindow && !mainWindow.isDestroyed()) {
390:     mainWindow.setMenuBarVisibility(true);
391:     mainWindow.setAutoHideMenuBar(false);
392:     mainWindow.setMenu(createAppMenu(origin));
393:   }
394: };
395: mainWindow.on("enter-full-screen", refreshFullscreenMenu);
396: mainWindow.on("leave-full-screen", refreshFullscreenMenu);
397:
398: mainWindow.webContents.on("before-input-event", (event, input) => {
399:   if (input.key === "F11") {
400:     event.preventDefault();
401:     setBuilderFullScreen();
402:     return;
403:   }
404:   if (input.key === "Escape" && mainWindow.isFullScreen()) {
405:     event.preventDefault();
406:     setBuilderFullScreen(false);
407:   }
408: });
409:
410: mainWindow.webContents.setWindowOpenHandler(({ url }) => {
411:   if (url.startsWith(origin)) return { action: "allow" };
412:   shell.openExternal(url);
413:   return { action: "deny" };
414: });
415:
416: mainWindow.webContents.on("will-navigate", (event, url) => {
417:   if (url.startsWith(origin)) return;
418:   event.preventDefault();
419:   shell.openExternal(url);
420: });
421:
422: mainWindow.loadURL(`${origin}/template-preview.html`);
423: }

```

## refreshFullscreenMenu - lines 388-394

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 388-394 (7 source lines).                                                                                                                                                                       |
| Purpose            | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.                                                                                                  |
| Declaration        | const refreshFullscreenMenu = () => {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isDestroyed, setMenuBarVisibility, setAutoHideMenuBar, setMenu, createAppMenu                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

388: const refreshFullscreenMenu = () => {
389:   if (mainWindow && !mainWindow.isDestroyed()) {
390:     mainWindow.setMenuBarVisibility(true);
391:     mainWindow.setAutoHideMenuBar(false);
392:     mainWindow.setMenu(createAppMenu(origin));
393:   }
394: };

```

## boot - lines 425-447

| Item            | Explanation                                                                                          |
|-----------------|------------------------------------------------------------------------------------------------------|
| Source location | Lines 425-447 (23 source lines).                                                                     |
| Purpose         | Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff. |
| Declaration     | async function boot() {                                                                              |
| Touches         | Mostly local calculations or local UI state.                                                         |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | requestSingleInstanceLock, quit, on, isMinimized, restore, focus, whenReady, resolveWorkspaceRoots, startBuilderServer, createWindow, showErrorBox                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

425: async function boot() {
426:   const gotLock = app.requestSingleInstanceLock();
427:   if (!gotLock) {
428:     app.quit();
429:     return;
430:   }
431:
432:   app.on("second-instance", () => {
433:     if (!mainWindow) return;
434:     if (mainWindow.isMinimized()) mainWindow.restore();
435:     mainWindow.focus();
436:   });
437:
438:   await app.whenReady();
439:   try {
440:     const { workspaceRoot, portfolioRoot } = await resolveWorkspaceRoots();
441:     builderOrigin = await startBuilderServer(workspaceRoot, portfolioRoot);
442:     createWindow(workspaceRoot, builderOrigin);
443:   } catch (error) {
444:     dialog.showErrorBox("OMB Portfolio Builder could not start", error?.message || String(error));
445:     app.quit();
446:   }
447: }

```

## Representative opening snippet

```

1: const { app, Browserwindow, Menu, dialog, nativeTheme, shell } = require("electron");
2: const { execFile } = require("node:child_process");
3: const fs = require("node:fs");
4: const fsp = require("node:fs/promises");
5: const net = require("node:net");
6: const path = require("node:path");
7: const { pathToFileURL } = require("node:url");
8: const { promisify } = require("node:util");
9:
10: let mainWindow = null;
11: let builderOrigin = "";
12: let updateShutdownStarted = false;
13: const execFileAsync = promisify(execFile);
14: const defaultRepositoryUrl = process.env.OMB_BUILDER_REPOSITORY || "";
15: const defaultWorkspaceHomeName = "OMB Portfolio Builder";
16:
17: const rootFilesToRefresh = [
18:   "server.mjs",
19:   "index.html",
20:   "styles.css",
21:   "script.js",
22:   "electronics-search.js",
23:   "builder-rich-future-sections.js",
24:   "template-preview.html",
25:   "template-preview.css",
26:   "template-preview.js",
27:   "_headers"
28: ];
29:
30: const rootFilesToSeed = [
31:   "projects.json",
32:   "project-templates.json",
33:   "README.md",
34:   ".nojekyll",
35:   "robots.txt"
36: ];
37:
38: const directoryRefreshes = ["cloudflare"];
39: const directorySeeds = ["assets", "Backgrounds", "docs", ".well-known"];
40: const portfolioFilesToSeed = [
41:   "projects.json",
42:   "index.html",
43:   "styles.css",
44:   "script.js",
45:   "electronics-search.js",
46:   ".nojekyll",

```

```

47:   "robots.txt",
48:   "_headers",
49:   "sitemap.xml"
50: ];
51: const portfolioDirectoriesToSeed = ["assets", "Backgrounds", "docs", ".well-known"];
52: const gitCandidates = [
53:   process.env.GIT_EXE,
54:   "git",
55:   "C:\\Program Files\\Git\\cmd\\git.exe",
56:   "C:\\Program Files\\Git\\bin\\git.exe",
57:   "C:\\Program Files (x86)\\Git\\cmd\\git.exe",
58:   "C:\\Program Files (x86)\\Git\\bin\\git.exe"
59: ].filter(Boolean);
60:
61: function dispatchBuilderMenuAction(action) {
62:   if (!mainWindow || mainWindow.isDestroyed()) return;
63:   const payload = JSON.stringify(action);
64:   mainWindow.webContents.executeJavaScript(
65:     `window.dispatchEvent(new CustomEvent("builder-menu-action", { detail: ${payload} }));`,
66:     true
67:   ).catch(() => {});
68: }
69:
70: function quitForBuilderUpdate() {
71:   if (updateShutdownStarted) return;
72:   updateShutdownStarted = true;

```



# Source File: package.json

| Item                | Explanation                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------------------|
| File                | package.json                                                                                                 |
| Purpose             | Node/Electron package manifest with scripts, dependencies, version, installer settings, and extra resources. |
| Lines               | 85                                                                                                           |
| Size                | 2,412 bytes                                                                                                  |
| Talks to            | builder frontend, local backend, public website runtime, portfolio catalog, Cloudflare/AI layer              |
| Functions           | 0                                                                                                            |
| Variables/constants | 0                                                                                                            |
| API mentions        | 0                                                                                                            |
| Signals             | 16                                                                                                           |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                        | Explanation                                                                                                                      |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Security or auth at line 5  | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "author": "Maurice Otieno",               |
| Installer at line 10        | Windows setup, update, shortcut, or installation behavior. Evidence: "pack": "electron-builder --win portable --publish never",  |
| Installer at line 11        | Windows setup, update, shortcut, or installation behavior. Evidence: "installer": "electron-builder --win nsis --publish never", |
| Installer at line 12        | Windows setup, update, shortcut, or installation behavior. Evidence: "dist": "electron-builder --win --publish never"            |
| Installer at line 16        | Windows setup, update, shortcut, or installation behavior. Evidence: "electron-builder": "^26.0.12"                              |
| Cloudflare or AI at line 37 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "cloudflare/**",                          |
| Installer at line 50        | Windows setup, update, shortcut, or installation behavior. Evidence: "installer-notes.txt",                                      |
| Security or auth at line 54 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "_headers",                               |
| Installer at line 64        | Windows setup, update, shortcut, or installation behavior. Evidence: "nsis"                                                      |
| Installer at line 70        | Windows setup, update, shortcut, or installation behavior. Evidence: "nsis": {                                                   |
| Installer at line 76        | Windows setup, update, shortcut, or installation behavior. Evidence: "include": "build/installer.nsh",                           |
| Installer at line 77        | Windows setup, update, shortcut, or installation behavior. Evidence: "createDesktopShortcut": "always",                          |
| Installer at line 78        | Windows setup, update, shortcut, or installation behavior. Evidence: "createStartMenuShortcut": true,                            |
| Installer at line 79        | Windows setup, update, shortcut, or installation behavior. Evidence: "deleteAppDataOnUninstall": true,                           |
| Installer at line 81        | Windows setup, update, shortcut, or installation behavior. Evidence: "license": "installer-notes.txt",                           |
| Installer at line 82        | Windows setup, update, shortcut, or installation behavior. Evidence: "shortcutName": "OMB Portfolio Builder"                     |

## JSON/YAML-style keys detected

| Item    | Explanation                              |
|---------|------------------------------------------|
| name    | Line 2: "name": "omb-portfolio-builder", |
| version | Line 3: "version": "0.2.34",             |
| private | Line 4: "private": false,                |

| Item                               | Explanation                                                                                                      |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|
| author                             | Line 5: "author": "Maurice Otieno",                                                                              |
| description                        | Line 6: "description": "Standalone Windows desktop application for building and publishing portfolio websites.", |
| main                               | Line 7: "main": "main.cjs",                                                                                      |
| scripts                            | Line 8: "scripts": {                                                                                             |
| dev                                | Line 9: "dev": "electron .",                                                                                     |
| pack                               | Line 10: "pack": "electron-builder --win portable --publish never",                                              |
| installer                          | Line 11: "installer": "electron-builder --win nsis --publish never",                                             |
| dist                               | Line 12: "dist": "electron-builder --win --publish never"                                                        |
| devDependencies                    | Line 14: "devDependencies": {                                                                                    |
| electron                           | Line 15: "electron": "^39.2.7",                                                                                  |
| electron-builder                   | Line 16: "electron-builder": "^26.0.12"                                                                          |
| build                              | Line 18: "build": {                                                                                              |
| appId                              | Line 19: "appId": "com.mauriceotieno.portfoliobuilder",                                                          |
| productName                        | Line 20: "productName": "OMB Portfolio Builder",                                                                 |
| copyright                          | Line 21: "copyright": "Copyright (c) Maurice Otieno",                                                            |
| asar                               | Line 22: "asar": true,                                                                                           |
| directories                        | Line 23: "directories": {                                                                                        |
| output                             | Line 24: "output": "dist"                                                                                        |
| files                              | Line 26: "files": [                                                                                              |
| extraResources                     | Line 30: "extraResources": [                                                                                     |
| from                               | Line 32: "from": ".",                                                                                            |
| to                                 | Line 33: "to": "site",                                                                                           |
| filter                             | Line 34: "filter": [                                                                                             |
| win                                | Line 60: "win": {                                                                                                |
| icon                               | Line 61: "icon": "assets/omb-app-icon.ico",                                                                      |
| target                             | Line 62: "target": [                                                                                             |
| portable                           | Line 67: "portable": {                                                                                           |
| artifactName                       | Line 68: "artifactName": "OMB-Portfolio-Builder-Portable-\${version}-\${arch}.\${ext}"                           |
| nsis                               | Line 70: "nsis": {                                                                                               |
| artifactName                       | Line 71: "artifactName": "OMB-Portfolio-Builder-Setup-\${version}-\${arch}.\${ext}",                             |
| oneClick                           | Line 72: "oneClick": false,                                                                                      |
| perMachine                         | Line 73: "perMachine": false,                                                                                    |
| allowElevation                     | Line 74: "allowElevation": false,                                                                                |
| allowToChangeInstallationDirectory | Line 75: "allowToChangeInstallationDirectory": true,                                                             |
| include                            | Line 76: "include": "build/installer.nsh",                                                                       |
| createDesktopShortcut              | Line 77: "createDesktopShortcut": "always",                                                                      |
| createStartMenuShortcut            | Line 78: "createStartMenuShortcut": true,                                                                        |
| deleteAppDataOnUninstall           | Line 79: "deleteAppDataOnUninstall": true,                                                                       |
| runAfterFinish                     | Line 80: "runAfterFinish": true,                                                                                 |
| license                            | Line 81: "license": "installer-notes.txt",                                                                       |

| Item         | Explanation                                      |
|--------------|--------------------------------------------------|
| shortcutName | Line 82: "shortcutName": "OMB Portfolio Builder" |

## Representative opening snippet

```

1: {
2:   "name": "omb-portfolio-builder",
3:   "version": "0.2.34",
4:   "private": false,
5:   "author": "Maurice Otieno",
6:   "description": "Standalone Windows desktop application for building and publishing portfolio
websites.",
7:   "main": "main.cjs",
8:   "scripts": {
9:     "dev": "electron .",
10:    "pack": "electron-builder --win portable --publish never",
11:    "installer": "electron-builder --win nsis --publish never",
12:    "dist": "electron-builder --win --publish never"
13:  },
14:  "devDependencies": {
15:    "electron": "^39.2.7",
16:    "electron-builder": "^26.0.12"
17:  },
18:  "build": {
19:    "appId": "com.mauriceotieno.portfoliobuilder",
20:    "productName": "OMB Portfolio Builder",
21:    "copyright": "Copyright (c) Maurice Otieno",
22:    "asar": true,
23:    "directories": {
24:      "output": "dist"
25:    },
26:    "files": [
27:      "main.cjs",
28:      "package.json"
29:    ],
30:    "extraResources": [
31:      {
32:        "from": ".",
33:        "to": "site",
34:        "filter": [
35:          "assets/**",
36:          "Backgrounds/**",
37:          "cloudflare/**",
38:          "docs/**",
39:          "server.mjs",
40:          "index.html",
41:          "styles.css",
42:          "script.js",
43:          "electronics-search.js",
44:          "builder-rich-future-sections.js",
45:          "projects.json",
46:          "project-templates.json",
47:          "template-preview.html",
48:          "template-preview.css",
49:          "template-preview.js",
50:          "installer-notes.txt",
51:          "README.md",
52:          "portfolio-README.md",
53:          ".nojekyll",
54:          "_headers",
55:          ".well-known/**",
56:          "robots.txt"
57:        ]
58:      }
59:    ],
60:    "win": {
61:      "icon": "assets/omb-app-icon.ico",
62:      "target": [
63:        "portable",
64:        "nsis"
65:      ]
66:    },
67:    "portable": {
68:      "artifactName": "OMB-Portfolio-Builder-Portable-${version}-${arch}.${ext}"
69:    },
70:    "nsis": {
71:      "artifactName": "OMB-Portfolio-Builder-Setup-${version}-${arch}.${ext}",
72:      "oneClick": false,

```

# Source File: pnpm-lock.yaml

| Item                | Explanation                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------|
| File                | pnpm-lock.yaml                                                                              |
| Purpose             | Pinned dependency lockfile so installs and GitHub Actions use the same dependency versions. |
| Lines               | 2,233                                                                                       |
| Size                | 69,301 bytes                                                                                |
| Talks to            | Mostly local to its own feature area.                                                       |
| Functions           | 0                                                                                           |
| Variables/constants | 0                                                                                           |
| API mentions        | 0                                                                                           |
| Signals             | 285                                                                                         |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                             | Explanation                                                                                                                                                                                                      |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installer at line 14             | Windows setup, update, shortcut, or installation behavior. Evidence: electron-builder:                                                                                                                           |
| Installer at line 16             | Windows setup, update, shortcut, or installation behavior. Evidence: version: 26.15.3(electron-builder-squirrel-windows@26.15.3)                                                                                 |
| Security or auth at line 21      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: resolution: {integrity: sha512-i4/rNPRS84t0vSRa2HorerGRXWyF4vThfHesw0dmcWHP+cspK743UanA0suA5Q5y8kzY2y6YKrvbIUUn69BCAiA==} |
| Compiler or simulator at line 22 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=10.12.0'}                                                                                                            |
| Compiler or simulator at line 31 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=12'}                                                                                                                 |
| Compiler or simulator at line 35 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=14'}                                                                                                                 |
| Compiler or simulator at line 39 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 10.0.0'}                                                                                                            |
| Compiler or simulator at line 43 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=12.0.0'}                                                                                                             |
| Compiler or simulator at line 48 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=22.12.0'}                                                                                                            |
| Compiler or simulator at line 53 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=16.4'}                                                                                                               |
| Compiler or simulator at line 57 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=14.14'}                                                                                                              |
| Compiler or simulator at line 62 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=18.0.0'}                                                                                                             |
| Process execution at line 64     | Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: '@malept/cross-spawn-promise@2.0.0':                                                                                  |
| Compiler or simulator at line 66 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 12.13.0'}                                                                                                           |
| Compiler or simulator at line 70 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 10.0.0'}                                                                                                            |
| Compiler or simulator at line 74 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 16'}                                                                                                                |

| Item                              | Explanation                                                                                                                                                                                                            |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compiler or simulator at line 78  | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 20.19.0'}                                                                                                                 |
| Compiler or simulator at line 85  | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=8.0.0'}                                                                                                                    |
| Compiler or simulator at line 92  | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=14.18.0'}                                                                                                                  |
| Compiler or simulator at line 96  | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=10'}                                                                                                                       |
| Compiler or simulator at line 100 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=10'}                                                                                                                       |
| Local storage at line 102         | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: '@types/cacheable-request@6.0.3':                                                                                        |
| Local storage at line 111         | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: '@types/http-cache-semantics@4.2.0':                                                                                     |
| Compiler or simulator at line 120 | Part of the Compile Code workspace or language/tool detection. Evidence: '@types/node@22.20.0':                                                                                                                        |
| Compiler or simulator at line 123 | Part of the Compile Code workspace or language/tool detection. Evidence: '@types/node@26.0.0':                                                                                                                         |
| Compiler or simulator at line 134 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=10.0.0'}                                                                                                                   |
| Compiler or simulator at line 138 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '^20.17.0    >=22.9.0'}                                                                                                       |
| Compiler or simulator at line 142 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 14'}                                                                                                                      |
| Compiler or simulator at line 149 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=8'}                                                                                                                        |
| Compiler or simulator at line 153 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=8'}                                                                                                                        |
| Compiler or simulator at line 157 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=14.0.0'}                                                                                                                   |
| Installer at line 160             | Windows setup, update, shortcut, or installation behavior. Evidence: electron-builder-squirrel-windows: 26.15.3                                                                                                        |
| Compiler or simulator at line 167 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=12.0.0'}                                                                                                                   |
| Compiler or simulator at line 171 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=0.12.0'}                                                                                                                   |
| Compiler or simulator at line 179 | Part of the Compile Code workspace or language/tool detection. Evidence: at-least-node@1.0.0:                                                                                                                          |
| Compiler or simulator at line 181 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 4.0.0'}                                                                                                                   |
| Compiler or simulator at line 191 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: 18    20    >=22}                                                                                                             |
| Network request at line 201       | Frontend-backend communication or public web/API calls. Evidence: deprecated: Package no longer supported. Contact Support at <a href="https://www.npmjs.com/support">https://www.npmjs.com/support</a> for more info. |
| Compiler or simulator at line 211 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: 18    20    >=22}                                                                                                             |
| Compiler or simulator at line 221 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=12.0.0'}                                                                                                                   |
| Compiler or simulator at line 225 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=14.0.0'}                                                                                                                   |
| Compiler or simulator at line 229 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=6.0.0'}                                                                                                                    |
| Local storage at line 231         | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: cacheable-lookup@5.0.4:                                                                                                  |

| Item                              | Explanation                                                                                                            |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Compiler or simulator at line 233 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=10.6.0'}                   |
| Local storage at line 235         | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: cacheable-request@7.0.4: |
| Compiler or simulator at line 237 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=8'}                        |
| Compiler or simulator at line 241 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 0.4'}                     |
| Compiler or simulator at line 245 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=10'}                       |
| Compiler or simulator at line 249 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=18'}                       |
| Compiler or simulator at line 256 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=8'}                        |
| Compiler or simulator at line 260 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=8'}                        |
| Compiler or simulator at line 264 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=12'}                       |
| Rich text at line 269             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color-convert@2.0.1:              |
| Compiler or simulator at line 271 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=7.0.0'}                    |
| Rich text at line 273             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color-name@1.1.4:                 |
| Compiler or simulator at line 278 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 0.8'}                     |
| Compiler or simulator at line 282 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 6'}                       |
| Compiler or simulator at line 286 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: ^12.20.0    >=14}             |
| Compiler or simulator at line 290 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=0.10.0'}                   |
| Process execution at line 301     | Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: cross-spawn@7.0.6:          |
| Compiler or simulator at line 303 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 8'}                       |
| Compiler or simulator at line 307 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=6.0'}                      |
| Rich text at line 309             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: supports-color: '**               |
| Rich text at line 311             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: supports-color:                   |
| Compiler or simulator at line 316 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=10'}                       |
| Compiler or simulator at line 320 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=10'}                       |
| Compiler or simulator at line 324 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 0.4'}                     |
| Compiler or simulator at line 328 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 0.4'}                     |
| Compiler or simulator at line 332 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=0.4.0'}                    |
| Compiler or simulator at line 334 | Part of the Compile Code workspace or language/tool detection. Evidence: detect-node@2.1.0:                            |
| Compiler or simulator at line 345 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=12'}                       |
| Compiler or simulator at line 349 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=12'}                       |

| Item                              | Explanation                                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Compiler or simulator at line 353 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 0.4'}              |
| Compiler or simulator at line 360 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=0.10.0'}            |
| Installer at line 363             | Windows setup, update, shortcut, or installation behavior. Evidence: electron-builder-squirrel-windows@26.15.3: |
| Installer at line 366             | Windows setup, update, shortcut, or installation behavior. Evidence: electron-builder@26.15.3:                  |
| Compiler or simulator at line 368 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=14.0.0'}            |
| Installer at line 374             | Windows setup, update, shortcut, or installation behavior. Evidence: electron-winstaller@5.4.0:                 |
| Compiler or simulator at line 376 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>=8.0.0'}             |
| Compiler or simulator at line 380 | Part of the Compile Code workspace or language/tool detection. Evidence: engines: {node: '>= 12.20.55'}         |

## Representative opening snippet

```

1: lockfileVersion: '9.0'
2:
3: settings:
4:   autoInstallPeers: true
5:   excludeLinksFromLockfile: false
6:
7: importers:
8:
9:   ..
10:     devDependencies:
11:       electron:
12:         specifier: ^39.2.7
13:         version: 39.8.10
14:       electron-builder:
15:         specifier: ^26.0.12
16:         version: 26.15.3(electron-builder-squirrel-windows@26.15.3)
17:
18: packages:
19:
20:   '@electron/asar@3.4.1':
21:     resolution: {integrity:
sha512-i4/rNPRS84t0vSRa2HorerGRXWyF4vThfHesw0dmcwHp+cspK743UanA0suA5Q5y8kzY2y6YKrvbIU69BCAiA==}
22:     engines: {node: '>=10.12.0'}
23:     hasBin: true
24:
25:   '@electron/fuses@1.8.0':
26:     resolution: {integrity:
sha512-zx0EIq78WLY/lBb1uXlziZmDZI4ubcCXIMJ4uGjXzZW0nS19TjSPeXPAjzzTmKQlJUzm0SbmZhPKP7tuQ1SsEw==}
27:     hasBin: true
28:
29:   '@electron/get@2.0.3':
30:     resolution: {integrity:
sha512-Qkzpg2s9GnVV2I2BjRksUi43U5e6+zaQMcoJy0C+C5oxaKl+fmckGDQFtRpZpZV0NqekuZz+tGz7EA9TVnQtQ==}
31:     engines: {node: '>=12'}
32:
33:   '@electron/get@3.1.0':
34:     resolution: {integrity:
sha512-F+nKc0xw+kVbBRhFzaMgPy3KwmuNTYX1fx6+FxxoSnNgwYX6LD7AKBTWkU0MQ6IBoe7dz069CNkR673sPAgkCQ==}
35:     engines: {node: '>=14'}
36:
37:   '@electron/notarize@2.5.0':
38:     resolution: {integrity:
sha512-jNT8nwH1f9X5GEITXaQ8IF/KdskvIK0Fb2CvumsveVidzPSc+mvhhTmdAGSYF30+Nq49LJ7y+ss0DRXu06+A==}
39:     engines: {node: '>= 10.0.0'}
40:
41:   '@electron/osx-sign@1.3.3':
42:     resolution: {integrity:
sha512-KZ8mhXvWv2rIEgMbWZ4y33bDHyUKMXnx4M0sTyPNK/vcB81ImdeY9Ggdqy0SwbMDgmbqyQ+phgejh6V3R2QuSg==}
43:     engines: {node: '>=12.0.0'}
44:     hasBin: true
45:
46:   '@electron/rebuild@4.0.4':
47:     resolution: {integrity:
sha512-Rzc39XPdk/+wBG8MfwAHohXfleP0ITUfulb6Rgz3R0NeSB1noE+E9/M/cb8ftCAiyDD9PPHLuuWgE1GaInbKg==}
48:     engines: {node: '>=22.12.0'}
49:     hasBin: true
50:
51:   '@electron/universal@2.0.3':
52:     resolution: {integrity:
sha512-Wn9sPYIVFRFL5HmwMJKARCCF7rqQ/EurkfQ/rJZ14mHP3iYTjZSIO5VonEAnhWeAXwtw7z0ekGRlc6yTtZ0t+g==}
53:     engines: {node: '>=16.4'}
54:

```

```
55: '@electron/windows-sign@1.2.2':
56:   resolution: {integrity:
sha512-dfZeox66AvdPtb2LD80sIIQh12Tp0GNCRUdFBHIKGpbmopZto2/A8nSpYYLoedPIHpqkeblZ/k80V0Gy7PYuyQ==}
57:   engines: {node: '>=14.14'}
58:   hasBin: true
59:
60: '@isaacs/fs-minipass@4.0.1':
61:   resolution: {integrity:
sha512-wgm9Ehl2jpeqP3zw/7mo3kRHFp5MEDhqAdwy1fTGkHAwnkG0VsgpvQhL8B5n1qlb01jV3n/bI0ZfZp5lWA1k4w==}
62:   engines: {node: '>=18.0.0'}
63:
64: '@malept/cross-spawn-promise@2.0.0':
65:   resolution: {integrity:
sha512-1DpKU0Z5ThltBwjNySMC14g0CkbyhCaz9FkxqNsZI6uAPJXFS8cMXlBKo26FJ8ZuW6S9GCMcR9I05k2X5/9Fg==}
66:   engines: {node: '>= 12.13.0'}
67:
68: '@malept/flatpak-bundler@0.4.0':
69:   resolution: {integrity:
sha512-9Q0tNffcoF/c1seMCDnjckb3R9WHcG34tky+FHpnKKCW0wc/scYLwMt0+ptyGUfMW0/b/n4qRiAl laFhc90j7Q==}
70:   engines: {node: '>= 10.0.0'}
71:
72: '@noble/hashes@1.4.0':
```



# Source File: pnpm-workspace.yaml

| Item                | Explanation                                    |
|---------------------|------------------------------------------------|
| File                | pnpm-workspace.yaml                            |
| Purpose             | pnpm workspace declaration for the repository. |
| Lines               | 3                                              |
| Size                | 58 bytes                                       |
| Talks to            | Mostly local to its own feature area.          |
| Functions           | 0                                              |
| Variables/constants | 0                                              |
| API mentions        | 0                                              |
| Signals             | 1                                              |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                | Explanation                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------|
| Installer at line 3 | Windows setup, update, shortcut, or installation behavior. Evidence: electron-winstaller: true |

## Representative opening snippet

```
1: allowBuilds:
2:   electron: true
3:   electron-winstaller: true
```

# Source File: portfolio-README.md

| Item                | Explanation                                                                            |
|---------------------|----------------------------------------------------------------------------------------|
| File                | portfolio-README.md                                                                    |
| Purpose             | README for the generated public portfolio website rather than the builder application. |
| Lines               | 67                                                                                     |
| Size                | 2,893 bytes                                                                            |
| Talks to            | public website runtime, portfolio catalog, Cloudflare/AI layer, GitHub/release layer   |
| Functions           | 0                                                                                      |
| Variables/constants | 0                                                                                      |
| API mentions        | 1                                                                                      |
| Signals             | 10                                                                                     |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## API endpoints mentioned or called

| Item              | Explanation                              |
|-------------------|------------------------------------------|
| /api/portfolio-ai | Line 47. Handles AI assistant questions. |

## Important implementation signals

| Item                        | Explanation                                                                                                                                                                                                                                                                                                         |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Cloudflare or AI at line 5  | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: For the full beginner explanation of how the builder creates this website, how the files communicate, how GitHub/Cloudflare publishing works, and what each build tool does, open:                                           |
| Installer at line 26        | Windows setup, update, shortcut, or installation behavior. Evidence: - Installer scripts or app release artifacts.                                                                                                                                                                                                  |
| Security or auth at line 28 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: - Private credentials, tokens, or authentication cache files.                                                                                                                                                                |
| Cloudflare or AI at line 42 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: If the website uses the AI assistant, its backend belongs to the Cloudflare Worker in `cloudflare/`. That Worker is part of the website runtime. It is separate from the Windows builder application.                        |
| Cloudflare or AI at line 47 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <meta name="portfolio-ai-endpoint" content="/api/portfolio-ai" />                                                                                                                                                            |
| Cloudflare or AI at line 50 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: For `mauriceotieno.com`, deploy the Worker from `cloudflare/` and keep DNS proxied through Cloudflare so the route `mauriceotieno.com/api/portfolio-ai` reaches the Worker. Configure the Worker secret with:                |
| Cloudflare or AI at line 53 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: npx wrangler secret put OPENAI_API_KEY                                                                                                                                                                                       |
| Cloudflare or AI at line 59 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: cd cloudflare                                                                                                                                                                                                                |
| Cloudflare or AI at line 60 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: npx wrangler deploy                                                                                                                                                                                                          |
| Cloudflare or AI at line 63 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: If the AI widget falls back to local/canned answers on the public site, check three things first: the meta endpoint exists in `index.html`, the Cloudflare Worker route exists for `/api/portfolio-ai`, and the Worker has t |

## Representative opening snippet

```
1: # Portfolio Website
2:
3: This repository/folder contains the generated portfolio website files. It is the public site output, not
the Windows builder application source.
```

```

4:
5: For the full beginner explanation of how the builder creates this website, how the files communicate, how
GitHub/Cloudflare publishing works, and what each build tool does, open:
6:
7: ```text
8: docs/OMB_Portfolio_Builder_Complete_Guide.docx
9: ```
10:
11: ## What Belongs Here
12:
13: - `index.html` for the public page shell.
14: - `styles.css` for website styling.
15: - `script.js` and `electronics-search.js` for website interaction.
16: - `projects.json` for the saved public portfolio catalog.
17: - `assets/` for public images, icons, resumes, and profile media.
18: - `docs/` for public project documents and downloadable evidence.
19: - `Backgrounds/` for public background images.
20: - `CNAME` when a custom domain is configured.
21: - `robots.txt` and `sitemap.xml` for search engines.
22:
23: ## What Does Not Belong Here
24:
25: - Builder desktop app source files.
26: - Installer scripts or app release artifacts.
27: - Local draft files such as `projects.local.json`.
28: - Private credentials, tokens, or authentication cache files.
29:
30: ## Hosting
31:
32: The site is designed for static hosting such as GitHub Pages. A custom domain can be connected by placing
the domain name in `CNAME` and pointing DNS at the hosting provider.
33:
34: ## Updating Content
35:
36: The website files are generated from saved public catalog data and public assets. Edit content in the
local editing workspace, save the draft, preview the generated site, then publish the generated output into this
folder/repository.
37:
38: Manual edits inside this folder can be overwritten by the next publish. Keep lasting content changes in
the editing workspace.
39:
40: ## Portfolio AI
41:
42: If the website uses the AI assistant, its backend belongs to the Cloudflare Worker in `cloudflare/`. That
Worker is part of the website runtime. It is separate from the Windows builder application.
43:
44: The public page should include:
45:
46: ```html
47: <meta name="portfolio-ai-endpoint" content="/api/portfolio-ai" />
48: ```
49:
50: For `mauriceotieno.com`, deploy the Worker from `cloudflare/` and keep DNS proxied through Cloudflare so
the route `mauriceotieno.com/api/portfolio-ai` reaches the Worker. Configure the Worker secret with:
51:
52: ```powershell
53: npx wrangler secret put OPENAI_API_KEY
54: ```
55:
56: Then deploy:
57:
58: ```powershell
59: cd cloudflare
60: npx wrangler deploy
61: ```
62:
63: If the AI widget falls back to local/canned answers on the public site, check three things first: the
meta endpoint exists in `index.html`, the Cloudflare Worker route exists for `/api/portfolio-ai`, and the Worker
has the `OPENAI_API_KEY` secret.
64:
65: ## Public Safety
66:
67: Only public files should be stored here. Before publishing, confirm that resumes, documents, images,
project files, source links, and downloadable artifacts are intended for public viewing.

```

# Source File: project-templates.json

| Item                | Explanation                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------|
| File                | project-templates.json                                                                       |
| Purpose             | Appearance template definitions used by projects to control visual feel rather than content. |
| Lines               | 184                                                                                          |
| Size                | 6,428 bytes                                                                                  |
| Talks to            | Mostly local to its own feature area.                                                        |
| Functions           | 0                                                                                            |
| Variables/constants | 0                                                                                            |
| API mentions        | 0                                                                                            |
| Signals             | 0                                                                                            |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## JSON/YAML-style keys detected

| Item        | Explanation                                                                                                                   |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| templates   | Line 2: "templates": [                                                                                                        |
| id          | Line 4: "id": "appearance-light-blue-red-click",                                                                              |
| label       | Line 5: "label": "Light blue, navy hover, red click",                                                                         |
| description | Line 6: "description": "Light blue engineering background with calm navy hover states and a sharp red click flash.",          |
| visual      | Line 7: "visual": {                                                                                                           |
| style       | Line 8: "style": "schematic",                                                                                                 |
| palette     | Line 9: "palette": ["#e9f8ff", "#1f6ed4", "#ef4444", "#ffffff"],                                                              |
| background  | Line 10: "background": "#e9f8ff",                                                                                             |
| panel       | Line 11: "panel": "#ffffff",                                                                                                  |
| accent      | Line 12: "accent": "#1f6ed4",                                                                                                 |
| hover       | Line 13: "hover": "#1f6ed4",                                                                                                  |
| click       | Line 14: "click": "#ef4444",                                                                                                  |
| clickText   | Line 15: "clickText": "#ffffff",                                                                                              |
| line        | Line 16: "line": "#b8def3",                                                                                                   |
| text        | Line 17: "text": "#172636",                                                                                                   |
| interaction | Line 18: "interaction": "Navy-blue hover with a red active click flash."                                                      |
| id          | Line 22: "id": "appearance-white-blue-click",                                                                                 |
| label       | Line 23: "label": "Clean white, blue hover, white click",                                                                     |
| description | Line 24: "description": "Minimal white project panels with professional blue hover behavior and quiet white click feedback.", |
| visual      | Line 25: "visual": {                                                                                                          |
| style       | Line 26: "style": "device",                                                                                                   |
| palette     | Line 27: "palette": ["#ffffff", "#2563eb", "#f8fbff"],                                                                        |
| background  | Line 28: "background": "#ffffff",                                                                                             |

| Item        | Explanation                                                                                                              |
|-------------|--------------------------------------------------------------------------------------------------------------------------|
| panel       | Line 29: "panel": "#ffffff",                                                                                             |
| accent      | Line 30: "accent": "#2563eb",                                                                                            |
| hover       | Line 31: "hover": "#2563eb",                                                                                             |
| click       | Line 32: "click": "#ffffff",                                                                                             |
| clickText   | Line 33: "clickText": "#172636",                                                                                         |
| line        | Line 34: "line": "#d8e2ef",                                                                                              |
| text        | Line 35: "text": "#172636",                                                                                              |
| interaction | Line 36: "interaction": "Blue hover with a clean white click state."                                                     |
| id          | Line 40: "id": "appearance-deep-navy-cyan-click",                                                                        |
| label       | Line 41: "label": "Deep navy, cyan hover, white click",                                                                  |
| description | Line 42: "description": "Dark navy technical background with cyan hover states and high-contrast white click feedback.", |
| visual      | Line 43: "visual": {                                                                                                     |
| style       | Line 44: "style": "chip",                                                                                                |
| palette     | Line 45: "palette": ["#10263c", "#0f6f96", "#ffffff"],                                                                   |
| background  | Line 46: "background": "#10263c",                                                                                        |
| panel       | Line 47: "panel": "#f8fbff",                                                                                             |
| accent      | Line 48: "accent": "#0f6f96",                                                                                            |
| hover       | Line 49: "hover": "#0f6f96",                                                                                             |
| click       | Line 50: "click": "#ffffff",                                                                                             |
| clickText   | Line 51: "clickText": "#142536",                                                                                         |
| line        | Line 52: "line": "#8ccfe8",                                                                                              |
| text        | Line 53: "text": "#f8fbff",                                                                                              |
| interaction | Line 54: "interaction": "Cyan hover with a white active click state."                                                    |
| id          | Line 58: "id": "appearance-warm-red-pale-click",                                                                         |
| label       | Line 59: "label": "Warm red, wine hover, pale click",                                                                    |
| description | Line 60: "description": "Warm signal-lab styling with wine-red hover behavior and a soft pale-red click response.",      |
| visual      | Line 61: "visual": {                                                                                                     |
| style       | Line 62: "style": "waveform",                                                                                            |
| palette     | Line 63: "palette": ["#fff7f5", "#c2413b", "#fff4f2"],                                                                   |
| background  | Line 64: "background": "#fff7f5",                                                                                        |
| panel       | Line 65: "panel": "#ffffff",                                                                                             |
| accent      | Line 66: "accent": "#b9332e",                                                                                            |
| hover       | Line 67: "hover": "#c2413b",                                                                                             |
| click       | Line 68: "click": "#fff4f2",                                                                                             |
| clickText   | Line 69: "clickText": "#2f2020",                                                                                         |
| line        | Line 70: "line": "#efb0aa",                                                                                              |
| text        | Line 71: "text": "#2f2020",                                                                                              |
| interaction | Line 72: "interaction": "Wine-red hover with a pale red click state."                                                    |
| id          | Line 76: "id": "appearance-emerald-mint-click",                                                                          |
| label       | Line 77: "label": "Emerald bench, green hover, mint click",                                                              |
| description | Line 78: "description": "Instrumentation-style panels with emerald hover states and a light mint click response.",       |

| Item        | Explanation                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------|
| visual      | Line 79: "visual": {                                                                                               |
| style       | Line 80: "style": "instrument",                                                                                    |
| palette     | Line 81: "palette": ["#f0fdf6", "#16865a", "#ffffff"],                                                             |
| background  | Line 82: "background": "#f0fdf6",                                                                                  |
| panel       | Line 83: "panel": "#####",                                                                                         |
| accent      | Line 84: "accent": "#16865a",                                                                                      |
| hover       | Line 85: "hover": "#16865a",                                                                                       |
| click       | Line 86: "click": "#ffffff",                                                                                       |
| clickText   | Line 87: "clickText": "#182f26",                                                                                   |
| line        | Line 88: "line": "#9adfbf",                                                                                        |
| text        | Line 89: "text": "#182f26",                                                                                        |
| interaction | Line 90: "interaction": "Emerald hover with a soft mint click state."                                              |
| id          | Line 94: "id": "appearance-amber-cream-click",                                                                     |
| label       | Line 95: "label": "Amber rail, gold hover, cream click",                                                           |
| description | Line 96: "description": "Power-rail inspired appearance with amber hover behavior and warm cream click feedback.", |
| visual      | Line 97: "visual": {                                                                                               |

## Representative opening snippet

```

1: {
2:   "templates": [
3:     {
4:       "id": "appearance-light-blue-red-click",
5:       "label": "Light blue, navy hover, red click",
6:       "description": "Light blue engineering background with calm navy hover states and a sharp red click
flash.",
7:       "visual": {
8:         "style": "schematic",
9:         "palette": ["#e9f8ff", "#1f6ed4", "#ef4444", "#####"],
10:        "background": "#e9f8ff",
11:        "panel": "#####",
12:        "accent": "#1f6ed4",
13:        "hover": "#1f6ed4",
14:        "click": "#ef4444",
15:        "clickText": "#####",
16:        "line": "#b8def3",
17:        "text": "#172636",
18:        "interaction": "Navy-blue hover with a red active click flash."
19:      }
20:    },
21:    {
22:      "id": "appearance-white-blue-click",
23:      "label": "Clean white, blue hover, white click",
24:      "description": "Minimal white project panels with professional blue hover behavior and quiet white
click feedback.",
25:      "visual": {
26:        "style": "device",
27:        "palette": ["#####", "#2563eb", "#f8fbff"],
28:        "background": "#####",
29:        "panel": "#####",
30:        "accent": "#2563eb",
31:        "hover": "#2563eb",
32:        "click": "#####",
33:        "clickText": "#172636",
34:        "line": "#d8e2ef",
35:        "text": "#172636",
36:        "interaction": "Blue hover with a clean white click state."
37:      }
38:    },
39:    {
40:      "id": "appearance-deep-navy-cyan-click",
41:      "label": "Deep navy, cyan hover, white click",
42:      "description": "Dark navy technical background with cyan hover states and high-contrast white click
feedback.",
43:      "visual": {
44:        "style": "chip",
45:        "palette": ["#10263c", "#0f6f96", "#####"],
46:        "background": "#10263c",
47:        "panel": "#f8fbff",

```

```

48:         "accent": "#0f6f96",
49:         "hover": "#0f6f96",
50:         "click": "#ffffff",
51:         "clickText": "#142536",
52:         "line": "#8ccfe8",
53:         "text": "#f8fbff",
54:         "interaction": "Cyan hover with a white active click state."
55:     }
56: },
57: {
58:     "id": "appearance-warm-red-pale-click",
59:     "label": "Warm red, wine hover, pale click",
60:     "description": "Warm signal-lab styling with wine-red hover behavior and a soft pale-red click
response.",
61:     "visual": {
62:         "style": "waveform",
63:         "palette": ["#fff7f5", "#c2413b", "#fff4f2"],
64:         "background": "#fff7f5",
65:         "panel": "#ffffff",
66:         "accent": "#b9332e",
67:         "hover": "#c2413b",
68:         "click": "#fff4f2",
69:         "clickText": "#2f2020",
70:         "line": "#efb0aa",
71:         "text": "#2f2020",
72:         "interaction": "Wine-red hover with a pale red click state."

```

# Source File: projects.json

| Item                | Explanation                                                                |
|---------------------|----------------------------------------------------------------------------|
| File                | projects.json                                                              |
| Purpose             | Public project catalog consumed by the website after parsing builder data. |
| Lines               | 46                                                                         |
| Size                | 1,490 bytes                                                                |
| Talks to            | GitHub/release layer                                                       |
| Functions           | 0                                                                          |
| Variables/constants | 0                                                                          |
| API mentions        | 0                                                                          |
| Signals             | 0                                                                          |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## JSON/YAML-style keys detected

| Item         | Explanation                                                                                                                                                |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| categories   | Line 2: "categories": [                                                                                                                                    |
| id           | Line 4: "id": "analog-mixed-signal",                                                                                                                       |
| label        | Line 5: "label": "Analog and Mixed Signal",                                                                                                                |
| description  | Line 6: "description": "Circuit-level work including amplifiers, filters, converters, measurement front ends, power stages, and mixed-signal interfaces.", |
| accent       | Line 7: "accent": "#117c7a"                                                                                                                                |
| id           | Line 10: "id": "digital",                                                                                                                                  |
| label        | Line 11: "label": "Digital",                                                                                                                               |
| description  | Line 12: "description": "Digital design projects involving logic design, FPGA workflows, verification, timing, and ASIC-oriented architecture.",           |
| accent       | Line 13: "accent": "#52606d"                                                                                                                               |
| id           | Line 16: "id": "embedded",                                                                                                                                 |
| label        | Line 17: "label": "Embedded",                                                                                                                              |
| description  | Line 18: "description": "MCU-based systems with firmware, board bring-up, interfaces, sensing, control, and hardware/software integration.",               |
| accent       | Line 19: "accent": "#d18b21"                                                                                                                               |
| funFacts     | Line 22: "funFacts": [],                                                                                                                                   |
| funFactsRich | Line 23: "funFactsRich": null,                                                                                                                             |
| profile      | Line 24: "profile": {                                                                                                                                      |
| brandImage   | Line 25: "brandImage": "",                                                                                                                                 |
| brandText    | Line 26: "brandText": "Portfolio",                                                                                                                         |
| contactIntro | Line 27: "contactIntro": "",                                                                                                                               |
| displayName  | Line 28: "displayName": "",                                                                                                                                |
| email        | Line 29: "email": "",                                                                                                                                      |
| githubUrl    | Line 30: "githubUrl": "",                                                                                                                                  |



| Item           | Explanation                                                                                                                                                                                                 |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| heroImage      | Line 31: "heroImage": "",                                                                                                                                                                                   |
| linkedinUrl    | Line 32: "linkedinUrl": "",                                                                                                                                                                                 |
| phone          | Line 33: "phone": "",                                                                                                                                                                                       |
| portfolioLabel | Line 34: "portfolioLabel": "Portfolio",                                                                                                                                                                     |
| profileImage   | Line 35: "profileImage": "",                                                                                                                                                                                |
| resumeUrl      | Line 36: "resumeUrl": "",                                                                                                                                                                                   |
| websiteUrl     | Line 37: "websiteUrl": ""                                                                                                                                                                                   |
| siteContent    | Line 39: "siteContent": {                                                                                                                                                                                   |
| heroCopy       | Line 40: "heroCopy": "Add your projects, documents, diagrams, source code, images, profile details, and links.\nSave drafts locally, preview the site, then publish when your target repository is ready.", |
| heroEyebrow    | Line 41: "heroEyebrow": "Engineering portfolio",                                                                                                                                                            |
| heroTitle      | Line 42: "heroTitle": "Build a portfolio that presents your work clearly."                                                                                                                                  |
| projects       | Line 44: "projects": [],                                                                                                                                                                                    |
| siteSections   | Line 45: "siteSections": []                                                                                                                                                                                 |

## Representative opening snippet

```

1: {
2:   "categories": [
3:     {
4:       "id": "analog-mixed-signal",
5:       "label": "Analog and Mixed Signal",
6:       "description": "Circuit-level work including amplifiers, filters, converters, measurement front
ends, power stages, and mixed-signal interfaces.",
7:       "accent": "#117c7a"
8:     },
9:     {
10:      "id": "digital",
11:      "label": "Digital",
12:      "description": "Digital design projects involving logic design, FPGA workflows, verification,
timing, and ASIC-oriented architecture.",
13:      "accent": "#52606d"
14:    },
15:    {
16:      "id": "embedded",
17:      "label": "Embedded",
18:      "description": "MCU-based systems with firmware, board bring-up, interfaces, sensing, control, and
hardware/software integration.",
19:      "accent": "#d18b21"
20:    }
21:  ],
22:  "funFacts": [],
23:  "funFactsRich": null,
24:  "profile": {
25:    "brandImage": "",
26:    "brandText": "Portfolio",
27:    "contactIntro": "",
28:    "displayName": "",
29:    "email": "",
30:    "githubUrl": "",
31:    "heroImage": "",
32:    "linkedinUrl": "",
33:    "phone": "",
34:    "portfolioLabel": "Portfolio",
35:    "profileImage": "",
36:    "resumeUrl": "",
37:    "websiteUrl": ""
38:  },
39:  "siteContent": {
40:    "heroCopy": "Add your projects, documents, diagrams, source code, images, profile details, and
links.\nSave drafts locally, preview the site, then publish when your target repository is ready.",
41:    "heroEyebrow": "Engineering portfolio",
42:    "heroTitle": "Build a portfolio that presents your work clearly."
43:  },
44:  "projects": [],
45:  "siteSections": []
46: }
```

# Source File: robots.txt

| Item                | Explanation                                  |
|---------------------|----------------------------------------------|
| File                | robots.txt                                   |
| Purpose             | Crawler instruction file for search engines. |
| Lines               | 2                                            |
| Size                | 23 bytes                                     |
| Talks to            | Mostly local to its own feature area.        |
| Functions           | 0                                            |
| Variables/constants | 0                                            |
| API mentions        | 0                                            |
| Signals             | 0                                            |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Representative opening snippet

```
1: User-agent: *
2: Allow: /
```

# Source File: script.js

| Item                | Explanation                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------|
| File                | script.js                                                                                                      |
| Purpose             | Public website JavaScript for navigation, portfolio rendering, search, AI UI, and interactive project windows. |
| Lines               | 3,710                                                                                                          |
| Size                | 158,556 bytes                                                                                                  |
| Talks to            | portfolio catalog, GitHub/release layer                                                                        |
| Functions           | 236                                                                                                            |
| Variables/constants | 140                                                                                                            |
| API mentions        | 2                                                                                                              |
| Signals             | 380                                                                                                            |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## API endpoints mentioned or called

| Item              | Explanation                                |
|-------------------|--------------------------------------------|
| /api/portfolio-ai | Line 1492. Handles AI assistant questions. |
| /api/portfolio-ai | Line 1493. Handles AI assistant questions. |

## Important implementation signals

| Item                   | Explanation                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser DOM at line 1  | Builder or website interface behavior in the browser/Electron renderer. Evidence: const grid = document.querySelector("#project-grid");                       |
| Browser DOM at line 2  | Builder or website interface behavior in the browser/Electron renderer. Evidence: const searchInput = document.querySelector("#project-search");              |
| Browser DOM at line 3  | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectFilters = document.querySelector("#project-filters");          |
| Browser DOM at line 4  | Builder or website interface behavior in the browser/Electron renderer. Evidence: let filterButtons = [...document.querySelectorAll(".filter-button")];       |
| Browser DOM at line 5  | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectCount = document.querySelector("#project-count");              |
| Browser DOM at line 6  | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectTrackCount = document.querySelector("#project-track-count");   |
| Browser DOM at line 7  | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectTrackLabels = document.querySelector("#project-track-labels"); |
| Browser DOM at line 8  | Builder or website interface behavior in the browser/Electron renderer. Evidence: const year = document.querySelector("#year");                               |
| Browser DOM at line 9  | Builder or website interface behavior in the browser/Electron renderer. Evidence: const funFactsCallout = document.querySelector("#fun-facts-callout");       |
| Browser DOM at line 10 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const heroEyebrow = document.querySelector(".hero-content .eyebrow");       |
| Browser DOM at line 11 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const heroTitle = document.querySelector("#hero-title");                    |
| Browser DOM at line 12 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const heroCopy = document.querySelector(".hero-copy");                      |

| Item                              | Explanation                                                                                                                                                                                |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser DOM at line 13            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandName = document.querySelector(".brand strong");</code>                                  |
| Browser DOM at line 14            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandSubtitle = document.querySelector(".brand small");</code>                               |
| Browser DOM at line 15            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandIcon = document.querySelector(".brand-icon");</code>                                    |
| Browser DOM at line 16            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandText = document.querySelector(".omb-engraving");</code>                                 |
| Browser DOM at line 17            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const headerAvatar = document.querySelector(".header-avatar");</code>                              |
| Browser DOM at line 18            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const headerAvatarImage = document.querySelector(".header-avatar img");</code>                     |
| Browser DOM at line 19            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const heroImage = document.querySelector(".hero-image");</code>                                    |
| Browser DOM at line 20            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const resumeSection = document.querySelector("#resume");</code>                                    |
| Browser DOM at line 21            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactBand = document.querySelector("#contact");</code>                                     |
| Browser DOM at line 22            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profilePhoto = document.querySelector(".profile-photo");</code>                              |
| Browser DOM at line 23            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactTitle = document.querySelector("#contact-title");</code>                              |
| Browser DOM at line 24            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactIntro = document.querySelector(".contact-intro");</code>                              |
| Browser DOM at line 25            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactDetails = document.querySelector(".contact-details");</code>                          |
| Browser DOM at line 26            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactLinks = document.querySelector(".contact-links");</code>                              |
| Browser DOM at line 27            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const footerOwner = document.querySelector(".site-footer span");</code>                            |
| Browser DOM at line 29            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantForm = document.querySelector("#ai-assistant-form");</code>                       |
| Browser DOM at line 30            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantPanel = document.querySelector("#ask-ai");</code>                                 |
| Browser DOM at line 31            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantInput = document.querySelector("#ai-assistant-input");</code>                     |
| Browser DOM at line 32            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantLog = document.querySelector("#ai-assistant-log");</code>                         |
| Browser DOM at line 33            | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantStatus = document.querySelector("#ai-assistant-status");</code>                   |
| Cloudflare or AI at line 34       | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>const aiClearChatButton = document.querySelector("#ai-clear-chat");</code>                    |
| Cloudflare or AI at line 57       | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>let assistantChatHistory = [];</code>                                                         |
| Compiler or simulator at line 73  | Part of the Compile Code workspace or language/tool detection. Evidence: <code>{ id: "verilog", label: "Verilog", aliases: ["verilog", "v"] },</code>                                      |
| Compiler or simulator at line 74  | Part of the Compile Code workspace or language/tool detection. Evidence: <code>{ id: "systemverilog", label: "SystemVerilog", aliases: ["systemverilog", "system verilog", "sv"] },</code> |
| Compiler or simulator at line 76  | Part of the Compile Code workspace or language/tool detection. Evidence: <code>{ id: "java", label: "Java", aliases: ["java"] },</code>                                                    |
| Compiler or simulator at line 78  | Part of the Compile Code workspace or language/tool detection. Evidence: <code>{ id: "python", label: "Python", aliases: ["python", "py"] },</code>                                        |
| Compiler or simulator at line 104 | Part of the Compile Code workspace or language/tool detection. Evidence: <code>"embedded-wireless-node": "appearance-deep-navy-cyan-click",</code>                                         |

| Item                    | Explanation                                                                                                                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser DOM at line 175 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>dialog.classList.add("is-draggable-dialog");</code>                                                                      |
| Browser DOM at line 184 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>if (dialog.querySelector("[data-section-dialog-resize-handle"])) return;</code>                                          |
| Browser DOM at line 186 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const handle = document.createElement("div");</code>                                                                     |
| Browser DOM at line 188 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>handle.dataset.sectionDialogResizeHandle = direction;</code>                                                             |
| Browser DOM at line 197 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>dialog.classList.add("is-resized-dialog");</code>                                                                        |
| Browser DOM at line 255 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>dialog.classList.add("is-dragging-dialog");</code>                                                                       |
| Browser DOM at line 273 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>activeSectionDialogDrag.dialog.classList.remove("is-dragging-dialog");</code>                                            |
| Browser DOM at line 281 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>if (!handle    !dialog?.open    dialog.classList.contains("is-minimized-dialog")) return;</code>                         |
| Browser DOM at line 285 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>direction: handle.dataset.sectionDialogResizeHandle,</code>                                                              |
| Browser DOM at line 293 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>dialog.classList.add("is-resizing-dialog");</code>                                                                       |
| Browser DOM at line 309 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>activeSectionDialogResize.dialog.classList.remove("is-resizing-dialog");</code>                                          |
| Browser DOM at line 314 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const minimized = dialog.classList.contains("is-minimized-dialog");</code>                                               |
| Browser DOM at line 315 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const button = dialog.querySelector(".section-view-minimize");</code>                                                    |
| Browser DOM at line 324 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>dialog.classList.toggle("is-minimized-dialog");</code>                                                                   |
| Browser DOM at line 331 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.addEventListener("pointerdown", beginSectionDialogResize);</code>                                               |
| Browser DOM at line 332 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.addEventListener("pointerdown", beginSectionDialogDrag);</code>                                                 |
| Browser DOM at line 333 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.addEventListener("pointermove", moveSectionDialogResize);</code>                                                |
| Browser DOM at line 334 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.addEventListener("pointermove", moveSectionDialogDrag);</code>                                                  |
| Browser DOM at line 335 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.addEventListener("pointerup", endSectionDialogResize);</code>                                                   |
| Browser DOM at line 336 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.addEventListener("pointerup", endSectionDialogDrag);</code>                                                     |
| Browser DOM at line 337 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.addEventListener("pointercancel", endSectionDialogResize);</code>                                               |
| Browser DOM at line 338 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.addEventListener("pointercancel", endSectionDialogDrag);</code>                                                 |
| Browser DOM at line 339 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>window.addEventListener("resize", () =&gt; {</code>                                                                      |
| Browser DOM at line 340 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>document.querySelectorAll(".section-view-dialog.is-draggable-dialog[open]").forEach((dialog) =&gt; {</code>              |
| Rich text at line 370   | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>function normalizeCodePasteMode(value = "") {</code>                                                                  |
| Rich text at line 374   | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>function normalizeCodeText(value = "", pasteMode = "source") {</code>                                                 |
| Rich text at line 376   | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>if (normalizeCodePasteMode(pasteMode) === "source") return normalized.replace(/t/g, " ").replace(/s+\$/g, "");</code> |

| Item                              | Explanation                                                                                                                                                                                                                                    |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compiler or simulator at line 400 | Part of the Compile Code workspace or language/tool detection. Evidence: if ( <b>(\b(always_ff always_comb always_latch logic interface covergroup assert s+property typedef s+enum class s+\w+)\b/.test(code))</b> return "systemverilog";    |
| Compiler or simulator at line 401 | Part of the Compile Code workspace or language/tool detection. Evidence: if ( <b>(\b(module endmodule always assign reg wire initial posedge negedge)\b/.test(code))</b> return "verilog";                                                     |
| Cloudflare or AI at line 402      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: if ( <b>(/^s*\.(?!(tran ac dc op model subckt ends param)\b/im.test(code)    \bV\w+\s+\w+\s+\w+\s+(?:DC SIN PULSE)?/i.test(code))</b> return "Itspice"; |
| Compiler or simulator at line 403 | Part of the Compile Code workspace or language/tool detection. Evidence: if ( <b>(\b(def elif import s+\w+ from s+\w+\s+import self None True False)\b/.test(code))</b> return "python";                                                       |
| Compiler or simulator at line 404 | Part of the Compile Code workspace or language/tool detection. Evidence: if ( <b>(\b(public s+class private s+ protected s+ static s+void s+main System\.out)\b/.test(code))</b> return "java";                                                |
| Security or auth at line 412      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: function tokenizedCodeHtml(code = "", language = "javascript") {                                                                                        |
| Security or auth at line 414      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const tokens = [];                                                                                                                                      |
| Security or auth at line 417      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const token = `@@CODE_TOKEN_\${tokens.length}@`;                                                                                                        |
| Security or auth at line 418      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: tokens.push(` <span &gt;\${match}&lt;="" class="\${className}" span&gt;`);<="" td=""></span>                                                            |
| Security or auth at line 419      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: return token;                                                                                                                                           |
| Security or auth at line 424      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: protect(/&lt;!--[\s\S]*?-->/g, "code-token-comment");                                                                                                   |
| Security or auth at line 425      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: protect(/(&lt;\/?)([a-zA-Z0-9:-]+)([\s\S]*?)(\/?&gt;)/g, "code-token-tag");                                                                             |
| Compiler or simulator at line 427 | Part of the Compile Code workspace or language/tool detection. Evidence: const commentPattern = ["python", "Itspice"].includes(normalizeCodeLanguage(language))                                                                                |
| Security or auth at line 430      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: protect(commentPattern, "code-token-comment");                                                                                                          |

## Important constants and variables

| Item               | Explanation                                                                         |
|--------------------|-------------------------------------------------------------------------------------|
| grid               | Line 1: const grid = document.querySelector("#project-grid");                       |
| searchInput        | Line 2: const searchInput = document.querySelector("#project-search");              |
| projectFilters     | Line 3: const projectFilters = document.querySelector("#project-filters");          |
| filterButtons      | Line 4: let filterButtons = [...document.querySelectorAll(".filter-button")];       |
| projectCount       | Line 5: const projectCount = document.querySelector("#project-count");              |
| projectTrackCount  | Line 6: const projectTrackCount = document.querySelector("#project-track-count");   |
| projectTrackLabels | Line 7: const projectTrackLabels = document.querySelector("#project-track-labels"); |
| year               | Line 8: const year = document.querySelector("#year");                               |
| funFactsCallout    | Line 9: const funFactsCallout = document.querySelector("#fun-facts-callout");       |
| heroEyebrow        | Line 10: const heroEyebrow = document.querySelector(".hero-content .eyebrow");      |
| heroTitle          | Line 11: const heroTitle = document.querySelector("#hero-title");                   |
| heroCopy           | Line 12: const heroCopy = document.querySelector(".hero-copy");                     |
| brandName          | Line 13: const brandName = document.querySelector(".brand strong");                 |
| brandSubtitle      | Line 14: const brandSubtitle = document.querySelector(".brand small");              |
| brandIcon          | Line 15: const brandIcon = document.querySelector(".brand-icon");                   |
| brandText          | Line 16: const brandText = document.querySelector(".omb-engraving");                |
| headerAvatar       | Line 17: const headerAvatar = document.querySelector(".header-avatar");             |

| Item                      | Explanation                                                                        |
|---------------------------|------------------------------------------------------------------------------------|
| headerAvatarImage         | Line 18: const headerAvatarImage = document.querySelector(".header-avatar img");   |
| heroImage                 | Line 19: const heroImage = document.querySelector(".hero-image");                  |
| resumeSection             | Line 20: const resumeSection = document.querySelector("#resume");                  |
| contactBand               | Line 21: const contactBand = document.querySelector("#contact");                   |
| profilePhoto              | Line 22: const profilePhoto = document.querySelector(".profile-photo");            |
| contactTitle              | Line 23: const contactTitle = document.querySelector("#contact-title");            |
| contactIntro              | Line 24: const contactIntro = document.querySelector(".contact-intro");            |
| contactDetails            | Line 25: const contactDetails = document.querySelector(".contact-details");        |
| contactLinks              | Line 26: const contactLinks = document.querySelector(".contact-links");            |
| footerOwner               | Line 27: const footerOwner = document.querySelector(".site-footer span");          |
| searchWrap                | Line 28: const searchWrap = searchInput?.closest(".search-wrap");                  |
| aiAssistantForm           | Line 29: const aiAssistantForm = document.querySelector("#ai-assistant-form");     |
| aiAssistantPanel          | Line 30: const aiAssistantPanel = document.querySelector("#ask-ai");               |
| aiAssistantInput          | Line 31: const aiAssistantInput = document.querySelector("#ai-assistant-input");   |
| aiAssistantLog            | Line 32: const aiAssistantLog = document.querySelector("#ai-assistant-log");       |
| aiAssistantStatus         | Line 33: const aiAssistantStatus = document.querySelector("#ai-assistant-status"); |
| aiClearChatButton         | Line 34: const aiClearChatButton = document.querySelector("#ai-clear-chat");       |
| categories                | Line 36: let categories = [];                                                      |
| projects                  | Line 37: let projects = [];                                                        |
| siteSections              | Line 38: let siteSections = [];                                                    |
| funFacts                  | Line 39: let funFacts = [];                                                        |
| funFactsRich              | Line 40: let funFactsRich = null;                                                  |
| fieldStyles               | Line 41: let fieldStyles = {};                                                     |
| siteContent               | Line 42: let siteContent = null;                                                   |
| siteContentRich           | Line 43: let siteContentRich = {};                                                 |
| profile                   | Line 44: let profile = null;                                                       |
| profileRich               | Line 45: let profileRich = {};                                                     |
| activeFilter              | Line 46: let activeFilter = "all";                                                 |
| activeSectionDialogDrag   | Line 47: let activeSectionDialogDrag = null;                                       |
| activeSectionDialogResize | Line 48: let activeSectionDialogResize = null;                                     |
| sectionDialogDragEnabled  | Line 49: let sectionDialogDragEnabled = false;                                     |
| isApplyingSectionRoute    | Line 50: let isApplyingSectionRoute = false;                                       |
| searchPanel               | Line 51: let searchPanel = null;                                                   |
| searchStatus              | Line 52: let searchStatus = null;                                                  |
| searchLimitSelect         | Line 53: let searchLimitSelect = null;                                             |
| currentSearchResults      | Line 54: let currentSearchResults = [];                                            |
| assistantSourceCounter    | Line 55: let assistantSourceCounter = 0;                                           |
| assistantSourceMap        | Line 56: let assistantSourceMap = new Map();                                       |
| assistantChatHistory      | Line 57: let assistantChatHistory = [];                                            |
| searchableEntries         | Line 58: let searchableEntries = [];                                               |
| searchResultLimit         | Line 59: let searchResultLimit = 10;                                               |

| Item                   | Explanation                                                                                |
|------------------------|--------------------------------------------------------------------------------------------|
| searchHighlightTimer   | Line 60: let searchHighlightTimer = 0;                                                     |
| searchHighlightName    | Line 61: const searchHighlightName = "portfolio-search-highlight";                         |
| sectionRouteKey        | Line 63: const sectionRouteKey = "portfolio-section";                                      |
| sectionRouteParams     | Line 64: const sectionRouteParams = {                                                      |
| supportedCodeLanguages | Line 70: const supportedCodeLanguages = [                                                  |
| legacyTemplateSkins    | Line 82: const legacyTemplateSkins = {                                                     |
| defaultSiteContent     | Line 108: const defaultSiteContent = {                                                     |
| defaultProfile         | Line 114: const defaultProfile = {                                                         |
| items                  | Line 137: const items = Array.isArray(value) ? value : String(value    "").split(/\r?\n/); |
| facts                  | Line 146: const facts = normalizeFunFacts(funFacts);                                       |
| hasRichFacts           | Line 147: const hasRichFacts = Boolean(funFactsRich?.blocks?.length);                      |
| rect                   | Line 162: const rect = dialog.getBoundingClientRect();                                     |
| margin                 | Line 163: const margin = 12;                                                               |
| maxLeft                | Line 164: const maxLeft = Math.max(margin, window.innerWidth - rect.width - margin);       |
| maxTop                 | Line 165: const maxTop = Math.max(margin, window.innerHeight - rect.height - margin);      |
| rect                   | Line 173: const rect = dialog.getBoundingClientRect();                                     |
| position               | Line 174: const position = clampSectionDialogPosition(rect.left, rect.top, dialog);        |
| handle                 | Line 186: const handle = document.createElement("div");                                    |
| rect                   | Line 196: const rect = dialog.getBoundingClientRect();                                     |
| margin                 | Line 204: const margin = 12;                                                               |
| minWidth               | Line 205: const minWidth = Math.min(320, Math.max(180, window.innerWidth - margin * 2));   |
| minHeight              | Line 206: const minHeight = Math.min(170, Math.max(120, window.innerHeight - margin * 2)); |

## JSON/YAML-style keys detected

| Item                       | Explanation                                                               |
|----------------------------|---------------------------------------------------------------------------|
| skin-light-blue            | Line 83: "skin-light-blue": "appearance-light-blue-red-click",            |
| skin-clean-white           | Line 84: "skin-clean-white": "appearance-white-blue-click",               |
| skin-deep-navy             | Line 85: "skin-deep-navy": "appearance-deep-navy-cyan-click",             |
| skin-red-warm              | Line 86: "skin-red-warm": "appearance-warm-red-pale-click",               |
| skin-emerald-instrument    | Line 87: "skin-emerald-instrument": "appearance-emerald-mint-click",      |
| skin-amber-power           | Line 88: "skin-amber-power": "appearance-amber-cream-click",              |
| skin-violet-mixed          | Line 89: "skin-violet-mixed": "appearance-violet-lilac-click",            |
| skin-graphite-asic         | Line 90: "skin-graphite-asic": "appearance-graphite-white-click",         |
| analog-opamp-topology      | Line 91: "analog-opamp-topology": "appearance-light-blue-red-click",      |
| analog-power-charger       | Line 92: "analog-power-charger": "appearance-amber-cream-click",          |
| analog-filter-front-end    | Line 93: "analog-filter-front-end": "appearance-light-blue-red-click",    |
| analog-sensor-interface    | Line 94: "analog-sensor-interface": "appearance-emerald-mint-click",      |
| analog-mixed-signal-timing | Line 95: "analog-mixed-signal-timing": "appearance-violet-lilac-click",   |
| digital-fpga-pipeline      | Line 96: "digital-fpga-pipeline": "appearance-deep-navy-cyan-click",      |
| digital-asic-block         | Line 97: "digital-asic-block": "appearance-graphite-white-click",         |
| digital-verification-suite | Line 98: "digital-verification-suite": "appearance-graphite-white-click", |



| Item                          | Explanation                                                                 |
|-------------------------------|-----------------------------------------------------------------------------|
| digital-interface-controller  | Line 99: "digital-interface-controller": "appearance-deep-navy-cyan-click", |
| digital-signal-processing     | Line 100: "digital-signal-processing": "appearance-violet-lilac-click",     |
| embedded-stm32-sensor         | Line 101: "embedded-stm32-sensor": "appearance-emerald-mint-click",         |
| embedded-rtos-control         | Line 102: "embedded-rtos-control": "appearance-graphite-white-click",       |
| embedded-power-monitor        | Line 103: "embedded-power-monitor": "appearance-amber-cream-click",         |
| embedded-wireless-node        | Line 104: "embedded-wireless-node": "appearance-deep-navy-cyan-click",      |
| embedded-motor-control        | Line 105: "embedded-motor-control": "appearance-violet-lilac-click"         |
| Brief                         | Line 792: "Brief": "Overview",                                              |
| Project Brief                 | Line 793: "Project Brief": "Overview",                                      |
| Design Brief                  | Line 794: "Design Brief": "Design Overview",                                |
| Simulation Brief              | Line 795: "Simulation Brief": "Simulation Overview"                         |
| --responsive-section-card-min | Line 1029: "--responsive-section-card-min": profile.sectionCardMin,         |
| --responsive-content-max      | Line 1030: "--responsive-content-max": profile.contentMax,                  |
| --responsive-media-max        | Line 1031: "--responsive-media-max": profile.mediaMax,                      |
| --responsive-touch-target     | Line 1032: "--responsive-touch-target": profile.touchTarget                 |
| --accent                      | Line 1047: "--accent": accent,                                              |
| --template-bg                 | Line 1048: "--template-bg": visual.background,                              |
| --template-panel              | Line 1049: "--template-panel": visual.panel,                                |
| --template-accent             | Line 1050: "--template-accent": visual.accent,                              |
| --template-hover              | Line 1051: "--template-hover": visual.hover,                                |
| --template-click              | Line 1052: "--template-click": visual.click,                                |
| --template-click-text         | Line 1053: "--template-click-text": visual.clickText,                       |
| --template-line               | Line 1054: "--template-line": visual.line,                                  |
| --template-text               | Line 1055: "--template-text": visual.text,                                  |
| --accent                      | Line 1080: "--accent": accent,                                              |
| --template-bg                 | Line 1081: "--template-bg": visual.background,                              |
| --template-panel              | Line 1082: "--template-panel": visual.panel,                                |
| --template-accent             | Line 1083: "--template-accent": visual.accent,                              |
| --template-hover              | Line 1084: "--template-hover": visual.hover,                                |
| --template-click              | Line 1085: "--template-click": visual.click,                                |
| --template-click-text         | Line 1086: "--template-click-text": visual.clickText,                       |
| --template-line               | Line 1087: "--template-line": visual.line,                                  |
| --template-text               | Line 1088: "--template-text": visual.text,                                  |

## Event handlers and user interactions

| Item     | Explanation                                                         |
|----------|---------------------------------------------------------------------|
| Line 331 | document.addEventListener("pointerdown", beginSectionDialogResize); |
| Line 332 | document.addEventListener("pointerdown", beginSectionDialogDrag);   |
| Line 333 | document.addEventListener("pointermove", moveSectionDialogResize);  |
| Line 334 | document.addEventListener("pointermove", moveSectionDialogDrag);    |
| Line 335 | document.addEventListener("pointerup", endSectionDialogResize);     |

| Item      | Explanation                                                                                                              |
|-----------|--------------------------------------------------------------------------------------------------------------------------|
| Line 336  | <code>document.addEventListener("pointerup", endSectionDialogDrag);</code>                                               |
| Line 337  | <code>document.addEventListener("pointercancel", endSectionDialogResize);</code>                                         |
| Line 338  | <code>document.addEventListener("pointercancel", endSectionDialogDrag);</code>                                           |
| Line 339  | <code>window.addEventListener("resize", () =&gt; {</code>                                                                |
| Line 1214 | <code>button.addEventListener("click", () =&gt; {</code>                                                                 |
| Line 3272 | <code>searchLimitSelect.addEventListener("change", () =&gt; {</code>                                                     |
| Line 3276 | <code>searchPanel.addEventListener("mousedown", (event) =&gt; event.preventDefault());</code>                            |
| Line 3277 | <code>searchPanel.addEventListener("click", (event) =&gt; {</code>                                                       |
| Line 3434 | <code>dialog.querySelector(".section-view-close").addEventListener("click", () =&gt; closeSectionDialog(dialog));</code> |
| Line 3435 | <code>dialog.addEventListener("cancel", () =&gt; clearSectionRouteInHistory());</code>                                   |
| Line 3436 | <code>dialog.addEventListener("close", () =&gt; {</code>                                                                 |
| Line 3445 | <code>dialog.querySelector(".section-view-back").addEventListener("click", () =&gt; {</code>                             |
| Line 3457 | <code>dialog.querySelector(".section-view-forward").addEventListener("click", () =&gt; {</code>                          |
| Line 3464 | <code>dialog.querySelector("#section-view-content").addEventListener("click", (event) =&gt; {</code>                     |
| Line 3562 | <code>window.addEventListener("popstate", (event) =&gt; {</code>                                                         |
| Line 3567 | <code>searchInput.addEventListener("input", handleSearchInput);</code>                                                   |
| Line 3568 | <code>searchInput.addEventListener("focus", showSearchPanelIfNeeded);</code>                                             |
| Line 3569 | <code>searchInput.addEventListener("keydown", (event) =&gt; {</code>                                                     |
| Line 3580 | <code>document.addEventListener("click", (event) =&gt; {</code>                                                          |
| Line 3586 | <code>document.addEventListener("click", (event) =&gt; {</code>                                                          |
| Line 3599 | <code>aiAssistantForm?.addEventListener("submit", (event) =&gt; {</code>                                                 |
| Line 3606 | <code>aiClearChatButton?.addEventListener("click", clearAssistantChat);</code>                                           |
| Line 3607 | <code>window.addEventListener("resize", updateAssistantPanelGrowth);</code>                                              |
| Line 3609 | <code>aiAssistantLog?.addEventListener("click", (event) =&gt; {</code>                                                   |
| Line 3616 | <code>grid.addEventListener("click", (event) =&gt; {</code>                                                              |

## Function-by-function detail

### *normalize - lines 132-134*

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 132-134 (3 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | <code>function normalize(value) {</code>                                                                                                                                                              |
| Touches            | <code>filesystem</code>                                                                                                                                                                               |
| Calls detected     | <code>toLowerCase</code>                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
132: function normalize(value) {
```

```

133:   return String(value || "").toLowerCase();
134: }

```

## normalizeFunFacts - lines 136-142

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 136-142 (7 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeFunFacts(value) {                                                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isArray, split, map, replace, trim, filter, slice                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

136: function normalizeFunFacts(value) {
137:   const items = Array.isArray(value) ? value : String(value || "").split(/\r?\n/);
138:   return items
139:     .map((item) => String(item || "").replace(/\s+/g, " ").trim())
140:     .filter(Boolean)
141:     .slice(0, 10);
142: }

```

## renderFunFacts - lines 144-159

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 144-159 (16 source lines).                                                                                                                                                                      |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderFunFacts() {                                                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeFunFacts, renderRichContent, slice, join, map, renderInlineMath                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

144: function renderFunFacts() {
145:   if (!funFactsCallout) return;
146:   const facts = normalizeFunFacts(funFacts);
147:   const hasRichFacts = Boolean(funFactsRich?.blocks?.length);
148:   funFactsCallout.hidden = !facts.length && !hasRichFacts;
149:   funFactsCallout.innerHTML = facts.length || hasRichFacts ? `
150:     <div class="fun-facts-window">
151:       <span class="fun-facts-label">Fun fact:</span>
152:       <div class="fun-facts-lines">
153:         ${hasRichFacts
154:           ? renderRichContent({ blocks: funFactsRich.blocks.slice(0, 3) }, facts.join("\n"))
155:           : facts.map((fact) => `<p class="fun-fact-line">${renderInlineMath(fact)}</p>`).join("")}
156:       </div>
157:     </div>
158:   ` : "";
159: }

```

## clampSectionDialogPosition - lines 161-170

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 161-170 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function clampSectionDialogPosition(left, top, dialog) {                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getBoundingClientRect, max, min                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

161: function clampSectionDialogPosition(left, top, dialog) {
162:   const rect = dialog.getBoundingClientRect();
163:   const margin = 12;
164:   const maxLeft = Math.max(margin, window.innerWidth - rect.width - margin);
165:   const maxTop = Math.max(margin, window.innerHeight - rect.height - margin);
166:   return {
167:     left: Math.min(Math.max(left, margin), maxLeft),
168:     top: Math.min(Math.max(top, margin), maxTop)
169:   };
170: }

```

## ***anchorSectionDialog - lines 172-181***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 172-181 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function anchorSectionDialog(dialog) {                                                                                                                                                                |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | getBoundingClientRect, clampSectionDialogPosition, add                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

172: function anchorSectionDialog(dialog) {
173:   const rect = dialog.getBoundingClientRect();
174:   const position = clampSectionDialogPosition(rect.left, rect.top, dialog);
175:   dialog.classList.add("is-draggable-dialog");
176:   dialog.style.left = `${position.left}px`;
177:   dialog.style.top = `${position.top}px`;
178:   dialog.style.right = "auto";
179:   dialog.style.bottom = "auto";
180:   dialog.style.margin = "0";
181: }

```

## ***ensureSectionResizeHandles - lines 183-192***

| Item            | Explanation                                   |
|-----------------|-----------------------------------------------|
| Source location | Lines 183-192 (10 source lines).              |
| Purpose         | Part of the public website runtime.           |
| Declaration     | function ensureSectionResizeHandles(dialog) { |
| Touches         | browser/Electron DOM                          |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | querySelector, forEach, createElement, setAttribute, append                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

183: function ensureSectionResizeHandles(dialog) {
184:   if (dialog.querySelector("[data-section-dialog-resize-handle]")) return;
185:   ["n", "e", "s", "w", "ne", "nw", "se", "sw"].forEach((direction) => {
186:     const handle = document.createElement("div");
187:     handle.className = `dialog-resize-handle resize-${direction}`;
188:     handle.dataset.sectionDialogResizeHandle = direction;
189:     handle.setAttribute("aria-hidden", "true");
190:     dialog.append(handle);
191:   });
192: }

```

## ***anchorSectionDialogForResize - lines 194-201***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 194-201 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function anchorSectionDialogForResize(dialog) {                                                                                                                                                       |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | anchorSectionDialog, getBoundingClientRect, add                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

194: function anchorSectionDialogForResize(dialog) {
195:   anchorSectionDialog(dialog);
196:   const rect = dialog.getBoundingClientRect();
197:   dialog.classList.add("is-resized-dialog");
198:   dialog.style.width = `${rect.width}px`;
199:   dialog.style.height = `${rect.height}px`;
200:   dialog.style.maxHeight = "none";
201: }

```

## ***sectionDialogResizeBounds - lines 203-230***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 203-230 (28 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sectionDialogResizeBounds(state, event) {                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | min, max, includes                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
203: function sectionDialogResizeBounds(state, event) {
204:   const margin = 12;
205:   const minWidth = Math.min(320, Math.max(180, window.innerWidth - margin * 2));
206:   const minHeight = Math.min(170, Math.max(120, window.innerHeight - margin * 2));
207:   const direction = state.direction;
208:   let { left, top, width, height } = state.rect;
209:   const dx = event.clientX - state.startX;
210:   const dy = event.clientY - state.startY;
211:
212:   if (direction.includes("e")) {
213:     width = Math.min(Math.max(state.rect.width + dx, minWidth), window.innerWidth - left - margin);
214:   }
215:   if (direction.includes("s")) {
216:     height = Math.min(Math.max(state.rect.height + dy, minHeight), window.innerHeight - top - margin);
217:   }
218:   if (direction.includes("w")) {
219:     const right = state.rect.left + state.rect.width;
220:     left = Math.min(Math.max(state.rect.left + dx, margin), right - minWidth);
221:     width = right - left;
222:   }
223:   if (direction.includes("n")) {
224:     const bottom = state.rect.top + state.rect.height;
225:     top = Math.min(Math.max(state.rect.top + dy, margin), bottom - minHeight);
226:     height = bottom - top;
227:   }
228:
229:   return { height, left, top, width };
230: }
```

### ***sectionDialogFromDragEvent - lines 232-239***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 232-239 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sectionDialogFromDragEvent(event) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closest                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
232: function sectionDialogFromDragEvent(event) {
233:   if (event.target.closest("[data-section-dialog-resize-handle]")) return null;
234:   const handle = event.target.closest("[data-section-dialog-drag='true']");
235:   if (!handle) return null;
236:   if (event.target.closest("button, a, input, textarea, select, label, summary")) return null;
237:   const dialog = handle.closest("dialog");
238:   return dialog?.open ? dialog : null;
239: }
```

### ***beginSectionDialogDrag - lines 241-257***

| Item               | Explanation                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------|
| Source location    | Lines 241-257 (17 source lines).                                                            |
| Purpose            | Part of the public website runtime.                                                         |
| Declaration        | function beginSectionDialogDrag(event) {                                                    |
| Touches            | browser/Electron DOM                                                                        |
| Calls detected     | sectionDialogFromDragEvent, anchorSectionDialog, getBoundingClientRect, add, preventDefault |
| API endpoints      | None detected inside this function body.                                                    |
| Storage keys       | No local/session storage keys detected.                                                     |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                               |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

241: function beginSectionDialogDrag(event) {
242:   if (event.button !== 0) return;
243:   const dialog = sectionDialogFromDragEvent(event);
244:   if (!dialog) return;
245:   anchorSectionDialog(dialog);
246:   const rect = dialog.getBoundingClientRect();
247:   activeSectionDialogDrag = {
248:     dialog,
249:     offsetX: event.clientX - rect.left,
250:     offsetY: event.clientY - rect.top,
251:     pointerId: event.pointerId,
252:     pointerTarget: event.target
253:   };
254:   event.target.setPointerCapture?.(event.pointerId);
255:   dialog.classList.add("is-dragging-dialog");
256:   event.preventDefault();
257: }

```

## ***moveSectionDialogDrag - lines 259-268***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 259-268 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function moveSectionDialogDrag(event) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | clampSectionDialogPosition                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

259: function moveSectionDialogDrag(event) {
260:   if (!activeSectionDialogDrag) return;
261:   const next = clampSectionDialogPosition(
262:     event.clientX - activeSectionDialogDrag.offsetX,
263:     event.clientY - activeSectionDialogDrag.offsetY,
264:     activeSectionDialogDrag.dialog
265:   );
266:   activeSectionDialogDrag.dialog.style.left = `${next.left}px`;
267:   activeSectionDialogDrag.dialog.style.top = `${next.top}px`;
268: }

```

## ***endSectionDialogDrag - lines 270-275***

| Item               | Explanation                                   |
|--------------------|-----------------------------------------------|
| Source location    | Lines 270-275 (6 source lines).               |
| Purpose            | Part of the public website runtime.           |
| Declaration        | function endSectionDialogDrag() {             |
| Touches            | browser/Electron DOM                          |
| Calls detected     | remove                                        |
| API endpoints      | None detected inside this function body.      |
| Storage keys       | No local/session storage keys detected.       |
| Async/return clues | 0 await expression(s), 1 return statement(s). |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

270: function endSectionDialogDrag() {
271:   if (!activeSectionDialogDrag) return;
272:   activeSectionDialogDrag.pointerTarget?.releasePointerCapture?.(activeSectionDialogDrag.pointerId);
273:   activeSectionDialogDrag.dialog.classList.remove("is-dragging-dialog");
274:   activeSectionDialogDrag = null;
275: }

```

## ***beginSectionDialogResize - lines 277-295***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 277-295 (19 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function beginSectionDialogResize(event) {                                                                                                                                                            |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | closest, contains, anchorSectionDialogForResize, getBoundingClientRect, add, preventDefault                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

277: function beginSectionDialogResize(event) {
278:   if (event.button !== 0) return;
279:   const handle = event.target.closest("[data-section-dialog-resize-handle]");
280:   const dialog = handle?.closest("dialog");
281:   if (!handle || !dialog?.open || dialog.classList.contains("is-minimized-dialog")) return;
282:   anchorSectionDialogForResize(dialog);
283:   activeSectionDialogResize = {
284:     dialog,
285:     direction: handle.dataset.sectionDialogResizeHandle,
286:     pointerId: event.pointerId,
287:     pointerTarget: event.target,
288:     rect: dialog.getBoundingClientRect(),
289:     startX: event.clientX,
290:     startY: event.clientY
291:   };
292:   event.target.setPointerCapture?.(event.pointerId);
293:   dialog.classList.add("is-resizing-dialog");
294:   event.preventDefault();
295: }

```

## ***moveSectionDialogResize - lines 297-304***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 297-304 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function moveSectionDialogResize(event) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | sectionDialogResizeBounds                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |



Source excerpt:

```
297: function moveSectionDialogResize(event) {
298:   if (!activeSectionDialogResize) return;
299:   const next = sectionDialogResizeBounds(activeSectionDialogResize, event);
300:   activeSectionDialogResize.dialog.style.left = `${next.left}px`;
301:   activeSectionDialogResize.dialog.style.top = `${next.top}px`;
302:   activeSectionDialogResize.dialog.style.width = `${next.width}px`;
303:   activeSectionDialogResize.dialog.style.height = `${next.height}px`;
304: }
```

## ***endSectionDialogResize - lines 306-311***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 306-311 (6 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function endSectionDialogResize() {                                                                                                                                                                   |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | remove                                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
306: function endSectionDialogResize() {
307:   if (!activeSectionDialogResize) return;
308:   activeSectionDialogResize.pointerTarget?.releasePointerCapture?.(activeSectionDialogResize.pointerId);
309:   activeSectionDialogResize.dialog.classList.remove("is-resizing-dialog");
310:   activeSectionDialogResize = null;
311: }
```

## ***updateSectionDialogMinimize - lines 313-320***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 313-320 (8 source lines).                                                                                                                                                                       |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateSectionDialogMinimize(dialog) {                                                                                                                                                        |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | contains, querySelector, setAttribute                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
313: function updateSectionDialogMinimize(dialog) {
314:   const minimized = dialog.classList.contains("is-minimized-dialog");
315:   const button = dialog.querySelector(".section-view-minimize");
316:   if (!button) return;
317:   button.textContent = minimized ? "+" : "-";
318:   button.title = minimized ? "Restore window" : "Minimize window";
319:   button.setAttribute("aria-label", minimized ? "Restore window" : "Minimize window");
320: }
```

## ***toggleSectionDialogMinimized - lines 322-326***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 322-326 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function toggleSectionDialogMinimized(dialog) {                                                                                                                                                       |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | anchorSectionDialog, toggle, updateSectionDialogMinimize                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

322: function toggleSectionDialogMinimized(dialog) {
323:   anchorSectionDialog(dialog);
324:   dialog.classList.toggle("is-minimized-dialog");
325:   updateSectionDialogMinimize(dialog);
326: }

```

## ***enableSectionDialogDrag - lines 328-347***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 328-347 (20 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function enableSectionDialogDrag() {                                                                                                                                                                  |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | addEventListener, querySelectorAll, forEach, getBoundingClientRect, clampSectionDialogPosition                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

328: function enableSectionDialogDrag() {
329:   if (sectionDialogDragEnabled) return;
330:   sectionDialogDragEnabled = true;
331:   document.addEventListener("pointerdown", beginSectionDialogResize);
332:   document.addEventListener("pointerdown", beginSectionDialogDrag);
333:   document.addEventListener("pointermove", moveSectionDialogResize);
334:   document.addEventListener("pointermove", moveSectionDialogDrag);
335:   document.addEventListener("pointerup", endSectionDialogResize);
336:   document.addEventListener("pointerup", endSectionDialogDrag);
337:   document.addEventListener("pointercancel", endSectionDialogResize);
338:   document.addEventListener("pointercancel", endSectionDialogDrag);
339:   window.addEventListener("resize", () => {
340:     document.querySelectorAll(".section-view-dialog.is-draggable-dialog[open]").forEach((dialog) => {
341:       const rect = dialog.getBoundingClientRect();
342:       const next = clampSectionDialogPosition(rect.left, rect.top, dialog);
343:       dialog.style.left = `${next.left}px`;
344:       dialog.style.top = `${next.top}px`;
345:     });
346:   });
347: }

```

## ***escapeHtml - lines 349-356***

| Item            | Explanation                         |
|-----------------|-------------------------------------|
| Source location | Lines 349-356 (8 source lines).     |
| Purpose         | Part of the public website runtime. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function escapeHtml(value) {                                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replaceAll                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

349: function escapeHtml(value) {
350:   return String(value || "")
351:     .replaceAll("&", "&amp;")
352:     .replaceAll("<", "&lt;")
353:     .replaceAll(">", "&gt;")
354:     .replaceAll("'", "&quot;")
355:     .replaceAll('"', "&#039;");
356: }

```

## ***normalizeCodeLanguage - lines 358-364***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 358-364 (7 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCodeLanguage(value = "") {                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim, toLowerCase, replace, find, includes                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

358: function normalizeCodeLanguage(value = "") {
359:   const clean = String(value || "").trim().toLowerCase().replace(/[_-]+/g, " ");
360:   const match = supportedCodeLanguages.find((language) =>
361:     language.id === clean || language.aliases.includes(clean)
362:   );
363:   return match?.id || "javascript";
364: }

```

## ***codeLanguageLabel - lines 366-368***

| Item               | Explanation                                   |
|--------------------|-----------------------------------------------|
| Source location    | Lines 366-368 (3 source lines).               |
| Purpose            | Part of the public website runtime.           |
| Declaration        | function codeLanguageLabel(value = "") {      |
| Touches            | filesystem                                    |
| Calls detected     | find, normalizeCodeLanguage                   |
| API endpoints      | None detected inside this function body.      |
| Storage keys       | No local/session storage keys detected.       |
| Async/return clues | 0 await expression(s), 1 return statement(s). |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

366: function codeLanguageLabel(value = "") {
367:   return supportedCodeLanguages.find((language) => language.id === normalizeCodeLanguage(value))?.label
  || "Code";
368: }

```

## ***normalizeCodePasteMode - lines 370-372***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 370-372 (3 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCodePasteMode(value = "") {                                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

370: function normalizeCodePasteMode(value = "") {
371:   return value === "source" ? "source" : "basic";
372: }

```

## ***normalizeCodeText - lines 374-383***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 374-383 (10 source lines).                                                                                                                                                                      |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCodeText(value = "", pasteMode = "source") {                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | replace, normalizeCodePasteMode, split, map, trimEnd, join, trim                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

374: function normalizeCodeText(value = "", pasteMode = "source") {
375:   const normalized = String(value || "").replace(/\r\n?/g, "\n").replace(/\u00a0/g, " ");
376:   if (normalizeCodePasteMode(pasteMode) === "source") return normalized.replace(/\t/g, "
").replace(/\s+$/g, "");
377:   return normalized
378:     .split("\n")
379:     .map((line) => line.trimEnd())
380:     .join("\n")
381:     .replace(/\n{4,}/g, "\n\n\n")
382:     .trim();
383: }

```

## ***sourceLooksCpp - lines 385-389***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 385-389 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sourceLooksCpp(source = "") {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | b                                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

385: function sourceLooksCpp(source = "") {
386:   const text = String(source || "");
387:   return
#include\s*<(?:iostream|string|vector|array|map|unordered_map|memory|algorithm|optional|variant)>/.test(text)
||
388:   /\b(std::|cout|cin|cerr|namespace|template\s*<|class\s+\w+|new\s+\w|delete\s+|constexpr|nullptr|using
\s+namespace)\b/.test(text);
389: }

```

## sourceLooksC - lines 391-395

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 391-395 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sourceLooksC(source = "") {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | b                                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

391: function sourceLooksC(source = "") {
392:   const text = String(source || "");
393:   return #include\s*<(?:stdio|stdlib|string|stdint|stdbool|math)\.h>/.test(text) ||
394:   /\b(printf|scanf|malloc|calloc|realloc|free|struct\s+\w+|typedef\s+struct)\b/.test(text);
395: }

```

## detectCodeLanguage - lines 397-410

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 397-410 (14 source lines).             |
| Purpose         | Part of the public website runtime.          |
| Declaration     | function detectCodeLanguage(value = "") {    |
| Touches         | Mostly local calculations or local UI state. |
| Calls detected  | b, sourceLooksCpp, sourceLooksC              |
| API endpoints   | None detected inside this function body.     |
| Storage keys    | No local/session storage keys detected.      |



```

ar32_t|class|compl|concept|const|consteval|constexpr|constinit|const_cast|continue|co_await|co_return|co_yield|d
ecltype|default|delete|do|double|dynamic_cast|else|enum|explicit|export|extern|false|float|for|friend|if|inline|
int|long|mutable|namespace|new|noexcept|not|not_eq|nullptr|operator|or|or_eq|private|protected|public|register|r
einterpret_cast|requires|return|short|signed|sizeof|static|static_assert|static_cast|struct|switch|template|this
|thread_local|throw|true|try|typedef|typeid|typename|union|unsigned|using|virtual|void|volatile|wchar_t|while|xor
r|xor_eq",
442:     verilog: "always|and|assign|begin|buf|case|casex|casez|deassign|default|defparam|disable|edge|else|en
d|endcase|endfunction|endmodule|endprimitive|endspecify|endtable|endtask|event|for|force|forever|fork|function|g
enerate|genvar|if|initial|inout|input|integer|join|module|nand|negedge|nor|not|or|output|parameter|posedge|primi
tive|reg|release|repeat|signed|specify|supply0|supply1|table|task|tri|wand|while|wire|wor|xnor|xor",
443:     systemverilog: "always|always_comb|always_ff|always_latch|assign|automatic|begin|bit|case|class|clock
ing|covergroup|default|disable|do|else|end|endcase|endclass|endclocking|endfunction|endmodule|endpackage|endtask
|enum|for|forever|function|generate|genvar|if|initial|input|int|interface|logic|module|negedge|output|package|pa
rameter|posedge|reg|return|signed|task|typedef|wire",
444:     ltspice: "ac|dc|end|ends|four|func|global|ic|include|lib|meas|model|nodeset|op|options|param|plot|pro
be|save|step|subckt|temp|tran",
445:     java: "abstract|assert|boolean|break|byte|case|catch|char|class|const|continue|default|do|double|else
|enum|extends|final|finally|float|for|if|implements|import|instanceof|int|interface|long|native|new|null|package
|private|protected|public|return|short|static|strictfp|super|switch|synchronized|this|throw|throws|transient|try
|void|volatile|while",
446:     javascript: "async|await|break|case|catch|class|const|continue|debugger|default|delete|do|else|export
|extends|false|finally|for|from|function|if|import|in|instanceof|let|new|null|of|return|static|super|switch|this
|throw|true|try|typeof|undefined|var|void|while|yield",
447:     python: "and|as|assert|async|await|break|class|continue|def|del|elif|else|except|False|finally|for|fr
om|global|if|import|in|is|lambda|None|nonlocal|not|or|pass|raise|return|True|try|while|with|yield",
448:     html: "html|head|body|main|section|article|div|span|script|style|link|meta|title|header|footer|nav|bu
tton|input|form|label|img|a|p|pre|code"
449: };
450:     const keywords = keywordMap[normalizeCodeLanguage(language)] || keywordMap.javascript;
451:     html = html.replace(new RegExp(`\\b(${keywords})\\b`, "g"), '<span class="code-token-
keyword">$1</span>');
452:     html = html.replace(/@@CODE_TOKEN_(\\d+)@@/g, (_match, index) => tokens[Number(index)] || "");
453:     return html;
454: }

```

## protect - lines 415-421

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 415-421 (7 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | const protect = (pattern, className) => {                                                                                                                                                             |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | replace, push                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

415:     const protect = (pattern, className) => {
416:         html = html.replace(pattern, (match) => {
417:             const token = `@@CODE_TOKEN_${tokens.length}@@`;
418:             tokens.push(`<span class="${className}">${match}</span>`);
419:             return token;
420:         });
421:     };

```

## renderRichCodeBlock - lines 456-465

| Item            | Explanation                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------|
| Source location | Lines 456-465 (10 source lines).                                                                               |
| Purpose         | Renders UI, markup, or display output.                                                                         |
| Declaration     | function renderRichCodeBlock(block = {}) {                                                                     |
| Touches         | filesystem, authentication/security data                                                                       |
| Calls detected  | normalizeCodeLanguage, detectCodeLanguage, normalizeCodeText, escapeHtml, codeLanguageLabel, tokenizedCodeHtml |
| API endpoints   | None detected inside this function body.                                                                       |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

456: function renderRichCodeBlock(block = {}) {
457:   const language = normalizeCodeLanguage(block.language || detectCodeLanguage(block.code || ""));
458:   const code = normalizeCodeText(block.code || "", block.pasteMode || "source");
459:   return
460:     <figure class="rich-code-block rich-code-language-${language}">
461:       <figcaption><span>&lt;/span> ${escapeHtml(codeLanguageLabel(language))}</figcaption>
462:       <pre><code>${tokenizedCodeHtml(code, language)}</code></pre>
463:     </figure>
464:   ;
465: }

```

## ***renderPlainMultiline - lines 467-469***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 467-469 (3 source lines).                                                                                                                                                                       |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPlainMultiline(value = "") {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | escapeHtml, replace                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

467: function renderPlainMultiline(value = "") {
468:   return escapeHtml(value).replace(/\r\n?|\n/g, "<br>");
469: }

```

## ***normalizeSiteContent - lines 471-477***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 471-477 (7 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeSiteContent(content = {}) {                                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

471: function normalizeSiteContent(content = {}) {
472:   return {
473:     heroCopy: String(content.heroCopy || "").trim() || defaultSiteContent.heroCopy,
474:     heroEyebrow: String(content.heroEyebrow || "").trim() || defaultSiteContent.heroEyebrow,
475:     heroTitle: String(content.heroTitle || "").trim() || defaultSiteContent.heroTitle
476:   };

```



477: }

## ***normalizeProfile - lines 479-495***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 479-495 (17 source lines).                                                                                                                                                                      |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeProfile(profileValue = {}) {                                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
479: function normalizeProfile(profileValue = {}) {
480:   return {
481:     brandImage: String(profileValue.brandImage || "").trim(),
482:     brandText: String(profileValue.brandText || "").trim() || String(profileValue.displayName ||
483:     "").trim() || defaultProfile.brandText,
484:     contactIntro: String(profileValue.contactIntro || "").trim(),
485:     displayName: String(profileValue.displayName || "").trim(),
486:     email: String(profileValue.email || "").trim(),
487:     githubUrl: String(profileValue.githubUrl || "").trim(),
488:     heroImage: String(profileValue.heroImage || "").trim(),
489:     linkedinUrl: String(profileValue.linkedinUrl || "").trim(),
490:     phone: String(profileValue.phone || "").trim(),
491:     portfolioLabel: String(profileValue.portfolioLabel || "").trim() || defaultProfile.portfolioLabel,
492:     profileImage: String(profileValue.profileImage || "").trim(),
493:     resumeUrl: String(profileValue.resumeUrl || "").trim(),
494:     websiteUrl: String(profileValue.websiteUrl || "").trim()
495:   };
```

## ***mailComposeLink - lines 497-500***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 497-500 (4 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function mailComposeLink(email = "") {                                                                                                                                                                |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
497: function mailComposeLink(email = "") {
498:   const clean = String(email || "").trim();
499:   return clean ? `https://mail.google.com/mail/?view=cm&fs=1&to=${encodeURIComponent(clean)}` : "";
500: }
```

## ***phoneLink - lines 502-505***

| Item            | Explanation                     |
|-----------------|---------------------------------|
| Source location | Lines 502-505 (4 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function phoneLink(phone = "") {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, replace                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

502: function phoneLink(phone = "") {
503:   const clean = String(phone || "").trim();
504:   return clean ? `tel:${clean.replace(/^[^d+]/g, "")}` : "";
505: }

```

## ***renderSiteContent - lines 507-515***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 507-515 (9 source lines).                                                                                                                                                                       |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderSiteContent() {                                                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeSiteContent, renderRichFieldContent, applyPlainFieldStyle                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

507: function renderSiteContent() {
508:   const content = normalizeSiteContent(siteContent || {});
509:   if (heroEyebrow) heroEyebrow.innerHTML = renderRichFieldContent(siteContentRich.heroEyebrow,
content.heroEyebrow);
510:   if (heroTitle) heroTitle.innerHTML = renderRichFieldContent(siteContentRich.heroTitle,
content.heroTitle);
511:   if (heroCopy) heroCopy.innerHTML = renderRichFieldContent(siteContentRich.heroCopy, content.heroCopy);
512:   applyPlainFieldStyle(heroEyebrow, "site-hero-eyebrow");
513:   applyPlainFieldStyle(heroTitle, "site-hero-title");
514:   applyPlainFieldStyle(heroCopy, "site-hero-copy");
515: }

```

## ***renderProfile - lines 517-627***

| Item            | Explanation                                                                                                                                                                          |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 517-627 (111 source lines).                                                                                                                                                    |
| Purpose         | Renders UI, markup, or display output.                                                                                                                                               |
| Declaration     | function renderProfile() {                                                                                                                                                           |
| Touches         | browser/Electron DOM, filesystem                                                                                                                                                     |
| Calls detected  | normalizeProfile, mailComposeLink, phoneLink, normalizeLinkTarget, filter, renderRichFieldContent, applyPlainFieldStyle, slice, toUpperCase, setAttribute, querySelectorAll, forEach |
| API endpoints   | None detected inside this function body.                                                                                                                                             |
| Storage keys    | No local/session storage keys detected.                                                                                                                                              |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

517: function renderProfile() {
518:   const current = normalizeProfile(profile || {});
519:   const displayName = current.displayName || "Portfolio";
520:   const label = current.portfolioLabel || "Portfolio";
521:   const emailLink = mailComposeLink(current.email);
522:   const callLink = phoneLink(current.phone);
523:   const links = [
524:     current.email ? { label: "Email", url: emailLink, external: true } : null,
525:     current.phone ? { label: "Call", url: callLink } : null,
526:     current.githubUrl ? { label: "GitHub", url: normalizeLinkTarget(current.githubUrl, { assumeWeb: true
  })), external: true } : null,
527:     current.linkedinUrl ? { label: "LinkedIn", url: normalizeLinkTarget(current.linkedinUrl, { assumeWeb:
true })), external: true } : null,
528:     current.websiteUrl ? { label: "Website", url: normalizeLinkTarget(current.websiteUrl, { assumeWeb:
true })), external: true } : null,
529:     current.resumeUrl ? { label: "Resume", url: current.resumeUrl, external: false } : null
530:   ].filter(Boolean);
531:
532:   document.title = `${displayName} | ${label}`;
533:   if (brandName) brandName.innerHTML = renderRichFieldContent(profileRich.displayName, displayName);
534:   if (brandSubtitle) brandSubtitle.innerHTML = renderRichFieldContent(profileRich.portfolioLabel, label);
535:   applyPlainFieldStyle(brandName, "profile-display-name");
536:   applyPlainFieldStyle(brandSubtitle, "profile-portfolio-label");
537:   if (brandText) brandText.textContent = current.brandText || displayName.slice(0, 3).toUpperCase() ||
"PORT";
538:   if (brandIcon) {
539:     if (current.brandImage) {
540:       brandIcon.src = current.brandImage;
541:       brandIcon.hidden = false;
542:     } else {
543:       brandIcon.hidden = true;
544:     }
545:   }
546:
547:   if (headerAvatar && headerAvatarImage) {
548:     headerAvatar.hidden = !current.profileImage;
549:     if (current.profileImage) {
550:       headerAvatarImage.src = current.profileImage;
551:       headerAvatarImage.alt = displayName;
552:       headerAvatar.setAttribute("aria-label", `Go to ${displayName} contact information`);
553:     }
554:   }
555:
556:   if (heroImage) {
557:     if (current.heroImage) {
558:       heroImage.hidden = false;
559:       heroImage.src = current.heroImage;
560:       heroImage.alt = `${displayName} portfolio background`;
561:     } else {
562:       heroImage.hidden = true;
563:     }
564:   }
565:
566:   document.querySelectorAll('a[href="#resume"]').forEach((link) => {
567:     link.hidden = !current.resumeUrl;
568:   });
569:   const heroGithubLink = [...document.querySelectorAll(".hero-actions a")]
570:     .find((link) => /github/i.test(link.textContent || ""));
571:   if (heroGithubLink) {
572:     heroGithubLink.hidden = !current.githubUrl;
573:     if (current.githubUrl) heroGithubLink.href = normalizeLinkTarget(current.githubUrl, { assumeWeb: true
  });
574:   }
575:
576:   if (resumeSection) {
577:     resumeSection.hidden = !current.resumeUrl;
578:     resumeSection.querySelectorAll("a").forEach((link) => {
579:       link.href = current.resumeUrl || "#";
580:     });
581:     const resumeObject = resumeSection.querySelector("object");
582:     if (resumeObject) resumeObject.data = current.resumeUrl || "";
583:     const fallbackLink = resumeSection.querySelector("object a");
584:     if (fallbackLink) fallbackLink.href = current.resumeUrl || "#";
585:   }
586:
587:   if (profilePhoto) {
588:     profilePhoto.hidden = !current.profileImage;
589:     if (current.profileImage) {
590:       profilePhoto.src = current.profileImage;
591:       profilePhoto.alt = `Portrait of ${displayName}`;
592:     }

```

```

593:     }
594:     if (contactTitle) contactTitle.innerHTML = renderRichFieldContent(profileRich.displayName,
displayName);
595:     applyPlainFieldStyle(contactTitle, "profile-display-name");
596:     if (contactIntro) {
597:         contactIntro.innerHTML = renderRichFieldContent(
598:             profileRich.contactIntro,
599:             current.contactIntro || (current.displayName ? `${displayName}'s contact information and public
links.` : "Add contact details in the builder.")
600:         );
601:     }
602:     applyPlainFieldStyle(contactIntro, "profile-contact-intro");
603:     if (contactDetails) {
604:         contactDetails.innerHTML = [
605:             current.email ? `` : "",
606:             current.phone ? `` : ""
607:         \\].filter\\(Boolean\\).join\\(""\\);
608:     }
609:     if \\(contactLinks\\) {
610:         contactLinks.innerHTML = links.map\\(\\(link\\) => {
611:             const styleId = link.label === "GitHub"
612:                 ? "profile-github"
613:                 : link.label === "LinkedIn"
614:                 ? "profile-linkedin"
615:                 : link.label === "Website"
616:                 ? "profile-website"
617:                 : link.label === "Email"
618:                 ? "profile-email"
619:                 : link.label === "Call"
620:                 ? "profile-phone"
621:                 : "";
622:             return ``;
623:         }\\\).join\\\(""\\\);
624:     }
625:     if \\\(footerOwner\\\) footerOwner.innerHTML = `&copy; <span id="year">\\\${new Date\\\(\\\).getFullYear\\\(\\\)}</span>
\\\${displayName}`;
626:     if \\\(contactBand\\\) contactBand.hidden = !links.length && !current.profileImage && !current.contactIntro
&& !current.displayName;
627: }

```

## textBlocksFromPlainText - lines 629-637

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 629-637 (9 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function textBlocksFromPlainText(text) {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

629: function textBlocksFromPlainText(text) {
630:     return [{
631:         align: "left",
632:         fontFamily: "Arial",
633:         fontSize: "normal",
634:         text: String(text || ""),
635:         type: "paragraph"
636:     }];
637: }

```

## unwrapFormula - lines 639-652

| Item            | Explanation                      |
|-----------------|----------------------------------|
| Source location | Lines 639-652 (14 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function unwrapFormula(value) {                                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim, match                                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

639: function unwrapFormula(value) {
640:   const text = String(value || "").trim();
641:   const wrappers = [
642:     [/^\$\$([\s\S]+\)\$\$\$/, 1],
643:     [/^\$\$([\s\S]+\)\$\$/, 1],
644:     [/^\$\$([\s\S]+\)\$/, 1],
645:     [/^\$([\s\S]+\)\$/, 1]
646:   ];
647:   for (const [pattern, group] of wrappers) {
648:     const match = text.match(pattern);
649:     if (match) return match[group].trim();
650:   }
651:   return text;
652: }

```

## ***renderInlineMath - lines 654-663***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 654-663 (10 source lines).                                                                                                                                                                      |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderInlineMath(text) {                                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | escapeHtml, replace, b, match, slice, normalizeLinkTarget                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

654: function renderInlineMath(text) {
655:   const escaped = escapeHtml(text);
656:   const withMath = escaped.replace(/\$([\^$]+\)\$/g, '<span class="rich-inline-formula">$1</span>');
657:   return withMath.replace(/(https?:\/\/[\^$<]+|www\.[\^$<]+|(?<github|linkedin|gitlab|bitbucket|drive|d
ocs)\.com\/[\^$<]+)/gi, (match) => {
658:     const trailing = match.match(/[(),.;:!?]+\$/)?.[0] || "";
659:     const clean = trailing ? match.slice(0, -trailing.length) : match;
660:     const href = normalizeLinkTarget(clean, { assumeWeb: true });
661:     return `<a href="${href}" target="_blank" rel="noreferrer">${clean}</a>${trailing}`;
662:   });
663: }

```

## ***renderMultilineInlineText - lines 665-667***

| Item            | Explanation                                     |
|-----------------|-------------------------------------------------|
| Source location | Lines 665-667 (3 source lines).                 |
| Purpose         | Renders UI, markup, or display output.          |
| Declaration     | function renderMultilineInlineText(text = "") { |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | renderInlineMath, replace                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
665: function renderMultilineInlineText(text = "") {
666:   return renderInlineMath(text).replace(/\r\n?|\n/g, "<br>");
667: }
```

## ***looksLikeBareWebAddress - lines 669-676***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 669-676 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function looksLikeBareWebAddress(value = "") {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, split, toLowerCase                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
669: function looksLikeBareWebAddress(value = "") {
670:   const clean = String(value || "").trim();
671:   if (!clean || /\s/.test(clean) || /\^(\.?\.\?\/|#|mailto:|tel:)/i.test(clean)) return false;
672:   if (/^[a-z][a-z0-9+.-]*:/i.test(clean)) return false;
673:   const host = clean.split(/[/?#]/)[0].toLowerCase();
674:   if (!/^[a-z0-9-]+\.(.+[a-z0-9-]+\.$)/.test(host)) return false;
675:   return !/\.(pdf|docx?|xlsx?|pptx?|zip|7z|rar|png|jpe?g|gif|svg|webp|txt|md|csv|json|xml|log|c|h|cpp|hpp
|py|js|mjs|ts|v|sv|vhdl?|spice|cir|net|asc|sch|kicad_sch|kicad_pcb)$/i.test(host);
676: }
```

## ***looksLikeWebOrContactText - lines 678-686***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 678-686 (9 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function looksLikeWebOrContactText(value = "") {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, b, looksLikeBareWebAddress, split, some, replace                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
678: function looksLikeWebOrContactText(value = "") {
```

```

679:   const clean = String(value || "").trim();
680:   if (!clean) return false;
681:   if (/^b(?:https?:\/\/|www\.)[^s<>'"]+/i.test(clean)) return true;
682:   if (/^b[a-z0-9._%+~]@[a-z0-9.-]+\.[a-z]{2,}\b/i.test(clean)) return true;
683:   if (/^b(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/[^s<>'"]+/i.test(clean)) return true;
684:   if (looksLikeBareWebAddress(clean)) return true;
685:   return clean.split(/\s+/).some((word) => looksLikeBareWebAddress(word.replace(/^[('"]+|['"]+$/g,
  "")));
686: }

```

## normalizeLinkTarget - lines 688-695

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 688-695 (8 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeLinkTarget(target = "", options = {}) {                                                                                                                                             |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | trim, looksLikeBareWebAddress                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

688: function normalizeLinkTarget(target = "", options = {}) {
689:   const clean = String(target || "").trim();
690:   if (!clean) return "";
691:   if (/^\/\//.test(clean)) return `https:${clean}`;
692:   if (/^www\./i.test(clean)) return `https://${clean}`;
693:   if (options.assumeWeb && looksLikeBareWebAddress(clean)) return `https://${clean}`;
694:   return clean;
695: }

```

## isWebsiteLinkItem - lines 697-705

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 697-705 (9 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function isWebsiteLinkItem(item = {}, target = "") {                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, filter, join, toLowerCase, b                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

697: function isWebsiteLinkItem(item = {}, target = "") {
698:   const clean = String(target || "").trim();
699:   if (/^(https?:)?\/\//i.test(clean) || /^www\./i.test(clean)) return true;
700:   const typeText = [item.kind, item.type, item.status, item.meta, item.label, item.title]
701:     .filter(Boolean)
702:     .join(" ")
703:     .toLowerCase();
704:   return /^b(link|website|web link|url|external|github|linkedin|drive|social|profile)\b/.test(typeText);
705: }

```

## linkifyRichTextNodes - lines 707-742

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 707-742 (36 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function linkifyRichTextNodes(root) {                                                                                                                                                                 |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | createTreeWalker, acceptNode, closest, nextNode, push, forEach, createDocumentFragment, replace, b, append, createTextNode, slice                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

#### Source excerpt:

```

707: function linkifyRichTextNodes(root) {
708:   const walker = document.createTreeWalker(root, NodeFilter.SHOW_TEXT, {
709:     acceptNode(node) {
710:       if (!node.textContent ||
!(https?:\//|www\.(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\)/i.test(node.textContent)) return
NodeFilter.FILTER_REJECT;
711:       return node.parentElement?.closest("a") ? NodeFilter.FILTER_REJECT : NodeFilter.FILTER_ACCEPT;
712:     }
713:   });
714:   const nodes = [];
715:   let node = walker.nextNode();
716:   while (node) {
717:     nodes.push(node);
718:     node = walker.nextNode();
719:   }
720:
721:   nodes.forEach((textNode) => {
722:     const text = textNode.textContent || "";
723:     const fragment = document.createDocumentFragment();
724:     let cursor = 0;
725:     text.replace(/\b(https?:\//|www\.[^\s<]+|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com
\//|www\.[^\s<]+)/gi, (match, _url, offset) => {
726:       if (offset > cursor) fragment.append(document.createTextNode(text.slice(cursor, offset)));
727:       const trailing = match.match(/[,.;!?:]+$/)?.[0] || "";
728:       const clean = trailing ? match.slice(0, -trailing.length) : match;
729:       const link = document.createElement("a");
730:       link.href = normalizeLinkTarget(clean, { assumeWeb: true });
731:       link.target = "_blank";
732:       link.rel = "noreferrer";
733:       link.textContent = clean;
734:       fragment.append(link);
735:       if (trailing) fragment.append(document.createTextNode(trailing));
736:       cursor = offset + match.length;
737:       return match;
738:     });
739:     if (cursor < text.length) fragment.append(document.createTextNode(text.slice(cursor)));
740:     textNode.replaceWith(fragment);
741:   });
742: }

```

### ***sanitizeRichInlineHtml - lines 744-787***

| Item               | Explanation                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 744-787 (44 source lines).                                                                                                                               |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                  |
| Declaration        | function sanitizeRichInlineHtml(value = "") {                                                                                                                  |
| Touches            | browser/Electron DOM, filesystem                                                                                                                               |
| Calls detected     | createElement, Set, querySelectorAll, forEach, has, replaceWith, getAttribute, trim, normalizeLinkTarget, cleanFontFamily, normalizeFontPx, normalizeTextColor |
| API endpoints      | None detected inside this function body.                                                                                                                       |
| Storage keys       | No local/session storage keys detected.                                                                                                                        |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                  |



| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

744: function sanitizeRichInlineHtml(value = "") {
745:   const template = document.createElement("template");
746:   template.innerHTML = String(value || "");
747:   const allowedTags = new Set(["A", "B", "BR", "EM", "I", "SPAN", "STRONG", "U"]);
748:
749:   [...template.content.querySelectorAll("*")].forEach((node) => {
750:     if (!allowedTags.has(node.tagName)) {
751:       node.replaceWith(...node.childNodes);
752:       return;
753:     }
754:
755:     const href = node.tagName === "A" ? String(node.getAttribute("href") || "").trim() : "";
756:     const normalizedHref = normalizeLinkTarget(href, { assumeWeb: true });
757:     const safeHref = /^((https?:|mailto:|tel:|#|\/)/i.test(normalizedHref) ? normalizedHref : "");
758:     const fontFamily = cleanFontFamily(node.style.fontFamily || "");
759:     const fontPx = normalizeFontPx(parseFloat(node.style.fontSize || ""));
760:     const color = normalizeTextColor(node.style.color || "");
761:     const rawFontWeight = node.style.fontWeight || "";
762:     const fontWeight = /^(bold|[6-9]00)$/i.test(rawFontWeight)
763:       ? "700"
764:       : /^(normal|[1-5]00)$/i.test(rawFontWeight)
765:         ? "400"
766:         : "";
767:     const fontStyle = node.style.fontStyle === "italic" ? "italic" : "";
768:     const textDecoration = node.style.textDecoration.includes("underline") ? "underline" : "";
769:
770:     [...node.attributes].forEach((attribute) => node.removeAttribute(attribute.name));
771:
772:     if (node.tagName === "A" && safeHref) {
773:       node.setAttribute("href", safeHref);
774:       node.setAttribute("target", "_blank");
775:       node.setAttribute("rel", "noreferrer");
776:     }
777:     if (fontFamily) node.style.fontFamily = fontFamily;
778:     if (fontPx) node.style.fontSize = `${fontPx}px`;
779:     if (color) node.style.color = color;
780:     if (fontWeight) node.style.fontWeight = fontWeight;
781:     if (fontStyle) node.style.fontStyle = fontStyle;
782:     if (textDecoration) node.style.textDecoration = textDecoration;
783:   });
784:
785:   linkifyRichTextNodes(template.content);
786:   return template.innerHTML;
787: }

```

## ***displayTitle - lines 789-798***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 789-798 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function displayTitle(value, fallback = "Untitled item") {                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

789: function displayTitle(value, fallback = "Untitled item") {
790:   const title = String(value || fallback).trim();
791:   const replacements = {
792:     "Brief": "Overview",
793:     "Project Brief": "Overview",
794:     "Design Brief": "Design Overview",
795:     "Simulation Brief": "Simulation Overview"
796:   };
797:   return replacements[title] || title;

```

798: }

## ***cleanFontFamily - lines 800-807***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 800-807 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function cleanFontFamily(value = "") {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, map, replace, trim, filter, slice, join                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
800: function cleanFontFamily(value = "") {
801:   return String(value)
802:     .split(",")
803:     .map((item) => item.replace(/^[^\\w\\s-]/g, "").trim())
804:     .filter(Boolean)
805:     .slice(0, 3)
806:     .join(", ");
807: }
```

## ***normalizeFontPx - lines 809-812***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 809-812 (4 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeFontPx(value) {                                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isFinite                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
809: function normalizeFontPx(value) {
810:   const size = Number(value);
811:   return Number.isFinite(size) && size >= 8 && size <= 24 ? String(size) : "";
812: }
```

## ***normalizeTextColor - lines 813-820***

| Item            | Explanation                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Source location | Lines 813-820 (8 source lines).                                               |
| Purpose         | Validates or normalizes input so later code receives safe predictable values. |
| Declaration     | function normalizeTextColor(value = "") {                                     |
| Touches         | filesystem                                                                    |
| Calls detected  | trim                                                                          |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

813: function normalizeTextColor(value = "") {
814:     const color = String(value || "").trim();
815:     return /^(#[0-9a-f]{3}|[0-9a-f]{6})$/i.test(color) ||
816:         /^rgba?\(/i.test(color) ||
817:         /^hsla?\(/i.test(color)
818:         ? color
819:         : "";
820: }

```

## ***rgbColorToHex - lines 822-830***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 822-830 (9 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function rgbColorToHex(value = "") {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, toLowerCase, match, map, max, min, toString, padStart, join                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

822: function rgbColorToHex(value = "") {
823:     const color = String(value || "").trim();
824:     if (/^[0-9a-f]{6}$/i.test(color)) return color.toLowerCase();
825:     const shortHex = color.match(/^[0-9a-f]([0-9a-f])([0-9a-f])$/i);
826:     if (shortHex) return
`#${shortHex[1]}${shortHex[1]}${shortHex[2]}${shortHex[2]}${shortHex[3]}${shortHex[3]}`.toLowerCase();
827:     const rgb = color.match(/^rgba?\((\d+),\s*(\d+),\s*(\d+)/i);
828:     if (!rgb) return color;
829:     return `#${[rgb[1], rgb[2], rgb[3]].map((part) => Math.max(0, Math.min(255,
Number(part))).toString(16).padStart(2, "0")).join("")}`;
830: }

```

## ***normalizePlainFieldStyle - lines 832-844***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 832-844 (13 source lines).                                                                                                                                                                      |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizePlainFieldStyle(style = {}) {                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | cleanFontFamily, normalizeFontPx, normalizeTextColor, rgbColorToHex                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
832: function normalizePlainFieldStyle(style = {}) {
833:   const normalized = {};
834:   const fontFamily = cleanFontFamily(style.fontFamily || "");
835:   const fontPx = normalizeFontPx(style.fontPx || style.fontSize || "");
836:   const color = normalizeTextColor(style.color || "");
837:   if (fontFamily) normalized.fontFamily = fontFamily;
838:   if (fontPx) normalized.fontPx = fontPx;
839:   if (color) normalized.color = rgbColorToHex(color);
840:   if (style.bold === true || style.bold === "true") normalized.bold = true;
841:   if (style.italic === true || style.italic === "true") normalized.italic = true;
842:   if (style.underline === true || style.underline === "true") normalized.underline = true;
843:   return normalized;
844: }
```

***normalizeFieldStyles - lines 846-852***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 846-852 (7 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeFieldStyles(styles = {}) {                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | fromEntries, entries, map, normalizePlainFieldStyle, filter, keys                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
846: function normalizeFieldStyles(styles = {}) {
847:   return Object.fromEntries(
848:     Object.entries(styles || {})
849:       .map(([fieldId, style]) => [fieldId, normalizePlainFieldStyle(style)])
850:       .filter(([, style]) => Object.keys(style).length)
851:   );
852: }
```

***plainFieldStyleToCss - lines 854-864***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 854-864 (11 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function plainFieldStyleToCss(style = {}) {                                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizePlainFieldStyle, push, join                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
854: function plainFieldStyleToCss(style = {}) {
855:   const normalized = normalizePlainFieldStyle(style);
856:   const rules = [];
857:   if (normalized.fontFamily) rules.push(`font-family: ${normalized.fontFamily}`);
858:   if (normalized.fontPx) rules.push(`font-size: ${normalized.fontPx}px`);
859:   if (normalized.color) rules.push(`color: ${normalized.color}`);
860:   if (normalized.bold) rules.push("font-weight: 700");
861:   if (normalized.italic) rules.push("font-style: italic");
862:   if (normalized.underline) rules.push("text-decoration: underline");
863:   return rules.join(" ");
}
```

864: }

## ***applyPlainFieldStyle - lines 866-875***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 866-875 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function applyPlainFieldStyle(element, fieldId = "") {                                                                                                                                                |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizePlainFieldStyle                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
866: function applyPlainFieldStyle(element, fieldId = "") {
867:   if (!element) return;
868:   const style = normalizePlainFieldStyle(fieldStyles[fieldId] || {});
869:   element.style.fontFamily = style.fontFamily || "";
870:   element.style.fontSize = style.fontPx ? `${style.fontPx}px` : "";
871:   element.style.color = style.color || "";
872:   element.style.fontWeight = style.bold ? "700" : "";
873:   element.style.fontStyle = style.italic ? "italic" : "";
874:   element.style.textDecoration = style.underline ? "underline" : "";
875: }
```

## ***plainFieldStyleAttribute - lines 877-880***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 877-880 (4 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function plainFieldStyleAttribute(fieldId = "") {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | plainFieldStyleToCss, escapeHtml                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
877: function plainFieldStyleAttribute(fieldId = "") {
878:   const css = plainFieldStyleToCss(fieldStyles[fieldId] || {});
879:   return css ? `style="${escapeHtml(css)}"` : "";
880: }
```

## ***richTextStyle - lines 882-896***

| Item            | Explanation                                                                  |
|-----------------|------------------------------------------------------------------------------|
| Source location | Lines 882-896 (15 source lines).                                             |
| Purpose         | Part of the public website runtime.                                          |
| Declaration     | function richTextStyle(block = {}) {                                         |
| Touches         | filesystem                                                                   |
| Calls detected  | cleanFontFamily, normalizeFontPx, normalizeTextColor, push, escapeHtml, join |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

882: function richTextStyle(block = {}) {
883:     const styles = [];
884:     const fontFamily = cleanFontFamily(block.fontFamily || "Arial") || "Arial";
885:     const fontPx = normalizeFontPx(block.fontPx);
886:     const color = normalizeTextColor(block.color || "");
887:
888:     if (fontFamily) styles.push(`font-family: ${fontFamily}`);
889:     if (fontPx) styles.push(`font-size: ${fontPx}px`);
890:     if (color) styles.push(`color: ${color}`);
891:     if (block.bold) styles.push("font-weight: 700");
892:     if (block.italic) styles.push("font-style: italic");
893:     if (block.underline) styles.push("text-decoration: underline");
894:
895:     return styles.length ? ` style="${escapeHtml(styles.join("; "))}"` : "";
896: }

```

### ***normalizeCropAspect - lines 898-900***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 898-900 (3 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCropAspect(value = "") {                                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | includes                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

898: function normalizeCropAspect(value = "") {
899:     return ["1 / 1", "4 / 3", "16 / 9", "3 / 4"].includes(value) ? value : "original";
900: }

```

### ***normalizeCropZoom - lines 902-905***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 902-905 (4 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCropZoom(value = 1) {                                                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isFinite, min, max                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

902: function normalizeCropZoom(value = 1) {
903:     const zoom = Number(value);
904:     return Number.isFinite(zoom) ? Math.min(3, Math.max(1, zoom)) : 1;
905: }

```

## ***normalizeCropPosition - lines 907-910***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 907-910 (4 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCropPosition(value = 50) {                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isFinite, min, max                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

907: function normalizeCropPosition(value = 50) {
908:     const position = Number(value);
909:     return Number.isFinite(position) ? Math.min(100, Math.max(0, position)) : 50;
910: }

```

## ***richImageCropStyle - lines 912-921***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 912-921 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function richImageCropStyle(block = {}) {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeCropAspect, normalizeCropZoom, normalizeCropPosition, push, escapeHtml, join                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

912: function richImageCropStyle(block = {}) {
913:     const aspect = normalizeCropAspect(block.cropAspect);
914:     const styles = [
915:         `--crop-zoom: ${normalizeCropZoom(block.cropZoom)}`,
916:         `--crop-x: ${normalizeCropPosition(block.cropX)}%`,
917:         `--crop-y: ${normalizeCropPosition(block.cropY)}%`
918:     ];
919:     if (aspect !== "original") styles.push(`--crop-aspect: ${aspect}`);
920:     return ` style="${escapeHtml(styles.join("; "))}"`;
921: }

```

## ***richImageDownloadLink - lines 923-940***

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 923-940 (18 source lines).             |
| Purpose         | Part of the public website runtime.          |
| Declaration     | function richImageDownloadLink(block = {}) { |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | escapeHtml, cleanRichImageTitle, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

923: function richImageDownloadLink(block = {}) {
924:   const label = escapeHtml(cleanRichImageTitle(block) || block.caption || "Image file");
925:   const url = block.url || "#";
926:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(block, url) });
927:
928:   return `
929:   <p class="rich-download-only">
930:     <a
931:       class="resource-link"
932:       href="${escapeHtml(target)}"
933:       ${linkAttributes(target, block)}
934:       ${downloadAttribute(target, block)}
935:     >
936:       ${label}
937:     </a>
938:   </p>
939:   `;
940: }

```

## ***cleanRichImageTitle - lines 942-945***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 942-945 (4 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function cleanRichImageTitle(block = {}) {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

942: function cleanRichImageTitle(block = {}) {
943:   const title = String(block.title || "").trim();
944:   return /^pasted image\b/i.test(title) ? "" : title;
945: }

```

## ***renderRichContent - lines 947-982***

| Item            | Explanation                                                                                                                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 947-982 (36 source lines).                                                                                                                                                                                         |
| Purpose         | Renders UI, markup, or display output.                                                                                                                                                                                   |
| Declaration     | function renderRichContent(rich, fallbackText = "") {                                                                                                                                                                    |
| Touches         | filesystem                                                                                                                                                                                                               |
| Calls detected  | textBlocksFromPlainText, map, includes, richImageDownloadLink, cleanRichImageTitle, normalizeLinkTarget, normalizeCropAspect, richImageCropStyle, escapeHtml, unwrapFormula, looksLikeWebOrContactText, renderInlineMath |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 7 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

947: function renderRichContent(rich, fallbackText = "") {
948:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
949:   return `
950:     <div class="rich-content">
951:       ${blocks.map((block) => {
952:         const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
953:         if (block.type === "image") {
954:           if (block.display === "download") return richImageDownloadLink(block);
955:           const title = cleanRichImageTitle(block);
956:           const imageSrc = normalizeLinkTarget(block.url, { assumeWeb: true });
957:           return `
958:             <figure class="rich-image justify-${align}">
959:               <span class="rich-image-viewport crop-${normalizeCropAspect(block.cropAspect)} ===
"original" ? "original" : "active"${richImageCropStyle(block)}>
960:                 
961:               </span>
962:               ${title || block.caption} ? `<figcaption>${title} ?
`<strong>${escapeHtml(title)}</strong>` : ""}${block.caption ? `<span>${escapeHtml(block.caption)}</span>` :
""}</figcaption>` : ""}
963:             </figure>
964:           `;
965:         }
966:         if (block.type === "formula") {
967:           const formula = unwrapFormula(block.formula);
968:           if (looksLikeWebOrContactText(formula)) {
969:             return `<p class="rich-paragraph rich-text-normal
text-${align}">${renderInlineMath(formula)}</p>`;
970:           }
971:           return `<div class="rich-formula justify-${align}">${escapeHtml(formula)}</div>`;
972:         }
973:         if (block.type === "code") return renderRichCodeBlock(block);
974:         const size = ["small", "normal", "large"].includes(block.fontSize) ? block.fontSize : "normal";
975:         const content = block.html
976:           ? sanitizeRichInlineHtml(block.html)
977:           : renderInlineMath(block.text || "");
978:         return `<p class="rich-paragraph rich-text-${size}
text-${align}"${richTextStyle(block)}>${content}</p>`;
979:       }).join("")}
980:     </div>
981:   `;
982: }

```

## ***renderRichFieldContent - lines 984-994***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 984-994 (11 source lines).                                                                                                                                                                      |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderRichFieldContent(rich, fallbackText = "") {                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | textBlocksFromPlainText, map, includes, sanitizeRichInlineHtml, renderInlineMath, richTextStyle, filter, join                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

984: function renderRichFieldContent(rich, fallbackText = "") {
985:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
986:   return blocks.map((block) => {
987:     if (block.type !== "paragraph") return "";
988:     const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
989:     const content = block.html

```

```

990:         ? sanitizeRichInlineHtml(block.html)
991:         : renderInlineMath(block.text || "");
992:     return `<span class="rich-field-line text-${align}"${richTextStyle(block)}>${content}</span>`;
993:   }).filter(Boolean).join("");
994: }

```

## ***slugLabel - lines 996-999***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 996-999 (4 source lines).                                                                                                                                                                       |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function slugLabel(value) {                                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

996: function slugLabel(value) {
997:   const category = categories.find((item) => item.id === value);
998:   return category ? category.label : value;
999: }

```

## ***canonicalTemplateId - lines 1001-1003***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1001-1003 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function canonicalTemplateId(id) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1001: function canonicalTemplateId(id) {
1002:   return legacyTemplateSkins[id] || id || "";
1003: }

```

## ***projectTemplateId - lines 1005-1007***

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 1005-1007 (3 source lines).            |
| Purpose         | Part of the public website runtime.          |
| Declaration     | function projectTemplateId(project) {        |
| Touches         | Mostly local calculations or local UI state. |
| Calls detected  | canonicalTemplateId                          |
| API endpoints   | None detected inside this function body.     |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1005: function projectTemplateId(project) {
1006:   return canonicalTemplateId(project?.portfolioView?.template?.id || project?.templateId || "");
1007: }
```

### ***projectTemplateClass - lines 1009-1012***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1009-1012 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function projectTemplateClass(project) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectTemplateId                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1009: function projectTemplateClass(project) {
1010:   const templateId = projectTemplateId(project);
1011:   return templateId ? `project-template project-template-${templateId}` : "project-template-white";
1012: }
```

### ***responsiveClassName - lines 1014-1019***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1014-1019 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function responsiveClassName(value, fallback) {                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | toLowerCase, replace                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1014: function responsiveClassName(value, fallback) {
1015:   return String(value || fallback || "")
1016:     .toLowerCase()
1017:     .replace(/[^\a-z0-9]+/g, "-")
1018:     .replace(/^-|-$/g, "") || fallback;
1019: }
```

### ***projectResponsiveClass - lines 1021-1024***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1021-1024 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function projectResponsiveClass(project) {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectResponsiveProfile, responsiveClassName                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1021: function projectResponsiveClass(project) {
1022:   const profile = projectResponsiveProfile(project);
1023:   return `responsive-project responsive-density-${responsiveClassName(profile.density, "balanced")}
responsive-card-${responsiveClassName(profile.cardLayout, "single")}`;
1024: }

```

### ***responsiveStyleValues - lines 1026-1034***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1026-1034 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function responsiveStyleValues(project) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectResponsiveProfile                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1026: function responsiveStyleValues(project) {
1027:   const profile = projectResponsiveProfile(project);
1028:   return {
1029:     "--responsive-section-card-min": profile.sectionCardMin,
1030:     "--responsive-content-max": profile.contentMax,
1031:     "--responsive-media-max": profile.mediaMax,
1032:     "--responsive-touch-target": profile.touchTarget
1033:   };
1034: }

```

### ***hasPublicTemplate - lines 1036-1038***

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 1036-1038 (3 source lines).            |
| Purpose         | Part of the public website runtime.          |
| Declaration     | function hasPublicTemplate(project) {        |
| Touches         | Mostly local calculations or local UI state. |
| Calls detected  | projectTemplateId                            |
| API endpoints   | None detected inside this function body.     |
| Storage keys    | No local/session storage keys detected.      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1036: function hasPublicTemplate(project) {
1037:   return Boolean(projectTemplateId(project));
1038: }

```

## ***projectTemplateVisual - lines 1040-1042***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1040-1042 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function projectTemplateVisual(project) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1040: function projectTemplateVisual(project) {
1041:   return project?.portfolioView?.template?.visual || project?.templateVisual || null;
1042: }

```

## ***projectTemplateStyle - lines 1044-1062***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1044-1062 (19 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function projectTemplateStyle(project, accent) {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectTemplateVisual, responsiveStyleValues, entries, filter, map, join                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1044: function projectTemplateStyle(project, accent) {
1045:   const visual = projectTemplateVisual(project) || {};
1046:   const values = {
1047:     "--accent": accent,
1048:     "--template-bg": visual.background,
1049:     "--template-panel": visual.panel,
1050:     "--template-accent": visual.accent,
1051:     "--template-hover": visual.hover,
1052:     "--template-click": visual.click,
1053:     "--template-click-text": visual.clickText,
1054:     "--template-line": visual.line,
1055:     "--template-text": visual.text,
1056:     ...responsiveStyleValues(project)
1057:   };
1058:   return Object.entries(values)

```

```

1059:     .filter(([, value]) => value)
1060:     .map(([key, value]) => `${key}: ${value}`)
1061:     .join("; ");
1062: }

```

## ***applyProjectTemplateToElement - lines 1064-1098***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1064-1098 (35 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function applyProjectTemplateToElement(element, project, accent) {                                                                                                                                    |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | filter, startsWith, forEach, remove, projectTemplateClass, split, add, projectResponsiveClass, projectTemplateVisual, responsiveStyleValues, entries, setProperty                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1064: function applyProjectTemplateToElement(element, project, accent) {
1065:   if (!element) return;
1066:   [...element.classList]
1067:   .filter((className) =>
1068:     className === "project-template" ||
1069:     className === "project-template-white" ||
1070:     className === "responsive-project" ||
1071:     className.startsWith("project-template-") ||
1072:     className.startsWith("responsive-density-") ||
1073:     className.startsWith("responsive-card-")
1074:   )
1075:   .forEach((className) => element.classList.remove(className));
1076:   projectTemplateClass(project).split(" ").forEach((className) => element.classList.add(className));
1077:   projectResponsiveClass(project).split(" ").forEach((className) => element.classList.add(className));
1078:   const visual = projectTemplateVisual(project) || {};
1079:   const values = {
1080:     "--accent": accent,
1081:     "--template-bg": visual.background,
1082:     "--template-panel": visual.panel,
1083:     "--template-accent": visual.accent,
1084:     "--template-hover": visual.hover,
1085:     "--template-click": visual.click,
1086:     "--template-click-text": visual.clickText,
1087:     "--template-line": visual.line,
1088:     "--template-text": visual.text,
1089:     ...responsiveStyleValues(project)
1090:   };
1091:   Object.entries(values).forEach(([key, value]) => {
1092:     if (value) {
1093:       element.style.setProperty(key, value);
1094:     } else {
1095:       element.style.removeProperty(key);
1096:     }
1097:   });
1098: }

```

## ***parsedItemTerms - lines 1100-1110***

| Item            | Explanation                              |
|-----------------|------------------------------------------|
| Source location | Lines 1100-1110 (11 source lines).       |
| Purpose         | Part of the public website runtime.      |
| Declaration     | function parsedItemTerms(item) {         |
| Touches         | filesystem                               |
| Calls detected  | flatMap                                  |
| API endpoints   | None detected inside this function body. |
| Storage keys    | No local/session storage keys detected.  |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1100: function parsedItemTerms(item) {
1101:   return [
1102:     item?.title,
1103:     item?.description,
1104:     item?.meta,
1105:     item?.url,
1106:     item?.kind,
1107:     ...(item?.rich?.blocks || []).flatMap((block) => [block.text, block.formula, block.title,
block.caption]),
1108:     ...(item?.children || []).flatMap(parsedItemTerms)
1109:   ];
1110: }

```

## ***flattenProject - lines 1112-1165***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1112-1165 (54 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function flattenProject(project) {                                                                                                                                                                    |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | slugLabel, electronicsSearchKeywords, flatMap, map, filter, join                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1112: function flattenProject(project) {
1113:   const categoryLabel = slugLabel(project.category);
1114:   const electronicsKeywords = window.electronicsSearchKeywords
1115:     ? window.electronicsSearchKeywords(project, categoryLabel)
1116:     : [];
1117:   const richSummaryTerms = (project.summaryRich?.blocks || []).flatMap((block) => [
1118:     block.text,
1119:     block.formula,
1120:     block.title,
1121:     block.caption
1122:   ]);
1123:   const parsedChildTerms = (project.portfolioView?.sections || []).flatMap((section) =>
1124:     (section.items || []).flatMap(parsedItemTerms)
1125:   );
1126:   const design = project.design || {};
1127:   const designTerms = [
1128:     design.brief?.summary,
1129:     ...(design.brief?.files || []).flatMap((item) => [item.title, item.description, item.url]),
1130:     design.documentation?.summary,
1131:     ...(design.documentation?.files || []).flatMap((item) => [item.title, item.description, item.url]),
1132:     ...(design.documentation?.references || []).flatMap((item) => [item.title, item.description,
item.url]),
1133:     ...(design.documentation?.mathAnalysis || []).flatMap((item) => [item.title, item.description]),
1134:     design.simulation?.summary,
1135:     ...(design.simulation?.files || []).flatMap((item) => [item.title, item.description, item.url]),
1136:     ...(design.simulation?.results || []).flatMap((item) => [item.title, item.description, item.url])
1137:   ];
1138:   return [
1139:     project.id,
1140:     project.title,
1141:     categoryLabel,
1142:     project.status,
1143:     project.summary,
1144:     ...richSummaryTerms,
1145:     ...parsedChildTerms,
1146:     ...designTerms,
1147:     ...(project.focus || []),
1148:     ...(project.highlights || []),
1149:     ...(project.tools || []).map((item) => typeof item === "string" ? item : [item.name, item.title,

```

```

item.label, item.description].filter(Boolean).join(" "),
1150:   ...(project.languages || []),
1151:   ...(project.documents || []).flatMap((item) => [item.title, item.type, item.status, item.url]),
1152:   ...(project.tests || []).flatMap((item) => [item.name, item.method, item.status, item.result,
item.artifact]),
1153:   ...(project.pcbss || []).flatMap((item) => [item.name, item.revision, item.status, item.artifact]),
1154:   ...(project.media || []).flatMap((item) => [item.title, item.caption, item.url]),
1155:   ...(project.sections || []).flatMap((section) => [
1156:     section.title,
1157:     section.description,
1158:     ...(section.items || []).flatMap((item) => [item.title, item.description, item.type, item.status,
item.url])
1159:   ]),
1160:   ...(project.links || []).flatMap((item) => [item.label, item.url]),
1161:   ...electronicsKeywords
1162: ]
1163:   .map(normalize)
1164:   .join(" ");
1165: }

```

## ***richSummaryTerms - lines 1117-1126***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1117-1126 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | const richSummaryTerms = (project.summaryRich?.blocks    []).flatMap((block) => [                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | flatMap                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1117:   const richSummaryTerms = (project.summaryRich?.blocks || []).flatMap((block) => [
1118:     block.text,
1119:     block.formula,
1120:     block.title,
1121:     block.caption
1122:   ]);
1123:   const parsedChildTerms = (project.portfolioView?.sections || []).flatMap((section) =>
1124:     (section.items || []).flatMap(parsedItemTerms)
1125:   );
1126:   const design = project.design || {};

```

## ***parsedChildTerms - lines 1123-1126***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1123-1126 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | const parsedChildTerms = (project.portfolioView?.sections    []).flatMap((section) =>                                                                                                                 |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | flatMap                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1123:   const parsedChildTerms = (project.portfolioView?.sections || []).flatMap((section) =>
1124:     (section.items || []).flatMap(parsedItemTerms)
1125:   );

```



```
1126:   const design = project.design || {};
```

## ***projectMatches - lines 1167-1169***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1167-1169 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function projectMatches(project, query) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | flattenProject, includes                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1167: function projectMatches(project, query) {  
1168:   return !query || flattenProject(project).includes(query);  
1169: }
```

## ***projectVisible - lines 1171-1174***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1171-1174 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function projectVisible(project, query) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectMatches                                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1171: function projectVisible(project, query) {  
1172:   const categoryMatch = activeFilter === "all" || project.category === activeFilter;  
1173:   return categoryMatch && projectMatches(project, query);  
1174: }
```

## ***normalizeCategory - lines 1176-1184***

| Item               | Explanation                                                                   |
|--------------------|-------------------------------------------------------------------------------|
| Source location    | Lines 1176-1184 (9 source lines).                                             |
| Purpose            | Validates or normalizes input so later code receives safe predictable values. |
| Declaration        | function normalizeCategory(category = {}) {                                   |
| Touches            | filesystem                                                                    |
| Calls detected     | trim, normalizeTextColor, toLowerCase, replace                                |
| API endpoints      | None detected inside this function body.                                      |
| Storage keys       | No local/session storage keys detected.                                       |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                 |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1176: function normalizeCategory(category = {}) {
1177:   const label = String(category.label || category.title || category.id || "Project category").trim() ||
"Project category";
1178:   return {
1179:     accent: normalizeTextColor(category.accent || "" || "#1677a8",
1180:     description: String(category.description || "").trim(),
1181:     id: String(category.id || label.toLowerCase().replace(/^[a-z0-9]+/g, "-").replace(/^-|-$/g, "")) ||
"category"),
1182:     label
1183:   };
1184: }

```

## ***hydrateProjectCategories - lines 1186-1201***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1186-1201 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function hydrateProjectCategories(rawCategories = [], rawProjects = []) {                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | map, Set, forEach, trim, has, split, filter, charAt, toUpperCase, slice, join, push                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1186: function hydrateProjectCategories(rawCategories = [], rawProjects = []) {
1187:   const normalized = rawCategories.map(normalizeCategory);
1188:   const knownIds = new Set(normalized.map((category) => category.id));
1189:   rawProjects.forEach((project) => {
1190:     const id = String(project?.category || "").trim();
1191:     if (!id || knownIds.has(id)) return;
1192:     const label = id
1193:       .split(/[-_\s]+/)
1194:       .filter(Boolean)
1195:       .map((part) => part.charAt(0).toUpperCase() + part.slice(1))
1196:       .join(" ") || "Project category";
1197:     normalized.push(normalizeCategory({ id, label, description: "Project category" }));
1198:     knownIds.add(id);
1199:   });
1200:   return normalized;
1201: }

```

## ***setActiveFilter - lines 1203-1209***

| Item               | Explanation                                   |
|--------------------|-----------------------------------------------|
| Source location    | Lines 1203-1209 (7 source lines).             |
| Purpose            | Part of the public website runtime.           |
| Declaration        | function setActiveFilter(filter = "all") {    |
| Touches            | browser/Electron DOM                          |
| Calls detected     | some, forEach, toggle                         |
| API endpoints      | None detected inside this function body.      |
| Storage keys       | No local/session storage keys detected.       |
| Async/return clues | 0 await expression(s), 0 return statement(s). |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1203: function setActiveFilter(filter = "all") {
1204:   activeFilter = filter;
1205:   if (activeFilter !== "all" && !categories.some((category) => category.id === activeFilter)) {
1206:     activeFilter = "all";
1207:   }
1208:   filterButtons.forEach((item) => item.classList.toggle("active", item.dataset.filter === activeFilter));
1209: }

```

## ***bindFilterButtons - lines 1211-1220***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1211-1220 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function bindFilterButtons() {                                                                                                                                                                        |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelectorAll, forEach, addEventListener, setActiveFilter, renderProjects, updateSearchDropdown                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1211: function bindFilterButtons() {
1212:   filterButtons = [...document.querySelectorAll(".filter-button")];
1213:   filterButtons.forEach((button) => {
1214:     button.addEventListener("click", () => {
1215:       setActiveFilter(button.dataset.filter || "all");
1216:       renderProjects();
1217:       updateSearchDropdown();
1218:     });
1219:   });
1220: }

```

## ***renderCategoryFilters - lines 1222-1241***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1222-1241 (20 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderCategoryFilters() {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, join, escapeHtml, bindFilterButtons, setActiveFilter                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1222: function renderCategoryFilters() {
1223:   if (projectTrackCount) {
1224:     projectTrackCount.textContent = `${categories.length} Track${categories.length === 1 ? "" : "s"}`;
1225:   }
1226:   if (projectTrackLabels) {
1227:     projectTrackLabels.textContent = categories.length

```

```

1228:         ? categories.map((category) => category.label).join(", ")
1229:         : "project categories";
1230:     }
1231:     if (projectFilters) {
1232:         projectFilters.innerHTML = `
1233:             <button class="filter-button active" type="button" data-filter="all">All</button>
1234:             ${categories.map((category) => `
1235:                 <button class="filter-button" type="button" data-
filter="${escapeHtml(category.id)}">${escapeHtml(category.label)}</button>
1236:             `).join("")}
1237:         `;
1238:     }
1239:     bindFilterButtons();
1240:     setActiveFilter(activeFilter);
1241: }

```

## ***flattenSearchText - lines 1243-1251***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1243-1251 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function flattenSearchText(values = []) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | flat, filter, map, stringify, join, replace, trim                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1243: function flattenSearchText(values = []) {
1244:     return values
1245:         .flat(Infinity)
1246:         .filter((value) => value !== null && value !== undefined)
1247:         .map((value) => typeof value === "object" ? JSON.stringify(value) : String(value))
1248:         .join(" ")
1249:         .replace(/\s+/g, " ")
1250:         .trim();
1251: }

```

## ***richTextTerms - lines 1253-1264***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1253-1264 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function richTextTerms(rich) {                                                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | flatMap                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1253: function richTextTerms(rich) {
1254:     return (rich?.blocks || []).flatMap((block) => [
1255:         block.text,
1256:         block.html,
1257:         block.formula,
1258:         block.code,
1259:         block.language,
1260:         block.title,

```

```

1261:     block.caption,
1262:     block.url
1263:   });
1264: }

```

## ***addSearchEntry - lines 1266-1276***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1266-1276 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function addSearchEntry(entries, entry) {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | flattenSearchText, normalize, push                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1266: function addSearchEntry(entries, entry) {
1267:   const text = flattenSearchText([entry.title, entry.type, entry.context, entry.text, entry.url]);
1268:   if (!entry.title && !text) return null;
1269:   const storedEntry = {
1270:     ...entry,
1271:     normalizedText: normalize(text),
1272:     text
1273:   };
1274:   entries.push(storedEntry);
1275:   return storedEntry;
1276: }

```

## ***searchSnippet - lines 1278-1286***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1278-1286 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function searchSnippet(text = "", query = "") {                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | replace, trim, normalize, indexOf, slice, max, min                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1278: function searchSnippet(text = "", query = "") {
1279:   const clean = String(text || "").replace(/\s+/g, " ").trim();
1280:   if (!clean) return "";
1281:   const index = normalize(clean).indexOf(normalize(query));
1282:   if (index < 0) return clean.slice(0, 150);
1283:   const start = Math.max(0, index - 54);
1284:   const end = Math.min(clean.length, index + query.length + 80);
1285:   return `${start ? "... " : ""}${clean.slice(start, end)}${end < clean.length ? "... " : ""}`;
1286: }

```

## ***entryScore - lines 1288-1314***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1288-1314 (27 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function entryScore(entry, query) {                                                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, trim, split, filter, startsWith, includes, forEach                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1288: function entryScore(entry, query) {
1289:   const normalizedQuery = normalize(query).trim();
1290:   if (!normalizedQuery) return 0;
1291:   const title = normalize(entry.title);
1292:   const type = normalize(entry.type);
1293:   const context = normalize(entry.context);
1294:   const text = entry.normalizedText || "";
1295:   const words = normalizedQuery.split(/\s+/).filter(Boolean);
1296:   let score = 0;
1297:
1298:   if (title === normalizedQuery) score += 160;
1299:   if (title.startsWith(normalizedQuery)) score += 120;
1300:   if (title.includes(normalizedQuery)) score += 90;
1301:   if (context.includes(normalizedQuery)) score += 55;
1302:   if (type.includes(normalizedQuery)) score += 35;
1303:   if (text.includes(normalizedQuery)) score += 45;
1304:   words.forEach((word) => {
1305:     if (title.includes(word)) score += 16;
1306:     if (context.includes(word)) score += 10;
1307:     if (text.includes(word)) score += 6;
1308:   });
1309:
1310:   if (score > 0 && entry.kind === "file") score += 18;
1311:   if (score > 0 && entry.kind === "section") score += 14;
1312:   if (score > 0 && entry.kind === "project") score += 12;
1313:   return score;
1314: }

```

## ***entryMatches - lines 1316-1318***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1316-1318 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function entryMatches(entry, query) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | entryScore                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1316: function entryMatches(entry, query) {
1317:   return entryScore(entry, query) > 0;
1318: }

```

## ***fileIsBrowserSearchable - lines 1320-1323***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1320-1323 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function fileIsBrowserSearchable(url = "") {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, toLowerCase                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1320: function fileIsBrowserSearchable(url = "") {
1321:   const clean = String(url || "").split(/[?#]/)[0].toLowerCase();
1322:   return /\.(txt|md|markdown|csv|json|xml|log|c|h|cpp|hpp|py|js|mjs|ts|v|sv|vhd|?|spice|cir|net|asc|sch|k
icad_sch|kicad_pcb)$/i.test(clean);
1323: }

```

## ***indexSearchableFileText - lines 1325-1338***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1325-1338 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | async function indexSearchableFileText(entry) {                                                                                                                                                       |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | fileIsBrowserSearchable, startsWith, fetch, text, flattenSearchText, slice, normalize, updateSearchDropdown                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1325: async function indexSearchableFileText(entry) {
1326:   if (!entry.url || !fileIsBrowserSearchable(entry.url)) return;
1327:   if (/^(https?:)?\/\//i.test(entry.url) && !entry.url.startsWith(window.location.origin)) return;
1328:   try {
1329:     const response = await fetch(entry.url);
1330:     if (!response.ok) return;
1331:     const text = await response.text();
1332:     entry.text = flattenSearchText([entry.text, text.slice(0, 60000)]);
1333:     entry.normalizedText = normalize(entry.text);
1334:     updateSearchDropdown();
1335:   } catch {
1336:     // Files remain searchable by title, path, caption, and catalog text if direct text fetch is
unavailable.
1337:   }
1338: }

```

## ***addProjectSearchEntries - lines 1340-1414***

| Item            | Explanation                                          |
|-----------------|------------------------------------------------------|
| Source location | Lines 1340-1414 (75 source lines).                   |
| Purpose         | Part of the public website runtime.                  |
| Declaration     | function addProjectSearchEntries(entries, project) { |
| Touches         | filesystem                                           |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | slugLabel, richTextTerms, electronicsSearchKeywords, addSearchEntry, filter, sectionHasRenderableContent, forEach, displayTitle, parsedItemTerms, fileNameFromUrl, itemUrl, join                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1340: function addProjectSearchEntries(entries, project) {
1341:   const categoryLabel = slugLabel(project.category);
1342:   const baseContext = `${categoryLabel} / ${project.title}`;
1343:   const projectTerms = [
1344:     project.id,
1345:     project.summary,
1346:     richTextTerms(project.summaryRich),
1347:     project.focus,
1348:     project.highlights,
1349:     project.tools,
1350:     project.languages,
1351:     project.links,
1352:     window.electronicsSearchKeywords ? window.electronicsSearchKeywords(project, categoryLabel) : []
1353:   ];
1354:
1355:   addSearchEntry(entries, {
1356:     context: categoryLabel,
1357:     kind: "project",
1358:     projectId: project.id,
1359:     title: project.title,
1360:     type: "Project",
1361:     text: projectTerms
1362:   });
1363:
1364:   const sections = (project.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
1365:   sections.forEach((section, sectionIndex) => {
1366:     addSearchEntry(entries, {
1367:       context: baseContext,
1368:       kind: "section",
1369:       projectId: project.id,
1370:       sectionIndex: String(sectionIndex),
1371:       title: displayTitle(section.title, "Section"),
1372:       type: "Project section",
1373:       text: [section.description, richTextTerms(section.rich), parsedItemTerms(section)]
1374:     });
1375:
1376:     const walkItems = (items = [], path = [], parentTitles = []) => {
1377:       items.forEach((item, index) => {
1378:         const nextPath = [...path, index];
1379:         const title = displayTitle(item.title || item.name || item.label || fileNameFromUrl(item.url ||
""), "Project item");
1380:         const url = itemUrl(item);
1381:         const context = [baseContext, displayTitle(section.title, "Section"),
...parentTitles].filter(Boolean).join(" / ");
1382:         const entry = {
1383:           context,
1384:           kind: url ? "file" : "subsection",
1385:           projectId: project.id,
1386:           resourcePath: pathToString(nextPath),
1387:           sectionIndex: String(sectionIndex),
1388:           title,
1389:           type: url ? "File or uploaded asset" : "Subsection",
1390:           url,
1391:           text: [item.description, item.meta, item.kind, item.type, url, richTextTerms(item.rich),
parsedItemTerms(item)]
1392:         };
1393:         const storedEntry = addSearchEntry(entries, entry);
1394:         if (url && storedEntry) indexSearchableFileText(storedEntry);
1395:         walkItems(nodeChildren(item), nextPath, [...parentTitles, title]);
1396:       });
1397:     };
1398:     walkItems(section.items || []);
1399:   });
1400:
1401:   uniqueDownloads(collectDownloadsFromValue(project, "Project file", [])).forEach((download) => {
1402:     const entry = {
1403:       context: baseContext,
1404:       kind: "file",
1405:       projectId: project.id,
1406:       title: download.title || fileNameFromUrl(download.url),
1407:       type: download.type || "Uploaded file",
1408:       url: download.url,
1409:       text: [download.description, download.status, download.url]

```



```

1410:     };
1411:     const storedEntry = addSearchEntry(entries, entry);
1412:     if (storedEntry) indexSearchableFileText(storedEntry);
1413:   });
1414: }

```

## sections - lines 1364-1399

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1364-1399 (36 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | const sections = (project.portfolioView?.sections    []).filter((section) => section.id !== "brief" && sectionHasRenderableContent(section));                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | filter, sectionHasRenderableContent, forEach, addSearchEntry, displayTitle, richTextTerms, parsedItemTerms, fileNameFromUrl, itemUrl, join, pathToString, indexSearchableFileText                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1364:   const sections = (project.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
1365:   sections.forEach((section, sectionIndex) => {
1366:     addSearchEntry(entries, {
1367:       context: baseContext,
1368:       kind: "section",
1369:       projectId: project.id,
1370:       sectionIndex: String(sectionIndex),
1371:       title: displayTitle(section.title, "Section"),
1372:       type: "Project section",
1373:       text: [section.description, richTextTerms(section.rich), parsedItemTerms(section)]
1374:     });
1375:
1376:     const walkItems = (items = [], path = [], parentTitles = []) => {
1377:       items.forEach((item, index) => {
1378:         const nextPath = [...path, index];
1379:         const title = displayTitle(item.title || item.name || item.label || fileNameFromUrl(item.url ||
""), "Project item");
1380:         const url = itemUrl(item);
1381:         const context = [baseContext, displayTitle(section.title, "Section"),
...parentTitles].filter(Boolean).join(" / ");
1382:         const entry = {
1383:           context,
1384:           kind: url ? "file" : "subsection",
1385:           projectId: project.id,
1386:           resourcePath: pathToString(nextPath),
1387:           sectionIndex: String(sectionIndex),
1388:           title,
1389:           type: url ? "File or uploaded asset" : "Subsection",
1390:           url,
1391:           text: [item.description, item.meta, item.kind, item.type, url, richTextTerms(item.rich),
parsedItemTerms(item)]
1392:         };
1393:         const storedEntry = addSearchEntry(entries, entry);
1394:         if (url && storedEntry) indexSearchableFileText(storedEntry);
1395:         walkItems(nodeChildren(item), nextPath, [...parentTitles, title]);
1396:       });
1397:     };
1398:     walkItems(section.items || []);
1399:   });

```

## addSiteSectionSearchEntries - lines 1416-1463

| Item            | Explanation                                     |
|-----------------|-------------------------------------------------|
| Source location | Lines 1416-1463 (48 source lines).              |
| Purpose         | Part of the public website runtime.             |
| Declaration     | function addSiteSectionSearchEntries(entries) { |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeProfile, splice, forEach, addSearchEntry, indexSearchableFileText, filter, richTextTerms                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1416: function addSiteSectionSearchEntries(entries) {
1417:   const current = normalizeProfile(profile || {});
1418:   const pageSections = [
1419:     { id: "top", title: heroTitle?.textContent || "Front page", type: "Page section", text:
[heroEyebrow?.textContent, heroCopy?.textContent] },
1420:     { id: "projects", title: "Engineering Projects", type: "Directory", text: "project directory
categories files sections" },
1421:     { id: "process", title: "Process", type: "Page section", text: "engineering process design testing
documentation" },
1422:     { id: "contact", title: "Contact", type: "Page section", text: [current.displayName, current.email,
current.phone, current.githubUrl, current.linkedinUrl, current.websiteUrl, current.contactIntro] }
1423:   ];
1424:   if (current.resumeUrl) {
1425:     pageSections.splice(2, 0, { id: "resume", title: "Resume", type: "Page section", text: "resume
professional profile document" });
1426:   }
1427:   pageSections.forEach((section) => addSearchEntry(entries, {
1428:     domTarget: section.id,
1429:     kind: "page",
1430:     title: section.title,
1431:     type: section.type,
1432:     text: section.text
1433:   }));
1434:
1435:   if (current.resumeUrl) {
1436:     const resumeTextEntry = addSearchEntry(entries, {
1437:       context: "Professional Profile / Resume",
1438:       kind: "file",
1439:       title: "Resume",
1440:       type: "Resume",
1441:       url: current.resumeUrl,
1442:       text: [current.displayName, current.portfolioLabel, current.contactIntro, current.resumeUrl]
1443:     });
1444:     if (resumeTextEntry) indexSearchableFileText(resumeTextEntry);
1445:   }
1446:
1447:   (siteSections || []).filter(siteSectionRenderable).forEach((section) => {
1448:     addSearchEntry(entries, {
1449:       domTarget: section.id,
1450:       kind: "page",
1451:       title: section.title || "Portfolio section",
1452:       type: "Portfolio section",
1453:       text: [section.description, richTextTerms(section.richDescription), section.links, section.assets]
1454:     });
1455:     (section.subsections || []).forEach((item) => addSearchEntry(entries, {
1456:       domTarget: section.id,
1457:       kind: "page",
1458:       title: item.title || "Portfolio subsection",
1459:       type: "Portfolio subsection",
1460:       text: [item.description, richTextTerms(item.richDescription), item.links]
1461:     }));
1462:   });
1463: }

```

## rebuildSearchIndex - lines 1465-1470

| Item            | Explanation                                                   |
|-----------------|---------------------------------------------------------------|
| Source location | Lines 1465-1470 (6 source lines).                             |
| Purpose         | Part of the public website runtime.                           |
| Declaration     | function rebuildSearchIndex() {                               |
| Touches         | Mostly local calculations or local UI state.                  |
| Calls detected  | addSiteSectionSearchEntries, forEach, addProjectSearchEntries |
| API endpoints   | None detected inside this function body.                      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1465: function rebuildSearchIndex() {
1466:   const entries = [];
1467:   addSiteSectionSearchEntries(entries);
1468:   projects.forEach((project) => addProjectSearchEntries(entries, project));
1469:   searchableEntries = entries;
1470: }

```

## ***searchResultsFor - lines 1472-1486***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1472-1486 (15 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function searchResultsFor(query) {                                                                                                                                                                    |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, trim, Set, map, entryScore, filter, sort, localeCompare, join, has, add                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1472: function searchResultsFor(query) {
1473:   const normalizedQuery = normalize(query).trim();
1474:   if (!normalizedQuery) return [];
1475:   const seen = new Set();
1476:   return searchableEntries
1477:     .map((entry) => ({ ...entry, score: entryScore(entry, normalizedQuery) }))
1478:     .filter((entry) => entry.score > 0)
1479:     .sort((a, b) => b.score - a.score || a.title.localeCompare(b.title))
1480:     .filter((entry) => {
1481:       const key = [entry.kind, entry.projectId, entry.sectionIndex, entry.resourcePath, entry.domTarget,
entry.url, entry.title].join("|");
1482:       if (seen.has(key)) return false;
1483:       seen.add(key);
1484:       return true;
1485:     });
1486: }

```

## ***assistantEndpoint - lines 1488-1494***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1488-1494 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantEndpoint() {                                                                                                                                                                        |
| Touches            | browser/Electron DOM, network/API requests                                                                                                                                                            |
| Calls detected     | trim, querySelector, includes                                                                                                                                                                         |
| API endpoints      | /api/portfolio-ai                                                                                                                                                                                     |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1488: function assistantEndpoint() {
1489:   if (typeof window.OMB_AI_ENDPOINT === "string") return window.OMB_AI_ENDPOINT.trim();
1490:   const metaEndpoint = document.querySelector('meta[name="portfolio-ai-endpoint"]')?.content || "";
1491:   if (metaEndpoint.trim()) return String(metaEndpoint).trim();
1492:   if (["localhost", "127.0.0.1"].includes(window.location.hostname)) return "/api/portfolio-ai";
1493:   return "/api/portfolio-ai";
1494: }
```

## ***assistantSourceLabel - lines 1496-1505***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1496-1505 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantSourceLabel(result = {}) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, map, trim, filter, slice, join                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1496: function assistantSourceLabel(result = {}) {
1497:   const title = result.title || "Portfolio item";
1498:   if (result.kind === "project" || result.kind === "page") return title;
1499:   const contextParts = String(result.context || "")
1500:     .split("/")
1501:     .map((item) => item.trim())
1502:     .filter(Boolean);
1503:   const specificContext = contextParts.slice(-2).join(" / ");
1504:   return [title, specificContext].filter(Boolean).join(" - ");
1505: }
```

## ***assistantProjectSummary - lines 1507-1522***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1507-1522 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantProjectSummary(project = {}) {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | slugLabel, filter, sectionHasRenderableContent, map, displayTitle                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1507: function assistantProjectSummary(project = {}) {
1508:   const categoryLabel = slugLabel(project.category);
1509:   const sections = (project.portfolioView?.sections || [])
1510:     .filter((section) => section.id !== "brief" && sectionHasRenderableContent(section))
1511:     .map((section) => displayTitle(section.title, "Section"));
1512:   return {
1513:     category: categoryLabel,
1514:     focus: project.focus || [],
1515:     id: project.id,
1516:     links: project.links || [],
1517:     sections,
1518:     summary: project.summary || "",
1519:     title: project.portfolioView?.title || project.title,
```

```

1520:     tools: project.tools || []
1521:   };
1522: }

```

## assistantProjectTitleAcronyms - lines 1524-1535

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1524-1535 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantProjectTitleAcronyms(title = "") {                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, split, filter, Set, min, add, slice, map, join                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1524: function assistantProjectTitleAcronyms(title = "") {
1525:   const words = normalize(title)
1526:   .split(/\s+/)
1527:   .filter((word) => word.length > 2);
1528:   const acronyms = new Set();
1529:   for (let start = 0; start < words.length; start += 1) {
1530:     for (let length = 2; length <= Math.min(5, words.length - start); length += 1) {
1531:       acronyms.add(words.slice(start, start + length).map((word) => word[0]).join(""));
1532:     }
1533:   }
1534:   return [...acronyms].filter((item) => item.length >= 2);
1535: }

```

## assistantProjectIsNamedInQuestion - lines 1537-1553

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1537-1553 (17 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantProjectIsNamedInQuestion(question = "", project = {}) {                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, Set, includes, split, assistantProjectTitleAcronyms, some, has, filter, min                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1537: function assistantProjectIsNamedInQuestion(question = "", project = {}) {
1538:   const clean = normalize(question);
1539:   if (!clean) return false;
1540:   const fillerWords = new Set(["with", "from", "into", "using", "project", "design", "system",
"systems"]);
1541:   const rawTitle = project.portfolioView?.title || project.title || project.id;
1542:   const title = normalize(rawTitle);
1543:   if (!title) return false;
1544:   if (clean.includes(title)) return true;
1545:   const cleanWords = new Set(clean.split(/\s+/));
1546:   const acronymMatch = assistantProjectTitleAcronyms(rawTitle)
1547:   .some((acronym) => cleanWords.has(acronym));
1548:   if (acronymMatch) return true;
1549:   const words = title.split(/\s+/).filter((word) => word.length > 3 && !fillerWords.has(word));
1550:   if (!words.length) return false;
1551:   const matches = words.filter((word) => clean.includes(word)).length;
1552:   return words.length <= 2 ? matches === words.length : matches >= Math.min(3, words.length);

```

1553: }

### ***assistantNamedProjectIds - lines 1555-1559***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1555-1559 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantNamedProjectIds(question = "") {                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, assistantProjectIsNamedInQuestion, map                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1555: function assistantNamedProjectIds(question = "") {
1556:   return projects
1557:     .filter((project) => assistantProjectIsNamedInQuestion(question, project))
1558:     .map((project) => project.id);
1559: }
```

### ***assistantProjectTitleMatches - lines 1561-1563***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1561-1563 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantProjectTitleMatches(question = "") {                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | assistantNamedProjectIds                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1561: function assistantProjectTitleMatches(question = "") {
1562:   return assistantNamedProjectIds(question).length > 0;
1563: }
```

### ***assistantUrlKind - lines 1565-1577***

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 1565-1577 (13 source lines).           |
| Purpose         | Part of the public website runtime.          |
| Declaration     | function assistantUrlKind(url = "") {        |
| Touches         | Mostly local calculations or local UI state. |
| Calls detected  | split, toLowerCase, fileIsBrowserSearchable  |
| API endpoints   | None detected inside this function body.     |
| Storage keys    | No local/session storage keys detected.      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 10 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1565: function assistantUrlKind(url = "") {
1566:   const clean = String(url || "").split(/[?#]/)[0].toLowerCase();
1567:   if (!clean) return "unknown";
1568:   if (/github\.com|raw\.githubusercontent\.com|gist\.githubusercontent\.com/i.test(clean)) return
"github";
1569:   if (/linkedin\.com/i.test(clean)) return "linkedin";
1570:   if (/\. (png|jpe?g|gif|webp|svg)$/i.test(clean)) return "image";
1571:   if (/\.pdf$/i.test(clean)) return "pdf";
1572:   if (/\. (zip|7z|rar)$/i.test(clean)) return "archive";
1573:   if (/\. (xlsx?|csv)$/i.test(clean)) return "spreadsheet";
1574:   if (fileIsBrowserSearchable(clean)) return "text";
1575:   if (/^https?:\/\/\//i.test(clean)) return "webpage";
1576:   return "file";
1577: }

```

## ***assistantAbsoluteUrl - lines 1579-1658***

| Item               | Explanation                                                                                                                                                                                                                   |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1579-1658 (80 source lines).                                                                                                                                                                                            |
| Purpose            | Part of the public website runtime.                                                                                                                                                                                           |
| Declaration        | function assistantAbsoluteUrl(url = "") {                                                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                                                    |
| Calls detected     | normalizeLinkTarget, looksLikeBareWebAddress, URL, assistantLinkRecordsFromValue, uniqueDownloads, collectDownloadsFromValue, forEach, push, assistantUrlKind, fileNameFromUrl, assistantUrlsFromTextValue, flattenSearchText |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                      |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                       |
| Async/return clues | 0 await expression(s), 9 return statement(s).                                                                                                                                                                                 |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                         |

Source excerpt:

```

1579: function assistantAbsoluteUrl(url = "") {
1580:   const target = normalizeLinkTarget(url, { assumeWeb: looksLikeBareWebAddress(url) ||
/^https?:\/\/\//i.test(url) });
1581:   if (!target) return "";
1582:   try {
1583:     return new URL(target, window.location.href).href;
1584:   } catch {
1585:     return target;
1586:   }
1587: }
1588:
1589: function assistantLinkRecordsFromValue(value, context = "") {
1590:   const records = [];
1591:   const downloads = uniqueDownloads(collectDownloadsFromValue(value, "Portfolio asset", []));
1592:   downloads.forEach((download) => {
1593:     const rawUrl = download.url || download.href || download.path || "";
1594:     if (!rawUrl) return;
1595:     records.push({
1596:       context,
1597:       description: download.description || "",
1598:       kind: download.kind || assistantUrlKind(rawUrl),
1599:       label: download.title || download.label || fileNameFromUrl(rawUrl),
1600:       url: assistantAbsoluteUrl(rawUrl)
1601:     });
1602:   });
1603:   return records;
1604: }
1605:
1606: function assistantUrlsFromTextValue(value, context = "") {
1607:   const text = flattenSearchText([value]);
1608:   const matches = text.match(/\b(?:https?:\/\/\|www\.)[^s<>"])+/gi) || [];
1609:   return [...new Set(matches.map((url) => url.replace(/[(),.;:!?]+$/g, "")))]
1610:     .slice(0, 16)
1611:     .map((url) => ({
1612:       context,

```

```

1613:     description: "",
1614:     kind: assistantUrlKind(url),
1615:     label: /github\.com/i.test(url) ? "GitHub repository or profile" : fileNameFromUrl(url),
1616:     type: "Public link",
1617:     url: assistantAbsoluteUrl(url)
1618:   }));
1619: }
1620:
1621: function assistantPublicProfileLinks() {
1622:   const records = [];
1623:   const add = (item = {}, context = "") => {
1624:     const rawUrl = item.url || item.href || item.path || "";
1625:     if (!rawUrl) return;
1626:     const label = item.label || item.title || fileNameFromUrl(rawUrl);
1627:     records.push({
1628:       context,
1629:       kind: item.kind || assistantUrlKind(rawUrl),
1630:       label,
1631:       text: item.description || "",
1632:       url: assistantAbsoluteUrl(rawUrl)
1633:     });
1634:   };
1635:
1636:   const current = normalizeProfile(profile || {});
1637:   if (current.resumeUrl) add({ label: "Resume", kind: "resume", url: current.resumeUrl }, "Professional
Profile");
1638:   if (current.githubUrl) add({ label: "GitHub", kind: "github", url: current.githubUrl }, "Professional
Profile");
1639:   if (current.linkedinUrl) add({ label: "LinkedIn", kind: "linkedin", url: current.linkedinUrl },
"Professional Profile");
1640:   if (current.websiteUrl) add({ label: "Website", kind: "webpage", url: current.websiteUrl },
"Professional Profile");
1641:
1642:   (siteSections || []).forEach((section) => {
1643:     (section.links || []).forEach((link) => add(link, section.title || "Portfolio section"));
1644:   });
1645:
1646:   return records.filter((record, index, array) => (
1647:     record.url && array.findIndex((item) => item.url === record.url && item.label === record.label) ===
index
1648:   ));
1649: }
1650:
1651: function assistantProjectEvidence(project = {}) {
1652:   const downloads = uniqueDownloads(collectDownloadsFromValue(project, "Project asset", []));
1653:   const downloadRecords = downloads.slice(0, 26).map((download) => {
1654:     const rawUrl = download.url || "";
1655:     return {
1656:       description: download.description || "",
1657:       kind: assistantUrlKind(rawUrl),
1658:       label: download.title || download.label || fileNameFromUrl(rawUrl),

```

## ***assistantLinkRecordsFromValue - lines 1589-1604***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1589-1604 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantLinkRecordsFromValue(value, context = "") {                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | uniqueDownloads, collectDownloadsFromValue, forEach, push, assistantUrlKind, fileNameFromUrl, assistantAbsoluteUrl                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1589: function assistantLinkRecordsFromValue(value, context = "") {
1590:   const records = [];
1591:   const downloads = uniqueDownloads(collectDownloadsFromValue(value, "Portfolio asset", []));
1592:   downloads.forEach((download) => {
1593:     const rawUrl = download.url || download.href || download.path || "";
1594:     if (!rawUrl) return;
1595:     records.push({
1596:       context,
1597:       description: download.description || "",
1598:       kind: download.kind || assistantUrlKind(rawUrl),

```



```

1599:         label: download.title || download.label || fileNameFromUrl(rawUrl),
1600:         url: assistantAbsoluteUrl(rawUrl)
1601:     });
1602: });
1603: return records;
1604: }

```

## ***assistantUrlsFromTextValue - lines 1606-1619***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1606-1619 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantUrlsFromTextValue(value, context = "") {                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | flattenSearchText, match, b, Set, map, replace, slice, assistantUrlKind, fileNameFromUrl, assistantAbsoluteUrl                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1606: function assistantUrlsFromTextValue(value, context = "") {
1607:   const text = flattenSearchText([value]);
1608:   const matches = text.match(/\b(?:https?:\/\/|www\.)[^s<>"])+/gi) || [];
1609:   return [...new Set(matches.map((url) => url.replace(/[,.;!]?+$/g, "")))]
1610:     .slice(0, 16)
1611:     .map((url) => ({
1612:       context,
1613:       description: "",
1614:       kind: assistantUrlKind(url),
1615:       label: /github\.com/i.test(url) ? "GitHub repository or profile" : fileNameFromUrl(url),
1616:       type: "Public link",
1617:       url: assistantAbsoluteUrl(url)
1618:     }));
1619: }

```

## ***assistantPublicProfileLinks - lines 1621-1649***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1621-1649 (29 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantPublicProfileLinks() {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | fileNameFromUrl, push, assistantUrlKind, assistantAbsoluteUrl, normalizeProfile, add, forEach, filter, findIndex                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1621: function assistantPublicProfileLinks() {
1622:   const records = [];
1623:   const add = (item = {}, context = "") => {
1624:     const rawUrl = item.url || item.href || item.path || "";
1625:     if (!rawUrl) return;
1626:     const label = item.label || item.title || fileNameFromUrl(rawUrl);
1627:     records.push({
1628:       context,
1629:       kind: item.kind || assistantUrlKind(rawUrl),
1630:       label,
1631:       text: item.description || "",
1632:       url: assistantAbsoluteUrl(rawUrl)

```

```

1633:     });
1634:   });
1635:
1636:   const current = normalizeProfile(profile || {});
1637:   if (current.resumeUrl) add({ label: "Resume", kind: "resume", url: current.resumeUrl }, "Professional
Profile");
1638:   if (current.githubUrl) add({ label: "GitHub", kind: "github", url: current.githubUrl }, "Professional
Profile");
1639:   if (current.linkedinUrl) add({ label: "LinkedIn", kind: "linkedin", url: current.linkedinUrl },
"Professional Profile");
1640:   if (current.websiteUrl) add({ label: "Website", kind: "webpage", url: current.websiteUrl },
"Professional Profile");
1641:
1642:   (siteSections || []).forEach((section) => {
1643:     (section.links || []).forEach((link) => add(link, section.title || "Portfolio section"));
1644:   });
1645:
1646:   return records.filter((record, index, array) => (
1647:     record.url && array.findIndex((item) => item.url === record.url && item.label === record.label) ===
index
1648:   ));
1649: }

```

## ***assistantProjectEvidence - lines 1651-1667***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1651-1667 (17 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantProjectEvidence(project = {}) {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | uniqueDownloads, collectDownloadsFromValue, slice, map, assistantUrlKind, fileNameFromUrl, assistantAbsoluteUrl, assistantUrlsFromTextValue, filter, findIndex                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1651: function assistantProjectEvidence(project = {}) {
1652:   const downloads = uniqueDownloads(collectDownloadsFromValue(project, "Project asset", []));
1653:   const downloadRecords = downloads.slice(0, 26).map((download) => {
1654:     const rawUrl = download.url || "";
1655:     return {
1656:       description: download.description || "",
1657:       kind: assistantUrlKind(rawUrl),
1658:       label: download.title || download.label || fileNameFromUrl(rawUrl),
1659:       type: download.type || "",
1660:       url: assistantAbsoluteUrl(rawUrl)
1661:     };
1662:   });
1663:   const textLinkRecords = assistantUrlsFromTextValue([project.summary, project.links,
project.portfolioView, project.portfolioView?.title || project.title];
1664:   return [...downloadRecords, ...textLinkRecords].filter((record, index, array) => (
1665:     record.url && array.findIndex((item) => item.url === record.url) === index
1666:   ));
1667: }

```

## ***assistantQuestionIsCasual - lines 1669-1675***

| Item            | Explanation                                         |
|-----------------|-----------------------------------------------------|
| Source location | Lines 1669-1675 (7 source lines).                   |
| Purpose         | Part of the public website runtime.                 |
| Declaration     | function assistantQuestionIsCasual(question = "") { |
| Touches         | filesystem                                          |
| Calls detected  | normalize                                           |
| API endpoints   | None detected inside this function body.            |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1669: function assistantQuestionIsCasual(question = "") {
1670:   const clean = normalize(question);
1671:   return
1672:   /^(hi|hello|hey|good\s+(morning|afternoon|evening)|yo|sup|thanks|thank\s+you|ok|okay)\b/.test(clean)
1673:   || /\^(how\s+are\s+you|what\s+can\s+you\s+help.*|what\s+do\s+you\s+do|who\s+are\s+you|help|help\s+me|c
1674:   an\s+you\s+help|nice\s+to\s+meet\s+you)\??$/ .test(clean)
1675:   || /\^(what'\s+is|do\s+you\s+know)\s+my\s+name\??$/ .test(clean)
1676:   || /\^who\s+am\s+i\??$/ .test(clean);
1677: }

```

## ***assistantQuestionHasPortfolioIntent - lines 1677-1682***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1677-1682 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantQuestionHasPortfolioIntent(question = "") {                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, assistantProjectTitleMatches, b                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1677: function assistantQuestionHasPortfolioIntent(question = "") {
1678:   const clean = normalize(question);
1679:   return assistantProjectTitleMatches(question)
1680:   || /\b(maurice|otieno|portfolio|resume|github|linkedin|contact|email|phone|project|projects|repo|repo
1681:   sitory|repositories|file|files|document|documents|artifact|artifacts|link|links|download|open|show|where|built|b
1682:   uild|created|designed|implemented)\b/.test(clean)
1683:   || /\b(your|his|maurice's)\s+(project|projects|resume|github|linkedin|portfolio|work|email|phone|cont
1684:   act|repo|repository|files?|documents?|links?)\b/.test(clean);
1685: }

```

## ***assistantConversationSuggestsPortfolioContext - lines 1684-1688***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1684-1688 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantConversationSuggestsPortfolioContext(history = []) {                                                                                                                                |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | slice, some, b, normalize                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1684: function assistantConversationSuggestsPortfolioContext(history = []) {

```

```

1685:   return history
1686:   .slice(-4)
1687:   .some((item) => /\b(maurice|otieno|portfolio|project|resume|github|linkedin|electronics|engineering)\
b/.test(normalize(item?.content)));
1688: }

```

## ***assistantQuestionAllowsPublicLookup - lines 1690-1695***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1690-1695 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantQuestionAllowsPublicLookup(question = "", intent = assistantQuestionIntent(question)) {                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | assistantQuestionIntent, normalize, b                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1690: function assistantQuestionAllowsPublicLookup(question = "", intent = assistantQuestionIntent(question)) {
1691:   const clean = normalize(question);
1692:   return intent === "general_engineering"
1693:     || intent === "general_knowledge"
1694:     || /\b(github|linkedin|repo|repository|repositories|public\s+link|website|web\s+page|profile|resume|u
ploadeds+file|document|source\s+code)\b/.test(clean);
1695: }

```

## ***assistantQuestionLooksConceptual - lines 1697-1702***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1697-1702 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantQuestionLooksConceptual(question = "") {                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, assistantQuestionIsEngineeringRelated, b                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1697: function assistantQuestionLooksConceptual(question = "") {
1698:   const clean = normalize(question);
1699:   if (!assistantQuestionIsEngineeringRelated(question)) return false;
1700:   return
/^(what|what\s+is|what\s+are|what's|define|explain|describe|how|why|compare|differentiate)\b/.test(clean)
1701:     || /\b(definition|meaning|basics|overview|introduction|difference\s+between|how\s+does|how\s+do|why\s
+does|why\s+do)\b/.test(clean);
1702: }

```

## ***assistantQuestionIntent - lines 1704-1717***

| Item            | Explanation                         |
|-----------------|-------------------------------------|
| Source location | Lines 1704-1717 (14 source lines).  |
| Purpose         | Part of the public website runtime. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function assistantQuestionIntent(question = "", history = assistantChatHistory) {                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | assistantQuestionIsCasual, normalize, assistantQuestionHasPortfolioIntent, assistantConversationSuggestsPortfolioContext, b, assistantQuestionIsEngineeringRelated, assistantQuestionLooksConceptual  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 7 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1704: function assistantQuestionIntent(question = "", history = assistantChatHistory) {
1705:   if (assistantQuestionIsCasual(question)) return "general_conversation";
1706:   if (/^(what'?s|what\s+is|do\s+you\s+know)\s+my\s+name\??$/i.test(normalize(question)) ||
/who\s+am\s+i\??$/i.test(normalize(question))) {
1707:     return "general_conversation";
1708:   }
1709:   const hasPortfolioIntent = assistantQuestionHasPortfolioIntent(question);
1710:   if (!hasPortfolioIntent && assistantConversationSuggestsPortfolioContext(history) &&
/b(name|you|your|he|his|him|that|this|it)\b/i.test(normalize(question))) {
1711:     return "general_conversation";
1712:   }
1713:   if (!hasPortfolioIntent && assistantQuestionIsEngineeringRelated(question)) return
"general_engineering";
1714:   if (!hasPortfolioIntent && assistantQuestionLooksConceptual(question)) return "general_engineering";
1715:   if (hasPortfolioIntent) return "portfolio_specific";
1716:   return "general_knowledge";
1717: }

```

## ***assistantQuestionTokens - lines 1719-1730***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1719-1730 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantQuestionTokens(question = "") {                                                                                                                                                     |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | Set, normalize, split, map, trim, filter, has                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1719: function assistantQuestionTokens(question = "") {
1720:   const stopWords = new Set([
1721:     "about", "again", "also", "and", "are", "can", "could", "does", "for", "from", "give", "have", "his",
1722:     "into", "link", "links", "maurice", "me", "my", "open", "otieno", "please", "portfolio", "project",
1723:     "projects", "show", "tell", "that", "the", "their", "this", "what", "where", "which", "who", "with",
1724:     "work", "your"
1725:   ]);
1726:   return normalize(question)
1727:     .split(/[\a-z0-9+#.]+/i)
1728:     .map((token) => token.trim())
1729:     .filter((token) => token.length > 2 && !stopWords.has(token));
1730: }

```

## ***assistantSpecificSourceTokens - lines 1732-1739***

| Item            | Explanation                       |
|-----------------|-----------------------------------|
| Source location | Lines 1732-1739 (8 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantSpecificSourceTokens(question = "") {                                                                                                                                               |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | Set, assistantQuestionTokens, filter, has                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1732: function assistantSpecificSourceTokens(question = "") {
1733:   const genericTokens = new Set([
1734:     "built", "build", "created", "designed", "did", "done", "engineering", "file", "files", "hardware",
1735:     "item", "items", "section", "sections", "system", "systems", "thing", "things", "tool", "tools",
1736:     "use", "used", "using", "website"
1737:   ]);
1738:   return assistantQuestionTokens(question).filter((token) => !genericTokens.has(token));
1739: }

```

### ***assistantPromptTargetsProjectLanding - lines 1741-1746***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1741-1746 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantPromptTargetsProjectLanding(question = "") {                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, b                                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1741: function assistantPromptTargetsProjectLanding(question = "") {
1742:   const clean = normalize(question);
1743:   return /\b(open|show|view|go\s+to|take\s+me\s+to)\b/.test(clean)
1744:     && /\b(project|overview|page)\b/.test(clean)
1745:     && /\b(code|source|repo|repository|github|file|files|document|documents|test|tests|result|results|pc
b|schematic|simulation|tool|tools|resume|linkedin)\b/.test(clean);
1746: }

```

### ***assistantEntryRelationScore - lines 1748-1762***

| Item            | Explanation                                                         |
|-----------------|---------------------------------------------------------------------|
| Source location | Lines 1748-1762 (15 source lines).                                  |
| Purpose         | Part of the public website runtime.                                 |
| Declaration     | function assistantEntryRelationScore(result = {}, question = "") {  |
| Touches         | filesystem, authentication/security data                            |
| Calls detected  | assistantSpecificSourceTokens, normalize, filter, join, includes, b |
| API endpoints   | None detected inside this function body.                            |
| Storage keys    | No local/session storage keys detected.                             |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1748: function assistantEntryRelationScore(result = {}, question = "") {
1749:   const tokens = assistantSpecificSourceTokens(question);
1750:   const haystack = normalize([result.title, result.context, result.type, result.text,
result.url].filter(Boolean).join(" "));
1751:   const tokenHits = tokens.filter((token) => haystack.includes(token)).length;
1752:   let relation = tokenHits * 30;
1753:
1754:   if (tokens.length && tokenHits === tokens.length) relation += 25;
1755:   if (Number(result.score || 0) >= 120) relation += 15;
1756:   if (Number(result.score || 0) >= 80) relation += 8;
1757:   if (result.kind === "project") relation += 12;
1758:   if (result.kind === "file" && /\b(file|files|download|document|documents|code|source|github|repo|reposit
ory|test|tests|result|results|pcb|schematic|simulation)\b/.test(normalize(question))) {
1759:     relation += 16;
1760:   }
1761:   return relation;
1762: }

```

## ***assistantNamedProjectTopicTokens - lines 1764-1776***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1764-1776 (13 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantNamedProjectTopicTokens(question = "", namedIds = assistantNamedProjectIds(question)) {                                                                                             |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | assistantNamedProjectIds, Set, map, find, filter, forEach, normalize, split, add, assistantProjectTitleAcronyms, assistantSpecificSourceTokens, has                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1764: function assistantNamedProjectTopicTokens(question = "", namedIds = assistantNamedProjectIds(question)) {
1765:   const projectVocabulary = new Set();
1766:   namedIds
1767:     .map((id) => projects.find((project) => project.id === id))
1768:     .filter(Boolean)
1769:     .forEach((project) => {
1770:       const title = project.portfolioView?.title || project.title || project.id;
1771:       normalize(title).split(/\s+/).filter((word) => word.length > 2).forEach((word) =>
projectVocabulary.add(word));
1772:       assistantProjectTitleAcronyms(title).forEach((acronym) => projectVocabulary.add(acronym));
1773:       projectVocabulary.add(normalize(project.id));
1774:     });
1775:   return assistantSpecificSourceTokens(question).filter((token) => !projectVocabulary.has(token));
1776: }

```

## ***assistantSourcesForDisplay - lines 1778-1813***

| Item            | Explanation                                                                                                                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 1778-1813 (36 source lines).                                                                                                                                                                                         |
| Purpose         | Part of the public website runtime.                                                                                                                                                                                        |
| Declaration     | function assistantSourcesForDisplay(question = "", results = [], intent = assistantQuestionIntent(question)) {                                                                                                             |
| Touches         | filesystem, authentication/security data                                                                                                                                                                                   |
| Calls detected  | assistantQuestionIntent, assistantSpecificSourceTokens, assistantNamedProjectIds, Set, assistantNamedProjectTopicTokens, b, normalize, assistantPromptTargetsProjectLanding, map, assistantEntryRelationScore, filter, has |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 14 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1778: function assistantSourcesForDisplay(question = "", results = [], intent =
assistantQuestionIntent(question)) {
1779:   if (intent !== "portfolio_specific") return [];
1780:   const tokens = assistantSpecificSourceTokens(question);
1781:   const namedProjectIdList = assistantNamedProjectIds(question);
1782:   const namedProjectIds = new Set(namedProjectIdList);
1783:   const namedProjectTopicTokens = assistantNamedProjectTopicTokens(question, namedProjectIdList);
1784:   const asksForLinks = /\b(open|show|where|link|links|github|linkedin|resume|download|file|files|repo|rep
ository|source\s+code)\b/.test(normalize(question));
1785:   const asksForProjectLanding = assistantPromptTargetsProjectLanding(question);
1786:   const pageLinkIntent = /\b(resume|github|linkedin|contact|email|phone)\b/.test(normalize(question));
1787:   const asksForSpecificArtifact = /\b(code|source|repo|repository|github|file|files|document|documents|te
st|tests|result|results|pcb|schematic|simulation|tool|tools)\b/.test(normalize(question));
1788:
1789:   if (!namedProjectIds.size && !pageLinkIntent && tokens.length === 0) return [];
1790:
1791:   const filtered = results
1792:     .map((result) => ({ ...result, relation: assistantEntryRelationScore(result, question) })))
1793:     .filter((result) => {
1794:       if (!result) return false;
1795:       if (namedProjectIds.size && !namedProjectIds.has(result.projectId)) return false;
1796:       if (asksForProjectLanding) return result.kind === "project";
1797:       const haystack = normalize([result.title, result.context, result.type, result.text,
result.url].filter(Boolean).join(" "));
1798:       const tokenHits = tokens.filter((token) => haystack.includes(token)).length;
1799:       const topicHits = namedProjectTopicTokens.filter((token) => haystack.includes(token)).length;
1800:       if (namedProjectIds.size && namedProjectTopicTokens.length && result.kind !== "project" &&
topicHits === 0) return false;
1801:       if (namedProjectIds.size && namedProjectTopicTokens.length && result.kind === "project" &&
asksForSpecificArtifact) return false;
1802:       if (namedProjectIds.size && result.kind === "project") return true;
1803:       if (pageLinkIntent && tokenHits >= 1) return true;
1804:       if (!tokens.length) return false;
1805:       if (tokenHits >= Math.min(2, tokens.length)) return true;
1806:       if (asksForLinks && tokenHits >= 1 && result.relation >= 35) return true;
1807:       return result.relation >= 55;
1808:     });
1809:
1810:   return filtered
1811:     .sort((a, b) => b.relation - a.relation || Number(b.score || 0) - Number(a.score || 0))
1812:     .slice(0, asksForLinks ? 4 : 2);
1813: }

```

## ***assistantContextForQuestion - lines 1815-1890***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1815-1890 (76 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantContextForQuestion(question = "", intent = assistantQuestionIntent(question), knownResults = null) {                                                                                |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | assistantQuestionIntent, includes, searchResultsFor, slice, Set, map, filter, assistantNamedProjectIds, find, b, normalize, assistantPublicProfileLinks                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1815: function assistantContextForQuestion(question = "", intent = assistantQuestionIntent(question),
knownResults = null) {
1816:   const directResults = ["general_engineering", "general_knowledge",
"general_conversation"].includes(intent)

```



```

1817:     ? []
1818:     : (knownResults || searchResultsFor(question)).slice(0, 8);
1819: const projectIds = [...new Set([
1820:   ...directResults.map((item) => item.projectId).filter(Boolean),
1821:   ...assistantNamedProjectIds(question)
1822: ])]);
1823: const matchedProjects = projectIds
1824:   .map((id) => projects.find((project) => project.id === id))
1825:   .filter(Boolean)
1826:   .slice(0, 5);
1827: const includePortfolioLinkContext = intent === "portfolio_specific";
1828: const wantsPublicCode = /\b(github|repo|repository|repositories|source\s|code|code|snippet|firmware|ver
1829: iolog|systemverilog|python|javascript)\b/.test(normalize(question));
1829: const publicProfileFetches = includePortfolioLinkContext
1830:   ? assistantPublicProfileLinks()
1831:   .map((item) => ({ ...item, question }));
1832:   .sort((a, b) => {
1833:     if (!wantsPublicCode) return 0;
1834:     const aGithub = /github/i.test(`${a.kind || ""} ${a.url || ""}`) ? 1 : 0;
1835:     const bGithub = /github/i.test(`${b.kind || ""} ${b.url || ""}`) ? 1 : 0;
1836:     return bGithub - aGithub;
1837:   })
1838:   : [];
1839: const resultFetches = directResults.map((result) => ({
1840:   context: result.context,
1841:   kind: assistantUrlKind(result.url),
1842:   title: result.title,
1843:   type: result.type,
1844:   url: assistantAbsoluteUrl(result.url || ""),
1845:   question
1846: }));
1847: const projectFetches = matchedProjects.flatMap((project) => assistantProjectEvidence(project)
1848:   .filter((item) => ["text", "webpage", "github", "linkedin", "resume_text"].includes(item.kind))
1849:   .slice(0, 6)
1850:   .map((item) => ({ ...item, question })));
1851: const sourceFetches = [
1852:   ...(wantsPublicCode ? publicProfileFetches : []),
1853:   ...resultFetches,
1854:   ...(wantsPublicCode ? [] : publicProfileFetches),
1855:   ...projectFetches
1856: ].filter((item) => item.url);
1857: const uniqueSourceFetches = sourceFetches.filter((item, index, array) => (
1858:   array.findIndex((candidate) => candidate.url === item.url) === index
1859: )).slice(0, 14);
1860: return {
1861:   categories: categories.map((category) => ({ id: category.id, label: category.label })),
1862:   intent,
1863:   profile: normalizeProfile(profile || {}),
1864:   knowledgeManifest: {
1865:     publicProfiles: includePortfolioLinkContext ? assistantPublicProfileLinks() : [],
1866:     publicSourcePolicy: "Only public profile and project links shown in the portfolio are used. GitHub
repository links can be expanded into repository metadata, README text, and selected public source files when
the visitor asks for code.",
1867:     matchedProjectEvidence: matchedProjects.map((project) => ({
1868:       evidence: assistantProjectEvidence(project),
1869:       projectId: project.id,
1870:       title: project.portfolioView?.title || project.title
1871:     })),
1872:     note: "Image records include filenames, captions, descriptions, and neighboring portfolio text. Use
a vision-capable backend before claiming visual details not present in captions or text."
1873:   },
1874:   portfolioContextPolicy: intent !== "portfolio_specific"
1875:     ? "Answer from general knowledge first. Do not lead with the portfolio owner's project context
unless the visitor explicitly asks to connect the concept to the portfolio."
1876:     : "Answer from the configured portfolio context first. Do not invent project details outside the
supplied context.",
1877:   projects: matchedProjects.map(assistantProjectSummary),
1878:   question,
1879:   results: directResults.map((result) => ({
1880:     context: result.context,
1881:     kind: result.kind,
1882:     text: searchSnippet(result.text, question),
1883:     title: result.title,
1884:     type: result.type,
1885:     url: result.url
1886: })),
1887:   sourceFetches: uniqueSourceFetches,
1888:   siteSections: siteSections.filter(siteSectionRenderable).map((section) => ({ id: section.id, title:
section.title })))
1889:   };
1890: }

```

## assistantRemember - lines 1892-1901

| Item            | Explanation                         |
|-----------------|-------------------------------------|
| Source location | Lines 1892-1901 (10 source lines).  |
| Purpose         | Part of the public website runtime. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function assistantRemember(role, content = "") {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, push, slice                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1892: function assistantRemember(role, content = "") {
1893:   const cleanRole = role === "assistant" ? "assistant" : "user";
1894:   const cleanContent = String(content || "").trim();
1895:   if (!cleanContent) return;
1896:   assistantChatHistory.push({
1897:     role: cleanRole,
1898:     content: cleanContent.slice(0, 1400)
1899:   });
1900:   assistantChatHistory = assistantChatHistory.slice(-10);
1901: }

```

## ***assistantConversationForRequest - lines 1903-1912***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1903-1912 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantConversationForRequest(currentQuestion = "", priorConversation = assistantChatHistory) {                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | slice, map, push                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1903: function assistantConversationForRequest(currentQuestion = "", priorConversation = assistantChatHistory)
{
1904:   const history = priorConversation.slice(-8).map((item) => ({
1905:     role: item.role === "assistant" ? "assistant" : "user",
1906:     content: String(item.content || "").slice(0, 1200)
1907:   }));
1908:   if (currentQuestion) {
1909:     history.push({ role: "user", content: String(currentQuestion).slice(0, 1200) });
1910:   }
1911:   return history;
1912: }

```

## ***assistantFallbackResults - lines 1914-1972***

| Item            | Explanation                                                                                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 1914-1972 (59 source lines).                                                                                                                     |
| Purpose         | Part of the public website runtime.                                                                                                                    |
| Declaration     | function assistantFallbackResults(question = "", intent = assistantQuestionIntent(question)) {                                                         |
| Touches         | filesystem                                                                                                                                             |
| Calls detected  | assistantQuestionIntent, normalize, trim, assistantNamedProjectIds, searchResultsFor, filter, includes, slice, map, find, slugLabel, flattenSearchText |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 9 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1914: function assistantFallbackResults(question = "", intent = assistantQuestionIntent(question)) {
1915:   if (intent !== "portfolio_specific") return [];
1916:   const clean = normalize(question).trim();
1917:   if (!clean) return [];
1918:   const namedProjectIds = assistantNamedProjectIds(question);
1919:   let results = searchResultsFor(clean);
1920:   if (namedProjectIds.length && results.length) {
1921:     const focusedResults = results.filter((result) => namedProjectIds.includes(result.projectId));
1922:     if (focusedResults.length) return focusedResults.slice(0, 6);
1923:   }
1924:   if (results.length) return results.slice(0, 6);
1925:
1926:   if (namedProjectIds.length) {
1927:     return namedProjectIds
1928:       .map((id) => projects.find((project) => project.id === id))
1929:       .filter(Boolean)
1930:       .map((project) => ({
1931:         context: slugLabel(project.category),
1932:         kind: "project",
1933:         projectId: project.id,
1934:         text: flattenSearchText([project.summary, project.focus, project.tools, project.languages]),
1935:         title: project.portfolioView?.title || project.title,
1936:         type: "Project"
1937:       })))
1938:     .slice(0, 6);
1939:   }
1940:
1941:   const category = categories.find((item) => clean.includes(normalize(item.label)) ||
clean.includes(normalize(item.id)));
1942:   if (category) {
1943:     return projects
1944:       .filter((project) => project.category === category.id)
1945:       .map((project) => ({
1946:         context: category.label,
1947:         kind: "project",
1948:         projectId: project.id,
1949:         text: flattenSearchText([project.summary, project.focus, project.tools, project.languages]),
1950:         title: project.portfolioView?.title || project.title,
1951:         type: "Project"
1952:       })))
1953:     .slice(0, 6);
1954:   }
1955:
1956:   if (/resume|professional|contact|email|phone|github|linkedin|profile/.test(clean)) {
1957:     return searchResultsFor("resume contact github professional profile").slice(0, 6);
1958:   }
1959:
1960:   if (/project|work|portfolio|hardware|design/.test(clean)) {
1961:     return projects.slice(0, 6).map((project) => ({
1962:       context: slugLabel(project.category),
1963:       kind: "project",
1964:       projectId: project.id,
1965:       text: project.summary || "",
1966:       title: project.portfolioView?.title || project.title,
1967:       type: "Project"
1968:     }));
1969:   }
1970:
1971:   return [];
1972: }

```

### ***assistantQuestionIsEngineeringRelated - lines 1974-1977***

| Item            | Explanation                                                     |
|-----------------|-----------------------------------------------------------------|
| Source location | Lines 1974-1977 (4 source lines).                               |
| Purpose         | Part of the public website runtime.                             |
| Declaration     | function assistantQuestionIsEngineeringRelated(question = "") { |
| Touches         | filesystem                                                      |
| Calls detected  | normalize, b                                                    |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1974: function assistantQuestionIsEngineeringRelated(question = "") {
1975:   const clean = normalize(question);
1976:   return /\b(op\s*amp|amplifier|analog|mixed\s*signal|adc|dac|filter|vco|oscillator|pwm|charger|rectifier
|regulator|buck|boost|ldo|mosfet|bjt|transistor|diode|pcb|schematic|layout|ground|noise|frequency|gain|phase|lts
pice|kicad|vivado|verilog|vhdl|fpga|asic|stm32|mcu|microcontroller|embedded|firmware|rtos|i2c|spi|uart|sensor|co
ntrol|signal|circuit|electronics|hardware|power)\b/.test(clean);
1977: }

```

## ***assistantGeneralEngineeringAnswer - lines 1979-2091***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1979-2091 (113 source lines).                                                                                                                                                                   |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantGeneralEngineeringAnswer(question = "") {                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, assistantQuestionIsEngineeringRelated, join, b                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 9 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1979: function assistantGeneralEngineeringAnswer(question = "") {
1980:   const clean = normalize(question);
1981:   if (!assistantQuestionIsEngineeringRelated(question)) return "";
1982:
1983:   if (/bembedded\s+systems?b/.test(clean)) {
1984:     return [
1985:       "Embedded systems is a field of engineering focused on computers that are built into larger
products, machines, instruments, or control systems.",
1986:       "",
1987:       "Unlike a laptop or desktop computer, an embedded system is usually designed for a specific job. It
combines hardware, firmware, sensors, actuators, communication interfaces, and timing constraints so the
product can make measurements, control outputs, or respond to events reliably.",
1988:       "",
1989:       "Common examples include:",
1990:       "- A microcontroller controlling a motor drive.",
1991:       "- A sensor node collecting temperature, pressure, or motion data.",
1992:       "- A medical device running real-time measurement firmware.",
1993:       "- An automotive controller reading sensors and driving actuators.",
1994:       "",
1995:       "The important engineering ideas are timing, interrupts, power use, peripheral interfaces,
reliability, and how the firmware proves the hardware is behaving correctly."
1996:     ].join("\n");
1997:   }
1998:
1999:   if (/b(op\s*amp|amplifier|fully differential|differential)b/.test(clean)) {
2000:     return [
2001:       "A useful way to think about an op amp is as a high-gain error-correcting block. With negative
feedback, the circuit around it sets the useful behavior: gain, filtering, buffering, summing, integration, or
differential conversion.",
2002:       "",
2003:       "For a fully differential op amp:",
2004:       "- The signal is carried on two opposite-phase outputs instead of one output referenced to
ground.",
2005:       "- The differential output improves noise immunity and dynamic range.",
2006:       "- A common-mode feedback loop is normally needed to hold the average output voltage at the desired
common-mode level.",
2007:       "",
2008:       "In portfolio terms, this connects most naturally to analog and mixed-signal design, especially
filters, VCO control paths, ADC front ends, and precision measurement circuits."
2009:     ].join("\n");

```

```

2010: }
2011:
2012: if (/b(pwm|vco|oscillator|frequency)\b/.test(clean)) {
2013:     return [
2014:         "PWM-driven VCO work is about converting a digital timing signal into an analog control quantity
and then into a frequency.",
2015:         "",
2016:         "The usual signal chain is:",
2017:         "- PWM duty cycle encodes the control value.",
2018:         "- A low-pass filter turns the PWM waveform into an average voltage.",
2019:         "- The VCO maps that voltage to oscillation frequency.",
2020:         "- Tests compare duty cycle, control voltage, frequency range, ripple, and stability.",
2021:         "",
2022:         "That is a mixed-signal problem because digital timing, analog filtering, oscillator behavior, and
measurement all interact."
2023:     ].join("\n");
2024: }
2025:
2026: if (/b(fpga|asic|verilog|vhdl|vivado|digital)\b/.test(clean)) {
2027:     return [
2028:         "FPGA and ASIC work both implement digital hardware, but the engineering tradeoffs are different.",
2029:         "",
2030:         "- FPGA design is reconfigurable, fast to test, and useful for prototyping or deployment when
flexibility matters.",
2031:         "- ASIC design targets a fixed silicon implementation, so verification, timing closure, power,
area, and design-for-test become much more permanent decisions.",
2032:         "- Verilog or VHDL describes hardware structure and behavior; it is not software running line by
line.",
2033:         "",
2034:         "For a recruiter, strong evidence includes block diagrams, RTL, simulation waveforms, timing
reports, verification plans, and notes explaining design tradeoffs."
2035:     ].join("\n");
2036: }
2037:
2038: if (/b(stm32|mcu|microcontroller|embedded|firmware|rtos|i2c|spi|uart)\b/.test(clean)) {
2039:     return [
2040:         "Embedded work is strongest when the firmware is tied clearly to hardware behavior.",
2041:         "",
2042:         "A good explanation usually covers:",
2043:         "- The MCU role in the system.",
2044:         "- The peripherals used, such as ADC, PWM, timers, I2C, SPI, UART, or GPIO.",
2045:         "- How timing, interrupts, sampling, power, and fault handling were managed.",
2046:         "- What was measured to prove the system worked.",
2047:         "",
2048:         "STM32CubeIDE is useful because it helps configure peripherals, generate startup code, debug
firmware, and manage the build flow around STM32 devices."
2049:     ].join("\n");
2050: }
2051:
2052: if (/b(pcb|layout|kicad|schematic|ground|noise)\b/.test(clean)) {
2053:     return [
2054:         "PCB design is where the schematic becomes a physical electrical system.",
2055:         "",
2056:         "The important ideas are:",
2057:         "- Current returns matter as much as current paths.",
2058:         "- Grounding, decoupling, trace width, component placement, and loop area affect noise and
stability.",
2059:         "- Analog, power, and switching sections should be arranged so noisy currents do not corrupt
sensitive measurements.",
2060:         "- A good portfolio PCB section should show schematic intent, layout decisions, board files, bring-
up notes, and test results.",
2061:         "",
2062:         "KiCad evidence is strongest when the design files are paired with clear measurements or bring-up
photos."
2063:     ].join("\n");
2064: }
2065:
2066: if (/b(charger|rectifier|ac\s*\to\s*\dc|regulator|buck|boost|ldo|power)\b/.test(clean)) {
2067:     return [
2068:         "An AC-to-DC charger or power supply usually has a few major stages.",
2069:         "",
2070:         "- Input protection and filtering handle transients and noise.",
2071:         "- Rectification converts AC into pulsating DC.",
2072:         "- Bulk capacitance smooths the rectified waveform.",
2073:         "- Regulation sets the desired output voltage or current.",
2074:         "- Load testing checks ripple, thermal behavior, regulation, and fault response.",
2075:         "",
2076:         "For portfolio presentation, the best proof is a clear schematic, expected waveforms, measured
output behavior, and a short explanation of component choices."
2077:     ].join("\n");
2078: }
2079:
2080: return [
2081:     "I can answer this as a general electronics or hardware-engineering question, even if it is not
directly tied to one saved project yet.",
2082:     "",
2083:     "The best way to approach it is:",
2084:     "- Define the signal or energy flow.",
2085:     "- Identify the active devices, passive networks, and control points.",
2086:     "- State what should be measured to prove the design works.",
2087:     "- Connect the explanation back to a project artifact: schematic, code, test data, PCB, or
simulation."

```

```

2088:     "",
2089:     "If you ask about a specific circuit, tool, waveform, or failure mode, I can make the explanation
more concrete."
2090:   ].join("\n");
2091: }

```

## assistantLocalAnswer - lines 2093-2188

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2093-2188 (96 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantLocalAnswer(question = "", results = [], intent = assistantQuestionIntent(question)) {                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | assistantQuestionIntent, normalizeProfile, normalize, b, join, assistantGeneralEngineeringAnswer, replace, Set, map, filter, find, slice                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 11 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2093: function assistantLocalAnswer(question = "", results = [], intent = assistantQuestionIntent(question)) {
2094:   const ownerName = normalizeProfile(profile || {}).displayName || "the portfolio owner";
2095:   if (intent === "general_conversation") {
2096:     const clean = normalize(question);
2097:     if (/^(hi|hello|hey|yo|sup)\b/.test(clean)) {
2098:       return "Hi, what can I do for you?";
2099:     }
2100:     if (/^(what'?s|what\s+is|do\s+you\s+know)\s+my\s+name\??$/i.test(clean) ||
/who\s+am\s+i\??$/i.test(clean)) {
2101:       return `I am an AI agent for ${ownerName}'s portfolio. I know the portfolio owner is ${ownerName},
but I do not know a visitor's personal name unless they tell me.`;
2102:     }
2103:     if (/^(who are you|what are you)\b/.test(clean)) {
2104:       return `I am an AI agent for ${ownerName}'s portfolio. I can help with projects, resume links,
public links, files, and related engineering questions.`;
2105:     }
2106:     return [
2107:       `I am here with you. I can help explore ${ownerName}'s portfolio, explain projects, summarize
files, open relevant sections, or answer related engineering questions.`,
2108:       "",
2109:       "You can ask things like:",
2110:       "- What is embedded systems?",
2111:       "- Show me a project.",
2112:       "- What tools were used?",
2113:       "- Explain the difference between FPGA and ASIC design."
2114:     ].join("\n");
2115:   }
2116:   if (intent === "general_engineering") {
2117:     const generalAnswer = assistantGeneralEngineeringAnswer(question);
2118:     if (generalAnswer) return generalAnswer;
2119:     return [
2120:       "I can answer that as a general engineering question, but I need a little more detail to make the
answer useful.",
2121:       "",
2122:       "Try asking about a specific circuit, device, tool, protocol, waveform, or failure mode."
2123:     ].join("\n");
2124:   }
2125:   if (intent === "general_knowledge") {
2126:     const generalAnswer = assistantGeneralEngineeringAnswer(question);
2127:     if (generalAnswer) return generalAnswer;
2128:     return [
2129:       "I can answer that as a general question from the local fallback, but the live model will usually
give a deeper answer when it is reachable.",
2130:       "",
2131:       "Short answer:",
2132:       `~ ${question.replace(/\?+$/, "")} is a topic I can discuss by breaking it into definition,
purpose, main parts, and practical examples.`
2133:     ].join("\n");
2134:     if (question.toLowerCase().includes(ownerName.toLowerCase())) {
2135:       return `If the question connects to Maurice's portfolio, I will use the saved project context and links
as evidence.`;
2136:     }
2137:     return `If it is broader than the portfolio, I will answer it as a general engineering or technology
concept first.`;
2138:   }
2139: }

```

```

2140:   if (!projects.length) return "The project catalog is still loading. Try again in a moment.";
2141:   if (!results.length) {
2142:     return [
2143:       `I could not find a strong portfolio match for "${question}".`,
2144:       "",
2145:       "Try asking about:",
2146:       "- A project name or category.",
2147:       "- A tool such as LTspice, KiCad, Vivado, STM32CubeIDE, or GitHub.",
2148:       "- A hardware topic such as op amps, PCB testing, VCOs, embedded firmware, FPGA, ASIC, or power
circuits."
2149:     ].join("\n");
2150:   }
2151:
2152:   const projectIds = [...new Set(results.map((item) => item.projectId).filter(Boolean))];
2153:   const matchedProjects = projectIds
2154:     .map((id) => projects.find((project) => project.id === id))
2155:     .filter(Boolean);
2156:   const projectNames = matchedProjects.map((project) => project.portfolioView?.title ||
project.title).slice(0, 3);
2157:   const fileCount = results.filter((item) => item.kind === "file").length;
2158:   const sectionCount = results.filter((item) => item.kind === "section" || item.kind ===
"subsection").length;
2159:   const pageCount = results.filter((item) => item.kind === "page").length;
2160:   const parts = [];
2161:
2162:   if (projectNames.length) {
2163:     parts.push(`The strongest portfolio match is ${projectNames.join(projectNames.length > 2 ? ", " : "
and ")}.`);
2164:   } else if (pageCount) {
2165:     parts.push("The strongest matches are in the profile, resume, contact, or portfolio information
sections.");
2166:   } else {
2167:     parts.push(`I found ${results.length} relevant portfolio item${results.length === 1 ? "" : "s"}.`);
2168:   }
2169:
2170:   parts.push("");
2171:   parts.push("What I found:");
2172:   if (sectionCount || fileCount) {
2173:     parts.push(`- ${sectionCount} section match${sectionCount === 1 ? "" : "es"}.`);
2174:     if (fileCount) parts.push(`- ${fileCount} file or link match${fileCount === 1 ? "" : "es"}.`);
2175:   }
2176:   if (pageCount) parts.push(`- ${pageCount} page or profile match${pageCount === 1 ? "" : "es"}.`);
2177:   if (projectNames.length) {
2178:     parts.push(`- Best project candidate${projectNames.length === 1 ? "" : "s"}: ${projectNames.join(",
")}.`);
2179:   }
2180:
2181:   const firstSnippet = searchSnippet(results[0].text, question);
2182:   if (firstSnippet) {
2183:     parts.push("");
2184:     parts.push("Why it matched:");
2185:     parts.push(firstSnippet);
2186:   }
2187:   return parts.join("\n");
2188: }

```

## assistantSourcesMarkup - lines 2190-2208

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2190-2208 (19 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantSourcesMarkup(results = []) {                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | slice, map, set, escapeHtml, assistantSourceLabel, join                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2190: function assistantSourcesMarkup(results = []) {
2191:   const sources = results.slice(0, 5);
2192:   if (!sources.length) return "";
2193:   const sourceButtons = sources.map((source) => {
2194:     const id = String(assistantSourceCounter++);
2195:     assistantSourceMap.set(id, source);
2196:     return `

```

```

2197:         <button type="button" data-ai-source-
id="${id}">${escapeHtml(assistantSourceLabel(source))}</button>
2198:     `;
2199:   }).join("");
2200:   return `
2201:     <div class="ai-source-panel">
2202:       <p class="ai-source-heading">Relevant portfolio links</p>
2203:       <div class="ai-source-list" aria-label="Assistant sources">
2204:         ${sourceButtons}
2205:       </div>
2206:     </div>
2207:   `;
2208: }

```

## setAssistantStatus - lines 2210-2214

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2210-2214 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function setAssistantStatus(message, state = "ready") {                                                                                                                                               |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2210: function setAssistantStatus(message, state = "ready") {
2211:   if (!aiAssistantStatus) return;
2212:   aiAssistantStatus.textContent = message;
2213:   aiAssistantStatus.dataset.state = state;
2214: }

```

## assistantDesktopDockedHeight - lines 2216-2221

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2216-2221 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function assistantDesktopDockedHeight() {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getComputedStyle, getPropertyValue, trim, endsWith, min, max                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2216: function assistantDesktopDockedHeight() {
2217:   if (!aiAssistantPanel) return 360;
2218:   const value = getComputedStyle(aiAssistantPanel).getPropertyValue("--ai-console-rest-height").trim();
2219:   if (value.endsWith("px")) return Number.parseFloat(value) || 360;
2220:   return Math.min(Math.max(window.innerHeight * 0.38, 340), 390);
2221: }

```

## updateAssistantPanelGrowth - lines 2223-2246



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2223-2246 (24 source lines).                                                                                                                                                                    |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateAssistantPanelGrowth() {                                                                                                                                                               |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | matchMedia, removeProperty, remove, assistantDesktopDockedHeight, getBoundingClientRect, querySelector, querySelectorAll, reduce, min, round, max, setProperty                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2223: function updateAssistantPanelGrowth() {
2224:   if (!aiAssistantPanel || !aiAssistantLog || !aiAssistantForm) return;
2225:   const isDesktop = window.matchMedia("(min-width: 901px)").matches;
2226:   if (!isDesktop) {
2227:     aiAssistantPanel.style.removeProperty("--ai-console-live-height");
2228:     aiAssistantPanel.classList.remove("ai-assistant-panel-expanded");
2229:     return;
2230:   }
2231:
2232:   const dockedHeight = assistantDesktopDockedHeight();
2233:   const formHeight = aiAssistantForm.getBoundingClientRect().height || 46;
2234:   const clearHeight = aiClearChatButton?.getBoundingClientRect().height || 38;
2235:   const titleHeight = document.querySelector("#ai-assistant-title")?.getBoundingClientRect().height ||
34;
2236:   const chromeHeight = titleHeight + formHeight + clearHeight + 92;
2237:   const messages = [...aiAssistantLog.querySelectorAll(".ai-message")];
2238:   const messageGap = messages.length > 1 ? (messages.length - 1) * 10 : 0;
2239:   const messageHeight = messages.reduce((sum, message) => sum + message.getBoundingClientRect().height,
0) + messageGap;
2240:   const contentHeight = messageHeight + chromeHeight;
2241:   const maxHeight = Math.min(window.innerHeight * 0.76, 720);
2242:   const nextHeight = Math.round(Math.max(dockedHeight, Math.min(contentHeight, maxHeight)));
2243:
2244:   aiAssistantPanel.style.setProperty("--ai-console-live-height", `${nextHeight}px`);
2245:   aiAssistantPanel.classList.toggle("ai-assistant-panel-expanded", nextHeight > dockedHeight + 12);
2246: }

```

## ***renderAssistantInline - lines 2248-2250***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2248-2250 (3 source lines).                                                                                                                                                                     |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderAssistantInline(value = "") {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | renderInlineMath, replace                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2248: function renderAssistantInline(value = "") {
2249:   return renderInlineMath(value).replace(/\^*\([^^\)]*\)/g, "<strong>$1</strong>");
2250: }

```

## ***renderAssistantAnswerContent - lines 2252-2304***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2252-2304 (53 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderAssistantAnswerContent(content = "") {                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replace, split, push, map, renderAssistantInline, join, escapeHtml, forEach, trim, match, flushCode, flushBullets                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 7 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2252: function renderAssistantAnswerContent(content = "") {
2253:   const lines = String(content || "").replace(/\r\n?/g, "\n").split("\n");
2254:   const html = [];
2255:   let bulletItems = [];
2256:   let codeBlock = null;
2257:   const flushBullets = () => {
2258:     if (!bulletItems.length) return;
2259:     html.push(`<ul>${bulletItems.map((item) =>
`<li>${renderAssistantInline(item)}</li>`).join("")}</ul>`);
2260:     bulletItems = [];
2261:   };
2262:   const flushCode = () => {
2263:     if (!codeBlock) return;
2264:     const language = codeBlock.language ? ` data-language="${escapeHtml(codeBlock.language)}" ` : "";
2265:     html.push(`<pre class="ai-code-
block"${language}><code>${escapeHtml(codeBlock.lines.join("\n"))}</code></pre>`);
2266:     codeBlock = null;
2267:   };
2268:
2269:   lines.forEach((line) => {
2270:     const trimmed = line.trim();
2271:     const fence = trimmed.match(/^````([a-z0-9_+.#-]*)\s*$/i);
2272:     if (fence) {
2273:       if (codeBlock) {
2274:         flushCode();
2275:       } else {
2276:         flushBullets();
2277:         codeBlock = { language: fence[1] || "", lines: [] };
2278:       }
2279:       return;
2280:     }
2281:     if (codeBlock) {
2282:       codeBlock.lines.push(line);
2283:       return;
2284:     }
2285:     if (!trimmed) {
2286:       flushBullets();
2287:       return;
2288:     }
2289:     const bullet = trimmed.match(/^[-*\u2022]\s+(.+)/u);
2290:     if (bullet) {
2291:       bulletItems.push(bullet[1]);
2292:       return;
2293:     }
2294:     flushBullets();
2295:     if (/^[^!?]{2,64}:$/.test(trimmed)) {
2296:       html.push(`<p><strong>${renderAssistantInline(trimmed.slice(0, -1))}</strong></p>`);
2297:     } else {
2298:       html.push(`<p>${renderAssistantInline(trimmed)}</p>`);
2299:     }
2300:   });
2301:   flushCode();
2302:   flushBullets();
2303:   return `<div class="ai-answer-content">${html.join("")} || "<p>I am ready.</p>"</div>`;
2304: }

```

## ***flushBullets - lines 2257-2261***

| Item            | Explanation                         |
|-----------------|-------------------------------------|
| Source location | Lines 2257-2261 (5 source lines).   |
| Purpose         | Part of the public website runtime. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | const flushBullets = () => {                                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | push, map, renderAssistantInline, join                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2257:   const flushBullets = () => {
2258:     if (!bulletItems.length) return;
2259:     html.push(`<ul>${bulletItems.map((item) =>
`<li>${renderAssistantInline(item)}</li>`).join("")}</ul>`);
2260:     bulletItems = [];
2261:   };

```

## ***flushCode - lines 2262-2267***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2262-2267 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | const flushCode = () => {                                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | escapeHtml, push, join                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2262:   const flushCode = () => {
2263:     if (!codeBlock) return;
2264:     const language = codeBlock.language ? ` data-language="${escapeHtml(codeBlock.language)}"` : "";
2265:     html.push(`<pre class="ai-code-
block"${language}><code>${escapeHtml(codeBlock.lines.join("\n"))}</code></pre>`);
2266:     codeBlock = null;
2267:   };

```

## ***appendAssistantMessage - lines 2306-2317***

| Item               | Explanation                                                                                                                        |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2306-2317 (12 source lines).                                                                                                 |
| Purpose            | Part of the public website runtime.                                                                                                |
| Declaration        | function appendAssistantMessage(role, content, sources = []) {                                                                     |
| Touches            | browser/Electron DOM, filesystem                                                                                                   |
| Calls detected     | createElement, renderAssistantAnswerContent, assistantSourcesMarkup, renderMultilineInlineText, append, updateAssistantPanelGrowth |
| API endpoints      | None detected inside this function body.                                                                                           |
| Storage keys       | No local/session storage keys detected.                                                                                            |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                      |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2306: function appendAssistantMessage(role, content, sources = []) {
2307:   if (!aiAssistantLog) return;
2308:   const article = document.createElement("article");
2309:   article.className = `ai-message ai-message-${role}`;
2310:   article.innerHTML = role === "assistant"
2311:     ? `${renderAssistantAnswerContent(content)}${assistantSourcesMarkup(sources)}`
2312:     : `<p>${renderMultilineInlineText(content)}</p>`;
2313:   aiAssistantLog.append(article);
2314:   updateAssistantPanelGrowth();
2315:   aiAssistantLog.scrollTop = aiAssistantLog.scrollHeight;
2316:   return article;
2317: }

```

## ***scrollAssistantAnswerToStart - lines 2319-2329***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2319-2329 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function scrollAssistantAnswerToStart(article) {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | max, requestAnimationFrame                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2319: function scrollAssistantAnswerToStart(article) {
2320:   if (!aiAssistantLog || !article) return;
2321:   const scrollToAnswer = () => {
2322:     aiAssistantLog.scrollTop = Math.max(0, article.offsetTop - aiAssistantLog.offsetTop - 4);
2323:   };
2324:   if (typeof window.requestAnimationFrame === "function") {
2325:     window.requestAnimationFrame(scrollToAnswer);
2326:   } else {
2327:     window.setTimeout(scrollToAnswer, 0);
2328:   }
2329: }

```

## ***scrollToAnswer - lines 2321-2323***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2321-2323 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | const scrollToAnswer = () => {                                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | max                                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2321:   const scrollToAnswer = () => {
2322:     aiAssistantLog.scrollTop = Math.max(0, article.offsetTop - aiAssistantLog.offsetTop - 4);
2323:   };

```

## ***appendAssistantPendingMessage - lines 2331-2335***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2331-2335 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function appendAssistantPendingMessage() {                                                                                                                                                            |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | appendAssistantMessage, add                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2331: function appendAssistantPendingMessage() {
2332:   const article = appendAssistantMessage("assistant", "Thinking", []);
2333:   article?.classList.add("ai-message-pending");
2334:   return article;
2335: }

```

## ***replaceAssistantMessage - lines 2337-2346***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2337-2346 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function replaceAssistantMessage(article, content, sources = []) {                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | appendAssistantMessage, renderAssistantAnswerContent, assistantSourcesMarkup, updateAssistantPanelGrowth, scrollAssistantAnswerToStart                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2337: function replaceAssistantMessage(article, content, sources = []) {
2338:   if (!article) {
2339:     appendAssistantMessage("assistant", content, sources);
2340:     return;
2341:   }
2342:   article.className = "ai-message ai-message-assistant";
2343:   article.innerHTML = `${renderAssistantAnswerContent(content)}${assistantSourcesMarkup(sources)}`;
2344:   updateAssistantPanelGrowth();
2345:   scrollAssistantAnswerToStart(article);
2346: }

```

## ***setAssistantBusy - lines 2348-2352***

| Item            | Explanation                         |
|-----------------|-------------------------------------|
| Source location | Lines 2348-2352 (5 source lines).   |
| Purpose         | Part of the public website runtime. |
| Declaration     | function setAssistantBusy(isBusy) { |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | querySelector                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2348: function setAssistantBusy(isBusy) {
2349:   const submitButton = aiAssistantForm?.querySelector('button[type="submit"]');
2350:   if (aiAssistantInput) aiAssistantInput.disabled = isBusy;
2351:   if (submitButton) submitButton.disabled = isBusy;
2352: }

```

### ***clearAssistantChat - lines 2354-2363***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2354-2363 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function clearAssistantChat() {                                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Map, setAssistantStatus, updateAssistantPanelGrowth, focus                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2354: function clearAssistantChat() {
2355:   if (!aiAssistantLog) return;
2356:   assistantSourceCounter = 0;
2357:   assistantSourceMap = new Map();
2358:   assistantChatHistory = [];
2359:   aiAssistantLog.innerHTML = "";
2360:   setAssistantStatus("Chat cleared");
2361:   updateAssistantPanelGrowth();
2362:   aiAssistantInput?.focus();
2363: }

```

### ***assistantRemoteTimeoutMs - lines 2365-2374***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2365-2374 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function assistantRemoteTimeoutMs(question = "", context = {}, options = {}) {                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, isArray, some, toLowerCase, b                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2365: function assistantRemoteTimeoutMs(question = "", context = {}, options = {}) {
2366:   const clean = normalize(question);
2367:   const sourceFetches = Array.isArray(context.sourceFetches) ? context.sourceFetches : [];
2368:   const usesGitHub = sourceFetches.some((source) => String(source.kind || "").toLowerCase() === "github"
|| /github\.com/i.test(source.url || ""));
2369:   if (usesGitHub || /\b(github|repo|repository|repositories|source\s+code|code|snippet|firmware|verilog|s
ystemverilog|python|javascript|circuit\s+file)\b/.test(clean)) {
2370:     return 45000;
2371:   }
2372:   if (options.allowWebSearch) return 30000;
2373:   return 18000;
2374: }
```

## ***askRemoteAssistant - lines 2376-2411***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2376-2411 (36 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function askRemoteAssistant(question, context, options = {}) {                                                                                                                                  |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | assistantEndpoint, AbortController, abort, assistantRemoteTimeoutMs, fetch, stringify, Error, json, trim                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2376: async function askRemoteAssistant(question, context, options = {}) {
2377:   const endpoint = assistantEndpoint();
2378:   if (!endpoint) return null;
2379:   const controller = new AbortController();
2380:   let didTimeout = false;
2381:   const timer = setTimeout(() => {
2382:     didTimeout = true;
2383:     controller.abort();
2384:   }, assistantRemoteTimeoutMs(question, context, options));
2385:   let response;
2386:   try {
2387:     response = await fetch(endpoint, {
2388:       method: "POST",
2389:       headers: { "Content-Type": "application/json" },
2390:       signal: controller.signal,
2391:       body: JSON.stringify({
2392:         context,
2393:         conversation: options.conversation || [],
2394:         intent: options.intent || context.intent || "portfolio_specific",
2395:         allowWebSearch: Boolean(options.allowWebSearch),
2396:         question,
2397:         mode: "portfolio-plus-general-knowledge",
2398:         responseType: "chatgpt-like-answer-first-links-second"
2399:       })
2400:     });
2401:   } catch (error) {
2402:     if (didTimeout) throw new Error("AI request timed out while reading public sources.");
2403:     throw error;
2404:   } finally {
2405:     clearTimeout(timer);
2406:   }
2407:   if (!response.ok) throw new Error("AI service unavailable");
2408:   const data = await response.json();
2409:   const answer = String(data.answer || data.message || "").trim();
2410:   return answer ? { answer, model: data.model || "", usedWebSearch: Boolean(data.usedWebSearch) } : null;
2411: }
```

## ***answerAssistantQuestion - lines 2413-2458***

| Item            | Explanation                         |
|-----------------|-------------------------------------|
| Source location | Lines 2413-2458 (46 source lines).  |
| Purpose         | Part of the public website runtime. |

| Item               | Explanation                                                                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | async function answerAssistantQuestion(question = "") {                                                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                                                                               |
| Calls detected     | trim, slice, appendAssistantMessage, assistantRemember, appendAssistantPendingMessage, setAssistantBusy, setAssistantStatus, assistantEndpoint, assistantQuestionIntent, assistantFallbackResults, assistantSourcesForDisplay, assistantContextForQuestion |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                                   |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                                    |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                                                                              |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                                      |

Source excerpt:

```

2413: async function answerAssistantQuestion(question = "") {
2414:   const cleanQuestion = String(question || "").trim();
2415:   if (!cleanQuestion) return;
2416:   const priorConversation = assistantChatHistory.slice();
2417:   appendAssistantMessage("user", cleanQuestion);
2418:   assistantRemember("user", cleanQuestion);
2419:   const pendingMessage = appendAssistantPendingMessage();
2420:   setAssistantBusy(true);
2421:   try {
2422:     setAssistantStatus(assistantEndpoint() ? "Composing AI answer" : "Composing local answer",
"working");
2423:
2424:     const intent = assistantQuestionIntent(cleanQuestion, priorConversation);
2425:     const rawSources = assistantFallbackResults(cleanQuestion, intent);
2426:     const sources = assistantSourcesForDisplay(cleanQuestion, rawSources, intent);
2427:     const context = assistantContextForQuestion(cleanQuestion, intent, rawSources);
2428:     const conversation = assistantConversationForRequest(cleanQuestion, priorConversation);
2429:     try {
2430:       const remoteAnswer = await askRemoteAssistant(cleanQuestion, context, {
2431:         allowWebSearch: assistantQuestionAllowsPublicLookup(cleanQuestion, intent),
2432:         conversation,
2433:         intent
2434:       });
2435:       if (remoteAnswer?.answer) {
2436:         replaceAssistantMessage(pendingMessage, remoteAnswer.answer, sources);
2437:         assistantRemember("assistant", remoteAnswer.answer);
2438:         setAssistantStatus(remoteAnswer.model ? `AI answer ready: ${remoteAnswer.model}` : "AI answer
ready");
2439:         return;
2440:       }
2441:     } catch (error) {
2442:       setAssistantStatus(error?.message || "Local answer used", "error");
2443:     }
2444:
2445:     const fallbackAnswer = assistantLocalAnswer(cleanQuestion, sources, intent);
2446:     replaceAssistantMessage(pendingMessage, fallbackAnswer, sources);
2447:     assistantRemember("assistant", fallbackAnswer);
2448:     if (!assistantEndpoint()) setAssistantStatus("Local portfolio intelligence");
2449:   } catch {
2450:     const errorAnswer = "I had trouble processing that question. Try rephrasing it with a project name,
tool name, or electronics topic.";
2451:     replaceAssistantMessage(pendingMessage, errorAnswer, []);
2452:     assistantRemember("assistant", errorAnswer);
2453:     setAssistantStatus("Assistant error", "error");
2454:   } finally {
2455:     setAssistantBusy(false);
2456:     aiAssistantInput?.focus();
2457:   }
2458: }

```

## ***linkAttributes - lines 2459-2462***

| Item            | Explanation                               |
|-----------------|-------------------------------------------|
| Source location | Lines 2459-2462 (4 source lines).         |
| Purpose         | Part of the public website runtime.       |
| Declaration     | function linkAttributes(url, item = {}) { |
| Touches         | filesystem                                |
| Calls detected  | normalizeLinkTarget, isWebsiteLinkItem    |
| API endpoints   | None detected inside this function body.  |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2459: function linkAttributes(url, item = {}) {
2460:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(item, url) });
2461:   return /^https?:\/\//i.test(target) ? ' target="_blank" rel="noreferrer"' : "";
2462: }

```

## ***isLocalDownloadTarget - lines 2464-2467***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2464-2467 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function isLocalDownloadTarget(target = "") {                                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeLinkTarget                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2464: function isLocalDownloadTarget(target = "") {
2465:   const value = normalizeLinkTarget(target, { assumeWeb: true });
2466:   return Boolean(value) && !/^(https?:)?\/\//i.test(value) && !/^(mailto:|tel:|#)/i.test(value);
2467: }

```

## ***downloadAttribute - lines 2469-2472***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2469-2472 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function downloadAttribute(target = "", item = {}) {                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeLinkTarget, isWebsiteLinkItem, isLocalDownloadTarget                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2469: function downloadAttribute(target = "", item = {}) {
2470:   const value = normalizeLinkTarget(target, { assumeWeb: isWebsiteLinkItem(item, target) });
2471:   return isLocalDownloadTarget(value) ? " download" : "";
2472: }

```

## ***fileNameFromUrl - lines 2474-2481***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2474-2481 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function fileNameFromUrl(url = "") {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, pop                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2474: function fileNameFromUrl(url = "") {
2475:     const clean = String(url || "").split(/[?#]/)[0].split("/").pop() || "Download file";
2476:     try {
2477:         return decodeURIComponent(clean);
2478:     } catch {
2479:         return clean;
2480:     }
2481: }

```

## ***resourceLink - lines 2483-2501***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2483-2501 (19 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function resourceLink(item = {}, label = item.label    item.title    item.name    fileNameFromUrl(item.url    item.artifact)) {                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | fileNameFromUrl, normalizeLinkTarget, isWebsiteLinkItem, escapeHtml, linkAttributes, downloadAttribute                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2483: function resourceLink(item = {}, label = item.label || item.title || item.name ||
fileNameFromUrl(item.url || item.artifact)) {
2484:     const rawTarget = item.url || item.artifact || item.href || item.file || item.path || item.src || "";
2485:     const target = normalizeLinkTarget(rawTarget, { assumeWeb: isWebsiteLinkItem(item, rawTarget) });
2486:
2487:     if (!target || item.status === "planned") {
2488:         return `<span class="resource-link muted">${escapeHtml(label || "Planned file")}</span>`;
2489:     }
2490:
2491:     return `
2492:     <a
2493:         class="resource-link"
2494:         href="${escapeHtml(target)}"
2495:         ${linkAttributes(target, item)}
2496:         ${downloadAttribute(target, item)}
2497:     >
2498:         ${escapeHtml(label || fileNameFromUrl(target))}
2499:     </a>
2500: `;
2501: }

```

## ***pillList - lines 2503-2510***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2503-2510 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function pillList(items, className = "") {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, join                                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2503: function pillList(items, className = "") {
2504:   return (items || [])
2505:     .map((item) => {
2506:       const label = typeof item === "string" ? item : item.name || item.title || item.label || "";
2507:       return label ? `

```

## ***evidenceList - lines 2512-2518***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2512-2518 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function evidenceList(items, renderItem, emptyMessage) {                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, join                                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2512: function evidenceList(items, renderItem, emptyMessage) {
2513:   if (!items || !items.length) {
2514:     return `
```

## ***detailBlock - lines 2520-2527***

| Item            | Explanation                                       |
|-----------------|---------------------------------------------------|
| Source location | Lines 2520-2527 (8 source lines).                 |
| Purpose         | Part of the public website runtime.               |
| Declaration     | function detailBlock(title, className, content) { |
| Touches         | Mostly local calculations or local UI state.      |
| Calls detected  | No obvious named calls detected by the generator. |
| API endpoints   | None detected inside this function body.          |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2520: function detailBlock(title, className, content) {
2521:   return `
2522:     <details class="${className}">
2523:       <summary>${title}</summary>
2524:       ${content}
2525:     </details>
2526:   `;
2527: }

```

## mediaGrid - lines 2529-2549

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2529-2549 (21 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function mediaGrid(items) {                                                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | map, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, join                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2529: function mediaGrid(items) {
2530:   if (!items || !items.length) {
2531:     return `<p class="evidence-empty">No project images have been added yet.</p>`;
2532:   }
2533:
2534:   return `
2535:     <div class="media-grid">
2536:       ${items.map((item) => `
2537:         <figure>
2538:           <a href="${escapeHtml(normalizeLinkTarget(item.url, { assumeweb: isWebsiteLinkItem(item,
item.url) })))}"${linkAttributes(item.url, item)}>
2539:             
2540:           </a>
2541:           <figcaption>
2542:             <strong>${escapeHtml(item.title)}</strong>
2543:             <span>${escapeHtml(item.caption || "")}</span>
2544:           </figcaption>
2545:         </figure>
2546:       `).join("")}
2547:     </div>
2548:   `;
2549: }

```

## siteSectionHasContent - lines 2551-2560

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 2551-2560 (10 source lines).           |
| Purpose         | Part of the public website runtime.          |
| Declaration     | function siteSectionHasContent(section) {    |
| Touches         | Mostly local calculations or local UI state. |
| Calls detected  | some                                         |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2551: function siteSectionHasContent(section) {
2552:   return Boolean(
2553:     section?.description ||
2554:     section?.richDescription?.blocks?.length ||
2555:     section?.backgroundImage ||
2556:     (section?.links || []).some((link) => link.label || link.url) ||
2557:     (section?.assets || []).some((asset) => asset.title || asset.url) ||
2558:     (section?.subsections || []).some((item) => item.title || item.description ||
item.richDescription?.blocks?.length || (item.links || []).length)
2559:   );
2560: }

```

### ***siteSectionRenderable - lines 2562-2564***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2562-2564 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function siteSectionRenderable(section) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | siteSectionHasContent                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2562: function siteSectionRenderable(section) {
2563:   return section?.visible !== false && siteSectionHasContent(section);
2564: }

```

### ***renderDynamicLinks - lines 2566-2570***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2566-2570 (5 source lines).                                                                                                                                                                     |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderDynamicLinks(items = []) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, map, resourceLink, join                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2566: function renderDynamicLinks(items = []) {
2567:   const links = items.filter((item) => item?.url && (item.label || item.title));
2568:   if (!links.length) return "";

```

```

2569:   return `<div class="resource-list dynamic-link-list">${links.map((item) => resourceLink(item,
item.label || item.title || "Open")).join("")}</div>`;
2570: }

```

## renderSiteSections - lines 2572-2609

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2572-2609 (38 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderSiteSections() {                                                                                                                                                                       |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, filter, map, url, escapeHtml, renderRichContent, renderDynamicLinks, join                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2572: function renderSiteSections() {
2573:   const mount = document.querySelector("#dynamic-sections");
2574:   if (!mount) return;
2575:   const visibleSections = (siteSections || []).filter(siteSectionRenderable);
2576:   mount.innerHTML = visibleSections.map((section) => {
2577:     const style = section.backgroundImage ? ` style="--section-bg:
url('${escapeHtml(section.backgroundImage)}')` : "";
2578:     const links = [...(section.links || []), ...(section.assets || [])];
2579:     const subsections = (section.subsections || []).filter((item) => item.title || item.description ||
item.richDescription?.blocks?.length || (item.links || []).length);
2580:     return `
2581:       <section class="section dynamic-section" id="${escapeHtml(section.id || "")}"${style}>
2582:         <div class="dynamic-section-surface">
2583:           <div class="section-heading">
2584:             <div>
2585:               <h2>${escapeHtml(section.title || "Untitled section")}</h2>
2586:             </div>
2587:           </div>
2588:           ${section.description || section.richDescription?.blocks?.length
? `<div class="dynamic-section-copy">${renderRichContent(section.richDescription,
section.description || "")}</div>`
2590:             : ""}
2591:           ${subsections.length ? `
2592:             <div class="dynamic-section-grid">
2593:               ${subsections.map((item) => `
2594:                 <article class="dynamic-section-card">
2595:                   <h3>${escapeHtml(item.title || "Untitled")}</h3>
2596:                   ${item.description || item.richDescription?.blocks?.length
? renderRichContent(item.richDescription, item.description || "")
2598:                     : ""}
2599:                   ${renderDynamicLinks(item.links || [])}
2600:                 </article>
2601:               `).join("")}
2602:             </div>
2603:             : ""}
2604:             ${renderDynamicLinks(links)}
2605:           </div>
2606:         </section>
2607:       `;
2608:   }).join("");
2609: }

```

## subsections - lines 2579-2581

| Item            | Explanation                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 2579-2581 (3 source lines).                                                                                                                                      |
| Purpose         | Part of the public website runtime.                                                                                                                                    |
| Declaration     | const subsections = (section.subsections    []).filter((item) => item.title    item.description    item.richDescription?.blocks?.length    (item.links    []).length); |
| Touches         | Mostly local calculations or local UI state.                                                                                                                           |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | filter, escapeHtml                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2579:     const subsections = (section.subsections || []).filter((item) => item.title || item.description ||
item.richDescription?.blocks?.length || (item.links || []).length);
2580:     return `
2581:       <section class="section dynamic-section" id="${escapeHtml(section.id || "")}"${style}>

```

## ***customSectionBlocks - lines 2611-2622***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2611-2622 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function customSectionBlocks(project) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, detailBlock, evidenceList, resourceLink, join                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2611: function customSectionBlocks(project) {
2612:   return (project.sections || []).map((section) => detailBlock(section.title, "evidence-block evidence-
wide", `
2613:     ${section.description ? `<p class="evidence-empty">${section.description}</p>` : ""}
2614:     ${evidenceList(section.items || [], (item) => `
2615:       <li>
2616:         ${item.url ? resourceLink(item, item.title) : `<strong>${item.title}</strong>`}
2617:         <span>${item.type || "Section item"} &middot; ${item.status || "tracked"}</span>
2618:         ${item.description ? `<p>${item.description}</p>` : ""}
2619:       </li>
2620:     `, "No content has been added yet.")]
2621:   `)).join("");
2622: }

```

## ***itemUrl - lines 2623-2625***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2623-2625 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function itemUrl(item = {}) {                                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2623: function itemUrl(item = {}) {  
2624:     return item.url || item.artifact || item.href || item.file || item.path || item.src || "";  
2625: }
```

### ***itemLabel - lines 2627-2630***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2627-2630 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function itemLabel(item = {}, fallback = "Download file") {                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | itemUrl, fileNameFromUrl                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2627: function itemLabel(item = {}, fallback = "Download file") {  
2628:     const url = itemUrl(item);  
2629:     return item.title || item.label || item.name || item.caption || fileNameFromUrl(url) || fallback;  
2630: }
```

### ***normalizeDownloadAsset - lines 2632-2643***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2632-2643 (12 source lines).                                                                                                                                                                    |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeDownloadAsset(item = {}, fallbackType = "File") {                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | itemUrl, itemLabel                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2632: function normalizeDownloadAsset(item = {}, fallbackType = "File") {  
2633:     const url = itemUrl(item);  
2634:     if (!url || item.status === "planned") return null;  
2635:     return {  
2636:         title: itemLabel(item),  
2637:         url,  
2638:         status: item.status || "uploaded",  
2639:         type: item.type || item.kind || item.meta || fallbackType,  
2640:         description: item.description || item.caption || item.summary || ""  
2641:     };  
2642: }  
2643: }
```

### ***collectRichDownloads - lines 2645-2659***

| Item            | Explanation                         |
|-----------------|-------------------------------------|
| Source location | Lines 2645-2659 (15 source lines).  |
| Purpose         | Part of the public website runtime. |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function collectRichDownloads(rich, fallbackType = "File", output = []) {                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | forEach, push, cleanRichImageTitle, fileNameFromUrl                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2645: function collectRichDownloads(rich, fallbackType = "File", output = []) {
2646:   (rich?.blocks || []).forEach((block) => {
2647:     if (!block?.url) return;
2648:
2649:     output.push({
2650:       title: cleanRichImageTitle(block) || block.title || block.caption ||
fileNameFromUrl(block.url),
2651:       url: block.url,
2652:       status: "uploaded",
2653:       type: block.type === "image" ? "Image" : fallbackType,
2654:       description: block.caption || ""
2655:     });
2656:   });
2657:
2658:   return output;
2659: }

```

## ***collectDownloadsFromValue - lines 2661-2689***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2661-2689 (29 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function collectDownloadsFromValue(value, fallbackType = "File", output = []) {                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isArray, forEach, normalizeDownloadAsset, push, collectRichDownloads                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2661: function collectDownloadsFromValue(value, fallbackType = "File", output = []) {
2662:   if (!value) return output;
2663:
2664:   if (Array.isArray(value)) {
2665:     value.forEach((item) => collectDownloadsFromValue(item, fallbackType, output));
2666:     return output;
2667:   }
2668:
2669:   if (typeof value !== "object") return output;
2670:
2671:   const asset = normalizeDownloadAsset(value, fallbackType);
2672:   if (asset) output.push(asset);
2673:
2674:   collectRichDownloads(value.rich, fallbackType, output);
2675:   collectRichDownloads(value.summaryRich, fallbackType, output);
2676:
2677:   [
2678:     "files",
2679:     "assets",
2680:     "items",
2681:     "children",
2682:     "links",
2683:     "results",
2684:     "references",

```

```

2685:         "mathAnalysis"
2686:     ].forEach((key) => collectDownloadsFromValue(value[key], fallbackType, output));
2687:
2688:     return output;
2689: }

```

## uniqueDownloads - lines 2691-2700

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2691-2700 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function uniqueDownloads(items = []) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Set, filter, has, add                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2691: function uniqueDownloads(items = []) {
2692:     const seen = new Set();
2693:
2694:     return items.filter((item) => {
2695:         const key = `${item.url}:${item.title}`;
2696:         if (!item.url || seen.has(key)) return false;
2697:         seen.add(key);
2698:         return true;
2699:     });
2700: }

```

## renderDownloadBlock - lines 2702-2717

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2702-2717 (16 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderDownloadBlock(title, items = []) {                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | uniqueDownloads, detailBlock, escapeHtml, map, resourceLink, renderMultilineInlineText, join                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2702: function renderDownloadBlock(title, items = []) {
2703:     const downloads = uniqueDownloads(items);
2704:     if (!downloads.length) return "";
2705:
2706:     return detailBlock(escapeHtml(title), "evidence-block evidence-wide download-evidence-block", `
2707:     <ul class="download-list">
2708:         ${downloads.map((item) => `
2709:             <li>
2710:                 ${resourceLink(item, item.title)}
2711:                 <span>${escapeHtml(item.type || "File")} &middot; ${escapeHtml(item.status ||
"uploaded")}</span>
2712:                 ${item.description ? `<p>${renderMultilineInlineText(item.description)}</p>` : ""}
2713:             </li>
2714:         `).join("")}
2715:     </ul>
2716:     `);
2717: }

```

## ***renderCollectedDownloadSections - lines 2719-2760***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2719-2760 (42 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderCollectedDownloadSections(project) {                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | uniqueDownloads, collectDownloadsFromValue, push, renderDownloadBlock, addBlock, forEach, displayTitle, filter, join                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2719: function renderCollectedDownloadSections(project) {
2720:   const blocks = [];
2721:
2722:   const addBlock = (title, source, fallbackType = "File") => {
2723:     const downloads = uniqueDownloads(collectDownloadsFromValue(source, fallbackType, []));
2724:     if (downloads.length) blocks.push(renderDownloadBlock(title, downloads));
2725:   };
2726:
2727:   addBlock("Overview downloads", [
2728:     { summaryRich: project.summaryRich },
2729:     project.design?.brief?.files,
2730:     project.design?.brief?.summaryRich
2731:   ], "Overview file");
2732:
2733:   addBlock("Design downloads", [
2734:     project.design?.documentation?.files,
2735:     project.design?.documentation?.references,
2736:     project.design?.documentation?.mathAnalysis
2737:   ], "Design file");
2738:
2739:   addBlock("Simulation downloads", [
2740:     project.design?.simulation?.files,
2741:     project.design?.simulation?.results
2742:   ], "Simulation file");
2743:
2744:   addBlock("Project documents", project.documents, "Document");
2745:   addBlock("Test artifacts", project.tests, "Test artifact");
2746:   addBlock("PCB artifacts", project.pcb, "PCB artifact");
2747:   addBlock("Images and media", project.media, "Image");
2748:
2749:   (project.sections || []).forEach((section) => {
2750:     addBlock(`${displayTitle(section.title, "Custom section")} downloads`, section.items ||
section.files || [], section.title || "Section file");
2751:   });
2752:
2753:   (project.portfolioView?.sections || []).
2754:     .filter((section) => section.id !== "brief")
2755:     .forEach((section) => {
2756:       addBlock(`${displayTitle(section.title, "Section")} downloads`, section.items || [],
section.title || "Section file");
2757:     });
2758:
2759:   return blocks.join("");
2760: }
```

## ***renderParsedBriefBlock - lines 2761-2771***

| Item            | Explanation                                                      |
|-----------------|------------------------------------------------------------------|
| Source location | Lines 2761-2771 (11 source lines).                               |
| Purpose         | Renders UI, markup, or display output.                           |
| Declaration     | function renderParsedBriefBlock(section, fallbackSummary = "") { |
| Touches         | Mostly local calculations or local UI state.                     |
| Calls detected  | renderRichContent                                                |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2761: function renderParsedBriefBlock(section, fallbackSummary = "") {
2762:   const briefItem = section?.items?.[0] || {};
2763:   const briefText = briefItem.description || fallbackSummary || "";
2764:   if (!briefItem.rich?.blocks?.length && !briefText) return "";
2765:   return
2766:     <details class="project-brief-default parsed-summary" open>
2767:       <summary>Overview</summary>
2768:       ${renderRichContent(briefItem.rich, briefText)}
2769:     </details>
2770:   `;
2771: }

```

## ***pathToString - lines 2773-2775***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2773-2775 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function pathToString(path = []) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | join                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2773: function pathToString(path = []) {
2774:   return path.join(".");
2775: }

```

## ***pathFromString - lines 2777-2780***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2777-2780 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function pathFromString(value = "") {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, map, filter, isInteger                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2777: function pathFromString(value = "") {
2778:   if (!value) return [];
2779:   return value.split(".").map((item) => Number(item)).filter((item) => Number.isInteger(item));

```

```
2780: }
```

## ***sectionRouteState - lines 2782-2789***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2782-2789 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sectionRouteState(projectId, sectionIndex, resourcePath = "") {                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2782: function sectionRouteState(projectId, sectionIndex, resourcePath = "") {
2783:   return {
2784:     appwindow: sectionRouteKey,
2785:     projectId: String(projectId || ""),
2786:     resourcePath: String(resourcePath || ""),
2787:     sectionIndex: String(sectionIndex ?? "")
2788:   };
2789: }
```

## ***canUseSectionHistory - lines 2791-2793***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2791-2793 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function canUseSectionHistory() {                                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2791: function canUseSectionHistory() {
2792:   return Boolean(window.history && typeof window.history.pushState === "function" && typeof
window.history.replaceState === "function");
2793: }
```

## ***sectionBaseUrl - lines 2795-2799***

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 2795-2799 (5 source lines).            |
| Purpose         | Part of the public website runtime.          |
| Declaration     | function sectionBaseUrl() {                  |
| Touches         | Mostly local calculations or local UI state. |
| Calls detected  | URL, values, forEach, delete                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2795: function sectionBaseUrl() {
2796:   const url = new URL(window.location.href);
2797:   Object.values(sectionRouteParams).forEach((param) => url.searchParams.delete(param));
2798:   return url;
2799: }

```

## ***sectionRouteUrl - lines 2801-2809***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2801-2809 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sectionRouteUrl(projectId, sectionIndex, resourcePath = "") {                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | sectionBaseUrl, set, delete                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2801: function sectionRouteUrl(projectId, sectionIndex, resourcePath = "") {
2802:   const url = sectionBaseUrl();
2803:   url.searchParams.set(sectionRouteParams.project, String(projectId || ""));
2804:   url.searchParams.set(sectionRouteParams.section, String(sectionIndex ?? ""));
2805:   if (resourcePath) url.searchParams.set(sectionRouteParams.path, String(resourcePath));
2806:   else url.searchParams.delete(sectionRouteParams.path);
2807:   url.hash = "projects";
2808:   return url;
2809: }

```

## ***sectionRouteFromState - lines 2811-2819***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2811-2819 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sectionRouteFromState(state) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2811: function sectionRouteFromState(state) {
2812:   if (!state || state.appWindow !== sectionRouteKey) return null;
2813:   if (!state.projectId || state.sectionIndex === "") return null;

```

```

2814:   return {
2815:     projectId: String(state.projectId),
2816:     resourcePath: String(state.resourcePath || ""),
2817:     sectionIndex: String(state.sectionIndex)
2818:   };
2819: }

```

## sectionRouteFromLocation - lines 2821-2831

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2821-2831 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sectionRouteFromLocation() {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | URLSearchParams, get                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2821: function sectionRouteFromLocation() {
2822:   const params = new URLSearchParams(window.location.search);
2823:   const projectId = params.get(sectionRouteParams.project);
2824:   const sectionIndex = params.get(sectionRouteParams.section);
2825:   if (!projectId || sectionIndex === null) return null;
2826:   return {
2827:     projectId,
2828:     resourcePath: params.get(sectionRouteParams.path) || "",
2829:     sectionIndex
2830:   };
2831: }

```

## sameSectionRoute - lines 2833-2838

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2833-2838 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sameSectionRoute(left, right) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2833: function sameSectionRoute(left, right) {
2834:   return Boolean(left && right &&
2835:     left.projectId === right.projectId &&
2836:     String(left.sectionIndex) === String(right.sectionIndex) &&
2837:     String(left.resourcePath || "") === String(right.resourcePath || ""));
2838: }

```

## syncSectionRouteToHistory - lines 2840-2851

| Item            | Explanation                        |
|-----------------|------------------------------------|
| Source location | Lines 2840-2851 (12 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Keeps two places aligned, such as local data and target files.                                                                                                                                        |
| Declaration        | function syncSectionRouteToHistory(projectId, sectionIndex, resourcePath = "", mode = "push") {                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | canUseSectionHistory, sectionRouteState, sectionRouteUrl, sectionRouteFromState, sectionRouteFromLocation, sameSectionRoute                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2840: function syncSectionRouteToHistory(projectId, sectionIndex, resourcePath = "", mode = "push") {
2841:   if (isApplyingSectionRoute || mode === "none" || !canUseSectionHistory()) return;
2842:   const route = sectionRouteState(projectId, sectionIndex, resourcePath);
2843:   const url = sectionRouteUrl(projectId, sectionIndex, resourcePath);
2844:   const currentRoute = sectionRouteFromState(history.state) || sectionRouteFromLocation();
2845:   const method = mode === "replace" || sameSectionRoute(currentRoute, route) ? "replaceState" :
"pushState";
2846:   try {
2847:     history[method](route, "", url);
2848:   } catch {
2849:     // Preview iframes and restricted browser contexts may reject history writes.
2850:   }
2851: }

```

### ***clearSectionRouteInHistory - lines 2853-2860***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2853-2860 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function clearSectionRouteInHistory() {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | canUseSectionHistory, replaceState, sectionBaseUrl                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2853: function clearSectionRouteInHistory() {
2854:   if (isApplyingSectionRoute || !canUseSectionHistory()) return;
2855:   try {
2856:     history.replaceState({ appWindow: "portfolio-base" }, "", sectionBaseUrl());
2857:   } catch {
2858:     // The visible dialog is already closed; URL cleanup is best effort.
2859:   }
2860: }

```

### ***initializeSectionHistoryState - lines 2862-2871***

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 2862-2871 (10 source lines).           |
| Purpose         | Part of the public website runtime.          |
| Declaration     | function initializeSectionHistoryState() {   |
| Touches         | Mostly local calculations or local UI state. |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | canUseSectionHistory, sectionRouteFromLocation, sectionRouteFromState, replaceState, sectionRouteState, sectionRouteUrl                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2862: function initializeSectionHistoryState() {
2863:   if (!canUseSectionHistory()) return;
2864:   const route = sectionRouteFromLocation() || sectionRouteFromState(history.state);
2865:   try {
2866:     if (route) history.replaceState(sectionRouteState(route.projectId, route.sectionIndex,
route.resourcePath), "", sectionRouteUrl(route.projectId, route.sectionIndex, route.resourcePath));
2867:     else if (!history.state?.appwindow) history.replaceState({ appwindow: "portfolio-base" }, "",
window.location.href);
2868:   } catch {
2869:     // Browser history state is progressive enhancement for the portfolio windows.
2870:   }
2871: }

```

## ***nodeAtPath - lines 2873-2882***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2873-2882 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function nodeAtPath(section, path = []) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2873: function nodeAtPath(section, path = []) {
2874:   let node = section;
2875:   let children = section?.items || [];
2876:   for (const index of path) {
2877:     node = children[index];
2878:     if (!node) return null;
2879:     children = node.children || [];
2880:   }
2881:   return node;
2882: }

```

## ***nodeChildren - lines 2884-2886***

| Item            | Explanation                                       |
|-----------------|---------------------------------------------------|
| Source location | Lines 2884-2886 (3 source lines).                 |
| Purpose         | Part of the public website runtime.               |
| Declaration     | function nodeChildren(node) {                     |
| Touches         | Mostly local calculations or local UI state.      |
| Calls detected  | No obvious named calls detected by the generator. |
| API endpoints   | None detected inside this function body.          |
| Storage keys    | No local/session storage keys detected.           |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2884: function nodeChildren(node) {
2885:   return node?.items || node?.children || [];
2886: }

```

## ***richHasRenderableContent - lines 2888-2897***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2888-2897 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function richHasRenderableContent(rich) {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | some                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2888: function richHasRenderableContent(rich) {
2889:   return Boolean(rich?.blocks?.some((block) =>
2890:     block.type === "image" && block.url ||
2891:     block.type === "formula" && block.formula ||
2892:     block.type === "code" && block.code ||
2893:     block.text ||
2894:     block.title ||
2895:     block.caption
2896:   ));
2897: }

```

## ***nodeHasRenderableContent - lines 2899-2910***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2899-2910 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function nodeHasRenderableContent(node) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | richHasRenderableContent, nodeChildren, some, includes                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2899: function nodeHasRenderableContent(node) {
2900:   if (!node) return false;
2901:   if (node.kind === "summary") return Boolean(node.description || richHasRenderableContent(node.rich));
2902:   const children = nodeChildren(node);
2903:   if (node.kind === "subsection") {
2904:     return Boolean(node.description || node.url || richHasRenderableContent(node.rich) ||
children.some(nodeHasRenderableContent));
2905:   }

```

```

2906:   if (["tool", "language"].includes(node.kind)) {
2907:     return Boolean(node.title || node.description || node.url || richHasRenderableContent(node.rich) ||
children.some(nodeHasRenderableContent));
2908:   }
2909:   return Boolean(node.description || node.url || richHasRenderableContent(node.rich) ||
children.some(nodeHasRenderableContent));
2910: }

```

## ***sectionHasRenderableContent - lines 2912-2914***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2912-2914 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sectionHasRenderableContent(section) {                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | richHasRenderableContent, some                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2912: function sectionHasRenderableContent(section) {
2913:   return Boolean(section?.description || richHasRenderableContent(section?.rich) || (section?.items ||
[])).some(nodeHasRenderableContent));
2914: }

```

## ***responsiveChildren - lines 2916-2918***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2916-2918 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function responsiveChildren(node) {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2916: function responsiveChildren(node) {
2917:   return node?.items || node?.children || [];
2918: }

```

## ***richContainsMedia - lines 2920-2922***

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 2920-2922 (3 source lines).            |
| Purpose         | Part of the public website runtime.          |
| Declaration     | function richContainsMedia(rich) {           |
| Touches         | Mostly local calculations or local UI state. |
| Calls detected  | some                                         |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2920: function richContainsMedia(rich) {
2921:   return Boolean(rich?.blocks?.some((block) => block.type === "image" && block.url));
2922: }

```

## ***responsiveNodeHasMedia - lines 2924-2930***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2924-2930 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function responsiveNodeHasMedia(node) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | richContainsMedia, responsiveChildren, some                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2924: function responsiveNodeHasMedia(node) {
2925:   if (!node) return false;
2926:   if (node.kind === "image" || richContainsMedia(node.rich)) return true;
2927:   const url = String(node.url || "");
2928:   if (/\. (apng|avif|gif|jpe?g|png|svg|webp)(\?|#|$)/i.test(url)) return true;
2929:   return responsiveChildren(node).some(responsiveNodeHasMedia);
2930: }

```

## ***responsiveNodeCount - lines 2932-2935***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2932-2935 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function responsiveNodeCount(node) {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | nodeHasRenderableContent, responsiveChildren, reduce                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2932: function responsiveNodeCount(node) {
2933:   if (!node || !nodeHasRenderableContent(node)) return 0;
2934:   return 1 + responsiveChildren(node).reduce((count, child) => count + responsiveNodeCount(child), 0);
2935: }

```

## ***responsiveNodeDepth - lines 2937-2942***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2937-2942 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function responsiveNodeDepth(node) {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | nodeHasRenderableContent, responsiveChildren, filter, max, map                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2937: function responsiveNodeDepth(node) {
2938:   if (!node || !nodeHasRenderableContent(node)) return 0;
2939:   const children = responsiveChildren(node).filter(nodeHasRenderableContent);
2940:   if (!children.length) return 1;
2941:   return 1 + Math.max(...children.map(responsiveNodeDepth));
2942: }

```

## ***inferResponsiveProfile - lines 2944-2976***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2944-2976 (33 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function inferResponsiveProfile(project) {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, sectionHasRenderableContent, reduce, responsiveNodeCount, max, map, some, responsiveNodeHasMedia, hasPublicTemplate                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2944: function inferResponsiveProfile(project) {
2945:   const sections = project?.portfolioView?.sections || [];
2946:   const visibleSections = sections.filter((section) => section?.id !== "brief" &&
sectionHasRenderableContent(section));
2947:   const itemCount = visibleSections.reduce(
2948:     (count, section) => count + (section.items || []).reduce((sum, item) => sum +
responsiveNodeCount(item), 0),
2949:     0
2950:   );
2951:   const maxDepth = visibleSections.reduce(
2952:     (depth, section) => Math.max(depth, ...(section.items || []).map(responsiveNodeDepth), 1),
2953:     1
2954:   );
2955:   const hasMedia = visibleSections.some((section) => responsiveNodeHasMedia(section));
2956:   const density = itemCount >= 18 || visibleSections.length >= 6 || maxDepth >= 4
2957:     ? "dense"
2958:     : itemCount <= 6 && visibleSections.length <= 2
2959:     ? "simple"
2960:     : "balanced";
2961:   return {
2962:     version: 1,
2963:     breakpoints: {
2964:       mobile: 640,
2965:       tablet: 900,
2966:       desktop: 1180
2967:     },
2968:     cardLayout: hasPublicTemplate(project) ? "single" : "split",
2969:     contentMax: hasMedia || maxDepth >= 3 ? "1180px" : "980px",
2970:     density,

```

```

2971:     mediaMax: hasMedia ? "860px" : "720px",
2972:     sectionCardMin: density === "dense" ? "210px" : density === "simple" ? "280px" : "240px",
2973:     touchTarget: "44px",
2974:     windowMode: "full-screen"
2975:   };
2976: }

```

## projectResponsiveProfile - lines 2978-2982

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2978-2982 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function projectResponsiveProfile(project) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | inferResponsiveProfile                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2978: function projectResponsiveProfile(project) {
2979:   const savedProfile = project?.portfolioView?.responsive;
2980:   if (savedProfile?.version) return { ...inferResponsiveProfile(project), ...savedProfile };
2981:   return inferResponsiveProfile(project);
2982: }

```

## nodeSummary - lines 2984-2992

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2984-2992 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function nodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | escapeHtml, renderRichContent                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2984: function nodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {
2985:   if (!rich?.blocks?.length && !text) return "";
2986:   return `
2987:     <details class="parsed-summary" open>
2988:       <summary>${escapeHtml(title)}</summary>
2989:       ${renderRichContent(rich, text || "")}
2990:     </details>
2991:   `;
2992: }

```

## nodeSummary - lines 2994-2997

| Item            | Explanation                         |
|-----------------|-------------------------------------|
| Source location | Lines 2994-2997 (4 source lines).   |
| Purpose         | Part of the public website runtime. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function nodeIsOverview(item = {}) {                                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, displayTitle, endsWith                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2994: function nodeIsOverview(item = {}) {
2995:   const title = normalize(displayTitle(item.title || item.label || ""));
2996:   return item.kind === "summary" || title === "overview" || title.endsWith(" overview");
2997: }

```

## ***nodeOverviewDetails - lines 2999-3017***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2999-3017 (19 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function nodeOverviewDetails(node, children = []) {                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, push, renderRichContent, forEach, renderInlineSectionFiles, nodeChildren, join                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2999: function nodeOverviewDetails(node, children = []) {
3000:   const overviewItems = children.filter(nodeIsOverview);
3001:   const blocks = [];
3002:   if (node?.rich?.blocks?.length || node?.description) {
3003:     blocks.push(renderRichContent(node.rich, node.description || ""));
3004:   }
3005:   overviewItems.forEach((item) => {
3006:     if (item.rich?.blocks?.length || item.description) blocks.push(renderRichContent(item.rich,
item.description || ""));
3007:     const itemFiles = renderInlineSectionFiles(nodeChildren(item));
3008:     if (itemFiles) blocks.push(itemFiles);
3009:   });
3010:   if (!blocks.length) return "";
3011:   return `
3012:     <details class="parsed-summary section-overview-default" open>
3013:       <summary>Overview</summary>
3014:       ${blocks.join("")}
3015:     </details>
3016:   `;
3017: }

```

## ***parsedNodeCard - lines 3019-3032***

| Item            | Explanation                                                    |
|-----------------|----------------------------------------------------------------|
| Source location | Lines 3019-3032 (14 source lines).                             |
| Purpose         | Part of the public website runtime.                            |
| Declaration     | function parsedNodeCard(node, projectId, sectionIndex, path) { |
| Touches         | Mostly local calculations or local UI state.                   |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | displayTitle, pathToString, escapeHtml                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3019: function parsedNodeCard(node, projectId, sectionIndex, path) {
3020:   const title = displayTitle(node.title);
3021:   return `
3022:     <button
3023:       class="section-open-card subsection-open-card"
3024:       type="button"
3025:       data-section-project="${projectId}"
3026:       data-section-index="${sectionIndex}"
3027:       data-resource-path="${pathToString(path)}"
3028:     >
3029:       <span>${escapeHtml(title)}</span>
3030:     </button>
3031:   `;
3032: }

```

## ***parsedChildCards - lines 3034-3042***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3034-3042 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function parsedChildCards(children, projectId, sectionIndex, basePath = []) {                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, nodeHasRenderableContent, nodeIsOverview, map, parsedNodeCard, join                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3034: function parsedChildCards(children, projectId, sectionIndex, basePath = []) {
3035:   const visibleChildren = children.filter((child) => nodeHasRenderableContent(child) && !child.url &&
!nodeIsOverview(child));
3036:   if (!visibleChildren.length) return "";
3037:   return `
3038:     <div class="subsection-grid section-content-grid">
3039:       ${children.map((child, index) => nodeHasRenderableContent(child) && !child.url &&
!nodeIsOverview(child) ? parsedNodeCard(child, projectId, sectionIndex, [...basePath, index]) : "").join("")}
3040:     </div>
3041:   `;
3042: }

```

## ***parsedNodeContent - lines 3044-3058***

| Item            | Explanation                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------------------|
| Source location | Lines 3044-3058 (15 source lines).                                                                    |
| Purpose         | Part of the public website runtime.                                                                   |
| Declaration     | function parsedNodeContent(node, projectId, sectionIndex, path = []) {                                |
| Touches         | Mostly local calculations or local UI state.                                                          |
| Calls detected  | renderInlineSectionFiles, nodeChildren, filter, nodeIsOverview, nodeOverviewDetails, parsedChildCards |
| API endpoints   | None detected inside this function body.                                                              |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3044: function parsedNodeContent(node, projectId, sectionIndex, path = []) {
3045:   const isContainer = !node.kind || node.kind === "subsection";
3046:   if (!isContainer && node.url) return renderInlineSectionFiles([node]);
3047:
3048:   const children = nodeChildren(node);
3049:   const contentChildren = children.filter((child) => !nodeIsOverview(child));
3050:
3051:   return `
3052:   <article class="parsed-window-panel section-directory">
3053:     ${nodeOverviewDetails(node, children)}
3054:     ${parsedChildCards(children, projectId, sectionIndex, path)}
3055:     ${renderInlineSectionFiles(contentChildren)}
3056:   </article>
3057: `;
3058: }

```

## ***parsedSectionContent - lines 3060-3064***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3060-3064 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function parsedSectionContent(section, projectId, sectionIndex, path = []) {                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | nodeAtPath, parsedNodeContent                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3060: function parsedSectionContent(section, projectId, sectionIndex, path = []) {
3061:   const node = path.length ? nodeAtPath(section, path) : section;
3062:   if (!node) return `<p class="evidence-empty">This section could not be found.</p>`;
3063:   return parsedNodeContent(node, projectId, sectionIndex, path);
3064: }

```

## ***parsedSection - lines 3066-3080***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3066-3080 (15 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function parsedSection(section, index, project) {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, some, escapeHtml, displayTitle                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3066: function parsedSection(section, index, project) {
3067:   const visibleItems = (section.items || []).filter(nodeHasRenderableContent);
3068:   const hasImages = visibleItems.some((item) => item.kind === "image");
3069:
3070:   return `
3071:     <button
3072:       class="section-open-card evidence-block ${hasImages ? "evidence-wide" : ""}"
3073:       type="button"
3074:       data-section-project="${project.id}"
3075:       data-section-index="${index}"
3076:     >
3077:       <span>${escapeHtml(displayTitle(section.title, "Section"))}</span>
3078:     </button>
3079:   `;
3080: }

```

## projectCard - lines 3082-3164

| Item               | Explanation                                                                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3082-3164 (83 source lines).                                                                                                                                                                      |
| Purpose            | Part of the public website runtime.                                                                                                                                                                     |
| Declaration        | function projectCard(project) {                                                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                              |
| Calls detected     | find, filter, sectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, renderParsedBriefBlock, map, parsedSection, join, renderMultilineInlineText, detailBlock |
| API endpoints      | None detected inside this function body.                                                                                                                                                                |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                 |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                           |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.   |

Source excerpt:

```

3082: function projectCard(project) {
3083:   if (project.portfolioView) {
3084:     const category = categories.find((item) => item.id === project.category) || {};
3085:     const accent = category.accent || "#117c7a";
3086:     const sections = project.portfolioView.sections || [];
3087:     const briefSection = sections.find((section) => section.id === "brief");
3088:     const otherSections = sections.filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
3089:
3090:     return `
3091:       <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" id="${project.id}" style="${projectTemplateStyle(project, accent)}">
3092:         <div class="project-body">
3093:           <h3>${project.portfolioView.title || project.title}</h3>
3094:           ${renderParsedBriefBlock(briefSection, project.summary)}
3095:           <div class="evidence-grid" aria-label="${project.title} parsed project content">
3096:             ${otherSections.map((section, index) => parsedSection(section, index, project)).join("")}
3097:           </div>
3098:         </div>
3099:       </article>
3100:     `;
3101:   }
3102:
3103:   const category = categories.find((item) => item.id === project.category) || {};
3104:   const accent = category.accent || "#117c7a";
3105:
3106:   return `
3107:     <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" id="${project.id}" style="${projectTemplateStyle(project, accent)}">
3108:       <div class="project-body">
3109:         <h3>${project.title}</h3>
3110:         <p class="rich-paragraph">${renderMultilineInlineText(project.summary)}</p>
3111:
3112:         ${project.highlights && project.highlights.length ? detailBlock("Project highlights", "project-
drawer", `
3113:           <ul class="highlight-list">
3114:             ${project.highlights.map((item) => `<li>${item}</li>`).join("")}
3115:           </ul>
3116:         `) : ""}
3117:
3118:         <div class="evidence-grid" aria-label="${project.title} evidence blocks">
3119:           ${detailBlock("Documents", "evidence-block", evidenceList(project.documents, (item) => `
3120:             <li>
3121:               ${resourceLink(item, item.title)}
3122:               <span>${item.type || "Document"} &mdot; ${item.status || "tracked"}</span>
3123:             </li>

```

```

3124:         ` , "No document artifact has been added yet.")]})
3125:
3126:     ${detailBlock("Tests and results", "evidence-block", evidenceList(project.tests, (item) => `
3127:         <li>
3128:             ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3129:             <span>${item.method || "Validation"} &middot; ${item.status || "tracked"}</span>
3130:             ${item.result ? `<p>${item.result}</p>` : ""}
3131:         </li>
3132:     ` , "No test artifact has been added yet.")]})
3133:
3134:     ${detailBlock("PCBs built", "evidence-block", evidenceList(project.pcb, (item) => `
3135:         <li>
3136:             ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3137:             <span>${item.revision || "Revision"} &middot; ${item.status || "tracked"}</span>
3138:         </li>
3139:     ` , "No PCB build has been added yet.")]})
3140:
3141:     ${detailBlock("Images", "evidence-block evidence-wide", mediaGrid(project.media))}
3142:     ${customSectionBlocks(project)}
3143: </div>
3144:
3145:     ${detailBlock("Tools and implementation files", "project-drawer", `
3146:         <div class="project-tooling">
3147:             <div>
3148:                 <h4>Tools Used</h4>
3149:                 <div class="project-meta">${pillList(project.tools, "tool-tag")}</div>
3150:             </div>
3151:             <div>
3152:                 <h4>Languages</h4>
3153:                 <div class="project-meta">${pillList(project.languages, "language-tag")}</div>
3154:             </div>
3155:         </div>
3156:     `)}
3157:
3158:     <div class="resource-list">
3159:         ${project.links || []}.map((item) => resourceLink(item)).join("")
3160:     </div>
3161: </div>
3162: </article>
3163: `;
3164: }

```

## categorySection - lines 3166-3181

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3166-3181 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function categorySection(category, visibleProjects) {                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | escapeHtml, map, join                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3166: function categorySection(category, visibleProjects) {
3167:     return `
3168:         <section class="category-section ${visibleProjects.length ? "" : "empty-category-section"}" aria-
3169:             labelledby="${escapeHtml(category.id)}-title">
3170:             <div class="category-heading">
3171:                 <div>
3172:                     <h3 id="${escapeHtml(category.id)}-title">${escapeHtml(category.label)}</h3>
3173:                 </div>
3174:                 ${visibleProjects.length ? `<span>${visibleProjects.length} project${visibleProjects.length === 1
3175: ? "" : "s"}</span>` : ""}
3176:             </div>
3177:             ${visibleProjects.length && category.description ? `<p class="category-
3178: description">${escapeHtml(category.description)}</p>` : ""}
3179:             <div class="category-projects">
3180:                 ${visibleProjects.map(projectCard).join("")}
3181:             </div>
3182:         </section>
3183:     `;
3184: }

```

## ***escapeRegExp - lines 3183-3185***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3183-3185 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function escapeRegExp(value = "") {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replace                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3183: function escapeRegExp(value = "") {
3184:   return String(value).replace(/[*+?^${}()|[\]\\]/g, "\\$&");
3185: }
```

## ***highlightQueryHtml - lines 3187-3193***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3187-3193 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function highlightQueryHtml(value = "", query = "") {                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | escapeHtml, trim, escapeRegExp, replace                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3187: function highlightQueryHtml(value = "", query = "") {
3188:   const escaped = escapeHtml(value);
3189:   const cleanQuery = String(query || "").trim();
3190:   if (!cleanQuery) return escaped;
3191:   const pattern = new RegExp(`${escapeRegExp(escapeHtml(cleanQuery))}`, "ig");
3192:   return escaped.replace(pattern, '<mark class="search-result-mark">$1</mark>');
3193: }
```

## ***clearSearchHighlights - lines 3195-3197***

| Item               | Explanation                                       |
|--------------------|---------------------------------------------------|
| Source location    | Lines 3195-3197 (3 source lines).                 |
| Purpose            | Part of the public website runtime.               |
| Declaration        | function clearSearchHighlights() {                |
| Touches            | Mostly local calculations or local UI state.      |
| Calls detected     | No obvious named calls detected by the generator. |
| API endpoints      | None detected inside this function body.          |
| Storage keys       | No local/session storage keys detected.           |
| Async/return clues | 0 await expression(s), 0 return statement(s).     |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3195: function clearSearchHighlights() {
3196:   window.CSS?.highlights?.delete?.(searchHighlightName);
3197: }

```

## ***textNodeSearchTargets - lines 3199-3204***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3199-3204 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function textNodeSearchTargets() {                                                                                                                                                                    |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, filter                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3199: function textNodeSearchTargets() {
3200:   return [
3201:     document.querySelector("main"),
3202:     document.querySelector("#section-view-dialog[open] .section-view-content")
3203:   ].filter(Boolean);
3204: }

```

## ***applySearchHighlights - lines 3206-3238***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3206-3238 (33 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function applySearchHighlights(query) {                                                                                                                                                               |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | clearSearchHighlights, trim, normalize, textNodeSearchTargets, forEach, createTreeWalker, acceptNode, includes, closest, nextNode, indexOf, createRange                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3206: function applySearchHighlights(query) {
3207:   clearSearchHighlights();
3208:   const cleanQuery = String(query || "").trim();
3209:   if (!cleanQuery || !window.CSS?.highlights || typeof window.Highlight !== "function") return 0;
3210:   const ranges = [];
3211:   const needle = normalize(cleanQuery);
3212:
3213:   textNodeSearchTargets().forEach((root) => {
3214:     const walker = document.createTreeWalker(root, NodeFilter.SHOW_TEXT, {
3215:       acceptNode(node) {
3216:         if (!node.textContent || !normalize(node.textContent).includes(needle)) return
NodeFilter.FILTER_REJECT;
3217:         if (node.parentElement?.closest("script, style, input, textarea, select, button, .search-engine,

```

```

.site-header")) return NodeFilter.FILTER_REJECT;
3218:     return NodeFilter.FILTER_ACCEPT;
3219:   }
3220: });
3221: let node = walker.nextNode();
3222: while (node) {
3223:   const haystack = normalize(node.textContent);
3224:   let index = haystack.indexOf(needle);
3225:   while (index >= 0) {
3226:     const range = document.createRange();
3227:     range.setStart(node, index);
3228:     range.setEnd(node, index + cleanQuery.length);
3229:     ranges.push(range);
3230:     index = haystack.indexOf(needle, index + Math.max(1, cleanQuery.length));
3231:   }
3232:   node = walker.nextNode();
3233: }
3234: });
3235:
3236: if (ranges.length) window.CSS.highlights.set(searchHighlightName, new window.Highlight(...ranges));
3237: return ranges.length;
3238: }

```

## queueSearchHighlights - lines 3240-3243

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3240-3243 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function queueSearchHighlights() {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | applySearchHighlights                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3240: function queueSearchHighlights() {
3241:   clearTimeout(searchHighlightTimer);
3242:   searchHighlightTimer = setTimeout(() => applySearchHighlights(searchInput?.value || ""), 30);
3243: }

```

## ensureSearchPanel - lines 3245-3283

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3245-3283 (39 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function ensureSearchPanel() {                                                                                                                                                                        |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | createElement, replaceWith, append, querySelector, addEventListener, updateSearchDropdown, preventDefault, closest, goToSearchResult                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3245: function ensureSearchPanel() {
3246:   if (!searchInput || !searchWrap || searchPanel) return;
3247:   const shell = document.createElement("div");
3248:   shell.className = "search-engine";
3249:   searchWrap.replaceWith(shell);

```

```

3250: shell.append(searchWrap);
3251: searchPanel = document.createElement("div");
3252: searchPanel.className = "search-results-panel";
3253: searchPanel.hidden = true;
3254: searchPanel.innerHTML = `
3255:   <div class="search-results-tools">
3256:     <span id="search-status">Type to search projects, files, sections, and phrases.</span>
3257:     <label>
3258:       <span>Show</span>
3259:       <select id="search-result-limit" aria-label="Number of search results">
3260:         <option value="5">5</option>
3261:         <option value="10" selected>10</option>
3262:         <option value="15">15</option>
3263:         <option value="20">20</option>
3264:       </select>
3265:     </label>
3266:   </div>
3267:   <div class="search-results-list" role="listbox" aria-label="Search results"></div>
3268: `;
3269: shell.append(searchPanel);
3270: searchStatus = searchPanel.querySelector("#search-status");
3271: searchLimitSelect = searchPanel.querySelector("#search-result-limit");
3272: searchLimitSelect.addEventListener("change", () => {
3273:   searchResultLimit = Number(searchLimitSelect.value) || 10;
3274:   updateSearchDropdown();
3275: });
3276: searchPanel.addEventListener("mousedown", (event) => event.preventDefault());
3277: searchPanel.addEventListener("click", (event) => {
3278:   const button = event.target.closest("[data-search-result-index]");
3279:   if (!button) return;
3280:   const result = currentSearchResults[Number(button.dataset.searchResultIndex)];
3281:   if (result) goToSearchResult(result);
3282: });
3283: }

```

## setSearchMessage - lines 3285-3289

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3285-3289 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function setSearchMessage(message, state = "neutral") {                                                                                                                                               |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3285: function setSearchMessage(message, state = "neutral") {
3286:   if (!searchStatus) return;
3287:   searchStatus.textContent = message;
3288:   searchStatus.dataset.state = state;
3289: }

```

## renderSearchResults - lines 3291-3327

| Item            | Explanation                                                                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 3291-3327 (37 source lines).                                                                                                                            |
| Purpose         | Renders UI, markup, or display output.                                                                                                                        |
| Declaration     | function renderSearchResults(query, visibleCount) {                                                                                                           |
| Touches         | browser/Electron DOM                                                                                                                                          |
| Calls detected  | ensureSearchPanel, querySelector, trim, searchResultsFor, slice, setAttribute, setSearchMessage, removeAttribute, map, highlightQueryHtml, escapeHtml, filter |
| API endpoints   | None detected inside this function body.                                                                                                                      |
| Storage keys    | No local/session storage keys detected.                                                                                                                       |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3291: function renderSearchResults(query, visibleCount) {
3292:   ensureSearchPanel();
3293:   if (!searchPanel) return;
3294:   const list = searchPanel.querySelector(".search-results-list");
3295:   const cleanQuery = String(query || "").trim();
3296:   currentSearchResults = searchResultsFor(cleanQuery);
3297:   const limitedResults = currentSearchResults.slice(0, searchResultLimit);
3298:   searchPanel.hidden = !cleanQuery;
3299:   searchInput.setAttribute("aria-expanded", String(Boolean(cleanQuery)));
3300:
3301:   if (!cleanQuery) {
3302:     list.innerHTML = "";
3303:     setSearchMessage("Type to search projects, files, sections, and phrases.");
3304:     searchInput.removeAttribute("aria-invalid");
3305:     return;
3306:   }
3307:
3308:   if (!currentSearchResults.length) {
3309:     list.innerHTML = "";
3310:     setSearchMessage(`"${cleanQuery}" was not found on this portfolio.`, "not-found");
3311:     searchInput.setAttribute("aria-invalid", "true");
3312:     return;
3313:   }
3314:
3315:   searchInput.removeAttribute("aria-invalid");
3316:   const visibleText = visibleCount
3317:     ? `${visibleCount} visible match${visibleCount === 1 ? "" : "es"} highlighted.`
3318:     : "Matches found elsewhere. Choose a result below.";
3319:   setSearchMessage(`${visibleText} Showing ${limitedResults.length} of ${currentSearchResults.length}.`,
3320: "found");
3321:   list.innerHTML = limitedResults.map((result, index) => `
3322:     <button class="search-result-item" type="button" role="option" data-search-result-index="${index}">
3323:       <span class="search-result-title">${highlightQueryHtml(result.title || "Untitled result",
3324: cleanQuery)}</span>
3325:       <span class="search-result-meta">${escapeHtml([result.type, result.context].filter(Boolean).join("
3326: / ")).join("<br>")}</span>
3327:       ${searchSnippet(result.text, cleanQuery)} ? <span class="search-result-
3328: snippet">${highlightQueryHtml(searchSnippet(result.text, cleanQuery), cleanQuery)}</span>` : ""}
3329:     </button>
3330:   `).join("");
3331: }

```

## updateSearchDropdown - lines 3329-3333

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3329-3333 (5 source lines).                                                                                                                                                                     |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateSearchDropdown() {                                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | applySearchHighlights, renderSearchResults                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3329: function updateSearchDropdown() {
3330:   const query = searchInput?.value || "";
3331:   const visibleCount = applySearchHighlights(query);
3332:   renderSearchResults(query, visibleCount);
3333: }

```

## showSearchPanelIfNeeded - lines 3335-3338



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3335-3338 (4 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function showSearchPanelIfNeeded() {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, updateSearchDropdown                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3335: function showSearchPanelIfNeeded() {
3336:   if (!searchInput?.value?.trim()) return;
3337:   updateSearchDropdown();
3338: }

```

## ***scrollToDomTarget - lines 3340-3352***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3340-3352 (13 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function scrollToDomTarget(id) {                                                                                                                                                                      |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | getElementById, closeSectionDialogForRoute, scrollIntoView, replaceState, queueSearchHighlights                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3340: function scrollToDomTarget(id) {
3341:   const target = id ? document.getElementById(id) : null;
3342:   if (!target) return false;
3343:   closeSectionDialogForRoute();
3344:   target.scrollIntoView({ behavior: "smooth", block: "start" });
3345:   try {
3346:     history.replaceState(history.state || { appWindow: "portfolio-base" }, "", `#${id}`);
3347:   } catch {
3348:     // Scrolling is still the important behavior.
3349:   }
3350:   queueSearchHighlights();
3351:   return true;
3352: }

```

## ***goToSearchResult - lines 3354-3377***

| Item            | Explanation                                                                                                                        |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 3354-3377 (24 source lines).                                                                                                 |
| Purpose         | Part of the public website runtime.                                                                                                |
| Declaration     | function goToSearchResult(result) {                                                                                                |
| Touches         | browser/Electron DOM                                                                                                               |
| Calls detected  | scrollToDomTarget, openParsedSection, queueSearchHighlights, getElementById, setActiveFilter, renderProjects, scrollIntoView, open |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3354: function goToSearchResult(result) {
3355:   if (!result) return;
3356:   searchPanel.hidden = true;
3357:   if (result.domTarget && scrollToDomTarget(result.domTarget)) return;
3358:
3359:   if (result.projectId && result.sectionIndex !== undefined) {
3360:     openParsedSection(result.projectId, result.sectionIndex, result.resourcePath || "");
3361:     queueSearchHighlights();
3362:     return;
3363:   }
3364:
3365:   if (result.projectId) {
3366:     if (!document.getElementById(result.projectId)) {
3367:       setActiveFilter("all");
3368:       renderProjects();
3369:     }
3370:     const target = document.getElementById(result.projectId);
3371:     target?.scrollIntoView({ behavior: "smooth", block: "center" });
3372:     queueSearchHighlights();
3373:     return;
3374:   }
3375:
3376:   if (result.url) window.open(result.url, "_blank", "noopener");
3377: }

```

## ***handleSearchInput - lines 3379-3382***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3379-3382 (4 source lines).                                                                                                                                                                     |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handleSearchInput() {                                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | renderProjects, updateSearchDropdown                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3379: function handleSearchInput() {
3380:   renderProjects();
3381:   updateSearchDropdown();
3382: }

```

## ***renderProjects - lines 3384-3409***

| Item            | Explanation                                                                                |
|-----------------|--------------------------------------------------------------------------------------------|
| Source location | Lines 3384-3409 (26 source lines).                                                         |
| Purpose         | Renders UI, markup, or display output.                                                     |
| Declaration     | function renderProjects() {                                                                |
| Touches         | filesystem                                                                                 |
| Calls detected  | normalize, trim, filter, projectVisible, queueSearchHighlights, map, categorySection, join |
| API endpoints   | None detected inside this function body.                                                   |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3384: function renderProjects() {
3385:   const query = normalize(searchInput.value).trim();
3386:   const visible = projects.filter((project) => projectVisible(project, query));
3387:
3388:   projectCount.textContent = projects.length;
3389:
3390:   if (!visible.length) {
3391:     grid.innerHTML = `
3392:       <div class="empty-state">
3393:         <h3>No projects match that view.</h3>
3394:         <p>Search covers categories, project names, documents, tests, PCB builds, tools, languages, and
links.</p>
3395:       </div>
3396:     `;
3397:     queueSearchHighlights();
3398:     return;
3399:   }
3400:
3401:   grid.innerHTML = categories
3402:     .map((category) => {
3403:       const visibleProjects = visible.filter((project) => project.category === category.id);
3404:       const shouldShowEmptyCategory = !query && (activeFilter === "all" || activeFilter === category.id);
3405:       return visibleProjects.length || shouldShowEmptyCategory ? categorySection(category,
visibleProjects) : "";
3406:     })
3407:     .join("");
3408:   queueSearchHighlights();
3409: }

```

## ***ensureSectionDialog - lines 3411-3470***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3411-3470 (60 source lines).                                                                                                                                                                    |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function ensureSectionDialog() {                                                                                                                                                                      |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, createElement, append, addEventListener, closeSectionDialog, clearSectionRouteInHistory, remove, updateSectionDialogMinimize, focus, pathFromString, sectionForwardStack, push         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3411: function ensureSectionDialog() {
3412:   let dialog = document.querySelector("#section-view-dialog");
3413:   if (dialog) return dialog;
3414:
3415:   dialog = document.createElement("dialog");
3416:   dialog.id = "section-view-dialog";
3417:   dialog.className = "section-view-dialog";
3418:   dialog.innerHTML = `
3419:     <div class="section-view-shell">
3420:       <div class="section-view-heading">
3421:         <div class="section-view-titlebar">
3422:           <button class="section-view-back" type="button" title="Previous view" aria-label="Previous
view" hidden>&larr;</button>
3423:           <button class="section-view-forward" type="button" title="Next view" aria-label="Next view"
hidden>&rarr;</button>
3424:           <h2 id="section-view-title">Section</h2>
3425:         </div>
3426:         <div class="section-view-actions">
3427:           <button class="section-view-close" type="button" aria-label="Close section">&times;</button>
3428:         </div>

```

```

3429:         </div>
3430:         <div class="section-view-content" id="section-view-content"></div>
3431:     </div>
3432: `;
3433:     document.body.append(dialog);
3434:     dialog.querySelector(".section-view-close").addEventListener("click", () =>
closeSectionDialog(dialog));
3435:     dialog.addEventListener("cancel", () => clearSectionRouteInHistory());
3436:     dialog.addEventListener("close", () => {
3437:         document.body.classList.remove("full-window-open");
3438:         dialog.classList.remove("is-minimized-dialog");
3439:         updateSectionDialogMinimize(dialog);
3440:         const returnTarget = document.querySelector(
3441:             `[data-section-project="${dialog.dataset.projectId}"][data-section-
index="${dialog.dataset.sectionIndex}"]`
3442:         );
3443:         returnTarget?.focus({ preventScroll: true });
3444:     });
3445:     dialog.querySelector(".section-view-back").addEventListener("click", () => {
3446:         const path = pathFromString(dialog.dataset.resourcePath || "");
3447:         if (!path.length) {
3448:             closeSectionDialog(dialog);
3449:             return;
3450:         }
3451:         const stack = sectionForwardStack(dialog);
3452:         stack.push(pathToString(path));
3453:         dialog.dataset.forwardStack = JSON.stringify(stack);
3454:         path.pop();
3455:         openParsedSection(dialog.dataset.projectId, dialog.dataset.sectionIndex, pathToString(path), {
historyMode: "replace", preserveForward: true });
3456:     });
3457:     dialog.querySelector(".section-view-forward").addEventListener("click", () => {
3458:         const stack = sectionForwardStack(dialog);
3459:         const nextPath = stack.pop();
3460:         if (!nextPath) return;
3461:         dialog.dataset.forwardStack = JSON.stringify(stack);
3462:         openParsedSection(dialog.dataset.projectId, dialog.dataset.sectionIndex, nextPath, { historyMode:
"replace", preserveForward: true });
3463:     });
3464:     dialog.querySelector("#section-view-content").addEventListener("click", (event) => {
3465:         const nodeButton = event.target.closest("[data-resource-path]");
3466:         if (!nodeButton) return;
3467:         openParsedSection(nodeButton.dataset.sectionProject, nodeButton.dataset.sectionIndex,
nodeButton.dataset.resourcePath);
3468:     });
3469:     return dialog;
3470: }

```

## sectionForwardStack - lines 3472-3478

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3472-3478 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function sectionForwardStack(dialog) {                                                                                                                                                                |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | parse                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3472: function sectionForwardStack(dialog) {
3473:     try {
3474:         return JSON.parse(dialog.dataset.forwardStack || "[]");
3475:     } catch {
3476:         return [];
3477:     }
3478: }

```

## updateSectionNavigation - lines 3480-3486

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3480-3486 (7 source lines).                                                                                                                                                                     |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateSectionNavigation(dialog, path) {                                                                                                                                                      |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, setAttribute, sectionForwardStack                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3480: function updateSectionNavigation(dialog, path) {
3481:   const back = dialog.querySelector(".section-view-back");
3482:   back.hidden = false;
3483:   back.title = path.length ? "Previous view" : "Back to projects";
3484:   back.setAttribute("aria-label", path.length ? "Previous view" : "Back to projects");
3485:   dialog.querySelector(".section-view-forward").hidden = !sectionForwardStack(dialog).length;
3486: }

```

## ***closeSectionDialog - lines 3488-3491***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3488-3491 (4 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function closeSectionDialog(dialog) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | close, clearSectionRouteInHistory                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3488: function closeSectionDialog(dialog) {
3489:   dialog.close();
3490:   clearSectionRouteInHistory();
3491: }

```

## ***closeOrStepBackSectionDialog - lines 3493-3504***

| Item               | Explanation                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3493-3504 (12 source lines).                                                                             |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                      |
| Declaration        | function closeOrStepBackSectionDialog(dialog) {                                                                |
| Touches            | browser/Electron DOM                                                                                           |
| Calls detected     | pathFromString, sectionForwardStack, push, pathToString, stringify, pop, openParsedSection, closeSectionDialog |
| API endpoints      | None detected inside this function body.                                                                       |
| Storage keys       | No local/session storage keys detected.                                                                        |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                  |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3493: function closeOrStepBackSectionDialog(dialog) {
3494:   const path = pathFromString(dialog.dataset.resourcePath || "");
3495:   if (path.length) {
3496:     const stack = sectionForwardStack(dialog);
3497:     stack.push(pathToString(path));
3498:     dialog.dataset.forwardStack = JSON.stringify(stack);
3499:     path.pop();
3500:     openParsedSection(dialog.dataset.projectId, dialog.dataset.sectionIndex, pathToString(path), {
historyMode: "replace", preserveForward: true });
3501:     return;
3502:   }
3503:   closeSectionDialog(dialog);
3504: }

```

## ***openParsedSection - lines 3506-3532***

| Item               | Explanation                                                                                                                                                                                                               |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3506-3532 (27 source lines).                                                                                                                                                                                        |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                                                 |
| Declaration        | function openParsedSection(projectId, sectionIndex, resourcePath = "", options = {}) {                                                                                                                                    |
| Touches            | browser/Electron DOM                                                                                                                                                                                                      |
| Calls detected     | find, filter, sectionHasRenderableContent, pathFromString, nodeAtPath, nodeHasRenderableContent, ensureSectionDialog, applyProjectTemplateToElement, pathToString, syncSectionRouteToHistory, querySelector, displayTitle |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                  |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                   |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                                             |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                     |

Source excerpt:

```

3506: function openParsedSection(projectId, sectionIndex, resourcePath = "", options = {}) {
3507:   const project = projects.find((item) => item.id === projectId);
3508:   const sections = (project?.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
3509:   const section = sections[Number(sectionIndex)];
3510:   if (!section) return false;
3511:   const path = pathFromString(resourcePath);
3512:   const node = path.length ? nodeAtPath(section, path) : section;
3513:   if (!node) return false;
3514:   if (path.length ? !nodeHasRenderableContent(node) : !sectionHasRenderableContent(section)) return
false;
3515:
3516:   const dialog = ensureSectionDialog();
3517:   const category = categories.find((item) => item.id === project.category) || {};
3518:   applyProjectTemplateToElement(dialog, project, category.accent || "#117c7a");
3519:   dialog.dataset.projectId = projectId;
3520:   dialog.dataset.sectionIndex = String(sectionIndex);
3521:   dialog.dataset.resourcePath = pathToString(path);
3522:   if (!options.preserveForward) dialog.dataset.forwardStack = "[]";
3523:   syncSectionRouteToHistory(projectId, sectionIndex, pathToString(path), options.historyMode || "push");
3524:   dialog.querySelector("#section-view-title").textContent = displayTitle(node.title || section.title,
"Section");
3525:   updateSectionNavigation(dialog, path);
3526:   dialog.querySelector("#section-view-content").innerHTML = parsedSectionContent(section, projectId,
Number(sectionIndex), path);
3527:   document.body.classList.add("full-window-open");
3528:   if (!dialog.open) dialog.showModal();
3529:   dialog.scrollTop = 0;
3530:   queueSearchHighlights();
3531:   return true;
3532: }

```

## ***sections - lines 3508-3514***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3508-3514 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | <code>const sections = (project?.portfolioView?.sections    []).filter((section) =&gt; section.id !== "brief" &amp;&amp; sectionHasRenderableContent(section));</code>                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>filter</code> , <code>sectionHasRenderableContent</code> , <code>pathFromString</code> , <code>nodeAtPath</code> , <code>nodeHasRenderableContent</code>                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 <code>await</code> expression(s), 3 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3508:   const sections = (project?.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
3509:   const section = sections[Number(sectionIndex)];
3510:   if (!section) return false;
3511:   const path = pathFromString(resourcePath);
3512:   const node = path.length ? nodeAtPath(section, path) : section;
3513:   if (!node) return false;
3514:   if (path.length ? !nodeHasRenderableContent(node) : !sectionHasRenderableContent(section)) return
false;

```

## ***closeSectionDialogForRoute - lines 3534-3537***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3534-3537 (4 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | <code>function closeSectionDialogForRoute() {</code>                                                                                                                                                  |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | <code>querySelector</code> , <code>close</code>                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 <code>await</code> expression(s), 0 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3534: function closeSectionDialogForRoute() {
3535:   const dialog = document.querySelector("#section-view-dialog");
3536:   if (dialog?.open) dialog.close();
3537: }

```

## ***applySectionRoute - lines 3539-3555***

| Item            | Explanation                                                              |
|-----------------|--------------------------------------------------------------------------|
| Source location | Lines 3539-3555 (17 source lines).                                       |
| Purpose         | Part of the public website runtime.                                      |
| Declaration     | <code>function applySectionRoute(route) {</code>                         |
| Touches         | Mostly local calculations or local UI state.                             |
| Calls detected  | <code>closeSectionDialogForRoute</code> , <code>openParsedSection</code> |
| API endpoints   | None detected inside this function body.                                 |
| Storage keys    | No local/session storage keys detected.                                  |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3539: function applySectionRoute(route) {
3540:   isApplyingSectionRoute = true;
3541:   try {
3542:     if (!route) {
3543:       closeSectionDialogForRoute();
3544:       return;
3545:     }
3546:
3547:     const opened = openParsedSection(route.projectId, route.sectionIndex, route.resourcePath, {
3548:       historyMode: "none",
3549:       preserveForward: true
3550:     });
3551:     if (!opened) closeSectionDialogForRoute();
3552:   } finally {
3553:     isApplyingSectionRoute = false;
3554:   }
3555: }

```

### ***restoreSectionRouteAfterCatalogLoad - lines 3557-3560***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3557-3560 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the public website runtime.                                                                                                                                                                   |
| Declaration        | function restoreSectionRouteAfterCatalogLoad() {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | initializeSectionHistoryState, applySectionRoute, sectionRouteFromLocation, sectionRouteFromState                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3557: function restoreSectionRouteAfterCatalogLoad() {
3558:   initializeSectionHistoryState();
3559:   applySectionRoute(sectionRouteFromLocation() || sectionRouteFromState(history.state));
3560: }

```

### ***renderInlineSectionFiles - lines 3621-3639***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3621-3639 (19 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderInlineSectionFiles(items = []) {                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | uniqueDownloads, map, normalizeDownloadAsset, filter, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute, fileNameFromUrl, join                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:



```

3621: function renderInlineSectionFiles(items = []) {
3622:   const downloads = uniqueDownloads(items.map((item) => normalizeDownloadAsset(item,
"File")).filter(Boolean));
3623:   if (!downloads.length) return "";
3624:
3625:   return `
3626:     <div class="section-file-list">
3627:       ${downloads.map((item) => `
3628:         <a
3629:           class="section-file-link"
3630:           href="${escapeHtml(normalizeLinkTarget(item.url, { assumeWeb: isWebsiteLinkItem(item, item.url)
3631:         }})}"
3632:           ${linkAttributes(item.url, item)}
3633:           ${downloadAttribute(item.url, item)}
3634:         >
3635:           ${escapeHtml(item.title || fileNameFromUrl(item.url))}
3636:         </a>
3637:       `).join("")}
3638:     </div>
3639:   `;

```

## normalizeTextColor - lines 3640-3647

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3640-3647 (8 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeTextColor(value = "") {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3640: function normalizeTextColor(value = "") {
3641:   const color = String(value || "").trim();
3642:   return /^[0-9a-f]{3}|[0-9a-f]{6}$/i.test(color) ||
3643:     /^rgba?\(/i.test(color) ||
3644:     /^hsla?\(/i.test(color)
3645:     ? color
3646:     : "";
3647: }

```

## loadProjectCatalog - lines 3648-3708

| Item               | Explanation                                                                                                                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3648-3708 (61 source lines).                                                                                                                                                                                                         |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                                                              |
| Declaration        | function loadProjectCatalog() {                                                                                                                                                                                                            |
| Touches            | filesystem, network/API requests                                                                                                                                                                                                           |
| Calls detected     | hydrateProjectCategories, normalizeFieldStyles, normalizeSiteContent, normalizeProfile, normalizeFunFacts, renderProfile, renderSiteContent, renderFunFacts, renderSiteSections, renderCategoryFilters, renderProjects, rebuildSearchIndex |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                   |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                    |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                                                              |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                      |

Source excerpt:

```

3648: function loadProjectCatalog() {

```

```

3649:   if (window.__PORTFOLIO_CATALOG__) {
3650:     projects = window.__PORTFOLIO_CATALOG__.projects || [];
3651:     categories = hydrateProjectCategories(window.__PORTFOLIO_CATALOG__.categories || [], projects);
3652:     siteSections = window.__PORTFOLIO_CATALOG__.siteSections || [];
3653:     fieldStyles = normalizeFieldStyles(window.__PORTFOLIO_CATALOG__.fieldStyles || {});
3654:     siteContent = normalizeSiteContent(window.__PORTFOLIO_CATALOG__.siteContent || {});
3655:     siteContentRich = window.__PORTFOLIO_CATALOG__.siteContentRich || {};
3656:     profile = normalizeProfile(window.__PORTFOLIO_CATALOG__.profile || {});
3657:     profileRich = window.__PORTFOLIO_CATALOG__.profileRich || {};
3658:     funFacts = normalizeFunFacts(window.__PORTFOLIO_CATALOG__.funFacts || []);
3659:     funFactsRich = window.__PORTFOLIO_CATALOG__.funFactsRich || null;
3660:     renderProfile();
3661:     renderSiteContent();
3662:     renderFunFacts();
3663:     renderSiteSections();
3664:     renderCategoryFilters();
3665:     renderProjects();
3666:     rebuildSearchIndex();
3667:     updateSearchDropdown();
3668:     restoreSectionRouteAfterCatalogLoad();
3669:     return;
3670:   }
3671:
3672:   const catalogUrl = new URL("projects.json", window.location.href);
3673:   catalogUrl.searchParams.set("v", String(Date.now()));
3674:   fetch(catalogUrl.href, { cache: "no-store" })
3675:     .then((response) => {
3676:       if (!response.ok) throw new Error("Could not load projects.json");
3677:       return response.json();
3678:     })
3679:     .then((data) => {
3680:       projects = data.projects || [];
3681:       categories = hydrateProjectCategories(data.categories || [], projects);
3682:       siteSections = data.siteSections || [];
3683:       fieldStyles = normalizeFieldStyles(data.fieldStyles || {});
3684:       siteContent = normalizeSiteContent(data.siteContent || {});
3685:       siteContentRich = data.siteContentRich || {};
3686:       profile = normalizeProfile(data.profile || {});
3687:       profileRich = data.profileRich || {};
3688:       funFacts = normalizeFunFacts(data.funFacts || []);
3689:       funFactsRich = data.funFactsRich || null;
3690:       renderProfile();
3691:       renderSiteContent();
3692:       renderFunFacts();
3693:       renderSiteSections();
3694:       renderCategoryFilters();
3695:       renderProjects();
3696:       rebuildSearchIndex();
3697:       updateSearchDropdown();
3698:       restoreSectionRouteAfterCatalogLoad();
3699:     })
3700:     .catch(() => {
3701:       grid.innerHTML = `
3702:         <div class="empty-state">
3703:           <h3>Project data did not load.</h3>
3704:           <p>The project catalog is temporarily unavailable.</p>
3705:         </div>
3706:       `;
3707:     });
3708: }

```

## Representative opening snippet

```

1: const grid = document.querySelector("#project-grid");
2: const searchInput = document.querySelector("#project-search");
3: const projectFilters = document.querySelector("#project-filters");
4: let filterButtons = [...document.querySelectorAll(".filter-button")];
5: const projectCount = document.querySelector("#project-count");
6: const projectTrackCount = document.querySelector("#project-track-count");
7: const projectTrackLabels = document.querySelector("#project-track-labels");
8: const year = document.querySelector("#year");
9: const funFactsCallout = document.querySelector("#fun-facts-callout");
10: const heroEyebrow = document.querySelector(".hero-content .eyebrow");
11: const heroTitle = document.querySelector("#hero-title");
12: const heroCopy = document.querySelector(".hero-copy");
13: const brandName = document.querySelector(".brand strong");
14: const brandSubtitle = document.querySelector(".brand small");
15: const brandIcon = document.querySelector(".brand-icon");
16: const brandText = document.querySelector(".omb-engraving");
17: const headerAvatar = document.querySelector(".header-avatar");
18: const headerAvatarImage = document.querySelector(".header-avatar img");
19: const heroImage = document.querySelector(".hero-image");
20: const resumeSection = document.querySelector("#resume");
21: const contactBand = document.querySelector("#contact");
22: const profilePhoto = document.querySelector(".profile-photo");
23: const contactTitle = document.querySelector("#contact-title");
24: const contactIntro = document.querySelector(".contact-intro");
25: const contactDetails = document.querySelector(".contact-details");
26: const contactLinks = document.querySelector(".contact-links");
27: const footerOwner = document.querySelector(".site-footer span");

```

```

28: const searchWrap = searchInput?.closest(".search-wrap");
29: const aiAssistantForm = document.querySelector("#ai-assistant-form");
30: const aiAssistantPanel = document.querySelector("#ask-ai");
31: const aiAssistantInput = document.querySelector("#ai-assistant-input");
32: const aiAssistantLog = document.querySelector("#ai-assistant-log");
33: const aiAssistantStatus = document.querySelector("#ai-assistant-status");
34: const aiClearChatButton = document.querySelector("#ai-clear-chat");
35:
36: let categories = [];
37: let projects = [];
38: let siteSections = [];
39: let funFacts = [];
40: let funFactsRich = null;
41: let fieldStyles = {};
42: let siteContent = null;
43: let siteContentRich = {};
44: let profile = null;
45: let profileRich = {};
46: let activeFilter = "all";
47: let activeSectionDialogDrag = null;
48: let activeSectionDialogResize = null;
49: let sectionDialogDragEnabled = false;
50: let isApplyingSectionRoute = false;
51: let searchPanel = null;
52: let searchStatus = null;
53: let searchLimitSelect = null;
54: let currentSearchResults = [];
55: let assistantSourceCounter = 0;
56: let assistantSourceMap = new Map();
57: let assistantChatHistory = [];
58: let searchableEntries = [];
59: let searchResultLimit = 10;
60: let searchHighlightTimer = 0;
61: const searchHighlightName = "portfolio-search-highlight";
62:
63: const sectionRouteKey = "portfolio-section";
64: const sectionRouteParams = {
65:   path: "portfolioPath",
66:   project: "portfolioProject",
67:   section: "portfolioSection"
68: };
69:
70: const supportedCodeLanguages = [
71:   { id: "c", label: "C", aliases: ["c"] },
72:   { id: "cpp", label: "C++", aliases: ["cpp", "c++", "cplusplus"] },

```

# Source File: server.mjs

| Item                | Explanation                                                                                                                               |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| File                | server.mjs                                                                                                                                |
| Purpose             | Local backend server used by the builder for drafts, parsing, publishing, compiler execution, authentication checks, and file operations. |
| Lines               | 3,632                                                                                                                                     |
| Size                | 147,116 bytes                                                                                                                             |
| Talks to            | public website runtime, portfolio catalog, Cloudflare/AI layer, GitHub/release layer                                                      |
| Functions           | 141                                                                                                                                       |
| Variables/constants | 140                                                                                                                                       |
| API mentions        | 21                                                                                                                                        |
| Signals             | 566                                                                                                                                       |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Imports, requires, and external dependencies

| Item   | Explanation                                                                                                         |
|--------|---------------------------------------------------------------------------------------------------------------------|
| Line 1 | import { createServer } from "node:http";                                                                           |
| Line 2 | import { execFile, spawn } from "node:child_process";                                                               |
| Line 3 | import { createHash } from "node:crypto";                                                                           |
| Line 4 | import { access as fsAccess, chmod, cp, mkdir, mkdtemp, readFile, readdir, rm, writeFile } from "node:fs/promises"; |
| Line 5 | import os from "node:os";                                                                                           |
| Line 6 | import path from "node:path";                                                                                       |
| Line 7 | import { promisify } from "node:util";                                                                              |
| Line 8 | import { fileURLToPath } from "node:url";                                                                           |

## API endpoints mentioned or called

| Item                    | Explanation                                                                   |
|-------------------------|-------------------------------------------------------------------------------|
| /api/catalog            | Line 3343. Loads project/catalog data for the builder.                        |
| /api/templates          | Line 3353. Loads appearance template definitions.                             |
| /api/publish-target     | Line 3358. Reads or writes the Git publishing target and authorization state. |
| /api/system-check       | Line 3367. Checks local tool/system readiness.                                |
| /api/app-update         | Line 3372. Checks or starts builder app update behavior.                      |
| /api/security-report    | Line 3377. Returns security/download/auth reporting for the builder.          |
| /api/code/tools         | Line 3386. Checks compiler/runtime availability.                              |
| /api/portfolio-ai       | Line 3402. Handles AI assistant questions.                                    |
| /api/code/beautify      | Line 3407. Formats source code for cleaner display and editing.               |
| /api/code/save          | Line 3422. Saves source code into the local compile workspace.                |
| /api/code/compile       | Line 3433. Compiles or runs a source file and returns terminal output.        |
| /api/code/install-tools | Line 3444. Attempts to install missing compiler tools.                        |

| Item                          | Explanation                                                                                                 |
|-------------------------------|-------------------------------------------------------------------------------------------------------------|
| /api/publish-target           | Line 3455. Reads or writes the Git publishing target and authorization state.                               |
| /api/install-git              | Line 3465. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior. |
| /api/app-update/install       | Line 3479. Checks or starts builder app update behavior.                                                    |
| /api/github-authenticate      | Line 3493. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior. |
| /api/sync-from-publish-target | Line 3510. Reads or writes the Git publishing target and authorization state.                               |
| /api/save-draft               | Line 3525. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior. |
| /api/apply-catalog            | Line 3525. Loads project/catalog data for the builder.                                                      |
| /api/apply-catalog            | Line 3534. Loads project/catalog data for the builder.                                                      |
| /api/upload                   | Line 3577. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior. |

## Important implementation signals

| Item                            | Explanation                                                                                                                                                                                                                |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compiler or simulator at line 1 | Part of the Compile Code workspace or language/tool detection. Evidence: import { createServer } from "node:http";                                                                                                         |
| Process execution at line 2     | Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: import { execFile, spawn } from "node:child_process";                                                                           |
| Compiler or simulator at line 3 | Part of the Compile Code workspace or language/tool detection. Evidence: import { createHash } from "node:crypto";                                                                                                         |
| File read at line 4             | Reads local builder state, project files, website assets, or configuration. Evidence: import { access as fsAccess, chmod, cp, mkdir, mkdtemp, readFile, readdir, rm, writeFile } from "node:fs/promises";                  |
| Compiler or simulator at line 5 | Part of the Compile Code workspace or language/tool detection. Evidence: import os from "node:os";                                                                                                                         |
| Compiler or simulator at line 6 | Part of the Compile Code workspace or language/tool detection. Evidence: import path from "node:path";                                                                                                                     |
| Compiler or simulator at line 7 | Part of the Compile Code workspace or language/tool detection. Evidence: import { promisify } from "node:util";                                                                                                            |
| Compiler or simulator at line 8 | Part of the Compile Code workspace or language/tool detection. Evidence: import { fileURLToPath } from "node:url";                                                                                                         |
| Process execution at line 15    | Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: const execFileAsync = promisify(execFile);                                                                                      |
| File read at line 16            | Reads local builder state, project files, website assets, or configuration. Evidence: const packageJson = JSON.parse(await readFile(path.join(root, "package.json"), "utf8").catch(() => "{}"));                           |
| Browser DOM at line 34          | Builder or website interface behavior in the browser/Electron renderer. Evidence: ".xlsx": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",                                                            |
| Security or auth at line 40     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthCachePath = path.join(root, ".omb-publish-session.json");                                                          |
| Security or auth at line 51     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "_headers",                                                                                                                         |
| Security or auth at line 57     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthCacheTtlMs = 24 * 60 * 60 * 1000;                                                                                  |
| Security or auth at line 58     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthExtendedTtlMs = 30 * 24 * 60 * 60 * 1000;                                                                          |
| Security or auth at line 59     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthHistoryWindowMs = 7 * 24 * 60 * 60 * 1000;                                                                         |
| Security or auth at line 60     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthExtendedThreshold = 3;                                                                                             |
| Security or auth at line 63     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthorizationHelp = [                                                                                                  |
| Security or auth at line 65     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Open Publishing target, enter the repository URL, then click Authenticate target.",                                                |
| Git command at line 66          | Repository state, publishing, branch checks, or release automation. Evidence: "A GitHub/Git Credential Manager browser sign-in may open. Sign in with an account that has write access to the selected Pages repository.", |
| Security or auth at line 67     | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Authenticate target only verifies write access and remembers it for the matching repository and branch.",                          |

| Item                              | Explanation                                                                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security or auth at line 68       | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Load from target only imports compatible website files from the authenticated repository into the local builder workspace.", |
| Git command at line 72            | Repository state, publishing, branch checks, or release automation. Evidence: process.env.GIT_EXE,                                                                                                                   |
| Git command at line 73            | Repository state, publishing, branch checks, or release automation. Evidence: "git",                                                                                                                                 |
| Git command at line 74            | Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files\\Git\\cmd\\git.exe",                                                                                                |
| Git command at line 75            | Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files\\Git\\bin\\git.exe",                                                                                                |
| Git command at line 76            | Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files (x86)\\Git\\cmd\\git.exe",                                                                                          |
| Git command at line 77            | Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files (x86)\\Git\\bin\\git.exe",                                                                                          |
| Git command at line 78            | Repository state, publishing, branch checks, or release automation. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Programs", "Git", "cmd", "git.exe") : "",                              |
| Git command at line 79            | Repository state, publishing, branch checks, or release automation. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Programs", "Git", "bin", "git.exe") : ""                               |
| Compiler or simulator at line 87  | Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["gcc"],                                                                                                                      |
| Compiler or simulator at line 97  | Part of the Compile Code workspace or language/tool detection. Evidence: verilog: {                                                                                                                                  |
| Compiler or simulator at line 100 | Part of the Compile Code workspace or language/tool detection. Evidence: label: "Verilog",                                                                                                                           |
| Compiler or simulator at line 101 | Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["iverilog", "vvp"],                                                                                                          |
| Compiler or simulator at line 102 | Part of the Compile Code workspace or language/tool detection. Evidence: winget: ["Icarus.Verilog"]                                                                                                                  |
| Compiler or simulator at line 104 | Part of the Compile Code workspace or language/tool detection. Evidence: systemverilog: {                                                                                                                            |
| Compiler or simulator at line 107 | Part of the Compile Code workspace or language/tool detection. Evidence: label: "SystemVerilog",                                                                                                                     |
| Compiler or simulator at line 108 | Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["iverilog", "vvp"],                                                                                                          |
| Compiler or simulator at line 109 | Part of the Compile Code workspace or language/tool detection. Evidence: winget: ["Icarus.Verilog"]                                                                                                                  |
| Compiler or simulator at line 118 | Part of the Compile Code workspace or language/tool detection. Evidence: java: {                                                                                                                                     |
| Compiler or simulator at line 119 | Part of the Compile Code workspace or language/tool detection. Evidence: defaultFile: "Main.java",                                                                                                                   |
| Compiler or simulator at line 120 | Part of the Compile Code workspace or language/tool detection. Evidence: extensions: [".java"],                                                                                                                      |
| Compiler or simulator at line 121 | Part of the Compile Code workspace or language/tool detection. Evidence: label: "Java",                                                                                                                              |
| Compiler or simulator at line 122 | Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["javac", "java"],                                                                                                            |
| Compiler or simulator at line 129 | Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["node"],                                                                                                                     |
| Compiler or simulator at line 132 | Part of the Compile Code workspace or language/tool detection. Evidence: python: {                                                                                                                                   |
| Compiler or simulator at line 135 | Part of the Compile Code workspace or language/tool detection. Evidence: label: "Python",                                                                                                                            |
| Compiler or simulator at line 136 | Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["python"],                                                                                                                   |
| Compiler or simulator at line 149 | Part of the Compile Code workspace or language/tool detection. Evidence: gcc: [                                                                                                                                      |

| Item                              | Explanation                                                                                                                                                                                                                                                                            |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compiler or simulator at line 150 | Part of the Compile Code workspace or language/tool detection. Evidence: "gcc",                                                                                                                                                                                                        |
| Compiler or simulator at line 151 | Part of the Compile Code workspace or language/tool detection. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Microsoft", "WinGet", "Packages", "BrechtSanders.WinLibs.POSIX.UCRT_Microsoft.Winget.Source_8wekyb3d8bbwe", "mingw64", "bin", "gcc.exe") : "" |
| Installer at line 156             | Windows setup, update, shortcut, or installation behavior. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Microsoft", "WinGet", "Packages", "BrechtSanders.WinLibs.POSIX.UCRT_Microsoft.Winget.Source_8wekyb3d8bbwe", "mingw64", "bin", "g++.exe") : ""     |
| Compiler or simulator at line 161 | Part of the Compile Code workspace or language/tool detection. Evidence: javac: [                                                                                                                                                                                                      |
| Compiler or simulator at line 162 | Part of the Compile Code workspace or language/tool detection. Evidence: "javac",                                                                                                                                                                                                      |
| Compiler or simulator at line 163 | Part of the Compile Code workspace or language/tool detection. Evidence: "C:\\Program Files\\Eclipse Adoptium\\jdk-21.0.11.10-hotspot\\bin\\javac.exe"                                                                                                                                 |
| Compiler or simulator at line 165 | Part of the Compile Code workspace or language/tool detection. Evidence: java: [                                                                                                                                                                                                       |
| Compiler or simulator at line 166 | Part of the Compile Code workspace or language/tool detection. Evidence: "java",                                                                                                                                                                                                       |
| Compiler or simulator at line 167 | Part of the Compile Code workspace or language/tool detection. Evidence: "C:\\Program Files\\Eclipse Adoptium\\jdk-21.0.11.10-hotspot\\bin\\java.exe"                                                                                                                                  |
| Compiler or simulator at line 169 | Part of the Compile Code workspace or language/tool detection. Evidence: node: [                                                                                                                                                                                                       |
| Compiler or simulator at line 170 | Part of the Compile Code workspace or language/tool detection. Evidence: /(?:^[\\])?node(?:\\.exe)?\$/i.test(process.execPath    "") ? process.execPath : "",                                                                                                                          |
| Compiler or simulator at line 171 | Part of the Compile Code workspace or language/tool detection. Evidence: "node",                                                                                                                                                                                                       |
| Compiler or simulator at line 172 | Part of the Compile Code workspace or language/tool detection. Evidence: "C:\\Program Files\\nodejs\\node.exe",                                                                                                                                                                        |
| Compiler or simulator at line 173 | Part of the Compile Code workspace or language/tool detection. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Programs", "nodejs", "node.exe") : ""                                                                                                         |
| Compiler or simulator at line 175 | Part of the Compile Code workspace or language/tool detection. Evidence: python: [process.env.PYTHON, "python", "py"],                                                                                                                                                                 |
| Compiler or simulator at line 176 | Part of the Compile Code workspace or language/tool detection. Evidence: iverilog: ["iverilog", "C:\\iverilog\\bin\\iverilog.exe"],                                                                                                                                                    |
| Compiler or simulator at line 177 | Part of the Compile Code workspace or language/tool detection. Evidence: vvp: ["vvp", "C:\\iverilog\\bin\\vvp.exe"],                                                                                                                                                                   |
| Installer at line 180             | Windows setup, update, shortcut, or installation behavior. Evidence: "C:\\Program Files\\ADI\\LTspice\\LTspice.exe",                                                                                                                                                                   |
| Installer at line 181             | Windows setup, update, shortcut, or installation behavior. Evidence: "C:\\Program Files\\LTC\\LTspiceXVII\\XVIIx64.exe",                                                                                                                                                               |
| Installer at line 182             | Windows setup, update, shortcut, or installation behavior. Evidence: "C:\\Program Files\\Analog Devices\\LTspice\\LTspice.exe",                                                                                                                                                        |
| Installer at line 183             | Windows setup, update, shortcut, or installation behavior. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Programs", "ADI", "LTspice", "LTspice.exe") : ""                                                                                                  |
| Local storage at line 186         | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: const compileToolCache = new Map();                                                                                                                                                      |
| Local storage at line 187         | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: const compileToolVersionCache = new Map();                                                                                                                                               |
| Cloudflare or AI at line 192      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "Answer the visitor's question first in a concise ChatGPT-like format, then add portfolio links or context only when they help.",                                                               |
| Security or auth at line 210      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Do not invent portfolio projects, credentials, employers, files, or test results that are not in the context.",                                                                                |
| Security or auth at line 218      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: function securityHeaders(extra = {}) {                                                                                                                                                          |
| Security or auth at line 222      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Permissions-Policy": "camera=(), microphone=(), geolocation=(), payment=(), usb=()",                                                                                                           |

| Item                              | Explanation                                                                                                                                          |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security or auth at line 230      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ...securityHeaders(),                                         |
| Local storage at line 232         | Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: "Cache-Control": "no-store, no-cache, must-revalidate" |
| Compiler or simulator at line 263 | Part of the Compile Code workspace or language/tool detection. Evidence: systemverilog: "systemverilog",                                             |
| Compiler or simulator at line 264 | Part of the Compile Code workspace or language/tool detection. Evidence: sv: "systemverilog",                                                        |

## Important constants and variables

| Item                         | Explanation                                                                                                                 |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| root                         | Line 10: const root = path.dirname(fileURLToPath(import.meta.url));                                                         |
| portfolioRoot                | Line 11: const portfolioRoot = path.resolve(process.env.OMB_PORTFOLIO_WORKSPACE    root);                                   |
| compileRoot                  | Line 12: const compileRoot = path.resolve(process.env.OMB_CODE_WORKSPACE    path.join(path.dirname(root), "compile-code")); |
| port                         | Line 13: const port = Number(process.env.PORT    8080);                                                                     |
| host                         | Line 14: const host = process.env.HOST    "0.0.0.0";                                                                        |
| execFileAsync                | Line 15: const execFileAsync = promisify(execFile);                                                                         |
| packageJson                  | Line 16: const packageJson = JSON.parse(await readFile(path.join(root, "package.json"), "utf8").catch(() => "{}"));         |
| types                        | Line 18: const types = {                                                                                                    |
| draftPath                    | Line 38: const draftPath = path.join(root, "projects.local.json");                                                          |
| catalogPath                  | Line 39: const catalogPath = path.join(root, "projects.json");                                                              |
| publishAuthCachePath         | Line 40: const publishAuthCachePath = path.join(root, ".omb-publish-session.json");                                         |
| publishPaths                 | Line 41: const publishPaths = [                                                                                             |
| publishAuthCacheTtlMs        | Line 57: const publishAuthCacheTtlMs = 24 * 60 * 60 * 1000;                                                                 |
| publishAuthExtendedTtlMs     | Line 58: const publishAuthExtendedTtlMs = 30 * 24 * 60 * 60 * 1000;                                                         |
| publishAuthHistoryWindowMs   | Line 59: const publishAuthHistoryWindowMs = 7 * 24 * 60 * 60 * 1000;                                                        |
| publishAuthExtendedThreshold | Line 60: const publishAuthExtendedThreshold = 3;                                                                            |
| defaultSiteRepository        | Line 61: const defaultSiteRepository = process.env.OMB_BUILDER_REPOSITORY    "";                                            |
| blockedAppUpdateVersions     | Line 62: const blockedAppUpdateVersions = new Set(["0.2.16"]);                                                              |
| publishAuthorizationHelp     | Line 63: const publishAuthorizationHelp = [                                                                                 |
| gitCandidates                | Line 71: const gitCandidates = [                                                                                            |
| compileLanguageProfiles      | Line 82: const compileLanguageProfiles = {                                                                                  |
| compileToolCandidates        | Line 148: const compileToolCandidates = {                                                                                   |
| compileToolCache             | Line 186: const compileToolCache = new Map();                                                                               |
| compileToolVersionCache      | Line 187: const compileToolVersionCache = new Map();                                                                        |
| portfolioAilInstructions     | Line 189: const portfolioAilInstructions = [                                                                                |
| clean                        | Line 257: const clean = String(value    "").trim().toLowerCase().replace(/[_\s-]+/g, "");                                   |
| aliases                      | Line 258: const aliases = {                                                                                                 |
| text                         | Line 283: const text = String(source    "");                                                                                |
| text                         | Line 289: const text = String(source    "");                                                                                |
| ext                          | Line 295: const ext = path.extname(String(fileName    "").toLowerCase());                                                   |
| byName                       | Line 305: const byName = languageFromFileName(fileName, code);                                                              |



| Item           | Explanation                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------------------|
| source         | Line 307: const source = String(code    "");                                                                                    |
| profile        | Line 322: const profile = compileLanguageProfiles[normalizeCodeLanguage(language)]    compileLanguageProfiles.javascript;       |
| fallback       | Line 323: const fallback = profile.defaultFile;                                                                                 |
| parsed         | Line 324: const parsed = path.parse(String(value    fallback));                                                                 |
| safeName       | Line 325: const safeName = safeSegment(parsed.name, path.parse(fallback).name);                                                 |
| ext            | Line 326: let ext = String(parsed.ext    path.extname(fallback)).toLowerCase();                                                 |
| depth          | Line 332: let depth = 0;                                                                                                        |
| line           | Line 337: const line = rawLine.trim();                                                                                          |
| formatted      | Line 340: const formatted = `\${" ".repeat(depth)}\${line}`;                                                                    |
| opens          | Line 341: const opens = (line.match(/[{}()]/g)    []).length;                                                                   |
| closes         | Line 342: const closes = (line.match(/[\[\]]/g)    []).length;                                                                  |
| normalized     | Line 351: const normalized = String(code    "").replace(/\r\n?/g, "\n").replace(/\t/g, " ");                                    |
| lang           | Line 352: const lang = normalizeCodeLanguage(language);                                                                         |
| entries        | Line 376: let entries = [];                                                                                                     |
| fullPath       | Line 383: const fullPath = path.join(folder, entry.name);                                                                       |
| found          | Line 388: const found = await findExecutableUnder(path.join(folder, entry.name), names, maxDepth - 1);                          |
| candidates     | Line 396: const candidates = (compileToolCandidates[toolName]    [toolName]).filter(Boolean);                                   |
| command        | Line 406: const command = process.platform === "win32" ? "where" : "command";                                                   |
| args           | Line 407: const args = process.platform === "win32" ? ["-v", candidate];                                                        |
| result         | Line 408: const result = await execFileAsync(command, args, { timeout: 5000, windowsHide: true });                              |
| found          | Line 409: const found = String(result.stdout    "").split(/\r?\n/).map((line) => line.trim()).find(Boolean);                    |
| found          | Line 420: const found = await findExecutableUnder(                                                                              |
| found          | Line 430: const found = await findExecutableUnder("C:\\Program Files\\Eclipse Adoptium", [`\${toolName}.exe`], 5);              |
| elapsedSeconds | Line 444: const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";                 |
| elapsed        | Line 445: const elapsed = elapsedSeconds ? `in \${elapsedSeconds}s` : "";                                                       |
| status         | Line 446: const status = result.timedOut                                                                                        |
| output         | Line 449: const output = [result.stdout, result.stderr].map((part) => String(part    "").trimEnd()).filter(Boolean).join("\n"); |
| clean          | Line 454: const clean = String(value    "");                                                                                    |
| output         | Line 464: let output = String(text    "");                                                                                      |
| from           | Line 466: const from = replacement?.from    "";                                                                                 |
| to             | Line 467: const to = replacement?.to    "";                                                                                     |
| isCpp          | Line 481: const isCpp = language === "cpp";                                                                                     |
| ext            | Line 482: const ext = path.extname(String(fileName    "").toLowerCase());                                                       |
| header         | Line 483: const header = [".h", ".hpp", ".hh", ".hxx"].includes(ext);                                                           |
| parsed         | Line 504: const parsed = path.parse(String(fileName    "program"));                                                             |
| result         | Line 511: const result = await runProcess(toolPath, ["--version"], { timeoutMs: 7000 });                                        |
| version        | Line 512: const version = [result.stdout, result.stderr]                                                                        |
| elapsedSeconds | Line 524: const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";                 |
| status         | Line 525: const status = result.timedOut                                                                                        |
| output         | Line 528: const output = [result.stdout, result.stderr].map((part) => String(part    "").trimEnd()).filter(Boolean).join("\n"); |

| Item           | Explanation                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------|
| raw            | Line 533: const raw = [result.stdout, result.stderr].map((part) => String(part    "").trimEnd()).filter(Boolean).join("\n"); |
| lines          | Line 534: const lines = raw.split(/\r?\n/).map((line) => line.trimEnd()).filter(Boolean);                                    |
| finishLines    | Line 535: const finishLines = [];                                                                                            |
| signalLines    | Line 536: const signalLines = [];                                                                                            |
| elapsedSeconds | Line 544: const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";              |
| status         | Line 545: const status = result.timedOut                                                                                     |
| body           | Line 548: const body = signalLines.length                                                                                    |
| suffix         | Line 555: const suffix = signalLines.length && finishLines.length ? `\\n\${finishLines.join("\\n")}` : "";                   |
| timeoutMs      | Line 560: const timeoutMs = options.timeoutMs    20000;                                                                      |

## JSON/YAML-style keys detected

| Item                       | Explanation                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------|
| .css                       | Line 19: ".css": "text/css",                                                                    |
| .csv                       | Line 20: ".csv": "text/csv",                                                                    |
| .html                      | Line 21: ".html": "text/html",                                                                  |
| .ico                       | Line 22: ".ico": "image/x-icon",                                                                |
| .js                        | Line 23: ".js": "application/javascript",                                                       |
| .json                      | Line 24: ".json": "application/json",                                                           |
| .md                        | Line 25: ".md": "text/markdown",                                                                |
| .pdf                       | Line 26: ".pdf": "application/pdf",                                                             |
| .png                       | Line 27: ".png": "image/png",                                                                   |
| .jpg                       | Line 28: ".jpg": "image/jpeg",                                                                  |
| .jpeg                      | Line 29: ".jpeg": "image/jpeg",                                                                 |
| .svg                       | Line 30: ".svg": "image/svg+xml",                                                               |
| .txt                       | Line 31: ".txt": "text/plain",                                                                  |
| .webp                      | Line 32: ".webp": "image/webp",                                                                 |
| .xml                       | Line 33: ".xml": "application/xml",                                                             |
| .xlsx                      | Line 34: ".xlsx": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",          |
| .zip                       | Line 35: ".zip": "application/zip"                                                              |
| g++                        | Line 153: "g++": [                                                                              |
| clang++                    | Line 159: "clang++": ["clang++"],                                                               |
| X-Content-Type-Options     | Line 220: "X-Content-Type-Options": "nosniff",                                                  |
| Referrer-Policy            | Line 221: "Referrer-Policy": "strict-origin-when-cross-origin",                                 |
| Permissions-Policy         | Line 222: "Permissions-Policy": "camera=(), microphone=(), geolocation=(), payment=(), usb=()", |
| Cross-Origin-Opener-Policy | Line 223: "Cross-Origin-Opener-Policy": "same-origin",                                          |
| Content-Type               | Line 231: "Content-Type": "application/json",                                                   |
| Cache-Control              | Line 232: "Cache-Control": "no-store, no-cache, must-revalidate"                                |
| c++                        | Line 262: "c++": "cpp",                                                                         |
| Accept                     | Line 1386: "Accept": accept,                                                                    |
| User-Agent                 | Line 1387: "User-Agent": "OMB-Portfolio-Builder-AI"                                             |
| Authorization              | Line 3314: "Authorization": `Bearer \${apiKey}`,                                                |

| Item          | Explanation                                                                             |
|---------------|-----------------------------------------------------------------------------------------|
| Content-Type  | Line 3315: "Content-Type": "application/json"                                           |
| Content-Type  | Line 3621: "Content-Type": types[path.extname(filePath)]    "application/octet-stream", |
| Cache-Control | Line 3622: "Cache-Control": "no-store, no-cache, must-revalidate"                       |

## Function-by-function detail

### *securityHeaders - lines 218-226*

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 218-226 (9 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function securityHeaders(extra = {}) {                                                                                                                                                                |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

218: function securityHeaders(extra = {}) {
219:   return {
220:     "X-Content-Type-Options": "nosniff",
221:     "Referrer-Policy": "strict-origin-when-cross-origin",
222:     "Permissions-Policy": "camera=(), microphone=(), geolocation=(), payment=(), usb=()",
223:     "Cross-Origin-Opener-Policy": "same-origin",
224:     ...extra
225:   };
226: }

```

### *sendJson - lines 228-235*

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 228-235 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function sendJson(response, status, data) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | writeHead, securityHeaders, end, stringify                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

228: function sendJson(response, status, data) {
229:   response.writeHead(status, {
230:     ...securityHeaders(),
231:     "Content-Type": "application/json",
232:     "Cache-Control": "no-store, no-cache, must-revalidate"
233:   });
234:   response.end(JSON.stringify(data, null, 2));
235: }

```

***clampText - lines 237-239***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 237-239 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function clampText(value, maxLength = 12000) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | slice                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
237: function clampText(value, maxLength = 12000) {
238:   return String(value || "").slice(0, maxLength);
239: }
```

***stripHtmlToText - lines 241-254***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 241-254 (14 source lines).                                                                                                                                                                      |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function stripHtmlToText(value = "") {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replace, trim                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
241: function stripHtmlToText(value = "") {
242:   return String(value || "")
243:     .replace(/<script[\s\S]*?<\script>/gi, " ")
244:     .replace(/<style[\s\S]*?<\style>/gi, " ")
245:     .replace(/[<^>]+>/g, " ")
246:     .replace(/&nbsp;/gi, " ")
247:     .replace(/&amp;/gi, "&")
248:     .replace(/&lt;/gi, "<")
249:     .replace(/&gt;/gi, ">")
250:     .replace(/&quot;/gi, "'")
251:     .replace(/&#39;/g, "'")
252:     .replace(/\\s+/g, " ")
253:     .trim();
254: }
```

***normalizeCodeLanguage - lines 256-280***

| Item            | Explanation                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Source location | Lines 256-280 (25 source lines).                                              |
| Purpose         | Validates or normalizes input so later code receives safe predictable values. |
| Declaration     | function normalizeCodeLanguage(value = "") {                                  |
| Touches         | filesystem                                                                    |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | trim, toLowerCase, replace                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

256: function normalizeCodeLanguage(value = "") {
257:   const clean = String(value || "").trim().toLowerCase().replace(/[_\s-]+/g, "");
258:   const aliases = {
259:     c: "c",
260:     cpp: "cpp",
261:     cplusplus: "cpp",
262:     "c++": "cpp",
263:     systemverilog: "systemverilog",
264:     sv: "systemverilog",
265:     verilog: "verilog",
266:     v: "verilog",
267:     ltspice: "ltspice",
268:     spice: "ltspice",
269:     cir: "ltspice",
270:     java: "java",
271:     javascript: "javascript",
272:     js: "javascript",
273:     node: "javascript",
274:     python: "python",
275:     py: "python",
276:     html: "html",
277:     htm: "html"
278:   };
279:   return aliases[clean] || (compileLanguageProfiles[clean] ? clean : "javascript");
280: }

```

## sourceLooksCpp - lines 282-286

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 282-286 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function sourceLooksCpp(source = "") {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | b                                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

282: function sourceLooksCpp(source = "") {
283:   const text = String(source || "");
284:   return
285:   /#include\s*<(?:iostream|string|vector|array|map|unordered_map|memory|algorithm|optional|variant)>/.test(text)
  ||
  285:   /\b(std::|cout|cin|cerr|namespace|template\s*<|class\s+\w+|new\s+\w+|delete\s+|constexpr|nullptr|using
\s+namespace)\b/.test(text);
286: }

```

## sourceLooksC - lines 288-292

| Item            | Explanation                                                                                |
|-----------------|--------------------------------------------------------------------------------------------|
| Source location | Lines 288-292 (5 source lines).                                                            |
| Purpose         | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function sourceLooksC(source = "") {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | b                                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

288: function sourceLooksC(source = "") {
289:   const text = String(source || "");
290:   return /#include\s*<(?:stdio|stdlib|string|stdint|stdbool|math)\.h>/ .test(text) ||
291:     /\b(printf|scanf|malloc|calloc|realloc|free|struct\s+\w+|typedef\s+struct)\b/ .test(text);
292: }

```

## ***languageFromFileName - lines 294-302***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 294-302 (9 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function languageFromFileName(fileName = "", code = "") {                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | extname, toLowerCase, sourceLooksCpp, sourceLooksC, entries, find, includes                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

294: function languageFromFileName(fileName = "", code = "") {
295:   const ext = path.extname(String(fileName || "").toLowerCase());
296:   if (ext === ".h") {
297:     if (sourceLooksCpp(code)) return "cpp";
298:     if (sourceLooksC(code)) return "c";
299:     return "c";
300:   }
301:   return Object.entries(compileLanguageProfiles).find(([ , profile]) =>
profile.extensions.includes(ext))?.[0] || "";
302: }

```

## ***detectCodeLanguageFromSource - lines 304-319***

| Item               | Explanation                                                                                |
|--------------------|--------------------------------------------------------------------------------------------|
| Source location    | Lines 304-319 (16 source lines).                                                           |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow. |
| Declaration        | function detectCodeLanguageFromSource(code = "", fileName = "") {                          |
| Touches            | Mostly local calculations or local UI state.                                               |
| Calls detected     | languageFromFileName, b, sourceLooksCpp, sourceLooksC                                      |
| API endpoints      | None detected inside this function body.                                                   |
| Storage keys       | No local/session storage keys detected.                                                    |
| Async/return clues | 0 await expression(s), 10 return statement(s).                                             |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

304: function detectCodeLanguageFromSource(code = "", fileName = "") {
305:   const byName = languageFromFileName(fileName, code);
306:   if (byName) return byName;
307:   const source = String(code || "");
308:   if (/<\/?[a-z][\s\S]*?>/i.test(source) || /<!doctype\s+html/i.test(source)) return "html";
309:   if (/b(always_ff|always_comb|always_latch|logic|interface|covergroup|assert\s+property|typedef\s+enum|
class\s+w+)\b/.test(source)) return "systemverilog";
310:   if (/b(module|endmodule|always|assign|reg|wire|initial|posedge|negedge)\b/.test(source)) return
"verilog";
311:   if (/^s*(?!(tran|ac|dc|op|model|subckt|ends|param)\b/im.test(source) ||
/bV\w+\s+\w+\s+\w+(?:DC|SIN|PULSE)?/i.test(source)) return "ltspice";
312:   if (/b(def|elif|from\s+w+\s+import|self|None|True|False)\b/.test(source)) return "python";
313:   if (/b(public\s+class|static\s+void\s+main|System\.out)\b/.test(source)) return "java";
314:   if (sourceLooksCpp(source) || sourceLooksC(source) ||
/b(#include|main\s*\(|puts\s*\(|fprintf|sizeof\s*\(|)\b/.test(source)) {
315:     return sourceLooksCpp(source) ? "cpp" : "c";
316:   }
317:   if (/b(const|let|var|function|=>|console\.log|document\.|window\.)\b/.test(source)) return
"javascript";
318:   return "javascript";
319: }

```

## safeCodeFileName - lines 321-329

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 321-329 (9 source lines).                                                                                                                                                                       |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function safeCodeFileName(value = "", language = "javascript") {                                                                                                                                      |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeCodeLanguage, parse, safeSegment, extname, toLowerCase, includes                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

321: function safeCodeFileName(value = "", language = "javascript") {
322:   const profile = compileLanguageProfiles[normalizeCodeLanguage(language)] ||
compileLanguageProfiles.javascript;
323:   const fallback = profile.defaultFile;
324:   const parsed = path.parse(String(value || fallback));
325:   const safeName = safeSegment(parsed.name, path.parse(fallback).name);
326:   let ext = String(parsed.ext || path.extname(fallback)).toLowerCase();
327:   if (!profile.extensions.includes(ext)) ext = path.extname(fallback);
328:   return `${safeName}${ext}`;
329: }

```

## indentBraceCode - lines 331-345

| Item            | Explanation                                                                                |
|-----------------|--------------------------------------------------------------------------------------------|
| Source location | Lines 331-345 (15 source lines).                                                           |
| Purpose         | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow. |
| Declaration     | function indentBraceCode(code = "") {                                                      |
| Touches         | filesystem                                                                                 |
| Calls detected  | replace, split, map, trim, max, repeat, match                                              |
| API endpoints   | None detected inside this function body.                                                   |
| Storage keys    | No local/session storage keys detected.                                                    |





| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | pathExists, readdir, join, isFile, some, toLowerCase, isDirectory                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

374: async function findExecutableUnder(folder, names = [], maxDepth = 5) {
375:   if (!folder || maxDepth < 0 || !(await pathExists(folder))) return "";
376:   let entries = [];
377:   try {
378:     entries = await readdir(folder, { withFileTypes: true });
379:   } catch {
380:     return "";
381:   }
382:   for (const entry of entries) {
383:     const fullPath = path.join(folder, entry.name);
384:     if (entry.isFile() && names.some((name) => entry.name.toLowerCase() === name.toLowerCase())) return
fullPath;
385:   }
386:   for (const entry of entries) {
387:     if (!entry.isDirectory()) continue;
388:     const found = await findExecutableUnder(path.join(folder, entry.name), names, maxDepth - 1);
389:     if (found) return found;
390:   }
391:   return "";
392: }

```

## ***findTool - lines 394-437***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 394-437 (44 source lines).                                                                                                                                                                      |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function findTool(toolName) {                                                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | has, get, filter, isAbsolute, pathExists, set, execFileAsync, split, map, trim, find, includes                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 4 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

394: async function findTool(toolName) {
395:   if (compileToolCache.has(toolName)) return compileToolCache.get(toolName);
396:   const candidates = (compileToolCandidates[toolName] || [toolName]).filter(Boolean);
397:   for (const candidate of candidates) {
398:     if (path.isAbsolute(candidate)) {
399:       if (await pathExists(candidate)) {
400:         compileToolCache.set(toolName, candidate);
401:         return candidate;
402:       }
403:       continue;
404:     }
405:     try {
406:       const command = process.platform === "win32" ? "where" : "command";
407:       const args = process.platform === "win32" ? [candidate] : ["-v", candidate];
408:       const result = await execFileAsync(command, args, { timeout: 5000, windowsHide: true });
409:       const found = String(result.stdout || "").split(/\r?\n/).map((line) => line.trim()).find(Boolean);
410:       if (found) {
411:         compileToolCache.set(toolName, found);
412:         return found;
413:       }
414:     } catch {
415:       // Try the next candidate.
416:     }

```

```

417:   }
418:
419:   if (["gcc", "g++"].includes(toolName) && process.env.LOCALAPPDATA) {
420:     const found = await findExecutableUnder(
421:       path.join(process.env.LOCALAPPDATA, "Microsoft", "winGet", "Packages"),
422:       [toolName === "gcc" ? "gcc.exe" : "g++.exe"],
423:       7
424:     );
425:     compileToolCache.set(toolName, found || "");
426:     return found || "";
427:   }
428:
429:   if (["javac", "java"].includes(toolName)) {
430:     const found = await findExecutableUnder("C:\\Program Files\\Eclipse Adoptium", [`${toolName}.exe`],
5);
431:     compileToolCache.set(toolName, found || "");
432:     return found || "";
433:   }
434:
435:   compileToolCache.set(toolName, "");
436:   return "";
437: }

```

## ***terminalLine - lines 439-441***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 439-441 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function terminalLine(label, text = "") {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim, filter, join                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

439: function terminalLine(label, text = "") {
440:   return [`$ ${label}`, String(text || "").trim()].filter(Boolean).join("\n");
441: }

```

## ***processTerminalText - lines 443-451***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 443-451 (9 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function processTerminalText(result = {}) {                                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isFinite, toFixed, map, trimEnd, filter, join                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

443: function processTerminalText(result = {}) {
444:   const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";
445:   const elapsed = elapsedSeconds ? ` in ${elapsedSeconds}s` : "";
446:   const status = result.timedOut
447:     ? `Timed out after ${elapsedSeconds} ? ${elapsedSeconds}s` : "the configured timeout"`
448:     : `Exit code ${result.code ?? "unknown"}${elapsed}`;
449:   const output = [result.stdout, result.stderr].map((part) => String(part ||

```

```

    "").trimEnd()).filter(Boolean).join("\n");
    450:   return [status, output || (result.ok ? "Completed without diagnostic output." : "No compiler output was
    returned.")]].join("\n");
    451: }

```

## ***pathVariantsForReplacement - lines 453-461***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 453-461 (9 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function pathVariantsForReplacement(value = "") {                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | from, Set, normalize, replace, filter                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

    453: function pathVariantsForReplacement(value = "") {
    454:   const clean = String(value || "");
    455:   return Array.from(new Set([
    456:     clean,
    457:     path.normalize(clean),
    458:     clean.replace(/\\\/g, "/"),
    459:     path.normalize(clean).replace(/\\\/g, "/")
    460:   ]).filter(Boolean));
    461: }

```

## ***replacePathReferences - lines 463-474***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 463-474 (12 source lines).                                                                                                                                                                      |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function replacePathReferences(text = "", replacements = []) {                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | pathVariantsForReplacement, split, join                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

    463: function replacePathReferences(text = "", replacements = []) {
    464:   let output = String(text || "");
    465:   for (const replacement of replacements) {
    466:     const from = replacement?.from || "";
    467:     const to = replacement?.to || "";
    468:     if (!from || !to) continue;
    469:     for (const variant of pathVariantsForReplacement(from)) {
    470:       output = output.split(variant).join(to);
    471:     }
    472:   }
    473:   return output;
    474: }

```

## ***processTerminalTextWithPaths - lines 476-478***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 476-478 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function processTerminalTextWithPaths(result = {}, replacements = []) {                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | replacePathReferences, processTerminalText                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

476: function processTerminalTextWithPaths(result = {}, replacements = []) {
477:   return replacePathReferences(processTerminalText(result), replacements);
478: }

```

## ***cFamilyCompileProfile - lines 480-501***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 480-501 (22 source lines).                                                                                                                                                                      |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function cFamilyCompileProfile(language = "c", fileName = "main.c") {                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | extname, toLowerCase, includes                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

480: function cFamilyCompileProfile(language = "c", fileName = "main.c") {
481:   const isCpp = language === "cpp";
482:   const ext = path.extname(String(fileName || "")).toLowerCase();
483:   const header = [".h", ".hpp", ".hh", ".hxx"].includes(ext);
484:   return {
485:     compiler: isCpp ? "g++" : "gcc",
486:     standard: isCpp ? "-std=c++20" : "-std=c17",
487:     languageFlag: isCpp ? "c++" : "c",
488:     label: isCpp ? "C++" : "C",
489:     header,
490:     flags: [
491:       isCpp ? "-std=c++20" : "-std=c17",
492:       "-Wall",
493:       "-Wextra",
494:       "-Wpedantic",
495:       "-Wshadow",
496:       "-O0",
497:       "-g3",
498:       "-fdiagnostics-color=never"
499:     ]
500:   };
501: }

```

## ***cFamilyBinaryName - lines 503-506***

| Item            | Explanation                                                                                |
|-----------------|--------------------------------------------------------------------------------------------|
| Source location | Lines 503-506 (4 source lines).                                                            |
| Purpose         | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function cFamilyBinaryName(fileName = "program") {                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | parse, safeSegment                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

503: function cFamilyBinaryName(fileName = "program") {
504:   const parsed = path.parse(String(fileName || "program"));
505:   return `${safeSegment(parsed.name, "program")}.exe`;
506: }

```

## ***toolVersionLine - lines 508-521***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 508-521 (14 source lines).                                                                                                                                                                      |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function toolVersionLine(toolPath = "") {                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | has, get, runProcess, map, trim, filter, join, split, find, basename, set                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

508: async function toolVersionLine(toolPath = "") {
509:   if (!toolPath) return "";
510:   if (compileToolVersionCache.has(toolPath)) return compileToolVersionCache.get(toolPath);
511:   const result = await runProcess(toolPath, ["--version"], { timeoutMs: 7000 });
512:   const version = [result.stdout, result.stderr]
513:     .map((part) => String(part || "").trim())
514:     .filter(Boolean)
515:     .join("\n")
516:     .split(/\r?\n/)
517:     .map((line) => line.trim())
518:     .find(Boolean) || path.basename(toolPath);
519:   compileToolVersionCache.set(toolPath, version);
520:   return version;
521: }

```

## ***cFamilyRunOutput - lines 523-530***

| Item            | Explanation                                                                                |
|-----------------|--------------------------------------------------------------------------------------------|
| Source location | Lines 523-530 (8 source lines).                                                            |
| Purpose         | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow. |
| Declaration     | function cFamilyRunOutput(result = {}) {                                                   |
| Touches         | Mostly local calculations or local UI state.                                               |
| Calls detected  | isFinite, toFixed, map, trimEnd, filter, join                                              |
| API endpoints   | None detected inside this function body.                                                   |
| Storage keys    | No local/session storage keys detected.                                                    |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem, external processes                                                                                                                                                                        |
| Calls detected     | now, spawn, execFile, kill, on, toString, call, write, end, resolve, trim                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

559: async function runProcess(command, args = [], options = {}) {
560:   const timeoutMs = options.timeoutMs || 20000;
561:   const cwd = options.cwd || compileRoot;
562:   return new Promise((resolve) => {
563:     const startedAt = Date.now();
564:     const child = spawn(command, args, {
565:       cwd,
566:       env: { ...process.env, ...(options.env || {}) },
567:       shell: false,
568:       windowsHide: true
569:     });
570:     let stdout = "";
571:     let stderr = "";
572:     let timedOut = false;
573:     const timer = setTimeout(() => {
574:       timedOut = true;
575:       if (process.platform === "win32" && child.pid) {
576:         execFile("taskkill", ["pid", String(child.pid), "/T", "/F"], { windowsHide: true }, () => {});
577:       }
578:       child.kill("SIGKILL");
579:     }, timeoutMs);
580:     child.stdout?.on("data", (chunk) => {
581:       stdout += chunk.toString();
582:     });
583:     child.stderr?.on("data", (chunk) => {
584:       stderr += chunk.toString();
585:     });
586:     if (Object.prototype.hasOwnProperty.call(options, "input")) {
587:       child.stdin?.write(String(options.input || ""));
588:     }
589:     child.stdin?.end();
590:     child.on("error", (error) => {
591:       clearTimeout(timer);
592:       resolve({ ok: false, code: -1, stdout, stderr: `${stderr}\n${error.message}`.trim(), timedOut,
elapsedMs: Date.now() - startedAt });
593:     });
594:     child.on("close", (code) => {
595:       clearTimeout(timer);
596:       resolve({ ok: !timedOut && code === 0, code, stdout, stderr, timedOut, elapsedMs: Date.now() -
startedAt });
597:     });
598:   });
599: }

```

## ***compileToolStatus - lines 601-620***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 601-620 (20 source lines).                                                                                                                                                                      |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | async function compileToolStatus() {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Set, values, flatMap, findTool, entries, every, fromEntries, map                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

### *isHdlLanguage* - lines 622-624

***inferCompileFileRole* - lines 626-633**

```

626: function inferCompileFileRole(fileName = "", code = "", language = "") {
627:   if (!isHdlLanguage(language)) return "source";
628:   const haystack = `${fileName}\n${code}`.toLowerCase();
629:   if (/^\b(tb|testbench)\b|(^|[_-])tb(?:_|\.|\\)|test(?:_|\\)?bench|\\$dumpfile|\\$dumpvars|module\s+tb(?:_a-z0-9)*\/i.test(haystack)) {
630:     return "testbench";
631:   }
632:   return "design";

```



633: }

## ***normalizeCompileFileRole - lines 635-639***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 635-639 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function normalizeCompileFileRole(value = "", language = "") {                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isHdlLanguage, trim, toLowerCase, replace, includes                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
635: function normalizeCompileFileRole(value = "", language = "") {
636:   if (!isHdlLanguage(language)) return "source";
637:   const clean = String(value || "").trim().toLowerCase().replace(/[_\s-]+/g, "");
638:   return ["tb", "testbench", "bench"].includes(clean) ? "testbench" : "design";
639: }
```

## ***saveCompileSource - lines 641-662***

| Item               | Explanation                                                                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 641-662 (22 source lines).                                                                                                                                                                              |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                          |
| Declaration        | async function saveCompileSource({ projectId, fileId, title, fileName, language, role = "", code, stdin = "" }) {                                                                                             |
| Touches            | filesystem                                                                                                                                                                                                    |
| Calls detected     | normalizeCodeLanguage, detectCodeLanguageFromSource, safeSegment, safeCodeFileName, parse, resolveInsideCompileRoot, mkdir, writeFile, clampText, normalizeCompileFileRole, inferCompileFileRole, toISOString |
| API endpoints      | None detected inside this function body.                                                                                                                                                                      |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                       |
| Async/return clues | 4 await expression(s), 1 return statement(s).                                                                                                                                                                 |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.         |

Source excerpt:

```
641: async function saveCompileSource({ projectId, fileId, title, fileName, language, role = "", code, stdin =
"" }) {
642:   const lang = normalizeCodeLanguage(language || detectCodeLanguageFromSource(code, fileName));
643:   const projectFolder = safeSegment(projectId, "project");
644:   const codeFileName = safeCodeFileName(fileName, lang);
645:   const fileFolder = safeSegment(fileId || path.parse(codeFileName).name, path.parse(codeFileName).name);
646:   const folder = resolveInsideCompileRoot(projectFolder, fileFolder);
647:   const sourcePath = resolveInsideCompileRoot(projectFolder, fileFolder, codeFileName);
648:   await mkdir(folder, { recursive: true });
649:   await writeFile(sourcePath, String(code || ""), "utf8");
650:   if (stdin) await writeFile(resolveInsideCompileRoot(projectFolder, fileFolder, "stdin.txt"),
String(stdin), "utf8");
651:   const metadata = {
652:     id: fileFolder,
653:     title: clampText(title || codeFileName, 180),
654:     fileName: codeFileName,
655:     language: lang,
656:     role: normalizeCompileFileRole(role || inferCompileFileRole(codeFileName, code, lang), lang),
657:     sourcePath,
658:     savedAt: new Date().toISOString()
659:   };
660:   await writeFile(resolveInsideCompileRoot(projectFolder, fileFolder, "compile-meta.json"),
`${JSON.stringify(metadata, null, 2)}\n`, "utf8");
661:   return metadata;
662: }
```

## ***javaMainClassName - lines 664-670***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 664-670 (7 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function javaMainClassName(code = "", fileName = "Main.java") {                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | match, parse                                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
664: function javaMainClassName(code = "", fileName = "Main.java") {
665:   const source = String(code || "");
666:   const publicMatch = source.match(/bpublic\s+class\s+([A-Za-z_$][\w$]*)/);
667:   if (publicMatch) return publicMatch[1];
668:   const classMatch = source.match(/bclass\s+([A-Za-z_$][\w$]*)/);
669:   return classMatch?.[1] || path.parse(fileName).name || "Main";
670: }
```

## ***compileCacheKey - lines 672-677***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 672-677 (6 source lines).                                                                                                                                                                       |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function compileCacheKey(parts = {}) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | createHash, update, stringify, digest, slice                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
672: function compileCacheKey(parts = {}) {
673:   return createHash("sha256")
674:     .update(JSON.stringify(parts))
675:     .digest("hex")
676:     .slice(0, 24);
677: }
```

## ***compileCacheDirectory - lines 679-681***

| Item            | Explanation                                                |
|-----------------|------------------------------------------------------------|
| Source location | Lines 679-681 (3 source lines).                            |
| Purpose         | Part of the Compile Code language/tool/build/run flow.     |
| Declaration     | function compileCacheDirectory(projectId, language, key) { |
| Touches         | Mostly local calculations or local UI state.               |
| Calls detected  | resolveInsideCompileRoot, safeSegment                      |
| API endpoints   | None detected inside this function body.                   |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

679: function compileCacheDirectory(projectId, language, key) {
680:   return resolveInsideCompileRoot(safeSegment(projectId, "project"), ".build-cache",
safeSegment(language, "language"), key);
681: }

```

## ***cachedBuildLine - lines 683-685***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 683-685 (3 source lines).                                                                                                                                                                       |
| Purpose            | Reads, writes, or validates cached state.                                                                                                                                                             |
| Declaration        | function cachedBuildLine(type, outputPath) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

683: function cachedBuildLine(type, outputPath) {
684:   return `${type} cache hit: using existing artifact\n${outputPath}`;
685: }

```

## ***hdlModuleNames - lines 687-689***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 687-689 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function hdlModuleNames(code = "") {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | matchAll, map                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

687: function hdlModuleNames(code = "") {
688:   return [...String(code || "").matchAll(/bmodule\s+([A-Za-z_$][\w$]*)/g)].map((match) => match[1]);
689: }

```

## ***hdlHasWaveDump - lines 691-694***

| Item            | Explanation                     |
|-----------------|---------------------------------|
| Source location | Lines 691-694 (4 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function hdlHasWaveDump(code = "") {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

691: function hdlHasWaveDump(code = "") {
692:   const source = String(code || "");
693:   return /\$dumpfile\s*\(/.test(source) && /\$dumpvars\s*\(/.test(source);
694: }

```

## ***hdlFilesFromPayload - lines 696-727***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 696-727 (32 source lines).                                                                                                                                                                      |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function hdlFilesFromPayload(payload = {}, activeFileName = "", activeLanguage = "verilog") {                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isArray, Map, normalizeCodeLanguage, detectCodeLanguageFromSource, isHdlLanguage, trim, safeCodeFileName, safeSegment, parse, set, clampText, normalizeCompileFileRole                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

696: function hdlFilesFromPayload(payload = {}, activeFileName = "", activeLanguage = "verilog") {
697:   const incoming = Array.isArray(payload.workspaceFiles) ? payload.workspaceFiles : [];
698:   const byId = new Map();
699:   const addFile = (item = {}, fallbackId = "") => {
700:     const code = String(item.code || "");
701:     const language = normalizeCodeLanguage(item.language || detectCodeLanguageFromSource(code,
item.fileName));
702:     if (!isHdlLanguage(language) || !code.trim()) return;
703:     const fileName = safeCodeFileName(item.fileName || activeFileName ||
compileLanguageProfiles[language].defaultFile, language);
704:     const id = safeSegment(item.id || fallbackId || fileName, path.parse(fileName).name);
705:     byId.set(id, {
706:       id,
707:       title: clampText(item.title || fileName, 180),
708:       fileName,
709:       language,
710:       role: normalizeCompileFileRole(item.role || inferCompileFileRole(fileName, code, language),
language),
711:       code
712:     });
713:   };
714:   incoming.forEach((item, index) => addFile(item, `hdl-${index + 1}`));
715:   addFile({
716:     id: payload.fileId,
717:     title: payload.title,
718:     fileName: activeFileName,
719:     language: activeLanguage,
720:     role: payload.role,
721:     code: payload.code
722:   }, "active");
723:   return [...byId.values()].sort((a, b) => {
724:     if (a.role === b.role) return a.fileName.localeCompare(b.fileName);
725:     return a.role === "testbench" ? 1 : -1;

```

```

726:   });
727: }

```

## ***compileActionFromPayload - lines 729-733***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 729-733 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function compileActionFromPayload(value = "", language = "") {                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, toLowerCase, replace, includes, isHdlLanguage                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

729: function compileActionFromPayload(value = "", language = "") {
730:   const clean = String(value || "").trim().toLowerCase().replace(/[_\s-]+/g, "-");
731:   if (["compile", "build", "run", "simulate"].includes(clean)) return clean;
732:   return isHdlLanguage(language) ? "simulate" : "run";
733: }

```

## ***compileActionLabel - lines 735-740***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 735-740 (6 source lines).                                                                                                                                                                       |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function compileActionLabel(action = "run", language = "") {                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isHdlLanguage                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

735: function compileActionLabel(action = "run", language = "") {
736:   if (action === "compile") return "Compile";
737:   if (action === "build") return "Build project";
738:   if (action === "simulate") return "Simulate";
739:   return isHdlLanguage(language) ? "Simulate" : "Run";
740: }

```

## ***compileWorkspaceFilesFromPayload - lines 742-770***

| Item            | Explanation                                                                                                                                                                   |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 742-770 (29 source lines).                                                                                                                                              |
| Purpose         | Part of the Compile Code language/tool/build/run flow.                                                                                                                        |
| Declaration     | function compileWorkspaceFilesFromPayload(payload = {}, activeFileName = "", activeLanguage = "javascript") {                                                                 |
| Touches         | filesystem                                                                                                                                                                    |
| Calls detected  | isArray, Map, trim, normalizeCodeLanguage, detectCodeLanguageFromSource, safeCodeFileName, safeSegment, parse, set, clampText, normalizeCompileFileRole, inferCompileFileRole |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

742: function compileWorkspaceFilesFromPayload(payload = {}, activeFileName = "", activeLanguage =
"javascript") {
743:   const incoming = Array.isArray(payload.workspaceFiles) ? payload.workspaceFiles : [];
744:   const byId = new Map();
745:   const addFile = (item = {}, fallbackId = "") => {
746:     const code = String(item.code || "");
747:     if (!code.trim()) return;
748:     const language = normalizeCodeLanguage(item.language || detectCodeLanguageFromSource(code,
item.fileName));
749:     const fileName = safeCodeFileName(item.fileName || activeFileName ||
compileLanguageProfiles[language]?.defaultFile || "source.txt", language);
750:     const id = safeSegment(item.id || fallbackId || fileName, path.parse(fileName).name);
751:     byId.set(id, {
752:       id,
753:       title: clampText(item.title || fileName, 180),
754:       fileName,
755:       language,
756:       role: normalizeCompileFileRole(item.role || inferCompileFileRole(fileName, code, language),
language),
757:       code
758:     });
759:   };
760:   incoming.forEach((item, index) => addFile(item, `workspace-${index + 1}`));
761:   addFile({
762:     id: payload.fileId,
763:     title: payload.title,
764:     fileName: activeFileName,
765:     language: activeLanguage,
766:     role: payload.role,
767:     code: payload.code
768:   }, "active");
769:   return [...byId.values()].sort((a, b) => a.fileName.localeCompare(b.fileName));
770: }

```

## ***writeCompileWorkspaceSources - lines 772-788***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 772-788 (17 source lines).                                                                                                                                                                      |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | async function writeCompileWorkspaceSources(files = [], targetDir = "", options = {}) {                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | mkdir, Map, includes, normalizeCodeLanguage, extname, toLowerCase, parse, get, set, join, writeFile, push                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

772: async function writeCompileWorkspaceSources(files = [], targetDir = "", options = {}) {
773:   await mkdir(targetDir, { recursive: true });
774:   const seen = new Map();
775:   const written = [];
776:   for (const file of files) {
777:     if (options.languages?.length && !options.languages.includes(normalizeCodeLanguage(file.language)))
continue;
778:     if (options.extensions?.length &&
!options.extensions.includes(path.extname(file.fileName).toLowerCase())) continue;
779:     const parsed = path.parse(file.fileName);
780:     const count = seen.get(file.fileName) || 0;
781:     seen.set(file.fileName, count + 1);
782:     const uniqueName = count ? `${parsed.name}_${count}${parsed.ext}` : file.fileName;
783:     const sourcePath = path.join(targetDir, uniqueName);

```

```

784:     await writeFile(sourcePath, file.code, "utf8");
785:     written.push({ ...file, uniqueName, sourcePath });
786:   }
787:   return written;
788: }

```

## sourceDisplayName - lines 790-792

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 790-792 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function sourceDisplayName(file = {}) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | basename                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

790: function sourceDisplayName(file = {}) {
791:   return file.uniqueName || file.fileName || path.basename(file.sourcePath || "source");
792: }

```

## cFamilyWorkspaceSources - lines 794-803

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 794-803 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function cFamilyWorkspaceSources(files = [], language = "c", activeFileName = "") {                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Set, filter, has, extname, toLowerCase, some, push                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

794: function cFamilyWorkspaceSources(files = [], language = "c", activeFileName = "") {
795:   const exts = language === "cpp"
796:     ? new Set([".cpp", ".cc", ".cxx", ".c++", ".hpp", ".hh", ".hxx", ".h"])
797:     : new Set([".c", ".h"]);
798:   const selected = files.filter((file) => exts.has(path.extname(file.fileName).toLowerCase()));
799:   if (!selected.some((file) => file.fileName === activeFileName)) {
800:     selected.push(...files.filter((file) => file.fileName === activeFileName));
801:   }
802:   return selected;
803: }

```

## writeHdlSimulationSources - lines 805-821

| Item            | Explanation                                                           |
|-----------------|-----------------------------------------------------------------------|
| Source location | Lines 805-821 (17 source lines).                                      |
| Purpose         | Persists data to local state, disk, credentials, or browser storage.  |
| Declaration     | async function writeHdlSimulationSources(files = [], cacheDir = "") { |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | join, rm, mkdir, Map, parse, get, set, writeFile, push                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

805: async function writeHdlSimulationSources(files = [], cacheDir = "") {
806:   const sourceDir = path.join(cacheDir, "sources");
807:   await rm(sourceDir, { recursive: true, force: true });
808:   await mkdir(sourceDir, { recursive: true });
809:   const seen = new Map();
810:   const written = [];
811:   for (const file of files) {
812:     const parsed = path.parse(file.fileName);
813:     const count = seen.get(file.fileName) || 0;
814:     seen.set(file.fileName, count + 1);
815:     const uniqueName = count ? `${parsed.name}_${count}${parsed.ext}` : file.fileName;
816:     const sourcePath = path.join(sourceDir, uniqueName);
817:     await writeFile(sourcePath, file.code, "utf8");
818:     written.push({ ...file, uniqueName, sourcePath });
819:   }
820:   return written;
821: }

```

## ***findFilesByExtension - lines 823-836***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 823-836 (14 source lines).                                                                                                                                                                      |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function findFilesByExtension(folder = "", extension = ".vcd", maxDepth = 3) {                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | pathExists, readdir, join, isDirectory, push, isFile, toLowerCase, endsWith                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

823: async function findFilesByExtension(folder = "", extension = ".vcd", maxDepth = 3) {
824:   if (!folder || maxDepth < 0 || !(await pathExists(folder))) return [];
825:   const entries = await readdir(folder, { withFileTypes: true }).catch(() => []);
826:   const matches = [];
827:   for (const entry of entries) {
828:     const fullPath = path.join(folder, entry.name);
829:     if (entry.isDirectory()) {
830:       matches.push(...await findFilesByExtension(fullPath, extension, maxDepth - 1));
831:     } else if (entry.isFile() && entry.name.toLowerCase().endsWith(extension.toLowerCase())) {
832:       matches.push(fullPath);
833:     }
834:   }
835:   return matches;
836: }

```

## ***normalizeVcdValue - lines 838-844***

| Item            | Explanation                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Source location | Lines 838-844 (7 source lines).                                               |
| Purpose         | Validates or normalizes input so later code receives safe predictable values. |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function normalizeVcdValue(raw = "") {                                                                                                                                                                |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim, toLowerCase, slice, replace                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

838: function normalizeVcdValue(raw = "") {
839:   const clean = String(raw || "").trim();
840:   if (!clean) return "x";
841:   if (/^[01xz]$/i.test(clean)) return clean.toLowerCase();
842:   if (/^[br][01xz_]+$/i.test(clean)) return clean.slice(1).replace(/_/g, "").toLowerCase();
843:   return clean.toLowerCase();
844: }

```

## ***parseVcdScopeText - lines 846-933***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 846-933 (88 source lines).                                                                                                                                                                      |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function parseVcdScopeText(text = "", source = "waveform.vcd") {                                                                                                                                      |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | split, Map, trim, startsWith, replace, push, includes, match, pop, has, set, filter                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

846: function parseVcdScopeText(text = "", source = "waveform.vcd") {
847:   const lines = String(text || "").split(/\r?\n/);
848:   const scopes = [];
849:   const signalsByCode = new Map();
850:   const timeScaleParts = [];
851:   let inTimescale = false;
852:   let time = 0;
853:   let maxTime = 0;
854:   let definitionDone = false;
855:   for (const rawLine of lines) {
856:     const line = rawLine.trim();
857:     if (!line) continue;
858:     if (line.startsWith("$timescale")) {
859:       inTimescale = true;
860:       const inline = line.replace("$timescale", "").replace("$end", "").trim();
861:       if (inline) timeScaleParts.push(inline);
862:       continue;
863:     }
864:     if (inTimescale) {
865:       if (line === "$end") {
866:         inTimescale = false;
867:       } else {
868:         timeScaleParts.push(line.replace("$end", "").trim());
869:         if (line.includes("$end")) inTimescale = false;
870:       }
871:       continue;
872:     }
873:     const scopeMatch = line.match(/^\$scope\s+\$s+(.?)\s+\$end$/);
874:     if (scopeMatch) {
875:       scopes.push(scopeMatch[1]);
876:       continue;
877:     }
878:     if (/^\$scope\b/.test(line)) {

```

```

879:     scopes.pop();
880:     continue;
881: }
882: const varMatch = line.match(/^$var\s+\S+\s+(\d+)\s+(\S+)\s+(.?)\s+\$end$/);
883: if (varMatch) {
884:     const [, width, code, reference] = varMatch;
885:     if (!signalsByCode.has(code) && signalsByCode.size < 64) {
886:         const cleanReference = reference.replace(/\s+\[^\s\]+\$/ , "");
887:         signalsByCode.set(code, {
888:             code,
889:             name: [...scopes, cleanReference].filter(Boolean).join("."),
890:             reference: cleanReference,
891:             width: Number(width) || 1,
892:             changes: []
893:         });
894:     }
895:     continue;
896: }
897: if (line.startsWith("$enddefinitions")) {
898:     definitionDone = true;
899:     continue;
900: }
901: if (!definitionDone) continue;
902: if (line.startsWith("#")) {
903:     time = Number(line.slice(1)) || 0;
904:     maxTime = Math.max(maxTime, time);
905:     continue;
906: }
907: let code = "";
908: let value = "";
909: const vectorMatch = line.match(/^(br)[01xz_]+\s+(\S+)\$/i);
910: if (vectorMatch) {
911:     value = normalizeVcdValue(vectorMatch[1]);
912:     code = vectorMatch[2];
913: } else if (/^[01xz]/i.test(line)) {
914:     value = normalizeVcdValue(line[0]);
915:     code = line.slice(1).trim();
916: }
917: const signal = signalsByCode.get(code);
918: if (!signal || !value) continue;
919: const previous = signal.changes[signal.changes.length - 1];
920: if (!previous || previous.value !== value || previous.time !== time) {
921:     if (signal.changes.length < 600) signal.changes.push({ time, value });
922: }
923: }
924: const signals = [...signalsByCode.values()]
925:     .filter((signal) => signal.changes.length)
926:     .slice(0, 32);
927: return {
928:     source: path.basename(source),
929:     timeScale: timeScaleParts.join(" ").replace(/\s+/g, " ").trim() || "ticks",
930:     maxTime,
931:     signals
932: };
933: }

```

## readHdlWaveform - lines 935-942

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 935-942 (8 source lines).                                                                                                                                                                       |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function readHdlWaveform(cacheDir = "") {                                                                                                                                                       |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | findFilesByExtension, sort, readFile, slice, parseVcdScopeText                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

935: async function readHdlWaveform(cacheDir = "") {
936:     const vcdFiles = await findFilesByExtension(cacheDir, ".vcd", 4);
937:     if (!vcdFiles.length) return null;
938:     const selected = vcdFiles.sort((a, b) => a.length - b.length)[0];
939:     const text = await readFile(selected, "utf8");
940:     const limited = text.length > 6_000_000 ? text.slice(0, 6_000_000) : text;
941:     return parseVcdScopeText(limited, selected);

```

942: }

## clearHdlWaveforms - lines 944-947

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 944-947 (4 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function clearHdlWaveforms(cacheDir = "") {                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | findFilesByExtension, all, map, rm                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
944: async function clearHdlWaveforms(cacheDir = "") {
945:   const vcdFiles = await findFilesByExtension(cacheDir, ".vcd", 4);
946:   await Promise.all(vcdFiles.map((filePath) => rm(filePath, { force: true })).catch(() => {}));
947: }
```

## compileAndRunCode - lines 949-1319

| Item               | Explanation                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 949-1319 (371 source lines).                                                                                                                                                                                         |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                                     |
| Declaration        | async function compileAndRunCode(payload = {}) {                                                                                                                                                                           |
| Touches            | filesystem, authentication/security data                                                                                                                                                                                   |
| Calls detected     | trim, toLowerCase, normalizeCodeLanguage, detectCodeLanguageFromSource, safeCodeFileName, saveCompileSource, safeSegment, resolveInsideCompileRoot, compileActionFromPayload, compileWorkspaceFilesFromPayload, async, now |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                   |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                    |
| Async/return clues | 44 await expression(s), 15 return statement(s).                                                                                                                                                                            |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                      |

Source excerpt:

```
949: async function compileAndRunCode(payload = {}) {
950:   const requestedLanguage = String(payload.language || "").trim().toLowerCase();
951:   const language = requestedLanguage && requestedLanguage !== "auto"
952:     ? normalizeCodeLanguage(requestedLanguage)
953:     : detectCodeLanguageFromSource(payload.code, payload.fileName);
954:   const profile = compileLanguageProfiles[language] || compileLanguageProfiles.javascript;
955:   const fileName = safeCodeFileName(payload.fileName, language);
956:   const saved = await saveCompileSource({ ...payload, language, fileName });
957:   const projectFolder = safeSegment(payload.projectId, "project");
958:   const sourcePath = resolveInsideCompileRoot(projectFolder, safeSegment(saved.id), saved.fileName);
959:   const sourceCode = String(payload.code || "");
960:   const action = compileActionFromPayload(payload.action, language);
961:   const allWorkspaceFiles = compileWorkspaceFilesFromPayload(payload, saved.fileName, language);
962:   let runDir = "";
963:   let runSourcePath = "";
964:   let runWorkspaceSources = [];
965:   const ensureRunSource = async (options = {}) => {
966:     if (runDir && runSourcePath) return { runDir, runSourcePath, writtenSources: runWorkspaceSources };
967:     const runId = `${Date.now()}-${Math.random().toString(16).slice(2)}`;
968:     runDir = resolveInsideCompileRoot(projectFolder, "runs", runId);
969:     await mkdir(runDir, { recursive: true });
970:     const filesToWrite = Array.isArray(options.files) && options.files.length ? options.files :
allWorkspaceFiles;
971:     runWorkspaceSources = await writeCompileWorkspaceSources(filesToWrite, runDir, {
972:       languages: options.languages,
```

```

973:     extensions: options.extensions
974:   });
975:   const active = runWorkspaceSources.find((file) => file.id === saved.id || file.fileName ===
saved.fileName);
976:   runSourcePath = active?.sourcePath || path.join(runDir, saved.fileName);
977:   if (!active) {
978:     await cp(sourcePath, runSourcePath, { force: true });
979:     runWorkspaceSources.push({
980:       id: saved.id,
981:       title: saved.title,
982:       fileName: saved.fileName,
983:       language,
984:       role: saved.role,
985:       code: sourceCode,
986:       uniqueName: saved.fileName,
987:       sourcePath: runSourcePath
988:     });
989:   }
990:   return { runDir, runSourcePath, writtenSources: runWorkspaceSources };
991: };
992: const stdin = String(payload.stdin || "");
993: const forceRebuild = payload.forceRebuild === true;
994: const terminal = [];
995: terminal.push(`${compileActionLabel(action, language)} request for ${saved.fileName}`);
996: const append = (label, result) => {
997:   terminal.push(terminalLine(label, processTerminalText(result)));
998: };
999: const missing = async (tools) => {
1000:   const found = {};
1001:   for (const tool of tools) found[tool] = await findTool(tool);
1002:   const missingTools = tools.filter((tool) => !found[tool]);
1003:   return { found, missingTools };
1004: };
1005:
1006: if (language === "html") {
1007:   const openTags = (String(payload.code || "").match(/<([a-z][\w:-]*)\b(?:^|>)*>/gi) || []).length;
1008:   const closeTags = (String(payload.code || "").match(/<\/([a-z][\w:-]*)>/gi) || []).length;
1009:   const ok = openTags >= closeTags;
1010:   return {
1011:     ok,
1012:     language,
1013:     saved,
1014:     terminal: [
1015:       "HTML validation pass",
1016:       ok ? "No obvious tag-balance issue was detected." : "The file may have more closing tags than
opening tags.",
1017:       `Saved source: ${saved.sourcePath}`
1018:     ].join("\n")
1019:   };
1020: }
1021:
1022: const stdinOptions = stdin ? { input: stdin } : {};
1023: let ok = false;
1024:
1025: if (language === "javascript") {
1026:   const node = await findTool("node");
1027:   if (!node) return { ok: false, language, saved, terminal: "Node.js was not found. Install Node.js to
run JavaScript." };
1028:   const jsFiles = allWorkspaceFiles.filter((file) => normalizeCodeLanguage(file.language) ===
"javascript");
1029:   const runSource = await ensureRunSource({ files: jsFiles.length ? jsFiles : allWorkspaceFiles,
languages: ["javascript"] });
1030:   const targets = action === "build"
? runSource.writtenSources.filter((file) => normalizeCodeLanguage(file.language) === "javascript")
: runSource.writtenSources.filter((file) => file.sourcePath === runSource.runSourcePath);
1031:   ok = true;
1032:   for (const target of targets.length ? targets : [{ sourcePath: runSource.runSourcePath, uniqueName:
saved.fileName }]) {
1033:     const check = await runProcess(node, ["--check", target.sourcePath], { cwd: runSource.runDir,
timeoutMs: 15000 });
1034:     terminal.push(terminalLine(`${path.basename(node)} --check ${sourceDisplayName(target)}`,
processTerminalText(check)));
1035:     ok = ok && check.ok;
1036:   }
1037:   if (action === "compile" || action === "build") {
1038:     terminal.push(ok ? "JavaScript syntax check completed." : "JavaScript syntax check found errors.");
1039:   } else if (ok) {
1040:     const run = await runProcess(node, [runSource.runSourcePath], { cwd: runSource.runDir, timeoutMs:
20000, ...stdinOptions });
1041:     append(`${path.basename(node)} ${saved.fileName}`, run);
1042:     ok = run.ok;
1043:   }
1044: } else if (language === "python") {
1045:   const python = await findTool("python");
1046:   if (!python) return { ok: false, language, saved, terminal: "Python was not found. Install Python to
run Python code." };
1047:   const pyFiles = allWorkspaceFiles.filter((file) => normalizeCodeLanguage(file.language) ===
"python");
1048:   const runSource = await ensureRunSource({ files: pyFiles.length ? pyFiles : allWorkspaceFiles,
languages: ["python"] });
1049:   const targets = action === "build"
? runSource.writtenSources.filter((file) => normalizeCodeLanguage(file.language) === "python")

```

```

1053:         : runSource.writtenSources.filter((file) => file.sourcePath === runSource.runSourcePath);
1054:         ok = true;
1055:         for (const target of targets.length ? targets : [{ sourcePath: runSource.runSourcePath, uniqueName:
saved.fileName }]) {
1056:             const check = await runProcess(python, ["-m", "py_compile", target.sourcePath], { cwd:
runSource.runDir, timeoutMs: 15000 });
1057:             terminal.push(terminalLine(`${path.basename(python)} -m py_compile ${sourceDisplayName(target)}`,
processTerminalText(check)));
1058:             ok = ok && check.ok;
1059:         }
1060:         if (action === "compile" || action === "build") {
1061:             terminal.push(ok ? "Python bytecode compile check completed." : "Python compile check found
errors.");
1062:         } else if (ok) {
1063:             const run = await runProcess(python, [runSource.runSourcePath], { cwd: runSource.runDir, timeoutMs:
20000, ...stdinOptions });
1064:             append(`${path.basename(python)} ${saved.fileName}`, run);
1065:             ok = run.ok;
1066:         }
1067:         } else if (language === "c" || language === "cpp") {
1068:             const cProfile = cFamilyCompileProfile(language, saved.fileName);
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

## append - lines 996-998

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 996-998 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | const append = (label, result) => {                                                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | push, terminalLine, processTerminalText                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

996:     const append = (label, result) => {
997:         terminal.push(terminalLine(label, processTerminalText(result)));
998:     };

```

## missing - lines 999-1004

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 999-1004 (6 source lines).                                                                                                                                                                      |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | const missing = async (tools) => {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | async, findTool, filter                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

999:     const missing = async (tools) => {
1000:         const found = {};
1001:         for (const tool of tools) found[tool] = await findTool(tool);
1002:         const missingTools = tools.filter((tool) => !found[tool]);
1003:         return { found, missingTools };
1004:     };

```

## installCompilerTools - lines 1321-1352

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1321-1352 (32 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | async function installCompilerTools(language = "") {                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeCodeLanguage, findTool, runProcess, push, terminalLine, join, processTerminalText, clear, compileToolStatus                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1321: async function installCompilerTools(language = "") {
1322:   const lang = normalizeCodeLanguage(language);
1323:   const profile = compileLanguageProfiles[lang] || compileLanguageProfiles.javascript;
1324:   if (!profile.winget?.length) {
1325:     return {
1326:       ok: false,
1327:       language: lang,
1328:       terminal: `${profile.label} does not have an automatic Winget compiler install configured. Install
the required tools manually, then restart the builder.`
1329:     };
1330:   }
1331:   const winget = await findTool("winget") || "winget";
1332:   const terminal = [];
1333:   let ok = true;
1334:   for (const packageId of profile.winget) {
1335:     const args = [
1336:       "install",
1337:       "--source",
1338:       "winget",
1339:       "--accept-source-agreements",
1340:       "--accept-package-agreements",
1341:       "--silent",
1342:       "--id",
1343:       packageId,
1344:       "--exact"
1345:     ];
1346:     const result = await runProcess(winget, args, { cwd: root, timeoutMs: 600000 });
1347:     terminal.push(terminalLine(`winget ${args.join(" ")}`), processTerminalText(result));
1348:     ok = ok && result.ok;
1349:   }
1350:   compileToolCache.clear();
1351:   return { ok, language: lang, terminal: terminal.join("\n\n"), tools: await compileToolStatus() };
1352: }
```

## sourceLooksTextual - lines 1354-1360

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1354-1360 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function sourceLooksTextual(source = {}) {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | toLowerCase, includes                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1354: function sourceLooksTextual(source = {}) {
1355:   const url = String(source.url || "");
1356:   const kind = String(source.kind || "").toLowerCase();
1357:   if ([ "text", "webpage", "github", "linkedin", "public_profile", "resume_text", "code" ].includes(kind))
return true;
1358:   return /\. (txt|md|markdown|csv|json|xml|log|c|h|cpp|hpp|py|js|mjs|ts|tsx|v|sv|vhd|?|spice|cir|net|asc|s
ch|kicad_sch|kicad_pcb|ino|xdc|sdc|tcl)$/i.test(url)
1359:   || /github\.com|raw\.githubusercontent\.com/i.test(url);
1360: }

```

## ***sourceUrlAllowed - lines 1362-1379***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1362-1379 (18 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function sourceUrlAllowed(url) {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | URL, toLowerCase, includes                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1362: function sourceUrlAllowed(url) {
1363:   try {
1364:     const parsed = new URL(url);
1365:     const hostName = parsed.hostname.toLowerCase();
1366:     return [
1367:       "localhost",
1368:       "127.0.0.1",
1369:       "github.com",
1370:       "api.github.com",
1371:       "raw.githubusercontent.com",
1372:       "gist.githubusercontent.com",
1373:       "linkedin.com",
1374:       "www.linkedin.com"
1375:     ].includes(hostName);
1376:   } catch {
1377:     return false;
1378:   }
1379: }

```

## ***gitHubHeaders - lines 1384-1389***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1384-1389 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function gitHubHeaders(accept = "application/vnd.github+json") {                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1384: function gitHubHeaders(accept = "application/vnd.github+json") {
1385:   return {
1386:     "Accept": accept,
1387:     "User-Agent": "OMB-Portfolio-Builder-AI"
1388:   };

```

1389: }

## ***parseGitHubSourceUrl - lines 1391-1418***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1391-1418 (28 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function parseGitHubSourceUrl(url) {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | URL, toLowerCase, split, filter, slice, join                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 7 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1391: function parseGitHubSourceUrl(url) {
1392:   try {
1393:     const parsed = new URL(url);
1394:     const host = parsed.hostname.toLowerCase();
1395:     const parts = parsed.pathname.split("/").filter(Boolean);
1396:     if (host === "raw.githubusercontent.com" && parts.length >= 4) {
1397:       return {
1398:         type: "file",
1399:         owner: parts[0],
1400:         repo: parts[1],
1401:         branch: parts[2],
1402:         filePath: parts.slice(3).join("/")
1403:       };
1404:     }
1405:     if (host !== "github.com" || !parts.length) return null;
1406:     if (parts.length === 1) return { type: "profile", owner: parts[0] };
1407:     const base = { owner: parts[0], repo: parts[1] };
1408:     if (parts[2] === "blob" && parts.length >= 5) {
1409:       return { ...base, type: "file", branch: parts[3], filePath: parts.slice(4).join("/") };
1410:     }
1411:     if (parts[2] === "tree" && parts.length >= 4) {
1412:       return { ...base, type: "repo", branch: parts[3] };
1413:     }
1414:     return { ...base, type: "repo" };
1415:   } catch {
1416:     return null;
1417:   }
1418: }
```

## ***fetchGitHubJson - lines 1420-1424***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1420-1424 (5 source lines).                                                                                                                                                                     |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function fetchGitHubJson(url) {                                                                                                                                                                 |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | fetch, gitHubHeaders, json                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1420: async function fetchGitHubJson(url) {
1421:   const response = await fetch(url, { headers: gitHubHeaders() });
1422:   if (!response.ok) return null;
```



```

1423:   return response.json().catch(() => null);
1424: }

```

## ***fetchLimitedText - lines 1426-1434***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1426-1434 (9 source lines).                                                                                                                                                                     |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function fetchLimitedText(url, maxLength = 12000) {                                                                                                                                             |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | fetch, gitHubHeaders, clampText, text, get, stripHtmlToText                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1426: async function fetchLimitedText(url, maxLength = 12000) {
1427:   const response = await fetch(url, {
1428:     headers: gitHubHeaders("text/plain,text/markdown,text/html,application/json;q=0.8,*/*;q=0.1")
1429:   });
1430:   if (!response.ok) return "";
1431:   const rawText = clampText(await response.text(), maxLength);
1432:   const contentType = response.headers.get("Content-Type") || "";
1433:   return /html/i.test(contentType) ? stripHtmlToText(rawText) : rawText;
1434: }

```

## ***githubQuestionTokens - lines 1436-1442***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1436-1442 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function githubQuestionTokens(question = "") {                                                                                                                                                        |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | Set, toLowerCase, replace, split, filter, includes                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1436: function githubQuestionTokens(question = "") {
1437:   return [...new Set(String(question || "").
1438:     .toLowerCase()
1439:     .replace(/[\^a-z0-9_#\+\.\s-]+/g, " ")
1440:     .split(/\s+/)
1441:     .filter((token) => token.length >= 3 && !["the", "and", "with", "from", "show", "code", "file",
"github"].includes(token))]];
1442: }

```

## ***scoreGitHubFile - lines 1444-1455***

| Item            | Explanation                                                            |
|-----------------|------------------------------------------------------------------------|
| Source location | Lines 1444-1455 (12 source lines).                                     |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior. |
| Declaration     | function scoreGitHubFile(filePath = "", question = "") {               |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | toLowerCase, githubQuestionTokens, b, forEach, includes                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1444: function scoreGitHubFile(filePath = "", question = "") {
1445:   const cleanPath = filePath.toLowerCase();
1446:   const tokens = githubQuestionTokens(question);
1447:   let score = githubTextFilePattern.test(cleanPath) ? 10 : 0;
1448:   if (/readme\.md$/i.test(cleanPath)) score += 55;
1449:   if (/\. (c|h|cpp|hpp|py|js|mjs|ts|v|sv|vhd|?|ino|tcl|xdc|sdc)$/i.test(cleanPath)) score += 28;
1450:   if (/b(test|tb|bench|sim|simulation|src|firmware|hardware|rtl|driver|main)\b/i.test(cleanPath)) score
+= 12;
1451:   tokens.forEach((token) => {
1452:     if (cleanPath.includes(token)) score += 18;
1453:   });
1454:   return score;
1455: }

```

## wantsGitHubCode - lines 1457-1459

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1457-1459 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function wantsGitHubCode(question = "") {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | b                                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1457: function wantsGitHubCode(question = "") {
1458:   return /\b(code|source|snippet|implementation|firmware|driver|module|verilog|vhd|python|javascript|typ
escript|c\+\+|cpp|c\s+code|pull|show|display)\b/i.test(question);
1459: }

```

## scoreGitHubRepo - lines 1461-1481

| Item               | Explanation                                                            |
|--------------------|------------------------------------------------------------------------|
| Source location    | Lines 1461-1481 (21 source lines).                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior. |
| Declaration        | function scoreGitHubRepo(repo = {}, question = "") {                   |
| Touches            | filesystem, authentication/security data                               |
| Calls detected     | githubQuestionTokens, isArray, join, toLowerCase, forEach, includes, b |
| API endpoints      | None detected inside this function body.                               |
| Storage keys       | No local/session storage keys detected.                                |
| Async/return clues | 0 await expression(s), 1 return statement(s).                          |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1461: function scoreGitHubRepo(repo = {}, question = "") {
1462:   const tokens = githubQuestionTokens(question);
1463:   const haystack = [
1464:     repo.full_name,
1465:     repo.name,
1466:     repo.description,
1467:     repo.language,
1468:     ...(Array.isArray(repo.topics) ? repo.topics : [])
1469:   ].join(" ").toLowerCase();
1470:   let score = repo.fork ? -10 : 12;
1471:   if (!repo.archived) score += 6;
1472:   if (repo.language) score += 4;
1473:   if (repo.name && !/\.github\.io$/i.test(repo.name)) score += 3;
1474:   tokens.forEach((token) => {
1475:     if (haystack.includes(token)) score += 22;
1476:   });
1477:   if (/\\b(vco|oscillator|pwm|analog|mixed|pcb|firmware|stm32|fpga|asic|verilog|embedded|signal|design)\\b/
i.test(haystack)) {
1478:     score += 12;
1479:   }
1480:   return score;
1481: }

```

## fetchGitHubProfileSource - lines 1483-1543

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1483-1543 (61 source lines).                                                                                                                                                                    |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function fetchGitHubProfileSource(source = {}, parsed = {}) {                                                                                                                                   |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | all, fetchGitHubJson, isArray, wantsGitHubCode, sort, scoreGitHubRepo, slice, fetchGitHubRepositorySource, trim, push, map, join                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1483: async function fetchGitHubProfileSource(source = {}, parsed = {}) {
1484:   const owner = parsed.owner;
1485:   const [profile, repos] = await Promise.all([
1486:     fetchGitHubJson(`https://api.github.com/users/${encodeURIComponent(owner)}`),
1487:     fetchGitHubJson(`https://api.github.com/users/${encodeURIComponent(owner)}/repos?per_page=30&sort=updated`)
1488:   ]);
1489:   const repoList = Array.isArray(repos) ? repos : [];
1490:   const includeCode = wantsGitHubCode(source.question || "");
1491:   const selectedRepos = includeCode
1492:     ? [...repoList]
1493:       .sort((a, b) => scoreGitHubRepo(b, source.question || "") - scoreGitHubRepo(a, source.question || ""))
1494:       .slice(0, 3)
1495:       : [];
1496:   const repoBlocks = [];
1497:   for (const repo of selectedRepos) {
1498:     if (!repo?.name || !repo?.html_url) continue;
1499:     const repoSource = await fetchGitHubRepositorySource({
1500:       ...source,
1501:       context: `Public GitHub repository selected from ${owner}'s profile`,
1502:       label: repo.full_name,
1503:       maxCodeFiles: 3,
1504:       maxFileTextLength: 3600,
1505:       maxRepositoryTextLength: 9000,
1506:       url: repo.html_url
1507:     }, {
1508:       owner,
1509:       repo: repo.name,
1510:     }

```

```

1511:         type: "repo",
1512:         branch: repo.default_branch
1513:     });
1514:     if (repoSource?.text?.trim()) repoBlocks.push(repoSource.text.trim());
1515: }
1516:
1517: const lines = [
1518:     `GitHub public profile: ${owner}`,
1519:     profile?.name ? `Name: ${profile.name}` : "",
1520:     profile?.bio ? `Bio: ${profile.bio}` : "",
1521:     profile?.html_url ? `Profile URL: ${profile.html_url}` : source.url,
1522:     "",
1523:     "Public repositories visible from the profile:",
1524:     ...repoList.slice(0, 20).map((repo) => [
1525:         `- ${repo.full_name}`,
1526:         repo.description ? `: ${repo.description}` : "",
1527:         repo.language ? ` | Language: ${repo.language}` : "",
1528:         repo.html_url ? ` | URL: ${repo.html_url}` : "",
1529:         repo.updated_at ? ` | Updated: ${repo.updated_at}` : ""
1530:     ]).join("")),
1531:     includeCode && selectedRepos.length ? "" : "",
1532:     includeCode && selectedRepos.length ? "Selected public repositories inspected for source code:" : "",
1533:     ...selectedRepos.map((repo) => `- ${repo.full_name}${repo.language ? ` | Language: ${repo.language}` : ""}${repo.description ? ` | ${repo.description}` : ""}`),
1534:     ...repoBlocks.flatMap((block) => ["", "---", block])
1535: ].filter(Boolean);
1536: return {
1537:     context: source.context || "Public GitHub profile",
1538:     title: source.title || source.label || `GitHub profile: ${owner}`,
1539:     type: includeCode ? "public GitHub profile and selected repository code" : "public GitHub profile",
1540:     url: source.url,
1541:     text: clampText(lines.join("\n"), includeCode ? 30000 : 9000)
1542: };
1543: }

```

## fetchGitHubRepositorySource - lines 1545-1612

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1545-1612 (68 source lines).                                                                                                                                                                    |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function fetchGitHubRepositorySource(source = {}, parsed = {}) {                                                                                                                                |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | fetchGitHubJson, wantsGitHubCode, max, min, isFinite, filter, fetchLimitedText, trim, push, clampText, join, isArray                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 5 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1545: async function fetchGitHubRepositorySource(source = {}, parsed = {}) {
1546:     const { owner, repo } = parsed;
1547:     const metadata = await
fetchGitHubJson(`https://api.github.com/repos/${encodeURIComponent(owner)}/${encodeURIComponent(repo)}`);
1548:     const branch = parsed.branch || metadata?.default_branch || "main";
1549:     const question = source.question || "";
1550:     const requestedMaxCodeFiles = Number(source.maxCodeFiles || (wantsGitHubCode(question) ? 6 : 3));
1551:     const maxCodeFiles = Math.max(1, Math.min(8, Number.isFinite(requestedMaxCodeFiles) ?
requestedMaxCodeFiles : 3));
1552:     const requestedFileTextLength = Number(source.maxFileTextLength || (wantsGitHubCode(question) ? 7000 :
3000));
1553:     const maxFileTextLength = Math.max(1000, Math.min(10000, Number.isFinite(requestedFileTextLength) ?
requestedFileTextLength : 3000));
1554:     const requestedRepoTextLength = Number(source.maxRepositoryTextLength || (wantsGitHubCode(question) ?
22000 : 14000));
1555:     const maxRepositoryTextLength = Math.max(5000, Math.min(26000, Number.isFinite(requestedRepoTextLength)
? requestedRepoTextLength : 14000));
1556:     const lines = [
1557:         `GitHub repository: ${owner}/${repo}`,
1558:         metadata?.description ? `Description: ${metadata.description}` : "",
1559:         metadata?.language ? `Primary language: ${metadata.language}` : "",
1560:         metadata?.html_url ? `Repository URL: ${metadata.html_url}` : source.url,
1561:         metadata?.default_branch ? `Default branch: ${metadata.default_branch}` : "",
1562:         ""
1563:     ].filter(Boolean);
1564:
1565:     if (parsed.type === "file" && parsed.filePath) {
1566:         const rawUrl = `https://raw.githubusercontent.com/${owner}/${repo}/${branch}/${parsed.filePath}`;

```

```

1567:     const text = await fetchLimitedText(rawUrl, 14000);
1568:     if (text.trim()) lines.push(`Source file: ${parsed.filePath}`, text.trim());
1569:     return {
1570:       context: source.context || "Public GitHub source file",
1571:       title: source.title || source.label || `${owner}/${repo}/${parsed.filePath}`,
1572:       type: "public GitHub source file",
1573:       url: source.url,
1574:       text: clampText(lines.join("\n"), 14000)
1575:     };
1576:   }
1577:
1578:   const tree = await fetchGitHubJson(`https://api.github.com/repos/${encodeURIComponent(owner)}/${encodeURIComponent(repo)}/git/trees/${encodeURIComponent(branch)}?recursive=1`);
1579:   const files = Array.isArray(tree?.tree)
1580:     ? tree.tree
1581:     : [];
1582:   .filter((item) => item.type === "blob" && item.path && githubTextFilePattern.test(item.path) && !githubSkipFilePattern.test(item.path))
1583:   .sort((a, b) => scoreGitHubFile(b.path, question) - scoreGitHubFile(a.path, question))
1584:   : [];
1585:   const readmePath = files.find((file) => /^(^|\/)readme\.md$/i.test(file.path))?.path || "README.md";
1586:   const readmeText = await
1587:   fetchLimitedText(`https://raw.githubusercontent.com/${owner}/${repo}/${branch}/${readmePath}`, 9000).catch(() =>
1588:   "");
1589:   if (readmeText.trim()) lines.push(`README excerpt from ${readmePath}:`, readmeText.trim(), "");
1590:   if (files.length) {
1591:     lines.push("Repository text/code file list:", ...files.slice(0, 30).map((file) => `- ${file.path}`),
1592:     "");
1593:   }
1594:   const includeCode = wantsGitHubCode(question);
1595:   const selectedFiles = files
1596:     .filter((file) => !/readme\.md$/i.test(file.path))
1597:     .slice(0, maxCodeFiles);
1598:   for (const file of selectedFiles) {
1599:     const rawUrl = `https://raw.githubusercontent.com/${owner}/${repo}/${branch}/${file.path}`;
1600:     const fileText = await fetchLimitedText(rawUrl, maxFileTextLength).catch(() => "");
1601:     if (!fileText.trim()) continue;
1602:     lines.push(`Source file: ${file.path}`, fileText.trim(), "");
1603:   }
1604:
1605:   return {
1606:     context: source.context || "Public GitHub repository",
1607:     title: source.title || source.label || `${owner}/${repo}`,
1608:     type: includeCode ? "public GitHub repository and code" : "public GitHub repository",
1609:     url: source.url,
1610:     text: clampText(lines.join("\n"), maxRepositoryTextLength)
1611:   };
1612: }

```

## ***fetchGitHubSourceText - lines 1614-1619***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1614-1619 (6 source lines).                                                                                                                                                                     |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function fetchGitHubSourceText(source = {}) {                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | parseGitHubSourceUrl, fetchGitHubProfileSource, fetchGitHubRepositorySource                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1614: async function fetchGitHubSourceText(source = {}) {
1615:   const parsed = parseGitHubSourceUrl(source.url || "");
1616:   if (!parsed) return null;
1617:   if (parsed.type === "profile") return fetchGitHubProfileSource(source, parsed);
1618:   return fetchGitHubRepositorySource(source, parsed);
1619: }

```

## ***readLocalSourceText - lines 1621-1639***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1621-1639 (19 source lines).                                                                                                                                                                    |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function readLocalSourceText(url) {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | URL, Set, has, toLowerCase, replace, includes, extname, readFile, resolveInsideRoot                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1621: async function readLocalSourceText(url) {
1622:   let parsed;
1623:   try {
1624:     parsed = new URL(url);
1625:   } catch {
1626:     return "";
1627:   }
1628:   const localHosts = new Set(["localhost", "127.0.0.1"]);
1629:   if (!localHosts.has(parsed.hostname.toLowerCase())) return "";
1630:   const pathname = decodeURIComponent(parsed.pathname.replace(/^\/+/, ""));
1631:   if (!pathname || pathname.includes("..")) return "";
1632:   const ext = path.extname(pathname).toLowerCase();
1633:   if (![".txt", ".md", ".json", ".csv", ".log", ".xml", ".js", ".mjs", ".css", ".c", ".h", ".cpp",
".hpp", ".py", ".v", ".sv", ".spice", ".cir", ".net", ".asc", ".sch", ".kicad_sch", ".kicad_pcb"].includes(ext))
return "";
1634:   try {
1635:     return await readFile(resolveInsideRoot(pathname), "utf8");
1636:   } catch {
1637:     return "";
1638:   }
1639: }

```

## ***fetchSourceText - lines 1641-1679***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1641-1679 (39 source lines).                                                                                                                                                                    |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function fetchSourceText(source = {}) {                                                                                                                                                         |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | sourceLooksTextual, sourceUrlAllowed, fetchGitHubSourceText, trim, readLocalSourceText, clampText, replace, fetch, get, text, stripHtmlToText                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 4 await expression(s), 8 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1641: async function fetchSourceText(source = {}) {
1642:   const url = String(source.url || "");
1643:   if (!url || !sourceLooksTextual(source) || !sourceUrlAllowed(url)) return null;
1644:   if (/github\.com|raw\.githubusercontent\.com/i.test(url)) {
1645:     const githubText = await fetchGitHubSourceText(source);
1646:     if (githubText?.text?.trim()) return githubText;
1647:   }
1648:   const localText = await readLocalSourceText(url);
1649:   if (localText.trim()) {
1650:     return {
1651:       context: source.context || "",
1652:       title: source.title || source.label || url,
1653:       type: source.type || source.kind || "text source",
1654:       url,

```

```

1655:         text: clampText(localText.replace(/\s+\n/g, "\n").trim(), 9000)
1656:     };
1657: }
1658:
1659: try {
1660:     const response = await fetch(url, {
1661:         headers: { "Accept": "text/plain,text/markdown,text/html,application/json;q=0.8,*/*;q=0.1" }
1662:     });
1663:     if (!response.ok) return null;
1664:     const contentType = response.headers.get("Content-Type") || "";
1665:     if (!/text|json|xml|html|markdown/i.test(contentType)) return null;
1666:     const rawText = clampText(await response.text(), 12000);
1667:     const text = /html/i.test(contentType) ? stripHtmlToText(rawText) : rawText;
1668:     if (!text.trim()) return null;
1669:     return {
1670:         context: source.context || "",
1671:         title: source.title || source.label || url,
1672:         type: source.type || source.kind || "public source",
1673:         url,
1674:         text: clampText(text.trim(), 9000)
1675:     };
1676: } catch {
1677:     return null;
1678: }
1679: }

```

## enrichPortfolioContext - lines 1681-1692

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1681-1692 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function enrichPortfolioContext(context = {}) {                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isArray, slice, map, all, filter                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1681: async function enrichPortfolioContext(context = {}) {
1682:     const question = String(context.question || "");
1683:     const sources = Array.isArray(context.sourceFetches)
1684:       ? context.sourceFetches.slice(0, 10).map((source) => ({ ...source, question: source.question ||
question }))
1685:       : [];
1686:     const sourceExcerpts = (await Promise.all(sources.map(fetchSourceText))).filter(Boolean);
1687:     return {
1688:       ...context,
1689:       sourceExcerpts,
1690:       sourceFetchPolicy: "Fetched excerpts are limited to safe text-like same-site, GitHub/raw GitHub,
LinkedIn, and public portfolio URLs. GitHub repository links can be expanded into repository metadata, README
text, and selected public source files when a question asks for code. PDFs/images are represented by captions,
metadata, filenames, and companion text files when available."
1691:     };
1692: }

```

## cleanConversationHistory - lines 1694-1703

| Item            | Explanation                                                                                |
|-----------------|--------------------------------------------------------------------------------------------|
| Source location | Lines 1694-1703 (10 source lines).                                                         |
| Purpose         | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow. |
| Declaration     | function cleanConversationHistory(value) {                                                 |
| Touches         | Mostly local calculations or local UI state.                                               |
| Calls detected  | isArray, slice, map, clampText, trim, filter                                               |
| API endpoints   | None detected inside this function body.                                                   |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1694: function cleanConversationHistory(value) {
1695:   if (!Array.isArray(value)) return [];
1696:   return value
1697:     .slice(-8)
1698:     .map((item) => ({
1699:       role: item?.role === "assistant" ? "assistant" : "user",
1700:       content: clampText(item?.content, 1400).trim()
1701:     }))
1702:     .filter((item) => item.content);
1703: }

```

### ***extractOpenAiText - lines 1705-1717***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1705-1717 (13 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function extractOpenAiText(data) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, isArray, forEach, push, join                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1705: function extractOpenAiText(data) {
1706:   if (typeof data?.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
1707:   const output = Array.isArray(data?.output) ? data.output : [];
1708:   const chunks = [];
1709:
1710:   output.forEach((item) => {
1711:     (item.content || []).forEach((content) => {
1712:       if (typeof content.text === "string" && content.text.trim()) chunks.push(content.text.trim());
1713:     });
1714:   });
1715:
1716:   return chunks.join("\n\n").trim();
1717: }

```

### ***extractOllamaText - lines 1719-1723***

| Item               | Explanation                                                                                |
|--------------------|--------------------------------------------------------------------------------------------|
| Source location    | Lines 1719-1723 (5 source lines).                                                          |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow. |
| Declaration        | function extractOllamaText(data = {}) {                                                    |
| Touches            | Mostly local calculations or local UI state.                                               |
| Calls detected     | trim                                                                                       |
| API endpoints      | None detected inside this function body.                                                   |
| Storage keys       | No local/session storage keys detected.                                                    |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                              |



| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1719: function extractOllamaText(data = {}) {
1720:   if (typeof data.message?.content === "string") return data.message.content.trim();
1721:   if (typeof data.response === "string") return data.response.trim();
1722:   return "";
1723: }

```

## ***callOllamaPortfolioAi - lines 1725-1773***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1725-1773 (49 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function callOllamaPortfolioAi({ question, intent, conversation, context, allowWebSearch }) {                                                                                                   |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | AbortController, abort, fetch, replace, stringify, clampText, join, json, extractOllamaText                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1725: async function callOllamaPortfolioAi({ question, intent, conversation, context, allowWebSearch }) {
1726:   const host = process.env.OLLAMA_HOST || "http://127.0.0.1:11434";
1727:   const model = process.env.OLLAMA_MODEL || "llama3.2";
1728:   const controller = new AbortController();
1729:   const timeout = setTimeout(() => controller.abort(), Number(process.env.OLLAMA_TIMEOUT_MS || 45000));
1730:   try {
1731:     const response = await fetch(`${host.replace(/\/+$/, "").}/api/chat`, {
1732:       method: "POST",
1733:       headers: { "Content-Type": "application/json" },
1734:       signal: controller.signal,
1735:       body: JSON.stringify({
1736:         model,
1737:         stream: false,
1738:         messages: [
1739:           { role: "system", content: portfolioAiInstructions },
1740:           {
1741:             role: "user",
1742:             content: [
1743:               `Visitor question: ${question}`,
1744:               `Question intent: ${intent}`,
1745:               `Web/public lookup requested by browser: ${allowWebSearch ? "yes" : "no"}`,
1746:               "",
1747:               "Recent conversation JSON:",
1748:               clampText(JSON.stringify(conversation, null, 2), 8000),
1749:               "",
1750:               "Portfolio context JSON:",
1751:               clampText(JSON.stringify(context, null, 2), 18000)
1752:             ].join("\n")
1753:           }
1754:         ],
1755:         options: {
1756:           temperature: Number(process.env.OLLAMA_TEMPERATURE || 0.35)
1757:         }
1758:       })
1759:     });
1760:     const data = await response.json().catch(() => ({}));
1761:     if (!response.ok) {
1762:       return { ok: false, error: data?.error || `Ollama returned HTTP ${response.status}` };
1763:     }
1764:     const answer = extractOllamaText(data);
1765:     return answer
1766:       ? { ok: true, answer, model }
1767:       : { ok: false, error: "Ollama did not return an answer." };
1768:   } catch (error) {
1769:     return { ok: false, error: error?.name === "AbortError" ? "Ollama timed out." : "Ollama is not reachable on this machine." };
1770:   } finally {
1771:     clearTimeout(timeout);

```

```

1772:   }
1773: }

```

## ***ruleBasedConversationAnswer - lines 1775-1786***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1775-1786 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function ruleBasedConversationAnswer(question = "") {                                                                                                                                                 |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | toLowerCase, b                                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1775: function ruleBasedConversationAnswer(question = "") {
1776:   const clean = String(question || "").toLowerCase();
1777:   if (/^(\b(thanks|thank you|appreciate it)\b/.test(clean)) {
1778:     return "You're welcome. I can keep helping with this portfolio, project evidence, resume links, or
related electronics and embedded-systems questions.";
1779:   }
1780:
1781:   if (/^(\b(who are you|what are you)\b/.test(clean)) {
1782:     return "I am this portfolio's assistant. I can help visitors explore the saved engineering work,
explain project details, summarize portfolio evidence, and answer related electronics, embedded, analog,
digital, FPGA, ASIC, PCB, and firmware questions.";
1783:   }
1784:
1785:   return "Hi. I can help you explore this portfolio, explain projects, summarize project evidence, open
relevant sections, or answer related electronics and embedded-systems questions. You can ask about a specific
project, a tool like KiCad or STM32CubeIDE, or a general topic such as embedded systems, op amps, FPGA design,
ASICs, or PCB testing.";
1786: }

```

## ***isLocalRequest - lines 1788-1791***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1788-1791 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function isLocalRequest(request) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1788: function isLocalRequest(request) {
1789:   const address = request.socket.remoteAddress || "";
1790:   return address === "127.0.0.1" || address === "::1" || address === "::ffff:127.0.0.1";
1791: }

```

## ***readJsonFile - lines 1793-1795***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1793-1795 (3 source lines).                                                                                                                                                                     |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function readJsonFile(filePath) {                                                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | parse, readFile                                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1793: async function readJsonFile(filePath) {
1794:   return JSON.parse(await readFile(filePath, "utf8"));
1795: }
```

## ***readRequestJson - lines 1797-1810***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1797-1810 (14 source lines).                                                                                                                                                                    |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function readRequestJson(request) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | Error, push, parse, concat, toString, replace                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1797: async function readRequestJson(request) {
1798:   const chunks = [];
1799:   let size = 0;
1800:
1801:   for await (const chunk of request) {
1802:     size += chunk.length;
1803:     if (size > 60 * 1024 * 1024) {
1804:       throw new Error("Request body is too large.");
1805:     }
1806:     chunks.push(chunk);
1807:   }
1808:
1809:   return JSON.parse((Buffer.concat(chunks).toString("utf8") || "{}").replace(/^\uFEFF/, ""));
1810: }
```

## ***safeSegment - lines 1812-1819***

| Item            | Explanation                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Source location | Lines 1812-1819 (8 source lines).                                             |
| Purpose         | Validates or normalizes input so later code receives safe predictable values. |
| Declaration     | function safeSegment(value, fallback = "item") {                              |
| Touches         | Mostly local calculations or local UI state.                                  |
| Calls detected  | toLowerCase, trim, replace                                                    |
| API endpoints   | None detected inside this function body.                                      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1812: function safeSegment(value, fallback = "item") {
1813:   const segment = String(value || fallback)
1814:     .toLowerCase()
1815:     .trim()
1816:     .replace(/^[^a-z0-9._-]+/g, "-")
1817:     .replace(/^-+|-+$/g, "");
1818:   return segment || fallback;
1819: }

```

## ***safeFileName - lines 1821-1826***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1821-1826 (6 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function safeFileName(value) {                                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | parse, safeSegment, replace                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1821: function safeFileName(value) {
1822:   const parsed = path.parse(String(value || "file"));
1823:   const name = safeSegment(parsed.name, "file");
1824:   const ext = safeSegment(parsed.ext.replace(".", ""), "");
1825:   return ext ? `${name}.${ext}` : name;
1826: }

```

## ***resolveInsideRoot - lines 1828-1834***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1828-1834 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function resolveInsideRoot(...segments) {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, join, startsWith, Error                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1828: function resolveInsideRoot(...segments) {
1829:   const target = path.normalize(path.join(root, ...segments));
1830:   if (!target.startsWith(root)) {
1831:     throw new Error("Resolved path is outside the workspace.");
1832:   }

```

```

1833:   return target;
1834: }

```

## ***resolveInsidePortfolioRoot - lines 1836-1843***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1836-1843 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function resolveInsidePortfolioRoot(...segments) {                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | resolve, relative, startsWith, isAbsolute, Error                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1836: function resolveInsidePortfolioRoot(...segments) {
1837:   const target = path.resolve(portfolioRoot, ...segments);
1838:   const relative = path.relative(portfolioRoot, target);
1839:   if (relative.startsWith(".") || path.isAbsolute(relative)) {
1840:     throw new Error("Resolved path is outside the portfolio workspace.");
1841:   }
1842:   return target;
1843: }

```

## ***resolveInsideCompileRoot - lines 1845-1852***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1845-1852 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function resolveInsideCompileRoot(...segments) {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | resolve, relative, startsWith, isAbsolute, Error                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1845: function resolveInsideCompileRoot(...segments) {
1846:   const target = path.resolve(compileRoot, ...segments);
1847:   const relative = path.relative(compileRoot, target);
1848:   if (relative.startsWith(".") || path.isAbsolute(relative)) {
1849:     throw new Error("Resolved path is outside the compile workspace.");
1850:   }
1851:   return target;
1852: }

```

## ***samePath - lines 1854-1856***

| Item            | Explanation                                                                                |
|-----------------|--------------------------------------------------------------------------------------------|
| Source location | Lines 1854-1856 (3 source lines).                                                          |
| Purpose         | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow. |
| Declaration     | function samePath(left, right) {                                                           |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | resolve, toLowerCase                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1854: function samePath(left, right) {
1855:   return path.resolve(left).toLowerCase() === path.resolve(right).toLowerCase();
1856: }

```

## ***pathExists - lines 1858-1865***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1858-1865 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function pathExists(filePath) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | fsAccess                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1858: async function pathExists(filePath) {
1859:   try {
1860:     await fsAccess(filePath);
1861:     return true;
1862:   } catch {
1863:     return false;
1864:   }
1865: }

```

## ***publishAccessError - lines 1867-1877***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1867-1877 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function publishAccessError(message, details = "", extra = {}) {                                                                                                                                      |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | Error                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1867: function publishAccessError(message, details = "", extra = {}) {
1868:   const error = new Error(message);

```

```

1869:   error.code = "PUBLISH_AUTHORIZATION_REQUIRED";
1870:   error.details = details || publishAuthorizationHelp;
1871:   error.publishAccess = {
1872:     authorizationRequired: true,
1873:     help: publishAuthorizationHelp,
1874:     ...extra
1875:   };
1876:   return error;
1877: }

```

## ***gitFailureText - lines 1879-1885***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1879-1885 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function gitFailureText(error) {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, join, trim                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1879: function gitFailureText(error) {
1880:   return [
1881:     error?.stderr,
1882:     error?.stdout,
1883:     error?.message
1884:   ].filter(Boolean).join("\n").trim();
1885: }

```

## ***remoteUrlForDisplay - lines 1887-1891***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1887-1891 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function remoteUrlForDisplay(remoteUrl = "") {                                                                                                                                                        |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | replace                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1887: function remoteUrlForDisplay(remoteUrl = "") {
1888:   return String(remoteUrl || "")
1889:     .replace(/^https:\/\/([^\:\/]+):[^\:\/]+@/i, "https://$1:***@")
1890:     .replace(/^https:\/\/[^\:\/]+@/i, "https://");
1891: }

```

## ***parseGitHubRemote - lines 1893-1907***

| Item            | Explanation                                                            |
|-----------------|------------------------------------------------------------------------|
| Source location | Lines 1893-1907 (15 source lines).                                     |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function parseGitHubRemote(remoteUrl = "") {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, match, replace, URL, split                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1893: function parseGitHubRemote(remoteUrl = "") {
1894:   const trimmed = String(remoteUrl || "").trim();
1895:   const sshMatch = trimmed.match(/^git@github\.com:([^\s]+)\.(.+)?(?:\.git)?$/i);
1896:   if (sshMatch) return { owner: sshMatch[1], repo: sshMatch[2].replace(/\.git$/i, "") };
1897:
1898:   try {
1899:     const parsed = new URL(trimmed.replace(/^git\+/, ""));
1900:     if (!/github\.com$/i.test(parsed.hostname)) return null;
1901:     const parts = parsed.pathname.replace(/^\/+/, "").split("/");
1902:     if (parts.length < 2) return null;
1903:     return { owner: parts[0], repo: parts[1].replace(/\.git$/i, "") };
1904:   } catch {
1905:     return null;
1906:   }
1907: }

```

## ***validatePublishRemoteUrl - lines 1909-1927***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1909-1927 (19 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function validatePublishRemoteUrl(value = "") {                                                                                                                                                       |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | trim, URL, toLowerCase, Error, replace, split, includes                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1909: function validatePublishRemoteUrl(value = "") {
1910:   const remoteUrl = String(value || "").trim();
1911:   if (!remoteUrl) return "";
1912:   if (/^git@github\.com:[A-Za-z0-9_.-]+\.[A-Za-z0-9_.-]+(?:\.git)?$/i.test(remoteUrl)) return remoteUrl;
1913:   try {
1914:     const parsed = new URL(remoteUrl);
1915:     if (parsed.protocol !== "https:" || parsed.hostname.toLowerCase() !== "github.com") {
1916:       throw new Error("Only GitHub HTTPS repository URLs are supported by the guided setup.");
1917:     }
1918:     const parts = parsed.pathname.replace(/^\/+/, "").split("/");
1919:     if (parts.length < 2 || !parts[0] || !parts[1]) {
1920:       throw new Error("GitHub repository URL must include owner and repository name.");
1921:     }
1922:     return `https://github.com/${parts[0]}/${parts[1].replace(/\.git$/i, "")}.git`;
1923:   } catch (error) {
1924:     if (error.message?.includes("Only GitHub") || error.message?.includes("owner")) throw error;
1925:     throw new Error("Enter a GitHub repository URL such as https://github.com/USERNAME/USERNAME.github.io.git.");
1926:   }
1927: }

```

## ***validateCustomDomain - lines 1929-1936***



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1929-1936 (8 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function validateCustomDomain(value = "") {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, toLowerCase, Error                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1929: function validateCustomDomain(value = "") {
1930:   const domain = String(value || "").trim().toLowerCase();
1931:   if (!domain) return "";
1932:   if (!/^(?:[a-z0-9]{0,61}[a-z0-9]?\.\.)+[a-z]{2,63}$/i.test(domain)) {
1933:     throw new Error("Custom domain must look like example.com or portfolio.example.com.");
1934:   }
1935:   return domain;
1936: }

```

## ***compareVersions - lines 1938-1947***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1938-1947 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function compareVersions(left = "", right = "") {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, map, max                                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1938: function compareVersions(left = "", right = "") {
1939:   const leftParts = String(left || "0").split(/[.-]/).map((part) => Number.parseInt(part, 10) || 0);
1940:   const rightParts = String(right || "0").split(/[.-]/).map((part) => Number.parseInt(part, 10) || 0);
1941:   const length = Math.max(leftParts.length, rightParts.length);
1942:   for (let index = 0; index < length; index += 1) {
1943:     const delta = (leftParts[index] || 0) - (rightParts[index] || 0);
1944:     if (delta !== 0) return delta;
1945:   }
1946:   return 0;
1947: }

```

## ***bumpPublishedAssetVersions - lines 1949-1965***

| Item            | Explanation                                                            |
|-----------------|------------------------------------------------------------------------|
| Source location | Lines 1949-1965 (17 source lines).                                     |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior. |
| Declaration     | async function bumpPublishedAssetVersions(baseRoot = portfolioRoot) {  |
| Touches         | filesystem                                                             |
| Calls detected  | join, readFile, now, toString, replace, writeFile                      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1949: async function bumpPublishedAssetVersions(baseRoot = portfolioRoot) {
1950:   let html = "";
1951:   const indexPath = path.join(baseRoot, "index.html");
1952:   try {
1953:     html = await readFile(indexPath, "utf8");
1954:   } catch {
1955:     return false;
1956:   }
1957:   const version = Date.now().toString();
1958:   const next = html
1959:     .replace(/(href="styles\.css)(?:\?v=[^"]*)?"/g, ` $1?v=${version}`)
1960:     .replace(/(src="script\.js)(?:\?v=[^"]*)?"/g, ` $1?v=${version}`)
1961:     .replace(/(src="electronics-search\.js)(?:\?v=[^"]*)?"/g, ` $1?v=${version}`);
1962:   if (next === html) return false;
1963:   await writeFile(indexPath, next, "utf8");
1964:   return true;
1965: }

```

## ***workspaceHasCompatibleSiteFiles - lines 1967-1976***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1967-1976 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function workspaceHasCompatibleSiteFiles(baseRoot = portfolioRoot) {                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | readFile, join                                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1967: async function workspaceHasCompatibleSiteFiles(baseRoot = portfolioRoot) {
1968:   for (const fileName of ["index.html", "styles.css", "script.js"]) {
1969:     try {
1970:       await readFile(path.join(baseRoot, fileName));
1971:     } catch {
1972:       return false;
1973:     }
1974:   }
1975:   return true;
1976: }

```

## ***getPublishTargetInfo - lines 1978-2044***

| Item            | Explanation                                                                                                                                                                                                            |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 1978-2044 (67 source lines).                                                                                                                                                                                     |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                                 |
| Declaration     | async function getPublishTargetInfo() {                                                                                                                                                                                |
| Touches         | filesystem, authentication/security data                                                                                                                                                                               |
| Calls detected  | samePath, workspaceHasCompatibleSiteFiles, runPublishGit, trim, remoteUrlForDisplay, parseGitHubRemote, readFile, resolveInsidePortfolioRoot, readPublishAuthCache, publishAuthCachelsFresh, publishAuthCacheExpiresAt |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 6 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1978: async function getPublishTargetInfo() {
1979:   const info = {
1980:     workspace: root,
1981:     portfolioWorkspace: portfolioRoot,
1982:     separatedWorkspace: !samePath(root, portfolioRoot),
1983:     defaultRepository: defaultSiteRepository,
1984:     gitBacked: false,
1985:     compatible: await workspaceHasCompatibleSiteFiles(),
1986:     remote: "",
1987:     branch: "",
1988:     repository: "",
1989:     customDomain: ""
1990:   };
1991:
1992:   try {
1993:     info.gitBacked = (await runPublishGit(["rev-parse", "--is-inside-work-tree"])).stdout.trim() ===
"true";
1994:   } catch {
1995:     return info;
1996:   }
1997:
1998:   try {
1999:     info.remote = remoteUrlForDisplay((await runPublishGit(["remote", "get-url",
"origin"])).stdout.trim());
2000:   } catch {
2001:     info.remote = "";
2002:   }
2003:
2004:   try {
2005:     info.branch = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2006:   } catch {
2007:     info.branch = "";
2008:   }
2009:
2010:   const parsedRemote = parseGitHubRemote(info.remote);
2011:   info.repository = parsedRemote ? `${parsedRemote.owner}/${parsedRemote.repo}` : info.remote;
2012:
2013:   try {
2014:     info.customDomain = (await readFile(resolveInsidePortfolioRoot("CNAME"), "utf8")).trim();
2015:   } catch {
2016:     info.customDomain = "";
2017:   }
2018:
2019:   const accessShape = {
2020:     branch: info.branch,
2021:     remote: info.remote,
2022:     repository: info.repository
2023:   };
2024:   const cached = await readPublishAuthCache();
2025:   if (publishAuthCacheIsFresh(cached, accessShape)) {
2026:     info.authorizationChecked = true;
2027:     info.authorizationCached = true;
2028:     info.checkedAt = cached.checkedAt;
2029:     info.expiresAt = publishAuthCacheExpiresAt(cached);
2030:     info.extendedTrust = Boolean(cached.extendedTrust);
2031:     info.successCountLastWeek = cached.successCountLastWeek || 0;
2032:     info.trustDays = cached.trustDays || (cached.extendedTrust ? 30 : 1);
2033:   } else {
2034:     info.authorizationChecked = false;
2035:     info.authorizationCached = false;
2036:     info.checkedAt = "";
2037:     info.expiresAt = "";
2038:     info.extendedTrust = false;
2039:     info.successCountLastWeek = cached?.successCountLastWeek || 0;
2040:     info.trustDays = 0;
2041:   }
2042:
2043:   return info;
2044: }

```

***runGit - lines 2046-2064***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2046-2064 (19 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function runGit(args, options = {}) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | execFileAsync, Error                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2046: async function runGit(args, options = {}) {
2047:   let lastError = null;
2048:   for (const candidate of gitCandidates) {
2049:     try {
2050:       const result = await execFileAsync(candidate, args, {
2051:         cwd: options.cwd || root,
2052:         maxBuffer: options.maxBuffer || 10 * 1024 * 1024,
2053:         timeout: options.timeout || 0,
2054:         windowsHide: options.windowsHide !== false
2055:       });
2056:       return { ...result, git: candidate };
2057:     } catch (error) {
2058:       lastError = error;
2059:       if (error.code === "ENOENT") continue;
2060:       throw error;
2061:     }
2062:   }
2063:   throw lastError || new Error("Git executable was not found.");
2064: }

```

## ***runPublishGit - lines 2066-2068***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2066-2068 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function runPublishGit(args, options = {}) {                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | runGit                                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2066: async function runPublishGit(args, options = {}) {
2067:   return runGit(args, { ...options, cwd: options.cwd || portfolioRoot });
2068: }

```

## ***runOptionalCommand - lines 2070-2087***

| Item            | Explanation                                                                                |
|-----------------|--------------------------------------------------------------------------------------------|
| Source location | Lines 2070-2087 (18 source lines).                                                         |
| Purpose         | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow. |
| Declaration     | async function runOptionalCommand(command, args = [], options = {}) {                      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | execFileAsync                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2070: async function runOptionalCommand(command, args = [], options = {}) {
2071:   try {
2072:     const result = await execFileAsync(command, args, {
2073:       cwd: options.cwd || root,
2074:       maxBuffer: options.maxBuffer || 2 * 1024 * 1024,
2075:       timeout: options.timeout || 15000,
2076:       windowsHide: options.windowsHide !== false
2077:     });
2078:     return { ok: true, stdout: result.stdout || "", stderr: result.stderr || "" };
2079:   } catch (error) {
2080:     return {
2081:       ok: false,
2082:       stdout: error.stdout || "",
2083:       stderr: error.stderr || "",
2084:       error: error.message || String(error)
2085:     };
2086:   }
2087: }

```

## ***getGitStatus - lines 2089-2136***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2089-2136 (48 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function getGitStatus() {                                                                                                                                                                       |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | runGit, trim                                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2089: async function getGitStatus() {
2090:   let git = { ok: false, stdout: "", stderr: "", error: "Git for Windows was not found." };
2091:   try {
2092:     const gitResult = await runGit(["--version"]);
2093:     git = { ok: true, stdout: gitResult.stdout || "", stderr: gitResult.stderr || "" };
2094:   } catch (error) {
2095:     git = { ok: false, stdout: error.stdout || "", stderr: error.stderr || "", error: error.message ||
"Git for Windows was not found." };
2096:   }
2097:   let credentialManager = { ok: false, version: "", error: "Git Credential Manager was not found." };
2098:   if (git.ok) {
2099:     let gcm = { ok: false, stdout: "", stderr: "", error: "Git Credential Manager was not found." };
2100:     try {
2101:       const gcmResult = await runGit(["credential-manager", "--version"]);
2102:       gcm = { ok: true, stdout: gcmResult.stdout || "", stderr: gcmResult.stderr || "" };
2103:     } catch (error) {
2104:       gcm = { ok: false, stdout: error.stdout || "", stderr: error.stderr || "", error: error.message ||
"Git Credential Manager was not found." };
2105:     }
2106:     credentialManager = {
2107:       ok: gcm.ok,
2108:       version: (gcm.stdout || gcm.stderr || "").trim(),
2109:       error: gcm.ok ? "" : (gcm.stderr || gcm.error || "Git Credential Manager was not found.")
2110:     };
2111:   }

```

```

2112:
2113:   return {
2114:     git: {
2115:       ok: git.ok,
2116:       version: (git.stdout || git.stderr || "").trim(),
2117:       error: git.ok ? "" : (git.stderr || git.error || "Git for Windows was not found.")
2118:     },
2119:     credentialManager,
2120:     nodeRuntime: {
2121:       ok: true,
2122:       version: process.versions.node,
2123:       bundled: true,
2124:       note: "Bundled inside the desktop app. No manual Node.js install is required."
2125:     },
2126:     pnpm: {
2127:       ok: true,
2128:       required: false,
2129:       note: "pnpm is only needed by developers building the installer from source."
2130:     },
2131:     app: {
2132:       version: process.env.OMB_APP_VERSION || packageJson.version || "0.0.0",
2133:       desktopApp: process.env.OMB_DESKTOP_APP === "1"
2134:     }
2135:   };
2136: }

```

## ***publishPathIsStageable - lines 2138-2152***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2138-2152 (15 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function publishPathIsStageable(relativePath) {                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | fsAccess, resolveInsidePortfolioRoot, runPublishGit                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2138: async function publishPathIsStageable(relativePath) {
2139:   try {
2140:     await fsAccess(resolveInsidePortfolioRoot(relativePath));
2141:     return true;
2142:   } catch {
2143:     // Missing files can still be stageable if Git already tracks them and they were deleted.
2144:   }
2145:
2146:   try {
2147:     await runPublishGit(["ls-files", "--error-unmatch", "--", relativePath]);
2148:     return true;
2149:   } catch {
2150:     return false;
2151:   }
2152: }

```

## ***stageablePublishPaths - lines 2154-2160***

| Item            | Explanation                                                            |
|-----------------|------------------------------------------------------------------------|
| Source location | Lines 2154-2160 (7 source lines).                                      |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior. |
| Declaration     | async function stageablePublishPaths(pathsToCheck) {                   |
| Touches         | Mostly local calculations or local UI state.                           |
| Calls detected  | publishPathIsStageable, push                                           |
| API endpoints   | None detected inside this function body.                               |
| Storage keys    | No local/session storage keys detected.                                |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2154: async function stageablePublishPaths(pathsToCheck) {
2155:   const stageable = [];
2156:   for (const relativePath of pathsToCheck) {
2157:     if (await publishPathIsStageable(relativePath)) stageable.push(relativePath);
2158:   }
2159:   return stageable;
2160: }

```

## ***runGitWithInput - lines 2162-2204***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2162-2204 (43 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function runGitWithInput(args, input, options = {}) {                                                                                                                                           |
| Touches            | external processes                                                                                                                                                                                    |
| Calls detected     | spawn, on, toString, once, resolve, Error, reject, end                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2162: async function runGitWithInput(args, input, options = {}) {
2163:   let lastError = null;
2164:   for (const candidate of gitCandidates) {
2165:     try {
2166:       return await new Promise((resolve, reject) => {
2167:         const child = spawn(candidate, args, {
2168:           cwd: options.cwd || portfolioRoot,
2169:           windowsHide: true,
2170:           stdio: ["pipe", "pipe", "pipe"]
2171:         });
2172:         let stdout = "";
2173:         let stderr = "";
2174:
2175:         child.stdout.on("data", (chunk) => {
2176:           stdout += chunk.toString();
2177:         });
2178:         child.stderr.on("data", (chunk) => {
2179:           stderr += chunk.toString();
2180:         });
2181:         child.once("error", reject);
2182:         child.once("close", (code) => {
2183:           if (code === 0) {
2184:             resolve({ stdout, stderr, git: candidate });
2185:             return;
2186:           }
2187:           const error = new Error(stderr || stdout || `Git exited with code ${code}.`);
2188:           error.stdout = stdout;
2189:           error.stderr = stderr;
2190:           error.code = code;
2191:           error.git = candidate;
2192:           reject(error);
2193:         });
2194:         child.stdin.end(input);
2195:       });
2196:     } catch (error) {
2197:       lastError = error;
2198:       if (error.code === "ENOENT") continue;
2199:       throw error;
2200:     }
2201:   }
2202: }
2203: throw lastError || new Error("Git executable was not found.");
2204: }

```

## storeGitCredentials - lines 2206-2237

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2206-2237 (32 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | async function storeGitCredentials(remoteUrl, username, password) {                                                                                                                                   |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | trim, Error, URL, toLowerCase, replace, runPublishGit, join, runGitWithInput                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2206: async function storeGitCredentials(remoteUrl, username, password) {
2207:   const cleanUsername = String(username || "").trim();
2208:   const cleanPassword = String(password || "");
2209:   if (!cleanUsername && !cleanPassword) return { stored: false };
2210:   if (!cleanUsername || !cleanPassword) {
2211:     throw new Error("Both username and password/token are required when credentials are provided.");
2212:   }
2213:
2214:   let parsed = null;
2215:   try {
2216:     parsed = new URL(remoteUrl);
2217:   } catch {
2218:     throw new Error("Username and password/token credentials require a GitHub HTTPS repository URL.");
2219:   }
2220:
2221:   if (parsed.protocol !== "https:" || parsed.hostname.toLowerCase() !== "github.com") {
2222:     throw new Error("Username and password/token credentials require a GitHub HTTPS repository URL.");
2223:   }
2224:
2225:   const pathName = parsed.pathname.replace(/^\/+/, "");
2226:   await runPublishGit(["config", "--local", "credential.useHttpPath", "true"]);
2227:   const credentialInput = [
2228:     "protocol=https",
2229:     "host=github.com",
2230:     `path=${pathName}`,
2231:     `username=${cleanUsername}`,
2232:     `password=${cleanPassword}`,
2233:     ""
2234:   ].join("\n");
2235:   await runGitWithInput(["credential", "approve"], credentialInput, { cwd: portfolioRoot });
2236:   return { stored: true, username: cleanUsername };
2237: }
```

## validateCredentialPair - lines 2239-2246

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2239-2246 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of authentication, authorization, or credential checking.                                                                                                                                        |
| Declaration        | function validateCredentialPair(username, password) {                                                                                                                                                 |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | trim, Error                                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2239: function validateCredentialPair(username, password) {
```



```

2240:   const cleanUsername = String(username || "").trim();
2241:   const cleanPassword = String(password || "");
2242:   if ((cleanUsername && !cleanPassword) || (!cleanUsername && cleanPassword)) {
2243:     throw new Error("Both username and password/token are required when credentials are provided.");
2244:   }
2245:   return { cleanUsername, cleanPassword };
2246: }

```

## normalizeGitBranchName - lines 2248-2327

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2248-2327 (80 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function normalizeGitBranchName(value = "") {                                                                                                                                                         |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | trim, includes, endsWith, startsWith, detectRemoteDefaultBranch, runPublishGit, match, originRemoteExists, setOriginRemote, checkoutPublishBranch, verifyPublishWriteAccess, safeSegment              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 10 await expression(s), 10 return statement(s).                                                                                                                                                       |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2248: function normalizeGitBranchName(value = "") {
2249:   const branch = String(value || "").trim();
2250:   if (
2251:     !branch ||
2252:     branch.length > 180 ||
2253:     branch.includes(".") ||
2254:     branch.endsWith(".") ||
2255:     branch.includes("@") ||
2256:     /[\\:\s~^?*[\]\x00-\x1f]/.test(branch) ||
2257:     branch.startsWith("/") ||
2258:     branch.endsWith("/") ||
2259:     branch.includes("//")
2260:   ) {
2261:     return "";
2262:   }
2263:   return branch;
2264: }
2265:
2266: async function detectRemoteDefaultBranch(remoteUrl = "") {
2267:   if (!remoteUrl) return "";
2268:   try {
2269:     const result = await runPublishGit(["ls-remote", "--symref", remoteUrl, "HEAD"], { timeout: 45000 });
2270:     const match = String(result.stdout || "").match(/^ref:\s+refs\/heads\/(.+)\s+HEAD$/m);
2271:     return normalizeGitBranchName(match?.[1] || "");
2272:   } catch {
2273:     return "";
2274:   }
2275: }
2276:
2277: async function originRemoteExists() {
2278:   try {
2279:     await runPublishGit(["remote", "get-url", "origin"]);
2280:     return true;
2281:   } catch {
2282:     return false;
2283:   }
2284: }
2285:
2286: async function setOriginRemote(remoteUrl) {
2287:   if (await originRemoteExists()) {
2288:     await runPublishGit(["remote", "set-url", "origin", remoteUrl]);
2289:   } else {
2290:     await runPublishGit(["remote", "add", "origin", remoteUrl]);
2291:   }
2292: }
2293:
2294: async function checkoutPublishBranch(branch) {
2295:   const cleanBranch = normalizeGitBranchName(branch) || "main";
2296:   const current = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2297:   if (current === cleanBranch) return cleanBranch;
2298:   try {
2299:     await runPublishGit(["checkout", cleanBranch]);
2300:   } catch {
2301:     await runPublishGit(["checkout", "-B", cleanBranch]);

```

```

2302:   }
2303:   return cleanBranch;
2304: }
2305:
2306: async function verifyPublishWriteAccess(access = {}) {
2307:   const checkBranch = `omb-auth-check-${safeSegment(os.hostname(), "device")}-${Date.now()}`;
2308:   try {
2309:     await runPublishGit(["push", "--dry-run", "origin", `HEAD:refs/heads/${checkBranch}`], {
2310:       timeout: 60 * 1000
2311:     });
2312:     return {
2313:       method: "temporary-dry-run-ref",
2314:       ref: checkBranch
2315:     };
2316:   } catch (error) {
2317:     throw publishAccessError(
2318:       "GitHub authorization is required before this website can be changed.",
2319:       gitFailureText(error),
2320:       access
2321:     );
2322:   }
2323: }
2324:
2325: async function ensurePublishHeadForWriteCheck() {
2326:   try {
2327:     await runPublishGit(["rev-parse", "--verify", "HEAD"]);

```

## ***detectRemoteDefaultBranch - lines 2266-2275***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2266-2275 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function detectRemoteDefaultBranch(remoteUrl = "") {                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | runPublishGit, match, normalizeGitBranchName                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2266: async function detectRemoteDefaultBranch(remoteUrl = "") {
2267:   if (!remoteUrl) return "";
2268:   try {
2269:     const result = await runPublishGit(["ls-remote", "--symref", remoteUrl, "HEAD"], { timeout: 45000 });
2270:     const match = String(result.stdout || "").match(/^ref:\s+refs\/heads\/(.+)\s+HEAD$/m);
2271:     return normalizeGitBranchName(match?.[1] || "");
2272:   } catch {
2273:     return "";
2274:   }
2275: }

```

## ***originRemoteExists - lines 2277-2284***

| Item               | Explanation                                                            |
|--------------------|------------------------------------------------------------------------|
| Source location    | Lines 2277-2284 (8 source lines).                                      |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior. |
| Declaration        | async function originRemoteExists() {                                  |
| Touches            | Mostly local calculations or local UI state.                           |
| Calls detected     | runPublishGit                                                          |
| API endpoints      | None detected inside this function body.                               |
| Storage keys       | No local/session storage keys detected.                                |
| Async/return clues | 1 await expression(s), 2 return statement(s).                          |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2277: async function originRemoteExists() {
2278:   try {
2279:     await runPublishGit(["remote", "get-url", "origin"]);
2280:     return true;
2281:   } catch {
2282:     return false;
2283:   }
2284: }

```

## ***setOriginRemote - lines 2286-2292***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2286-2292 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function setOriginRemote(remoteUrl) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | originRemoteExists, runPublishGit                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2286: async function setOriginRemote(remoteUrl) {
2287:   if (await originRemoteExists()) {
2288:     await runPublishGit(["remote", "set-url", "origin", remoteUrl]);
2289:   } else {
2290:     await runPublishGit(["remote", "add", "origin", remoteUrl]);
2291:   }
2292: }

```

## ***checkoutPublishBranch - lines 2294-2304***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2294-2304 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function checkoutPublishBranch(branch) {                                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeGitBranchName, runPublishGit, trim                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2294: async function checkoutPublishBranch(branch) {
2295:   const cleanBranch = normalizeGitBranchName(branch) || "main";
2296:   const current = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2297:   if (current === cleanBranch) return cleanBranch;
2298:   try {
2299:     await runPublishGit(["checkout", cleanBranch]);
2300:   } catch {
2301:     await runPublishGit(["checkout", "-B", cleanBranch]);

```

```

2302:   }
2303:   return cleanBranch;
2304: }

```

## verifyPublishWriteAccess - lines 2306-2323

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2306-2323 (18 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function verifyPublishWriteAccess(access = {}) {                                                                                                                                                |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | safeSegment, hostname, now, runPublishGit, publishAccessError, gitFailureText                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2306: async function verifyPublishWriteAccess(access = {}) {
2307:   const checkBranch = `omb-auth-check-${safeSegment(os.hostname(), "device")}-${Date.now()}`;
2308:   try {
2309:     await runPublishGit(["push", "--dry-run", "origin", `HEAD:refs/heads/${checkBranch}`], {
2310:       timeout: 60 * 1000
2311:     });
2312:     return {
2313:       method: "temporary-dry-run-ref",
2314:       ref: checkBranch
2315:     };
2316:   } catch (error) {
2317:     throw publishAccessError(
2318:       "GitHub authorization is required before this website can be changed.",
2319:       gitFailureText(error),
2320:       access
2321:     );
2322:   }
2323: }

```

## ensurePublishHeadForWriteCheck - lines 2325-2342

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2325-2342 (18 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function ensurePublishHeadForWriteCheck() {                                                                                                                                                     |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | runPublishGit                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2325: async function ensurePublishHeadForWriteCheck() {
2326:   try {
2327:     await runPublishGit(["rev-parse", "--verify", "HEAD"]);
2328:     return { created: false };
2329:   } catch {
2330:     await runPublishGit([
2331:       "-c",
2332:       "user.name=OMB Portfolio Builder",
2333:       "-c",
2334:       "user.email=omb-builder@localhost",
2335:       "commit",

```

```

2336:         "--allow-empty",
2337:         "-m",
2338:         "Initialize publish authorization check"
2339:     });
2340:     return { created: true };
2341: }
2342: }

```

## ***synchronizePublishBranchFromRemote - lines 2344-2373***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2344-2373 (30 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function synchronizePublishBranchFromRemote(branch, access = {}) {                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeGitBranchName, runPublishGit, trim, checkoutPublishBranch, publishAccessError, gitFailureText                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 4 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2344: async function synchronizePublishBranchFromRemote(branch, access = {}) {
2345:     const cleanBranch = normalizeGitBranchName(branch) || "main";
2346:     try {
2347:         const remoteBranch = await runPublishGit(["ls-remote", "--heads", "origin", cleanBranch], {
2348:             timeout: 45 * 1000
2349:         });
2350:         if (!String(remoteBranch.stdout || "").trim()) {
2351:             return {
2352:                 branch: cleanBranch,
2353:                 remoteBranchExists: false,
2354:                 synchronized: false
2355:             };
2356:         }
2357:
2358:         await runPublishGit(["fetch", "origin", cleanBranch], { timeout: 2 * 60 * 1000 });
2359:         await checkoutPublishBranch(cleanBranch);
2360:         await runPublishGit(["reset", "--hard", `origin/${cleanBranch}`], { timeout: 60 * 1000 });
2361:         return {
2362:             branch: cleanBranch,
2363:             remoteBranchExists: true,
2364:             synchronized: true
2365:         };
2366:     } catch (error) {
2367:         throw publishAccessError(
2368:             "The selected website target could not be synchronized from GitHub.",
2369:             gitFailureText(error),
2370:             { ...access, branch: cleanBranch }
2371:         );
2372:     }
2373: }

```

## ***capturePublishTargetState - lines 2375-2403***

| Item               | Explanation                                                            |
|--------------------|------------------------------------------------------------------------|
| Source location    | Lines 2375-2403 (29 source lines).                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior. |
| Declaration        | async function capturePublishTargetState() {                           |
| Touches            | filesystem                                                             |
| Calls detected     | runPublishGit, trim, readFile, resolveInsidePortfolioRoot              |
| API endpoints      | None detected inside this function body.                               |
| Storage keys       | No local/session storage keys detected.                                |
| Async/return clues | 3 await expression(s), 1 return statement(s).                          |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2375: async function capturePublishTargetState() {
2376:   const snapshot = {
2377:     branch: "",
2378:     customDomainExists: false,
2379:     customDomainText: "",
2380:     remote: "",
2381:     remoteExists: false
2382:   };
2383:   try {
2384:     snapshot.remote = (await runPublishGit(["remote", "get-url", "origin"])).stdout.trim();
2385:     snapshot.remoteExists = true;
2386:   } catch {
2387:     snapshot.remote = "";
2388:     snapshot.remoteExists = false;
2389:   }
2390:   try {
2391:     snapshot.branch = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2392:   } catch {
2393:     snapshot.branch = "";
2394:   }
2395:   try {
2396:     snapshot.customDomainText = await readFile(resolveInsidePortfolioRoot("CNAME"), "utf8");
2397:     snapshot.customDomainExists = true;
2398:   } catch {
2399:     snapshot.customDomainText = "";
2400:     snapshot.customDomainExists = false;
2401:   }
2402:   return snapshot;
2403: }

```

## ***restorePublishTargetState - lines 2405-2429***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2405-2429 (25 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function restorePublishTargetState(snapshot = {}) {                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | setOriginRemote, originRemoteExists, runPublishGit, checkoutPublishBranch, writeFile, resolveInsidePortfolioRoot, rm                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 6 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2405: async function restorePublishTargetState(snapshot = {}) {
2406:   try {
2407:     if (snapshot.remoteExists && snapshot.remote) {
2408:       await setOriginRemote(snapshot.remote);
2409:     } else if (await originRemoteExists()) {
2410:       await runPublishGit(["remote", "remove", "origin"]);
2411:     }
2412:   } catch {
2413:     // Best-effort rollback; preserve the original error for the caller.
2414:   }
2415:   try {
2416:     if (snapshot.branch) await checkoutPublishBranch(snapshot.branch);
2417:   } catch {
2418:     // Best-effort rollback; preserve the original error for the caller.
2419:   }
2420:   try {
2421:     if (snapshot.customDomainExists) {
2422:       await writeFile(resolveInsidePortfolioRoot("CNAME"), snapshot.customDomainText, "utf8");
2423:     } else {
2424:       await rm(resolveInsidePortfolioRoot("CNAME"), { force: true });
2425:     }
2426:   } catch {
2427:     // Best-effort rollback; preserve the original error for the caller.
2428:   }

```

2429: }

## ***writeTargetCustomDomain - lines 2431-2439***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2431-2439 (9 source lines).                                                                                                                                                                     |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | async function writeTargetCustomDomain(domain, customDomainProvided) {                                                                                                                                |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | resolveInsidePortfolioRoot, samePath, push, resolveInsideRoot, writeFile, rm                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2431: async function writeTargetCustomDomain(domain, customDomainProvided) {
2432:   const targetPaths = [resolveInsidePortfolioRoot("CNAME")];
2433:   if (!samePath(root, portfolioRoot)) targetPaths.push(resolveInsideRoot("CNAME"));
2434:   if (customDomainProvided && domain) {
2435:     for (const targetPath of targetPaths) await writeFile(targetPath, `${domain}\n`, "utf8");
2436:   } else if (customDomainProvided && !domain) {
2437:     for (const targetPath of targetPaths) await rm(targetPath, { force: true });
2438:   }
2439: }
```

## ***ensureGitRepository - lines 2441-2454***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2441-2454 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function ensureGitRepository() {                                                                                                                                                                |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | mkdir, runPublishGit, trim                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 5 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2441: async function ensureGitRepository() {
2442:   await mkdir(portfolioRoot, { recursive: true });
2443:   try {
2444:     const insideWorkTree = (await runPublishGit(["rev-parse", "--is-inside-work-tree"])).stdout.trim();
2445:     if (insideWorkTree === "true") return;
2446:   } catch {
2447:     await runPublishGit(["init"]);
2448:   }
2449:
2450:   const branch = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2451:   if (!branch) {
2452:     await runPublishGit(["checkout", "-B", "main"]);
2453:   }
2454: }
```

## ***configurePublishTarget - lines 2456-2486***

| Item               | Explanation                                                                                                                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2456-2486 (31 source lines).                                                                                                                                                                                                       |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                                                   |
| Declaration        | async function configurePublishTarget(options = {}) {                                                                                                                                                                                    |
| Touches            | authentication/security data                                                                                                                                                                                                             |
| Calls detected     | call, validatePublishRemoteUrl, validateCustomDomain, validateCredentialPair, ensureGitRepository, setOriginRemote, detectRemoteDefaultBranch, checkoutPublishBranch, writeTargetCustomDomain, getPublishTargetInfo, storeGitCredentials |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                 |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                  |
| Async/return clues | 8 await expression(s), 1 return statement(s).                                                                                                                                                                                            |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                    |

Source excerpt:

```

2456: async function configurePublishTarget(options = {}) {
2457:   const {
2458:     repositoryUrl = "",
2459:     customDomain = "",
2460:     authUsername = "",
2461:     authPassword = ""
2462:   } = options;
2463:   const customDomainProvided = Object.prototype.hasOwnProperty.call(options, "customDomain");
2464:   const remoteUrl = validatePublishRemoteUrl(repositoryUrl);
2465:   const domain = validateCustomDomain(customDomain);
2466:   validateCredentialPair(authUsername, authPassword);
2467:
2468:   await ensureGitRepository();
2469:
2470:   if (remoteUrl) {
2471:     await setOriginRemote(remoteUrl);
2472:     const defaultBranch = await detectRemoteDefaultBranch(remoteUrl);
2473:     if (defaultBranch) await checkoutPublishBranch(defaultBranch);
2474:   }
2475:
2476:   await writeTargetCustomDomain(domain, customDomainProvided);
2477:
2478:   const currentRemote = remoteUrl || (await getPublishTargetInfo()).remote;
2479:   const credentials = await storeGitCredentials(currentRemote, authUsername, authPassword);
2480:   const target = await getPublishTargetInfo();
2481:   return {
2482:     ...target,
2483:     credentialsStored: credentials.stored,
2484:     credentialUsername: credentials.username || ""
2485:   };
2486: }

```

## ***readPublishAuthCache - lines 2488-2494***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2488-2494 (7 source lines).                                                                                                                                                                     |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function readPublishAuthCache() {                                                                                                                                                               |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | parse, readFile                                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2488: async function readPublishAuthCache() {
2489:   try {
2490:     return JSON.parse(await readFile(publishAuthCachePath, "utf8"));

```



```

2491:   } catch {
2492:     return null;
2493:   }
2494: }

```

## ***writePublishAuthCache - lines 2496-2535***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2496-2535 (40 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | async function writePublishAuthCache(access = {}) {                                                                                                                                                   |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | readPublishAuthCache, getTime, isArray, map, parsePublishAuthTimestamp, filter, isFinite, toISOString, publishAuthCacheScope, writeFile, stringify, chmod                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2496: async function writePublishAuthCache(access = {}) {
2497:   const previousCache = await readPublishAuthCache();
2498:   const checkedAt = new Date();
2499:   const checkedAtMs = checkedAt.getTime();
2500:   const sameTarget = previousCache
2501:     && previousCache.remote === access.remote
2502:     && previousCache.branch === access.branch
2503:     && previousCache.repository === access.repository;
2504:   const previousHistory = sameTarget && Array.isArray(previousCache.successHistory)
2505:     ? previousCache.successHistory
2506:     : [];
2507:   const successTimestamps = [
2508:     ...previousHistory
2509:       .map((entry) => parsePublishAuthTimestamp(entry))
2510:       .filter((timestamp) => Number.isFinite(timestamp))
2511:       .filter((timestamp) => checkedAtMs - timestamp < publishAuthExtendedTtlMs),
2512:     checkedAtMs
2513:   ];
2514:   const successHistory = successTimestamps.map((timestamp) => new Date(timestamp).toISOString());
2515:   const successCountLastWeek = successTimestamps
2516:     .filter((timestamp) => checkedAtMs - timestamp < publishAuthHistoryWindowMs)
2517:     .length;
2518:   const extendedTrust = successCountLastWeek > publishAuthExtendedThreshold;
2519:   const ttlMs = extendedTrust ? publishAuthExtendedTtlMs : publishAuthCacheTtlMs;
2520:   const cache = {
2521:     branch: access.branch || "",
2522:     checkedAt: checkedAt.toISOString(),
2523:     expiresAt: new Date(checkedAtMs + ttlMs).toISOString(),
2524:     extendedTrust,
2525:     remote: access.remote || "",
2526:     repository: access.repository || "",
2527:     scope: publishAuthCacheScope(),
2528:     successCountLastWeek,
2529:     successHistory,
2530:     trustDays: extendedTrust ? 30 : 1
2531:   };
2532:   await writeFile(publishAuthCachePath, `${JSON.stringify(cache, null, 2)}\n`, "utf8");
2533:   await chmod(publishAuthCachePath, 0o600).catch(() => {});
2534:   return cache;
2535: }

```

## ***publishAuthCacheScope - lines 2537-2552***

| Item            | Explanation                                                            |
|-----------------|------------------------------------------------------------------------|
| Source location | Lines 2537-2552 (16 source lines).                                     |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior. |
| Declaration     | function publishAuthCacheScope() {                                     |
| Touches         | authentication/security data                                           |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | userInfo, createHash, update, hostname, join, digest                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2537: function publishAuthCacheScope() {
2538:   let username = process.env.USERNAME || process.env.USER || "";
2539:   try {
2540:     username = os.userInfo().username || username;
2541:   } catch {
2542:     // Environment username is a sufficient fallback for cache scoping.
2543:   }
2544:   return createHash("sha256")
2545:     .update([
2546:       os.hostname(),
2547:       username,
2548:       root,
2549:       portfolioRoot
2550:     ].join("\n"))
2551:     .digest("hex");
2552: }

```

### ***parsePublishAuthTimestamp - lines 2554-2559***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2554-2559 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function parsePublishAuthTimestamp(value = "") {                                                                                                                                                      |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | parse, isFinite, replace                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2554: function parsePublishAuthTimestamp(value = "") {
2555:   const direct = Date.parse(String(value || ""));
2556:   if (Number.isFinite(direct)) return direct;
2557:   const normalized = String(value || "").replace(/(\.d{3})\d+(Z|[-]\d{2}:?\d{2})$/i, "$1$2");
2558:   return Date.parse(normalized);
2559: }

```

### ***publishAuthCacheExpiresAt - lines 2561-2567***

| Item            | Explanation                                                            |
|-----------------|------------------------------------------------------------------------|
| Source location | Lines 2561-2567 (7 source lines).                                      |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior. |
| Declaration     | function publishAuthCacheExpiresAt(cache) {                            |
| Touches         | authentication/security data                                           |
| Calls detected  | parsePublishAuthTimestamp, isFinite, toISOString                       |
| API endpoints   | None detected inside this function body.                               |
| Storage keys    | No local/session storage keys detected.                                |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2561: function publishAuthCacheExpiresAt(cache) {
2562:   const explicitExpiresAt = parsePublishAuthTimestamp(cache?.expiresAt || "");
2563:   if (Number.isFinite(explicitExpiresAt)) return new Date(explicitExpiresAt).toISOString();
2564:   const checkedAt = parsePublishAuthTimestamp(cache?.checkedAt || "");
2565:   if (!Number.isFinite(checkedAt)) return "";
2566:   return new Date(checkedAt + publishAuthCacheTtlMs).toISOString();
2567: }

```

## ***publishAuthCachelsFresh - lines 2569-2576***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2569-2576 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function publishAuthCachelsFresh(cache, access) {                                                                                                                                                     |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | publishAuthCacheScope, parse, publishAuthCacheExpiresAt, isFinite, now                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2569: function publishAuthCacheIsFresh(cache, access) {
2570:   if (!cache || !access) return false;
2571:   if (cache.remote !== access.remote || cache.branch !== access.branch || cache.repository !==
access.repository) return false;
2572:   if (cache.scope !== publishAuthCacheScope()) return false;
2573:   const expiresAt = Date.parse(publishAuthCacheExpiresAt(cache));
2574:   if (!Number.isFinite(expiresAt)) return false;
2575:   return Date.now() < expiresAt;
2576: }

```

## ***assertPublishAccess - lines 2578-2669***

| Item               | Explanation                                                                                                                                                                                                                                                                    |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2578-2669 (92 source lines).                                                                                                                                                                                                                                             |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                                                                                         |
| Declaration        | async function assertPublishAccess(options = {}) {                                                                                                                                                                                                                             |
| Touches            | authentication/security data                                                                                                                                                                                                                                                   |
| Calls detected     | runPublishGit, trim, publishAccessError, gitFailureText, remoteUrlForDisplay, workspaceHasCompatibleSiteFiles, parseGitHubRemote, readPublishAuthCache, publishAuthCachelsFresh, publishAuthCacheExpiresAt, synchronizePublishBranchFromRemote, ensurePublishHeadForWriteCheck |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                                                       |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                                                        |
| Async/return clues | 9 await expression(s), 2 return statement(s).                                                                                                                                                                                                                                  |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                                                          |

Source excerpt:

```

2578: async function assertPublishAccess(options = {}) {
2579:   let insideWorkTree = "";
2580:   try {

```

```

2581:     insideWorkTree = (await runPublishGit(["rev-parse", "--is-inside-work-tree"])).stdout.trim();
2582:   } catch (error) {
2583:     throw publishAccessError(
2584:       "This workspace is not connected to a Git repository.",
2585:       gitFailureText(error),
2586:       { repository: defaultSiteRepository }
2587:     );
2588:   }
2589:
2590:   if (insideWorkTree !== "true") {
2591:     throw publishAccessError(
2592:       "This workspace is local-only and cannot publish yet.",
2593:       "Clone or associate a GitHub Pages/static-site repository before applying to site.",
2594:       { repository: defaultSiteRepository }
2595:     );
2596:   }
2597:
2598:   let remote = "";
2599:   let branch = "";
2600:   try {
2601:     remote = (await runPublishGit(["remote", "get-url", "origin"])).stdout.trim();
2602:     branch = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2603:   } catch (error) {
2604:     throw publishAccessError(
2605:       "A publish remote or branch is missing.",
2606:       gitFailureText(error),
2607:       { repository: defaultSiteRepository }
2608:     );
2609:   }
2610:
2611:   if (!remote || !branch) {
2612:     throw publishAccessError(
2613:       "A publish remote or branch is missing.",
2614:       "Set the origin remote and use a named branch before applying to site.",
2615:       { remote: remoteUrlForDisplay(remote), branch }
2616:     );
2617:   }
2618:
2619:   if (options.requireCompatible !== false && !await workspaceHasCompatibleSiteFiles()) {
2620:     throw publishAccessError(
2621:       "This repository does not look like a compatible static portfolio website.",
2622:       "The workspace must include index.html, styles.css, and script.js before it can be used as a
publish target.",
2623:       { remote: remoteUrlForDisplay(remote), branch }
2624:     );
2625:   }
2626:
2627:   const parsedRemote = parseGitHubRemote(remote);
2628:   const repository = parsedRemote ? `${parsedRemote.owner}/${parsedRemote.repo}` :
remoteUrlForDisplay(remote);
2629:
2630:   const access = {
2631:     branch,
2632:     remote: remoteUrlForDisplay(remote),
2633:     repository,
2634:     authorizationChecked: false
2635:   };
2636:
2637:   const cached = await readPublishAuthCache();
2638:   if (!options.force && publishAuthCacheIsFresh(cached, access)) {
2639:     return {
2640:       ...access,
2641:       authorizationCached: true,
2642:       authorizationChecked: true,
2643:       checkedAt: cached.checkedAt,
2644:       expiresAt: publishAuthCacheExpiresAt(cached),
2645:       extendedTrust: Boolean(cached.extendedTrust),
2646:       successCountLastWeek: cached.successCountLastWeek || 0,
2647:       trustDays: cached.trustDays || (cached.extendedTrust ? 30 : 1)
2648:     };
2649:   }
2650:
2651:   const remoteSync = await synchronizePublishBranchFromRemote(branch, access);
2652:   const localHead = await ensurePublishHeadForWriteCheck();
2653:   const writeCheck = await verifyPublishWriteAccess(access);
2654:
2655:   const authCache = await writePublishAuthCache(access);
2656:   return {
2657:     ...access,
2658:     remoteSync,
2659:     localHead,
2660:     writeCheck,
2661:     authorizationChecked: true,
2662:     authorizationCached: false,
2663:     checkedAt: authCache.checkedAt,
2664:     expiresAt: authCache.expiresAt,
2665:     extendedTrust: Boolean(authCache.extendedTrust),
2666:     successCountLastWeek: authCache.successCountLastWeek || 0,
2667:     trustDays: authCache.trustDays || 1
2668:   };
2669: }

```

## syncFromPublishTarget - lines 2671-2745

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2671-2745 (75 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function syncFromPublishTarget() {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | assertPublishAccess, runPublishGit, trim, mkdtemp, join, tmpdir, runGit, fsAccess, push, includes, publishAccessError, resolveInsideRoot                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 14 await expression(s), 1 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2671: async function syncFromPublishTarget() {
2672:   const publishAccess = await assertPublishAccess({ requireCompatible: false });
2673:   const branch = publishAccess.branch || "main";
2674:   const remote = (await runPublishGit(["remote", "get-url", "origin])).stdout.trim();
2675:   const importRoot = await mkdtemp(path.join(os.tmpdir(), "omb-publish-target-"));
2676:   const cloneRoot = path.join(importRoot, "target");
2677:   const importPaths = [
2678:     "projects.json",
2679:     "assets",
2680:     "docs",
2681:     "Backgrounds",
2682:     "CNAME",
2683:     ".nojekyll",
2684:     "robots.txt",
2685:     "sitemap.xml"
2686:   ];
2687:
2688:   try {
2689:     await runGit(["clone", "--depth", "1", "--branch", branch, remote, cloneRoot], {
2690:       cwd: importRoot,
2691:       timeout: 3 * 60 * 1000
2692:     });
2693:
2694:     const availablePaths = [];
2695:     for (const importPath of importPaths) {
2696:       try {
2697:         await fsAccess(path.join(cloneRoot, importPath));
2698:         availablePaths.push(importPath);
2699:       } catch {
2700:         // Optional path is absent in the selected site repository.
2701:       }
2702:     }
2703:
2704:     if (!availablePaths.includes("projects.json")) {
2705:       throw publishAccessError(
2706:         "The selected repository does not contain a projects.json portfolio catalog.",
2707:         "The target repository must contain a compatible OMB Portfolio Builder catalog before it can be
imported.",
2708:         publishAccess
2709:       );
2710:     }
2711:
2712:     const backupRoot = resolveInsideRoot(
2713:       ".omb-backups",
2714:       `load-target-${new Date().toISOString().replace(/[:.]/g, "-")}`
2715:     );
2716:     await mkdir(backupRoot, { recursive: true });
2717:
2718:     for (const importPath of availablePaths) {
2719:       const sourcePath = path.join(cloneRoot, importPath);
2720:       const targetPath = resolveInsideRoot(importPath);
2721:       try {
2722:         await fsAccess(targetPath);
2723:         await cp(targetPath, path.join(backupRoot, importPath), { recursive: true, force: true });
2724:         await rm(targetPath, { recursive: true, force: true });
2725:       } catch {
2726:         // Nothing local to back up for this path.
2727:       }
2728:       await cp(sourcePath, targetPath, { recursive: true, force: true });
2729:     }
2730:
2731:     const importedCatalog = await readFile(path.join(cloneRoot, "projects.json"), "utf8");
```

```

2732:     await writeFile(draftPath, importedCatalog, "utf8");
2733:     const portfolioSync = await syncPortfolioPublishFiles({ removeMissing: false });
2734:
2735:     return {
2736:       ...publishAccess,
2737:       imported: availablePaths,
2738:       portfolioSync,
2739:       backup: path.relative(root, backupRoot),
2740:       draftUpdated: true
2741:     };
2742:   } finally {
2743:     await rm(importRoot, { recursive: true, force: true });
2744:   }
2745: }

```

## ***authenticateGitHubForTarget - lines 2747-2843***

| Item               | Explanation                                                                                                                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2747-2843 (97 source lines).                                                                                                                                                                                                                      |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                                                                  |
| Declaration        | async function authenticateGitHubForTarget(options = {}) {                                                                                                                                                                                              |
| Touches            | authentication/security data                                                                                                                                                                                                                            |
| Calls detected     | call, validatePublishRemoteUrl, validateCustomDomain, validateCredentialPair, publishAccessError, getPublishTargetInfo, ensureGitRepository, capturePublishTargetState, setOriginRemote, getGitStatus, detectRemoteDefaultBranch, checkoutPublishBranch |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                                |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                                 |
| Async/return clues | 19 await expression(s), 2 return statement(s).                                                                                                                                                                                                          |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                                   |

Source excerpt:

```

2747: async function authenticateGitHubForTarget(options = {}) {
2748:   const {
2749:     repositoryUrl = "",
2750:     customDomain = "",
2751:     authUsername = "",
2752:     authPassword = ""
2753:   } = options;
2754:   const customDomainProvided = Object.prototype.hasOwnProperty.call(options, "customDomain");
2755:   const remoteUrl = validatePublishRemoteUrl(repositoryUrl);
2756:   const domain = validateCustomDomain(customDomain);
2757:   const credentials = validateCredentialPair(authUsername, authPassword);
2758:   if (!remoteUrl) {
2759:     throw publishAccessError(
2760:       "A GitHub repository URL is required before authentication.",
2761:       "Enter the GitHub Pages or compatible static-site repository URL, then click Authenticate target.",
2762:       await getPublishTargetInfo()
2763:     );
2764:   }
2765:
2766:   await ensureGitRepository();
2767:   const snapshot = await capturePublishTargetState();
2768:
2769:   try {
2770:     await setOriginRemote(remoteUrl);
2771:     const status = await getGitStatus();
2772:     const preliminaryTarget = await getPublishTargetInfo();
2773:     if (!status.git.ok) {
2774:       throw publishAccessError(
2775:         "Git for Windows is required for publishing.",
2776:         "Install Git for Windows, then reopen the builder and authenticate again.",
2777:         { ...preliminaryTarget, system: status }
2778:       );
2779:     }
2780:
2781:     const targetBranch = await detectRemoteDefaultBranch(remoteUrl) || "main";
2782:     await checkoutPublishBranch(targetBranch);
2783:
2784:     const targetBeforeAuth = await getPublishTargetInfo();
2785:     const cached = await readPublishAuthCache();
2786:     const cachedAccess = {
2787:       branch: targetBeforeAuth.branch,
2788:       remote: targetBeforeAuth.remote,
2789:       repository: targetBeforeAuth.repository
2790:     };
2791:     if (!credentials.cleanUsername && publishAuthCacheIsFresh(cached, cachedAccess)) {
2792:       await writeTargetCustomDomain(domain, customDomainProvided);

```

```

2793:     const target = await getPublishTargetInfo();
2794:     return {
2795:       ...cachedAccess,
2796:       branch: target.branch || cachedAccess.branch,
2797:       authorizationCached: true,
2798:       authorizationChecked: true,
2799:       checkedAt: cached.checkedAt,
2800:       expiresAt: publishAuthCacheExpiresAt(cached),
2801:       extendedTrust: Boolean(cached.extendedTrust),
2802:       successCountLastWeek: cached.successCountLastWeek || 0,
2803:       trustDays: cached.trustDays || (cached.extendedTrust ? 30 : 1),
2804:       credentialsStored: false,
2805:       credentialUsername: "",
2806:       system: status,
2807:       target
2808:     };
2809:   }
2810:
2811:   if (!status.credentialManager.ok && !credentials.cleanUsername) {
2812:     throw publishAccessError(
2813:       "Git Credential Manager is required for guided GitHub sign-in.",
2814:       "Install or repair Git for Windows with Git Credential Manager enabled, or provide a GitHub
username and personal access token.",
2815:       { ...targetBeforeAuth, system: status }
2816:     );
2817:   }
2818:
2819:   const storedCredentials = await storeGitCredentials(remoteUrl, authUsername, authPassword);
2820:   if (!storedCredentials.stored) {
2821:     await runGit(["credential-manager", "github", "login"], {
2822:       cwd: portfolioRoot,
2823:       timeout: 5 * 60 * 1000,
2824:       windowsHide: false
2825:     });
2826:   }
2827:
2828:   const access = await assertPublishAccess({ force: true, requireCompatible: false });
2829:   await writeTargetCustomDomain(domain, customDomainProvided);
2830:   const target = await getPublishTargetInfo();
2831:   return {
2832:     ...access,
2833:     branch: target.branch || access.branch,
2834:     credentialsStored: storedCredentials.stored,
2835:     credentialUsername: storedCredentials.username || "",
2836:     system: await getGitStatus(),
2837:     target
2838:   };
2839: } catch (error) {
2840:   await restorePublishTargetState(snapshot);
2841:   throw error;
2842: }
2843: }

```

## ***installGitForWindows - lines 2845-2868***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2845-2868 (24 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function installGitForWindows() {                                                                                                                                                               |
| Touches            | external processes, network/API requests, authentication/security data                                                                                                                                |
| Calls detected     | runOptionalCommand, spawn,_unref                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2845: async function installGitForWindows() {
2846:   const winget = await runOptionalCommand("winget", ["--version"]);
2847:   const downloadUrl = "https://git-scm.com/download/win";
2848:   if (!winget.ok) {
2849:     return {
2850:       launched: false,
2851:       downloadUrl,
2852:       message: "winget was not found. Open the Git for Windows download page and install Git with Git
Credential Manager enabled."
2853:     };

```

```

2854:   }
2855:
2856:   const child = spawn("winget", ["install", "--id", "Git.Git", "-e", "--source", "winget"], {
2857:     detached: true,
2858:     shell: true,
2859:     stdio: "ignore",
2860:     windowsHide: false
2861:   });
2862:   child.unref();
2863:   return {
2864:     launched: true,
2865:     downloadUrl,
2866:     message: "The Git for Windows installer was launched through winget. Follow the installer prompts,
then reopen the builder."
2867:   };
2868: }

```

## getUpdateInfo - lines 2870-2906

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2870-2906 (37 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function getUpdateInfo() {                                                                                                                                                                      |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | fetch, Error, json, replace, find, has, compareVersions                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2870: async function getUpdateInfo() {
2871:   const currentVersion = process.env.OMB_APP_VERSION || packageJson.version || "0.0.0";
2872:   const owner = "otienomaurice";
2873:   const repo = "omb-portfolio-builder";
2874:   try {
2875:     const response = await fetch(`https://api.github.com/repos/${owner}/${repo}/releases/latest`, {
2876:       headers: { "Accept": "application/vnd.github+json", "User-Agent": "OMB-Portfolio-Builder" }
2877:     });
2878:     if (!response.ok) throw new Error(`GitHub returned ${response.status}`);
2879:     const release = await response.json();
2880:     const latestVersion = String(release.tag_name || "").replace(/^builder-v/i, "");
2881:     const installer = (release.assets || []).find((asset) => /Setup-.*\.exe$/i.test(asset.name));
2882:     const portable = (release.assets || []).find((asset) => /Portable-.*\.exe$/i.test(asset.name));
2883:     const updateBlocked = blockedAppUpdateVersions.has(latestVersion);
2884:     return {
2885:       ok: true,
2886:       currentVersion,
2887:       latestVersion,
2888:       updateAvailable: !updateBlocked && compareVersions(latestVersion, currentVersion) > 0,
2889:       updateBlocked,
2890:       blockedReason: updateBlocked
? `Builder ${latestVersion} is skipped because its installer can hang while running the old
uninstaller during in-app updates. Wait for builder 0.2.17 or newer, then run Update again.`
2892:       : "",
2893:       releaseUrl: release.html_url || "",
2894:       installerUrl: installer?.browser_download_url || "",
2895:       installerName: installer?.name || "",
2896:       portableUrl: portable?.browser_download_url || ""
2897:     };
2898:   } catch (error) {
2899:     return {
2900:       ok: false,
2901:       currentVersion,
2902:       latestVersion: currentVersion,
2903:       updateAvailable: false,
2904:       error: error.message || "Could not check for updates."
2905:     };
2906:   }

```

## installer - lines 2881-2897



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2881-2897 (17 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | const installer = (release.assets    []).find((asset) => /Setup-.*\.exe\$/i.test(asset.name));                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, has, compareVersions                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2881:     const installer = (release.assets || []).find((asset) => /Setup-.*\.exe$/i.test(asset.name));
2882:     const portable = (release.assets || []).find((asset) => /Portable-.*\.exe$/i.test(asset.name));
2883:     const updateBlocked = blockedAppUpdateVersions.has(latestVersion);
2884:     return {
2885:       ok: true,
2886:       currentVersion,
2887:       latestVersion,
2888:       updateAvailable: !updateBlocked && compareVersions(latestVersion, currentVersion) > 0,
2889:       updateBlocked,
2890:       blockedReason: updateBlocked
2891:       ? `Builder ${latestVersion} is skipped because its installer can hang while running the old
uninstaller during in-app updates. Wait for builder 0.2.17 or newer, then run Update again.`
2892:       : "",
2893:       releaseUrl: release.html_url || "",
2894:       installerUrl: installer?.browser_download_url || "",
2895:       installerName: installer?.name || "",
2896:       portableUrl: portable?.browser_download_url || ""
2897:     };

```

## ***portable - lines 2882-2897***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2882-2897 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | const portable = (release.assets    []).find((asset) => /Portable-.*\.exe\$/i.test(asset.name));                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, has, compareVersions                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2882:     const portable = (release.assets || []).find((asset) => /Portable-.*\.exe$/i.test(asset.name));
2883:     const updateBlocked = blockedAppUpdateVersions.has(latestVersion);
2884:     return {
2885:       ok: true,
2886:       currentVersion,
2887:       latestVersion,
2888:       updateAvailable: !updateBlocked && compareVersions(latestVersion, currentVersion) > 0,
2889:       updateBlocked,
2890:       blockedReason: updateBlocked
2891:       ? `Builder ${latestVersion} is skipped because its installer can hang while running the old
uninstaller during in-app updates. Wait for builder 0.2.17 or newer, then run Update again.`
2892:       : "",
2893:       releaseUrl: release.html_url || "",
2894:       installerUrl: installer?.browser_download_url || "",
2895:       installerName: installer?.name || "",
2896:       portableUrl: portable?.browser_download_url || ""
2897:     };

```

## ***getBuilderReleaseDownloadReport - lines 2909-2914***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2909-2914 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function getBuilderReleaseDownloadReport() {                                                                                                                                                    |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | fetch                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2909: async function getBuilderReleaseDownloadReport() {
2910:   const owner = "otienomaurice";
2911:   const repo = "omb-portfolio-builder";
2912:   const response = await fetch(`https://api.github.com/repos/${owner}/${repo}/releases/latest`, {
2913:     headers: { "Accept": "application/vnd.github+json", "User-Agent": "OMB-Portfolio-Builder" }
2914:   });
```

## ***assets - lines 2917-2923***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2917-2923 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | const assets = (release.assets    []).map((asset) => ({                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map                                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2917:   const assets = (release.assets || []).map((asset) => ({
2918:     name: asset.name || "",
2919:     downloadCount: Number(asset.download_count || 0),
2920:     sizeBytes: Number(asset.size || 0),
2921:     updatedAt: asset.updated_at || "",
2922:     url: asset.browser_download_url || ""
2923:   }));
```

## ***getSecurityReport - lines 2933-2977***

| Item            | Explanation                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 2933-2977 (45 source lines).                                                                                                                      |
| Purpose         | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                              |
| Declaration     | async function getSecurityReport() {                                                                                                                    |
| Touches         | authentication/security data                                                                                                                            |
| Calls detected  | allSettled, getBuilderReleaseDownloadReport, readPublishAuthCache, getPublishTargetInfo, toISOString, publishAuthCachesFresh, publishAuthCacheExpiresAt |
| API endpoints   | None detected inside this function body.                                                                                                                |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2933: async function getSecurityReport() {
2934:   const [downloadResult, authCacheResult, targetResult] = await Promise.allSettled([
2935:     getBuilderReleaseDownloadReport(),
2936:     readPublishAuthCache(),
2937:     getPublishTargetInfo()
2938:   ]);
2939:   const authCache = authCacheResult.status === "fulfilled" ? authCacheResult.value : null;
2940:   const target = targetResult.status === "fulfilled" ? targetResult.value : null;
2941:   return {
2942:     ok: true,
2943:     generatedAt: new Date().toISOString(),
2944:     appDownloads: downloadResult.status === "fulfilled"
2945:       ? { ok: true, ...downloadResult.value }
2946:       : { ok: false, error: downloadResult.reason?.message || "GitHub release download counts were not
available." },
2947:     publishingAuth: {
2948:       targetRepository: target?.repository || "",
2949:       branch: target?.branch || "",
2950:       authenticated: Boolean(authCache && publishAuthCacheIsFresh(authCache, {
2951:         remote: target?.remote || "",
2952:         repository: target?.repository || "",
2953:         branch: target?.branch || ""
2954:       })),
2955:       expiresAt: publishAuthCacheExpiresAt(authCache) || "",
2956:       trustDays: Number(authCache?.trustDays || 0),
2957:       successCountLastWeek: Number(authCache?.successCountLastWeek || 0)
2958:     },
2959:     websiteAccess: {
2960:       available: false,
2961:       summary: "Visitor counts and visitor IP addresses are not available from a static GitHub Pages site
or GitHub release download counts.",
2962:       recommendedSetup: [
2963:         "Proxy the custom domain through Cloudflare instead of DNS-only.",
2964:         "Enable Cloudflare Web Analytics for aggregate page-view counts.",
2965:         "Use Cloudflare Logpush, Analytics Engine, or a Worker-backed endpoint if you need IP-level
security logs.",
2966:         "Publish a privacy notice before collecting raw IP addresses or persistent identifiers."
2967:       ]
2968:     },
2969:     securityControls: [
2970:       "Local builder write APIs are restricted to local requests.",
2971:       "Publishing target details are only returned to the local machine.",
2972:       "GitHub publishing authorization is cached for one day by default, or longer after repeated
successful authorizations.",
2973:       "The desktop app runs with Electron nodeIntegration disabled, contextIsolation enabled, and sandbox
enabled.",
2974:       "Deployable website security headers and security.txt metadata are included in the publishable site
files."
2975:     ]
2976:   };
2977: }

```

## ***safeUpdateFileSegment - lines 2979-2984***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2979-2984 (6 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function safeUpdateFileSegment(value = "") {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replace, slice                                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2979: function safeUpdateFileSegment(value = "") {
2980:   return String(value || "")
2981:   .replace(/[^a-z0-9._-]+/gi, "-")
2982:   .replace(/^-+|-+$/g, "")
2983:   .slice(0, 80) || "latest";
2984: }
```

## ***powershellSingleQuoted - lines 2986-2988***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2986-2988 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | function powershellSingleQuoted(value = "") {                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replaceAll                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2986: function powershellSingleQuoted(value = "") {
2987:   return `${String(value).replaceAll("'", "'')}';
2988: }
```

## ***downloadAndLaunchAppUpdate - lines 2990-3165***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2990-3165 (176 source lines).                                                                                                                                                                   |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | async function downloadAndLaunchAppUpdate() {                                                                                                                                                         |
| Touches            | filesystem, external processes, network/API requests                                                                                                                                                  |
| Calls detected     | getUpdateInfo, Error, join, tmpdir, mkdir, safeUpdateFileSegment, fetch, from, arrayBuffer, writeFile, dirname, homedir                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 7 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2990: async function downloadAndLaunchAppUpdate() {
2991:   const update = await getUpdateInfo();
2992:   if (!update.ok) throw new Error(update.error || "Could not check for updates.");
2993:   if (update.updateBlocked) {
2994:     throw new Error(update.blockedReason || `Builder ${update.latestVersion} is temporarily blocked from
automatic updates.`);
2995:   }
2996:   if (!update.updateAvailable) {
2997:     throw new Error(`OMB Portfolio Builder ${update.currentVersion || ""} is already up to date.`);
2998:   }
2999:   if (!update.installerUrl) {
3000:     throw new Error("The latest GitHub Release does not include a Windows setup installer.");
3001:   }
3002:   if (process.platform !== "win32") {
3003:     throw new Error("Automatic installer updates are currently available only on Windows.");
3004:   }
3005:
3006:   const updateRoot = path.join(os.tmpdir(), "omb-portfolio-builder-updates");
3007:   await mkdir(updateRoot, { recursive: true });
3008:   const installerName = update.installerName && /\.exe$/i.test(update.installerName)
3009:     ? update.installerName
```

```

3010:     : `OMB-Portfolio-Builder-Setup-${safeUpdateFileSegment(update.latestVersion)}-x64.exe`;
3011:     const installerPath = path.join(updateRoot, installerName);
3012:
3013:     const response = await fetch(update.installerUrl, {
3014:       headers: { "Accept": "application/octet-stream", "User-Agent": "OMB-Portfolio-Builder" }
3015:     });
3016:     if (!response.ok) throw new Error(`GitHub returned ${response.status} while downloading the
installer.`);
3017:     const installerBuffer = Buffer.from(await response.arrayBuffer());
3018:     if (installerBuffer.length < 1024 * 1024) {
3019:       throw new Error("The downloaded installer was unexpectedly small, so the update was stopped.");
3020:     }
3021:     await writeFile(installerPath, installerBuffer);
3022:
3023:     const updateLogPath = path.join(updateRoot,
`update-${safeUpdateFileSegment(update.latestVersion)}.log`);
3024:     const currentExecutable = /OMB Portfolio Builder\.exe$/i.test(process.execPath || "") ?
process.execPath : "";
3025:     const currentInstallDirectory = currentExecutable ? path.dirname(currentExecutable) : "";
3026:     const localAppDataExecutable = process.env.LOCALAPPDATA
3027:       ? path.join(process.env.LOCALAPPDATA, "Programs", "OMB Portfolio Builder", "OMB Portfolio
Builder.exe")
3028:       : "";
3029:     const legacyManagedApplicationExecutable = path.join(os.homedir(), "OMB", "application", "OMB Portfolio
Builder", "OMB Portfolio Builder.exe");
3030:     const legacyFlatApplicationExecutable = path.join(os.homedir(), "OMB", "application", "OMB Portfolio
Builder.exe");
3031:     const relaunchCandidates = Array.from(new Set([
3032:       localAppDataExecutable,
3033:       currentExecutable,
3034:       currentInstallDirectory ? path.join(currentInstallDirectory, "OMB Portfolio Builder.exe") : "",
3035:       legacyFlatApplicationExecutable,
3036:       legacyManagedApplicationExecutable,
3037:       process.env.ProgramFiles ? path.join(process.env.ProgramFiles, "OMB Portfolio Builder", "OMB
Portfolio Builder.exe") : "",
3038:       process.env["ProgramFiles(x86)"] ? path.join(process.env["ProgramFiles(x86)"], "OMB Portfolio
Builder", "OMB Portfolio Builder.exe") : ""
3039:     ]).filter(Boolean));
3040:     const relaunchCandidatesPs = `@(${relaunchCandidates.map(powershellSingleQuoted).join(", ")});`;
3041:     const launcherPath = path.join(updateRoot, `run-
update-${safeUpdateFileSegment(update.latestVersion)}.ps1`);
3042:     const launcherCommandPath = path.join(updateRoot, `run-
update-${safeUpdateFileSegment(update.latestVersion)}.cmd`);
3043:     const launcherScript = [
3044:       "$ErrorActionPreference = 'Continue'",
3045:       "`$parentPid = ${process.pid}`",
3046:       "`$installer = ${powershellSingleQuoted(installerPath)}",
3047:       "`$log = ${powershellSingleQuoted(updateLogPath)}",
3048:       "`$candidates = ${relaunchCandidatesPs}",
3049:       "function Write-UpdateLog([string]$Message) {",
3050:       "  try { Add-Content -LiteralPath $log -Value ((Get-Date).ToString('s') + ' ' + $Message) } catch
{}",
3051:       "}",
3052:       "function Wait-ForExecutable([string[]]$Paths, [int]$Seconds) {",
3053:       "  $deadline = (Get-Date).AddSeconds($Seconds)",
3054:       "  do {",
3055:       "    foreach ($candidate in $Paths) {",
3056:       "      if ($candidate -and (Test-Path -LiteralPath $candidate)) { return $candidate }",
3057:       "    }",
3058:       "    Start-Sleep -Seconds 1",
3059:       "  } while ((Get-Date) -lt $deadline)",
3060:       "  return $null",
3061:       "}",
3062:       "function Stop-BUILDERProcesses([int]$ProcessId, [string[]]$Paths) {",
3063:       "  try {",
3064:       "    $normalizedPaths = @($Paths | Where-Object { $_ } | ForEach-Object {",
3065:       "      try { [System.IO.Path]::GetFullPath($_).ToLowerInvariant() } catch { $_.ToLowerInvariant() }",
3066:       "    }",
3067:       "    $processes = Get-CimInstance Win32_Process | Where-Object {",
3068:       "      $_.ProcessId -eq $ProcessId -or",
3069:       "      $_.Name -eq 'OMB Portfolio Builder.exe' -or",
3070:       "      ($_.ExecutablePath -and ($normalizedPaths -contains
([System.IO.Path]::GetFullPath($_.ExecutablePath).ToLowerInvariant()))",
3071:       "    } | Sort-Object ProcessId -Unique",
3072:       "    foreach ($process in $processes) {",
3073:       "      try {",
3074:       "        Write-UpdateLog ('Stopping previous builder process PID ' + $process.ProcessId + ' at ' +
$process.ExecutablePath)",
3075:       "        Stop-Process -Id $process.ProcessId -Force -ErrorAction SilentlyContinue",
3076:       "      } catch { Write-UpdateLog ('Could not stop PID ' + $process.ProcessId + ': ' +
$_.Exception.Message) }",
3077:       "    }",
3078:       "    Start-Sleep -Seconds 2",
3079:       "  } catch { Write-UpdateLog ('Could not stop previous builder processes: ' + $_.Exception.Message)
}",
3080:       "}",
3081:       "function Run-Installer([switch]$Elevated) {",
3082:       "  $process = $null",
3083:       "  if ($Elevated) {",
3084:       "    Write-UpdateLog 'Starting installer with Windows elevation prompt.'",
3085:       "    $process = Start-Process -FilePath $installer -ArgumentList @('/S') -Verb RunAs -PassThru",
3086:       "  } else {",

```

```

3087:      "      Write-UpdateLog 'Starting installer silently.'",
3088:      "      $process = Start-Process -FilePath $installer -ArgumentList @('/S') -WindowStyle Hidden
-PassThru",
3089:      "    }",
3090:      "    if (-not $process) { return 0 }",
3091:      "    if (-not $process.WaitForExit(900000)) {",
3092:      "      Write-UpdateLog ('Installer timed out after 15 minutes. PID: ' + $process.Id)",
3093:      "      try { Stop-Process -Id $process.Id -Force -ErrorAction SilentlyContinue } catch { Write-
UpdateLog ('Could not stop timed-out installer: ' + $_.Exception.Message) }",
3094:      "      return 124",
3095:      "    }",
3096:      "    return $process.ExitCode",
3097:      "  }",
3098:      "Write-UpdateLog 'Update launcher started.'",
3099:      "Start-Sleep -Seconds 3",
3100:      "try { Wait-Process -Id $parentPid -Timeout 8; Write-UpdateLog 'Previous app process exited.' } catch
{ Write-UpdateLog ('Previous app wait finished: ' + $_.Exception.Message) }",
3101:      "Stop-BuilderProcesses $parentPid $candidates",
3102:      "Start-Sleep -Seconds 1",
3103:      "$installerProcess = $null",
3104:      "$installerExit = 1",
3105:      "try {",
3106:      "    Write-UpdateLog ('Starting installer: ' + $installer)",
3107:      "    $installerExit = Run-Installer",
3108:      "    Write-UpdateLog ('Installer exit code: ' + $installerExit)",
3109:      "} catch {",
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

## syncPortfolioPublishFiles - lines 3167-3200

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3167-3200 (34 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function syncPortfolioPublishFiles(options = {}) {                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | samePath, mkdir, resolveInsideRoot, resolveInsidePortfolioRoot, pathExists, rm, dirname, cp, push                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 6 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3167: async function syncPortfolioPublishFiles(options = {}) {
3168:   if (samePath(root, portfolioRoot)) {
3169:     return { copied: [], skipped: [], workspace: portfolioRoot, separatedWorkspace: false };
3170:   }
3171:
3172:   const { removeMissing = false } = options;
3173:   const copied = [];
3174:   const skipped = [];
3175:
3176:   await mkdir(portfolioRoot, { recursive: true });
3177:
3178:   for (const relativePath of publishPaths) {
3179:     const sourcePath = resolveInsideRoot(relativePath);
3180:     const targetPath = resolveInsidePortfolioRoot(relativePath);
3181:     if (await pathExists(sourcePath)) {
3182:       await rm(targetPath, { recursive: true, force: true });
3183:       await mkdir(path.dirname(targetPath), { recursive: true });
3184:       await cp(sourcePath, targetPath, { recursive: true, force: true });
3185:       copied.push(relativePath);
3186:     } else if (removeMissing) {
3187:       await rm(targetPath, { recursive: true, force: true });
3188:       skipped.push(relativePath);
3189:     } else {
3190:       skipped.push(relativePath);
3191:     }
3192:   }
3193:
3194:   return {
3195:     copied,
3196:     skipped,
3197:     workspace: portfolioRoot,
3198:     separatedWorkspace: true
3199:   };
3200: }

```

## ***publishSiteChanges - lines 3202-3238***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3202-3238 (37 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function publishSiteChanges(publishAccess = null) {                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | assertPublishAccess, runPublishGit, trim, synchronizePublishBranchFromRemote, syncPortfolioPublishFiles, bumpPublishedAssetVersions, stageablePublishPaths, Error, toISOString, slice                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 10 await expression(s), 1 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3202: async function publishSiteChanges(publishAccess = null) {
3203:   const access = publishAccess || await assertPublishAccess();
3204:   const branch = access.branch || (await runPublishGit(["branch", "--show-current"])).stdout.trim();
3205:   const remoteSync = await synchronizePublishBranchFromRemote(branch, access);
3206:   const portfolioSync = await syncPortfolioPublishFiles({ removeMissing: true });
3207:   await bumpPublishedAssetVersions(portfolioRoot);
3208:
3209:   const stageablePaths = await stageablePublishPaths(publishPaths);
3210:   if (!stageablePaths.length) {
3211:     throw new Error("No compatible site files were available to publish.");
3212:   }
3213:
3214:   await runPublishGit(["add", "-A", "--", ...stageablePaths]);
3215:   const status = await runPublishGit(["status", "--porcelain", "--", ...stageablePaths]);
3216:   const hasChanges = status.stdout.trim().length > 0;
3217:
3218:   let commit = null;
3219:   if (hasChanges) {
3220:     const message = `Update portfolio site ${new Date().toISOString().slice(0, 10)}`;
3221:     commit = await runPublishGit(["commit", "-m", message]);
3222:   }
3223:
3224:   const pushArgs = branch ? ["push", "origin", branch] : ["push"];
3225:   const push = await runPublishGit(pushArgs);
3226:
3227:   return {
3228:     ...access,
3229:     portfolioWorkspace: portfolioRoot,
3230:     remoteSync,
3231:     portfolioSync,
3232:     branch: branch || "current branch",
3233:     committed: hasChanges,
3234:     commitOutput: commit?.stdout || commit?.stderr || "",
3235:     pushed: true,
3236:     pushOutput: push.stdout || push.stderr || ""
3237:   };
3238: }
```

## ***handlePortfolioAi - lines 3240-3322***

| Item               | Explanation                                                                                                                                                 |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3240-3322 (83 source lines).                                                                                                                          |
| Purpose            | Handles an event, request, command, or user action.                                                                                                         |
| Declaration        | async function handlePortfolioAi(request, response) {                                                                                                       |
| Touches            | network/API requests, authentication/security data                                                                                                          |
| Calls detected     | readRequestJson, clampText, trim, cleanConversationHistory, Set, has, sendJson, enrichPortfolioContext, callOllamaPortfolioAi, toLowerCase, stringify, join |
| API endpoints      | None detected inside this function body.                                                                                                                    |
| Storage keys       | No local/session storage keys detected.                                                                                                                     |
| Async/return clues | 5 await expression(s), 4 return statement(s).                                                                                                               |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3240: async function handlePortfolioAi(request, response) {
3241:   const body = await readRequestJson(request);
3242:   const question = clampText(body.question, 1200).trim();
3243:   const conversation = cleanConversationHistory(body.conversation);
3244:   const validIntents = new Set(["general_conversation", "general_engineering", "general_knowledge",
"portfolio_specific"]);
3245:   const intent = validIntents.has(body.intent) ? body.intent : "portfolio_specific";
3246:   const allowWebSearch = body.allowWebSearch === true;
3247:
3248:   if (!question) {
3249:     sendJson(response, 400, { error: "Question is required." });
3250:     return;
3251:   }
3252:
3253:   const context = await enrichPortfolioContext({ ...(body.context || {}), question });
3254:
3255:   const apiKey = process.env.OPENAI_API_KEY || "";
3256:   if (!apiKey) {
3257:     const ollama = await callOllamaPortfolioAi({ question, intent, conversation, context, allowWebSearch
});
3258:     if (ollama.ok) {
3259:       sendJson(response, 200, {
3260:         answer: ollama.answer,
3261:         model: ollama.model,
3262:         provider: "ollama",
3263:         usedWebSearch: false
3264:       });
3265:       return;
3266:     }
3267:     sendJson(response, 503, { error: ollama.error || "No local Ollama model or OPENAI_API_KEY is
available for the local backend." });
3268:     return;
3269:   }
3270:
3271:   const model = process.env.OPENAI_MODEL || "gpt-5.4";
3272:   const fallbackModel = process.env.OPENAI_FALLBACK_MODEL || "gpt-4.1";
3273:   const webSearchMode = String(process.env.OPENAI_ENABLE_WEB_SEARCH || "auto").toLowerCase();
3274:   const enableWebSearch = webSearchMode === "true" || (webSearchMode !== "false" && allowWebSearch);
3275:   const buildPayload = (selectedModel) => ({
3276:     model: selectedModel,
3277:     input: [
3278:       {
3279:         role: "developer",
3280:         content: [{ type: "input_text", text: portfolioAiInstructions }]
3281:       },
3282:       {
3283:         role: "user",
3284:         content: [{
3285:           type: "input_text",
3286:           text: [
3287:             `Visitor question: ${question}`,
3288:             `Question intent: ${intent}`,
3289:             `Web search allowed for this question: ${enableWebSearch ? "yes" : "no"}`,
3290:             "",
3291:             "Recent conversation JSON:",
3292:             clampText(JSON.stringify(conversation, null, 2), 8000),
3293:             "",
3294:             "Portfolio context JSON:",
3295:             clampText(JSON.stringify(context, null, 2), 18000)
3296:           ].join("\n")
3297:         ]
3298:       }
3299:     ],
3300:     max_output_tokens: Number(process.env.OPENAI_MAX_OUTPUT_TOKENS || 1100),
3301:     reasoning: { effort: process.env.OPENAI_REASONING_EFFORT || "medium" },
3302:     text: { verbosity: process.env.OPENAI_VERBOSITY || "medium" }
3303:   });
3304:
3305:   const callModel = async (selectedModel) => {
3306:     const payload = buildPayload(selectedModel);
3307:     if (enableWebSearch) {
3308:       payload.tools = [{ type: "web_search", search_context_size:
process.env.OPENAI_WEB_SEARCH_CONTEXT_SIZE || "low" }];
3309:     }
3310:
3311:     const openAiResponse = await fetch("https://api.openai.com/v1/responses", {
3312:       method: "POST",
3313:       headers: {
3314:         "Authorization": `Bearer ${apiKey}`,
3315:         "Content-Type": "application/json"
3316:       },
3317:       body: JSON.stringify(payload)
3318:     });

```



```

3319:
3320:     const data = await openAiResponse.json().catch(() => ({}));
3321:     return { data, openAiResponse, selectedModel };
3322: };

```

## buildPayload - lines 3275-3303

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3275-3303 (29 source lines).                                                                                                                                                                    |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | const buildPayload = (selectedModel) => ({                                                                                                                                                            |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | clampText, stringify, join                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3275: const buildPayload = (selectedModel) => ({
3276:   model: selectedModel,
3277:   input: [
3278:     {
3279:       role: "developer",
3280:       content: [{ type: "input_text", text: portfolioAiInstructions }]
3281:     },
3282:     {
3283:       role: "user",
3284:       content: [{
3285:         type: "input_text",
3286:         text: [
3287:           `Visitor question: ${question}`,
3288:           `Question intent: ${intent}`,
3289:           `Web search allowed for this question: ${enableWebSearch ? "yes" : "no"}`,
3290:           "",
3291:           "Recent conversation JSON:",
3292:           clampText(JSON.stringify(conversation, null, 2), 8000),
3293:           "",
3294:           "Portfolio context JSON:",
3295:           clampText(JSON.stringify(context, null, 2), 18000)
3296:         ].join("\n")
3297:       }
3298:     ]
3299:   ],
3300:   max_output_tokens: Number(process.env.OPENAI_MAX_OUTPUT_TOKENS || 1100),
3301:   reasoning: { effort: process.env.OPENAI_REASONING_EFFORT || "medium" },
3302:   text: { verbosity: process.env.OPENAI_VERBOSITY || "medium" }
3303: });

```

## callModel - lines 3305-3318

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3305-3318 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.                                                                                                            |
| Declaration        | const callModel = async (selectedModel) => {                                                                                                                                                          |
| Touches            | network/API requests, authentication/security data                                                                                                                                                    |
| Calls detected     | async, buildPayload, fetch, stringify                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3305:   const callModel = async (selectedModel) => {
3306:     const payload = buildPayload(selectedModel);
3307:     if (enableWebSearch) {
3308:       payload.tools = [{ type: "web_search", search_context_size:
process.env.OPENAI_WEB_SEARCH_CONTEXT_SIZE || "low" }];
3309:     }
3310:
3311:     const openAiResponse = await fetch("https://api.openai.com/v1/responses", {
3312:       method: "POST",
3313:       headers: {
3314:         "Authorization": `Bearer ${apiKey}`,
3315:         "Content-Type": "application/json"
3316:       },
3317:       body: JSON.stringify(payload)
3318:     });

```

## handleApi - lines 3342-3602

| Item               | Explanation                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3342-3602 (261 source lines).                                                                                                                                                                                                                                                                                                                                                   |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                                                                                                                                                                                                   |
| Declaration        | async function handleApi(request, response, url) {                                                                                                                                                                                                                                                                                                                                    |
| Touches            | filesystem, network/API requests, authentication/security data                                                                                                                                                                                                                                                                                                                        |
| Calls detected     | readJsonFile, sendJson, join, isLocalRequest, getPublishTargetInfo, getGitStatus, getUpdateInfo, getSecurityReport, compileToolStatus, handlePortfolioAi, readRequestJson, normalizeCodeLanguage                                                                                                                                                                                      |
| API endpoints      | /api/app-update, /api/app-update/install, /api/apply-catalog, /api/catalog, /api/code/beautify, /api/code/compile, /api/code/install-tools, /api/code/save, /api/code/tools, /api/github-authenticate, /api/install-git, /api/portfolio-ai, /api/publish-target, /api/save-draft, /api/security-report, /api/sync-from-publish-target, /api/system-check, /api/templates, /api/upload |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                                                                                                                                                               |
| Async/return clues | 35 await expression(s), 29 return statement(s).                                                                                                                                                                                                                                                                                                                                       |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                                                                                                                                                                 |

### Source excerpt:

```

3342: async function handleApi(request, response, url) {
3343:   if (request.method === "GET" && url.pathname === "/api/catalog") {
3344:     try {
3345:       const catalog = await readJsonFile(draftPath);
3346:       sendJson(response, 200, { source: "draft", catalog });
3347:     } catch {
3348:       sendJson(response, 200, { source: "published", catalog: await readJsonFile(catalogPath) });
3349:     }
3350:     return true;
3351:   }
3352:
3353:   if (request.method === "GET" && url.pathname === "/api/templates") {
3354:     sendJson(response, 200, await readJsonFile(path.join(root, "project-templates.json")));
3355:     return true;
3356:   }
3357:
3358:   if (request.method === "GET" && url.pathname === "/api/publish-target") {
3359:     if (!isLocalRequest(request)) {
3360:       sendJson(response, 403, { error: "Publishing target details are only available from this computer."
});
3361:       return true;
3362:     }
3363:     sendJson(response, 200, { ok: true, target: await getPublishTargetInfo() });
3364:     return true;
3365:   }
3366:
3367:   if (request.method === "GET" && url.pathname === "/api/system-check") {
3368:     sendJson(response, 200, { ok: true, system: await getGitStatus(), target: await
getPublishTargetInfo() });
3369:     return true;
3370:   }
3371:
3372:   if (request.method === "GET" && url.pathname === "/api/app-update") {
3373:     sendJson(response, 200, await getUpdateInfo());
3374:     return true;
3375:   }
3376:
3377:   if (request.method === "GET" && url.pathname === "/api/security-report") {
3378:     if (!isLocalRequest(request)) {
3379:       sendJson(response, 403, { error: "Security reports are only available from this computer." });
3380:       return true;
3381:     }

```

```

3382:     sendJson(response, 200, await getSecurityReport());
3383:     return true;
3384: }
3385:
3386: if (request.method === "GET" && url.pathname === "/api/code/tools") {
3387:   if (!isLocalRequest(request)) {
3388:     sendJson(response, 403, { error: "Compiler details are only available from this computer." });
3389:     return true;
3390:   }
3391:   sendJson(response, 200, { ok: true, tools: await compileToolStatus() });
3392:   return true;
3393: }
3394:
3395: if (request.method !== "POST") return false;
3396:
3397: if (!isLocalRequest(request)) {
3398:   sendJson(response, 403, { error: "Write actions are only allowed from this computer." });
3399:   return true;
3400: }
3401:
3402: if (url.pathname === "/api/portfolio-ai") {
3403:   await handlePortfolioAi(request, response);
3404:   return true;
3405: }
3406:
3407: if (url.pathname === "/api/code/beautify") {
3408:   try {
3409:     const body = await readRequestJson(request);
3410:     const language = normalizeCodeLanguage(body.language || detectCodeLanguageFromSource(body.code,
body.fileName));
3411:     sendJson(response, 200, {
3412:       ok: true,
3413:       language,
3414:       code: beautifyCode(body.code || "", language)
3415:     });
3416:   } catch (error) {
3417:     sendJson(response, 400, { ok: false, error: error.message || "Code could not be beautified." });
3418:   }
3419:   return true;
3420: }
3421:
3422: if (url.pathname === "/api/code/save") {
3423:   try {
3424:     const body = await readRequestJson(request);
3425:     const saved = await saveCompileSource(body);
3426:     sendJson(response, 200, { ok: true, saved });
3427:   } catch (error) {
3428:     sendJson(response, 400, { ok: false, error: error.message || "Code could not be saved." });
3429:   }
3430:   return true;
3431: }
3432:
3433: if (url.pathname === "/api/code/compile") {
3434:   try {
3435:     const body = await readRequestJson(request);
3436:     const result = await compileAndRunCode(body);
3437:     sendJson(response, 200, { ok: result.ok, result });
3438:   } catch (error) {
3439:     sendJson(response, 400, { ok: false, error: error.message || "Code could not be compiled.", result:
null });
3440:   }
3441:   return true;
3442: }
3443:
3444: if (url.pathname === "/api/code/install-tools") {
3445:   try {
3446:     const body = await readRequestJson(request);
3447:     const result = await installCompilerTools(body.language || "");
3448:     sendJson(response, 200, result);
3449:   } catch (error) {
3450:     sendJson(response, 400, { ok: false, error: error.message || "Compiler tools could not be
installed." });
3451:   }
3452:   return true;
3453: }
3454:
3455: if (url.pathname === "/api/publish-target") {
3456:   await readRequestJson(request).catch(() => ({}));
3457:   sendJson(response, 400, {
3458:     ok: false,
3459:     error: "Publishing target setup now requires GitHub authentication.",
3460:     details: "Use Authenticate target. The builder keeps the previous target until GitHub write access
is verified."
3461:   });
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

## Representative opening snippet

```

1: import { createServer } from "node:http";
2: import { execFile, spawn } from "node:child_process";

```

```

3: import { createHash } from "node:crypto";
4: import { access as fsAccess, chmod, cp, mkdir, mkdtemp, readFile, readdir, rm, writeFile } from
"node:fs/promises";
5: import os from "node:os";
6: import path from "node:path";
7: import { promisify } from "node:util";
8: import { fileURLToPath } from "node:url";
9:
10: const root = path.dirname(fileURLToPath(import.meta.url));
11: const portfolioRoot = path.resolve(process.env.OMB_PORTFOLIO_WORKSPACE || root);
12: const compileRoot = path.resolve(process.env.OMB_CODE_WORKSPACE || path.join(path.dirname(root),
"compile-code"));
13: const port = Number(process.env.PORT || 8080);
14: const host = process.env.HOST || "0.0.0.0";
15: const execFileAsync = promisify(execFile);
16: const packageJson = JSON.parse(await readFile(path.join(root, "package.json"), "utf8")).catch(() =>
"{}"));
17:
18: const types = {
19:   ".css": "text/css",
20:   ".csv": "text/csv",
21:   ".html": "text/html",
22:   ".ico": "image/x-icon",
23:   ".js": "application/javascript",
24:   ".json": "application/json",
25:   ".md": "text/markdown",
26:   ".pdf": "application/pdf",
27:   ".png": "image/png",
28:   ".jpg": "image/jpeg",
29:   ".jpeg": "image/jpeg",
30:   ".svg": "image/svg+xml",
31:   ".txt": "text/plain",
32:   ".webp": "image/webp",
33:   ".xml": "application/xml",
34:   ".xlsx": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",
35:   ".zip": "application/zip"
36: };
37:
38: const draftPath = path.join(root, "projects.local.json");
39: const catalogPath = path.join(root, "projects.json");
40: const publishAuthCachePath = path.join(root, ".omb-publish-session.json");
41: const publishPaths = [
42:   "projects.json",
43:   "docs",
44:   "assets",
45:   "index.html",
46:   "styles.css",
47:   "script.js",
48:   "electronics-search.js",
49:   "Backgrounds",
50:   ".nojekyll",
51:   "_headers",
52:   ".well-known",
53:   "robots.txt",
54:   "sitemap.xml",
55:   "CNAME"
56: ];
57: const publishAuthCacheTtlMs = 24 * 60 * 60 * 1000;
58: const publishAuthExtendedTtlMs = 30 * 24 * 60 * 60 * 1000;
59: const publishAuthHistoryWindowMs = 7 * 24 * 60 * 60 * 1000;
60: const publishAuthExtendedThreshold = 3;
61: const defaultSiteRepository = process.env.OMB_BUILDER_REPOSITORY || "";
62: const blockedAppUpdateVersions = new Set(["0.2.16"]);
63: const publishAuthorizationHelp = [
64:   "Publishing was blocked before live website files were applied.",
65:   "Open Publishing target, enter the repository URL, then click Authenticate target.",
66:   "A GitHub/Git Credential Manager browser sign-in may open. Sign in with an account that has write
access to the selected Pages repository.",
67:   "Authenticate target only verifies write access and remembers it for the matching repository and
branch.",
68:   "Load from target only imports compatible website files from the authenticated repository into the
local builder workspace.",
69:   "Until a compatible writable website repository is associated, the builder remains local-only."
70: ].join(" ");
71: const gitCandidates = [
72:   process.env.GIT_EXE,

```

# Source File: styles.css

| Item                | Explanation                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------|
| File                | styles.css                                                                                                          |
| Purpose             | Public website CSS for layout, contact area, projects, responsive behavior, AI panel, and mobile/desktop rendering. |
| Lines               | 2,735                                                                                                               |
| Size                | 53,685 bytes                                                                                                        |
| Talks to            | Mostly local to its own feature area.                                                                               |
| Functions           | 0                                                                                                                   |
| Variables/constants | 0                                                                                                                   |
| API mentions        | 0                                                                                                                   |
| Signals             | 260                                                                                                                 |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                  | Explanation                                                                                                                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rich text at line 2   | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color-scheme: light;                                                                                     |
| Rich text at line 27  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);                                                                                       |
| Rich text at line 29  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-family: Inter, ui-sans-serif, system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif; |
| Rich text at line 38  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: inherit;                                                                                          |
| Rich text at line 99  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #fff;                                                                                             |
| Rich text at line 104 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.64rem;                                                                                      |
| Rich text at line 105 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;                                                                                        |
| Rich text at line 116 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);                                                                                     |
| Rich text at line 117 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;                                                                                      |
| Rich text at line 132 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);                                                                                     |
| Rich text at line 133 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.95rem;                                                                                      |
| Rich text at line 134 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 700;                                                                                        |
| Rich text at line 138 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--teal);                                                                                      |
| Rich text at line 196 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--teal);                                                                                      |
| Rich text at line 197 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;                                                                                      |
| Rich text at line 198 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 800;                                                                                        |
| Rich text at line 213 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(2.45rem, 6vw, 5.75rem);                                                                 |
| Rich text at line 220 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(1.85rem, 4vw, 3.2rem);                                                                  |
| Rich text at line 227 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 1.1rem;                                                                                       |
| Rich text at line 233 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #2e4d62;                                                                                          |
| Rich text at line 234 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(1rem, 2vw, 1.18rem);                                                                    |
| Rich text at line 252 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 800;                                                                                        |

| Item                         | Explanation                                                                                                                                                                              |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rich text at line 260        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #fff;                                                                                        |
| Rich text at line 277        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #083c4c;                                                                                     |
| Rich text at line 279        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 850;                                                                                   |
| Rich text at line 282        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: transition: background 160ms ease, border-color 160ms ease, color 160ms ease, transform 160ms ease; |
| Rich text at line 289        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: stroke: currentColor;                                                                               |
| Rich text at line 297        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: rgba(11, 114, 136, 0.72);                                                             |
| Rich text at line 299        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #062f3b;                                                                                     |
| Rich text at line 334        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);                                                                                |
| Rich text at line 356        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);                                                                                  |
| Rich text at line 357        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(0.98rem, 1.4vw, 1.12rem);                                                          |
| Rich text at line 373        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border: 1px solid color-mix(in srgb, var(--line), #ffffff 18%);                                     |
| Rich text at line 385        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);                                                                                  |
| Rich text at line 386        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 1rem;                                                                                    |
| Rich text at line 390        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);                                                                                |
| Rich text at line 391        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.92rem;                                                                                 |
| Rich text at line 434        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #086a7c;                                                                                     |
| Rich text at line 436        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;                                                                                 |
| Rich text at line 437        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 950;                                                                                   |
| Rich text at line 448        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);                                                                                  |
| Rich text at line 449        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(0.95rem, 1.3vw, 1.08rem);                                                          |
| Rich text at line 450        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 720;                                                                                   |
| Rich text at line 468        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #06384f;                                                                                     |
| Rich text at line 470        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.94rem;                                                                                 |
| Rich text at line 471        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;                                                                                   |
| Rich text at line 478        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: transition: background 140ms ease, border-color 140ms ease, color 140ms ease, transform 140ms ease; |
| Rich text at line 483        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #041c2a;                                                                                     |
| Rich text at line 484        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #47b8df;                                                                              |
| Rich text at line 498        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: stroke: currentColor;                                                                               |
| Rich text at line 547        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #162636;                                                                                     |
| Rich text at line 548        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(1.45rem, 2vw, 1.85rem);                                                            |
| Rich text at line 553        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #0b778d;                                                                                     |
| Rich text at line 554        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;                                                                                 |
| Rich text at line 555        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;                                                                                   |
| Cloudflare or AI at line 559 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: .ai-clear-chat {                                                                                  |
| Rich text at line 568        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #9f2222;                                                                                     |
| Rich text at line 570        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: inherit;                                                                                      |
| Rich text at line 571        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.82rem;                                                                                 |

| Item                         | Explanation                                                                                                                                                                              |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rich text at line 572        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 850;                                                                                   |
| Rich text at line 574        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: transition: background 150ms ease, border-color 150ms ease, color 150ms ease, transform 150ms ease; |
| Cloudflare or AI at line 577 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: .ai-clear-chat:hover,                                                                             |
| Cloudflare or AI at line 578 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: .ai-clear-chat:focus-visible {                                                                    |
| Rich text at line 579        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #b42222;                                                                              |
| Rich text at line 580        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #ffffff;                                                                                     |
| Cloudflare or AI at line 585 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: .ai-clear-chat:active {                                                                           |
| Rich text at line 586        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #9f2222;                                                                                     |
| Rich text at line 592        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #8a5a00;                                                                                     |
| Rich text at line 596        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #a62929;                                                                                     |
| Rich text at line 625        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #23384b;                                                                                     |
| Rich text at line 639        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #23384b;                                                                                     |
| Rich text at line 644        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #102b3b;                                                                                     |
| Rich text at line 655        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #eaf6ff;                                                                                     |
| Rich text at line 656        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: 0.82rem/1.5 Consolas, "Liberation Mono", "Courier New", monospace;                            |
| Rich text at line 661        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: inherit;                                                                                     |
| Rich text at line 666        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: rgba(16, 121, 144, 0.36);                                                             |
| Rich text at line 706        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #526d82;                                                                                     |
| Rich text at line 707        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;                                                                                 |
| Rich text at line 708        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;                                                                                   |
| Rich text at line 716        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #102b3b;                                                                                     |

## CSS selectors and visual hooks

| Item     | Explanation           |
|----------|-----------------------|
| Line 1   | :root                 |
| Line 21  | html                  |
| Line 25  | body                  |
| Line 33  | body.full-window-open |
| Line 37  | a                     |
| Line 44  | input:focus-visible   |
| Line 49  | .site-header          |
| Line 64  | .brand                |
| Line 71  | .brand-lockup         |
| Line 79  | .brand-icon           |
| Line 89  | .omb-engraving        |
| Line 111 | .brand small          |
| Line 115 | .brand small          |
| Line 120 | .header-right         |
| Line 128 | .site-nav             |

| Item     | Explanation                 |
|----------|-----------------------------|
| Line 137 | .site-nav a:hover           |
| Line 141 | .header-avatar              |
| Line 152 | .header-avatar img          |
| Line 160 | .hero                       |
| Line 170 | .hero-content               |
| Line 174 | .hero-image                 |
| Line 180 | .hero-shade                 |
| Line 186 | .hero-content               |
| Line 194 | .eyebrow                    |
| Line 206 | p                           |
| Line 210 | h1                          |
| Line 218 | h2                          |
| Line 225 | h3                          |
| Line 230 | .hero-copy                  |
| Line 237 | .hero-actions               |
| Line 246 | .resource-link              |
| Line 255 | .button                     |
| Line 259 | .button.primary             |
| Line 264 | .button.secondary           |
| Line 269 | .ai-jump-link               |
| Line 285 | .ai-jump-link svg           |
| Line 296 | .ai-jump-link:focus-visible |
| Line 303 | .resume-section             |
| Line 308 | .resume-actions             |
| Line 314 | .resume-viewer              |
| Line 324 | .resume-viewer object       |
| Line 331 | .resume-viewer p            |
| Line 337 | .dynamic-section            |
| Line 347 | .dynamic-section-surface    |
| Line 354 | .dynamic-section-copy       |
| Line 362 | .dynamic-section-grid       |
| Line 368 | .dynamic-section-card       |
| Line 380 | .dynamic-section-card p     |
| Line 384 | .dynamic-section-card h3    |
| Line 389 | .dynamic-section-card p     |
| Line 396 | .dynamic-link-list          |
| Line 400 | .fun-facts-section          |
| Line 409 | .fun-facts-section[hidden]  |
| Line 413 | .fun-facts-window           |
| Line 427 | .fun-facts-label            |
| Line 441 | .fun-facts-lines            |



| Item     | Explanation                                |
|----------|--------------------------------------------|
| Line 446 | .fun-fact-line                             |
| Line 454 | .builder-download-section                  |
| Line 462 | .builder-download-link                     |
| Line 482 | .builder-download-link:focus-visible       |
| Line 489 | .builder-download-link:active              |
| Line 494 | .builder-download-link svg                 |
| Line 505 | .sr-only                                   |
| Line 517 | .ai-assistant-panel                        |
| Line 528 | .ai-assistant-console                      |
| Line 545 | .ai-assistant-console h2                   |
| Line 552 | .ai-assistant-status                       |
| Line 559 | .ai-clear-chat                             |
| Line 578 | .ai-clear-chat:focus-visible               |
| Line 585 | .ai-clear-chat:active                      |
| Line 591 | .ai-assistant-status[data-state="working"] |
| Line 595 | .ai-assistant-status[data-state="error"]   |
| Line 599 | .ai-assistant-log                          |
| Line 613 | .ai-message                                |
| Line 623 | .ai-message p                              |
| Line 629 | .ai-answer-content                         |
| Line 634 | .ai-answer-content ul                      |
| Line 643 | .ai-answer-content strong                  |
| Line 647 | .ai-code-block                             |
| Line 660 | .ai-code-block code                        |
| Line 664 | .ai-message-user                           |
| Line 670 | .ai-message-assistant                      |
| Line 674 | .ai-message-pending p::after               |
| Line 691 | .ai-source-list                            |
| Line 698 | .ai-source-panel                           |
| Line 704 | .ai-source-heading                         |
| Line 713 | .ai-assistant-form button                  |
| Line 724 | .ai-source-list button                     |
| Line 732 | .ai-source-list button:focus-visible       |
| Line 738 | .ai-assistant-form                         |
| Line 745 | .ai-assistant-form input                   |
| Line 757 | .ai-assistant-form input:focus-visible     |
| Line 762 | .ai-assistant-form button                  |
| Line 772 | .ai-assistant-form button:focus-visible    |
| Line 776 | .summary-band                              |
| Line 784 | .metric                                    |
| Line 790 | .metric strong                             |

| Item     | Explanation                                    |
|----------|------------------------------------------------|
| Line 796 | .metric span                                   |
| Line 803 | .section                                       |
| Line 807 | .section-heading                               |
| Line 816 | .search-wrap                                   |
| Line 825 | .search-engine                                 |
| Line 830 | .search-engine .search-wrap                    |
| Line 834 | .search-wrap input                             |
| Line 845 | ::highlight(portfolio-search-highlight)        |
| Line 850 | .search-results-panel                          |
| Line 867 | .search-results-panel[hidden]                  |
| Line 871 | .search-results-tools                          |
| Line 883 | .search-results-tools [data-state="not-found"] |
| Line 887 | .search-results-tools [data-state="found"]     |
| Line 891 | .search-results-tools label                    |
| Line 898 | .search-results-tools select                   |
| Line 909 | .search-results-list                           |
| Line 914 | .search-result-item                            |
| Line 928 | .search-result-item:focus-visible              |
| Line 934 | .search-result-title                           |
| Line 940 | .search-result-snippet                         |
| Line 947 | .search-result-snippet                         |
| Line 951 | .search-result-mark                            |
| Line 958 | .filters                                       |

## Representative opening snippet

```

1: :root {
2:   color-scheme: light;
3:   --ink: #17202a;
4:   --muted: #526d82;
5:   --line: #c9deea;
6:   --paper: #eef8fc;
7:   --panel: #ffffff;
8:   --teal: #0d7f91;
9:   --amber: #d18b21;
10:  --coral: #bc5648;
11:  --graphite: #223244;
12:  --omb-copper: #c7772d;
13:  --focus: #0b6fce;
14:  --shadow: 0 18px 48px rgba(23, 32, 42, 0.14);
15: }
16:
17: * {
18:   box-sizing: border-box;
19: }
20:
21: html {
22:   scroll-behavior: smooth;
23: }
24:
25: body {
26:   margin: 0;
27:   color: var(--ink);
28:   background: var(--paper);
29:   font-family: Inter, ui-sans-serif, system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
30:   line-height: 1.5;
31: }
32:
33: body.full-window-open {
34:   overflow: hidden;

```

```
35: }
36:
37: a {
38:   color: inherit;
39:   text-decoration: none;
40: }
41:
42: a:focus-visible,
43: button:focus-visible,
44: input:focus-visible {
45:   outline: 3px solid var(--focus);
46:   outline-offset: 3px;
47: }
48:
49: .site-header {
50:   position: sticky;
51:   top: 0;
52:   z-index: 20;
53:   display: flex;
54:   align-items: center;
55:   justify-content: space-between;
56:   gap: 24px;
57:   min-height: 72px;
58:   padding: 12px clamp(18px, 4vw, 56px);
59:   border-bottom: 1px solid rgba(201, 222, 234, 0.84);
60:   background: rgba(238, 248, 252, 0.92);
61:   backdrop-filter: blur(16px);
62: }
63:
64: .brand {
65:   display: inline-flex;
66:   align-items: center;
67:   gap: 12px;
68:   min-width: 0;
69: }
70:
71: .brand-lockup {
72:   position: relative;
```

# Source File: template-preview.css

| Item                | Explanation                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------|
| File                | template-preview.css                                                                                                    |
| Purpose             | Builder application CSS for dialogs, windows, rich editors, compile code, dark mode, guide windows, and local previews. |
| Lines               | 5,749                                                                                                                   |
| Size                | 131,453 bytes                                                                                                           |
| Talks to            | builder frontend                                                                                                        |
| Functions           | 0                                                                                                                       |
| Variables/constants | 0                                                                                                                       |
| API mentions        | 0                                                                                                                       |
| Signals             | 738                                                                                                                     |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                  | Explanation                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------|
| Rich text at line 34  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;              |
| Rich text at line 36  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;          |
| Rich text at line 37  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;            |
| Rich text at line 43  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #f5c56d;       |
| Rich text at line 44  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #5f4207;              |
| Rich text at line 50  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #93c5fd;       |
| Rich text at line 51  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #164780;              |
| Rich text at line 57  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #86efac;       |
| Rich text at line 58  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #14532d;              |
| Rich text at line 63  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #fca5a5;       |
| Rich text at line 64  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #7f1d1d;              |
| Rich text at line 70  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #d6effa;       |
| Rich text at line 71  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;              |
| Rich text at line 74  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color 140ms ease,            |
| Rich text at line 75  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: background-color 140ms ease, |
| Rich text at line 76  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color 140ms ease,     |
| Rich text at line 83  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #b8e6f7;       |
| Rich text at line 84  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;              |
| Rich text at line 91  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #ffffff;       |
| Rich text at line 92  | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;              |
| Rich text at line 148 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);           |
| Rich text at line 149 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 1.18rem;          |
| Rich text at line 173 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);         |

| Item                  | Explanation                                                                                                            |
|-----------------------|------------------------------------------------------------------------------------------------------------------------|
| Rich text at line 174 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.72rem;               |
| Rich text at line 175 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;                 |
| Rich text at line 184 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);                |
| Rich text at line 186 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: inherit;                    |
| Rich text at line 187 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 700;                 |
| Rich text at line 191 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #4b91db;            |
| Rich text at line 198 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);              |
| Rich text at line 199 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;               |
| Rich text at line 205 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: inherit;                   |
| Rich text at line 222 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);                |
| Rich text at line 223 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.96rem;               |
| Rich text at line 241 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;                   |
| Rich text at line 243 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.75rem;               |
| Rich text at line 244 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;                 |
| Rich text at line 266 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;                   |
| Rich text at line 278 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);                |
| Rich text at line 279 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 1rem;                  |
| Rich text at line 283 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);              |
| Rich text at line 284 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;               |
| Rich text at line 292 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border: 2px solid currentColor;   |
| Rich text at line 304 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: background: currentColor;         |
| Rich text at line 309 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: box-shadow: 0 7px 0 currentColor; |
| Rich text at line 338 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;                   |
| Rich text at line 340 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;               |
| Rich text at line 341 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;                 |
| Rich text at line 356 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);              |
| Rich text at line 357 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;               |
| Rich text at line 358 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;                 |
| Rich text at line 366 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);                |
| Rich text at line 368 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: inherit;                    |
| Rich text at line 369 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.9rem;                |
| Rich text at line 378 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;                   |
| Rich text at line 380 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: inherit;                    |
| Rich text at line 381 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.8rem;                |
| Rich text at line 382 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;                 |
| Rich text at line 388 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #fffff;                    |
| Rich text at line 390 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #1f6ed4;            |
| Rich text at line 395 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);              |
| Rich text at line 396 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.82rem;               |
| Rich text at line 433 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);                |
| Rich text at line 435 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.92rem;               |

| Item                  | Explanation                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------|
| Rich text at line 436 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 950;                       |
| Rich text at line 456 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #087f9b;                         |
| Rich text at line 457 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 950;                       |
| Rich text at line 471 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;                         |
| Rich text at line 473 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.7rem;                      |
| Rich text at line 474 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 950;                       |
| Rich text at line 490 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #ffffff;                         |
| Rich text at line 496 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #ffffff;                         |
| Rich text at line 500 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: rgba(255, 255, 255, 0.7); |
| Rich text at line 542 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);                    |
| Rich text at line 543 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.86rem;                     |
| Rich text at line 555 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #2b5064;                         |
| Rich text at line 556 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.86rem;                     |
| Rich text at line 582 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #fafdff;                         |
| Rich text at line 583 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;                     |
| Rich text at line 601 | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);                    |

## CSS selectors and visual hooks

| Item     | Explanation                                  |
|----------|----------------------------------------------|
| Line 1   | .template-preview-page                       |
| Line 5   | .template-header                             |
| Line 20  | .section-actions                             |
| Line 27  | .builder-save-state                          |
| Line 42  | .builder-save-state[data-state="dirty"]      |
| Line 49  | .builder-save-state[data-state="publishing"] |
| Line 56  | .builder-save-state[data-state="published"]  |
| Line 62  | .builder-save-state[data-state="error"]      |
| Line 69  | .template-preview-page button                |
| Line 82  | .template-preview-page button:hover          |
| Line 90  | .template-preview-page button:active         |
| Line 98  | .builder-workspace                           |
| Line 107 | .builder-inspector                           |
| Line 116 | .template-preview                            |
| Line 123 | .builder-panel                               |
| Line 129 | .project-library-heading                     |
| Line 137 | .project-library-heading > div               |
| Line 143 | .project-library-heading h2                  |
| Line 147 | .project-library-heading h2                  |
| Line 153 | .project-library-tools                       |
| Line 165 | .project-search-field                        |

| Item     | Explanation                                            |
|----------|--------------------------------------------------------|
| Line 172 | .workflow-card span                                    |
| Line 179 | .project-search-field input                            |
| Line 190 | .project-search-field input:focus                      |
| Line 196 | #project-search-status                                 |
| Line 203 | .tree-category-toggle mark                             |
| Line 210 | .workflow-card                                         |
| Line 219 | .workflow-card strong                                  |
| Line 228 | .workflow-actions                                      |
| Line 234 | .workflow-metrics span                                 |
| Line 247 | .workflow-actions .button                              |
| Line 253 | .template-preview > .section-heading                   |
| Line 257 | .builder-guide-panel                                   |
| Line 273 | .builder-guide-panel small                             |
| Line 277 | .builder-guide-panel strong                            |
| Line 282 | .builder-guide-panel small                             |
| Line 287 | .builder-guide-icon                                    |
| Line 297 | .builder-guide-icon::after                             |
| Line 307 | .builder-guide-icon::before                            |
| Line 312 | .builder-guide-icon::after                             |
| Line 318 | .builder-guide-dialog-content                          |
| Line 328 | .builder-guide-actions                                 |
| Line 334 | .builder-guide-live span                               |
| Line 344 | .builder-guide-tools                                   |
| Line 353 | .builder-guide-tools label                             |
| Line 361 | .builder-guide-tools input                             |
| Line 373 | .builder-guide-actions button                          |
| Line 387 | .builder-guide-actions button:hover                    |
| Line 393 | .builder-guide-results                                 |
| Line 399 | .builder-guide-sections                                |
| Line 416 | .builder-guide-section                                 |
| Line 424 | .builder-guide-section[hidden]                         |
| Line 428 | .builder-guide-section summary                         |
| Line 444 | .builder-guide-section summary::-webkit-details-marker |
| Line 448 | .builder-guide-section summary::before                 |
| Line 462 | .builder-guide-section summary::after                  |
| Line 480 | .builder-guide-section[open] summary                   |
| Line 485 | .builder-guide-section[open] summary::before           |
| Line 489 | .builder-guide-section summary:hover                   |
| Line 495 | .builder-guide-section summary:hover::after            |
| Line 499 | .builder-guide-section summary:hover::after            |
| Line 504 | .builder-guide-section > div                           |

| Item     | Explanation                             |
|----------|-----------------------------------------|
| Line 513 | .builder-guide-topic-dialog             |
| Line 517 | .builder-guide-topic-content            |
| Line 522 | .builder-guide-topic-body               |
| Line 534 | .builder-guide-topic-copy               |
| Line 540 | .builder-guide-topic-body p             |
| Line 550 | .builder-guide-topic-body ul            |
| Line 561 | .builder-guide-topic-body li            |
| Line 565 | .publish-result-box                     |
| Line 570 | .publish-result-box p                   |
| Line 574 | .publish-result-box pre                 |
| Line 588 | .publish-target-form                    |
| Line 595 | .publish-target-form label              |
| Line 600 | .publish-target-field-note              |
| Line 606 | .publish-target-form input              |
| Line 616 | .publish-target-current                 |
| Line 627 | .publish-target-current div             |
| Line 633 | .publish-target-current strong          |
| Line 637 | .publish-target-current span            |
| Line 641 | .publish-target-readiness               |
| Line 648 | .publish-target-readiness strong        |
| Line 653 | .publish-target-readiness ul            |
| Line 661 | .publish-target-readiness li.ready      |
| Line 665 | .publish-target-readiness li.blocked    |
| Line 669 | .publish-target-tool-actions            |
| Line 675 | .publish-target-note                    |
| Line 682 | .profile-setup-fields                   |
| Line 687 | .profile-setup-fields label:first-child |
| Line 691 | .profile-asset-actions                  |
| Line 697 | .profile-asset-status                   |
| Line 704 | .app-update-dialog                      |
| Line 708 | .security-report-dialog                 |
| Line 712 | .app-update-content                     |
| Line 717 | .security-report-body                   |
| Line 727 | .security-report-body p                 |
| Line 734 | .app-update-body p:first-child          |
| Line 739 | .security-report-grid                   |
| Line 745 | .security-report-card                   |
| Line 756 | .security-report-card h3                |
| Line 763 | .security-report-card strong            |
| Line 769 | .security-report-card ul                |
| Line 779 | .security-report-assets                 |



| Item     | Explanation                              |
|----------|------------------------------------------|
| Line 787 | .security-report-assets li               |
| Line 797 | .security-report-assets li:last-child    |
| Line 801 | .security-report-assets span:first-child |
| Line 808 | .app-update-release-meta                 |
| Line 816 | .app-update-dialog-actions               |
| Line 820 | .app-update-dialog-actions [hidden]      |
| Line 824 | .portfolio-preview-button                |
| Line 834 | .portfolio-preview-icon                  |
| Line 842 | .portfolio-preview-icon::before          |
| Line 854 | .portfolio-preview-icon::after           |
| Line 867 | .save-all-icon                           |
| Line 876 | .save-all-icon::after                    |
| Line 881 | .save-all-icon::before                   |
| Line 890 | .save-all-icon::after                    |
| Line 902 | .builder-panel p                         |
| Line 906 | .builder-panel h1                        |
| Line 911 | .builder-panel h2                        |

## Representative opening snippet

```

1: .template-preview-page {
2:   background: #eef8fc;
3: }
4:
5: .template-header {
6:   position: sticky;
7:   top: 0;
8:   z-index: 20;
9:   display: flex;
10:  align-items: center;
11:  justify-content: space-between;
12:  gap: 18px;
13:  padding: 12px clamp(18px, 4vw, 56px);
14:  border-bottom: 1px solid rgba(201, 222, 234, 0.84);
15:  background: rgba(214, 239, 250, 0.96);
16:  backdrop-filter: blur(16px);
17: }
18:
19: .builder-header-actions,
20: .section-actions {
21:   display: flex;
22:   flex-wrap: wrap;
23:   gap: 10px;
24:   align-items: center;
25: }
26:
27: .builder-save-state {
28:   display: inline-flex;
29:   align-items: center;
30:   min-height: 34px;
31:   padding: 0 12px;
32:   border: 1px solid #9bd4ea;
33:   border-radius: 999px;
34:   color: #16384a;
35:   background: #f5fbfe;
36:   font-size: 0.78rem;
37:   font-weight: 900;
38:   letter-spacing: 0.01em;
39:   white-space: nowrap;
40: }
41:
42: .builder-save-state[data-state="dirty"] {
43:   border-color: #f5c56d;
44:   color: #5f4207;
45:   background: #fff7dc;
46: }
47:
48: .builder-save-state[data-state="saving"],

```

```
49: .builder-save-state[data-state="publishing"] {
50:   border-color: #93c5fd;
51:   color: #164780;
52:   background: #dbeafe;
53: }
54:
55: .builder-save-state[data-state="saved"],
56: .builder-save-state[data-state="published"] {
57:   border-color: #86efac;
58:   color: #14532d;
59:   background: #dcfce7;
60: }
61:
62: .builder-save-state[data-state="error"] {
63:   border-color: #fca5a5;
64:   color: #ff1d1d;
65:   background: #fee2e2;
66: }
67:
68: .template-preview-page .button,
69: .template-preview-page button {
70:   border-color: #d6effa;
71:   color: #16384a;
72:   background: #d6effa;
```

# Source File: template-preview.html

| Item                | Explanation                                                                  |
|---------------------|------------------------------------------------------------------------------|
| File                | template-preview.html                                                        |
| Purpose             | Builder HTML shell loaded inside the Electron app and local preview runtime. |
| Lines               | 1,349                                                                        |
| Size                | 92,731 bytes                                                                 |
| Talks to            | builder frontend, Cloudflare/AI layer, GitHub/release layer                  |
| Functions           | 0                                                                            |
| Variables/constants | 0                                                                            |
| API mentions        | 0                                                                            |
| Signals             | 124                                                                          |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## Important implementation signals

| Item                         | Explanation                                                                                                                                                                                                                                                                          |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Installer at line 11         | Windows setup, update, shortcut, or installation behavior. Evidence: <link rel="shortcut icon" href="assets/favicon.ico" />                                                                                                                                                          |
| Rich text at line 64         | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Rich text at line 79         | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Rich text at line 94         | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Rich text at line 118        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Rich text at line 133        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Rich text at line 148        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Rich text at line 163        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Rich text at line 178        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Network request at line 192  | Frontend-backend communication or public web/API calls. Evidence: data-placeholder="https://github.com/username"                                                                                                                                                                     |
| Rich text at line 193        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Network request at line 207  | Frontend-backend communication or public web/API calls. Evidence: data-placeholder="https://linkedin.com/in/username"                                                                                                                                                                |
| Rich text at line 208        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Network request at line 222  | Frontend-backend communication or public web/API calls. Evidence: data-placeholder="https://example.com"                                                                                                                                                                             |
| Rich text at line 223        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Rich text at line 259        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"                                                                                                                                                                          |
| Browser DOM at line 391      | Builder or website interface behavior in the browser/Electron renderer. Evidence: <li>Resume should be a public-safe PDF, not a private editable document.</li>                                                                                                                      |
| Installer at line 420        | Windows setup, update, shortcut, or installation behavior. Evidence: <p>All project creation starts from the main Add project control. This keeps project creation consistent and prevents hidden category-specific add buttons from creating projects with incomplete metadata.</p> |
| Security or auth at line 432 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Only categories that contain projects show project rows. Empty categories stay as clean category headers until a project is added.</li>                                                   |

| Item                              | Explanation                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rich text at line 462             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <p>Rich text fields behave like document editors. Click where you want to type, highlight the exact words you want to edit, then apply formatting to that selection. Formatting should not change an entire paragraph unless   |
| Rich text at line 463             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <p>The builder stores the text as structured content so the portfolio can render it cleanly later. That means the editor may standardize pasted formatting, but it should preserve intentional choices such as bold text, fo   |
| Rich text at line 464             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <p>Front-page, profile, contact, fun facts, and project overview fields now use the same selection-level text behavior. Highlight the exact words you want to edit, then use the right-click menu or floating selection insp   |
| Rich text at line 466             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>Right-click highlighted text in any builder text field to copy, cut, paste, select all, change font, change size, change color, and apply bold, italic, or underline.</li>                                                 |
| Rich text at line 467             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>The default typing font is Arial.</li>                                                                                                                                                                                     |
| Rich text at line 468             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>If selected text has mixed font, size, or color, the inspector fields are blank so you know the selection is mixed.</li>                                                                                                   |
| Rich text at line 469             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>Use Enter for a new paragraph. Pasted line breaks remain line breaks.</li>                                                                                                                                                 |
| Rich text at line 470             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>Typing and wrapping should not create fake new paragraphs; only hard enters and pasted line breaks do.</li>                                                                                                                |
| Rich text at line 471             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>Use the Code block command or Paste as code for source snippets so they render differently from general text.</li>                                                                                                         |
| Rich text at line 477             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>Keep color changes readable on the selected project appearance.</li>                                                                                                                                                       |
| Rich text at line 478             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>If a pasted paragraph looks strange, clear the formatting and reapply only the formatting you need.</li>                                                                                                                   |
| Rich text at line 480             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <p>If the selection menu disappears, click back inside the rich editor and highlight the text again. Before saving, read the section in the project preview to make sure line breaks, lists, and emphasized words still look   |
| Rich text at line 484             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <summary>6. Links, formulas, code, and pasted content</summary>                                                                                                                                                                |
| Cloudflare or AI at line 487      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <p>Use this section when pasting content from ChatGPT, documentation, GitHub, LinkedIn, LTspice notes, equations, source code, or lab references. The builder tries to recognize the content, but you should still check the |
| Compiler or simulator at line 491 | Part of the Compile Code workspace or language/tool detection. Evidence: <li>Use Code block or Paste as code for C, C++, SystemVerilog, LTspice, Java, JavaScript, Python, and HTML snippets.</li>                                                                                                                  |
| Rich text at line 492             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>Choose Basic paste when you want the builder to clean extra trailing whitespace, or Keep source spacing when indentation is important.</li>                                                                                |
| Compiler or simulator at line 495 | Part of the Compile Code workspace or language/tool detection. Evidence: <li>For Verilog and SystemVerilog, use Simulate to run the simulator. Simulation requires a Testbench file; plain Compile can still check or build HDL design files.</li>                                                                  |
| Security or auth at line 496      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Use Add testbench to create a starter testbench with \$dumpfile and \$dumpvars. Those waveform calls feed the signal scope, where HDL signals can be viewed over time.</li>                                              |
| Network request at line 503       | Frontend-backend communication or public web/API calls. Evidence: <li>Use full links when possible: <a href="https://github.com/user/repo">https://github.com/user/repo</a> is clearer than only github.com.</li>                                                                                                   |
| Compiler or simulator at line 507 | Part of the Compile Code workspace or language/tool detection. Evidence: <li>Use Compile Code to test C, C++, Java, JavaScript, Python, HTML checks, and any installed Verilog, SystemVerilog, or LTspice command-line tools before deciding what to publish later.</li>                                            |
| Security or auth at line 508      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Use Show output for terminal logs and Show scope for HDL timing waveforms after a successful testbench simulation.</li>                                                                                                  |
| Rich text at line 510             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>If a link is accidentally treated as a formula, remove the formula block and paste it again as normal text or link content.</li>                                                                                           |
| Cloudflare or AI at line 512      | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <p>When pasting from ChatGPT, review formulas and code before saving. If the content is a link, it should render as a clickable link, not as a formula. If the content is source code, it should render inside a dark code b |

| Item                              | Explanation                                                                                                                                                                                                                                                                                                         |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security or auth at line 519      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>Images are best for visual evidence: schematics, PCB layouts, test benches, oscilloscope captures, simulation plots, block diagrams, timing diagrams, microscope photos, assembled boards, or setup photos. Files are bes |
| Rich text at line 521             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>Paste an image into a rich editor to place it below the cursor line.</li>                                                                                                                                                  |
| Rich text at line 522             | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <li>Use Add image when you need source selection, rename, caption, alignment, crop, or visible/download-only control.</li>                                                                                                     |
| Compiler or simulator at line 535 | Part of the Compile Code workspace or language/tool detection. Evidence: <p>Project evidence is saved under docs/project-folder/section-or-subsection/filename. For example, a SystemVerilog design file could be saved as docs/laser-link/design/filter.sv. The upload status shows the generated pa               |
| Security or auth at line 581      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <summary>10. Publishing target and authentication</summary>                                                                                                                                                                  |
| Security or auth at line 583      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>Publishing target controls where Apply to site can push. A target is saved only after write access is verified, because the builder should never publish to a website unless the current user has permission to update th |
| Security or auth at line 584      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>This protects your portfolio from accidental pushes and also lets another user install the builder for their own site without getting access to your website. The target describes where the public portfolio will live;  |
| Security or auth at line 588      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Click Authenticate target to verify write access for that repository and branch.</li>                                                                                                                                    |
| Git command at line 589           | Repository state, publishing, branch checks, or release automation. Evidence: <li>Complete GitHub/Git Credential Manager sign-in only when prompted.</li>                                                                                                                                                           |
| Security or auth at line 590      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Use Load from target only after authentication succeeds and only when you want to import that website repository into this builder.</li>                                                                                 |
| Security or auth at line 595      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Authenticate target does not import content; Load from target does not authenticate a new target.</li>                                                                                                                   |
| Git command at line 596           | Repository state, publishing, branch checks, or release automation. Evidence: <li>If authentication fails, read the message before retrying; it usually explains whether Git, credentials, branch state, or repository access caused the failure.</li>                                                              |
| Security or auth at line 598      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>GitHub authorization is cached per repository and branch. Daily reuse avoids repeated prompts, and repeated successful authorization can extend trust for that exact target. If the target changes, authenticate again be |
| Git command at line 605           | Repository state, publishing, branch checks, or release automation. Evidence: <p>The builder can be used offline for most editing work. Publishing is the only part that needs the network and a Git target with write access.</p>                                                                                  |
| Git command at line 611           | Repository state, publishing, branch checks, or release automation. Evidence: <li>If offline, keep editing and saving drafts. Publish when the network and Git target are available.</li>                                                                                                                           |
| Security or auth at line 617      | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Use Apply to site only after the preview looks correct and the target is authenticated.</li>                                                                                                                             |
| Installer at line 625             | Windows setup, update, shortcut, or installation behavior. Evidence: <p>Use Check updates to manually look for a newer builder. Updates come from GitHub Releases after changes are merged into the release branch and a new installer is built. Use the Preferences menu to switch between light                   |
| Installer at line 629             | Windows setup, update, shortcut, or installation behavior. Evidence: <li>The app downloads the installer, closes the current builder, updates the existing install, and reopens the app.</li>                                                                                                                       |
| Installer at line 631             | Windows setup, update, shortcut, or installation behavior. Evidence: <li>If an old uninstaller fails, setup stops the builder process and continues updating the same folder.</li>                                                                                                                                  |
| Installer at line 636             | Windows setup, update, shortcut, or installation behavior. Evidence: <li>If a release is paused, wait for the next version instead of forcing the old installer.</li>                                                                                                                                               |
| Installer at line 637             | Windows setup, update, shortcut, or installation behavior. Evidence: <li>If Windows blocks a process from closing, use Task Manager as Administrator to end the stuck installer or old builder process.</li>                                                                                                        |
| Installer at line 638             | Windows setup, update, shortcut, or installation behavior. Evidence: <li>If the app opens but looks stale, close it fully and reopen it from the desktop shortcut.</li>                                                                                                                                             |
| Installer at line 640             | Windows setup, update, shortcut, or installation behavior. Evidence: <p>If the app cannot update, close any old OMB Builder process in Task Manager and reopen the app from the desktop shortcut. Avoid installing another copy in a different folder; duplicate copies can make Windows shortcut                   |
| Browser DOM at line 646           | Builder or website interface behavior in the browser/Electron renderer. Evidence: <p>Anything pushed to the portfolio repository can become public. Treat the portfolio as a public engineering document. If a file would be unsafe in a public GitHub repository, it should not be uploaded or published thro      |

| Item                         | Explanation                                                                                                                                                                                                                                                                                                             |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Security or auth at line 647 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>Good portfolio evidence is public-safe, clearly labeled, and useful to a visitor. Private credentials, private lab data, personal identity documents, and unpublished confidential work should stay out of the portfolio.</p> |
| Security or auth at line 649 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Do not upload private keys, passwords, private tokens, or private lab data.</li>                                                                                                                                             |
| Security or auth at line 652 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Keep authentication local. The builder should not commit credentials.</li>                                                                                                                                                   |
| Security or auth at line 657 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>Check code files for tokens, API keys, Wi-Fi credentials, private comments, or machine-specific paths.</li>                                                                                                                  |
| Cloudflare or AI at line 660 | Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <li>Use Security report to review app download counts, current publishing authorization status, and whether visitor logging requires Cloudflare analytics.</li>                                                                  |
| Security or auth at line 662 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>If another user installs the builder, they should connect their own target unless you explicitly authorize access to your target. The builder should help people create portfolios, but it should not give anyone silent      |
| Git command at line 669      | Repository state, publishing, branch checks, or release automation. Evidence: <p>The routine is intentionally repetitive because publishing is safer when every project has been parsed and every section has been previewed before Git pushes anything.</p>                                                            |
| Security or auth at line 684 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <li>For publishing target changes, authenticate before loading or applying site data.</li>                                                                                                                                       |
| Rich text at line 755        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <span>Accent color</span>                                                                                                                                                                                                          |
| Rich text at line 756        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <input id="category-accent" type="color" value="#1677a8" />                                                                                                                                                                        |
| Rich text at line 780        | Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <button type="button" data-title-action="paste">Paste as title</button>                                                                                                                                                            |
| Git command at line 868      | Repository state, publishing, branch checks, or release automation. Evidence: <input id="publish-target-repository" type="text" placeholder="https://github.com/USERNAME/USERNAME.github.io.git" />                                                                                                                     |
| Security or auth at line 871 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Authenticate target verifies that this computer can write to the selected GitHub Pages repository. Load from target imports the latest compatible website data from that authenticated repository into this builder. Authent     |
| Security or auth at line 874 | Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <button class="button primary" id="publish-target-auth" type="submit">Authenticate target</button>                                                                                                                               |

## Representative opening snippet

```

1: <!DOCTYPE html>
2: <html lang="en">
3:   <head>
4:     <meta charset="UTF-8" />
5:     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6:     <meta name="robots" content="noindex, nofollow" />
7:     <title>Local Portfolio Builder</title>
8:     <link rel="icon" type="image/png" sizes="16x16" href="assets/favicon-16.png" />
9:     <link rel="icon" type="image/png" sizes="32x32" href="assets/favicon-32.png" />
10:    <link rel="icon" type="image/png" sizes="192x192" href="assets/favicon-192.png" />
11:    <link rel="shortcut icon" href="assets/favicon.ico" />
12:    <link rel="apple-touch-icon" sizes="180x180" href="assets/apple-touch-icon.png" />
13:    <link rel="stylesheet" href="styles.css" />
14:    <link rel="stylesheet" href="template-preview.css" />
15:  </head>
16:  <body class="template-preview-page">
17:    <header class="template-header">
18:      <a class="brand" href="index.html#top" aria-label="Back to portfolio">
19:        <span class="brand-lockup" aria-hidden="true">
20:          
21:          <span class="omb-engraving">OMB</span>
22:        </span>
23:        <span>
24:          <strong>Local Portfolio Builder</strong>
25:          <small>Build, preview, save locally, then publish when ready</small>
26:        </span>
27:      </a>
28:      <div class="builder-header-actions">
29:        <span class="builder-save-state" id="builder-save-state" data-state="loaded">Loaded</span>
30:        <a class="button secondary" href="index.html#projects" target="_blank"
rel="noreferrer">Portfolio</a>
31:        <button class="button secondary" id="publish-target-open" type="button">Publishing
target</button>
32:        <button class="button secondary" id="security-report-open" type="button">Security

```

```

report</button>
33:         <button class="button secondary" id="app-update-check" type="button">Check updates</button>
34:         <button class="button primary" id="save-draft" type="button">Save draft</button>
35:         <button class="button secondary" id="apply-catalog" type="button">Apply to site</button>
36:     </div>
37: </header>
38:
39: <main class="builder-workspace">
40:     <aside class="builder-sidebar" aria-labelledby="library-title">
41:         <button class="builder-guide-panel" id="builder-guide-open" type="button" aria-label="Open
builder guide">
42:             <span class="builder-guide-icon" aria-hidden="true"></span>
43:             <span>
44:                 <strong id="library-title">Portfolio Builder</strong>
45:                 <small>Click for a quick build guide</small>
46:             </span>
47:         </button>
48:
49:         <div class="builder-panel site-section-panel">
50:             <div class="panel-title-row">
51:                 <h2>Front page text</h2>
52:                 <button class="button secondary compact-button" id="save-site-content"
type="button">Save</button>
53:             </div>
54:             <div class="front-page-text-fields">
55:                 <label>
56:                     <span>Small label</span>
57:                     <div
58:                         class="rich-summary-editor front-rich-field"
59:                         id="site-hero-eyebrow"
60:                         data-rich-editor="site-field"
61:                         data-site-field="heroEyebrow"
62:                         data-rich-text-only="true"
63:                         data-placeholder="Engineering portfolio"
64:                         contenteditable="true"
65:                         spellcheck="true"
66:                         role="textbox"
67:                         aria-label="Small front page label"
68:                     ></div>
69:                 </label>
70:                 <label>
71:                     <span>Main headline</span>
72:                     <div

```

# Source File: template-preview.js

| Item                | Explanation                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| File                | template-preview.js                                                                                                                                    |
| Purpose             | Builder frontend brain: state handling, rich editors, project builder, previews, parser triggers, publishing UI, compile workspace, and guide windows. |
| Lines               | 10,826                                                                                                                                                 |
| Size                | 454,437 bytes                                                                                                                                          |
| Talks to            | builder frontend, public website runtime, portfolio catalog, Cloudflare/AI layer, GitHub/release layer                                                 |
| Functions           | 570                                                                                                                                                    |
| Variables/constants | 140                                                                                                                                                    |
| API mentions        | 13                                                                                                                                                     |
| Signals             | 1973                                                                                                                                                   |

## How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

## API endpoints mentioned or called

| Item               | Explanation                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------|
| /api/save-draft    | Line 1637. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.  |
| /api/upload        | Line 5430. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.  |
| /api/upload        | Line 8413. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.  |
| /api/upload        | Line 8473. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.  |
| /api/apply-catalog | Line 8613. Loads project/catalog data for the builder.                                                       |
| /api/apply-catalog | Line 8637. Loads project/catalog data for the builder.                                                       |
| /api/apply-catalog | Line 8647. Loads project/catalog data for the builder.                                                       |
| /api/apply-catalog | Line 8662. Loads project/catalog data for the builder.                                                       |
| /api/apply-catalog | Line 8674. Loads project/catalog data for the builder.                                                       |
| /api/save-draft    | Line 8685. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.  |
| /api/save-draft    | Line 10770. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior. |
| /api/apply-catalog | Line 10772. Loads project/catalog data for the builder.                                                      |
| /api/save-draft    | Line 10796. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior. |

## Important implementation signals

| Item                  | Explanation                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser DOM at line 1 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideOpen = document.querySelector("#builder-guide-open");     |
| Browser DOM at line 2 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideDialog = document.querySelector("#builder-guide-dialog"); |
| Browser DOM at line 3 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideClose = document.querySelector("#builder-guide-close");   |
| Browser DOM at line 4 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideSearch = document.querySelector("#builder-guide-search"); |



| Item                   | Explanation                                                                                                                                                                           |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser DOM at line 5  | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const builderGuideResults = document.querySelector("#builder-guide-results");</code>          |
| Browser DOM at line 6  | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const builderGuideSections = document.querySelector("#builder-guide-sections");</code>        |
| Browser DOM at line 7  | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const builderGuideTopicDialog = document.querySelector("#builder-guide-topic-dialog");</code> |
| Browser DOM at line 8  | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const builderGuideTopicTitle = document.querySelector("#builder-guide-topic-title");</code>   |
| Browser DOM at line 9  | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const builderGuideTopicBody = document.querySelector("#builder-guide-topic-body");</code>     |
| Browser DOM at line 10 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const builderGuideTopicClose = document.querySelector("#builder-guide-topic-close");</code>   |
| Browser DOM at line 11 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const templateSelect = document.querySelector("#template-select");</code>                     |
| Browser DOM at line 12 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const newCategorySelect = document.querySelector("#new-category-select");</code>              |
| Browser DOM at line 13 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const placementSelect = document.querySelector("#placement-select");</code>                   |
| Browser DOM at line 14 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const newProjectTitle = document.querySelector("#new-project-title");</code>                  |
| Browser DOM at line 15 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const addProjectButton = document.querySelector("#add-project");</code>                       |
| Browser DOM at line 16 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const addCategoryButton = document.querySelector("#add-category");</code>                     |
| Browser DOM at line 17 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const addSiteSectionButton = document.querySelector("#add-site-section");</code>              |
| Browser DOM at line 18 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const funFactsInput = document.querySelector("#fun-facts-input");</code>                      |
| Browser DOM at line 19 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const funFactsCount = document.querySelector("#fun-facts-count");</code>                      |
| Browser DOM at line 20 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const saveFunFactsButton = document.querySelector("#save-fun-facts");</code>                  |
| Browser DOM at line 21 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const clearFunFactsButton = document.querySelector("#clear-fun-facts");</code>                |
| Browser DOM at line 22 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const siteHeroEyebrowInput = document.querySelector("#site-hero-eyebrow");</code>             |
| Browser DOM at line 23 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const siteHeroTitleInput = document.querySelector("#site-hero-title");</code>                 |
| Browser DOM at line 24 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const siteHeroCopyInput = document.querySelector("#site-hero-copy");</code>                   |
| Browser DOM at line 25 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const saveSiteContentButton = document.querySelector("#save-site-content");</code>            |
| Browser DOM at line 26 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileDisplayNameInput = document.querySelector("#profile-display-name");</code>       |
| Browser DOM at line 27 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profilePortfolioLabelInput = document.querySelector("#profile-portfolio-label");</code> |
| Browser DOM at line 28 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileContactIntroInput = document.querySelector("#profile-contact-intro");</code>     |
| Browser DOM at line 29 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileEmailInput = document.querySelector("#profile-email");</code>                    |
| Browser DOM at line 30 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profilePhoneInput = document.querySelector("#profile-phone");</code>                    |
| Browser DOM at line 31 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileGithubInput = document.querySelector("#profile-github");</code>                  |

| Item                   | Explanation                                                                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser DOM at line 32 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileLinkedinInput = document.querySelector("#profile-linkedin");           |
| Browser DOM at line 33 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileWebsiteInput = document.querySelector("#profile-website");             |
| Browser DOM at line 34 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const saveProfileContentButton = document.querySelector("#save-profile-content");   |
| Browser DOM at line 35 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileImageUploadButton = document.querySelector("#profile-image-upload");   |
| Browser DOM at line 36 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileHeroUploadButton = document.querySelector("#profile-hero-upload");     |
| Browser DOM at line 37 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileResumeUploadButton = document.querySelector("#profile-resume-upload"); |
| Browser DOM at line 38 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileBrandUploadButton = document.querySelector("#profile-brand-upload");   |
| Browser DOM at line 39 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileAssetStatus = document.querySelector("#profile-asset-status");         |
| Browser DOM at line 40 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const categoryDropdown = document.querySelector("#category-dropdown");              |
| Browser DOM at line 41 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const siteSectionList = document.querySelector("#site-section-list");               |
| Browser DOM at line 42 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectCreateDialog = document.querySelector("#project-create-dialog");       |
| Browser DOM at line 43 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectCreateForm = document.querySelector("#project-create-form");           |
| Browser DOM at line 44 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectCreateCategory = document.querySelector("#project-create-category");   |
| Browser DOM at line 45 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectCreateCancel = document.querySelector("#project-create-cancel");       |
| Browser DOM at line 46 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const categoryDialog = document.querySelector("#category-dialog");                  |
| Browser DOM at line 47 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const categoryForm = document.querySelector("#category-form");                      |
| Browser DOM at line 48 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const categoryTitle = document.querySelector("#category-title");                    |
| Browser DOM at line 49 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const categoryDescription = document.querySelector("#category-description");        |
| Browser DOM at line 50 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const categoryAccent = document.querySelector("#category-accent");                  |
| Browser DOM at line 51 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const categoryCancel = document.querySelector("#category-cancel");                  |
| Browser DOM at line 52 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectTree = document.querySelector("#project-tree");                        |
| Browser DOM at line 53 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectSearchInput = document.querySelector("#project-search");               |
| Browser DOM at line 54 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectSearchClear = document.querySelector("#project-search-clear");         |
| Browser DOM at line 55 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectSearchStatus = document.querySelector("#project-search-status");       |
| Browser DOM at line 56 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectDialog = document.querySelector("#project-dialog");                    |
| Browser DOM at line 57 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectWindowTitle = document.querySelector("#project-window-title");         |
| Browser DOM at line 58 | Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectWindowClose = document.querySelector("#project-window-close");         |

| Item                   | Explanation                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Browser DOM at line 59 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectWindowDelete = document.querySelector("#project-window-delete");</code>     |
| Browser DOM at line 60 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const viewProjectPreviewButton = document.querySelector("#view-project-preview");</code> |
| Browser DOM at line 61 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const saveProjectButton = document.querySelector("#save-project");</code>                |
| Browser DOM at line 62 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const saveProjectCloseButton = document.querySelector("#save-project-close");</code>     |
| Browser DOM at line 63 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectTitleMenu = document.querySelector("#project-title-menu");</code>           |
| Browser DOM at line 64 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const titleRenameActions = document.querySelector("#title-rename-actions");</code>       |
| Browser DOM at line 65 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const titleRenameSave = document.querySelector("#title-rename-save");</code>             |
| Browser DOM at line 66 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const titleRenameCancel = document.querySelector("#title-rename-cancel");</code>         |
| Browser DOM at line 67 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectMetaDialog = document.querySelector("#project-meta-dialog");</code>         |
| Browser DOM at line 68 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectMetaTitle = document.querySelector("#project-meta-title");</code>           |
| Browser DOM at line 69 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectMetaGrid = document.querySelector("#project-meta-grid");</code>             |
| Browser DOM at line 70 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectMetaClose = document.querySelector("#project-meta-close");</code>           |
| Browser DOM at line 71 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectFields = document.querySelector("#project-fields");</code>                  |
| Browser DOM at line 72 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const sectionTabs = document.querySelector("#section-tabs");</code>                      |
| Browser DOM at line 73 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const sectionContent = document.querySelector("#section-content");</code>                |
| Browser DOM at line 74 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectWindowContent = document.querySelector(".project-window-content");</code>   |
| Browser DOM at line 75 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewDialog = document.querySelector("#project-preview-dialog");</code>   |
| Browser DOM at line 76 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewTitle = document.querySelector("#project-preview-title");</code>     |
| Browser DOM at line 77 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewContent = document.querySelector("#project-preview-content");</code> |
| Browser DOM at line 78 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewClose = document.querySelector("#project-preview-close");</code>     |
| Browser DOM at line 79 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewBack = document.querySelector("#project-preview-back");</code>       |
| Browser DOM at line 80 | Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewForward = document.querySelector("#project-preview-forward");</code> |

## Important constants and variables

| Item                            | Explanation                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------------------|
| <code>builderGuideOpen</code>   | Line 1: <code>const builderGuideOpen = document.querySelector("#builder-guide-open");</code>     |
| <code>builderGuideDialog</code> | Line 2: <code>const builderGuideDialog = document.querySelector("#builder-guide-dialog");</code> |
| <code>builderGuideClose</code>  | Line 3: <code>const builderGuideClose = document.querySelector("#builder-guide-close");</code>   |
| <code>builderGuideSearch</code> | Line 4: <code>const builderGuideSearch = document.querySelector("#builder-guide-search");</code> |

| Item                       | Explanation                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------|
| builderGuideResults        | Line 5: const builderGuideResults = document.querySelector("#builder-guide-results");           |
| builderGuideSections       | Line 6: const builderGuideSections = document.querySelector("#builder-guide-sections");         |
| builderGuideTopicDialog    | Line 7: const builderGuideTopicDialog = document.querySelector("#builder-guide-topic-dialog");  |
| builderGuideTopicTitle     | Line 8: const builderGuideTopicTitle = document.querySelector("#builder-guide-topic-title");    |
| builderGuideTopicBody      | Line 9: const builderGuideTopicBody = document.querySelector("#builder-guide-topic-body");      |
| builderGuideTopicClose     | Line 10: const builderGuideTopicClose = document.querySelector("#builder-guide-topic-close");   |
| templateSelect             | Line 11: const templateSelect = document.querySelector("#template-select");                     |
| newCategorySelect          | Line 12: const newCategorySelect = document.querySelector("#new-category-select");              |
| placementSelect            | Line 13: const placementSelect = document.querySelector("#placement-select");                   |
| newProjectTitle            | Line 14: const newProjectTitle = document.querySelector("#new-project-title");                  |
| addProjectButton           | Line 15: const addProjectButton = document.querySelector("#add-project");                       |
| addCategoryButton          | Line 16: const addCategoryButton = document.querySelector("#add-category");                     |
| addSiteSectionButton       | Line 17: const addSiteSectionButton = document.querySelector("#add-site-section");              |
| funFactsInput              | Line 18: const funFactsInput = document.querySelector("#fun-facts-input");                      |
| funFactsCount              | Line 19: const funFactsCount = document.querySelector("#fun-facts-count");                      |
| saveFunFactsButton         | Line 20: const saveFunFactsButton = document.querySelector("#save-fun-facts");                  |
| clearFunFactsButton        | Line 21: const clearFunFactsButton = document.querySelector("#clear-fun-facts");                |
| siteHeroEyebrowInput       | Line 22: const siteHeroEyebrowInput = document.querySelector("#site-hero-eyebrow");             |
| siteHeroTitleInput         | Line 23: const siteHeroTitleInput = document.querySelector("#site-hero-title");                 |
| siteHeroCopyInput          | Line 24: const siteHeroCopyInput = document.querySelector("#site-hero-copy");                   |
| saveSiteContentButton      | Line 25: const saveSiteContentButton = document.querySelector("#save-site-content");            |
| profileDisplayNameInput    | Line 26: const profileDisplayNameInput = document.querySelector("#profile-display-name");       |
| profilePortfolioLabelInput | Line 27: const profilePortfolioLabelInput = document.querySelector("#profile-portfolio-label"); |
| profileContactIntroInput   | Line 28: const profileContactIntroInput = document.querySelector("#profile-contact-intro");     |
| profileEmailInput          | Line 29: const profileEmailInput = document.querySelector("#profile-email");                    |
| profilePhoneInput          | Line 30: const profilePhoneInput = document.querySelector("#profile-phone");                    |
| profileGithubInput         | Line 31: const profileGithubInput = document.querySelector("#profile-github");                  |
| profileLinkedInInput       | Line 32: const profileLinkedInInput = document.querySelector("#profile-linkedin");              |
| profileWebsiteInput        | Line 33: const profileWebsiteInput = document.querySelector("#profile-website");                |
| saveProfileContentButton   | Line 34: const saveProfileContentButton = document.querySelector("#save-profile-content");      |
| profileImageUploadButton   | Line 35: const profileImageUploadButton = document.querySelector("#profile-image-upload");      |
| profileHeroUploadButton    | Line 36: const profileHeroUploadButton = document.querySelector("#profile-hero-upload");        |
| profileResumeUploadButton  | Line 37: const profileResumeUploadButton = document.querySelector("#profile-resume-upload");    |
| profileBrandUploadButton   | Line 38: const profileBrandUploadButton = document.querySelector("#profile-brand-upload");      |
| profileAssetStatus         | Line 39: const profileAssetStatus = document.querySelector("#profile-asset-status");            |
| categoryDropdown           | Line 40: const categoryDropdown = document.querySelector("#category-dropdown");                 |
| siteSectionList            | Line 41: const siteSectionList = document.querySelector("#site-section-list");                  |
| projectCreateDialog        | Line 42: const projectCreateDialog = document.querySelector("#project-create-dialog");          |
| projectcreateForm          | Line 43: const projectcreateForm = document.querySelector("#project-create-form");              |
| projectCreateCategory      | Line 44: const projectCreateCategory = document.querySelector("#project-create-category");      |
| projectCreateCancel        | Line 45: const projectCreateCancel = document.querySelector("#project-create-cancel");          |

| Item                     | Explanation                                                                                |
|--------------------------|--------------------------------------------------------------------------------------------|
| categoryDialog           | Line 46: const categoryDialog = document.querySelector("#category-dialog");                |
| categoryForm             | Line 47: const categoryForm = document.querySelector("#category-form");                    |
| categoryTitle            | Line 48: const categoryTitle = document.querySelector("#category-title");                  |
| categoryDescription      | Line 49: const categoryDescription = document.querySelector("#category-description");      |
| categoryAccent           | Line 50: const categoryAccent = document.querySelector("#category-accent");                |
| categoryCancel           | Line 51: const categoryCancel = document.querySelector("#category-cancel");                |
| projectTree              | Line 52: const projectTree = document.querySelector("#project-tree");                      |
| projectSearchInput       | Line 53: const projectSearchInput = document.querySelector("#project-search");             |
| projectSearchClear       | Line 54: const projectSearchClear = document.querySelector("#project-search-clear");       |
| projectSearchStatus      | Line 55: const projectSearchStatus = document.querySelector("#project-search-status");     |
| projectDialog            | Line 56: const projectDialog = document.querySelector("#project-dialog");                  |
| projectWindowTitle       | Line 57: const projectWindowTitle = document.querySelector("#project-window-title");       |
| projectWindowClose       | Line 58: const projectWindowClose = document.querySelector("#project-window-close");       |
| projectWindowDelete      | Line 59: const projectWindowDelete = document.querySelector("#project-window-delete");     |
| viewProjectPreviewButton | Line 60: const viewProjectPreviewButton = document.querySelector("#view-project-preview"); |
| saveProjectButton        | Line 61: const saveProjectButton = document.querySelector("#save-project");                |
| saveProjectCloseButton   | Line 62: const saveProjectCloseButton = document.querySelector("#save-project-close");     |
| projectTitleMenu         | Line 63: const projectTitleMenu = document.querySelector("#project-title-menu");           |
| titleRenameActions       | Line 64: const titleRenameActions = document.querySelector("#title-rename-actions");       |
| titleRenameSave          | Line 65: const titleRenameSave = document.querySelector("#title-rename-save");             |
| titleRenameCancel        | Line 66: const titleRenameCancel = document.querySelector("#title-rename-cancel");         |
| projectMetaDialog        | Line 67: const projectMetaDialog = document.querySelector("#project-meta-dialog");         |
| projectMetaTitle         | Line 68: const projectMetaTitle = document.querySelector("#project-meta-title");           |
| projectMetaGrid          | Line 69: const projectMetaGrid = document.querySelector("#project-meta-grid");             |
| projectMetaClose         | Line 70: const projectMetaClose = document.querySelector("#project-meta-close");           |
| projectFields            | Line 71: const projectFields = document.querySelector("#project-fields");                  |
| sectionTabs              | Line 72: const sectionTabs = document.querySelector("#section-tabs");                      |
| sectionContent           | Line 73: const sectionContent = document.querySelector("#section-content");                |
| projectWindowContent     | Line 74: const projectWindowContent = document.querySelector(".project-window-content");   |
| projectPreviewDialog     | Line 75: const projectPreviewDialog = document.querySelector("#project-preview-dialog");   |
| projectPreviewTitle      | Line 76: const projectPreviewTitle = document.querySelector("#project-preview-title");     |
| projectPreviewContent    | Line 77: const projectPreviewContent = document.querySelector("#project-preview-content"); |
| projectPreviewClose      | Line 78: const projectPreviewClose = document.querySelector("#project-preview-close");     |
| projectPreviewBack       | Line 79: const projectPreviewBack = document.querySelector("#project-preview-back");       |
| projectPreviewForward    | Line 80: const projectPreviewForward = document.querySelector("#project-preview-forward"); |

## JSON/YAML-style keys detected

| Item             | Explanation                                                     |
|------------------|-----------------------------------------------------------------|
| skin-light-blue  | Line 433: "skin-light-blue": "appearance-light-blue-red-click", |
| skin-clean-white | Line 434: "skin-clean-white": "appearance-white-blue-click",    |
| skin-deep-navy   | Line 435: "skin-deep-navy": "appearance-deep-navy-cyan-click",  |

| Item                          | Explanation                                                                  |
|-------------------------------|------------------------------------------------------------------------------|
| skin-red-warm                 | Line 436: "skin-red-warm": "appearance-warm-red-pale-click",                 |
| skin-emerald-instrument       | Line 437: "skin-emerald-instrument": "appearance-emerald-mint-click",        |
| skin-amber-power              | Line 438: "skin-amber-power": "appearance-amber-cream-click",                |
| skin-violet-mixed             | Line 439: "skin-violet-mixed": "appearance-violet-lilac-click",              |
| skin-graphite-asic            | Line 440: "skin-graphite-asic": "appearance-graphite-white-click",           |
| analog-opamp-topology         | Line 441: "analog-opamp-topology": "appearance-light-blue-red-click",        |
| analog-power-charger          | Line 442: "analog-power-charger": "appearance-amber-cream-click",            |
| analog-filter-front-end       | Line 443: "analog-filter-front-end": "appearance-light-blue-red-click",      |
| analog-sensor-interface       | Line 444: "analog-sensor-interface": "appearance-emerald-mint-click",        |
| analog-mixed-signal-timing    | Line 445: "analog-mixed-signal-timing": "appearance-violet-lilac-click",     |
| digital-fpga-pipeline         | Line 446: "digital-fpga-pipeline": "appearance-deep-navy-cyan-click",        |
| digital-asic-block            | Line 447: "digital-asic-block": "appearance-graphite-white-click",           |
| digital-verification-suite    | Line 448: "digital-verification-suite": "appearance-graphite-white-click",   |
| digital-interface-controller  | Line 449: "digital-interface-controller": "appearance-deep-navy-cyan-click", |
| digital-signal-processing     | Line 450: "digital-signal-processing": "appearance-violet-lilac-click",      |
| embedded-stm32-sensor         | Line 451: "embedded-stm32-sensor": "appearance-emerald-mint-click",          |
| embedded-rtos-control         | Line 452: "embedded-rtos-control": "appearance-graphite-white-click",        |
| embedded-power-monitor        | Line 453: "embedded-power-monitor": "appearance-amber-cream-click",          |
| embedded-wireless-node        | Line 454: "embedded-wireless-node": "appearance-deep-navy-cyan-click",       |
| embedded-motor-control        | Line 455: "embedded-motor-control": "appearance-violet-lilac-click"          |
| engineering-file              | Line 2512: "engineering-file": "Engineering file link",                      |
| google-drive                  | Line 2515: "google-drive": "Google Drive link",                              |
| Brief                         | Line 2656: "Brief": "Overview",                                              |
| Project Brief                 | Line 2657: "Project Brief": "Overview",                                      |
| Design Brief                  | Line 2658: "Design Brief": "Design Overview",                                |
| Simulation Brief              | Line 2659: "Simulation Brief": "Simulation Overview"                         |
| --responsive-section-card-min | Line 2901: "--responsive-section-card-min": profile.sectionCardMin,          |
| --responsive-content-max      | Line 2902: "--responsive-content-max": profile.contentMax,                   |
| --responsive-media-max        | Line 2903: "--responsive-media-max": profile.mediaMax,                       |
| --responsive-touch-target     | Line 2904: "--responsive-touch-target": profile.touchTarget                  |
| --accent                      | Line 2919: "--accent": accent,                                               |
| --template-bg                 | Line 2920: "--template-bg": visual.background,                               |
| --template-panel              | Line 2921: "--template-panel": visual.panel,                                 |
| --template-accent             | Line 2922: "--template-accent": visual.accent,                               |
| --template-hover              | Line 2923: "--template-hover": visual.hover,                                 |
| --template-click              | Line 2924: "--template-click": visual.click,                                 |
| --template-click-text         | Line 2925: "--template-click-text": visual.clickText,                        |
| --template-line               | Line 2926: "--template-line": visual.line,                                   |
| --template-text               | Line 2927: "--template-text": visual.text,                                   |
| --accent                      | Line 2952: "--accent": accent,                                               |
| --template-bg                 | Line 2953: "--template-bg": visual.background,                               |

| Item                  | Explanation                                                         |
|-----------------------|---------------------------------------------------------------------|
| --template-panel      | Line 2954: "--template-panel": visual.panel,                        |
| --template-accent     | Line 2955: "--template-accent": visual.accent,                      |
| --template-hover      | Line 2956: "--template-hover": visual.hover,                        |
| --template-click      | Line 2957: "--template-click": visual.click,                        |
| --template-click-text | Line 2958: "--template-click-text": visual.clickText,               |
| --template-line       | Line 2959: "--template-line": visual.line,                          |
| --template-text       | Line 2960: "--template-text": visual.text,                          |
| compile-code          | Line 7449: "compile-code": () => renderCompileCodeSection(project), |

## Event handlers and user interactions

| Item      | Explanation                                                               |
|-----------|---------------------------------------------------------------------------|
| Line 1048 | document.addEventListener("pointerdown", beginDialogResize);              |
| Line 1049 | document.addEventListener("pointerdown", beginDialogDrag);                |
| Line 1050 | document.addEventListener("pointermove", moveDialogResize);               |
| Line 1051 | document.addEventListener("pointermove", moveDialogDrag);                 |
| Line 1052 | document.addEventListener("pointerup", endDialogResize);                  |
| Line 1053 | document.addEventListener("pointerup", endDialogDrag);                    |
| Line 1054 | document.addEventListener("pointercancel", endDialogResize);              |
| Line 1055 | document.addEventListener("pointercancel", endDialogDrag);                |
| Line 1056 | document.addEventListener("click", (event) => {                           |
| Line 1061 | document.addEventListener("click", (event) => {                           |
| Line 1067 | dialog.addEventListener("close", () => {                                  |
| Line 1076 | window.addEventListener("resize", () => {                                 |
| Line 1319 | document.addEventListener("visibilitychange", () => {                     |
| Line 6166 | siteLinkDialog.addEventListener("close", onClose);                        |
| Line 8325 | dialog.addEventListener("close", onClose);                                |
| Line 8747 | templateSelect.addEventListener("change", () => {                         |
| Line 8752 | templatePreviewClose.addEventListener("click", () => {                    |
| Line 8880 | builderGuideOpen.addEventListener("click", () => {                        |
| Line 8887 | builderGuideClose.addEventListener("click", () => {                       |
| Line 8891 | builderGuideTopicClose?.addEventListener("click", () => {                 |
| Line 8895 | builderGuideTopicDialog?.addEventListener("close", () => {                |
| Line 8899 | builderGuideSearch?.addEventListener("input", filterBuilderGuide);        |
| Line 8900 | builderGuideDialog?.addEventListener("click", (event) => {                |
| Line 8915 | builderGuideDialog?.addEventListener("keydown", (event) => {              |
| Line 8923 | projectWindowTitle.addEventListener("dblclick", (event) => {              |
| Line 8929 | projectWindowTitle.addEventListener("keydown", (event) => {               |
| Line 8946 | projectWindowTitle.addEventListener("contextmenu", (event) => {           |
| Line 8954 | projectTitleMenu.addEventListener("click", async (event) => {             |
| Line 8991 | titleRenameSave.addEventListener("click", () => endTitleRename(true));    |
| Line 8992 | titleRenameCancel.addEventListener("click", () => endTitleRename(false)); |

| Item      | Explanation                                                                               |
|-----------|-------------------------------------------------------------------------------------------|
| Line 8993 | projectMetaClose.addListener("click", () => closeDialogElement(projectMetaDialog));       |
| Line 8995 | viewProjectPreviewButton.addListener("click", openProjectPortfolioPreview);               |
| Line 8996 | projectPreviewClose.addListener("click", closeProjectPreviewWindow);                      |
| Line 8997 | projectPreviewDialog.addListener("close", () => {                                         |
| Line 9001 | projectPreviewBack.addListener("click", projectPreviewBackStep);                          |
| Line 9002 | projectPreviewForward.addListener("click", projectPreviewForwardStep);                    |
| Line 9004 | projectPreviewContent.addListener("click", (event) => {                                   |
| Line 9012 | saveProjectButton.addListener("click", saveSelectedProjectToPortfolio);                   |
| Line 9013 | saveProjectCloseButton.addListener("click", saveSelectedProjectAndClose);                 |
| Line 9014 | openPortfolioPreviewButton.addListener("click", openPortfolioPreview);                    |
| Line 9015 | saveAllSectionsButton.addListener("click", saveAllSections);                              |
| Line 9016 | portfolioPreviewClose.addListener("click", () => {                                        |
| Line 9019 | portfolioPreviewDialog.addListener("close", () => {                                       |
| Line 9022 | publishTargetOpen?.addListener("click", openPublishTargetDialog);                         |
| Line 9023 | publishTargetClose?.addListener("click", () => {                                          |
| Line 9026 | publishTargetCancel?.addListener("click", () => {                                         |
| Line 9029 | publishTargetForm?.addListener("submit", savePublishTarget);                              |
| Line 9030 | publishTargetSync?.addListener("click", syncFromPublishTarget);                           |
| Line 9031 | publishTargetCheck?.addListener("click", loadSystemReadiness);                            |
| Line 9032 | publishTargetInstallGit?.addListener("click", installGitForPublishing);                   |
| Line 9033 | publishTargetRepository?.addListener("input", () => {                                     |
| Line 9036 | publishTargetDomain?.addListener("input", () => {                                         |
| Line 9040 | publishTargetUsername?.addListener("input", markPublishTargetNeedsAuthentication);        |
| Line 9041 | publishTargetPassword?.addListener("input", markPublishTargetNeedsAuthentication);        |
| Line 9042 | publishResultClose.addListener("click", () => {                                           |
| Line 9046 | appUpdateClose?.addListener("click", () => {                                              |
| Line 9050 | appUpdateLater?.addListener("click", () => {                                              |
| Line 9058 | appUpdateSkip?.addListener("click", () => {                                               |
| Line 9065 | appUpdateApply?.addListener("click", () => {                                              |
| Line 9069 | appUpdateCheckButton?.addListener("click", () => {                                        |
| Line 9073 | securityReportOpen?.addListener("click", openSecurityReport);                             |
| Line 9074 | securityReportRefresh?.addListener("click", openSecurityReport);                          |
| Line 9075 | securityReportClose?.addListener("click", () => {                                         |
| Line 9078 | securityReportOk?.addListener("click", () => {                                            |
| Line 9082 | addProjectButton.addListener("click", () => {                                             |
| Line 9086 | projectSearchInput?.addListener("input", () => {                                          |
| Line 9091 | projectSearchClear?.addListener("click", () => {                                          |
| Line 9100 | quickOpenSelected?.addListener("click", () => {                                           |
| Line 9106 | quickPreviewSelected?.addListener("click", () => {                                        |
| Line 9112 | addCategoryButton?.addListener("click", openCategoryDialog);                              |
| Line 9113 | categoryCancel?.addListener("click", () => closeDialogElement(categoryDialog, "cancel")); |



| Item      | Explanation                                                     |
|-----------|-----------------------------------------------------------------|
| Line 9114 | categoryForm?.addEventListener("submit", (event) => {           |
| Line 9120 | addSiteSectionButton.addEventListener("click", addSiteSection); |
| Line 9122 | funFactsInput?.addEventListener("input", () => {                |
| Line 9128 | saveFunFactsButton?.addEventListener("click", () => {           |
| Line 9135 | clearFunFactsButton?.addEventListener("click", () => {          |
| Line 9146 | input.addEventListener("input", () => {                         |
| Line 9154 | saveSiteContentButton?.addEventListener("click", () => {        |
| Line 9171 | input.addEventListener("input", () => {                         |
| Line 9179 | saveProfileContentButton?.addEventListener("click", () => {     |

## Function-by-function detail

### *isHdlLanguage* - lines 240-242

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 240-242 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function isHdlLanguage(language = "") {                                                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | includes, normalizeCodeLanguage                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

240: function isHdlLanguage(language = "") {
241:   return ["verilog", "systemverilog"].includes(normalizeCodeLanguage(language));
242: }

```

### *inferCompileFileRole* - lines 244-251

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 244-251 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function inferCompileFileRole(fileName = "", code = "", language = "") {                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isHdlLanguage, toLowerCase, b, tb                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

244: function inferCompileFileRole(fileName = "", code = "", language = "") {
245:   if (!isHdlLanguage(language)) return "source";
246:   const haystack = `${fileName}\n${code}`.toLowerCase();

```

```

247:   if (/^b(tb|testbench)\b|(^|[_-])tb(?:[_-]|\.)|test(?:[_-]?bench|\\$dumpfile|\\$dumpvars|module\\s+tb(?:_a-z0-
9)]*/i.test(haystack)) {
248:     return "testbench";
249:   }
250:   return "design";
251: }

```

## ***normalizeCompileFileRole - lines 253-257***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 253-257 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function normalizeCompileFileRole(value = "", language = "") {                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isHdlLanguage, trim, toLowerCase, replace, includes                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

253: function normalizeCompileFileRole(value = "", language = "") {
254:   if (!isHdlLanguage(language)) return "source";
255:   const clean = String(value || "").trim().toLowerCase().replace(/[_\s-]+/g, "");
256:   return ["tb", "testbench", "bench"].includes(clean) ? "testbench" : "design";
257: }

```

## ***compileFileRoleLabel - lines 259-263***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 259-263 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function compileFileRoleLabel(role = "source") {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

259: function compileFileRoleLabel(role = "source") {
260:   if (role === "testbench") return "Testbench";
261:   if (role === "design") return "Design";
262:   return "Program source";
263: }

```

## ***setStatus - lines 509-511***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 509-511 (3 source lines).                              |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function setStatus(message) {                                |
| Touches         | Mostly local calculations or local UI state.                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
509: function setStatus(message) {
510:   builderStatus.textContent = message;
511: }
```

## ***setSaveState - lines 513-517***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 513-517 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function setSaveState(state, message) {                                                                                                                                                               |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
513: function setSaveState(state, message) {
514:   if (!builderSaveState) return;
515:   builderSaveState.dataset.state = state || "loaded";
516:   builderSaveState.textContent = message || "Loaded";
517: }
```

## ***projectSearchText - lines 519-521***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 519-521 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectSearchText(value = "") {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | toLowerCase, replace, trim                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
519: function projectSearchText(value = "") {
520:   return String(value || "").toLowerCase().replace(/\s+/g, " ").trim();
521: }
```

## ***regexEscape - lines 523-525***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 523-525 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function regexEscape(value = "") {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replace                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

523: function regexEscape(value = "") {
524:   return String(value).replace(/[*+?^$()|[\]\\"/g, "\\$&");
525: }

```

## ***highlightSearchText - lines 527-533***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 527-533 (7 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function highlightSearchText(value = "", query = "") {                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectSearchText, escapeHtml, regexEscape, replace                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

527: function highlightSearchText(value = "", query = "") {
528:   const text = String(value || "");
529:   const cleanQuery = projectSearchText(query);
530:   if (!cleanQuery) return escapeHtml(text);
531:   const matcher = new RegExp(`(${regexEscape(cleanQuery)})`, "ig");
532:   return escapeHtml(text).replace(matcher, "<mark>$1</mark>");
533: }

```

## ***projectSearchHaystack - lines 535-555***

| Item               | Explanation                                                   |
|--------------------|---------------------------------------------------------------|
| Source location    | Lines 535-555 (21 source lines).                              |
| Purpose            | Part of the private builder frontend and editing experience.  |
| Declaration        | function projectSearchHaystack(project = {}, category = {}) { |
| Touches            | Mostly local calculations or local UI state.                  |
| Calls detected     | projectSearchText, stringify, filter, join                    |
| API endpoints      | None detected inside this function body.                      |
| Storage keys       | No local/session storage keys detected.                       |
| Async/return clues | 0 await expression(s), 1 return statement(s).                 |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

535: function projectSearchHaystack(project = {}, category = {}) {
536:   return projectSearchText([
537:     category.label,
538:     category.description,
539:     project.title,
540:     project.summary,
541:     project.status,
542:     project.category,
543:     project.templateLabel,
544:     ...(project.focus || []),
545:     ...(project.highlights || []),
546:     JSON.stringify(project.sections || []),
547:     JSON.stringify(project.design || {}),
548:     JSON.stringify(project.documents || []),
549:     JSON.stringify(project.tests || []),
550:     JSON.stringify(project.tools || []),
551:     JSON.stringify(project.links || []),
552:     JSON.stringify(project.media || []),
553:     JSON.stringify(project.compileCode || {}),
554:   ]).filter(Boolean).join(" ");
555: }

```

## ***projectMatchesSearch - lines 557-561***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 557-561 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectMatchesSearch(project, category, query) {                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectSearchText, projectSearchHaystack, includes                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

557: function projectMatchesSearch(project, category, query) {
558:   const cleanQuery = projectSearchText(query);
559:   if (!cleanQuery) return true;
560:   return projectSearchHaystack(project, category).includes(cleanQuery);
561: }

```

## ***updateProjectSearchStatus - lines 563-575***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 563-575 (13 source lines).                                                                                                                                                                      |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateProjectSearchStatus(matchCount = catalog.projects.length) {                                                                                                                            |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | projectSearchText                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
563: function updateProjectSearchStatus(matchCount = catalog.projects.length) {
564:   const cleanQuery = projectSearchText(projectSearchQuery);
565:   if (projectSearchInput && document.activeElement !== projectSearchInput) {
566:     projectSearchInput.value = projectSearchQuery;
567:   }
568:   if (projectSearchClear) projectSearchClear.hidden = !cleanQuery;
569:   if (!projectSearchStatus) return;
570:   if (!cleanQuery) {
571:     projectSearchStatus.textContent = "Showing all projects.";
572:     return;
573:   }
574:   projectSearchStatus.textContent = `${matchCount} matching project${matchCount === 1 ? "" : "s"} for
"${projectSearchQuery}"`;
575: }
```

## ***updateBuilderWorkflow - lines 577-597***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 577-597 (21 source lines).                                                                                                                                                                      |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateBuilderWorkflow() {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, Set, map, filter, has, forEach                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
577: function updateBuilderWorkflow() {
578:   const project = selectedProject();
579:   const savedIds = new Set((savedPortfolioCatalog.projects || []).map((item) => item.id));
580:   const savedCount = (catalog.projects || []).filter((projectItem) =>
savedIds.has(projectItem.id)).length;
581:   const visibleSections = (catalog.siteSections || []).filter(siteSectionRenderable).length;
582:   if (workflowSelectedProject) {
583:     workflowSelectedProject.textContent = project ? project.title || "Untitled project" : "No project
selected";
584:   }
585:   if (workflowTotalProjects) {
586:     workflowTotalProjects.textContent = `${catalog.projects.length} project${catalog.projects.length ===
1 ? "" : "s"}`;
587:   }
588:   if (workflowSavedProjects) {
589:     workflowSavedProjects.textContent = `${savedCount} parsed`;
590:   }
591:   if (workflowVisibleSections) {
592:     workflowVisibleSections.textContent = `${visibleSections} live section${visibleSections === 1 ? "" :
"s"}`;
593:   }
594:   [quickOpenSelected, quickPreviewSelected].forEach((button) => {
595:     if (button) button.disabled = !project;
596:   });
597: }
```

## ***savedCount - lines 580-584***

| Item            | Explanation                                                                                               |
|-----------------|-----------------------------------------------------------------------------------------------------------|
| Source location | Lines 580-584 (5 source lines).                                                                           |
| Purpose         | Persists data to local state, disk, credentials, or browser storage.                                      |
| Declaration     | const savedCount = (catalog.projects    []).filter((projectItem) => savedIds.has(projectItem.id)).length; |
| Touches         | Mostly local calculations or local UI state.                                                              |
| Calls detected  | filter, has                                                                                               |
| API endpoints   | None detected inside this function body.                                                                  |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

580:   const savedCount = (catalog.projects || []).filter((projectItem) =>
savedIds.has(projectItem.id)).length;
581:   const visibleSections = (catalog.siteSections || []).filter(siteSectionRenderable).length;
582:   if (workflowSelectedProject) {
583:     workflowSelectedProject.textContent = project ? project.title || "Untitled project" : "No project
selected";
584:   }

```

## ***dialogDragHandles - lines 599-604***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 599-604 (6 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function dialogDragHandles(dialog) {                                                                                                                                                                  |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | querySelectorAll                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

599: function dialogDragHandles(dialog) {
600:   return [
601:     ...dialog.querySelectorAll(".project-dialog-heading, .project-preview-heading, .template-dialog-
heading"),
602:     ...dialog.querySelectorAll(".asset-dialog form > div:first-child")
603:   ];
604: }

```

## ***dialogWindowActionTarget - lines 606-608***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 606-608 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function dialogWindowActionTarget(handle) {                                                                                                                                                           |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

606: function dialogWindowActionTarget(handle) {
607:   return handle.querySelector(".project-window-actions, .section-view-actions") || handle;
608: }

```

## ***updateDialogWindowButtons - lines 610-635***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 610-635 (26 source lines).                                                                                                                                                                      |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateDialogWindowButtons(dialog) {                                                                                                                                                          |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | contains, querySelectorAll, forEach, setAttribute, dialogCanStepBack, dialogCanStepForward                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
610: function updateDialogWindowButtons(dialog) {
611:   const isMinimized = dialog.classList.contains("is-minimized-dialog");
612:   const isHidden = dialog.classList.contains("is-hidden-dialog");
613:   const isMaximized = dialog.classList.contains("is-maximized-dialog");
614:   dialog.querySelectorAll("[data-dialog-minimize='true']").forEach((button) => {
615:     button.textContent = isMinimized ? "+" : "\u2212";
616:     button.title = isMinimized ? "Restore window" : "Minimize window";
617:     button.setAttribute("aria-label", isMinimized ? "Restore window" : "Minimize window");
618:   });
619:   dialog.querySelectorAll("[data-dialog-hide='true']").forEach((button) => {
620:     button.textContent = isHidden ? "\u25b4" : "\u25be";
621:     button.title = isHidden ? "Show window" : "Hide window";
622:     button.setAttribute("aria-label", isHidden ? "Show window" : "Hide window");
623:   });
624:   dialog.querySelectorAll("[data-dialog-maximize='true']").forEach((button) => {
625:     button.textContent = isMaximized ? "\u2750" : "\u25a1";
626:     button.title = isMaximized ? "Restore size" : "Maximize window";
627:     button.setAttribute("aria-label", isMaximized ? "Restore size" : "Maximize window");
628:   });
629:   dialog.querySelectorAll("[data-dialog-back='true']").forEach((button) => {
630:     button.disabled = !dialogCanStepBack(dialog);
631:   });
632:   dialog.querySelectorAll("[data-dialog-forward='true']").forEach((button) => {
633:     button.disabled = !dialogCanStepForward(dialog);
634:   });
635: }
```

## ***dialogTitleForDock - lines 637-639***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 637-639 (3 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function dialogTitleForDock(dialog) {                                                                                                                                                                 |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, trim                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
637: function dialogTitleForDock(dialog) {
638:   return dialog.querySelector("h2")?.textContent?.trim() || "Window";
639: }
```

## ***makeDialogControl - lines 641-650***



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 641-650 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function makeDialogControl(action, label, title) {                                                                                                                                                    |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | createElement, setAttribute                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

641: function makeDialogControl(action, label, title) {
642:   const button = document.createElement("button");
643:   button.type = "button";
644:   button.className = `dialog-window-control dialog-window-${action}-control`;
645:   button.dataset.dialogAction = action;
646:   button.textContent = label;
647:   button.title = title;
648:   button.setAttribute("aria-label", title);
649:   return button;
650: }

```

## ***ensureDialogLeftControls - lines 652-674***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 652-674 (23 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function ensureDialogLeftControls(dialog, handle) {                                                                                                                                                   |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, createElement, makeDialogControl, append, prepend, forEach, toUpperCase, slice                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

652: function ensureDialogLeftControls(dialog, handle) {
653:   if (handle.querySelector(".dialog-window-left-controls")) return;
654:   const titlebar = handle.querySelector(".section-view-titlebar");
655:   const controls = document.createElement("div");
656:   controls.className = "dialog-window-left-controls";
657:
658:   if (titlebar?.querySelector(".section-view-back, .section-view-forward")) {
659:     const refresh = makeDialogControl("refresh", "\u21bb", "Refresh window");
660:     refresh.dataset.dialogRefresh = "true";
661:     controls.append(refresh);
662:     titlebar.prepend(controls);
663:   } else {
664:     [
665:       makeDialogControl("back", "\u2190", "Back"),
666:       makeDialogControl("forward", "\u2192", "Forward"),
667:       makeDialogControl("refresh", "\u21bb", "Refresh window")
668:     ].forEach((button) => {
669:
670: button.dataset[`dialog${button.dataset.dialogAction[0].toUpperCase()}${button.dataset.dialogAction.slice(1)}`] =
"true";
670:       controls.append(button);
671:     });
672:     handle.prepend(controls);
673:   }
674: }

```

## ensureDialogWindowControls - lines 676-720

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 676-720 (45 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function ensureDialogWindowControls(dialog, handle) {                                                                                                                                                 |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | querySelectorAll, forEach, remove, ensureDialogLeftControls, dialogWindowActionTarget, querySelector, createElement, append, makeDialogControl, find, closest, add                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
676: function ensureDialogWindowControls(dialog, handle) {
677:   dialog.querySelectorAll(".dialog-minimize-button").forEach((button) => button.remove());
678:   ensureDialogLeftControls(dialog, handle);
679:   const target = dialogWindowActionTarget(handle);
680:   let cluster = target.querySelector(".dialog-window-control-cluster");
681:   if (!cluster) {
682:     cluster = document.createElement("div");
683:     cluster.className = "dialog-window-control-cluster";
684:     target.append(cluster);
685:   }
686:
687:   if (!cluster.querySelector("[data-dialog-hide='true']")) {
688:     const hide = makeDialogControl("hide", "\u25be", "Hide window");
689:     hide.dataset.dialogHide = "true";
690:     cluster.append(hide);
691:   }
692:   if (!cluster.querySelector("[data-dialog-minimize='true']")) {
693:     const minimize = makeDialogControl("minimize", "\u2212", "Minimize window");
694:     minimize.dataset.dialogMinimize = "true";
695:     cluster.append(minimize);
696:   }
697:   if (!cluster.querySelector("[data-dialog-maximize='true']")) {
698:     const maximize = makeDialogControl("maximize", "\u25a1", "Maximize window");
699:     maximize.dataset.dialogMaximize = "true";
700:     cluster.append(maximize);
701:   }
702:
703:   let closeButton = cluster.querySelector(".project-window-close, .close-control, [data-dialog-
close='true']");
704:   if (!closeButton) {
705:     closeButton = [...dialog.querySelectorAll(".project-window-close, .close-control")]
706:       .find((button) => button.closest(".project-dialog-heading, .project-preview-heading, .template-
dialog-heading, .asset-dialog form > div:first-child") === handle);
707:   }
708:   if (closeButton) {
709:     closeButton.classList.add("dialog-window-control", "dialog-close-button");
710:     closeButton.dataset.dialogAction = "close";
711:     closeButton.dataset.dialogClose = "true";
712:     cluster.append(closeButton);
713:   } else {
714:     const cloneClose = makeDialogControl("close", "\u00d7", "Close window");
715:     cloneClose.classList.add("project-window-close", "dialog-close-button");
716:     cloneClose.dataset.dialogClose = "true";
717:     cluster.append(cloneClose);
718:   }
719:   updateDialogWindowButtons(dialog);
720: }
```

## ensureDialogResizeHandles - lines 722-731

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 722-731 (10 source lines).                             |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function ensureDialogResizeHandles(dialog) {                 |
| Touches         | browser/Electron DOM                                         |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | querySelector, forEach, createElement, setAttribute, append                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

722: function ensureDialogResizeHandles(dialog) {
723:   if (dialog.querySelector("[data-dialog-resize-handle]")) return;
724:   ["n", "e", "s", "w", "ne", "nw", "se", "sw"].forEach((direction) => {
725:     const handle = document.createElement("div");
726:     handle.className = `dialog-resize-handle resize-${direction}`;
727:     handle.dataset.dialogResizeHandle = direction;
728:     handle.setAttribute("aria-hidden", "true");
729:     dialog.append(handle);
730:   });
731: }

```

## ***markDialogDragHandles - lines 733-740***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 733-740 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function markDialogDragHandles(dialog) {                                                                                                                                                              |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | dialogDragHandles, forEach, ensureDialogWindowControls, ensureDialogResizeHandles                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

733: function markDialogDragHandles(dialog) {
734:   dialogDragHandles(dialog).forEach((handle) => {
735:     handle.dataset.dialogDragHandle = "true";
736:     handle.title = handle.title || "Drag to move this window";
737:     ensureDialogWindowControls(dialog, handle);
738:   });
739:   ensureDialogResizeHandles(dialog);
740: }

```

## ***draggableDialogFromEvent - lines 742-751***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 742-751 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function draggableDialogFromEvent(event) {                                                                                                                                                            |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | closest, contains                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
742: function draggableDialogFromEvent(event) {
743:   if (event.target.closest("[data-dialog-resize-handle]")) return null;
744:   const handle = event.target.closest("[data-dialog-drag-handle='true']");
745:   if (!handle) return null;
746:   const dialog = handle.closest("dialog");
747:   if (!dialog?.open) return null;
748:   if (dialog.classList.contains("is-hidden-dialog")) return null;
749:   if (event.target.closest("button, a, input, textarea, select, label, summary,
[contenteditable='true']")) return null;
750:   return dialog;
751: }
```

## ***clampDialogPosition - lines 753-762***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 753-762 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function clampDialogPosition(left, top, dialog) {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getBoundingClientRect, max, min                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
753: function clampDialogPosition(left, top, dialog) {
754:   const rect = dialog.getBoundingClientRect();
755:   const margin = 12;
756:   const maxLeft = Math.max(margin, window.innerWidth - rect.width - margin);
757:   const maxTop = Math.max(margin, window.innerHeight - rect.height - margin);
758:   return {
759:     left: Math.min(Math.max(left, margin), maxLeft),
760:     top: Math.min(Math.max(top, margin), maxTop)
761:   };
762: }
```

## ***anchorDialogForDrag - lines 764-773***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 764-773 (10 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function anchorDialogForDrag(dialog) {                                                                                                                                                                |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | getBoundingClientRect, clampDialogPosition, add                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
764: function anchorDialogForDrag(dialog) {
765:   const rect = dialog.getBoundingClientRect();
766:   const position = clampDialogPosition(rect.left, rect.top, dialog);
767:   dialog.classList.add("is-draggable-dialog");
768:   dialog.style.left = `${position.left}px`;
769:   dialog.style.top = `${position.top}px`;
770:   dialog.style.right = "auto";
771:   dialog.style.bottom = "auto";
772:   dialog.style.margin = "0";
```

773: }

## ***saveDialogRestoreState - lines 775-788***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 775-788 (14 source lines).                                                                                                                                                                      |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | function saveDialogRestoreState(dialog) {                                                                                                                                                             |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | getBoundingClientRect, stringify                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
775: function saveDialogRestoreState(dialog) {
776:   if (dialog.dataset.restoreWindowState) return;
777:   const rect = dialog.getBoundingClientRect();
778:   dialog.dataset.restoreWindowState = JSON.stringify({
779:     bottom: dialog.style.bottom || "",
780:     height: dialog.style.height || `${rect.height}px`,
781:     left: dialog.style.left || `${rect.left}px`,
782:     margin: dialog.style.margin || "",
783:     maxHeight: dialog.style.maxHeight || "",
784:     right: dialog.style.right || "",
785:     top: dialog.style.top || `${rect.top}px`,
786:     width: dialog.style.width || `${rect.width}px`
787:   });
788: }
```

## ***restoreDialogWindowState - lines 790-805***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 790-805 (16 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function restoreDialogWindowState(dialog) {                                                                                                                                                           |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | parse, removeAttribute, entries, forEach                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
790: function restoreDialogWindowState(dialog) {
791:   let state = null;
792:   try {
793:     state = JSON.parse(dialog.dataset.restoreWindowState || "null");
794:   } catch {
795:     state = null;
796:   }
797:   delete dialog.dataset.restoreWindowState;
798:   if (!state) {
799:     dialog.removeAttribute("style");
800:     return;
801:   }
802:   Object.entries(state).forEach(([key, value]) => {
803:     dialog.style[key] = value;
804:   });
805: }
```

## ***anchorDialogForResize - lines 807-814***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 807-814 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function anchorDialogForResize(dialog) {                                                                                                                                                              |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | anchorDialogForDrag, getBoundingClientRect, add                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
807: function anchorDialogForResize(dialog) {
808:   anchorDialogForDrag(dialog);
809:   const rect = dialog.getBoundingClientRect();
810:   dialog.classList.add("is-resized-dialog");
811:   dialog.style.width = `${rect.width}px`;
812:   dialog.style.height = `${rect.height}px`;
813:   dialog.style.maxHeight = "none";
814: }
```

## ***resizeDialogBounds - lines 816-843***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 816-843 (28 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function resizeDialogBounds(state, event) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | min, max, includes                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
816: function resizeDialogBounds(state, event) {
817:   const margin = 12;
818:   const direction = state.direction;
819:   const minWidth = Math.min(320, Math.max(180, window.innerWidth - margin * 2));
820:   const minHeight = Math.min(170, Math.max(120, window.innerHeight - margin * 2));
821:   let { left, top, width, height } = state.rect;
822:   const dx = event.clientX - state.startX;
823:   const dy = event.clientY - state.startY;
824:
825:   if (direction.includes("e")) {
826:     width = Math.min(Math.max(state.rect.width + dx, minWidth), window.innerWidth - left - margin);
827:   }
828:   if (direction.includes("s")) {
829:     height = Math.min(Math.max(state.rect.height + dy, minHeight), window.innerHeight - top - margin);
830:   }
831:   if (direction.includes("w")) {
832:     const right = state.rect.left + state.rect.width;
833:     left = Math.min(Math.max(state.rect.left + dx, margin), right - minWidth);
834:     width = right - left;
835:   }
836:   if (direction.includes("n")) {
837:     const bottom = state.rect.top + state.rect.height;
838:     top = Math.min(Math.max(state.rect.top + dy, margin), bottom - minHeight);
839:     height = bottom - top;
840: }
```

```

841:
842:   return { height, left, top, width };
843: }

```

## ***beginDialogResize - lines 845-863***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 845-863 (19 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function beginDialogResize(event) {                                                                                                                                                                   |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | closest, contains, anchorDialogForResize, getBoundingClientRect, add, preventDefault                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

845: function beginDialogResize(event) {
846:   if (event.button !== 0) return;
847:   const handle = event.target.closest("[data-dialog-resize-handle]");
848:   const dialog = handle?.closest("dialog");
849:   if (!handle || !dialog?.open || dialog.classList.contains("is-minimized-dialog") ||
dialog.classList.contains("is-hidden-dialog")) return;
850:   anchorDialogForResize(dialog);
851:   activeDialogResize = {
852:     dialog,
853:     direction: handle.dataset.dialogResizeHandle,
854:     pointerId: event.pointerId,
855:     pointerTarget: event.target,
856:     rect: dialog.getBoundingClientRect(),
857:     startX: event.clientX,
858:     startY: event.clientY
859:   };
860:   event.target.setPointerCapture?.(event.pointerId);
861:   dialog.classList.add("is-resizing-dialog");
862:   event.preventDefault();
863: }

```

## ***moveDialogResize - lines 865-872***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 865-872 (8 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function moveDialogResize(event) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | resizeDialogBounds                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

865: function moveDialogResize(event) {
866:   if (!activeDialogResize) return;
867:   const next = resizeDialogBounds(activeDialogResize, event);
868:   activeDialogResize.dialog.style.left = `${next.left}px`;
869:   activeDialogResize.dialog.style.top = `${next.top}px`;
870:   activeDialogResize.dialog.style.width = `${next.width}px`;
871:   activeDialogResize.dialog.style.height = `${next.height}px`;
872: }

```

## ***endDialogResize - lines 874-879***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 874-879 (6 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function endDialogResize() {                                                                                                                                                                          |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | remove                                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
874: function endDialogResize() {
875:   if (!activeDialogResize) return;
876:   activeDialogResize.pointerTarget?.releasePointerCapture?.(activeDialogResize.pointerId);
877:   activeDialogResize.dialog.classList.remove("is-resizing-dialog");
878:   activeDialogResize = null;
879: }
```

## ***beginDialogDrag - lines 881-897***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 881-897 (17 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function beginDialogDrag(event) {                                                                                                                                                                     |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | draggableDialogFromEvent, anchorDialogForDrag, getBoundingClientRect, add, preventDefault                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
881: function beginDialogDrag(event) {
882:   if (event.button !== 0) return;
883:   const dialog = draggableDialogFromEvent(event);
884:   if (!dialog) return;
885:   anchorDialogForDrag(dialog);
886:   const rect = dialog.getBoundingClientRect();
887:   activeDialogDrag = {
888:     dialog,
889:     offsetX: event.clientX - rect.left,
890:     offsetY: event.clientY - rect.top,
891:     pointerId: event.pointerId,
892:     pointerTarget: event.target
893:   };
894:   event.target.setPointerCapture?.(event.pointerId);
895:   dialog.classList.add("is-dragging-dialog");
896:   event.preventDefault();
897: }
```

## ***moveDialogDrag - lines 899-908***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 899-908 (10 source lines).                             |
| Purpose         | Part of the private builder frontend and editing experience. |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function moveDialogDrag(event) {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | clampDialogPosition                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

899: function moveDialogDrag(event) {
900:   if (!activeDialogDrag) return;
901:   const next = clampDialogPosition(
902:     event.clientX - activeDialogDrag.offsetX,
903:     event.clientY - activeDialogDrag.offsetY,
904:     activeDialogDrag.dialog
905:   );
906:   activeDialogDrag.dialog.style.left = `${next.left}px`;
907:   activeDialogDrag.dialog.style.top = `${next.top}px`;
908: }

```

## ***endDialogDrag - lines 910-915***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 910-915 (6 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function endDialogDrag() {                                                                                                                                                                            |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | remove                                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

910: function endDialogDrag() {
911:   if (!activeDialogDrag) return;
912:   activeDialogDrag.pointerTarget?.releasePointerCapture?.(activeDialogDrag.pointerId);
913:   activeDialogDrag.dialog.classList.remove("is-dragging-dialog");
914:   activeDialogDrag = null;
915: }

```

## ***toggleDialogMinimized - lines 917-927***

| Item               | Explanation                                                                                  |
|--------------------|----------------------------------------------------------------------------------------------|
| Source location    | Lines 917-927 (11 source lines).                                                             |
| Purpose            | Part of the private builder frontend and editing experience.                                 |
| Declaration        | function toggleDialogMinimized(dialog) {                                                     |
| Touches            | browser/Electron DOM                                                                         |
| Calls detected     | contains, toggleDialogHidden, anchorDialogForDrag, remove, toggle, updateDialogWindowButtons |
| API endpoints      | None detected inside this function body.                                                     |
| Storage keys       | No local/session storage keys detected.                                                      |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

917: function toggleDialogMinimized(dialog) {
918:   if (!dialog) return;
919:   if (dialog.classList.contains("is-hidden-dialog")) {
920:     toggleDialogHidden(dialog);
921:     return;
922:   }
923:   anchorDialogForDrag(dialog);
924:   dialog.classList.remove("is-maximized-dialog");
925:   dialog.classList.toggle("is-minimized-dialog");
926:   updateDialogWindowButtons(dialog);
927: }

```

## ***toggleDialogMaximized - lines 929-949***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 929-949 (21 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function toggleDialogMaximized(dialog) {                                                                                                                                                              |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | contains, remove, restoreDialogWindowState, saveDialogRestoreState, add, updateDialogWindowButtons                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

929: function toggleDialogMaximized(dialog) {
930:   if (!dialog) return;
931:   if (dialog.classList.contains("is-hidden-dialog")) dialog.classList.remove("is-hidden-dialog");
932:   if (dialog.classList.contains("is-maximized-dialog")) {
933:     dialog.classList.remove("is-maximized-dialog");
934:     restoreDialogWindowState(dialog);
935:   } else {
936:     saveDialogRestoreState(dialog);
937:     dialog.classList.remove("is-minimized-dialog");
938:     dialog.classList.add("is-draggable-dialog", "is-maximized-dialog");
939:     dialog.style.left = "0";
940:     dialog.style.top = "0";
941:     dialog.style.right = "auto";
942:     dialog.style.bottom = "auto";
943:     dialog.style.margin = "0";
944:     dialog.style.width = "100vw";
945:     dialog.style.height = "100dvh";
946:     dialog.style.maxHeight = "none";
947:   }
948:   updateDialogWindowButtons(dialog);
949: }

```

## ***toggleDialogHidden - lines 951-970***

| Item            | Explanation                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------------|
| Source location | Lines 951-970 (20 source lines).                                                                              |
| Purpose         | Part of the private builder frontend and editing experience.                                                  |
| Declaration     | function toggleDialogHidden(dialog) {                                                                         |
| Touches         | browser/Electron DOM                                                                                          |
| Calls detected  | contains, remove, restoreDialogWindowState, saveDialogRestoreState, add, min, calc, updateDialogWindowButtons |
| API endpoints   | None detected inside this function body.                                                                      |
| Storage keys    | No local/session storage keys detected.                                                                       |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

951: function toggleDialogHidden(dialog) {
952:   if (!dialog) return;
953:   if (dialog.classList.contains("is-hidden-dialog")) {
954:     dialog.classList.remove("is-hidden-dialog");
955:     restoreDialogWindowState(dialog);
956:   } else {
957:     saveDialogRestoreState(dialog);
958:     dialog.classList.remove("is-minimized-dialog", "is-maximized-dialog");
959:     dialog.classList.add("is-draggable-dialog", "is-hidden-dialog");
960:     dialog.style.left = "auto";
961:     dialog.style.top = "auto";
962:     dialog.style.right = "12px";
963:     dialog.style.bottom = "10px";
964:     dialog.style.margin = "0";
965:     dialog.style.width = "min(300px, calc(100vw - 24px))";
966:     dialog.style.height = "32px";
967:     dialog.style.maxHeight = "32px";
968:   }
969:   updateDialogWindowButtons(dialog);
970: }

```

### ***dialogCanStepBack - lines 972-976***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 972-976 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function dialogCanStepBack(dialog) {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

972: function dialogCanStepBack(dialog) {
973:   if (dialog === projectDialog) return projectWindowBackStack.length > 0;
974:   if (dialog === projectPreviewDialog) return activeProjectPreviewSectionIndex >= 0;
975:   return false;
976: }

```

### ***dialogCanStepForward - lines 978-982***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 978-982 (5 source lines).                              |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function dialogCanStepForward(dialog) {                      |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | No obvious named calls detected by the generator.            |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 3 return statement(s).                |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

978: function dialogCanStepForward(dialog) {
979:   if (dialog === projectDialog) return projectWindowForwardStack.length > 0;
980:   if (dialog === projectPreviewDialog) return activeProjectPreviewForwardStack.length > 0;
981:   return false;
982: }

```

## ***renderProjectWindowSectionFromHistory - lines 984-990***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 984-990 (7 source lines).                                                                                                                                                                       |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderProjectWindowSectionFromHistory(sectionId) {                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | renderAll, updateDialogWindowButtons                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

984: function renderProjectWindowSectionFromHistory(sectionId) {
985:   suppressProjectWindowHistory = true;
986:   activeSectionId = sectionId || "brief";
987:   renderAll();
988:   suppressProjectWindowHistory = false;
989:   updateDialogWindowButtons(projectDialog);
990: }

```

## ***stepProjectWindowBack - lines 992-996***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 992-996 (5 source lines).                                                                                                                                                                       |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function stepProjectWindowBack() {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | push, renderProjectWindowSectionFromHistory, pop                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

992: function stepProjectWindowBack() {
993:   if (!projectWindowBackStack.length) return;
994:   projectWindowForwardStack.push(activeSectionId);
995:   renderProjectWindowSectionFromHistory(projectWindowBackStack.pop());
996: }

```

## ***stepProjectWindowForward - lines 998-1002***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 998-1002 (5 source lines).                                                                                                                                                                      |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function stepProjectWindowForward() {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | push, renderProjectWindowSectionFromHistory, pop                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

998: function stepProjectWindowForward() {
999:   if (!projectWindowForwardStack.length) return;
1000:   projectWindowBackStack.push(activeSectionId);
1001:   renderProjectWindowSectionFromHistory(projectWindowForwardStack.pop());
1002: }

```

## ***refreshDialogWindow - lines 1004-1015***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1004-1015 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function refreshDialogWindow(dialog) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | saveVisibleRichEditors, refreshOpenPreviews, renderPreview, renderAll, setStatus, updateDialogWindowButtons                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1004: function refreshDialogWindow(dialog) {
1005:   saveVisibleRichEditors();
1006:   if (dialog === projectPreviewDialog) {
1007:     refreshOpenPreviews();
1008:   } else if (dialog === portfolioPreviewDialog) {
1009:     renderPreview();
1010:   } else if (dialog === projectDialog) {
1011:     renderAll();
1012:   }
1013:   setStatus("Window refreshed.");
1014:   updateDialogWindowButtons(dialog);
1015: }

```

## ***closeDialogFromControl - lines 1017-1025***

| Item            | Explanation                                                 |
|-----------------|-------------------------------------------------------------|
| Source location | Lines 1017-1025 (9 source lines).                           |
| Purpose         | Controls a window, dialog, panel, menu, or visible state.   |
| Declaration     | function closeDialogFromControl(dialog, button) {           |
| Touches         | browser/Electron DOM                                        |
| Calls detected  | querySelectorAll, find, contains, click, closeDialogElement |
| API endpoints   | None detected inside this function body.                    |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1017: function closeDialogFromControl(dialog, button) {
1018:   const originalClose = [...dialog.querySelectorAll(".project-window-close, .close-control")]
1019:     .find((item) => item !== button && !item.dataset.dialogClose);
1020:   if (originalClose && !button.classList.contains("close-control") && !button.id) {
1021:     originalClose.click();
1022:     return;
1023:   }
1024:   closeDialogElement(dialog, "close");
1025: }

```

## ***handleDialogWindowAction - lines 1027-1044***

| Item               | Explanation                                                                                                                                                                                                                |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1027-1044 (18 source lines).                                                                                                                                                                                         |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                                        |
| Declaration        | function handleDialogWindowAction(button) {                                                                                                                                                                                |
| Touches            | browser/Electron DOM                                                                                                                                                                                                       |
| Calls detected     | closest, toggleDialogHidden, toggleDialogMinimized, toggleDialogMaximized, refreshDialogWindow, stepProjectWindowBack, projectPreviewBackStep, stepProjectWindowForward, projectPreviewForwardStep, closeDialogFromControl |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                   |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                    |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                                              |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                      |

Source excerpt:

```

1027: function handleDialogWindowAction(button) {
1028:   const dialog = button.closest("dialog");
1029:   if (!dialog?.open) return;
1030:   const action = button.dataset.dialogAction;
1031:   if (action === "hide") toggleDialogHidden(dialog);
1032:   if (action === "minimize") toggleDialogMinimized(dialog);
1033:   if (action === "maximize") toggleDialogMaximized(dialog);
1034:   if (action === "refresh") refreshDialogWindow(dialog);
1035:   if (action === "back") {
1036:     if (dialog === projectDialog) stepProjectWindowBack();
1037:     else if (dialog === projectPreviewDialog) projectPreviewBackStep();
1038:   }
1039:   if (action === "forward") {
1040:     if (dialog === projectDialog) stepProjectWindowForward();
1041:     else if (dialog === projectPreviewDialog) projectPreviewForwardStep();
1042:   }
1043:   if (action === "close") closeDialogFromControl(dialog, button);
1044: }

```

## ***enableDraggableDialogs - lines 1046-1084***

| Item            | Explanation                                                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 1046-1084 (39 source lines).                                                                                                                                                |
| Purpose         | Part of the private builder frontend and editing experience.                                                                                                                      |
| Declaration     | function enableDraggableDialogs() {                                                                                                                                               |
| Touches         | browser/Electron DOM, filesystem                                                                                                                                                  |
| Calls detected  | querySelectorAll, forEach, addEventListener, closest, handleDialogWindowAction, toggleDialogHidden, remove, updateDialogWindowButtons, getBoundingClientRect, clampDialogPosition |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1046: function enableDraggableDialogs() {
1047:   document.querySelectorAll("dialog").forEach(markDialogDragHandles);
1048:   document.addEventListener("pointerdown", beginDialogResize);
1049:   document.addEventListener("pointerdown", beginDialogDrag);
1050:   document.addEventListener("pointermove", moveDialogResize);
1051:   document.addEventListener("pointermove", moveDialogDrag);
1052:   document.addEventListener("pointerup", endDialogResize);
1053:   document.addEventListener("pointerup", endDialogDrag);
1054:   document.addEventListener("pointercancel", endDialogResize);
1055:   document.addEventListener("pointercancel", endDialogDrag);
1056:   document.addEventListener("click", (event) => {
1057:     const button = event.target.closest("[data-dialog-action]");
1058:     if (!button) return;
1059:     handleDialogWindowAction(button);
1060:   });
1061:   document.addEventListener("click", (event) => {
1062:     const hiddenDialog = event.target.closest("dialog.is-hidden-dialog");
1063:     if (!hiddenDialog || event.target.closest("button")) return;
1064:     toggleDialogHidden(hiddenDialog);
1065:   });
1066:   document.querySelectorAll("dialog").forEach((dialog) => {
1067:     dialog.addEventListener("close", () => {
1068:       dialog.classList.remove("is-minimized-dialog");
1069:       dialog.classList.remove("is-hidden-dialog");
1070:       dialog.classList.remove("is-maximized-dialog");
1071:       dialog.classList.remove("is-resizing-dialog");
1072:       delete dialog.dataset.restoreWindowState;
1073:       updateDialogWindowButtons(dialog);
1074:     });
1075:   });
1076:   window.addEventListener("resize", () => {
1077:     document.querySelectorAll("dialog.is-draggable-dialog[open]").forEach((dialog) => {
1078:       const rect = dialog.getBoundingClientRect();
1079:       const next = clampDialogPosition(rect.left, rect.top, dialog);
1080:       dialog.style.left = `${next.left}px`;
1081:       dialog.style.top = `${next.top}px`;
1082:     });
1083:   });
1084: }

```

## ***markDraftNeedsSave - lines 1086-1089***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1086-1089 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function markDraftNeedsSave() {                                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | setSaveState                                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1086: function markDraftNeedsSave() {
1087:   draftSavedSinceChanges = false;
1088:   setSaveState("dirty", "Save draft needed");
1089: }

```

## ***showBuilderError - lines 1091-1097***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1091-1097 (7 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function showBuilderError(title, message, details = "") {                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | showModal                                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1091: function showBuilderError(title, message, details = "") {
1092:   publishResultEyebrow.textContent = "Builder check";
1093:   publishResultTitle.textContent = title;
1094:   publishResultMessage.textContent = message;
1095:   publishResultOutput.textContent = details;
1096:   publishResultDialog.showModal();
1097: }

```

## ***appUpdateWasSnoozed - lines 1105-1110***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1105-1110 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function appUpdateWasSnoozed(version = "") {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getItem, now                                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | omb-snoozed-update-at, omb-snoozed-update-version                                                                                                                                                     |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1105: function appUpdateWasSnoozed(version = "") {
1106:   if (!version) return false;
1107:   const snoozedVersion = localStorage.getItem("omb-snoozed-update-version") || "";
1108:   const snoozedAt = Number(localStorage.getItem("omb-snoozed-update-at") || 0);
1109:   return snoozedVersion === version && Date.now() - snoozedAt < APP_UPDATE_SNOOZE_MS;
1110: }

```

## ***shouldShowUpdateDialog - lines 1112-1117***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 1112-1117 (6 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function shouldShowUpdateDialog(update = {}) {               |
| Touches         | Mostly local calculations or local UI state.                 |
| Calls detected  | getItem, appUpdateWasSnoozed                                 |
| API endpoints   | None detected inside this function body.                     |
| Storage keys    | omb-dismissed-update-version, omb-skipped-update-version     |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1112: function shouldShowUpdateDialog(update = {}) {
1113:   if (!update.updateAvailable || !update.latestVersion) return false;
1114:   const skippedVersion = localStorage.getItem("omb-skipped-update-version") || localStorage.getItem("omb-
dismissed-update-version") || "";
1115:   if (skippedVersion === update.latestVersion) return false;
1116:   return !appUpdateWasSnoozed(update.latestVersion);
1117: }

```

## ***showUpdateDialog - lines 1119-1158***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1119-1158 (40 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function showUpdateDialog(update = {}, options = {}) {                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | shouldShowUpdateDialog, escapeHtml, filter, join, showModal, show                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1119: function showUpdateDialog(update = {}, options = {}) {
1120:   const force = Boolean(options.force);
1121:   if (!appUpdateDialog || (!force && !shouldShowUpdateDialog(update))) return;
1122:   pendingAppUpdate = update;
1123:   const hasInstaller = Boolean(update.installerUrl);
1124:   const updateBlocked = Boolean(update.updateBlocked);
1125:   const hasUpdate = Boolean(update.updateAvailable && update.latestVersion);
1126:   appUpdateTitle.textContent = updateBlocked ? "Update paused" : hasUpdate ? "Update available" :
"Builder is up to date";
1127:   appUpdateMessage.textContent = updateBlocked
1128:     ? `OMB Portfolio Builder ${update.latestVersion || "this release"} is available, but this version is
paused for automatic updates.`
1129:     : hasUpdate
1130:     ? `OMB Portfolio Builder ${update.latestVersion} is available. You are running
${update.currentVersion || "an older version"}.`
1131:     : `You are running OMB Portfolio Builder ${update.currentVersion || "the current installed
version"}.`;
1132:   appUpdateDetails.textContent = updateBlocked
1133:     ? (update.blockedReason || "This release is temporarily blocked from automatic updates. Check again
after the next release is available.")
1134:     : hasUpdate
1135:     ? "Choose Update to download the latest installer, close this app, and install the new version
automatically. Remind me later postpones the prompt; Skip this version ignores this release."
1136:     : "No newer released builder was found on GitHub Releases.";
1137:   appUpdateReleaseMeta.innerHTML = (hasUpdate || updateBlocked)
1138:     ? [
1139:       `<span>Current version: ${escapeHtml(update.currentVersion || "unknown")}</span>`,
1140:       `<span>Available version: ${escapeHtml(update.latestVersion || "unknown")}</span>`,
1141:       `update.releaseUrl ? `<span>Release source: GitHub Releases</span>` : ""
1142:     ].filter(Boolean).join("")
1143:     : [
1144:       `<span>Current version: ${escapeHtml(update.currentVersion || "unknown")}</span>`,
1145:       `update.checkedAt ? `<span>Checked: ${escapeHtml(update.checkedAt)}</span>` : ""
1146:     ].filter(Boolean).join("");
1147:   appUpdateLater.textContent = hasUpdate ? "Remind me later" : "Close";
1148:   appUpdateSkip.hidden = !hasUpdate;
1149:   appUpdateApply.hidden = !hasUpdate;
1150:   appUpdateApply.disabled = hasUpdate && !hasInstaller;
1151:   appUpdateApply.textContent = hasInstaller ? "Update" : "Open release page";
1152:   if (appUpdateDialog.open) return;
1153:   try {
1154:     appUpdateDialog.showModal();
1155:   } catch {
1156:     appUpdateDialog.show();

```

```

1157:   }
1158: }

```

## startAppUpdateInstall - lines 1160-1193

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1160-1193 (34 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function startAppUpdateInstall() {                                                                                                                                                              |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | fetch, now, stringify, json, Error, showBuilderError                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1160: async function startAppUpdateInstall() {
1161:   const update = pendingAppUpdate || {};
1162:   appUpdateApply.disabled = true;
1163:   appUpdateLater.disabled = true;
1164:   appUpdateSkip.disabled = true;
1165:   appUpdateApply.textContent = "Updating...";
1166:   appUpdateDetails.textContent = "Downloading the installer. The builder will close automatically, then
the installer will update the app.";
1167:   try {
1168:     const response = await fetch(`/api/app-update/install?t=${Date.now()}`, {
1169:       method: "POST",
1170:       cache: "no-store",
1171:       headers: { "Content-Type": "application/json" },
1172:       body: JSON.stringify({ latestVersion: update.latestVersion || "" })
1173:     });
1174:     const result = await response.json();
1175:     if (!response.ok || !result.ok) throw new Error(result.error || "The update could not be started.");
1176:     appUpdateMessage.textContent = "Update started";
1177:     appUpdateDetails.textContent = "The installer is ready. This window will close so the new version can
be installed.";
1178:   } catch (error) {
1179:     appUpdateApply.disabled = false;
1180:     appUpdateLater.disabled = false;
1181:     appUpdateSkip.disabled = false;
1182:     appUpdateApply.textContent = "Update";
1183:     appUpdateDetails.textContent = error.message || "The update could not be started.";
1184:     const fallback = update.releaseUrl || update.installerUrl || update.portableUrl || "";
1185:     if (fallback) {
1186:       showBuilderError(
1187:         "Automatic update could not start",
1188:         "The app could not run the installer automatically. The release page can still be opened
manually.",
1189:         fallback
1190:       );
1191:     }
1192:   }
1193: }

```

## checkForAppUpdates - lines 1195-1235

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 1195-1235 (41 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | async function checkForAppUpdates(options = {}) {            |
| Touches         | network/API requests                                         |
| Calls detected  | now, fetch, json, showUpdateDialog, toLocaleString           |
| API endpoints   | None detected inside this function body.                     |
| Storage keys    | No local/session storage keys detected.                      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1195: async function checkForAppUpdates(options = {}) {
1196:   const force = Boolean(options.force);
1197:   const manual = Boolean(options.manual);
1198:   if (!force && Date.now() - lastAppUpdateCheckAt < 5 * 60 * 1000) return;
1199:   lastAppUpdateCheckAt = Date.now();
1200:   if (manual) {
1201:     if (appUpdateCheckButton) {
1202:       appUpdateCheckButton.disabled = true;
1203:       appUpdateCheckButton.textContent = "Checking...";
1204:     }
1205:   }
1206:   try {
1207:     const response = await fetch(`/api/app-update?t=${Date.now()}`, { cache: "no-store" });
1208:     const update = await response.json();
1209:     if (response.ok && update.ok !== false) {
1210:       showUpdateDialog({ ...update, checkedAt: new Date().toLocaleString() }, { force: manual });
1211:     } else if (manual) {
1212:       showUpdateDialog({
1213:         currentVersion: update.currentVersion || "unknown",
1214:         updateAvailable: false,
1215:         checkedAt: new Date().toLocaleString()
1216:       }, { force: true });
1217:     }
1218:   } catch {
1219:     if (manual) {
1220:       showUpdateDialog({
1221:         currentVersion: "unknown",
1222:         updateAvailable: false,
1223:         checkedAt: new Date().toLocaleString()
1224:       }, { force: true });
1225:     }
1226:     // Update checks should never interrupt builder work.
1227:   } finally {
1228:     if (manual) {
1229:       if (appUpdateCheckButton) {
1230:         appUpdateCheckButton.disabled = false;
1231:         appUpdateCheckButton.textContent = "Check updates";
1232:       }
1233:     }
1234:   }
1235: }

```

### ***formatReportDate - lines 1237-1240***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1237-1240 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function formatReportDate(value = "") {                                                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isNaN, getTime, toLocaleString                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1237: function formatReportDate(value = "") {
1238:   const date = new Date(value);
1239:   return Number.isNaN(date.getTime()) ? "" : date.toLocaleString();
1240: }

```

### ***renderSecurityReport - lines 1242-1294***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1242-1294 (53 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderSecurityReport(report = {}) {                                                                                                                                                          |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | isArray, map, escapeHtml, toLocaleString, join, formatReportDate                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

#### Source excerpt:

```

1242: function renderSecurityReport(report = {}) {
1243:   if (!securityReportBody) return;
1244:   const downloads = report.appDownloads || {};
1245:   const assets = Array.isArray(downloads.assets) ? downloads.assets : [];
1246:   const auth = report.publishingAuth || {};
1247:   const access = report.websiteAccess || {};
1248:   const controls = Array.isArray(report.securityControls) ? report.securityControls : [];
1249:   const assetRows = assets.length
1250:     ? assets.map((asset) => `
1251:       <li>
1252:         <span>${escapeHtml(asset.name || "Release asset")}</span>
1253:         <span>${Number(asset.downloadCount || 0).toLocaleString()} downloads</span>
1254:       </li>
1255:     `).join("")
1256:   : `<li><span>No release assets found</span><span>0 downloads</span></li>`;
1257:   securityReportBody.innerHTML = `
1258:     <div class="security-report-grid">
1259:       <section class="security-report-card">
1260:         <h3>Builder downloads</h3>
1261:         <strong>${(downloads.ok ? Number(downloads.totalDownloads || 0).toLocaleString() :
"Unavailable")}</strong>
1262:         <p>${downloads.ok
? `Latest release: ${escapeHtml(downloads.releaseTag || downloads.releaseName || "latest")}`
: escapeHtml(downloads.error || "GitHub release counts are not available.")}</p>
1265:         <ul class="security-report-assets">${assetRows}</ul>
1266:       </section>
1267:       <section class="security-report-card">
1268:         <h3>Publishing authorization</h3>
1269:         <strong>${(auth.authenticated ? "Active" : "Not active")}</strong>
1270:         <p>${auth.authenticated
? `This builder has a remembered GitHub publishing session until
${escapeHtml(formatReportDate(auth.expiresAt) || auth.expiresAt || "the cache expires")}`
: "No fresh GitHub publishing session is cached for the current target."}</p>
1273:         <ul>
1274:           <li>Target: ${escapeHtml(auth.targetRepository || "No publishing target saved")}</li>
1275:           <li>Branch: ${escapeHtml(auth.branch || "Not set")}</li>
1276:           <li>Successful authorizations this week: ${Number(auth.successCountLastWeek ||
0).toLocaleString()}</li>
1277:         </ul>
1278:       </section>
1279:       <section class="security-report-card">
1280:         <h3>Website access reporting</h3>
1281:         <strong>${(access.available ? "Configured" : "Needs Cloudflare")}</strong>
1282:         <p>${escapeHtml(access.summary || "Visitor and IP reporting needs a server-side analytics
source.")}</p>
1283:         <ul>${(access.recommendedSetup || []).map((item) =>
`<li>${escapeHtml(item)}</li>`).join("")}</ul>
1284:       </section>
1285:       <section class="security-report-card">
1286:         <h3>Security controls</h3>
1287:         <strong>${(controls.length.toLocaleString())}</strong>
1288:         <p>Controls currently enabled or prepared for publishing.</p>
1289:         <ul>${(controls.map((item) => `<li>${escapeHtml(item)}</li>`).join("")}</ul>
1290:       </section>
1291:     </div>
1292:     <p>Generated ${escapeHtml(formatReportDate(report.generatedAt) || "just now")}</p>
1293:   `;
1294: }

```

#### ***openSecurityReport - lines 1296-1314***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1296-1314 (19 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | async function openSecurityReport() {                                                                                                                                                                 |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | showModal, updateDialogWindowButtons, fetch, now, json, Error, renderSecurityReport, escapeHtml                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1296: async function openSecurityReport() {
1297:   if (!securityReportDialog || !securityReportBody) return;
1298:   securityReportBody.innerHTML = "<p>Loading security report...</p>";
1299:   if (!securityReportDialog.open) securityReportDialog.showModal();
1300:   updateDialogWindowButtons(securityReportDialog);
1301:   try {
1302:     const response = await fetch(`/api/security-report?t=${Date.now()}`, { cache: "no-store" });
1303:     const report = await response.json();
1304:     if (!response.ok || report.ok === false) throw new Error(report.error || "Security report could not
be loaded.");
1305:     renderSecurityReport(report);
1306:   } catch (error) {
1307:     securityReportBody.innerHTML = `
1308:       <div class="security-report-card">
1309:         <h3>Security report unavailable</h3>
1310:         <p>${escapeHtml(error.message || "The report could not be loaded.")}</p>
1311:       </div>
1312:     `;
1313:   }
1314: }

```

## ***scheduleAppUpdateChecks - lines 1316-1325***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1316-1325 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function scheduleAppUpdateChecks() {                                                                                                                                                                  |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | checkForAppUpdates, addEventListener, now                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1316: function scheduleAppUpdateChecks() {
1317:   checkForAppUpdates({ force: true });
1318:   window.setInterval(() => checkForAppUpdates({ force: true }), APP_UPDATE_CHECK_INTERVAL_MS);
1319:   document.addEventListener("visibilitychange", () => {
1320:     if (document.visibilityState !== "visible") return;
1321:     if (Date.now() - lastAppUpdateCheckAt >= APP_UPDATE_FOCUS_RECHECK_MS) {
1322:       checkForAppUpdates({ force: true });
1323:     }
1324:   });
1325: }

```

## ***renderPublishTargetInfo - lines 1327-1353***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1327-1353 (27 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPublishTargetInfo(target = {}) {                                                                                                                                                       |
| Touches            | browser/Electron DOM, authentication/security data                                                                                                                                                    |
| Calls detected     | toLocaleString, map, escapeHtml, join                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1327: function renderPublishTargetInfo(target = {}) {
1328:   if (!publishTargetCurrent) return;
1329:   currentPublishTarget = target;
1330:   const checkedText = target.checkedAt ? ` at ${new Date(target.checkedAt).toLocaleString()}` : "";
1331:   const expiresText = target.expiresAt ? `; remembered until ${new
Date(target.expiresAt).toLocaleString()}` : "";
1332:   const trustText = target.extendedTrust ? " (30-day trusted target)" : target.authorizationCached ? "
(daily cache)" : "";
1333:   const authorizationStatus = target.authorizationChecked
1334:     ? `Verified${checkedText}${trustText}${expiresText}`
1335:     : "Not verified for this repository and branch";
1336:   const rows = [
1337:     ["Builder workspace", target.workspace || "Not available"],
1338:     ["Portfolio workspace", target.portfolioWorkspace || target.workspace || "Not available"],
1339:     ["Repository", target.repository || target.remote || "No repository connected"],
1340:     ["Branch", target.branch || "No branch selected"],
1341:     ["GitHub authorization", authorizationStatus],
1342:     ["Compatible site", target.compatible ? "Yes" : "No"],
1343:     ["Git-backed", target.gitBacked ? "Yes" : "No"]
1344:   ];
1345:   publishTargetCurrent.innerHTML = rows
1346:     .map(([label, value]) =>
`<div><strong>${escapeHtml(label)}</strong><span>${escapeHtml(value)}</span></div>`)
1347:     .join("");
1348:   if (publishTargetRepository) {
1349:     publishTargetRepository.value = target.remote || "";
1350:     publishTargetRepository.dataset.loadedValue = target.remote || "";
1351:   }
1352:   if (publishTargetSync) publishTargetSync.disabled = !target.authorizationChecked;
1353: }

```

## ***renderSystemReadiness - lines 1355-1394***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1355-1394 (40 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderSystemReadiness(system = {}, target = {}) {                                                                                                                                            |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | toLocaleString, map, escapeHtml, join                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1355: function renderSystemReadiness(system = {}, target = {}) {
1356:   if (!publishTargetReadinessList) return;
1357:   const rows = [
1358:     {
1359:       ok: true,
1360:       text: `Electron/Node runtime: bundled with the app${system.nodeRuntime?.version ? `

```

```

({$system.nodeRuntime.version})` : "").`
1361:   },
1362:   {
1363:     ok: system.pnpm?.required === false,
1364:     text: "pnpm: not required for installed users."
1365:   },
1366:   {
1367:     ok: Boolean(system.git?.ok),
1368:     text: system.git?.ok
1369:       ? `Git for Windows: ${system.git.version || "installed"}`.
1370:       : "Git for Windows: missing. Install it before loading or publishing a target."
1371:   },
1372:   {
1373:     ok: Boolean(system.credentialManager?.ok),
1374:     text: system.credentialManager?.ok
1375:       ? `Git Credential Manager: ${system.credentialManager.version || "installed"}`.
1376:       : "Git Credential Manager: missing. Install Git for Windows with Git Credential Manager enabled."
1377:   },
1378:   {
1379:     ok: Boolean(target?.remote),
1380:     text: target?.remote
1381:       ? `Publishing target: ${target.repository || target.remote}`.
1382:       : "Publishing target: enter a repository URL and click Authenticate target."
1383:   },
1384:   {
1385:     ok: Boolean(target?.authorizationChecked),
1386:     text: target?.authorizationChecked
1387:       ? `GitHub authorization: verified for ${target.branch || "the selected branch"}${target.expiresAt
1388:       ? ` until ${new Date(target.expiresAt).toLocaleString()}` : ""}`.
1389:       : "GitHub authorization: required before Load from target or Apply to site."
1390:   }
1391: ];
1392: publishTargetReadinessList.innerHTML = rows
1393:   .map((row) => `- 

```

## markPublishTargetNeedsAuthentication - lines 1396-1405

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1396-1405 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function markPublishTargetNeedsAuthentication() {                                                                                                                                                     |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1396: function markPublishTargetNeedsAuthentication() {
1397:   if (publishTargetSync) publishTargetSync.disabled = true;
1398:   if (currentPublishTarget) {
1399:     currentPublishTarget = {
1400:       ...currentPublishTarget,
1401:       authorizationChecked: false,
1402:       authorizationCached: false
1403:     };
1404:   }
1405: }

```

## loadSystemReadiness - lines 1407-1418

| Item            | Explanation                                                   |
|-----------------|---------------------------------------------------------------|
| Source location | Lines 1407-1418 (12 source lines).                            |
| Purpose         | Loads data from disk, network, browser storage, or a service. |
| Declaration     | async function loadSystemReadiness() {                        |
| Touches         | network/API requests                                          |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | fetch, now, json, Error, renderSystemReadiness, escapeHtml                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1407: async function loadSystemReadiness() {
1408:   if (!publishTargetReadinessList) return;
1409:   publishTargetReadinessList.innerHTML = "<li>Checking local publishing tools...</li>";
1410:   try {
1411:     const response = await fetch(`/api/system-check?t=${Date.now()}`, { cache: "no-store" });
1412:     const result = await response.json();
1413:     if (!response.ok || !result.ok) throw new Error(result.error || "System check failed.");
1414:     renderSystemReadiness(result.system || {}, result.target || currentPublishTarget || {});
1415:   } catch (error) {
1416:     publishTargetReadinessList.innerHTML = `<li class="blocked">${escapeHtml(error.message || "System
check failed.")}</li>`;
1417:   }
1418: }

```

## ***setPublishTargetProgress - lines 1420-1426***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1420-1426 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function setPublishTargetProgress(title, steps = []) {                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | escapeHtml, toLocaleTimeString, map, join                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1420: function setPublishTargetProgress(title, steps = []) {
1421:   if (!publishTargetCurrent) return;
1422:   publishTargetCurrent.innerHTML = `
1423:     <div><strong>${escapeHtml(title)}</strong><span>${escapeHtml(new
Date().toLocaleTimeString())}</span></div>
1424:     ${steps.map((step) =>
`<div><strong>${escapeHtml(step.label)}</strong><span>${escapeHtml(step.value)}</span></div>`).join("")}
1425:   `;
1426: }

```

## ***loadPublishTargetInfo - lines 1428-1439***

| Item            | Explanation                                                   |
|-----------------|---------------------------------------------------------------|
| Source location | Lines 1428-1439 (12 source lines).                            |
| Purpose         | Loads data from disk, network, browser storage, or a service. |
| Declaration     | async function loadPublishTargetInfo() {                      |
| Touches         | network/API requests                                          |
| Calls detected  | fetch, now, json, Error, renderPublishTargetInfo              |
| API endpoints   | None detected inside this function body.                      |
| Storage keys    | No local/session storage keys detected.                       |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1428: async function loadPublishTargetInfo() {
1429:   if (!publishTargetDialog) return;
1430:   publishTargetCurrent.textContent = "Current target is loading.";
1431:   try {
1432:     const response = await fetch(`/api/publish-target?t=${Date.now()}`, { cache: "no-store" });
1433:     const result = await response.json();
1434:     if (!response.ok || !result.ok) throw new Error(result.error || "Publishing target could not be
loaded.");
1435:     renderPublishTargetInfo(result.target || {});
1436:   } catch (error) {
1437:     publishTargetCurrent.textContent = error.message || "Publishing target could not be loaded.";
1438:   }
1439: }

```

## ***openPublishTargetDialog - lines 1441-1445***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1441-1445 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function openPublishTargetDialog() {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | loadPublishTargetInfo, loadSystemReadiness, showModal                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1441: async function openPublishTargetDialog() {
1442:   await loadPublishTargetInfo();
1443:   await loadSystemReadiness();
1444:   publishTargetDialog.showModal();
1445: }

```

## ***savePublishTarget - lines 1447-1450***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1447-1450 (4 source lines).                                                                                                                                                                     |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | async function savePublishTarget(event) {                                                                                                                                                             |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | preventDefault, authenticatePublishTarget                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1447: async function savePublishTarget(event) {
1448:   event.preventDefault();
1449:   await authenticatePublishTarget({ fromSave: true });

```

```
1450: }
```

### ***currentPublishTargetPayload - lines 1452-1456***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1452-1456 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | function currentPublishTargetPayload() {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1452: function currentPublishTargetPayload() {
1453:   return {
1454:     repositoryUrl: publishTargetRepository?.value || ""
1455:   };
1456: }
```

### ***installGitForPublishing - lines 1458-1481***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1458-1481 (24 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function installGitForPublishing() {                                                                                                                                                            |
| Touches            | network/API requests, authentication/security data                                                                                                                                                    |
| Calls detected     | fetch, now, json, Error, open, loadSystemReadiness, showBuilderError                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1458: async function installGitForPublishing() {
1459:   publishTargetCurrent.textContent = "Starting Git for Windows setup...";
1460:   try {
1461:     const response = await fetch(`/api/install-git?t=${Date.now()}`, {
1462:       method: "POST",
1463:       cache: "no-store",
1464:       headers: { "Content-Type": "application/json" },
1465:       body: "{}"
1466:     });
1467:     const result = await response.json();
1468:     if (!response.ok || !result.ok) throw new Error(result.error || "Git installer could not be
started.");
1469:     if (!result.install?.launched && result.install?.downloadUrl) {
1470:       window.open(result.install.downloadUrl, "_blank", "noreferrer");
1471:     }
1472:     await loadSystemReadiness();
1473:     showBuilderError(
1474:       result.install?.launched ? "Git setup started" : "Open Git download",
1475:       result.install?.message || "Install Git for Windows with Git Credential Manager enabled, then
reopen the builder.",
1476:       result.install?.downloadUrl || ""
1477:     );
1478:   } catch (error) {
1479:     publishTargetCurrent.textContent = error.message || "Git installer could not be started.";
1480:   }
1481: }
```

## authenticatePublishTarget - lines 1483-1536

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1483-1536 (54 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function authenticatePublishTarget(options = {}) {                                                                                                                                              |
| Touches            | network/API requests, authentication/security data                                                                                                                                                    |
| Calls detected     | setPublishTargetProgress, fetch, now, stringify, currentPublishTargetPayload, json, Error, filter, join, renderPublishTargetInfo, renderSystemReadiness, showBuilderError                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1483: async function authenticatePublishTarget(options = {}) {
1484:   const fromSave = Boolean(options.fromSave);
1485:   if (publishTargetSync) publishTargetSync.disabled = true;
1486:   setPublishTargetProgress(
1487:     fromSave ? "Authenticating publishing target" : "Checking GitHub authorization",
1488:     [
1489:       { label: "Step 1", value: "Validating the repository URL and local Git tools." },
1490:       { label: "Step 2", value: "Using the saved daily authorization if it is still valid." },
1491:       { label: "Step 3", value: "Finding the target repository default branch." },
1492:       { label: "Step 4", value: "Opening GitHub sign-in only when a fresh authorization is required." }
1493:     ]
1494:   );
1495:   try {
1496:     const response = await fetch(`/api/github-authenticate?t=${Date.now()}`, {
1497:       method: "POST",
1498:       cache: "no-store",
1499:       headers: { "Content-Type": "application/json" },
1500:       body: JSON.stringify(currentPublishTargetPayload())
1501:     });
1502:     const result = await response.json();
1503:     if (!response.ok || !result.ok) {
1504:       const message = result.error || "GitHub authentication did not complete.";
1505:       throw new Error([message, result.details].filter(Boolean).join("\n\n"));
1506:     }
1507:     renderPublishTargetInfo(result.target || {});
1508:     renderSystemReadiness(result.auth?.system || {}, result.target || {});
1509:     if (publishTargetPassword) publishTargetPassword.value = "";
1510:     const usedCachedAuth = Boolean(result.auth?.authorizationCached ||
result.target?.authorizationCached);
1511:     setPublishTargetProgress(
1512:       usedCachedAuth ? "Target authenticated from saved authorization" : "Target authenticated",
1513:       [
1514:         { label: "Repository", value: result.target?.repository || result.auth?.repository || "Selected
target" },
1515:         { label: "Branch", value: result.target?.branch || result.auth?.branch || "Detected target
branch" },
1516:         { label: "Authorization", value: usedCachedAuth ? "A valid saved authorization was reused; no new
GitHub sign-in was needed." : "GitHub write access was verified now." },
1517:         { label: "Next step", value: "Click Load from target when you want to import the latest
compatible website files into this builder." }
1518:       ]
1519:     );
1520:     if (publishTargetSync) publishTargetSync.disabled = false;
1521:   } catch (error) {
1522:     if (publishTargetSync) publishTargetSync.disabled = true;
1523:     setPublishTargetProgress(
1524:       "Target was not saved",
1525:       [
1526:         { label: "Reason", value: error.message || "GitHub authentication did not complete." },
1527:         { label: "Status", value: "The previous publishing target was kept. Fix the repository or sign-
in, then try Authenticate target again." }
1528:       ]
1529:     );
1530:     showBuilderError(
1531:       "Target authentication failed",
1532:       "The builder did not save this publishing target because GitHub write access was not verified.",
1533:       error.message || "GitHub authentication did not complete."
1534:     );
1535:   }
1536: }
```

## syncFromPublishTarget - lines 1538-1584

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1538-1584 (47 source lines).                                                                                                                                                                    |
| Purpose            | Part of publishing, Git, target repository, or authorization behavior.                                                                                                                                |
| Declaration        | async function syncFromPublishTarget(options = {}) {                                                                                                                                                  |
| Touches            | network/API requests, authentication/security data                                                                                                                                                    |
| Calls detected     | fetch, now, json, Error, filter, join, renderPublishTargetInfo, closeDialogElement, loadData, showBuilderError                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1538: async function syncFromPublishTarget(options = {}) {
1539:   if (!publishTargetDialog) return;
1540:   const afterAuth = Boolean(options.afterAuth);
1541:   publishTargetCurrent.textContent = afterAuth
1542:     ? "GitHub authentication was accepted. Importing the target portfolio now..."
1543:     : "Checking GitHub authorization and loading compatible portfolio files from the target
repository...";
1544:   try {
1545:     const response = await fetch(`/api/sync-from-publish-target?t=${Date.now()}`, {
1546:       method: "POST",
1547:       cache: "no-store",
1548:       headers: { "Content-Type": "application/json" },
1549:       body: "{}"
1550:     });
1551:     const result = await response.json();
1552:     if (!response.ok || !result.ok) {
1553:       throw new Error([
1554:         result.error || "Publishing target could not be imported.",
1555:         result.details || "Click Authenticate target, complete the browser sign-in if prompted, then try
Load from target again."
1556:       ].filter(Boolean).join("\n\n"));
1557:     }
1558:     renderPublishTargetInfo(result.target || {});
1559:     closeDialogElement(publishTargetDialog);
1560:     await loadData();
1561:     draftSavedSinceChanges = true;
1562:     showBuilderError(
1563:       afterAuth ? "GitHub connected and target loaded" : "Target content loaded",
1564:       afterAuth
1565:         ? "Authentication succeeded, and compatible portfolio files were loaded into this local builder
workspace. The local draft now matches the imported target catalog."
1566:         : "Compatible portfolio files were loaded from the authenticated publishing target into this
local builder workspace. The local draft now matches the imported target catalog.",
1567:       [
1568:         `Imported: ${((result.sync?.imported || []).join(", "))}`,
1569:         result.sync?.backup ? `Backup: ${result.sync.backup}` : ""
1570:       ].filter(Boolean).join("\n")
1571:     );
1572:   } catch (error) {
1573:     publishTargetCurrent.textContent = afterAuth
1574:       ? `Authentication completed, but target loading failed. ${error.message || "Publishing target could
not be imported."}`
1575:       : error.message || "Publishing target could not be imported.";
1576:     if (afterAuth) {
1577:       showBuilderError(
1578:         "Target saved, loading failed",
1579:         "GitHub authorization succeeded, but the builder could not import compatible portfolio files from
that target.",
1580:         error.message || "Publishing target could not be imported."
1581:       );
1582:     }
1583:   }
1584: }
```

## showPublishResult - lines 1586-1619

| Item            | Explanation                                                            |
|-----------------|------------------------------------------------------------------------|
| Source location | Lines 1586-1619 (34 source lines).                                     |
| Purpose         | Part of publishing, Git, target repository, or authorization behavior. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function showPublishResult(result) {                                                                                                                                                                  |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | filter, join, showModal                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1586: function showPublishResult(result) {
1587:   const publish = result.publish || {};
1588:   const pushed = Boolean(publish.pushed);
1589:   const authorizationRequired = Boolean(publish.authorizationRequired);
1590:   publishResultEyebrow.textContent = "GitHub publish";
1591:   publishResultTitle.textContent = pushed
1592:     ? "Successfully pushed to GitHub"
1593:     : authorizationRequired
1594:       ? "Publishing access required"
1595:       : "Site applied, push failed";
1596:   publishResultMessage.textContent = pushed
1597:     ? `Your portfolio changes were committed and pushed to ${publish.branch || "GitHub"}`
1598:     : authorizationRequired
1599:       ? "No live website files were applied. Open Publishing target, click Authenticate with GitHub,
complete the browser sign-in, then try Apply to site again."
1600:       : "The site was applied locally, but Git push did not complete.";
1601:
1602:   const output = [
1603:     pushed ? "PUSH STATUS: SUCCESS" : "PUSH STATUS: FAILED",
1604:     `FILE: ${result.file || "projects.json"}`,
1605:     authorizationRequired ? "AUTHORIZATION: REQUIRED BEFORE WEBSITE CHANGES" : "",
1606:     publish.repository ? `REPOSITORY: ${publish.repository}` : "",
1607:     publish.remote ? `REMOTE: ${publish.remote}` : "",
1608:     publish.branch ? `BRANCH: ${publish.branch}` : "",
1609:     publish.committed === false ? "COMMIT: No file changes to commit." : "",
1610:     publish.commitOutput ? `COMMIT OUTPUT:\n${publish.commitOutput}` : "",
1611:     publish.pushOutput ? `PUSH OUTPUT:\n${publish.pushOutput}` : "",
1612:     publish.error ? `ERROR:\n${publish.error}` : "",
1613:     publish.details ? `DETAILS:\n${publish.details}` : "",
1614:     publish.help ? `NEXT STEP:\n${publish.help}` : ""
1615:   ].filter(Boolean).join("\n\n");
1616:
1617:   publishResultOutput.textContent = output || "No Git output was returned.";
1618:   publishResultDialog.showModal();
1619: }

```

## ***scheduleAutosave - lines 1621-1627***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1621-1627 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function scheduleAutosave(delay = 900) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | markDraftNeedsSave, autosaveDraft                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1621: function scheduleAutosave(delay = 900) {
1622:   markDraftNeedsSave();
1623:   clearTimeout(autosaveTimer);
1624:   autosaveTimer = setTimeout(() => {
1625:     autosaveDraft();

```

```

1626:   }, delay);
1627: }

```

## autosaveDraft - lines 1629-1659

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1629-1659 (31 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function autosaveDraft() {                                                                                                                                                                      |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | setSaveState, fetch, stringify, json, setStatus, scheduleAutosave                                                                                                                                     |
| API endpoints      | /api/save-draft                                                                                                                                                                                       |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1629: async function autosaveDraft() {
1630:   if (autosaveInFlight) {
1631:     autosaveQueued = true;
1632:     return;
1633:   }
1634:   autosaveInFlight = true;
1635:   setSaveState("saving", "Autosaving...");
1636:   try {
1637:     const response = await fetch("/api/save-draft", {
1638:       method: "POST",
1639:       headers: { "Content-Type": "application/json" },
1640:       body: JSON.stringify({ catalog })
1641:     });
1642:     const result = await response.json();
1643:     if (!response.ok) {
1644:       setStatus(result.error || "Autosave failed.");
1645:       setSaveState("error", "Autosave failed");
1646:     } else if (!draftSavedSinceChanges) {
1647:       setSaveState("dirty", "Autosaved locally");
1648:     }
1649:   } catch {
1650:     setStatus("Autosave failed. Make sure the local server is running.");
1651:     setSaveState("error", "Autosave failed");
1652:   } finally {
1653:     autosaveInFlight = false;
1654:     if (autosaveQueued) {
1655:       autosaveQueued = false;
1656:       scheduleAutosave(300);
1657:     }
1658:   }
1659: }

```

## schedulePreviewRender - lines 1661-1667

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1661-1667 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function schedulePreviewRender() {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | renderPreview                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1661: function schedulePreviewRender() {
1662:   if (!portfolioPreviewDialog?.open) return;
1663:   clearTimeout(previewRenderTimer);
1664:   previewRenderTimer = setTimeout(() => {
1665:     renderPreview();
1666:   }, 220);
1667: }

```

## ***scheduleChromeRender - lines 1669-1676***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1669-1676 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function scheduleChromeRender() {                                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | renderTree, renderSectionTabs, selectedProject, renderPreview                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1669: function scheduleChromeRender() {
1670:   clearTimeout(chromeRenderTimer);
1671:   chromeRenderTimer = setTimeout(() => {
1672:     renderTree();
1673:     renderSectionTabs(selectedProject());
1674:     if (portfolioPreviewDialog?.open) renderPreview();
1675:   }, 260);
1676: }

```

## ***updateAssetDialogVisibility - lines 1678-1691***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1678-1691 (14 source lines).                                                                                                                                                                    |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateAssetDialogVisibility() {                                                                                                                                                              |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, trim, labelForUrlAssetKind, inferUrlAssetKind                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1678: function updateAssetDialogVisibility() {
1679:   const isLocal = assetSource.value === "local";
1680:   const isCaptionFile = captionSource.value === "file";
1681:   document.querySelector(".asset-local-field").hidden = !isLocal;
1682:   document.querySelector(".asset-url-field").hidden = isLocal;
1683:   document.querySelector(".caption-text-field").hidden = isCaptionFile;
1684:   document.querySelector(".caption-file-field").hidden = !isCaptionFile;
1685:   const url = assetUrl?.value?.trim() || "";
1686:   if (assetDetectedNote) {
1687:     assetDetectedNote.textContent = isLocal || !url
1688:       ? ""
1689:       : `Detected: ${labelForUrlAssetKind(inferUrlAssetKind(url, assetSource.value))}`;
1690:   }
1691: }

```

## ***slugify - lines 1693-1699***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1693-1699 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function slugify(value) {                                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | toLowerCase, trim, replace                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1693: function slugify(value) {
1694:   return String(value || "")
1695:     .toLowerCase()
1696:     .trim()
1697:     .replace(/^[a-z0-9]+/g, "-")
1698:     .replace(/^-+|-+$|/g, "");
1699: }
```

## ***wordCount - lines 1701-1703***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1701-1703 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function wordCount(value) {                                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, split, filter                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1701: function wordCount(value) {
1702:   return String(value || "").trim().split(/\s+/).filter(Boolean).length;
1703: }
```

## ***limitWords - lines 1705-1708***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 1705-1708 (4 source lines).                            |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function limitWords(value, maxWords = 1000) {                |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | trim, split, filter, slice, join                             |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 1 return statement(s).                |



| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1705: function limitWords(value, maxWords = 1000) {
1706:   const words = String(value || "").trim().split(/\s+/).filter(Boolean);
1707:   return words.length > maxWords ? words.slice(0, maxWords).join(" ") : value;
1708: }

```

## ***normalizeFunFacts - lines 1710-1716***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1710-1716 (7 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeFunFacts(value) {                                                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isArray, split, map, replace, trim, filter, slice                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1710: function normalizeFunFacts(value) {
1711:   const items = Array.isArray(value) ? value : String(value || "").split(/\r?\n/);
1712:   return items
1713:     .map((item) => String(item || "").replace(/\s+/g, " ").trim())
1714:     .filter(Boolean)
1715:     .slice(0, 3);
1716: }

```

## ***syncFunFactsFromInput - lines 1718-1726***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1718-1726 (9 source lines).                                                                                                                                                                     |
| Purpose            | Keeps two places aligned, such as local data and target files.                                                                                                                                        |
| Declaration        | function syncFunFactsFromInput() {                                                                                                                                                                    |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | extractRichSummary, normalizeFunFacts, plainTextFromRich                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1718: function syncFunFactsFromInput() {
1719:   if (!funFactsInput) return [];
1720:   const rich = extractRichSummary(funFactsInput);
1721:   const facts = normalizeFunFacts(plainTextFromRich(rich));
1722:   catalog.funFacts = facts;
1723:   catalog.funFactsRich = rich;
1724:   if (funFactsCount) funFactsCount.textContent = `${facts.length}/3`;
1725:   return facts;
1726: }

```

## ***renderFunFactsEditor - lines 1728-1734***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1728-1734 (7 source lines).                                                                                                                                                                     |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderFunFactsEditor() {                                                                                                                                                                     |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | normalizeFunFacts, populateStandaloneRichEditor, join                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1728: function renderFunFactsEditor() {
1729:   if (!funFactsInput) return;
1730:   const facts = normalizeFunFacts(catalog.funFacts || []);
1731:   if (funFactsCount) funFactsCount.textContent = `${facts.length}/3`;
1732:   if (document.activeElement === funFactsInput) return;
1733:   populateStandaloneRichEditor(funFactsInput, catalog.funFactsRich, facts.join("\n"));
1734: }

```

## ***populateTextOnlyRichField - lines 1736-1740***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1736-1740 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function populateTextOnlyRichField(field, rich, fallbackText = "") {                                                                                                                                  |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | populateStandaloneRichEditor                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1736: function populateTextOnlyRichField(field, rich, fallbackText = "") {
1737:   if (!field) return;
1738:   populateStandaloneRichEditor(field, rich, fallbackText);
1739:   field.dataset.richTextOnly = "true";
1740: }

```

## ***richFieldValue - lines 1742-1754***

| Item               | Explanation                                                          |
|--------------------|----------------------------------------------------------------------|
| Source location    | Lines 1742-1754 (13 source lines).                                   |
| Purpose            | Part of the private builder frontend and editing experience.         |
| Declaration        | function richFieldValue(field, fallbackText = "") {                  |
| Touches            | Mostly local calculations or local UI state.                         |
| Calls detected     | trim, textBlocksFromPlainText, extractRichSummary, plainTextFromRich |
| API endpoints      | None detected inside this function body.                             |
| Storage keys       | No local/session storage keys detected.                              |
| Async/return clues | 0 await expression(s), 2 return statement(s).                        |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1742: function richFieldValue(field, fallbackText = "") {
1743:   if (!field) {
1744:     return {
1745:       plain: String(fallbackText || "").trim(),
1746:       rich: { blocks: textBlocksFromPlainText(fallbackText) }
1747:     };
1748:   }
1749:   const rich = extractRichSummary(field);
1750:   return {
1751:     plain: plainTextFromRich(rich, "\n").trim(),
1752:     rich
1753:   };
1754: }

```

## ***mergeRichFieldMaps - lines 1756-1761***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1756-1761 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function mergeRichFieldMaps(current = {}, fallback = {}) {                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | clone                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1756: function mergeRichFieldMaps(current = {}, fallback = {}) {
1757:   return {
1758:     ...clone(fallback || {}),
1759:     ...clone(current || {})
1760:   };
1761: }

```

## ***parsedSiteContentRich - lines 1763-1765***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1763-1765 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedSiteContentRich() {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | mergeRichFieldMaps                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1763: function parsedSiteContentRich() {
1764:   return mergeRichFieldMaps(catalog.siteContentRich, savedPortfolioCatalog.siteContentRich);
1765: }

```

## ***parsedProfileRich - lines 1767-1769***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1767-1769 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedProfileRich() {                                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | mergeRichFieldMaps                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1767: function parsedProfileRich() {
1768:   return mergeRichFieldMaps(catalog.profileRich, savedPortfolioCatalog.profileRich);
1769: }
```

## ***parsedFunFactsRich - lines 1771-1775***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1771-1775 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedFunFactsRich() {                                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | call, clone                                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
1771: function parsedFunFactsRich() {
1772:   return Object.prototype.hasOwnProperty.call(catalog, "funFactsRich")
1773:     ? clone(catalog.funFactsRich || null)
1774:     : clone(savedPortfolioCatalog.funFactsRich || null);
1775: }
```

## ***syncPortfolioTextInputsForParsing - lines 1777-1781***

| Item               | Explanation                                                             |
|--------------------|-------------------------------------------------------------------------|
| Source location    | Lines 1777-1781 (5 source lines).                                       |
| Purpose            | Keeps two places aligned, such as local data and target files.          |
| Declaration        | function syncPortfolioTextInputsForParsing() {                          |
| Touches            | Mostly local calculations or local UI state.                            |
| Calls detected     | syncFunFactsFromInput, syncSiteContentFromInputs, syncProfileFromInputs |
| API endpoints      | None detected inside this function body.                                |
| Storage keys       | No local/session storage keys detected.                                 |
| Async/return clues | 0 await expression(s), 0 return statement(s).                           |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1777: function syncPortfolioTextInputsForParsing() {
1778:   if (funFactsInput) syncFunFactsFromInput();
1779:   if (siteHeroTitleInput || siteHeroCopyInput || siteHeroEyebrowInput) syncSiteContentFromInputs();
1780:   if (profileDisplayNameInput || profileEmailInput || profilePhoneInput) syncProfileFromInputs();
1781: }

```

## ***parsedPortfolioGlobals - lines 1783-1796***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1783-1796 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedPortfolioGlobals() {                                                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | syncPortfolioTextInputsForParsing, clone, normalizeFieldStyles, normalizeFunFacts, parsedFunFactsRich, normalizeProfile, parsedProfileRich, normalizeSiteContent, parsedSiteContentRich, filter, map  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1783: function parsedPortfolioGlobals() {
1784:   syncPortfolioTextInputsForParsing();
1785:   return {
1786:     categories: clone(catalog.categories || savedPortfolioCatalog.categories || []),
1787:     fieldStyles: clone(normalizeFieldStyles(catalog.fieldStyles || savedPortfolioCatalog.fieldStyles || {})),
1788:     funFacts: normalizeFunFacts(catalog.funFacts || savedPortfolioCatalog.funFacts || []),
1789:     funFactsRich: parsedFunFactsRich(),
1790:     profile: clone(normalizeProfile(catalog.profile || savedPortfolioCatalog.profile || {})),
1791:     profileRich: parsedProfileRich(),
1792:     siteContent: clone(normalizeSiteContent(catalog.siteContent || savedPortfolioCatalog.siteContent || {})),
1793:     siteContentRich: parsedSiteContentRich(),
1794:     siteSections: (catalog.siteSections || savedPortfolioCatalog.siteSections || []).filter(siteSectionRenderable).map(clone)
1795:   };
1796: }

```

## ***syncParsedPortfolioGlobalsIntoSaved - lines 1798-1800***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1798-1800 (3 source lines).                                                                                                                                                                     |
| Purpose            | Keeps two places aligned, such as local data and target files.                                                                                                                                        |
| Declaration        | function syncParsedPortfolioGlobalsIntoSaved() {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | assign, parsedPortfolioGlobals                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1798: function syncParsedPortfolioGlobalsIntoSaved() {
1799:   Object.assign(savedPortfolioCatalog, parsedPortfolioGlobals());
1800: }

```

## ***siteContentRichValue - lines 1802-1804***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1802-1804 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function siteContentRichValue(key) {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1802: function siteContentRichValue(key) {
1803:   return catalog.siteContentRich?.[key] || savedPortfolioCatalog.siteContentRich?.[key] || null;
1804: }

```

## ***profileRichValue - lines 1806-1808***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1806-1808 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function profileRichValue(key) {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1806: function profileRichValue(key) {
1807:   return catalog.profileRich?.[key] || savedPortfolioCatalog.profileRich?.[key] || null;
1808: }

```

## ***syncSiteRichField - lines 1810-1823***

| Item            | Explanation                                                                                       |
|-----------------|---------------------------------------------------------------------------------------------------|
| Source location | Lines 1810-1823 (14 source lines).                                                                |
| Purpose         | Keeps two places aligned, such as local data and target files.                                    |
| Declaration     | function syncSiteRichField(editor) {                                                              |
| Touches         | browser/Electron DOM, filesystem                                                                  |
| Calls detected  | richFieldValue, normalizeSiteContent, markDraftNeedsSave, scheduleAutosave, schedulePreviewRender |
| API endpoints   | None detected inside this function body.                                                          |
| Storage keys    | No local/session storage keys detected.                                                           |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1810: function syncSiteRichField(editor) {
1811:   const key = editor?.dataset.siteField;
1812:   if (!key) return;
1813:   catalog.siteContentRich = catalog.siteContentRich || {};
1814:   const { plain, rich } = richFieldValue(editor, catalog.siteContent?.[key] || "");
1815:   catalog.siteContent = normalizeSiteContent({
1816:     ...(catalog.siteContent || {}),
1817:     [key]: plain
1818:   });
1819:   catalog.siteContentRich[key] = rich;
1820:   markDraftNeedsSave();
1821:   scheduleAutosave();
1822:   schedulePreviewRender();
1823: }

```

## ***syncProfileRichField - lines 1825-1838***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1825-1838 (14 source lines).                                                                                                                                                                    |
| Purpose            | Keeps two places aligned, such as local data and target files.                                                                                                                                        |
| Declaration        | function syncProfileRichField(editor) {                                                                                                                                                               |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | richFieldValue, normalizeProfile, markDraftNeedsSave, scheduleAutosave, schedulePreviewRender                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1825: function syncProfileRichField(editor) {
1826:   const key = editor?.dataset.profileField;
1827:   if (!key) return;
1828:   catalog.profileRich = catalog.profileRich || {};
1829:   const { plain, rich } = richFieldValue(editor, catalog.profile?.[key] || "");
1830:   catalog.profile = normalizeProfile({
1831:     ...(catalog.profile || {}),
1832:     [key]: plain
1833:   });
1834:   catalog.profileRich[key] = rich;
1835:   markDraftNeedsSave();
1836:   scheduleAutosave();
1837:   schedulePreviewRender();
1838: }

```

## ***renderSiteContentEditor - lines 1840-1851***

| Item               | Explanation                                                           |
|--------------------|-----------------------------------------------------------------------|
| Source location    | Lines 1840-1851 (12 source lines).                                    |
| Purpose            | Renders UI, markup, or display output.                                |
| Declaration        | function renderSiteContentEditor() {                                  |
| Touches            | browser/Electron DOM, filesystem                                      |
| Calls detected     | normalizeSiteContent, populateTextOnlyRichField, siteContentRichValue |
| API endpoints      | None detected inside this function body.                              |
| Storage keys       | No local/session storage keys detected.                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                         |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1840: function renderSiteContentEditor() {
1841:   const content = normalizeSiteContent(catalog.siteContent || {});
1842:   if (siteHeroEyebrowInput && document.activeElement !== siteHeroEyebrowInput) {
1843:     populateTextOnlyRichField(siteHeroEyebrowInput, siteContentRichValue("heroEyebrow"),
content.heroEyebrow);
1844:   }
1845:   if (siteHeroTitleInput && document.activeElement !== siteHeroTitleInput) {
1846:     populateTextOnlyRichField(siteHeroTitleInput, siteContentRichValue("heroTitle"), content.heroTitle);
1847:   }
1848:   if (siteHeroCopyInput && document.activeElement !== siteHeroCopyInput) {
1849:     populateTextOnlyRichField(siteHeroCopyInput, siteContentRichValue("heroCopy"), content.heroCopy);
1850:   }
1851: }

```

## ***renderProfileEditor - lines 1853-1881***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1853-1881 (29 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderProfileEditor() {                                                                                                                                                                      |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | normalizeProfile, forEach, populateTextOnlyRichField, profileRichValue, filter, join                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1853: function renderProfileEditor() {
1854:   const profile = normalizeProfile(catalog.profile || {});
1855:   const fields = [
1856:     [profileDisplayNameInput, profile.displayName],
1857:     [profilePortfolioLabelInput, profile.portfolioLabel],
1858:     [profileContactIntroInput, profile.contactIntro],
1859:     [profileEmailInput, profile.email],
1860:     [profilePhoneInput, profile.phone],
1861:     [profileGithubInput, profile.githubUrl],
1862:     [profileLinkedInInput, profile.linkedinUrl],
1863:     [profileWebsiteInput, profile.websiteUrl]
1864:   ];
1865:   fields.forEach(([field, value]) => {
1866:     if (field && document.activeElement !== field) {
1867:       populateTextOnlyRichField(field, profileRichValue(field.dataset.profileField), value);
1868:     }
1869:   });
1870:   if (profileAssetStatus) {
1871:     const assetLabels = [
1872:       profile.profileImage ? "profile photo" : "",
1873:       profile.heroImage ? "main background" : "",
1874:       profile.resumeUrl ? "resume" : "",
1875:       profile.brandImage ? "brand icon" : ""
1876:     ].filter(Boolean);
1877:     profileAssetStatus.textContent = assetLabels.length
1878:       ? `Added: ${assetLabels.join(", ")}`
1879:       : "No profile photo, resume, brand icon, or main background added yet.";
1880:   }
1881: }

```

## ***syncSiteContentFromInputs - lines 1883-1897***

| Item            | Explanation                                                    |
|-----------------|----------------------------------------------------------------|
| Source location | Lines 1883-1897 (15 source lines).                             |
| Purpose         | Keeps two places aligned, such as local data and target files. |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function syncSiteContentFromInputs() {                                                                                                                                                                |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | richFieldValue, normalizeSiteContent                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1883: function syncSiteContentFromInputs() {
1884:   catalog.siteContentRich = catalog.siteContentRich || {};
1885:   const eyebrow = richFieldValue(siteHeroEyebrowInput, catalog.siteContent?.heroEyebrow || "");
1886:   const title = richFieldValue(siteHeroTitleInput, catalog.siteContent?.heroTitle || "");
1887:   const copy = richFieldValue(siteHeroCopyInput, catalog.siteContent?.heroCopy || "");
1888:   catalog.siteContent = normalizeSiteContent({
1889:     heroCopy: copy.plain,
1890:     heroEyebrow: eyebrow.plain,
1891:     heroTitle: title.plain
1892:   });
1893:   catalog.siteContentRich.heroCopy = copy.rich;
1894:   catalog.siteContentRich.heroEyebrow = eyebrow.rich;
1895:   catalog.siteContentRich.heroTitle = title.rich;
1896:   return catalog.siteContent;
1897: }

```

## ***syncProfileFromInputs - lines 1899-1926***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1899-1926 (28 source lines).                                                                                                                                                                    |
| Purpose            | Keeps two places aligned, such as local data and target files.                                                                                                                                        |
| Declaration        | function syncProfileFromInputs() {                                                                                                                                                                    |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | richFieldValue, normalizeProfile, entries, forEach                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1899: function syncProfileFromInputs() {
1900:   catalog.profileRich = catalog.profileRich || {};
1901:   const fields = {
1902:     contactIntro: richFieldValue(profileContactIntroInput, catalog.profile?.contactIntro || ""),
1903:     displayName: richFieldValue(profileDisplayNameInput, catalog.profile?.displayName || ""),
1904:     email: richFieldValue(profileEmailInput, catalog.profile?.email || ""),
1905:     githubUrl: richFieldValue(profileGithubInput, catalog.profile?.githubUrl || ""),
1906:     linkedinUrl: richFieldValue(profileLinkedinInput, catalog.profile?.linkedinUrl || ""),
1907:     phone: richFieldValue(profilePhoneInput, catalog.profile?.phone || ""),
1908:     portfolioLabel: richFieldValue(profilePortfolioLabelInput, catalog.profile?.portfolioLabel || ""),
1909:     websiteUrl: richFieldValue(profileWebsiteInput, catalog.profile?.websiteUrl || "")
1910:   };
1911:   catalog.profile = normalizeProfile({
1912:     ...(catalog.profile || {}),
1913:     contactIntro: fields.contactIntro.plain,
1914:     displayName: fields.displayName.plain,
1915:     email: fields.email.plain,
1916:     githubUrl: fields.githubUrl.plain,
1917:     linkedinUrl: fields.linkedinUrl.plain,
1918:     phone: fields.phone.plain,
1919:     portfolioLabel: fields.portfolioLabel.plain,
1920:     websiteUrl: fields.websiteUrl.plain
1921:   });
1922:   Object.entries(fields).forEach(([key, value]) => {
1923:     catalog.profileRich[key] = value.rich;

```

```

1924:   });
1925:   return catalog.profile;
1926: }

```

## clone - lines 1928-1930

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1928-1930 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function clone(value) {                                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | parse, stringify                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1928: function clone(value) {
1929:   return JSON.parse(JSON.stringify(value));
1930: }

```

## closeDialogElement - lines 1932-1948

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1932-1948 (17 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function closeDialogElement(dialog, returnValue = "close") {                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | hasAttribute, close, removeAttribute, dispatchEvent, Event                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1932: function closeDialogElement(dialog, returnValue = "close") {
1933:   if (!dialog) return;
1934:   const closeValue = String(returnValue || "close");
1935:   const wasOpen = Boolean(dialog.open || dialog.hasAttribute("open"));
1936:   if (typeof dialog.close === "function") {
1937:     try {
1938:       if (dialog.open) dialog.close(closeValue);
1939:     } catch {
1940:       // Fall through to manual cleanup if the browser rejects the native close call.
1941:     }
1942:     if (!dialog.open && !dialog.hasAttribute("open")) return;
1943:   }
1944:   dialog.returnValue = closeValue;
1945:   dialog.open = false;
1946:   dialog.removeAttribute("open");
1947:   if (wasOpen) dialog.dispatchEvent(new Event("close"));
1948: }

```

## getByPath - lines 1950-1952

| Item            | Explanation                       |
|-----------------|-----------------------------------|
| Source location | Lines 1950-1952 (3 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function getByPath(target, pathValue) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, filter, reduce                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1950: function getByPath(target, pathValue) {
1951:   return String(pathValue || "").split(".").filter(Boolean).reduce((current, key) => current?.[key],
target);
1952: }

```

## ***setByPath - lines 1954-1963***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1954-1963 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function setByPath(target, pathValue, value) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, filter, pop, forEach                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1954: function setByPath(target, pathValue, value) {
1955:   const keys = String(pathValue || "").split(".").filter(Boolean);
1956:   const last = keys.pop();
1957:   let current = target;
1958:   keys.forEach((key) => {
1959:     current[key] = current[key] || {};
1960:     current = current[key];
1961:   });
1962:   if (last) current[last] = value;
1963: }

```

## ***defaultDesignModel - lines 1965-1971***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 1965-1971 (7 source lines).                            |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function defaultDesignModel() {                              |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | No obvious named calls detected by the generator.            |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 1 return statement(s).                |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1965: function defaultDesignModel() {
1966:     return {
1967:         brief: { summary: "", summaryRich: null, files: [] },
1968:         documentation: { summary: "", summaryRich: null, files: [], references: [], mathAnalysis: [] },
1969:         simulation: { summary: "", summaryRich: null, files: [], results: [] }
1970:     };
1971: }

```

## ***isAnalogProject - lines 1973-1975***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1973-1975 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function isAnalogProject(project) {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1973: function isAnalogProject(project) {
1974:     return project?.category === "analog-mixed-signal";
1975: }

```

## ***ensureDesignModel - lines 1977-1992***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1977-1992 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function ensureDesignModel(project) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | defaultDesignModel                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1977: function ensureDesignModel(project) {
1978:     if (!project) return null;
1979:     const defaults = defaultDesignModel();
1980:     project.design = project.design || {};
1981:     project.design.brief = { ...defaults.brief, ...(project.design.brief || {}) };
1982:     project.design.documentation = { ...defaults.documentation, ...(project.design.documentation || {}) };
1983:     project.design.documentation.summaryRich = project.design.documentation.summaryRich || null;
1984:     project.design.simulation = { ...defaults.simulation, ...(project.design.simulation || {}) };
1985:     project.design.brief.files = project.design.brief.files || [];
1986:     project.design.documentation.files = project.design.documentation.files || [];
1987:     project.design.documentation.references = project.design.documentation.references || [];
1988:     project.design.documentation.mathAnalysis = project.design.documentation.mathAnalysis || [];
1989:     project.design.simulation.files = project.design.simulation.files || [];

```

```

1990:   project.design.simulation.results = project.design.simulation.results || [];
1991:   return project.design;
1992: }

```

## ensureCompileCode - lines 1994-2034

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 1994-2034 (41 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function ensureCompileCode(project) {                                                                                                                                                                 |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isArray, map, normalizeCodeLanguage, detectCodeLanguage, safeClientCodeFileName, defaultCodeFileName, normalizeCompileFileRole, inferCompileFileRole, slugify, now, some, compileAppendDestinations   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

1994: function ensureCompileCode(project) {
1995:   if (!project) return null;
1996:   project.compileCode = project.compileCode || {};
1997:   project.compileCode.files = Array.isArray(project.compileCode.files) ? project.compileCode.files : [];
1998:   project.compileCode.files = project.compileCode.files.map((file, index) => {
1999:     const language = normalizeCodeLanguage(file.language || detectCodeLanguage(file.code || "",
file.fileName || ""));
2000:     const fileName = safeClientCodeFileName(file.fileName || defaultCodeFileName(language), language);
2001:     const role = normalizeCompileFileRole(file.role || inferCompileFileRole(fileName, file.code || "",
language), language);
2002:     return {
2003:       id: file.id || slugify(`${fileName}-${index}-${Date.now()}`),
2004:       title: file.title || fileName,
2005:       fileName,
2006:       language,
2007:       role,
2008:       code: String(file.code || ""),
2009:       stdin: String(file.stdin || ""),
2010:       savedPath: file.savedPath || file.workspacePath || "",
2011:       savedAt: file.savedAt || "",
2012:       waveform: file.waveform || file.lastResult?.waveform || null,
2013:       lastResult: file.lastResult || null,
2014:       dirty: Boolean(file.dirty)
2015:     };
2016:   });
2017:   if (!project.compileCode.activeFileId || !project.compileCode.files.some((file) => file.id ===
project.compileCode.activeFileId)) {
2018:     project.compileCode.activeFileId = project.compileCode.files[0]?.id || "";
2019:   }
2020:   project.compileCode.terminal = String(project.compileCode.terminal || "");
2021:   const appendDestinations = compileAppendDestinations(project);
2022:   if (!project.compileCode.appendTargetId || !appendDestinations.some((destination) => destination.id ===
project.compileCode.appendTargetId)) {
2023:     project.compileCode.appendTargetId = appendDestinations[0]?.id || "";
2024:   }
2025:   project.compileCode.messages = Array.isArray(project.compileCode.messages)
? project.compileCode.messages.map((message) => ({
2026:     id: message.id || slugify(`message-${Date.now()}-${Math.random().toString(16).slice(2)}`),
2027:     at: message.at || new Date().toISOString(),
2028:     level: ["info", "success", "warning", "error"].includes(message.level) ? message.level : "info",
2029:     text: String(message.text || "")
2030:   })).filter((message) => message.text)
2031:   : [];
2032:   return project.compileCode;
2033: }
2034: }

```

## escapeHtml - lines 2036-2043

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2036-2043 (8 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function escapeHtml(value) {                                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replaceAll                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2036: function escapeHtml(value) {
2037:   return String(value || "")
2038:     .replaceAll("&", "&amp;")
2039:     .replaceAll("<", "&lt;")
2040:     .replaceAll(">", "&gt;")
2041:     .replaceAll("'", "&quot;")
2042:     .replaceAll('"', "&#039;");
2043: }

```

## ***escapeCodeHtml - lines 2045-2051***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2045-2051 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function escapeCodeHtml(value) {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replaceAll                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2045: function escapeCodeHtml(value) {
2046:   return String(value || "")
2047:     .replaceAll("&", "&amp;")
2048:     .replaceAll("<", "&lt;")
2049:     .replaceAll(">", "&gt;")
2050:     .replaceAll("'", "&quot;");
2051: }

```

## ***normalizeCodeLanguage - lines 2053-2059***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2053-2059 (7 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCodeLanguage(value = "") {                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim, toLowerCase, replace, find, includes                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2053: function normalizeCodeLanguage(value = "") {
2054:   const clean = String(value || "").trim().toLowerCase().replace(/[_-]+/g, " ");
2055:   const match = supportedCodeLanguages.find((language) =>
2056:     language.id === clean || language.aliases.includes(clean)
2057:   );
2058:   return match?.id || "javascript";
2059: }
```

### ***codeLanguageProfile - lines 2061-2063***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2061-2063 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function codeLanguageProfile(value = "") {                                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | find, normalizeCodeLanguage                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2061: function codeLanguageProfile(value = "") {
2062:   return supportedCodeLanguages.find((language) => language.id === normalizeCodeLanguage(value)) ||
supportedCodeLanguages.find((language) => language.id === "javascript");
2063: }
```

### ***codeLanguageLabel - lines 2065-2067***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2065-2067 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function codeLanguageLabel(value = "") {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | find, normalizeCodeLanguage                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2065: function codeLanguageLabel(value = "") {
2066:   return supportedCodeLanguages.find((language) => language.id === normalizeCodeLanguage(value))?.label
|| "Code";
2067: }
```

### ***sourceLooksCpp - lines 2069-2073***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2069-2073 (5 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function sourceLooksCpp(source = "") {                       |
| Touches         | Mostly local calculations or local UI state.                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | b                                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2069: function sourceLooksCpp(source = "") {
2070:   const text = String(source || "");
2071:   return
  /#include\s*<(?:iostream|string|vector|array|map|unordered_map|memory|algorithm|optional|variant)>/.test(text)
  ||
2072:   /\b(std::|cout|cin|cerr|namespace|template\s*<|class\s+\w+|new\s+\w|delete\s+|constexpr|nullptr|using
\s+namespace)\b/.test(text);
2073: }

```

## ***sourceLooksC - lines 2075-2079***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2075-2079 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function sourceLooksC(source = "") {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | b                                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2075: function sourceLooksC(source = "") {
2076:   const text = String(source || "");
2077:   return /#include\s*<(?:stdio|stdlib|string|stdint|stdbool|math)\.h>/.test(text) ||
2078:   /\b(printf|scanf|malloc|calloc|realloc|free|struct\s+\w+|typedef\s+struct)\b/.test(text);
2079: }

```

## ***languageFromFileName - lines 2081-2090***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2081-2090 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function languageFromFileName(fileName = "", source = "") {                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | toLowerCase, includes, split, pop, sourceLooksCpp, sourceLooksC, find                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2081: function languageFromFileName(fileName = "", source = "") {
2082:   const clean = String(fileName || "").toLowerCase();

```



```

2083:   const extension = clean.includes(".") ? `.${clean.split(".").pop()}` : "";
2084:   if (extension === ".h") {
2085:     if (sourceLooksCpp(source)) return "cpp";
2086:     if (sourceLooksC(source)) return "c";
2087:     return "c";
2088:   }
2089:   return supportedCodeLanguages.find((language) => language.extensions?.includes(extension))?.id || "";
2090: }

```

## defaultCodeFileName - lines 2092-2094

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2092-2094 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function defaultCodeFileName(language = "javascript") {                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | codeLanguageProfile                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2092: function defaultCodeFileName(language = "javascript") {
2093:   return codeLanguageProfile(language)?.defaultFile || "main.js";
2094: }

```

## safeClientCodeFileName - lines 2096-2104

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2096-2104 (9 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function safeClientCodeFileName(fileName = "", language = "javascript") {                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | codeLanguageProfile, trim, replace, match, toLowerCase, includes                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2096: function safeClientCodeFileName(fileName = "", language = "javascript") {
2097:   const profile = codeLanguageProfile(language);
2098:   const fallback = profile.defaultFile || "main.js";
2099:   const clean = String(fileName || fallback).trim().replace(/[\\/:?*"<>|]+/g, "-");
2100:   const namePart = clean.replace(/\.[^.]+\$/, "").replace(/^[a-zA-Z0-9._-]+/g, "-").replace(/^-+|-+$/g, "") || fallback.replace(/\.[^.]+\$/, "");
2101:   const extMatch = clean.match(/(\.[a-zA-Z0-9_+-]+)$/);
2102:   const ext = (extMatch?.[1] || fallback.match(/(\.[^.]+\$)/)?.[1] || ".js").toLowerCase();
2103:   return `${namePart}${profile.extensions?.includes(ext) ? ext : fallback.match(/(\.[^.]+\$)/)?.[1] || ".js"}`;
2104: }

```

## normalizeCodePasteMode - lines 2106-2108

| Item            | Explanation                       |
|-----------------|-----------------------------------|
| Source location | Lines 2106-2108 (3 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCodePasteMode(value = "") {                                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2106: function normalizeCodePasteMode(value = "") {
2107:   return value === "source" ? "source" : "basic";
2108: }
```

## ***normalizeCodeText - lines 2110-2119***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2110-2119 (10 source lines).                                                                                                                                                                    |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCodeText(value = "", pasteMode = "source") {                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | replace, normalizeCodePasteMode, split, map, trimEnd, join, trim                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2110: function normalizeCodeText(value = "", pasteMode = "source") {
2111:   const normalized = String(value || "").replace(/\r\n?/g, "\n").replace(/\u00a0/g, " ");
2112:   if (normalizeCodePasteMode(pasteMode) === "source") return normalized.replace(/\t/g, " ");
2113:   return normalized
2114:     .split("\n")
2115:     .map((line) => line.trimEnd())
2116:     .join("\n")
2117:     .replace(/\n{4,}/g, "\n\n\n")
2118:     .trim();
2119: }
```

## ***detectCodeLanguage - lines 2121-2136***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 2121-2136 (16 source lines).                           |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function detectCodeLanguage(value = "", fileName = "") {     |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | languageFromFileName, b, sourceLooksCpp, sourceLooksC        |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 10 return statement(s).               |



```

2169:     protect(/^\s*\w+[\n\]*/gm, "code-token-preprocessor");
2170:   }
2171: }
2172:
2173: protect(/\b(?:0x[\da-f]+|\d+(?:\.\d+)?(?:e[+-]?[d+])?)\b/gi, "code-token-number");
2174:
2175: const keywordMap = {
2176:   c: "_Alignas|_Alignof|_Atomic|_Bool|_Complex|_Generic|_Imaginary|_Noreturn|_Static_assert|_Thread_loc
al|auto|break|case|char|const|continue|default|do|double|else|enum|extern|float|for|goto|if|inline|int|long|regi
ster|restrict|return|short|signed|sizeof|static|struct|switch|typedef|union|unsigned|void|volatile|while",
2177:   cpp: "alignas|alignof|and|and_eq|asm|auto|bitand|bitor|bool|break|case|catch|char|char8_t|char16_t|ch
ar32_t|class|compl|concept|const|consteval|constexpr|constinit|const_cast|continue|co_await|co_return|co_yield|d
ecltype|default|delete|do|double|dynamic_cast|else|enum|explicit|export|extern|false|float|for|friend|if|inline|
int|long|mutable|namespace|new|noexcept|not|not_eq|nullptr|operator|or|or_eq|private|protected|public|register|r
einterpret_cast|requires|return|short|signed|sizeof|static|static_assert|static_cast|struct|switch|template|this
|thread_local|throw|true|try|typeid|typename|union|unsigned|using|virtual|void|volatile|wchar_t|while|xor
r|xor_eq",
2178:   verilog: "always|and|assign|begin|buf|case|casex|casez|deassign|default|defparam|disable|edge|else|en
d|endcase|endfunction|endmodule|endprimitive|endspecify|endtable|endtask|event|for|force|forever|fork|function|g
enerate|genvar|if|initial|inout|input|integer|join|module|nand|negedge|nor|not|or|output|parameter|posedge|primiti
ve|reg|release|repeat|signed|specify|supply0|supply1|table|task|tri|wand|while|wire|wor|xnor|xor",
2179:   systemverilog: "always|always_comb|always_ff|always_latch|assign|automatic|begin|bit|case|class|clock
ing|covergroup|default|disable|do|else|end|endcase|endclass|endclocking|endfunction|endmodule|endpackage|endtask
|enum|for|forever|function|generate|genvar|if|initial|input|int|interface|logic|module|negedge|output|package|pa
rameter|posedge|reg|return|signed|task|typedef|wire",
2180:   ltspice: "ac|dc|end|ends|four|func|global|ic|include|lib|meas|model|nodeset|op|options|param|plot|pro
be|save|step|subckt|temp|tran",
2181:   java: "abstract|assert|boolean|break|byte|case|catch|char|class|const|continue|default|do|double|else
|enum|extends|final|finally|float|for|if|implements|import|instanceof|int|interface|long|native|new|null|package
|private|protected|public|return|short|static|strictfp|super|switch|synchronized|this|throw|throws|transient|try
|void|volatile|while",
2182:   javascript: "async|await|break|case|catch|class|const|continue|debugger|default|delete|do|else|export
|extends|false|finally|for|from|function|if|import|in|instanceof|let|new|null|of|return|static|super|switch|this
|throw|true|try|typeof|undefined|var|void|while|yield",
2183:   python: "and|as|assert|async|await|break|class|continue|def|del|elif|else|except|False|finally|for|fr
om|global|if|import|in|is|lambda|None|nonlocal|not|or|pass|raise|return|True|try|while|with|yield",
2184:   html: "html|head|body|main|section|article|div|span|script|style|link|meta|title|header|footer|nav|bu
tton|input|form|label|img|a|p|pre|code"
2185:   };
2186:   const keywords = keywordMap[normalizeCodeLanguage(language)] || keywordMap.javascript;
2187:   html = html.replace(new RegExp(`\\b(${keywords})\\b`, "g"), '<span class="code-token-
keyword">$1</span>');
2188:   tokens.forEach((token) => {
2189:     html = html.replaceAll(token.token, token.html);
2190:   });
2191:   return html;
2192: }

```

## tokenName - lines 2141-2150

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2141-2150 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const tokenName = (index) => {                                                                                                                                                                        |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | fromCharCode, floor                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2141:   const tokenName = (index) => {
2142:     let value = Number(index) + 1;
2143:     let output = "";
2144:     while (value > 0) {
2145:       value -= 1;
2146:       output = String.fromCharCode(65 + (value % 26)) + output;
2147:       value = Math.floor(value / 26);
2148:     }
2149:     return `uE000${output}uE001`;
2150:   };

```

## protect - lines 2151-2157

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2151-2157 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const protect = (pattern, className) => {                                                                                                                                                             |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | replace, tokenName, push                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2151:   const protect = (pattern, className) => {
2152:     html = html.replace(pattern, (match) => {
2153:       const token = tokenName(tokens.length);
2154:       tokens.push({ token, html: `` });
2155:       return token;
2156:     });
2157:   };

```

## ***renderRichCodeBlock - lines 2194-2204***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2194-2204 (11 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderRichCodeBlock(block = {}) {                                                                                                                                                            |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | normalizeCodeLanguage, detectCodeLanguage, normalizeCodeText, codeLanguageLabel, escapeHtml, tokenizedCodeHtml                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2194: function renderRichCodeBlock(block = {}) {
2195:   const language = normalizeCodeLanguage(block.language || detectCodeLanguage(block.code || ""));
2196:   const code = normalizeCodeText(block.code || "", block.pasteMode || "source");
2197:   const label = codeLanguageLabel(language);
2198:   return
2199:     <figure class="rich-code-block rich-code-language-${language}">
2200:       <figcaption><span>&lt;/span> ${escapeHtml(label)}</figcaption>
2201:       <pre><code>${tokenizedCodeHtml(code, language)}</code></pre>
2202:     </figure>
2203:   `;
2204: }

```

## ***renderPlainMultiline - lines 2206-2208***

| Item            | Explanation                                  |
|-----------------|----------------------------------------------|
| Source location | Lines 2206-2208 (3 source lines).            |
| Purpose         | Renders UI, markup, or display output.       |
| Declaration     | function renderPlainMultiline(value = "") {  |
| Touches         | Mostly local calculations or local UI state. |
| Calls detected  | escapeHtml, replace                          |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2206: function renderPlainMultiline(value = "") {
2207:   return escapeHtml(value).replace(/\r\n?|\n/g, "<br>");
2208: }

```

## ***normalizeSiteContent - lines 2210-2216***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2210-2216 (7 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeSiteContent(content = {}) {                                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2210: function normalizeSiteContent(content = {}) {
2211:   return {
2212:     heroCopy: String(content.heroCopy || "").trim() || defaultSiteContent.heroCopy,
2213:     heroEyebrow: String(content.heroEyebrow || "").trim() || defaultSiteContent.heroEyebrow,
2214:     heroTitle: String(content.heroTitle || "").trim() || defaultSiteContent.heroTitle
2215:   };
2216: }

```

## ***normalizeProfile - lines 2218-2234***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2218-2234 (17 source lines).                                                                                                                                                                    |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeProfile(profile = {}) {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2218: function normalizeProfile(profile = {}) {
2219:   return {
2220:     brandImage: String(profile.brandImage || "").trim(),
2221:     brandText: String(profile.brandText || "").trim() || String(profile.displayName || "").trim() ||
defaultProfile.brandText,
2222:     contactIntro: String(profile.contactIntro || "").trim(),
2223:     displayName: String(profile.displayName || "").trim(),

```

```

2224:     email: String(profile.email || "").trim(),
2225:     githubUrl: String(profile.githubUrl || "").trim(),
2226:     heroImage: String(profile.heroImage || "").trim(),
2227:     linkedinUrl: String(profile.linkedinUrl || "").trim(),
2228:     phone: String(profile.phone || "").trim(),
2229:     portfolioLabel: String(profile.portfolioLabel || "").trim() || defaultProfile.portfolioLabel,
2230:     profileImage: String(profile.profileImage || "").trim(),
2231:     resumeUrl: String(profile.resumeUrl || "").trim(),
2232:     websiteUrl: String(profile.websiteUrl || "").trim()
2233:   };
2234: }

```

## ***normalizePlainFieldStyle - lines 2236-2250***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2236-2250 (15 source lines).                                                                                                                                                                    |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizePlainFieldStyle(style = {}) {                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | cleanFontFamily, normalizeFontPx, normalizeTextColor, rgbColorToHex                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2236: function normalizePlainFieldStyle(style = {}) {
2237:   const fontFamily = cleanFontFamily(style.fontFamily || "");
2238:   const fontPx = normalizeFontPx(style.fontPx || style.fontSize || "");
2239:   const color = normalizeTextColor(style.color || "");
2240:   const normalized = {};
2241:
2242:   if (fontFamily) normalized.fontFamily = fontFamily;
2243:   if (fontPx) normalized.fontPx = fontPx;
2244:   if (color) normalized.color = rgbColorToHex(color);
2245:   if (style.bold === true || style.bold === "true") normalized.bold = true;
2246:   if (style.italic === true || style.italic === "true") normalized.italic = true;
2247:   if (style.underline === true || style.underline === "true") normalized.underline = true;
2248:
2249:   return normalized;
2250: }

```

## ***normalizeFieldStyles - lines 2252-2259***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2252-2259 (8 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeFieldStyles(styles = {}) {                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | fromEntries, entries, filter, has, map, normalizePlainFieldStyle, keys                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2252: function normalizeFieldStyles(styles = {}) {
2253:   return Object.fromEntries(
2254:     Object.entries(styles || {})
2255:       .filter(([fieldId]) => persistentPlainStyleFieldIds.has(fieldId))
2256:       .map(([fieldId, style]) => [fieldId, normalizePlainFieldStyle(style)])
2257:       .filter(([ , style]) => Object.keys(style).length)

```

```

2258:   });
2259: }

```

## loadBuilderPreferences - lines 2261-2273

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2261-2273 (13 source lines).                                                                                                                                                                    |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | function loadBuilderPreferences() {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | parse, getItem, includes, applyBuilderPreferences                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2261: function loadBuilderPreferences() {
2262:   try {
2263:     const stored = JSON.parse(localStorage.getItem(preferenceStorageKey) || "{}");
2264:     builderPreferences = {
2265:       ...defaultBuilderPreferences,
2266:       ...stored,
2267:       theme: ["light", "dark"].includes(stored.theme) ? stored.theme : defaultBuilderPreferences.theme
2268:     };
2269:   } catch {
2270:     builderPreferences = { ...defaultBuilderPreferences };
2271:   }
2272:   applyBuilderPreferences();
2273: }

```

## applyBuilderPreferences - lines 2275-2280

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2275-2280 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function applyBuilderPreferences() {                                                                                                                                                                  |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | includes                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2275: function applyBuilderPreferences() {
2276:   const theme = ["light", "dark"].includes(builderPreferences.theme) ? builderPreferences.theme :
"light";
2277:   document.documentElement.dataset.builderTheme = theme;
2278:   document.body.dataset.builderTheme = theme;
2279:   if (preferenceTheme) preferenceTheme.value = theme;
2280: }

```

## setBuilderTheme - lines 2282-2287

| Item            | Explanation                       |
|-----------------|-----------------------------------|
| Source location | Lines 2282-2287 (6 source lines). |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function setBuilderTheme(theme = "light") {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | includes, setItem, stringify, applyBuilderPreferences, setStatus                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2282: function setBuilderTheme(theme = "light") {
2283:   builderPreferences.theme = ["light", "dark"].includes(theme) ? theme : "light";
2284:   localStorage.setItem(preferenceStorageKey, JSON.stringify(builderPreferences));
2285:   applyBuilderPreferences();
2286:   setStatus( `Builder ${builderPreferences.theme} mode enabled.` );
2287: }

```

## ***openPreferencesDialog - lines 2289-2292***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2289-2292 (4 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function openPreferencesDialog() {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | applyBuilderPreferences, showModal                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2289: function openPreferencesDialog() {
2290:   applyBuilderPreferences();
2291:   if (!preferencesDialog?.open) preferencesDialog.showModal();
2292: }

```

## ***saveBuilderPreferencesFromDialog - lines 2294-2297***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2294-2297 (4 source lines).                                                                                                                                                                     |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | function saveBuilderPreferencesFromDialog() {                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | setBuilderTheme, setStatus                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2294: function saveBuilderPreferencesFromDialog() {
2295:   setBuilderTheme(preferenceTheme?.value || "light");
2296:   setStatus("Preferences saved.");
2297: }
```

## ***plainFieldStyleToCss - lines 2299-2309***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2299-2309 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function plainFieldStyleToCss(style = {}) {                                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizePlainFieldStyle, push, join                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2299: function plainFieldStyleToCss(style = {}) {
2300:   const normalized = normalizePlainFieldStyle(style);
2301:   const rules = [];
2302:   if (normalized.fontFamily) rules.push(`font-family: ${normalized.fontFamily}`);
2303:   if (normalized.fontPx) rules.push(`font-size: ${normalized.fontPx}px`);
2304:   if (normalized.color) rules.push(`color: ${normalized.color}`);
2305:   if (normalized.bold) rules.push("font-weight: 700");
2306:   if (normalized.italic) rules.push("font-style: italic");
2307:   if (normalized.underline) rules.push("text-decoration: underline");
2308:   return rules.join("; ");
2309: }
```

## ***plainFieldStyleAttribute - lines 2311-2314***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2311-2314 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function plainFieldStyleAttribute(styles = {}, fieldId = "") {                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | plainFieldStyleToCss, escapeHtml                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2311: function plainFieldStyleAttribute(styles = {}, fieldId = "") {
2312:   const css = plainFieldStyleToCss(styles[fieldId]);
2313:   return css ? `style="${escapeHtml(css)}"` : "";
2314: }
```

## ***plainStyleFromControl - lines 2316-2326***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2316-2326 (11 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function plainStyleFromControl(control) {                                                                                                                                                             |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | normalizePlainFieldStyle                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2316: function plainStyleFromControl(control) {
2317:   if (!control) return {};
2318:   return normalizePlainFieldStyle({
2319:     bold: control.dataset.builderBold,
2320:     color: control.dataset.builderColor,
2321:     fontFamily: control.dataset.builderFontFamily,
2322:     fontPx: control.dataset.builderFontPx,
2323:     italic: control.dataset.builderItalic,
2324:     underline: control.dataset.builderUnderline
2325:   });
2326: }

```

## ***applyPlainFieldStyleToControl - lines 2328-2343***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2328-2343 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function applyPlainFieldStyleToControl(control, style = {}) {                                                                                                                                         |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | normalizePlainFieldStyle                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2328: function applyPlainFieldStyleToControl(control, style = {}) {
2329:   if (!control) return;
2330:   const normalized = normalizePlainFieldStyle(style);
2331:   control.dataset.builderFontFamily = normalized.fontFamily || "";
2332:   control.dataset.builderFontPx = normalized.fontPx || "";
2333:   control.dataset.builderColor = normalized.color || "";
2334:   control.dataset.builderBold = normalized.bold ? "true" : "false";
2335:   control.dataset.builderItalic = normalized.italic ? "true" : "false";
2336:   control.dataset.builderUnderline = normalized.underline ? "true" : "false";
2337:   control.style.fontFamily = normalized.fontFamily || "";
2338:   control.style.fontSize = normalized.fontPx ? `${normalized.fontPx}px` : "";
2339:   control.style.color = normalized.color || "";
2340:   control.style.fontWeight = normalized.bold ? "700" : "";
2341:   control.style.fontStyle = normalized.italic ? "italic" : "";
2342:   control.style.textDecoration = normalized.underline ? "underline" : "";
2343: }

```

## ***applyStoredPlainFieldStyle - lines 2345-2349***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2345-2349 (5 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function applyStoredPlainFieldStyle(control) {               |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | has, normalizeFieldStyles, applyPlainFieldStyleToControl                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2345: function applyStoredPlainFieldStyle(control) {
2346:   if (!control?.id || !persistentPlainStyleFieldIds.has(control.id)) return;
2347:   const styles = normalizeFieldStyles(catalog.fieldStyles || {});
2348:   applyPlainFieldStyleToControl(control, styles[control.id] || {});
2349: }

```

## ***storePlainControlStyle - lines 2351-2365***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2351-2365 (15 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | function storePlainControlStyle(control) {                                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | has, normalizeFieldStyles, plainStyleFromControl, keys, clone, markDraftNeedsSave, scheduleAutosave, schedulePreviewRender                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2351: function storePlainControlStyle(control) {
2352:   if (!control?.id || !persistentPlainStyleFieldIds.has(control.id)) return;
2353:   const nextStyles = normalizeFieldStyles(catalog.fieldStyles || {});
2354:   const style = plainStyleFromControl(control);
2355:   if (Object.keys(style).length) {
2356:     nextStyles[control.id] = style;
2357:   } else {
2358:     delete nextStyles[control.id];
2359:   }
2360:   catalog.fieldStyles = nextStyles;
2361:   savedPortfolioCatalog.fieldStyles = clone(nextStyles);
2362:   markDraftNeedsSave();
2363:   scheduleAutosave();
2364:   schedulePreviewRender();
2365: }

```

## ***mailComposeLink - lines 2367-2370***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2367-2370 (4 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function mailComposeLink(email = "") {                       |
| Touches         | network/API requests                                         |
| Calls detected  | trim                                                         |
| API endpoints   | None detected inside this function body.                     |
| Storage keys    | No local/session storage keys detected.                      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2367: function mailComposeLink(email = "") {
2368:   const clean = String(email || "").trim();
2369:   return clean ? `https://mail.google.com/mail/?view=cm&fs=1&to=${encodeURIComponent(clean)}` : "";
2370: }
```

## ***phoneLink - lines 2372-2375***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2372-2375 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function phoneLink(phone = "") {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, replace                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2372: function phoneLink(phone = "") {
2373:   const clean = String(phone || "").trim();
2374:   return clean ? `tel:${clean.replace(/[\d+]/g, "")}` : "";
2375: }
```

## ***textBlocksFromPlainText - lines 2377-2385***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2377-2385 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function textBlocksFromPlainText(text) {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2377: function textBlocksFromPlainText(text) {
2378:   return [{
2379:     align: "left",
2380:     fontFamily: "Arial",
2381:     fontSize: "normal",
2382:     text: String(text || ""),
2383:     type: "paragraph"
2384:   }];
2385: }
```

## ***richBlocksForProject - lines 2387-2389***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2387-2389 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richBlocksForProject(project) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | clone, textBlocksFromPlainText                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2387: function richBlocksForProject(project) {
2388:   return project?.summaryRich?.blocks?.length ? clone(project.summaryRich.blocks) :
textBlocksFromPlainText(project?.summary || "");
2389: }

```

## ***plainTextFromRich - lines 2391-2402***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2391-2402 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function plainTextFromRich(rich, separator = "\n\n") {                                                                                                                                                |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | map, filter, join                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2391: function plainTextFromRich(rich, separator = "\n\n") {
2392:   return (rich?.blocks || [])
2393:     .map((block) => {
2394:       if (block.type === "paragraph") return block.text || "";
2395:       if (block.type === "formula") return block.formula || "";
2396:       if (block.type === "code") return block.code || "";
2397:       if (block.type === "image") return [block.title, block.caption].filter(Boolean).join(" ");
2398:       return "";
2399:     })
2400:     .filter(Boolean)
2401:     .join(separator);
2402: }

```

## ***unwrapFormula - lines 2404-2417***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2404-2417 (14 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function unwrapFormula(value) {                              |
| Touches         | filesystem                                                   |
| Calls detected  | trim, match                                                  |
| API endpoints   | None detected inside this function body.                     |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2404: function unwrapFormula(value) {
2405:   const text = String(value || "").trim();
2406:   const wrappers = [
2407:     [/^\$\$([\s\S]+\)\$\$$/, 1],
2408:     [/^\$([\s\S]+\)\$$/, 1],
2409:     [/^\$([\s\S]+\)\$$/, 1],
2410:     [/^\$([\s\S]+\)\$$/, 1]
2411:   ];
2412:   for (const [pattern, group] of wrappers) {
2413:     const match = text.match(pattern);
2414:     if (match) return match[group].trim();
2415:   }
2416:   return text;
2417: }

```

### ***isFormulaOnly - lines 2419-2426***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2419-2426 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function isFormulaOnly(value) {                                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim, looksLikeWebOrContactText                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2419: function isFormulaOnly(value) {
2420:   const text = String(value || "").trim();
2421:   if (!text) return false;
2422:   if (looksLikeWebOrContactText(text)) return false;
2423:   if (/^\$\$[\s\S]+\)\$\$/.test(text) || /^\$([\s\S]+\)\$/.test(text) || /^\$([\s\S]+\)\$/.test(text))
return true;
2424:   const mathSignals = /\frac|\sqrt|\sum|\int|\Delta|\omega|\pi|[\^_√ΣΠΩμ]/;
2425:   return text.length <= 180 && mathSignals.test(text) && !/[.!?]\s*\$/.test(text);
2426: }

```

### ***renderInlineMath - lines 2428-2437***

| Item               | Explanation                                               |
|--------------------|-----------------------------------------------------------|
| Source location    | Lines 2428-2437 (10 source lines).                        |
| Purpose            | Renders UI, markup, or display output.                    |
| Declaration        | function renderInlineMath(text) {                         |
| Touches            | filesystem                                                |
| Calls detected     | escapeHtml, replace, b, match, slice, normalizeLinkTarget |
| API endpoints      | None detected inside this function body.                  |
| Storage keys       | No local/session storage keys detected.                   |
| Async/return clues | 0 await expression(s), 2 return statement(s).             |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2428: function renderInlineMath(text) {
2429:   const escaped = escapeHtml(text);
2430:   const withMath = escaped.replace(/$([^\$]+)\$/g, '<span class="rich-inline-formula">$1</span>');
2431:   return withMath.replace(/(https?:\/\/[^\s<]+|www\.[^\s<]+|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/[^\s<]+)/gi, (match) => {
2432:     const trailing = match.match(/[.,;:!?]+\$/)?.[0] || "";
2433:     const clean = trailing ? match.slice(0, -trailing.length) : match;
2434:     const href = normalizeLinkTarget(clean, { assumeWeb: true });
2435:     return `<a href="${href}" target="_blank" rel="noreferrer">${clean}</a>${trailing}`;
2436:   });
2437: }

```

## ***renderMultilineInlineText - lines 2439-2441***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2439-2441 (3 source lines).                                                                                                                                                                     |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderMultilineInlineText(text = "") {                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | renderInlineMath, replace                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2439: function renderMultilineInlineText(text = "") {
2440:   return renderInlineMath(text).replace(/\r\n?|\n/g, "<br>");
2441: }

```

## ***looksLikeBareWebAddress - lines 2443-2450***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2443-2450 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function looksLikeBareWebAddress(value = "") {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, split, toLowerCase                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2443: function looksLikeBareWebAddress(value = "") {
2444:   const clean = String(value || "").trim();
2445:   if (!clean || /\s/.test(clean) || /^(.?.?\/|#|mailto:|tel:)/i.test(clean)) return false;
2446:   if (/^[a-z][a-z0-9+.-]*$/i.test(clean)) return false;
2447:   const host = clean.split(/[/?#]/)[0].toLowerCase();
2448:   if (!/^[a-z0-9-]+\.[a-z0-9-]+\.$/.test(host)) return false;
2449:   return !/\.(pdf|docx?|xlsx?|pptx?|zip|7z|rar|png|jpe?g|gif|svg|webp|txt|md|csv|json|xml|log|c|h|cpp|hpp|py|js|mjs|ts|v|sv|vhdl?|spice|cir|net|asc|sch|kicad_sch|kicad_pcb)$/i.test(host);
2450: }

```



## ***looksLikeWebOrContactText - lines 2452-2460***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2452-2460 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function looksLikeWebOrContactText(value = "") {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, b, looksLikeBareWebAddress, split, some, replace                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2452: function looksLikeWebOrContactText(value = "") {
2453:   const clean = String(value || "").trim();
2454:   if (!clean) return false;
2455:   if (/b(?:https?:\/\/|www\.)[\s<>'"]+/i.test(clean)) return true;
2456:   if (/b[a-z0-9._%+-]@[a-z0-9.-]+\.[a-z]{2,}/i.test(clean)) return true;
2457:   if (/b(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/[\s<>'"]+/i.test(clean)) return true;
2458:   if (looksLikeBareWebAddress(clean)) return true;
2459:   return clean.split(/\s+/).some((word) => looksLikeBareWebAddress(word.replace(/^[('"]+|['"]+$/g, '')));
2460: }
```

## ***normalizeLinkTarget - lines 2462-2469***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2462-2469 (8 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeLinkTarget(target = "", options = {}) {                                                                                                                                             |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | trim, looksLikeBareWebAddress                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2462: function normalizeLinkTarget(target = "", options = {}) {
2463:   const clean = String(target || "").trim();
2464:   if (!clean) return "";
2465:   if (/^\/\//.test(clean)) return `https:${clean}`;
2466:   if (/^www\./i.test(clean)) return `https://${clean}`;
2467:   if (options.assumeWeb && looksLikeBareWebAddress(clean)) return `https://${clean}`;
2468:   return clean;
2469: }
```

## ***isWebsiteLinkItem - lines 2471-2479***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2471-2479 (9 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function isWebsiteLinkItem(item = {}, target = "") {         |
| Touches         | Mostly local calculations or local UI state.                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | trim, filter, join, toLowerCase, b                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2471: function isWebsiteLinkItem(item = {}, target = "") {
2472:   const clean = String(target || "").trim();
2473:   if (/^(https?:)?\/\//i.test(clean) || /^www\./i.test(clean)) return true;
2474:   const typeText = [item.kind, item.type, item.status, item.meta, item.label, item.title]
2475:     .filter(Boolean)
2476:     .join(" ")
2477:     .toLowerCase();
2478:   return /\b(link|website|web link|url|external|github|linkedin|drive|social|profile)\b/.test(typeText);
2479: }

```

## ***extensionFromUrl - lines 2481-2485***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2481-2485 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function extensionFromUrl(value = "") {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, match, toLowerCase                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2481: function extensionFromUrl(value = "") {
2482:   const clean = String(value || "").split(/[?#]/)[0];
2483:   const match = clean.match(/(\.[a-z0-9_]+)$/i);
2484:   return match ? match[1].toLowerCase() : "";
2485: }

```

## ***urlLooksLikeDirectImage - lines 2487-2491***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2487-2491 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function urlLooksLikeDirectImage(value = "") {                                                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeLinkTarget                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2487: function urlLooksLikeDirectImage(value = "") {

```

```

2488:   const clean = normalizeLinkTarget(value, { assumeWeb: true });
2489:   return /^(data:image\/|blob:)/i.test(clean) ||
2490:     /\. (png|jpe?g|webp|gif|svg|bmp|avif)([?#].*)?$/i.test(clean);
2491: }

```

## inferUrlAssetKind - lines 2493-2506

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2493-2506 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function inferUrlAssetKind(value = "", source = "web") {                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeLinkTarget, extensionFromUrl, urlLooksLikeDirectImage, includes, looksLikeBareWebAddress                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 10 return statement(s).                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2493: function inferUrlAssetKind(value = "", source = "web") {
2494:   const clean = normalizeLinkTarget(value, { assumeWeb: true });
2495:   const extension = extensionFromUrl(clean);
2496:   if (source === "drive" || /^(https?:\/\/)?(drive|docs)\.google\.com\/$/i.test(clean)) return "google-
drive";
2497:   if (/github\.com|raw\.githubusercontent\.com/i.test(clean)) return "github";
2498:   if (/linkedin\.com/i.test(clean)) return "linkedin";
2499:   if (urlLooksLikeDirectImage(clean)) return "image";
2500:   if ([".pdf"].includes(extension)) return "pdf";
2501:   if ([".doc", ".docx", ".ppt", ".pptx", ".xls", ".xlsx", ".csv", ".txt", ".md"].includes(extension))
return "document";
2502:   if ([".zip", ".7z", ".rar"].includes(extension)) return "archive";
2503:   if ([".c", ".h", ".cpp", ".hpp", ".py", ".js", ".mjs", ".ts", ".v", ".sv", ".vhd", ".vhd1", ".spice",
".cir", ".net", ".asc", ".sch", ".kicad_sch", ".kicad_pcb"].includes(extension)) return "engineering-file";
2504:   if (/^(https?:\/\/)?/i.test(clean) || /^www\./i.test(value) || looksLikeBareWebAddress(value)) return
"webpage";
2505:   return extension ? "file" : "link";
2506: }

```

## labelForUrlAssetKind - lines 2508-2522

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2508-2522 (15 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function labelForUrlAssetKind(kind = "link") {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2508: function labelForUrlAssetKind(kind = "link") {
2509:   return {
2510:     archive: "Archive link",
2511:     document: "Document link",
2512:     "engineering-file": "Engineering file link",
2513:     file: "File link",
2514:     github: "GitHub link",
2515:     "google-drive": "Google Drive link",
2516:     image: "Image URL",
2517:     linkedin: "LinkedIn link",

```

```

2518:     link: "Link",
2519:     pdf: "PDF link",
2520:     webpage: "Website link"
2521:   }[kind] || "Link";
2522: }

```

## displayNameFromUrl - lines 2524-2533

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2524-2533 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function displayNameFromUrl(value = "", fallback = "Linked asset") {                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeLinkTarget, URL, split, filter, pop, replace                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2524: function displayNameFromUrl(value = "", fallback = "Linked asset") {
2525:   const clean = normalizeLinkTarget(value, { assumeWeb: true });
2526:   try {
2527:     const parsed = new URL(clean);
2528:     const lastPath = parsed.pathname.split("/").filter(Boolean).pop();
2529:     return decodeURIComponent(lastPath || parsed.hostname.replace(/^www\./, "")) || fallback;
2530:   } catch {
2531:     return String(value || "").split("/").filter(Boolean).pop() || fallback;
2532:   }
2533: }

```

## linkifyRichTextNodes - lines 2535-2570

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2535-2570 (36 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function linkifyRichTextNodes(root) {                                                                                                                                                                 |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | createTreeWalker, acceptNode, closest, nextNode, push, forEach, createDocumentFragment, replace, b, append, createTextNode, slice                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2535: function linkifyRichTextNodes(root) {
2536:   const walker = document.createTreeWalker(root, NodeFilter.SHOW_TEXT, {
2537:     acceptNode(node) {
2538:       if (!node.textContent ||
!/(https?:\/\/|www\.)|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/)/i.test(node.textContent)) return
NodeFilter.FILTER_REJECT;
2539:       return node.parentElement?.closest("a") ? NodeFilter.FILTER_REJECT : NodeFilter.FILTER_ACCEPT;
2540:     }
2541:   });
2542:   const nodes = [];
2543:   let node = walker.nextNode();
2544:   while (node) {
2545:     nodes.push(node);
2546:     node = walker.nextNode();
2547:   }
2548: }

```

```

2549:   nodes.forEach((textNode) => {
2550:     const text = textNode.textContent || "";
2551:     const fragment = document.createDocumentFragment();
2552:     let cursor = 0;
2553:     text.replace(/b(https?:\\\/[^\s<]+|www\.[^\s<]+|(?:(github|linkedin|gitlab|bitbucket|drive|docs)\.com
\\\/[^\s<]+)/gi, (match, _url, offset) => {
2554:       if (offset > cursor) fragment.appendChild(document.createTextNode(text.slice(cursor, offset)));
2555:       const trailing = match.match(/[,.;!?:]+$/)?.[0] || "";
2556:       const clean = trailing ? match.slice(0, -trailing.length) : match;
2557:       const link = document.createElement("a");
2558:       link.href = normalizeLinkTarget(clean, { assumeWeb: true });
2559:       link.target = "_blank";
2560:       link.rel = "noreferrer";
2561:       link.textContent = clean;
2562:       fragment.appendChild(link);
2563:       if (trailing) fragment.appendChild(document.createTextNode(trailing));
2564:       cursor = offset + match.length;
2565:       return match;
2566:     });
2567:     if (cursor < text.length) fragment.appendChild(document.createTextNode(text.slice(cursor)));
2568:     textNode.replaceWith(fragment);
2569:   });
2570: }

```

## inlineStyleSignature - lines 2572-2583

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2572-2583 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function inlineStyleSignature(element) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | join                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2572: function inlineStyleSignature(element) {
2573:   const style = element?.style;
2574:   if (!style) return "";
2575:   return [
2576:     style.fontFamily || "",
2577:     style.fontSize || "",
2578:     style.color || "",
2579:     style.fontWeight || "",
2580:     style.fontStyle || "",
2581:     style.textDecoration || ""
2582:   ].join("|");
2583: }

```

## normalizeInlineStyleSpans - lines 2585-2605

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2585-2605 (21 source lines).                                                                                                                                                                    |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeInlineStyleSpans(root) {                                                                                                                                                            |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | querySelectorAll, forEach, remove, getAttribute, replaceWith, normalize, matches, inlineStyleSignature, append                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2585: function normalizeInlineStyleSpans(root) {
2586:   if (!root) return;
2587:   root.querySelectorAll("span.rich-inline-style, span[style]").forEach((span) => {
2588:     span.style.display = "inline";
2589:     if (!span.textContent) {
2590:       span.remove();
2591:       return;
2592:     }
2593:     if (!span.getAttribute("style")) span.replaceWith(...span.childNodes);
2594:   });
2595:   root.normalize();
2596:   root.querySelectorAll("span.rich-inline-style, span[style]").forEach((span) => {
2597:     let next = span.nextSibling;
2598:     while (next?.nodeType === Node.ELEMENT_NODE && next.matches("span.rich-inline-style, span[style]") &&
inlineStyleSignature(next) === inlineStyleSignature(span)) {
2599:       span.append(...next.childNodes);
2600:       const remove = next;
2601:       next = next.nextSibling;
2602:       remove.remove();
2603:     }
2604:   });
2605: }
```

## ***sanitizeRichInlineHtml - lines 2607-2651***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2607-2651 (45 source lines).                                                                                                                                                                    |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function sanitizeRichInlineHtml(value = "") {                                                                                                                                                         |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | createElement, Set, querySelectorAll, forEach, has, replaceWith, getAttribute, trim, normalizeLinkTarget, cleanFontFamily, normalizeFontPx, normalizeTextColor                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2607: function sanitizeRichInlineHtml(value = "") {
2608:   const template = document.createElement("template");
2609:   template.innerHTML = String(value || "");
2610:   const allowedTags = new Set(["A", "B", "BR", "EM", "I", "SPAN", "STRONG", "U"]);
2611:
2612:   [...template.content.querySelectorAll("*")].forEach((node) => {
2613:     if (!allowedTags.has(node.tagName)) {
2614:       node.replaceWith(...node.childNodes);
2615:       return;
2616:     }
2617:
2618:     const href = node.tagName === "A" ? String(node.getAttribute("href") || "").trim() : "";
2619:     const normalizedHref = normalizeLinkTarget(href, { assumeWeb: true });
2620:     const safeHref = /^(https?:|mailto:|tel:|#|\/)/i.test(normalizedHref) ? normalizedHref : "";
2621:     const fontFamily = cleanFontFamily(node.style.fontFamily || "");
2622:     const fontPx = normalizeFontPx(parseFloat(node.style.fontSize || ""));
2623:     const color = normalizeTextColor(node.style.color || "");
2624:     const rawFontWeight = node.style.fontWeight || "";
2625:     const fontWeight = /^(bold|[6-9]00)$/i.test(rawFontWeight)
2626:       ? "700"
2627:       : /^(normal|[1-5]00)$/i.test(rawFontWeight)
2628:         ? "400"
2629:         : "";
2630:     const fontStyle = node.style.fontStyle === "italic" ? "italic" : "";
2631:     const textDecoration = node.style.textDecoration.includes("underline") ? "underline" : "";
2632:
2633:     [...node.attributes].forEach((attribute) => node.removeAttribute(attribute.name));
2634:
2635:     if (node.tagName === "A" && safeHref) {
2636:       node.setAttribute("href", safeHref);
2637:       node.setAttribute("target", "_blank");
2638:       node.setAttribute("rel", "noreferrer");
2639:     }
2640:     if (fontFamily) node.style.fontFamily = fontFamily;
2641:     if (fontPx) node.style.fontSize = `${fontPx}px`;
2642:     if (color) node.style.color = color;
2643:     if (fontWeight) node.style.fontWeight = fontWeight;
```

```

2644:     if (fontStyle) node.style.fontStyle = fontStyle;
2645:     if (textDecoration) node.style.textDecoration = textDecoration;
2646:   });
2647:
2648:   linkifyRichTextNodes(template.content);
2649:   normalizeInlineStyleSpans(template.content);
2650:   return template.innerHTML;
2651: }

```

## ***displayTitle - lines 2653-2662***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2653-2662 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function displayTitle(value, fallback = "Untitled item") {                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2653: function displayTitle(value, fallback = "Untitled item") {
2654:   const title = String(value || fallback).trim();
2655:   const replacements = {
2656:     "Brief": "Overview",
2657:     "Project Brief": "Overview",
2658:     "Design Brief": "Design Overview",
2659:     "Simulation Brief": "Simulation Overview"
2660:   };
2661:   return replacements[title] || title;
2662: }

```

## ***cleanFontFamily - lines 2664-2671***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2664-2671 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function cleanFontFamily(value = "") {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, map, replace, trim, filter, slice, join                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2664: function cleanFontFamily(value = "") {
2665:   return String(value)
2666:     .split(",")
2667:     .map((item) => item.replace(/[\w\s-]/g, "").trim())
2668:     .filter(Boolean)
2669:     .slice(0, 3)
2670:     .join(", ");
2671: }

```

## ***normalizeFontPx - lines 2673-2676***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2673-2676 (4 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeFontPx(value) {                                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isFinite                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2673: function normalizeFontPx(value) {
2674:   const size = Number(value);
2675:   return Number.isFinite(size) && size >= 8 && size <= 24 ? String(size) : "";
2676: }

```

## ***normalizeTextColor - lines 2677-2684***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2677-2684 (8 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeTextColor(value = "") {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2677: function normalizeTextColor(value = "") {
2678:   const color = String(value || "").trim();
2679:   return /^#([0-9a-f]{3}|[0-9a-f]{6})$/i.test(color) ||
2680:     /^rgba?\(/i.test(color) ||
2681:     /^hsla?\(/i.test(color) ||
2682:     ? color
2683:     : "";
2684: }

```

## ***summaryRichPathFromTextPath - lines 2686-2688***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 2686-2688 (3 source lines).                            |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function summaryRichPathFromTextPath(pathValue = "") {       |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | replace                                                      |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 1 return statement(s).                |



| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2686: function summaryRichPathFromTextPath(pathValue = "") {
2687:     return String(pathValue || "").replace(/\.summary$/, ".summaryRich");
2688: }

```

## ***overviewFolderFromPath - lines 2690-2695***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2690-2695 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function overviewFolderFromPath(pathValue = "") {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | includes                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2690: function overviewFolderFromPath(pathValue = "") {
2691:     if (pathValue.includes("design.brief")) return "design-overview";
2692:     if (pathValue.includes("design.documentation")) return "documentation-overview";
2693:     if (pathValue.includes("design.simulation")) return "simulation-overview";
2694:     return "overview";
2695: }

```

## ***richBlocksFromValue - lines 2697-2699***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2697-2699 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richBlocksFromValue(rich, fallbackText = "") {                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | clone, textBlocksFromPlainText                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2697: function richBlocksFromValue(rich, fallbackText = "") {
2698:     return rich?.blocks?.length ? clone(rich.blocks) : textBlocksFromPlainText(fallbackText);
2699: }

```

## ***richTextStyle - lines 2700-2714***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2700-2714 (15 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function richTextStyle(block = {}) {                                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | cleanFontFamily, normalizeFontPx, normalizeTextColor, push, escapeHtml, join                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2700: function richTextStyle(block = {}) {
2701:     const styles = [];
2702:     const fontFamily = cleanFontFamily(block.fontFamily || "Arial") || "Arial";
2703:     const fontPx = normalizeFontPx(block.fontPx);
2704:     const color = normalizeTextColor(block.color || "");
2705:
2706:     if (fontFamily) styles.push(`font-family: ${fontFamily}`);
2707:     if (fontPx) styles.push(`font-size: ${fontPx}px`);
2708:     if (color) styles.push(`color: ${color}`);
2709:     if (block.bold) styles.push("font-weight: 700");
2710:     if (block.italic) styles.push("font-style: italic");
2711:     if (block.underline) styles.push("text-decoration: underline");
2712:
2713:     return styles.length ? ` style="${escapeHtml(styles.join("; "))}"` : "";
2714: }

```

## ***normalizeCropAspect - lines 2716-2718***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2716-2718 (3 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCropAspect(value = "") {                                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | includes                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2716: function normalizeCropAspect(value = "") {
2717:     return ["1 / 1", "4 / 3", "16 / 9", "3 / 4"].includes(value) ? value : "original";
2718: }

```

## ***normalizeCropZoom - lines 2720-2723***

| Item            | Explanation                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Source location | Lines 2720-2723 (4 source lines).                                             |
| Purpose         | Validates or normalizes input so later code receives safe predictable values. |
| Declaration     | function normalizeCropZoom(value = 1) {                                       |
| Touches         | filesystem                                                                    |
| Calls detected  | isFinite, min, max                                                            |
| API endpoints   | None detected inside this function body.                                      |
| Storage keys    | No local/session storage keys detected.                                       |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2720: function normalizeCropZoom(value = 1) {
2721:   const zoom = Number(value);
2722:   return Number.isFinite(zoom) ? Math.min(3, Math.max(1, zoom)) : 1;
2723: }

```

## ***normalizeCropPosition - lines 2725-2728***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2725-2728 (4 source lines).                                                                                                                                                                     |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeCropPosition(value = 50) {                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | isFinite, min, max                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2725: function normalizeCropPosition(value = 50) {
2726:   const position = Number(value);
2727:   return Number.isFinite(position) ? Math.min(100, Math.max(0, position)) : 50;
2728: }

```

## ***richImageCropStyle - lines 2730-2738***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2730-2738 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richImageCropStyle(block = {}) {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeCropAspect, normalizeCropZoom, normalizeCropPosition, push, escapeHtml, join                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2730: function richImageCropStyle(block = {}) {
2731:   const aspect = normalizeCropAspect(block.cropAspect);
2732:   const zoom = normalizeCropZoom(block.cropZoom);
2733:   const x = normalizeCropPosition(block.cropX);
2734:   const y = normalizeCropPosition(block.cropY);
2735:   const styles = [ `--crop-zoom: ${zoom}`, `--crop-x: ${x}%,` , `--crop-y: ${y}%,` ];
2736:   if (aspect !== "original") styles.push(`--crop-aspect: ${aspect}`);
2737:   return ` style="${escapeHtml(styles.join("; "))}"`;
2738: }

```

## ***richImageDownloadLink - lines 2740-2745***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2740-2745 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richImageDownloadLink(block = {}) {                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | escapeHtml, cleanRichImageTitle, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2740: function richImageDownloadLink(block = {}) {
2741:   const label = escapeHtml(cleanRichImageTitle(block) || block.caption || "Image file");
2742:   const url = block.url || "#";
2743:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(block, url) });
2744:   return `
```

## ***cleanRichImageTitle - lines 2747-2750***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2747-2750 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function cleanRichImageTitle(block = {}) {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2747: function cleanRichImageTitle(block = {}) {
2748:   const title = String(block.title || "").trim();
2749:   return /^pasted image\b/i.test(title) ? "" : title;
2750: }

```

## ***renderRichContent - lines 2752-2787***

| Item            | Explanation                                                                                                                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 2752-2787 (36 source lines).                                                                                                                                                                                       |
| Purpose         | Renders UI, markup, or display output.                                                                                                                                                                                   |
| Declaration     | function renderRichContent(rich, fallbackText = "") {                                                                                                                                                                    |
| Touches         | filesystem                                                                                                                                                                                                               |
| Calls detected  | textBlocksFromPlainText, map, includes, richImageDownloadLink, cleanRichImageTitle, normalizeLinkTarget, normalizeCropAspect, richImageCropStyle, escapeHtml, unwrapFormula, looksLikeWebOrContactText, renderInlineMath |
| API endpoints   | None detected inside this function body.                                                                                                                                                                                 |
| Storage keys    | No local/session storage keys detected.                                                                                                                                                                                  |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 7 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2752: function renderRichContent(rich, fallbackText = "") {
2753:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
2754:   return `
2755:     <div class="rich-content">
2756:       ${blocks.map((block) => {
2757:         const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
2758:         if (block.type === "image") {
2759:           if (block.display === "download") return richImageDownloadLink(block);
2760:           const title = cleanRichImageTitle(block);
2761:           const imageSrc = normalizeLinkTarget(block.url, { assumeWeb: true });
2762:           return `
2763:             <figure class="rich-image justify-${align}">
2764:               <span class="rich-image-viewport crop-${normalizeCropAspect(block.cropAspect)} ===
"original" ? "original" : "active"${`"${richImageCropStyle(block)}`>
2765:                 
2766:               </span>
2767:               ${title || block.caption} ? `<figcaption>${title ? `<strong>${escapeHtml(title)}</strong>`
: ""}${block.caption} ? `<span>${escapeHtml(block.caption)}</span>` : ""}</figcaption>` : ""
2768:             </figure>
2769:           `;
2770:         }
2771:         if (block.type === "formula") {
2772:           const formula = unwrapFormula(block.formula);
2773:           if (looksLikeWebOrContactText(formula)) {
2774:             return `<p class="rich-paragraph rich-text-normal
text-${align}">${renderInlineMath(formula)}</p>`;
2775:           }
2776:           return `<div class="rich-formula justify-${align}">${escapeHtml(formula)}</div>`;
2777:         }
2778:         if (block.type === "code") return renderRichCodeBlock(block);
2779:         const size = ["small", "normal", "large"].includes(block.fontSize) ? block.fontSize : "normal";
2780:         const content = block.html
2781:           ? sanitizeRichInlineHtml(block.html)
2782:           : renderInlineMath(block.text || "");
2783:         return `<p class="rich-paragraph rich-text-${size}
text-${align}"${richTextStyle(block)}>${content}</p>`;
2784:       }).join("")}
2785:     </div>
2786:   `;
2787: }

```

## ***renderRichFieldContent - lines 2789-2799***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2789-2799 (11 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderRichFieldContent(rich, fallbackText = "") {                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | textBlocksFromPlainText, map, includes, sanitizeRichInlineHtml, renderInlineMath, richTextStyle, filter, join                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2789: function renderRichFieldContent(rich, fallbackText = "") {
2790:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
2791:   return blocks.map((block) => {
2792:     if (block.type !== "paragraph") return "";
2793:     const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
2794:     const content = block.html
2795:       ? sanitizeRichInlineHtml(block.html)
2796:       : renderInlineMath(block.text || "");
2797:     return `<span class="rich-field-line text-${align}"${richTextStyle(block)}>${content}</span>`;
2798:   }).filter(Boolean).join("");
2799: }

```

## ***saveSelectedProjectToPortfolio - lines 2801-2817***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2801-2817 (17 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | function saveSelectedProjectToPortfolio() {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, parseProjectForPortfolio, syncParsedPortfolioGlobalsIntoSaved, filter, Set, map, has, findIndex, max, splice, markDraftNeedsSave, refreshOpenPreviews                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2801: function saveSelectedProjectToPortfolio() {
2802:   const project = selectedProject();
2803:   if (!project) return false;
2804:   const parsedProject = parseProjectForPortfolio(project);
2805:   syncParsedPortfolioGlobalsIntoSaved();
2806:   const nextProjects = (savedPortfolioCatalog.projects || []).filter((item) => item.id !== project.id);
2807:   const workingIds = new Set((catalog.projects || []).map((item) => item.id));
2808:   savedPortfolioCatalog.projects = nextProjects.filter((item) => workingIds.has(item.id));
2809:   const workingIndex = catalog.projects.findIndex((item) => item.id === project.id);
2810:   const insertAt = Math.max(0, workingIndex);
2811:   savedPortfolioCatalog.projects.splice(insertAt, 0, parsedProject);
2812:   markDraftNeedsSave();
2813:   refreshOpenPreviews();
2814:   setStatus(`${project.title} saved into the portfolio preview.`);
2815:   updateBuilderWorkflow();
2816:   return true;
2817: }
```

## ***nextProjects - lines 2806-2814***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2806-2814 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const nextProjects = (savedPortfolioCatalog.projects    []).filter((item) => item.id !== project.id);                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, Set, map, has, findIndex, max, splice, markDraftNeedsSave, refreshOpenPreviews, setStatus                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2806:   const nextProjects = (savedPortfolioCatalog.projects || []).filter((item) => item.id !== project.id);
2807:   const workingIds = new Set((catalog.projects || []).map((item) => item.id));
2808:   savedPortfolioCatalog.projects = nextProjects.filter((item) => workingIds.has(item.id));
2809:   const workingIndex = catalog.projects.findIndex((item) => item.id === project.id);
2810:   const insertAt = Math.max(0, workingIndex);
2811:   savedPortfolioCatalog.projects.splice(insertAt, 0, parsedProject);
2812:   markDraftNeedsSave();
2813:   refreshOpenPreviews();
2814:   setStatus(`${project.title} saved into the portfolio preview.`);
```

## ***removeProjectFromPortfolio - lines 2819-2821***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2819-2821 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function removeProjectFromPortfolio(projectId) {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter                                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2819: function removeProjectFromPortfolio(projectId) {
2820:   savedPortfolioCatalog.projects = (savedPortfolioCatalog.projects || []).filter((item) => item.id !==
projectId);
2821: }

```

## ***deleteProjectById - lines 2823-2836***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2823-2836 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function deleteProjectById(projectId) {                                                                                                                                                               |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | find, filter, removeProjectFromPortfolio, closeDialogElement, setStatus, scheduleAutosave, renderAll                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2823: function deleteProjectById(projectId) {
2824:   const project = catalog.projects.find((item) => item.id === projectId);
2825:   if (!project) return;
2826:   catalog.projects = catalog.projects.filter((item) => item.id !== project.id);
2827:   removeProjectFromPortfolio(project.id);
2828:   selectedProjectId = catalog.projects[0]?.id || "";
2829:   activeSectionId = "brief";
2830:   if (projectDialog?.open && project.id === projectWindowTitle.dataset.projectId) {
2831:     closeDialogElement(projectDialog);
2832:   }
2833:   setStatus("Project deleted locally.");
2834:   scheduleAutosave();
2835:   renderAll();
2836: }

```

## ***openDeleteConfirm - lines 2838-2843***

| Item            | Explanation                                                                    |
|-----------------|--------------------------------------------------------------------------------|
| Source location | Lines 2838-2843 (6 source lines).                                              |
| Purpose         | Controls a window, dialog, panel, menu, or visible state.                      |
| Declaration     | function openDeleteConfirm({ title = "Delete project", message, onConfirm }) { |
| Touches         | filesystem                                                                     |
| Calls detected  | showModal                                                                      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2838: function openDeleteConfirm({ title = "Delete project", message, onConfirm }) {
2839:   pendingDeleteAction = onConfirm;
2840:   deleteConfirmTitle.textContent = title;
2841:   deleteConfirmMessage.textContent = message;
2842:   deleteConfirmDialog.showModal();
2843: }

```

## ***requestProjectDelete - lines 2845-2853***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2845-2853 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function requestProjectDelete(projectId) {                                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | find, openDeleteConfirm, deleteProjectById                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2845: function requestProjectDelete(projectId) {
2846:   const project = catalog.projects.find((item) => item.id === projectId);
2847:   if (!project) return;
2848:   openDeleteConfirm({
2849:     title: "Delete project",
2850:     message: "Are you sure you want to delete this project?",
2851:     onConfirm: () => deleteProjectById(project.id)
2852:   });
2853: }

```

## ***parserItem - lines 2855-2866***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2855-2866 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parserItem(title, description = "", url = "", meta = "", kind = "text", rich = null, children = []) {                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, clone                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2855: function parserItem(title, description = "", url = "", meta = "", kind = "text", rich = null, children =
[] ) {

```



```

2856:   const item = {
2857:     children: children.filter((child) => child && (child.title || child.description || child.url ||
child.children?.length)),
2858:     description: description || "",
2859:     kind,
2860:     meta: meta || "",
2861:     title: title || "Untitled item",
2862:     url: url || ""
2863:   };
2864:   if (rich?.blocks?.length) item.rich = clone(rich);
2865:   return item;
2866: }

```

## ***canonicalTemplateId - lines 2868-2870***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2868-2870 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function canonicalTemplateId(id) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2868: function canonicalTemplateId(id) {
2869:   return legacyTemplateSkins[id] || id || "";
2870: }

```

## ***projectTemplateId - lines 2872-2874***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2872-2874 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectTemplateId(project) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | canonicalTemplateId                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2872: function projectTemplateId(project) {
2873:   return canonicalTemplateId(project?.portfolioView?.template?.id || project?.templateId || "");
2874: }

```

## ***projectTemplateFor - lines 2876-2879***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2876-2879 (4 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function projectTemplateFor(project) {                       |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectTemplateId, find                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2876: function projectTemplateFor(project) {
2877:   const templateId = projectTemplateId(project);
2878:   return templates.find((template) => template.id === templateId) || {};
2879: }

```

## ***projectTemplateClass - lines 2881-2884***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2881-2884 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectTemplateClass(project) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectTemplateId                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2881: function projectTemplateClass(project) {
2882:   const templateId = projectTemplateId(project);
2883:   return templateId ? `project-template project-template-${templateId}` : "project-template-white";
2884: }

```

## ***responsiveClassName - lines 2886-2891***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2886-2891 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function responsiveClassName(value, fallback) {                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | toLowerCase, replace                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2886: function responsiveClassName(value, fallback) {
2887:   return String(value || fallback || "")
2888:     .toLowerCase()
2889:     .replace(/^[a-z0-9]+/g, "-")
2890:     .replace(/^-|-$/g, "") || fallback;

```

2891: }

## ***projectResponsiveClass - lines 2893-2896***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2893-2896 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectResponsiveClass(project) {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectResponsiveProfile, responsiveClassName                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2893: function projectResponsiveClass(project) {
2894:   const profile = projectResponsiveProfile(project);
2895:   return `responsive-project responsive-density-${responsiveClassName(profile.density, "balanced")}
responsive-card-${responsiveClassName(profile.cardLayout, "single")}`;
2896: }
```

## ***responsiveStyleValues - lines 2898-2906***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2898-2906 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function responsiveStyleValues(project) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectResponsiveProfile                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
2898: function responsiveStyleValues(project) {
2899:   const profile = projectResponsiveProfile(project);
2900:   return {
2901:     "--responsive-section-card-min": profile.sectionCardMin,
2902:     "--responsive-content-max": profile.contentMax,
2903:     "--responsive-media-max": profile.mediaMax,
2904:     "--responsive-touch-target": profile.touchTarget
2905:   };
2906: }
```

## ***hasPublicTemplate - lines 2908-2910***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2908-2910 (3 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function hasPublicTemplate(project) {                        |
| Touches         | Mostly local calculations or local UI state.                 |
| Calls detected  | projectTemplateId                                            |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2908: function hasPublicTemplate(project) {
2909:   return Boolean(projectTemplateId(project));
2910: }

```

## ***projectTemplateVisual - lines 2912-2914***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2912-2914 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectTemplateVisual(project) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectTemplateFor                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2912: function projectTemplateVisual(project) {
2913:   return project?.portfolioView?.template?.visual || project?.templateVisual ||
projectTemplateFor(project).visual || null;
2914: }

```

## ***projectTemplateStyle - lines 2916-2934***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2916-2934 (19 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectTemplateStyle(project, accent) {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectTemplateVisual, responsiveStyleValues, entries, filter, map, join                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2916: function projectTemplateStyle(project, accent) {
2917:   const visual = projectTemplateVisual(project) || {};
2918:   const values = {
2919:     "--accent": accent,
2920:     "--template-bg": visual.background,
2921:     "--template-panel": visual.panel,
2922:     "--template-accent": visual.accent,
2923:     "--template-hover": visual.hover,
2924:     "--template-click": visual.click,
2925:     "--template-click-text": visual.clickText,

```

```

2926:     "--template-line": visual.line,
2927:     "--template-text": visual.text,
2928:     ...responsiveStyleValues(project)
2929:   };
2930:   return Object.entries(values)
2931:     .filter(([key, value]) => value)
2932:     .map(([key, value]) => `${key}: ${value}`)
2933:     .join("; ");
2934: }

```

## applyProjectTemplateToElement - lines 2936-2970

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2936-2970 (35 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function applyProjectTemplateToElement(element, project, accent) {                                                                                                                                    |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | filter, startsWith, forEach, remove, projectTemplateClass, split, add, projectResponsiveClass, projectTemplateVisual, responsiveStyleValues, entries, setProperty                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2936: function applyProjectTemplateToElement(element, project, accent) {
2937:   if (!element) return;
2938:   [...element.classList]
2939:     .filter((className) =>
2940:       className === "project-template" ||
2941:       className === "project-template-white" ||
2942:       className === "responsive-project" ||
2943:       className.startsWith("project-template-") ||
2944:       className.startsWith("responsive-density-") ||
2945:       className.startsWith("responsive-card-")
2946:     )
2947:     .forEach((className) => element.classList.remove(className));
2948:   projectTemplateClass(project).split(" ").forEach((className) => element.classList.add(className));
2949:   projectResponsiveClass(project).split(" ").forEach((className) => element.classList.add(className));
2950:   const visual = projectTemplateVisual(project) || {};
2951:   const values = {
2952:     "--accent": accent,
2953:     "--template-bg": visual.background,
2954:     "--template-panel": visual.panel,
2955:     "--template-accent": visual.accent,
2956:     "--template-hover": visual.hover,
2957:     "--template-click": visual.click,
2958:     "--template-click-text": visual.clickText,
2959:     "--template-line": visual.line,
2960:     "--template-text": visual.text,
2961:     ...responsiveStyleValues(project)
2962:   };
2963:   Object.entries(values).forEach(([key, value]) => {
2964:     if (value) {
2965:       element.style.setProperty(key, value);
2966:     } else {
2967:       element.style.removeProperty(key);
2968:     }
2969:   });
2970: }

```

## resourcesFrom - lines 2972-2988

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 2972-2988 (17 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function resourcesFrom(project, key) {                       |
| Touches         | Mostly local calculations or local UI state.                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | map, parserItem                                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2972: function resourcesFrom(project, key) {
2973:   return (project[key] || []).map((item) => {
2974:     if (key === "tests") {
2975:       return parserItem(item.name, item.result || item.method, item.artifact, item.method || item.status
|| "Test", "test", item.richDescription);
2976:     }
2977:     if (key === "pcbs") {
2978:       return parserItem(item.name, item.revision || item.status, item.artifact, item.status || "PCB",
"pcb", item.richDescription);
2979:     }
2980:     if (key === "media") {
2981:       return parserItem(item.title, item.caption, item.url, item.status || "Image", "image",
item.richDescription);
2982:     }
2983:     if (key === "links") {
2984:       return parserItem(item.label, item.description || "", item.url, "Link", "link",
item.richDescription);
2985:     }
2986:     return parserItem(item.title || item.name || item.label, item.description || item.caption, item.url
|| item.artifact, item.type || item.status || "Document", "document", item.richDescription);
2987:   });
2988: }

```

## ***resourcesFromItems - lines 2990-3003***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 2990-3003 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function resourcesFromItems(items = [], fallbackKind = "document") {                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, parserItem                                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

2990: function resourcesFromItems(items = [], fallbackKind = "document") {
2991:   return (items || []).map((item) => {
2992:     const url = item.url || item.artifact || "";
2993:     const kind = item.kind || fallbackKind;
2994:     return parserItem(
2995:       item.title || item.name || item.label || "Untitled item",
2996:       item.description || item.caption || item.result || item.method || "",
2997:       url,
2998:       item.type || item.status || item.method || "",
2999:       kind,
3000:       item.richDescription
3001:     );
3002:   });
3003: }

```

## ***parsedSubsection - lines 3005-3007***

| Item            | Explanation                       |
|-----------------|-----------------------------------|
| Source location | Lines 3005-3007 (3 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedSubsection(title, description = "", items = [], rich = null) {                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | parserItem                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3005: function parsedSubsection(title, description = "", items = [], rich = null) {
3006:   return parserItem(title, description, "", "Subsection", "subsection", rich, items);
3007: }

```

## ***richHasContent - lines 3009-3018***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3009-3018 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richHasContent(rich) {                                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | some                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3009: function richHasContent(rich) {
3010:   return Boolean(rich?.blocks?.some((block) =>
3011:     block.type === "image" && block.url ||
3012:     block.type === "formula" && block.formula ||
3013:     block.type === "code" && block.code ||
3014:     block.text ||
3015:     block.title ||
3016:     block.caption
3017:   ));
3018: }

```

## ***parsedItemHasContent - lines 3020-3031***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 3020-3031 (12 source lines).                           |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function parsedItemHasContent(item) {                        |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | richHasContent, some, includes                               |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 5 return statement(s).                |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3020: function parsedItemHasContent(item) {
3021:   if (!item) return false;
3022:   if (item.kind === "summary") return Boolean(item.description || richHasContent(item.rich));
3023:   const children = item.children || item.items || [];
3024:   if (item.kind === "subsection") {
3025:     return Boolean(item.description || item.url || richHasContent(item.rich) ||
children.some(parsedItemHasContent));
3026:   }
3027:   if (["tool", "language"].includes(item.kind)) {
3028:     return Boolean(item.title || item.description || item.url || richHasContent(item.rich) ||
children.some(parsedItemHasContent));
3029:   }
3030:   return Boolean(item.description || item.url || richHasContent(item.rich) ||
children.some(parsedItemHasContent));
3031: }

```

### ***parsedSectionHasContent - lines 3033-3035***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3033-3035 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedSectionHasContent(section) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | richHasContent, some                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3033: function parsedSectionHasContent(section) {
3034:   return Boolean(section?.description || richHasContent(section?.rich) || (section?.items ||
[]).some(parsedItemHasContent));
3035: }

```

### ***responsiveChildren - lines 3037-3039***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3037-3039 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function responsiveChildren(node) {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3037: function responsiveChildren(node) {
3038:   return node?.items || node?.children || [];
3039: }

```



## ***richContainsMedia - lines 3041-3043***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3041-3043 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richContainsMedia(rich) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | some                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3041: function richContainsMedia(rich) {
3042:   return Boolean(rich?.blocks?.some((block) => block.type === "image" && block.url));
3043: }
```

## ***responsiveNodeHasMedia - lines 3045-3051***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3045-3051 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function responsiveNodeHasMedia(node) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | richContainsMedia, responsiveChildren, some                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3045: function responsiveNodeHasMedia(node) {
3046:   if (!node) return false;
3047:   if (node.kind === "image" || richContainsMedia(node.rich)) return true;
3048:   const url = String(node.url || "");
3049:   if (/\. (apng|avif|gif|jpe?g|png|svg|webp)(\?|#|$)/i.test(url)) return true;
3050:   return responsiveChildren(node).some(responsiveNodeHasMedia);
3051: }
```

## ***responsiveNodeCount - lines 3053-3056***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 3053-3056 (4 source lines).                            |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function responsiveNodeCount(node) {                         |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | parsedItemHasContent, responsiveChildren, reduce             |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 2 return statement(s).                |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3053: function responsiveNodeCount(node) {
3054:   if (!node || !parsedItemHasContent(node)) return 0;
3055:   return 1 + responsiveChildren(node).reduce((count, child) => count + responsiveNodeCount(child), 0);
3056: }

```

## ***responsiveNodeDepth - lines 3058-3063***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3058-3063 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function responsiveNodeDepth(node) {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | parsedItemHasContent, responsiveChildren, filter, max, map                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3058: function responsiveNodeDepth(node) {
3059:   if (!node || !parsedItemHasContent(node)) return 0;
3060:   const children = responsiveChildren(node).filter(parsedItemHasContent);
3061:   if (!children.length) return 1;
3062:   return 1 + Math.max(...children.map(responsiveNodeDepth));
3063: }

```

## ***buildResponsiveProfile - lines 3065-3096***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3065-3096 (32 source lines).                                                                                                                                                                    |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function buildResponsiveProfile(project, parsedSections = []) {                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, parsedSectionHasContent, reduce, responsiveNodeCount, max, map, some, responsiveNodeHasMedia, hasPublicTemplate                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3065: function buildResponsiveProfile(project, parsedSections = []) {
3066:   const visibleSections = (parsedSections || []).filter((section) => section?.id !== "brief" &&
parsedSectionHasContent(section));
3067:   const itemCount = visibleSections.reduce(
3068:     (count, section) => count + (section.items || []).reduce((sum, item) => sum +
responsiveNodeCount(item), 0),
3069:     0
3070:   );
3071:   const maxDepth = visibleSections.reduce(
3072:     (depth, section) => Math.max(depth, ...(section.items || []).map(responsiveNodeDepth), 1),
3073:     1
3074:   );

```

```

3075:   const hasMedia = visibleSections.some((section) => responsiveNodeHasMedia(section));
3076:   const density = itemCount >= 18 || visibleSections.length >= 6 || maxDepth >= 4
3077:   ? "dense"
3078:   : itemCount <= 6 && visibleSections.length <= 2
3079:   ? "simple"
3080:   : "balanced";
3081:   return {
3082:     version: 1,
3083:     breakpoints: {
3084:       mobile: 640,
3085:       tablet: 900,
3086:       desktop: 1180
3087:     },
3088:     cardLayout: hasPublicTemplate(project) ? "single" : "split",
3089:     contentMax: hasMedia || maxDepth >= 3 ? "1180px" : "980px",
3090:     density,
3091:     mediaMax: hasMedia ? "860px" : "720px",
3092:     sectionCardMin: density === "dense" ? "210px" : density === "simple" ? "280px" : "240px",
3093:     touchTarget: "44px",
3094:     windowMode: "full-screen"
3095:   };
3096: }

```

## visibleSections - lines 3066-3095

| Item               | Explanation                                                                                                                                                                                                                                          |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3066-3095 (30 source lines).                                                                                                                                                                                                                   |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                                                                         |
| Declaration        | <code>const visibleSections = (parsedSections    []).filter((section) =&gt; section?.id !== "brief" &amp;&amp; parsedSectionHasContent(section));</code>                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                                                                         |
| Calls detected     | <code>filter</code> , <code>parsedSectionHasContent</code> , <code>reduce</code> , <code>responsiveNodeCount</code> , <code>max</code> , <code>map</code> , <code>some</code> , <code>responsiveNodeHasMedia</code> , <code>hasPublicTemplate</code> |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                             |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                              |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                                                                        |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                                |

Source excerpt:

```

3066:   const visibleSections = (parsedSections || []).filter((section) => section?.id !== "brief" &&
parsedSectionHasContent(section));
3067:   const itemCount = visibleSections.reduce(
3068:     (count, section) => count + (section.items || []).reduce((sum, item) => sum +
responsiveNodeCount(item), 0),
3069:     0
3070:   );
3071:   const maxDepth = visibleSections.reduce(
3072:     (depth, section) => Math.max(depth, ...(section.items || []).map(responsiveNodeDepth), 1),
3073:     1
3074:   );
3075:   const hasMedia = visibleSections.some((section) => responsiveNodeHasMedia(section));
3076:   const density = itemCount >= 18 || visibleSections.length >= 6 || maxDepth >= 4
3077:   ? "dense"
3078:   : itemCount <= 6 && visibleSections.length <= 2
3079:   ? "simple"
3080:   : "balanced";
3081:   return {
3082:     version: 1,
3083:     breakpoints: {
3084:       mobile: 640,
3085:       tablet: 900,
3086:       desktop: 1180
3087:     },
3088:     cardLayout: hasPublicTemplate(project) ? "single" : "split",
3089:     contentMax: hasMedia || maxDepth >= 3 ? "1180px" : "980px",
3090:     density,
3091:     mediaMax: hasMedia ? "860px" : "720px",
3092:     sectionCardMin: density === "dense" ? "210px" : density === "simple" ? "280px" : "240px",
3093:     touchTarget: "44px",
3094:     windowMode: "full-screen"
3095:   };

```

## projectResponsiveProfile - lines 3098-3107

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3098-3107 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectResponsiveProfile(project) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | buildResponsiveProfile                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3098: function projectResponsiveProfile(project) {
3099:   const savedProfile = project?.portfolioView?.responsive;
3100:   if (savedProfile?.version) {
3101:     return {
3102:       ...buildResponsiveProfile(project, project?.portfolioView?.sections || []),
3103:       ...savedProfile
3104:     };
3105:   }
3106:   return buildResponsiveProfile(project, project?.portfolioView?.sections || []);
3107: }

```

## ***parsedSection - lines 3109-3118***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3109-3118 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedSection(id, title, items = [], description = "", rich = null) {                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, richHasContent, clone                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3109: function parsedSection(id, title, items = [], description = "", rich = null) {
3110:   const section = {
3111:     description,
3112:     id,
3113:     items: items.filter(parsedItemHasContent),
3114:     title
3115:   };
3116:   if (richHasContent(rich)) section.rich = clone(rich);
3117:   return section;
3118: }

```

## ***parsedDesignSection - lines 3120-3146***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 3120-3146 (27 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function parsedDesignSection(project) {                      |
| Touches         | Mostly local calculations or local UI state.                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | isAnalogProject, ensureDesignModel, resourcesFrom, resourcesFromItems, parsedSubsection, parsedSection                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3120: function parsedDesignSection(project) {
3121:   if (isAnalogProject(project)) {
3122:     const design = ensureDesignModel(project);
3123:     const documentationItems = [
3124:       ...resourcesFrom(project, "documents"),
3125:       ...resourcesFromItems(design.documentation.files, "document"),
3126:       parsedSubsection("References", "", resourcesFromItems(design.documentation.references,
"reference")),
3127:       parsedSubsection("Math / Analysis", "", resourcesFromItems(design.documentation.mathAnalysis,
"analysis"))
3128:     ];
3129:     const boardMediaItems = [
3130:       ...resourcesFrom(project, "pcbs"),
3131:       ...resourcesFrom(project, "media")
3132:     ];
3133:     return parsedSection("design", "Design", [
3134:       parsedSubsection("Design Overview", design.brief.summary || "", [
3135:         ...resourcesFromItems(design.brief.files, "document")
3136:       ], design.brief.summaryRich),
3137:       parsedSubsection("Documentation", design.documentation.summary || "", documentationItems,
design.documentation.summaryRich),
3138:       parsedSubsection("PCB and Visual Evidence", "", boardMediaItems)
3139:     ]);
3140:   }
3141:   return parsedSection("design", "Design", [
3142:     ...resourcesFrom(project, "documents"),
3143:     ...resourcesFrom(project, "pcbs"),
3144:     ...resourcesFrom(project, "media")
3145:   ]);
3146: }

```

## ***parsedSimulationSection - lines 3148-3156***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3148-3156 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedSimulationSection(project) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isAnalogProject, ensureDesignModel, parsedSection, parsedSubsection, resourcesFromItems                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3148: function parsedSimulationSection(project) {
3149:   if (!isAnalogProject(project)) return null;
3150:   const design = ensureDesignModel(project);
3151:   return parsedSection("simulation", "Simulation", [
3152:     parsedSubsection("Simulation Overview", design.simulation.summary || "", [],
design.simulation.summaryRich),
3153:     parsedSubsection("Simulation Files", "", resourcesFromItems(design.simulation.files, "simulation-
file")),
3154:     parsedSubsection("Simulation Results", "", resourcesFromItems(design.simulation.results, "simulation-
result"))
3155:   ]);
3156: }

```

## ***parsedToolsSection - lines 3158-3169***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3158-3169 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedToolsSection(project) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, parserItem, toolName, toolDescription, itemTitle, parsedSection, resourcesFrom                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3158: function parsedToolsSection(project) {
3159:   const toolItems = (project.tools || []).map((tool) => parserItem(toolName(tool), toolDescription(tool),
3160:   "", "Tool", "tool", tool?.richDescription));
3161:   const languageItems = (project.languages || []).map((language) => {
3162:     const label = typeof language === "string" ? language : itemTitle(language);
3163:     return parserItem(label, "", "", "Language", "language");
3164:   });
3165:   return parsedSection("tools", "Tools", [
3166:     ...toolItems,
3167:     ...languageItems,
3168:     ...resourcesFrom(project, "links")
3169:   ]);
3170: }
```

## ***toolItems - lines 3159-3163***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3159-3163 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const toolItems = (project.tools    []).map((tool) => parserItem(toolName(tool), toolDescription(tool), "", "Tool", "tool", tool?.richDescription));                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, parserItem, toolName, toolDescription, itemTitle                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3159:   const toolItems = (project.tools || []).map((tool) => parserItem(toolName(tool), toolDescription(tool),
3160:   "", "Tool", "tool", tool?.richDescription));
3161:   const languageItems = (project.languages || []).map((language) => {
3162:     const label = typeof language === "string" ? language : itemTitle(language);
3163:     return parserItem(label, "", "", "Language", "language");
3164:   });
```

## ***languageItems - lines 3160-3163***

| Item            | Explanation                                                         |
|-----------------|---------------------------------------------------------------------|
| Source location | Lines 3160-3163 (4 source lines).                                   |
| Purpose         | Part of the private builder frontend and editing experience.        |
| Declaration     | const languageItems = (project.languages    []).map((language) => { |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, itemTitle, parserItem                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3160:   const languageItems = (project.languages || []).map((language) => {
3161:     const label = typeof language === "string" ? language : itemTitle(language);
3162:     return parserItem(label, "", "", "Language", "language");
3163:   });

```

## ***parseProjectForPortfolio - lines 3171-3219***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3171-3219 (49 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parseProjectForPortfolio(project) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectTemplateFor, sectionOptions, flatMap, parsedSection, parserItem, parsedDesignSection, parsedSimulationSection, resourcesFrom, parsedToolsSection, startsWith, replace, find                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 8 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3171: function parseProjectForPortfolio(project) {
3172:   const template = projectTemplateFor(project);
3173:   const parsedSections = sectionOptions(project).flatMap((section) => {
3174:     if (section.id === "brief") {
3175:       return parsedSection("brief", "Overview", [
3176:         parserItem("Overview", project.summary || "", "", "Overview", "summary", project.summaryRich)
3177:       ]);
3178:     }
3179:     if (section.id === "design") return parsedDesignSection(project);
3180:     if (section.id === "simulation") return parsedSimulationSection(project);
3181:     if (section.id === "tests") return parsedSection("tests", "Tests", resourcesFrom(project, "tests"));
3182:     if (section.id === "tools") return parsedToolsSection(project);
3183:     if (section.id.startsWith("custom:")) {
3184:       const sectionId = section.id.replace("custom:", "");
3185:       const custom = (project.sections || []).find((item) => item.id === sectionId);
3186:       return parsedSection(
3187:         `custom:${sectionId}`,
3188:         custom?.title || section.label,
3189:         (custom?.items || []).map((item) => item.url
3190:           ? parserItem(item.title, item.description, item.url, item.type || item.status || "File",
"file", item.richDescription)
3191:           : parsedSubsection(
3192:             item.title,
3193:             item.description,
3194:             resourcesFromItems(item.children || item.items || item.files || [], "file"),
3195:             item.richDescription
3196:           )),
3197:         custom?.description || "",
3198:         custom?.richDescription
3199:       );
3200:     }
3201:     return null;
3202:   }).filter(parsedSectionHasContent);
3203:
3204:   return {
3205:     ...clone(project),
3206:     portfolioView: {

```

```

3207:     builtAt: new Date().toISOString(),
3208:     template: {
3209:       group: "Appearance",
3210:       id: projectTemplateId(project),
3211:       label: template.label || project.templateLabel || "",
3212:       visual: template.visual ? clone(template.visual) : projectTemplateVisual(project)
3213:     },
3214:     responsive: buildResponsiveProfile(project, parsedSections),
3215:     title: project.title,
3216:     sections: parsedSections
3217:   }
3218: };
3219: }

```

## custom - lines 3185-3187

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3185-3187 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>const custom = (project.sections    []).find((item) =&gt; item.id === sectionId);</code>                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>find</code> , <code>parsedSection</code>                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3185:     const custom = (project.sections || []).find((item) => item.id === sectionId);
3186:     return parsedSection(
3187:       custom: `${sectionId}`,

```

## refreshOpenPreviews - lines 3221-3230

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3221-3230 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>function refreshOpenPreviews() {</code>                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>selectedProject</code> , <code>find</code> , <code>renderProjectWebsitePreview</code> , <code>renderPreview</code>                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3221: function refreshOpenPreviews() {
3222:   if (projectPreviewDialog?.open) {
3223:     const project = selectedProject();
3224:     const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project?.id);
3225:     renderProjectWebsitePreview(savedProject || project);
3226:   }
3227:   if (portfolioPreviewDialog?.open) {
3228:     renderPreview();
3229:   }
3230: }

```

## savedProject - lines 3224-3227



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3224-3227 (4 source lines).                                                                                                                                                                     |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | const savedProject = (savedPortfolioCatalog.projects    []).find((item) => item.id === project?.id);                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, renderProjectWebsitePreview                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3224:     const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project?.id);
3225:     renderProjectWebsitePreview(savedProject || project);
3226:   }
3227:   if (portfolioPreviewDialog?.open) {

```

## ***saveSelectedProjectAndClose - lines 3232-3236***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3232-3236 (5 source lines).                                                                                                                                                                     |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | function saveSelectedProjectAndClose() {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | saveSelectedProjectToPortfolio, closeDialogElement                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3232: function saveSelectedProjectAndClose() {
3233:   if (saveSelectedProjectToPortfolio()) {
3234:     closeDialogElement(projectDialog);
3235:   }
3236: }

```

## ***commitActiveProjectEdits - lines 3238-3246***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3238-3246 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function commitActiveProjectEdits() {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | saveVisibleRichEditors, saveRichSummary                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3238: async function commitActiveProjectEdits() {
3239:   saveVisibleRichEditors();
3240:
3241:   if (summaryEditorProjectId) {
3242:     return saveRichSummary();
3243:   }
3244:
3245:   return true;
3246: }
```

## ***saveAllSections - lines 3248-3259***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3248-3259 (12 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | async function saveAllSections(options = {}) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | commitActiveProjectEdits, syncParsedPortfolioGlobalsIntoSaved, map, parseProjectForPortfolio, markDraftNeedsSave, closeDialogElement, refreshOpenPreviews, setStatus, updateBuilderWorkflow           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3248: async function saveAllSections(options = {}) {
3249:   const settings = options && options.currentTarget ? {} : options;
3250:   if (!(await commitActiveProjectEdits())) return false;
3251:   syncParsedPortfolioGlobalsIntoSaved();
3252:   savedPortfolioCatalog.projects = (catalog.projects || []).map((project) =>
parseProjectForPortfolio(project));
3253:   markDraftNeedsSave();
3254:   if (projectDialog?.open) closeDialogElement(projectDialog);
3255:   if (settings.refreshOpenPreviews !== false) refreshOpenPreviews();
3256:   setStatus(`Saved ${savedPortfolioCatalog.projects.length}
project${savedPortfolioCatalog.projects.length === 1 ? "" : "s"} into the portfolio preview.`);
3257:   updateBuilderWorkflow();
3258:   return true;
3259: }
```

## ***selectedProject - lines 3261-3263***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3261-3263 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function selectedProject() {                                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3261: function selectedProject() {
3262:   return catalog.projects.find((project) => project.id === selectedProjectId) || catalog.projects[0];
3263: }
```

## ***sectionWindowId - lines 3265-3269***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3265-3269 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function sectionWindowId(sectionId) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | includes                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3265: function sectionWindowId(sectionId) {
3266:   if (["documents", "pcbs", "media"].includes(sectionId)) return "design";
3267:   if (["links", "highlights", "tools", "languages"].includes(sectionId)) return "tools";
3268:   return sectionId || "brief";
3269: }
```

## ***resetProjectWindowScroll - lines 3271-3279***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3271-3279 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function resetProjectWindowScroll() {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | requestAnimationFrame                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3271: function resetProjectWindowScroll() {
3272:   projectWindowContent.scrollTop = 0;
3273:   requestAnimationFrame(() => {
3274:     projectWindowContent.scrollTop = 0;
3275:     requestAnimationFrame(() => {
3276:       projectWindowContent.scrollTop = 0;
3277:     });
3278:   });
3279: }
```

## ***openProjectWindow - lines 3281-3302***

| Item            | Explanation                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 3281-3302 (22 source lines).                                                                                       |
| Purpose         | Controls a window, dialog, panel, menu, or visible state.                                                                |
| Declaration     | function openProjectWindow(projectId, sectionId = "brief") {                                                             |
| Touches         | browser/Electron DOM                                                                                                     |
| Calls detected  | sectionWindowId, removeAttribute, remove, renderAll, add, showModal, resetProjectWindowScroll, updateDialogWindowButtons |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3281: function openProjectWindow(projectId, sectionId = "brief") {
3282:   selectedProjectId = projectId;
3283:   projectWindowTitle.dataset.projectId = projectId;
3284:   activeSectionId = sectionWindowId(sectionId);
3285:   projectWindowBackStack = [];
3286:   projectWindowForwardStack = [];
3287:   projectTitleMenu.hidden = true;
3288:   projectDialog.removeAttribute("style");
3289:   projectDialog.classList.remove(
3290:     "is-draggable-dialog",
3291:     "is-hidden-dialog",
3292:     "is-maximized-dialog",
3293:     "is-minimized-dialog",
3294:     "is-resized-dialog"
3295:   );
3296:   delete projectDialog.dataset.restoreWindowState;
3297:   renderAll();
3298:   document.body.classList.add("project-window-open");
3299:   if (!projectDialog.open) projectDialog.showModal();
3300:   resetProjectWindowScroll();
3301:   updateDialogWindowButtons(projectDialog);
3302: }

```

### ***categoryById - lines 3304-3306***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3304-3306 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function categoryById(id) {                                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3304: function categoryById(id) {
3305:   return catalog.categories.find((category) => category.id === id) || {};
3306: }

```

### ***normalizeCategory - lines 3308-3318***

| Item            | Explanation                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Source location | Lines 3308-3318 (11 source lines).                                            |
| Purpose         | Validates or normalizes input so later code receives safe predictable values. |
| Declaration     | function normalizeCategory(category = {}) {                                   |
| Touches         | filesystem                                                                    |
| Calls detected  | trim, slugify, normalizeTextColor                                             |
| API endpoints   | None detected inside this function body.                                      |
| Storage keys    | No local/session storage keys detected.                                       |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3308: function normalizeCategory(category = {}) {
3309:   const label = String(category.label || category.title || "New category").trim() || "New category";
3310:   const id = slugify(category.id || label) || "category";
3311:   const accent = normalizeTextColor(category.accent || "") || "#1677a8";
3312:   return {
3313:     accent,
3314:     description: String(category.description || "").trim(),
3315:     id,
3316:     label
3317:   };
3318: }

```

## ***refreshCategoryControls - lines 3320-3327***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3320-3327 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function refreshCategoryControls() {                                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | map, clone, escapeHtml, join, renderCategoryDropdown                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3320: function refreshCategoryControls() {
3321:   catalog.categories = (catalog.categories || []).map(normalizeCategory);
3322:   savedPortfolioCatalog.categories = clone(catalog.categories || []);
3323:   if (newCategorySelect) {
3324:     newCategorySelect.innerHTML = catalog.categories.map((category) => `<option
value="${escapeHtml(category.id)}">${escapeHtml(category.label)}</option>`).join("");
3325:   }
3326:   renderCategoryDropdown();
3327: }

```

## ***linkAttributes - lines 3329-3332***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3329-3332 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function linkAttributes(url, item = {}) {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeLinkTarget, isWebsiteLinkItem                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3329: function linkAttributes(url, item = {}) {

```

```

3330:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(item, url) });
3331:   return /^https?:\/\//i.test(target) ? ' target="_blank" rel="noreferrer"' : "";
3332: }

```

## ***isLocalDownloadTarget - lines 3334-3337***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3334-3337 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function isLocalDownloadTarget(target = "") {                                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeLinkTarget                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3334: function isLocalDownloadTarget(target = "") {
3335:   const value = normalizeLinkTarget(target, { assumeWeb: true });
3336:   return Boolean(value) && !/^(https?:)?\/\//i.test(value) && !/^(mailto:|tel:|#)/i.test(value);
3337: }

```

## ***downloadAttribute - lines 3339-3342***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3339-3342 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function downloadAttribute(target = "", item = {}) {                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeLinkTarget, isWebsiteLinkItem, isLocalDownloadTarget                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3339: function downloadAttribute(target = "", item = {}) {
3340:   const value = normalizeLinkTarget(target, { assumeWeb: isWebsiteLinkItem(item, target) });
3341:   return isLocalDownloadTarget(value) ? " download" : "";
3342: }

```

## ***resourceLink - lines 3344-3353***

| Item            | Explanation                                                                           |
|-----------------|---------------------------------------------------------------------------------------|
| Source location | Lines 3344-3353 (10 source lines).                                                    |
| Purpose         | Part of the private builder frontend and editing experience.                          |
| Declaration     | function resourceLink(item, label = item.label    item.title    item.name) {          |
| Touches         | filesystem                                                                            |
| Calls detected  | normalizeLinkTarget, isWebsiteLinkItem, escapeHtml, linkAttributes, downloadAttribute |
| API endpoints   | None detected inside this function body.                                              |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3344: function resourceLink(item, label = item.label || item.title || item.name) {
3345:   const rawTarget = item.url || item.artifact || item.href || item.file || item.path || item.src || "";
3346:   const target = normalizeLinkTarget(rawTarget, { assumeWeb: isWebsiteLinkItem(item, rawTarget) });
3347:
3348:   if (!target || item.status === "planned") {
3349:     return `<span class="resource-link muted">${label}</span>`;
3350:   }
3351:
3352:   return `<a class="resource-link" href="${escapeHtml(target)}"${linkAttributes(target,
item)}${downloadAttribute(target, item)}>${escapeHtml(label)}</a>`;
3353: }

```

## ***pillList - lines 3355-3360***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3355-3360 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function pillList(items, className = "") {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, join                                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3355: function pillList(items, className = "") {
3356:   return (items || []).map((item) => {
3357:     const label = typeof item === "string" ? item : item.name || item.title || item.label || "";
3358:     return `<span class="tag ${className}">${label}</span>`;
3359:   }).join("");
3360: }

```

## ***evidenceList - lines 3362-3365***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3362-3365 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function evidenceList(items, renderItem, emptyMessage) {                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, join                                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3362: function evidenceList(items, renderItem, emptyMessage) {
3363:   if (!items || !items.length) return `<p class="evidence-empty">${emptyMessage}</p>`;

```

```

3364:   return `

${items.map(renderItem).join("")}</ul>`;
3365: }

```

## detailBlock - lines 3367-3374

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3367-3374 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function detailBlock(title, className, content) {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3367: function detailBlock(title, className, content) {
3368:   return `
3369:     <details class="${className}">
3370:       <summary>${title}</summary>
3371:       ${content}
3372:     </details>
3373:   `;
3374: }

```

## mediaGrid - lines 3376-3398

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3376-3398 (23 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function mediaGrid(items) {                                                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | map, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, join                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3376: function mediaGrid(items) {
3377:   if (!items || !items.length) return `

No project images have been added
yet.</p>`;
3378:
3379:   return `
3380:     <div class="media-grid">
3381:       ${items.map((item) => `
3382:         <figure>
3383:           ${item.status === "planned" ? `
3384:             <div class="media-placeholder">${item.title}</div>
3385:             .
3386:             <a href="${escapeHtml(normalizeLinkTarget(item.url, { assumeWeb: isWebsiteLinkItem(item,
item.url) })))"${linkAttributes(item.url, item)}>
3387:               
3388:             </a>
3389:           `}
3390:         <figcaption>
3391:           <strong>${escapeHtml(item.title)}</strong>
3392:           <span>${escapeHtml(item.caption || "")}</span>
3393:         </figcaption>
3394:       `}


```



```

3395:         `).join("")}
3396:     </div>
3397: `;
3398: }

```

## ***itemTitle - lines 3400-3403***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3400-3403 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function itemTitle(item) {                                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3400: function itemTitle(item) {
3401:   if (typeof item === "string") return item;
3402:   return item?.title || item?.name || item?.label || "Untitled item";
3403: }

```

## ***itemUrl - lines 3405-3407***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3405-3407 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function itemUrl(item) {                                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3405: function itemUrl(item) {
3406:   return item?.url || item?.artifact || "";
3407: }

```

## ***itemDescription - lines 3409-3411***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 3409-3411 (3 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function itemDescription(item) {                             |
| Touches         | Mostly local calculations or local UI state.                 |
| Calls detected  | No obvious named calls detected by the generator.            |
| API endpoints   | None detected inside this function body.                     |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3409: function itemDescription(item) {
3410:   return item?.description || item?.caption || item?.result || item?.method || "";
3411: }
```

### ***isImageUrl - lines 3413-3415***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3413-3415 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function isImageUrl(url) {                                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3413: function isImageUrl(url) {
3414:   return /\. (png|jpe?g|webp|gif|svg)(\?.*)?$/i.test(url || "");
3415: }
```

### ***portfolioLink - lines 3417-3421***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3417-3421 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function portfolioLink(url, label) {                                                                                                                                                                  |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeLinkTarget, escapeHtml, linkAttributes, downloadAttribute                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
3417: function portfolioLink(url, label) {
3418:   if (!url) return `<span>${label}</span>`;
3419:   const target = normalizeLinkTarget(url, { assumeWeb: true });
3420:   return `<a
href="${escapeHtml(target)}"${linkAttributes(target)}${downloadAttribute(target)}>${escapeHtml(label)}</a>`;
3421: }
```

### ***renderPortfolioResourceList - lines 3423-3444***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3423-3444 (22 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPortfolioResourceList(title, items, emptyMessage) {                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, itemTitle, itemUrl, itemDescription, portfolioLink, join                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3423: function renderPortfolioResourceList(title, items, emptyMessage) {
3424:   return `
3425:     <section class="portfolio-preview-section">
3426:       <h3>${title}</h3>
3427:       ${items || []}.length ? `
3428:         <div class="portfolio-resource-list">
3429:           ${items.map(item => {
3430:             const titleText = itemTitle(item);
3431:             const url = itemUrl(item);
3432:             const description = itemDescription(item);
3433:             return `
3434:               <article class="portfolio-resource">
3435:                 <strong>${portfolioLink(url, titleText)}</strong>
3436:                 ${description ? `<p>${description}</p>` : ""}
3437:               </article>
3438:             `;
3439:           }).join("")}
3440:         </div>
3441:       ` : `<p class="evidence-empty">${emptyMessage}</p>`
3442:     </section>
3443:   `;
3444: }

```

## renderPortfolioMedia - lines 3446-3468

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3446-3468 (23 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPortfolioMedia(items) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, itemUrl, isImageUrl, itemTitle, itemDescription, join                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3446: function renderPortfolioMedia(items) {
3447:   return `
3448:     <section class="portfolio-preview-section">
3449:       <h3>Images</h3>
3450:       ${items || []}.length ? `
3451:         <div class="portfolio-media-grid">
3452:           ${items.map(item => {
3453:             const url = itemUrl(item);
3454:             return `
3455:               <figure>
3456:                 ${isImageUrl(url) ? `` : `<div class="media-
placeholder">${itemTitle(item)}</div>`}
3457:               <figcaption>
3458:                 <strong>${itemTitle(item)}</strong>

```

```

3459:         ${itemDescription(item) ? `<span>${itemDescription(item)}</span>` : ""}
3460:       </figcaption>
3461:     </figure>
3462:   `;
3463:   }).join("")}
3464: </div>
3465: ` : `<p class="evidence-empty">No images have been added yet.</p>`
3466: </section>
3467: `;
3468: }

```

## renderPortfolioTools - lines 3470-3486

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3470-3486 (17 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPortfolioTools(project) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, toolName, toolDescription, join                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3470: function renderPortfolioTools(project) {
3471:   return
3472:     <section class="portfolio-preview-section">
3473:       <h3>Tools</h3>
3474:       ${((project.tools || []).length ? `
3475:         <div class="portfolio-tool-list">
3476:           ${project.tools.map((tool) => `
3477:             <article>
3478:               <strong>${toolName(tool)}</strong>
3479:               ${toolDescription(tool) ? `<p>${toolDescription(tool)}</p>` : ""}
3480:             </article>
3481:           `).join("")}
3482:         </div>
3483:       ` : `<p class="evidence-empty">No tools have been added yet.</p>`}
3484:     </section>
3485:   `;
3486: }

```

## renderPortfolioCustomSections - lines 3488-3505

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3488-3505 (18 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPortfolioCustomSections(project) {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, portfolioLink, join                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3488: function renderPortfolioCustomSections(project) {
3489:   return (project.sections || []).map((section) => `
3490:     <section class="portfolio-preview-section">
3491:       <h3>${section.title}</h3>
3492:       ${section.description ? `<p>${section.description}</p>` : ""}

```

```

3493:     ${section.items || []}.length ? `
3494:     <div class="portfolio-resource-list">
3495:       ${section.items.map((item) => `
3496:         <article class="portfolio-resource">
3497:           <strong>${portfolioLink(item.url, item.title || "Untitled subsection")}</strong>
3498:           ${item.description ? `<p>${item.description}</p>` : ""}
3499:         </article>
3500:       `).join("")}
3501:     </div>
3502:     : `<p class="evidence-empty">No content has been added yet.</p>`
3503:   </section>
3504: `).join("");
3505: }

```

## publicCustomSectionBlocks - lines 3507-3518

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3507-3518 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function publicCustomSectionBlocks(project) {                                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | map, detailBlock, renderMultilineInlineText, evidenceList, resourceLink, join                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3507: function publicCustomSectionBlocks(project) {
3508:   return (project.sections || []).map((section) => detailBlock(section.title, "evidence-block evidence-
wide", `
3509:     ${section.description ? `<p class="evidence-
empty">${renderMultilineInlineText(section.description)}</p>` : ""}
3510:     ${evidenceList(section.items || [], (item) => `
3511:       <li>
3512:         ${item.url ? resourceLink(item, item.title) : `<strong>${item.title}</strong>`}
3513:         <span>${item.type || "Section item"} &middot; ${item.status || "tracked"}</span>
3514:         ${item.description ? `<p>${renderMultilineInlineText(item.description)}</p>` : ""}
3515:       </li>
3516:     `, "No content has been added yet.")}
3517:   `).join("");
3518: }

```

## renderProjectPortfolioPreview - lines 3520-3556

| Item               | Explanation                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3520-3556 (37 source lines).                                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                                |
| Declaration        | function renderProjectPortfolioPreview(project) {                                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                                            |
| Calls detected     | categoryById, find, filter, previewSectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, renderParsedBriefBlock, map, parsedPublicSection, join, renderMultilineInlineText |
| API endpoints      | None detected inside this function body.                                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                 |

Source excerpt:

```

3520: function renderProjectPortfolioPreview(project) {
3521:   if (!project) return `<p class="evidence-empty">Select a project first.</p>`;
3522:   const category = categoryById(project.category);
3523:   const accent = category.accent || "#117c7a";

```

```

3524:   if (project.portfolioView) {
3525:     const briefSection = (project.portfolioView.sections || []).find((section) => section.id ===
"brief");
3526:     const otherSections = (project.portfolioView.sections || []).filter((section) => section.id !==
"brief" && previewSectionHasRenderableContent(section));
3527:     return `
3528:       <article class="portfolio-preview-article ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" style="${projectTemplateStyle(project, accent)}">
3529:         <header>
3530:           ${renderParsedBriefBlock(briefSection, project.summary)}
3531:         </header>
3532:         <div class="evidence-grid">
3533:           ${otherSections.map((section, index) => parsedPublicSection(section, index)).join("")}
3534:         </div>
3535:       </article>
3536:     `;
3537:   }
3538:   return `
3539:     <article class="portfolio-preview-article ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" style="${projectTemplateStyle(project, accent)}">
3540:       <header>
3541:         <div class="project-brief-default">
3542:           <h4>Overview</h4>
3543:           ${project.summary ? `<p class="rich-
paragraph">${renderMultilineInlineText(project.summary)}</p>` : `<p class="evidence-empty">No project overview
has been added yet.</p>`}
3544:         </div>
3545:       </header>
3546:       ${renderPortfolioResourceList("Documents", project.documents, "No documents have been added yet.")}
3547:       ${renderPortfolioResourceList("Tests", project.tests, "No tests have been added yet.")}
3548:       ${renderPortfolioResourceList("PCBs Built", project.pcb, "No PCB files have been added yet.")}
3549:       ${renderPortfolioMedia(project.media)}
3550:       ${renderPortfolioTools(project)}
3551:       ${renderPortfolioResourceList("Languages", (project.languages || []).map((item) => ({ title: typeOf
item === "string" ? item : itemTitle(item) })), "No languages have been added yet.")}
3552:       ${renderPortfolioResourceList("Links", project.links, "No links have been added yet.")}
3553:       ${renderPortfolioCustomSections(project)}
3554:     </article>
3555:   `;
3556: }

```

## briefSection - lines 3525-3528

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3525-3528 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const briefSection = (project.portfolioView.sections    []).find((section) => section.id === "brief");                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, filter, previewSectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3525:   const briefSection = (project.portfolioView.sections || []).find((section) => section.id ===
"brief");
3526:   const otherSections = (project.portfolioView.sections || []).filter((section) => section.id !==
"brief" && previewSectionHasRenderableContent(section));
3527:   return `
3528:     <article class="portfolio-preview-article ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" style="${projectTemplateStyle(project, accent)}">

```

## otherSections - lines 3526-3528

| Item            | Explanation                                                                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 3526-3528 (3 source lines).                                                                                                                        |
| Purpose         | Part of the private builder frontend and editing experience.                                                                                             |
| Declaration     | const otherSections = (project.portfolioView.sections    []).filter((section) => section.id !== "brief" && previewSectionHasRenderableContent(section)); |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, previewSectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3526:     const otherSections = (project.portfolioView.sections || []).filter((section) => section.id !==
"brief" && previewSectionHasRenderableContent(section));
3527:     return `
3528:         <article class="portfolio-preview-article ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" style="${projectTemplateStyle(project, accent)}">

```

## ***fullProjectPreviewHtmlExact - lines 3558-3685***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3558-3685 (128 source lines).                                                                                                                                                                   |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function fullProjectPreviewHtmlExact(project) {                                                                                                                                                       |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | parseProjectForPortfolio, categoryById, normalizeCategory, displayTitle, parsedPortfolioGlobals, stringify, replaceAll, escapeHtml, var, min, calc, media                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3558: function fullProjectPreviewHtmlExact(project) {
3559:   const baseHref = `${window.location.origin}/`;
3560:   if (!project) {
3561:     return `<!DOCTYPE html>
3562: <html lang="en">
3563:   <head>
3564:     <meta charset="UTF-8" />
3565:     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
3566:     <base href="${baseHref}" />
3567:     <link rel="stylesheet" href="styles.css" />
3568:     <title>Project preview</title>
3569:   </head>
3570:   <body class="project-isolated-preview">
3571:     <main class="project-isolated-shell">
3572:       <div class="empty-state">
3573:         <h3>No saved project preview</h3>
3574:         <p>Click Save project before viewing the parsed project preview.</p>
3575:       </div>
3576:     </main>
3577:   </body>
3578: </html>`;
3579:   }
3580:   const parsedProject = project.portfolioView ? project : parseProjectForPortfolio(project);
3581:   const categorySource = categoryById(parsedProject.category);
3582:   const previewCategory = normalizeCategory(categorySource.id ? categorySource : {
3583:     id: parsedProject.category || "project",
3584:     label: displayTitle(parsedProject.category || "Project", "Project")
3585:   });
3586:   const parsedGlobals = parsedPortfolioGlobals();
3587:   const previewDataObject = {
3588:     categories: [previewCategory],
3589:     fieldStyles: parsedGlobals.fieldStyles,
3590:     funFacts: [],
3591:     funFactsRich: null,
3592:     profile: parsedGlobals.profile,
3593:     profileRich: parsedGlobals.profileRich,

```

```

3594:     projects: [parsedProject],
3595:     siteContent: parsedGlobals.siteContent,
3596:     siteContentRich: parsedGlobals.siteContentRich,
3597:     siteSections: []
3598:   };
3599:   const previewData = JSON.stringify(previewDataObject).replaceAll("</", "<\\");
3600:   const title = parsedProject.portfolioView?.title || parsedProject.title || "Project preview";
3601:   return `<!DOCTYPE html>
3602: <html lang="en">
3603:   <head>
3604:     <meta charset="UTF-8" />
3605:     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
3606:     <base href="${baseHref}" />
3607:     <title>${escapeHtml(title)} | Project preview</title>
3608:     <link rel="stylesheet" href="styles.css" />
3609:     <style>
3610:       body.project-isolated-preview {
3611:         min-height: 100vh;
3612:         margin: 0;
3613:         background: var(--paper, #f8fbff);
3614:         color: var(--ink, #172636);
3615:       }
3616:
3617:       .project-isolated-shell {
3618:         width: min(1180px, calc(100vw - 32px));
3619:         margin: 0 auto;
3620:         padding: 28px 0 36px;
3621:       }
3622:
3623:       .project-isolated-support {
3624:         display: none !important;
3625:       }
3626:
3627:       body.project-isolated-preview .project-grid {
3628:         display: block;
3629:       }
3630:
3631:       body.project-isolated-preview .category-section {
3632:         padding: 0;
3633:         border: 0;
3634:         background: transparent;
3635:       }
3636:
3637:       body.project-isolated-preview .category-heading,
3638:       body.project-isolated-preview .category-description {
3639:         display: none;
3640:       }
3641:
3642:       body.project-isolated-preview .category-projects {
3643:         display: block;
3644:       }
3645:
3646:       body.project-isolated-preview .project-card {
3647:         margin: 0;
3648:       }
3649:
3650:       @media (max-width: 720px) {
3651:         .project-isolated-shell {
3652:           width: min(100vw - 20px, 100%);
3653:           padding: 14px 0 24px;
3654:         }
3655:       }
3656:     </style>
3657:   </head>
3658:   <body class="project-isolated-preview">
3659:     <main class="project-isolated-shell" aria-label="Isolated project preview">
3660:       <div class="project-isolated-support" aria-hidden="true">
3661:         <label class="search-wrap" for="project-search">
3662:           <span>Search</span>
3663:           <input id="project-search" type="search" tabindex="-1" />
3664:         </label>
3665:         <div id="project-filters">
3666:           <button class="filter-button active" type="button" data-filter="all" tabindex="-1">All</button>
3667:         </div>
3668:         <span id="project-count">0</span>
3669:         <span id="project-track-count">0 Tracks</span>
3670:         <span id="project-track-labels">project categories</span>
3671:         <span id="year">${new Date().getFullYear()}</span>
3672:       </div>
3673:
3674:       <section class="section project-isolated-section" id="projects" aria-labelledby="project-isolated-
title">
3675:         <h1 class="sr-only" id="project-isolated-title">${escapeHtml(title)}</h1>
3676:         <div class="project-grid" id="project-grid" aria-live="polite"></div>
3677:       </section>

```

... excerpt truncated after 120 lines. Open the source file for the rest of this function.

## renderProjectWebsitePreview - lines 3687-3705



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3687-3705 (19 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderProjectWebsitePreview(project) {                                                                                                                                                       |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | categoryById, applyProjectTemplateToElement, updateProjectPreviewNavigation, add, escapeHtml, querySelector, fullProjectPreviewHtmlExact                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3687: function renderProjectWebsitePreview(project) {
3688:   const category = categoryById(project?.category);
3689:   applyProjectTemplateToElement(projectPreviewDialog, project, category.accent || "#117c7a");
3690:   projectPreviewTitle.textContent = project?.portfolioView?.title || project?.title || "Project preview";
3691:   activeProjectPreviewProject = project || null;
3692:   activeProjectPreviewSectionIndex = -1;
3693:   activeProjectPreviewPath = [];
3694:   activeProjectPreviewForwardStack = [];
3695:   updateProjectPreviewNavigation();
3696:   projectPreviewContent.classList.add("is-website-preview");
3697:   projectPreviewContent.innerHTML = `
3698:     <iframe
3699:       class="portfolio-preview-frame project-preview-frame"
3700:       title="${escapeHtml(project?.title || "Project website preview")}"
3701:     ></iframe>
3702:   `;
3703:   const frame = projectPreviewContent.querySelector("iframe");
3704:   if (frame) frame.srcdoc = fullProjectPreviewHtmlExact(project);
3705: }

```

## ***openProjectPortfolioPreview - lines 3707-3715***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3707-3715 (9 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function openProjectPortfolioPreview() {                                                                                                                                                              |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | selectedProject, find, renderProjectWebsitePreview, add, showModal                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3707: function openProjectPortfolioPreview() {
3708:   const project = selectedProject();
3709:   const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project?.id);
3710:   const previewProject = savedProject || project;
3711:   renderProjectWebsitePreview(previewProject);
3712:   document.body.classList.add("full-window-open");
3713:   projectPreviewDialog.showModal();
3714:   projectPreviewDialog.scrollTop = 0;
3715: }

```

## ***savedProject - lines 3709-3714***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3709-3714 (6 source lines).                                                                                                                                                                     |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | <code>const savedProject = (savedPortfolioCatalog.projects    []).find((item) =&gt; item.id === project?.id);</code>                                                                                  |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | <code>find</code> , <code>renderProjectWebsitePreview</code> , <code>add</code> , <code>showModal</code>                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3709: const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project?.id);
3710: const previewProject = savedProject || project;
3711: renderProjectWebsitePreview(previewProject);
3712: document.body.classList.add("full-window-open");
3713: projectPreviewDialog.showModal();
3714: projectPreviewDialog.scrollTop = 0;

```

## ***updateProjectPreviewNavigation - lines 3717-3721***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3717-3721 (5 source lines).                                                                                                                                                                     |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | <code>function updateProjectPreviewNavigation() {</code>                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>updateDialogWindowButtons</code>                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3717: function updateProjectPreviewNavigation() {
3718:   projectPreviewBack.hidden = activeProjectPreviewSectionIndex < 0;
3719:   projectPreviewForward.hidden = !activeProjectPreviewForwardStack.length;
3720:   updateDialogWindowButtons(projectPreviewDialog);
3721: }

```

## ***openProjectPreviewNode - lines 3723-3745***

| Item               | Explanation                                                                                                                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3723-3745 (23 source lines).                                                                                                                                                                                                                             |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                                                                                      |
| Declaration        | <code>function openProjectPreviewNode(sectionIndex, path = [], options = {}) {</code>                                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                                                                                   |
| Calls detected     | <code>filter</code> , <code>previewSectionHasRenderableContent</code> , <code>previewNodeAtPath</code> , <code>parsedItemHasContent</code> , <code>updateProjectPreviewNavigation</code> , <code>displayTitle</code> , <code>parsedPublicSectionContent</code> |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                                       |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                                        |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                                                                                  |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3723: function openProjectPreviewNode(sectionIndex, path = [], options = {}) {
3724:   if (!activeProjectPreviewProject?.portfolioView) return;
3725:   const sections = (activeProjectPreviewProject.portfolioView.sections || []).filter((section) =>
section.id !== "brief" && previewSectionHasRenderableContent(section));
3726:   const section = sections[Number(sectionIndex)];
3727:   if (!section) return;
3728:   const node = path.length ? previewNodeAtPath(section, path) : section;
3729:   if (!node) return;
3730:   if (path.length ? !parsedItemHasContent(node) : !previewSectionHasRenderableContent(section)) return;
3731:
3732:   activeProjectPreviewSectionIndex = Number(sectionIndex);
3733:   activeProjectPreviewPath = path;
3734:   if (!options.preserveForward) activeProjectPreviewForwardStack = [];
3735:   updateProjectPreviewNavigation();
3736:   projectPreviewTitle.textContent = displayTitle(node.title || section.title, "Section");
3737:   projectPreviewContent.innerHTML = `
3738:     <article class="portfolio-preview-article">
3739:       <section class="portfolio-preview-section">
3740:         ${parsedPublicSectionContent(section, Number(sectionIndex), path)}
3741:       </section>
3742:     </article>
3743:   `;
3744:   projectPreviewDialog.scrollTop = 0;
3745: }

```

## sections - lines 3725-3730

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3725-3730 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const sections = (activeProjectPreviewProject.portfolioView.sections    []).filter((section) => section.id !== "brief" && previewSectionHasRenderableContent(section));                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, previewSectionHasRenderableContent, previewNodeAtPath, parsedItemHasContent                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3725:   const sections = (activeProjectPreviewProject.portfolioView.sections || []).filter((section) =>
section.id !== "brief" && previewSectionHasRenderableContent(section));
3726:   const section = sections[Number(sectionIndex)];
3727:   if (!section) return;
3728:   const node = path.length ? previewNodeAtPath(section, path) : section;
3729:   if (!node) return;
3730:   if (path.length ? !parsedItemHasContent(node) : !previewSectionHasRenderableContent(section)) return;

```

## projectPreviewBackStep - lines 3747-3769

| Item            | Explanation                                                                                        |
|-----------------|----------------------------------------------------------------------------------------------------|
| Source location | Lines 3747-3769 (23 source lines).                                                                 |
| Purpose         | Part of the private builder frontend and editing experience.                                       |
| Declaration     | function projectPreviewBackStep() {                                                                |
| Touches         | Mostly local calculations or local UI state.                                                       |
| Calls detected  | push, slice, renderProjectPortfolioPreview, updateProjectPreviewNavigation, openProjectPreviewNode |
| API endpoints   | None detected inside this function body.                                                           |
| Storage keys    | No local/session storage keys detected.                                                            |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3747: function projectPreviewBackStep() {
3748:   if (!activeProjectPreviewProject) return;
3749:   if (!activeProjectPreviewPath.length) {
3750:     if (activeProjectPreviewSectionIndex >= 0) {
3751:       activeProjectPreviewForwardStack.push({
3752:         path: activeProjectPreviewPath.slice(),
3753:         sectionIndex: activeProjectPreviewSectionIndex
3754:       });
3755:     }
3756:     projectPreviewTitle.textContent = activeProjectPreviewProject.title || "Project preview";
3757:     projectPreviewContent.innerHTML = renderProjectPortfolioPreview(activeProjectPreviewProject);
3758:     activeProjectPreviewSectionIndex = -1;
3759:     activeProjectPreviewPath = [];
3760:     updateProjectPreviewNavigation();
3761:     return;
3762:   }
3763:   activeProjectPreviewForwardStack.push({
3764:     path: activeProjectPreviewPath.slice(),
3765:     sectionIndex: activeProjectPreviewSectionIndex
3766:   });
3767:   const nextPath = activeProjectPreviewPath.slice(0, -1);
3768:   openProjectPreviewNode(activeProjectPreviewSectionIndex, nextPath, { preserveForward: true });
3769: }

```

## ***projectPreviewForwardStep - lines 3771-3775***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3771-3775 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectPreviewForwardStep() {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | pop, openProjectPreviewNode                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3771: function projectPreviewForwardStep() {
3772:   if (!activeProjectPreviewProject || !activeProjectPreviewForwardStack.length) return;
3773:   const next = activeProjectPreviewForwardStack.pop();
3774:   openProjectPreviewNode(next.sectionIndex, next.path, { preserveForward: true });
3775: }

```

## ***closeOrStepBackProjectPreview - lines 3777-3783***

| Item               | Explanation                                               |
|--------------------|-----------------------------------------------------------|
| Source location    | Lines 3777-3783 (7 source lines).                         |
| Purpose            | Controls a window, dialog, panel, menu, or visible state. |
| Declaration        | function closeOrStepBackProjectPreview() {                |
| Touches            | Mostly local calculations or local UI state.              |
| Calls detected     | projectPreviewBackStep, closeProjectPreviewWindow         |
| API endpoints      | None detected inside this function body.                  |
| Storage keys       | No local/session storage keys detected.                   |
| Async/return clues | 0 await expression(s), 1 return statement(s).             |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3777: function closeOrStepBackProjectPreview() {
3778:   if (activeProjectPreviewProject && (activeProjectPreviewSectionIndex >= 0 ||
activeProjectPreviewPath.length)) {
3779:     projectPreviewBackStep();
3780:     return;
3781:   }
3782:   closeProjectPreviewWindow();
3783: }

```

## ***closeProjectPreviewWindow - lines 3785-3791***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3785-3791 (7 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function closeProjectPreviewWindow() {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closeDialogElement                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3785: function closeProjectPreviewWindow() {
3786:   activeProjectPreviewProject = null;
3787:   activeProjectPreviewSectionIndex = -1;
3788:   activeProjectPreviewPath = [];
3789:   activeProjectPreviewForwardStack = [];
3790:   closeDialogElement(projectPreviewDialog);
3791: }

```

## ***publicProjectCard - lines 3793-3856***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3793-3856 (64 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function publicProjectCard(project) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | parsedPublicProjectCard, categoryById, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, detailBlock, map, itemTitle, join, evidenceList, resourceLink, mediaGrid                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3793: function publicProjectCard(project) {
3794:   if (project.portfolioView) return parsedPublicProjectCard(project);
3795:   const category = categoryById(project.category);
3796:   const accent = category.accent || "#117c7a";
3797:
3798:   return `
3799:     <article class="project-card catalog-card ${projectTemplateClass(project)}

```

```

3800:     <div class="project-body">
3801:       <h3>${project.title}</h3>
3802:       <p>${project.summary || ""}</p>
3803:
3804:       ${project.highlights && project.highlights.length ? detailBlock("Project highlights", "project-
drawer", `
3805:         <ul class="highlight-list">
3806:           ${project.highlights.map((item) => `<li>${typeof item === "string" ? item :
itemTitle(item)}</li>`).join("")}
3807:         </ul>
3808:       `) : ""}
3809:
3810:       <div class="evidence-grid" aria-label="${project.title} evidence blocks">
3811:         ${detailBlock("Documents", "evidence-block", evidenceList(project.documents, (item) => `
3812:           <li>
3813:             ${resourceLink(item, item.title)}
3814:             <span>${item.type || "Document"} &middot; ${item.status || "tracked"}</span>
3815:           </li>
3816:         `, "No document artifact has been added yet.))}
3817:
3818:         ${detailBlock("Tests and results", "evidence-block", evidenceList(project.tests, (item) => `
3819:           <li>
3820:             ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3821:             <span>${item.method || "Validation"} &middot; ${item.status || "tracked"}</span>
3822:             ${item.result ? `<p>${item.result}</p>` : ""}
3823:           </li>
3824:         `, "No test artifact has been added yet.))}
3825:
3826:         ${detailBlock("PCBs built", "evidence-block", evidenceList(project.pcb, (item) => `
3827:           <li>
3828:             ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3829:             <span>${item.revision || "Revision"} &middot; ${item.status || "tracked"}</span>
3830:           </li>
3831:         `, "No PCB build has been added yet.))}
3832:
3833:         ${detailBlock("Images", "evidence-block evidence-wide", mediaGrid(project.media))}
3834:         ${publicCustomSectionBlocks(project)}
3835:       </div>
3836:
3837:       ${detailBlock("Tools and implementation files", "project-drawer", `
3838:         <div class="project-tooling">
3839:           <div>
3840:             <h4>Tools Used</h4>
3841:             <div class="project-meta">${pillList(project.tools, "tool-tag")}</div>
3842:           </div>
3843:           <div>
3844:             <h4>Languages</h4>
3845:             <div class="project-meta">${pillList(project.languages, "language-tag")}</div>
3846:           </div>
3847:         </div>
3848:       `)}
3849:
3850:       <div class="resource-list">
3851:         ${project.links || []}.map((item) => resourceLink(item)).join("")
3852:       </div>
3853:     </div>
3854:   </article>
3855: `;
3856: }

```

## renderParsedBriefBlock - lines 3858-3868

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3858-3868 (11 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderParsedBriefBlock(section, fallbackSummary = "") {                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | renderRichContent                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3858: function renderParsedBriefBlock(section, fallbackSummary = "") {
3859:   const briefItem = section?.items?.[0] || {};

```

```

3860:   const briefText = briefItem.description || fallbackSummary || "";
3861:   if (!briefItem.rich?.blocks?.length && !briefText) return "";
3862:   return `
3863:     <details class="project-brief-default parsed-summary" open>
3864:       <summary>Overview</summary>
3865:       ${renderRichContent(briefItem.rich, briefText)}
3866:     </details>
3867:   `;
3868: }

```

## ***previewPathToString - lines 3870-3872***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3870-3872 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function previewPathToString(path = []) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | join                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3870: function previewPathToString(path = []) {
3871:   return path.join(".");
3872: }

```

## ***previewPathFromString - lines 3874-3877***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3874-3877 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function previewPathFromString(value = "") {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, map, filter, isInteger                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3874: function previewPathFromString(value = "") {
3875:   if (!value) return [];
3876:   return value.split(".").map((item) => Number(item)).filter((item) => Number.isInteger(item));
3877: }

```

## ***previewNodeAtPath - lines 3879-3888***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 3879-3888 (10 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function previewNodeAtPath(section, path = []) {             |
| Touches         | Mostly local calculations or local UI state.                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3879: function previewNodeAtPath(section, path = []) {
3880:   let node = section;
3881:   let children = section?.items || [];
3882:   for (const index of path) {
3883:     node = children[index];
3884:     if (!node) return null;
3885:     children = node.children || [];
3886:   }
3887:   return node;
3888: }

```

## ***previewNodeChildren - lines 3890-3892***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3890-3892 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function previewNodeChildren(node) {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3890: function previewNodeChildren(node) {
3891:   return node?.items || node?.children || [];
3892: }

```

## ***previewSectionHasRenderableContent - lines 3894-3896***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3894-3896 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function previewSectionHasRenderableContent(section) {                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | richHasContent, some                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3894: function previewSectionHasRenderableContent(section) {
3895:   return Boolean(section?.description || richHasContent(section?.rich) || (section?.items ||

```



```

    []).some(parsedItemHasContent));
3896: }

```

## previewNodeSummary - lines 3898-3906

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3898-3906 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function previewNodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | escapeHtml, renderRichContent                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3898: function previewNodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {
3899:   if (!rich?.blocks?.length && !text) return "";
3900:   return `
3901:     <details class="parsed-summary" open>
3902:       <summary>${escapeHtml(title)}</summary>
3903:       ${renderRichContent(rich, text || "")}
3904:     </details>
3905:   `;
3906: }

```

## previewNodeIsOverview - lines 3908-3911

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3908-3911 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function previewNodeIsOverview(item = {}) {                                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalize, displayTitle, endsWith                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3908: function previewNodeIsOverview(item = {}) {
3909:   const title = normalize(displayTitle(item.title || item.label || ""));
3910:   return item.kind === "summary" || title === "overview" || title.endsWith(" overview");
3911: }

```

## previewNodeOverviewDetails - lines 3913-3931

| Item            | Explanation                                                                          |
|-----------------|--------------------------------------------------------------------------------------|
| Source location | Lines 3913-3931 (19 source lines).                                                   |
| Purpose         | Part of the private builder frontend and editing experience.                         |
| Declaration     | function previewNodeOverviewDetails(node, children = []) {                           |
| Touches         | Mostly local calculations or local UI state.                                         |
| Calls detected  | filter, push, renderRichContent, forEach, previewFileList, previewNodeChildren, join |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3913: function previewNodeOverviewDetails(node, children = []) {
3914:   const overviewItems = children.filter(previewNodeIsOverview);
3915:   const blocks = [];
3916:   if (node?.rich?.blocks?.length || node?.description) {
3917:     blocks.push(renderRichContent(node.rich, node.description || ""));
3918:   }
3919:   overviewItems.forEach((item) => {
3920:     if (item.rich?.blocks?.length || item.description) blocks.push(renderRichContent(item.rich,
item.description || ""));
3921:     const itemFiles = previewFileList(previewNodeChildren(item));
3922:     if (itemFiles) blocks.push(itemFiles);
3923:   });
3924:   if (!blocks.length) return "";
3925:   return `
3926:     <details class="parsed-summary section-overview-default" open>
3927:       <summary>Overview</summary>
3928:       ${blocks.join("")}
3929:     </details>
3930:   `;
3931: }

```

### ***previewNodeCard - lines 3933-3940***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3933-3940 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function previewNodeCard(node, sectionIndex, path) {                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | displayTitle, previewPathToString, escapeHtml                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3933: function previewNodeCard(node, sectionIndex, path) {
3934:   const title = displayTitle(node.title);
3935:   return `
3936:     <button class="section-open-card subsection-open-card" type="button" data-preview-section-
index="${sectionIndex}" data-preview-resource-path="${previewPathToString(path)}">
3937:       <span>${escapeHtml(title)}</span>
3938:     </button>
3939:   `;
3940: }

```

### ***previewChildCards - lines 3942-3950***

| Item            | Explanation                                                                     |
|-----------------|---------------------------------------------------------------------------------|
| Source location | Lines 3942-3950 (9 source lines).                                               |
| Purpose         | Part of the private builder frontend and editing experience.                    |
| Declaration     | function previewChildCards(children, sectionIndex, basePath = []) {             |
| Touches         | Mostly local calculations or local UI state.                                    |
| Calls detected  | filter, parsedItemHasContent, previewNodeIsOverview, map, previewNodeCard, join |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3942: function previewChildCards(children, sectionIndex, basePath = []) {
3943:   const visibleChildren = children.filter((child) => parsedItemHasContent(child) && !child.url &&
!previewNodeIsOverview(child));
3944:   if (!visibleChildren.length) return "";
3945:   return
3946:     <div class="subsection-grid section-content-grid">
3947:       ${children.map((child, index) => parsedItemHasContent(child) && !child.url &&
!previewNodeIsOverview(child) ? previewNodeCard(child, sectionIndex, [...basePath, index]) : "").join("")}
3948:     </div>
3949:   `;
3950: }

```

## ***previewFileList - lines 3952-3969***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3952-3969 (18 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function previewFileList(children = []) {                                                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | filter, map, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute, join                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3952: function previewFileList(children = []) {
3953:   const files = children.filter((child) => child?.url);
3954:   if (!files.length) return "";
3955:   return `
3956:     <div class="section-file-list">
3957:       ${files.map((file) => `
3958:         <a
3959:           class="section-file-link"
3960:           href="${escapeHtml(normalizeLinkTarget(file.url, { assumeWeb: isWebsiteLinkItem(file, file.url)
3961:         })}"
3962:           ${linkAttributes(file.url, file)}
3963:         >
3964:           ${escapeHtml(file.title || "Open link")}
3965:         </a>
3966:       `).join("")}
3967:     </div>
3968:   `;
3969: }

```

## ***previewNodeContent - lines 3971-3983***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 3971-3983 (13 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function previewNodeContent(node, sectionIndex, path = []) { |
| Touches         | Mostly local calculations or local UI state.                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | previewFileList, previewNodeChildren, filter, previewNodeIsOverview, previewNodeOverviewDetails, previewChildCards                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3971: function previewNodeContent(node, sectionIndex, path = []) {
3972:   const isContainer = !node.kind || node.kind === "subsection";
3973:   if (!isContainer && node.url) return previewFileList([node]);
3974:   const children = previewNodeChildren(node);
3975:   const contentChildren = children.filter((child) => !previewNodeIsOverview(child));
3976:   return `
3977:     <article class="parsed-window-panel section-directory">
3978:       ${previewNodeOverviewDetails(node, children)}
3979:       ${previewChildCards(children, sectionIndex, path)}
3980:       ${previewFileList(contentChildren)}
3981:     </article>
3982:   `;
3983: }

```

## ***parsedPublicSectionContent - lines 3985-3989***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 3985-3989 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedPublicSectionContent(section, sectionIndex = 0, path = []) {                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | previewNodeAtPath, previewNodeContent                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3985: function parsedPublicSectionContent(section, sectionIndex = 0, path = []) {
3986:   const node = path.length ? previewNodeAtPath(section, path) : section;
3987:   if (!node) return `<p class="evidence-empty">This section could not be found.</p>`;
3988:   return previewNodeContent(node, sectionIndex, path);
3989: }

```

## ***parsedPublicSection - lines 3991-3999***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 3991-3999 (9 source lines).                            |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function parsedPublicSection(section, index = 0) {           |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | filter, some, escapeHtml, displayTitle                       |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 1 return statement(s).                |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

3991: function parsedPublicSection(section, index = 0) {
3992:   const visibleItems = (section.items || []).filter(parsedItemHasContent);
3993:   const hasImages = visibleItems.some(item => item.kind === "image");
3994:   return `
3995:     <button class="section-open-card evidence-block ${hasImages ? "evidence-wide" : ""}" type="button"
data-preview-section-index="${index}">
3996:       <span>${escapeHtml(displayTitle(section.title, "Section"))}</span>
3997:     </button>
3998:   `;
3999: }

```

## ***parsedPublicProjectCard - lines 4001-4019***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4001-4019 (19 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function parsedPublicProjectCard(project) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | categoryById, find, filter, previewSectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, renderParsedBriefBlock, map, parsedPublicSection, join            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4001: function parsedPublicProjectCard(project) {
4002:   const category = categoryById(project.category);
4003:   const accent = category.accent || "#117c7a";
4004:   const view = project.portfolioView;
4005:   const briefSection = (view.sections || []).find((section) => section.id === "brief");
4006:   const otherSections = (view.sections || []).filter((section) => section.id !== "brief" &&
previewSectionHasRenderableContent(section));
4007:
4008:   return `
4009:     <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" id="${project.id}" style="${projectTemplateStyle(project, accent)}">
4010:       <div class="project-body">
4011:         <h3>${view.title || project.title}</h3>
4012:         ${renderParsedBriefBlock(briefSection, project.summary)}
4013:         <div class="evidence-grid" aria-label="${project.title} parsed project content">
4014:           ${otherSections.map((section, index) => parsedPublicSection(section, index)).join("")}
4015:         </div>
4016:       </div>
4017:     </article>
4018:   `;
4019: }

```

## ***briefSection - lines 4005-4006***

| Item            | Explanation                                                                           |
|-----------------|---------------------------------------------------------------------------------------|
| Source location | Lines 4005-4006 (2 source lines).                                                     |
| Purpose         | Part of the private builder frontend and editing experience.                          |
| Declaration     | const briefSection = (view.sections    []).find((section) => section.id === "brief"); |
| Touches         | Mostly local calculations or local UI state.                                          |
| Calls detected  | find, filter, previewSectionHasRenderableContent                                      |
| API endpoints   | None detected inside this function body.                                              |
| Storage keys    | No local/session storage keys detected.                                               |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4005:  const briefSection = (view.sections || []).find((section) => section.id === "brief");
4006:  const otherSections = (view.sections || []).filter((section) => section.id !== "brief" &&
previewSectionHasRenderableContent(section));
```

## ***otherSections - lines 4006-4006***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4006-4006 (1 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const otherSections = (view.sections    []).filter((section) => section.id !== "brief" && previewSectionHasRenderableContent(section));                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, previewSectionHasRenderableContent                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4006:  const otherSections = (view.sections || []).filter((section) => section.id !== "brief" &&
previewSectionHasRenderableContent(section));
```

## ***publicCategorySection - lines 4021-4036***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4021-4036 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function publicCategorySection(category, visibleProjects) {                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, join                                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4021: function publicCategorySection(category, visibleProjects) {
4022:   return `
4023:     <section class="category-section ${visibleProjects.length ? "" : "empty-category-section"}" aria-
labelledby="${category.id}-preview-title">
4024:       <div class="category-heading">
4025:         <div>
4026:           <h3 id="${category.id}-preview-title">${category.label}</h3>
4027:         </div>
4028:         ${visibleProjects.length ? `<span>${visibleProjects.length} project${visibleProjects.length === 1
? "" : "s"}</span>` : ""}
4029:       </div>
4030:       ${visibleProjects.length ? `<p class="category-description">${category.description}</p>` : ""}
4031:       <div class="category-projects">
4032:         ${visibleProjects.map(publicProjectCard).join("")}
4033:       </div>
```

```

4034:     </section>
4035:   `;
4036: }

```

## renderedPublicProjects - lines 4038-4051

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4038-4051 (14 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderedPublicProjects() {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, filter, publicCategorySection, join                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4038: function renderedPublicProjects() {
4039:   if (!savedPortfolioCatalog.projects.length) {
4040:     return `
4041:       <div class="empty-state">
4042:         <h3>No parsed projects saved yet.</h3>
4043:         <p>Open a project, edit it, then click Save to build it into this portfolio preview.</p>
4044:       </div>
4045:     `;
4046:   }
4047:   return savedPortfolioCatalog.categories.map((category) => {
4048:     const categoryProjects = savedPortfolioCatalog.projects.filter((project) => project.category ===
category.id);
4049:     return publicCategorySection(category, categoryProjects);
4050:   }).join("");
4051: }

```

## sectionControls - lines 4053-4060

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4053-4060 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function sectionControls(sectionId, uploadKey = "", folder = "") {                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4053: function sectionControls(sectionId, uploadKey = "", folder = "") {
4054:   return
4055:     <div class="local-section-controls">
4056:       <button type="button" title="Edit this section" data-preview-section="${sectionId}">Edit</button>
4057:       ${uploadKey ? `<button type="button" title="Add file or image" data-preview-upload="${uploadKey}"
data-folder="${folder}">+</button>` : ""}
4058:     </div>
4059:   `;
4060: }

```

## customSectionBlocks - lines 4062-4074

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4062-4074 (13 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function customSectionBlocks(project) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, detailBlock, sectionControls, evidenceList, resourceLink, join                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4062: function customSectionBlocks(project) {
4063:   return (project.sections || []).map((section) => detailBlock(section.title, "evidence-block evidence-
wide", `
4064:     ${sectionControls(`custom:${section.id}`, "sections", section.id)}
4065:     ${section.description ? `

${section.description}

` : ""}
4066:     ${evidenceList(section.items || [], (item) => `
4067:       <li>
4068:         ${item.url ? resourceLink(item, item.title) : `${item.title}</strong>`}
4069:         <span>${item.type || "Section item"} &middot; ${item.status || "tracked"</span>
4070:         ${item.description ? `

```

## projectCard - lines 4076-4154

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4076-4154 (79 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectCard(project, isSelected = false) {                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | categoryById, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, detailBlock, map, join, sectionControls, evidenceList, resourceLink, mediaGrid, customSectionBlocks                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4076: function projectCard(project, isSelected = false) {
4077:   const category = categoryById(project.category);
4078:   const accent = category.accent || "#117c7a";
4079:
4080:   return `
4081:     <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)} ${isSelected ? "preview-project-card" : ""} id="${project.id}"
style="${projectTemplateStyle(project, accent)}">
4082:       <div class="project-body">
4083:         <div class="local-card-controls">
4084:           <button type="button" title="Open project" data-preview-project="${project.id}">Open</button>
4085:           <button type="button" title="Delete project" data-preview-
delete="${project.id}">Delete</button>
4086:         </div>
4087:         <h3>${project.title}</h3>
4088:         <p>${project.summary}</p>
4089:
4090:         ${project.highlights && project.highlights.length ? detailBlock("Project highlights", "project-
drawer", `
4091:           <ul class="highlight-list">
4092:             ${project.highlights.map((item) => `

```



```

4093:         </ul>
4094:     `) : ""}
4095:
4096:     <div class="evidence-grid" aria-label="${project.title} evidence blocks">
4097:         ${detailBlock("Documents", "evidence-block", `
4098:             ${sectionControls("documents", "documents", "documents")}
4099:             ${evidenceList(project.documents, (item) => `
4100:                 <li>
4101:                     ${resourceLink(item, item.title)}
4102:                     <span>${item.type || "Document"} &middot; ${item.status || "tracked"}</span>
4103:                 </li>
4104:             `, "No document artifact has been added yet.")}}
4105:         `)}
4106:
4107:         ${detailBlock("Tests and results", "evidence-block", `
4108:             ${sectionControls("tests", "tests", "tests")}
4109:             ${evidenceList(project.tests, (item) => `
4110:                 <li>
4111:                     ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
4112:                     <span>${item.method || "Validation"} &middot; ${item.status || "tracked"}</span>
4113:                     ${item.result ? `<p>${item.result}</p>` : ""}
4114:                 </li>
4115:             `, "No test artifact has been added yet.")}}
4116:         `)}
4117:
4118:         ${detailBlock("PCBs built", "evidence-block", `
4119:             ${sectionControls("pcbs", "pcbs", "pcbs")}
4120:             ${evidenceList(project.pcbs, (item) => `
4121:                 <li>
4122:                     ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
4123:                     <span>${item.revision || "Revision"} &middot; ${item.status || "tracked"}</span>
4124:                 </li>
4125:             `, "No PCB build has been added yet.")}}
4126:         `)}
4127:
4128:         ${detailBlock("Images", "evidence-block evidence-wide", `
4129:             ${sectionControls("media", "media", "images")}
4130:             ${mediaGrid(project.media)}
4131:         `)}
4132:         ${customSectionBlocks(project)}
4133:     </div>
4134:
4135:     ${detailBlock("Tools and implementation files", "project-drawer", `
4136:         <div class="project-tooling">
4137:             <div>
4138:                 <h4>Tools Used</h4>
4139:                 <div class="project-meta">${pillList(project.tools, "tool-tag")}</div>
4140:             </div>
4141:             <div>
4142:                 <h4>Languages</h4>
4143:                 <div class="project-meta">${pillList(project.languages, "language-tag")}</div>
4144:             </div>
4145:         </div>
4146:     `)}
4147:
4148:     <div class="resource-list">
4149:         ${project.links || []}.map((item) => resourceLink(item)).join("")
4150:     </div>
4151: </div>
4152: </article>
4153: `;
4154: }

```

## buildTemplateProject - lines 4156-4186

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4156-4186 (31 source lines).                                                                                                                                                                    |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function buildTemplateProject() {                                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | find, trim, slugify, clone, defaultDesignModel                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4156: function buildTemplateProject() {
4157:   const template = templates.find((item) => item.id === templateSelect.value) || null;
4158:   const title = newProjectTitle.value.trim() || "New Hardware Project";
4159:   const id = slugify(title);
4160:   const category = newCategorySelect.value || pendingCreateCategoryId || "analog-mixed-signal";
4161:
4162:   return {
4163:     id,
4164:     title,
4165:     templateId: template?.id || "",
4166:     templateLabel: template?.label || "",
4167:     templateGroup: "Appearance",
4168:     templateVisual: template?.visual ? clone(template.visual) : null,
4169:     category,
4170:     status: "Draft",
4171:     summary: "",
4172:     summaryRich: null,
4173:     focus: [],
4174:     highlights: [],
4175:     documents: [],
4176:     tests: [],
4177:     pcbs: [],
4178:     media: [],
4179:     tools: [],
4180:     languages: [],
4181:     links: [],
4182:     sections: [],
4183:     compileCode: { files: [], activeFileId: "", terminal: "" },
4184:     design: category === "analog-mixed-signal" ? defaultDesignModel() : undefined
4185:   };
4186: }

```

## ***insertProject - lines 4188-4204***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4188-4204 (17 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function insertProject(project) {                                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | filter, map, splice                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4188: function insertProject(project) {
4189:   const nextProjects = catalog.projects.filter((item) => item.id !== project.id);
4190:   const placement = placementSelect.value;
4191:
4192:   if (placement === "site-start") return [project, ...nextProjects];
4193:   if (placement === "site-end") return [...nextProjects, project];
4194:
4195:   const indexes = nextProjects
4196:     .map((item, index) => ({ item, index }))
4197:     .filter(({ item }) => item.category === project.category)
4198:     .map(({ index }) => index);
4199:
4200:   if (!indexes.length) return [...nextProjects, project];
4201:   const targetIndex = placement === "category-start" ? indexes[0] : indexes[indexes.length - 1] + 1;
4202:   nextProjects.splice(targetIndex, 0, project);
4203:   return nextProjects;
4204: }

```

## ***renderCategoryDropdown - lines 4206-4213***

| Item            | Explanation                            |
|-----------------|----------------------------------------|
| Source location | Lines 4206-4213 (8 source lines).      |
| Purpose         | Renders UI, markup, or display output. |
| Declaration     | function renderCategoryDropdown() {    |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, escapeHtml, join                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4206: function renderCategoryDropdown() {
4207:   categoryDropdown.innerHTML = (catalog.categories || []).map((category) => `
4208:     <button type="button" data-create-category="${escapeHtml(category.id)}" style="--category-accent:
${escapeHtml(category.accent || "#117c7a")}">
4209:       <span>${escapeHtml(category.label)}</span>
4210:       <small>${escapeHtml(category.description || "Project category")}</small>
4211:     </button>
4212:   `).join("");
4213: }

```

## ***toggleCategoryDropdown - lines 4215-4219***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4215-4219 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function toggleCategoryDropdown(forceOpen = null) {                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | setAttribute                                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4215: function toggleCategoryDropdown(forceOpen = null) {
4216:   const open = forceOpen === null ? categoryDropdown.hidden : forceOpen;
4217:   categoryDropdown.hidden = !open;
4218:   addProjectButton.setAttribute("aria-expanded", String(open));
4219: }

```

## ***createProjectInCategory - lines 4221-4233***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4221-4233 (13 source lines).                                                                                                                                                                    |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function createProjectInCategory(categoryId) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | buildTemplateProject, insertProject, add, toggleCategoryDropdown, setStatus, categoryById, scheduleAutosave, renderAll, openProjectWindow                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4221: function createProjectInCategory(categoryId) {
4222:   newCategorySelect.value = categoryId;
4223:   const project = buildTemplateProject();
4224:   catalog.projects = insertProject(project);
4225:   selectedProjectId = project.id;
4226:   activeSectionId = "brief";
4227:   expandedCategories.add(categoryId);
4228:   toggleCategoryDropdown(false);
4229:   setStatus(`Project added locally in ${categoryById(categoryId).label} || categoryId`. Click Save project
to include it in the portfolio preview.`);
4230:   scheduleAutosave();
4231:   renderAll();
4232:   openProjectWindow(project.id, "brief");
4233: }
```

## ***openProjectCreateDialog - lines 4235-4245***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4235-4245 (11 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function openProjectCreateDialog(categoryId) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | categoryById, groupedTemplateOptions, toggleCategoryDropdown, showModal                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4235: function openProjectCreateDialog(categoryId) {
4236:   pendingCreateCategoryId = categoryId;
4237:   newCategorySelect.value = categoryId;
4238:   const category = categoryById(categoryId);
4239:   projectCreateCategory.textContent = category.label || "Project category";
4240:   templateSelect.innerHTML = groupedTemplateOptions();
4241:   templateSelect.value = "";
4242:   newProjectTitle.value = "";
4243:   toggleCategoryDropdown(false);
4244:   projectCreateDialog.showModal();
4245: }
```

## ***groupedTemplateOptions - lines 4247-4253***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4247-4253 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function groupedTemplateOptions() {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, join                                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4247: function groupedTemplateOptions() {
4248:   const emptyOption = `<option value="">No appearance - white project layout</option>`;
4249:   const options = templates
4250:     .map((template) => `<option value="${template.id}">${template.label}</option>`)
4251:     .join("");
```

```

4252:   return `${emptyOption}${options}`;
4253: }

```

## showTemplatePreview - lines 4255-4276

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4255-4276 (22 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function showTemplatePreview(template) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | setProperty, map, join, showModal                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4255: function showTemplatePreview(template) {
4256:   if (!template) return;
4257:   const visual = template.visual || {};
4258:   const palette = visual.palette || ["#0f4c5c", "#7dd3c7", "#ffffff"];
4259:
4260:   templatePreviewGroup.textContent = visual.style ? `Appearance / ${visual.style}` : "Appearance";
4261:   templatePreviewTitle.textContent = template.label;
4262:   templatePreviewDescription.textContent = template.description || "";
4263:   templatePreviewCardTitle.textContent = template.label;
4264:   templatePreviewInteraction.textContent = visual.interaction || "Hover and click styles are shown in the
local builder.";
4265:   templatePreviewCard.style.setProperty("--template-primary", palette[0]);
4266:   templatePreviewCard.style.setProperty("--template-hover", palette[1]);
4267:   templatePreviewCard.style.setProperty("--template-paper", palette[2] || "#ffffff");
4268:   templatePreviewCard.style.setProperty("--template-bg", visual.background || palette[0]);
4269:   templatePreviewCard.style.setProperty("--template-panel", visual.panel || palette[2] || "#ffffff");
4270:   templatePreviewCard.style.setProperty("--template-accent", visual.accent || palette[1]);
4271:   templatePreviewCard.style.setProperty("--template-click", visual.click || palette[2] || "#ffffff");
4272:   templatePreviewCard.style.setProperty("--template-click-text", visual.clickText || visual.text ||
"#172636");
4273:   templateVisual.className = `template-visual template-visual-${visual.style || "schematic"}`;
4274:   templatePalette.innerHTML = palette.map((color) => `<span style="background:
${color}">${color}</span>`).join("");
4275:   templateDialog.showModal();
4276: }

```

## renderTree - lines 4278-4319

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4278-4319 (42 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderTree() {                                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | projectSearchText, map, filter, includes, projectMatchesSearch, has, escapeHtml, highlightSearchText, join, updateProjectSearchStatus                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4278: function renderTree() {
4279:   const cleanQuery = projectSearchText(projectSearchQuery);
4280:   let matchCount = 0;
4281:   const treeMarkup = catalog.categories.map((category) => {

```

```

4282:     const allCategoryProjects = catalog.projects.filter((project) => project.category === category.id);
4283:     const categoryHit = cleanQuery && projectSearchText(`${category.label}
${category.description}`).includes(cleanQuery);
4284:     const categoryProjects = cleanQuery
4285:       ? allCategoryProjects.filter((project) => categoryHit || projectMatchesSearch(project, category,
cleanQuery))
4286:       : allCategoryProjects;
4287:     if (cleanQuery && !categoryProjects.length && !categoryHit) return "";
4288:     matchCount += categoryProjects.length;
4289:     const isExpanded = cleanQuery ? true : expandedCategories.has(category.id);
4290:     return `
4291:       <section class="tree-category">
4292:         <div class="tree-category-heading">
4293:           <button class="tree-category-toggle" type="button" data-toggle-
category="${escapeHtml(category.id)}" aria-expanded="${isExpanded}">
4294:             <span>${highlightSearchText(category.label, projectSearchQuery)}</span>
4295:             <small>${allCategoryProjects.length} project${allCategoryProjects.length === 1 ? "" :
"s"}</small>
4296:           </button>
4297:         </div>
4298:         <div class="tree-projects" ${isExpanded ? "" : "hidden"}>
4299:           ${categoryProjects.length ? categoryProjects.map((project) => `
4300:             <div class="tree-project-row ${project.id === selectedProjectId ? "active" : ""}>
4301:               <button type="button" data-project-id="${escapeHtml(project.id)}">
4302:                 <span>${highlightSearchText(project.title, projectSearchQuery)}</span>
4303:               </button>
4304:               <button class="tree-project-delete" type="button" data-delete-
project="${escapeHtml(project.id)}" aria-label="Delete ${escapeHtml(project.title)}" title="Delete project">
4305:                 &times;
4306:               </button>
4307:             </div>
4308:           `).join("") : cleanQuery ? `<p class="tree-empty">Category matches, but no projects are in it
yet.</p>` : ""}
4309:         </div>
4310:       </section>
4311:     `;
4312:   }).join("");
4313:   projectTree.innerHTML = treeMarkup || `
4314:     <section class="tree-category tree-search-empty">
4315:       <p class="tree-empty">No project, file, tool, or section matched this search.</p>
4316:     </section>
4317:   `;
4318:   updateProjectSearchStatus(cleanQuery ? matchCount : catalog.projects.length);
4319: }

```

## summaryWordCount - lines 4321-4323

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4321-4323 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function summaryWordCount(project) {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | wordCount, plainTextFromRich                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4321: function summaryWordCount(project) {
4322:   return wordCount(project?.summaryRich?.blocks?.length ? plainTextFromRich(project.summaryRich) :
project?.summary || "");
4323: }

```

## renderSummaryPreview - lines 4325-4329

| Item            | Explanation                              |
|-----------------|------------------------------------------|
| Source location | Lines 4325-4329 (5 source lines).        |
| Purpose         | Renders UI, markup, or display output.   |
| Declaration     | function renderSummaryPreview(project) { |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | renderRichContent, renderMultilineInlineText                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4325: function renderSummaryPreview(project) {
4326:   if (project.summaryRich?.blocks?.length) return renderRichContent(project.summaryRich, project.summary
|| "");
4327:   if (project.summary) return `<p class="rich-
paragraph">${renderMultilineInlineText(project.summary)}</p>`;
4328:   return `<p class="evidence-empty">No project overview has been added yet.</p>`;
4329: }

```

## renderSummaryBuilder - lines 4331-4359

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4331-4359 (29 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderSummaryBuilder(project) {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | summaryWordCount, renderSummaryPreview                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4331: function renderSummaryBuilder(project) {
4332:   const isEditing = summaryEditorProjectId === project.id;
4333:   const summaryWords = summaryWordCount(project);
4334:   const hasSummary = summaryWords > 0 || Boolean(project.summaryRich?.blocks?.length);
4335:   return `
4336:     <div class="summary-builder wide-field">
4337:       <div class="summary-builder-heading">
4338:         <div>
4339:           <span>Project overview <small>${summaryWords}/1000 words</small></span>
4340:         </div>
4341:         <div class="summary-builder-actions">
4342:           ${isEditing
4343:             ? `<button class="button secondary" type="button" data-summary-cancel="true">Cancel
overview</button>`
4344:             : `<button class="button primary" type="button" data-summary-create="true">${hasSummary ?
"Edit overview" : "Create overview"}</button>`}
4345:           ${hasSummary && !isEditing ? `<button class="button danger-control" type="button" data-summary-
delete="true">Delete overview</button>` : ""}
4346:         </div>
4347:       </div>
4348:       ${isEditing ? `
4349:         <div class="rich-summary-panel">
4350:           <p class="rich-editor-hint">Right-click text or an inserted block for formatting, images,
formulas, code blocks, alignment, and delete controls.</p>
4351:           <div class="rich-summary-editor" data-rich-editor="summary" data-placeholder="Click anywhere
and start writing the project overview." contenteditable="true" spellcheck="true" aria-label="Rich overview
editor"></div>
4352:           <div class="summary-save-actions">
4353:             <button class="button primary" type="button" data-summary-save="true">Save overview</button>
4354:           </div>
4355:         </div>
4356:       : `<div class="summary-rendered-preview">${renderSummaryPreview(project)}</div>`}
4357:     </div>
4358:   `;
4359: }

```

## ***applyRichTextBlockStyle - lines 4361-4372***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4361-4372 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function applyRichTextBlockStyle(block) {                                                                                                                                                             |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | cleanFontFamily, normalizeFontPx, normalizeTextColor                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4361: function applyRichTextBlockStyle(block) {
4362:   const fontFamily = cleanFontFamily(block.dataset.fontFamily || "Arial") || "Arial";
4363:   const fontPx = normalizeFontPx(block.dataset.fontPx || "");
4364:   const color = normalizeTextColor(block.dataset.color || "");
4365:
4366:   block.style.fontFamily = fontFamily || "";
4367:   block.style.fontSize = fontPx ? `${fontPx}px` : "";
4368:   block.style.color = color || "";
4369:   block.style.fontWeight = block.dataset.bold === "true" ? "700" : "";
4370:   block.style.fontStyle = block.dataset.italic === "true" ? "italic" : "";
4371:   block.style.textDecoration = block.dataset.underline === "true" ? "underline" : "";
4372: }
```

## ***createRichTextBlock - lines 4374-4391***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4374-4391 (18 source lines).                                                                                                                                                                    |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function createRichTextBlock(text = "", fontSize = "normal", align = "left", fontFamily = "Arial", fontPx = "", color = "", bold = false, html = "", italic = false, underline = false) {             |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | createElement, cleanFontFamily, normalizeFontPx, normalizeTextColor, sanitizeRichInlineHtml, applyRichTextBlockStyle                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4374: function createRichTextBlock(text = "", fontSize = "normal", align = "left", fontFamily = "Arial", fontPx
= "", color = "", bold = false, html = "", italic = false, underline = false) {
4375:   const block = document.createElement("p");
4376:   block.className = `rich-block rich-text-block rich-text-${fontSize} text-${align}`;
4377:   block.dataset.type = "paragraph";
4378:   block.dataset.fontSize = fontSize;
4379:   block.dataset.align = align;
4380:   block.dataset.fontFamily = cleanFontFamily(fontFamily || "Arial") || "Arial";
4381:   block.dataset.fontPx = normalizeFontPx(fontPx);
4382:   block.dataset.color = normalizeTextColor(color);
4383:   block.dataset.bold = bold ? "true" : "false";
4384:   block.dataset.italic = italic ? "true" : "false";
4385:   block.dataset.underline = underline ? "true" : "false";
4386:   block.spellcheck = true;
4387:   if (html) block.innerHTML = sanitizeRichInlineHtml(html);
4388:   else block.textContent = text;
4389:   applyRichTextBlockStyle(block);
4390:   return block;
4391: }
```



## ***richBlockActions - lines 4393-4401***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4393-4401 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richBlockActions(label = "block", options = {}) {                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | escapeHtml                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4393: function richBlockActions(label = "block", options = {}) {
4394:   return `
4395:     <div class="rich-block-actions">
4396:       ${options.movable ? `<button class="rich-drag-handle" type="button" draggable="true" data-rich-
drag-handle aria-label="Move ${escapeHtml(label)}">Move</button>` : ""}
4397:       <button type="button" data-rich-block-action="edit" aria-label="Edit
${escapeHtml(label)}">Edit</button>
4398:       <button class="danger-icon" type="button" data-rich-block-action="delete" aria-label="Delete
${escapeHtml(label)}">Delete</button>
4399:     </div>
4400:   `;
4401: }
```

## ***createRichImageBlock - lines 4403-4431***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4403-4431 (29 source lines).                                                                                                                                                                    |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function createRichImageBlock(blockData) {                                                                                                                                                            |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | createElement, cleanRichImageTitle, normalizeCropAspect, normalizeCropZoom, normalizeCropPosition, richBlockActions, richImageCropStyle, escapeHtml, normalizeLinkTarget                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4403: function createRichImageBlock(blockData) {
4404:   const figure = document.createElement("figure");
4405:   const align = blockData.align || "center";
4406:   const title = cleanRichImageTitle(blockData);
4407:   figure.className = `rich-block rich-image-block justify-${align}`;
4408:   figure.dataset.type = "image";
4409:   figure.dataset.url = blockData.url || "";
4410:   figure.dataset.title = blockData.title || "";
4411:   figure.dataset.caption = blockData.caption || "";
4412:   figure.dataset.align = align;
4413:   figure.dataset.display = blockData.display === "download" ? "download" : "show";
4414:   figure.dataset.source = blockData.source || "local";
4415:   figure.dataset.cropAspect = normalizeCropAspect(blockData.cropAspect);
4416:   figure.dataset.cropZoom = String(normalizeCropZoom(blockData.cropZoom));
4417:   figure.dataset.cropX = String(normalizeCropPosition(blockData.cropX));
4418:   figure.dataset.cropY = String(normalizeCropPosition(blockData.cropY));
4419:   figure.contentEditable = "false";
4420:   figure.draggable = true;
4421:   figure.title = "Drag image to move it between text.";
4422:   figure.innerHTML = `
```

```

4423:     ${richBlockActions("overview image", { movable: true })}
4424:     ${figure.dataset.display === "download" ? `<span class="rich-download-badge">Download only</span>` :
""}
4425:     <span class="rich-image-viewport crop-${figure.dataset.cropAspect === "original" ? "original" :
"active"}">${richImageCropStyle(blockData)}>
4426:         
4427:     </span>
4428:     ${title || blockData.caption ? `<figcaption>${title ? `<strong>${escapeHtml(title)}</strong>` :
""}${blockData.caption ? `<span>${escapeHtml(blockData.caption)}</span>` : ""}</figcaption>` : ""}
4429: `;
4430: return figure;
4431: }

```

## createRichFormulaBlock - lines 4433-4446

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4433-4446 (14 source lines).                                                                                                                                                                    |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function createRichFormulaBlock(blockData) {                                                                                                                                                          |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | createElement, unwrapFormula, richBlockActions, escapeHtml                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4433: function createRichFormulaBlock(blockData) {
4434:     const block = document.createElement("div");
4435:     const align = blockData.align || "center";
4436:     block.className = `rich-block rich-formula-block justify-${align}`;
4437:     block.dataset.type = "formula";
4438:     block.dataset.formula = unwrapFormula(blockData.formula || "");
4439:     block.dataset.align = align;
4440:     block.contentEditable = "false";
4441:     block.innerHTML = `
4442:         ${richBlockActions("formula")}
4443:         <span class="formula-text">${escapeHtml(unwrapFormula(blockData.formula || ""))}</span>
4444:     `;
4445:     return block;
4446: }

```

## createRichCodeBlock - lines 4448-4465

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4448-4465 (18 source lines).                                                                                                                                                                    |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | function createRichCodeBlock(blockData = {}) {                                                                                                                                                        |
| Touches            | browser/Electron DOM, filesystem, authentication/security data                                                                                                                                        |
| Calls detected     | createElement, normalizeCodePasteMode, normalizeCodeText, normalizeCodeLanguage, detectCodeLanguage, richBlockActions, escapeHtml, codeLanguageLabel, tokenizedCodeHtml                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4448: function createRichCodeBlock(blockData = {}) {
4449:     const block = document.createElement("figure");
4450:     const pasteMode = normalizeCodePasteMode(blockData.pasteMode || "source");
4451:     const code = normalizeCodeText(blockData.code || "", pasteMode);

```

```

4452:   const language = normalizeCodeLanguage(blockData.language || detectCodeLanguage(code));
4453:   block.className = `rich-block rich-code-block rich-code-language-${language}`;
4454:   block.dataset.type = "code";
4455:   block.dataset.language = language;
4456:   block.dataset.pasteMode = pasteMode;
4457:   block.dataset.code = code;
4458:   block.contentEditable = "false";
4459:   block.innerHTML = `
4460:     ${richBlockActions("code block")}
4461:     <figcaption><span>&lt;/&gt;</span> ${escapeHtml(codeLanguageLabel(language))}<button type="button"
data-rich-block-action="copy-code">Copy code</button></figcaption>
4462:     <pre><code>${tokenizedCodeHtml(code, language)}</code></pre>
4463:   `;
4464:   return block;
4465: }

```

## refreshRichImageBlock - lines 4467-4497

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4467-4497 (31 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function refreshRichImageBlock(block, blockData) {                                                                                                                                                    |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | normalizeCropAspect, normalizeCropZoom, normalizeCropPosition, remove, add, cleanRichImageTitle, richBlockActions, richImageCropStyle, escapeHtml, normalizeLinkTarget                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4467: function refreshRichImageBlock(block, blockData) {
4468:   const align = blockData.align || block.dataset.align || "center";
4469:   block.dataset.url = blockData.url || block.dataset.url || "";
4470:   block.dataset.title = blockData.title || "";
4471:   block.dataset.caption = blockData.caption || "";
4472:   block.dataset.align = align;
4473:   block.dataset.display = blockData.display === "download" ? "download" : "show";
4474:   block.dataset.source = blockData.source || block.dataset.source || "local";
4475:   block.dataset.cropAspect = normalizeCropAspect(blockData.cropAspect || block.dataset.cropAspect);
4476:   block.dataset.cropZoom = String(normalizeCropZoom(blockData.cropZoom || block.dataset.cropZoom));
4477:   block.dataset.cropX = String(normalizeCropPosition(blockData.cropX ?? block.dataset.cropX));
4478:   block.dataset.cropY = String(normalizeCropPosition(blockData.cropY ?? block.dataset.cropY));
4479:   block.draggable = true;
4480:   block.title = "Drag image to move it between text.";
4481:   block.classList.remove("justify-left", "justify-center", "justify-right");
4482:   block.classList.add(`justify-${align}`);
4483:   const title = cleanRichImageTitle({ title: block.dataset.title });
4484:   block.innerHTML = `
4485:     ${richBlockActions("overview image", { movable: true })}
4486:     ${block.dataset.display === "download" ? `<span class="rich-download-badge">Download only</span>` :
""}
4487:     <span class="rich-image-viewport crop-${block.dataset.cropAspect === "original" ? "original" :
"active"}">${richImageCropStyle({
4488:       cropAspect: block.dataset.cropAspect,
4489:       cropZoom: block.dataset.cropZoom,
4490:       cropX: block.dataset.cropX,
4491:       cropY: block.dataset.cropY
4492:     })}>
4493:     
4494:     </span>
4495:     ${!(title || block.dataset.caption) ? `<figcaption>${title ? `<strong>${escapeHtml(title)}</strong>` :
""}${block.dataset.caption ? `<span>${escapeHtml(block.dataset.caption)}</span>` : ""}</figcaption>` : ""}
4496:   `;
4497: }

```

## refreshRichFormulaBlock - lines 4499-4510

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 4499-4510 (12 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function refreshRichFormulaBlock(block, blockData) {                                                                                                                                                  |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | unwrapFormula, remove, add, richBlockActions, escapeHtml                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4499: function refreshRichFormulaBlock(block, blockData) {
4500:   const align = blockData.align || block.dataset.align || "center";
4501:   const formula = unwrapFormula(blockData.formula || block.dataset.formula || "");
4502:   block.dataset.formula = formula;
4503:   block.dataset.align = align;
4504:   block.classList.remove("justify-left", "justify-center", "justify-right");
4505:   block.classList.add(`justify-${align}`);
4506:   block.innerHTML = `
4507:     ${richBlockActions("formula")}
4508:     <span class="formula-text">${escapeHtml(formula)}</span>
4509:   `;
4510: }
```

## ***refreshRichCodeBlock - lines 4512-4525***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4512-4525 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function refreshRichCodeBlock(block, blockData = {}) {                                                                                                                                                |
| Touches            | browser/Electron DOM, filesystem, authentication/security data                                                                                                                                        |
| Calls detected     | normalizeCodePasteMode, normalizeCodeText, normalizeCodeLanguage, detectCodeLanguage, richBlockActions, escapeHtml, codeLanguageLabel, tokenizedCodeHtml                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4512: function refreshRichCodeBlock(block, blockData = {}) {
4513:   const pasteMode = normalizeCodePasteMode(blockData.pasteMode || block.dataset.pasteMode || "source");
4514:   const code = normalizeCodeText(blockData.code || block.dataset.code || "", pasteMode);
4515:   const language = normalizeCodeLanguage(blockData.language || block.dataset.language ||
detectCodeLanguage(code));
4516:   block.dataset.language = language;
4517:   block.dataset.pasteMode = pasteMode;
4518:   block.dataset.code = code;
4519:   block.className = `rich-block rich-code-block rich-code-language-${language}`;
4520:   block.innerHTML = `
4521:     ${richBlockActions("code block")}
4522:     <figcaption><span><lt;/></span> ${escapeHtml(codeLanguageLabel(language))}<button type="button"
data-rich-block-action="copy-code">Copy code</button></figcaption>
4523:     <pre><code>${tokenizedCodeHtml(code, language)}</code></pre>
4524:   `;
4525: }
```

## ***createRichBlockElement - lines 4527-4544***

| Item            | Explanation                                                               |
|-----------------|---------------------------------------------------------------------------|
| Source location | Lines 4527-4544 (18 source lines).                                        |
| Purpose         | Creates an object, UI structure, process, payload, or generated artifact. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function createRichBlockElement(block) {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | createRichImageBlock, createRichFormulaBlock, createRichCodeBlock, createRichTextBlock                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4527: function createRichBlockElement(block) {
4528:   if (block.type === "image") return createRichImageBlock(block);
4529:   if (block.type === "formula") return createRichFormulaBlock(block);
4530:   if (block.type === "code") return createRichCodeBlock(block);
4531:
4532:   return createRichTextBlock(
4533:     block.text || "",
4534:     block.fontSize || "normal",
4535:     block.align || "left",
4536:     block.fontFamily || "Arial",
4537:     block.fontPx || "",
4538:     block.color || "",
4539:     Boolean(block.bold),
4540:     block.html || "",
4541:     Boolean(block.italic),
4542:     Boolean(block.underline)
4543:   );
4544: }
```

### ***populateSummaryEditor - lines 4546-4550***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4546-4550 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function populateSummaryEditor(project) {                                                                                                                                                             |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, populateRichEditor                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4546: function populateSummaryEditor(project) {
4547:   const editor = projectFields.querySelector("[data-rich-editor='summary']");
4548:   if (!editor) return;
4549:   populateRichEditor(editor, project);
4550: }
```

### ***populateStandaloneRichEditor - lines 4552-4560***

| Item            | Explanation                                                                       |
|-----------------|-----------------------------------------------------------------------------------|
| Source location | Lines 4552-4560 (9 source lines).                                                 |
| Purpose         | Part of the private builder frontend and editing experience.                      |
| Declaration     | function populateStandaloneRichEditor(editor, rich, fallbackText = "") {          |
| Touches         | Mostly local calculations or local UI state.                                      |
| Calls detected  | richBlocksFromValue, forEach, append, createRichBlockElement, createRichTextBlock |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4552: function populateStandaloneRichEditor(editor, rich, fallbackText = "") {
4553:   if (!editor) return;
4554:   editor.contentEditable = "true";
4555:   editor.tabIndex = 0;
4556:   editor.spellcheck = true;
4557:   editor.innerHTML = "";
4558:   richBlocksFromValue(rich, fallbackText).forEach((block) =>
editor.append(createRichBlockElement(block)));
4559:   if (!editor.children.length) editor.append(createRichTextBlock(""));
4560: }

```

## ***populateRichEditor - lines 4562-4592***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4562-4592 (31 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function populateRichEditor(editor, project = selectedProject()) {                                                                                                                                    |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | selectedProject, richBlocksFromValue, forEach, append, createRichBlockElement, createRichTextBlock, richBlocksForProject, getByPath                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4562: function populateRichEditor(editor, project = selectedProject()) {
4563:   if (!editor) return;
4564:   editor.contentEditable = "true";
4565:   editor.tabIndex = 0;
4566:   editor.spellcheck = true;
4567:
4568:   if (editor.dataset.richEditor === "pending-description") {
4569:     editor.innerHTML = "";
4570:     richBlocksFromValue(pendingEditor?.richDescription, pendingEditor?.description || "")
4571:       .forEach((block) => editor.append(createRichBlockElement(block)));
4572:     if (!editor.children.length) editor.append(createRichTextBlock(""));
4573:     return;
4574:   }
4575:
4576:   if (!project) return;
4577:   editor.innerHTML = "";
4578:
4579:   let blocks = [];
4580:
4581:   if (editor.dataset.richEditor === "summary" && !editor.dataset.richPath) {
4582:     blocks = richBlocksForProject(project);
4583:   } else {
4584:     const textPath = editor.dataset.richTextPath || "";
4585:     const richPath = editor.dataset.richPath || "";
4586:     blocks = richBlocksFromValue(getByPath(project, richPath), getByPath(project, textPath) || "");
4587:   }
4588:
4589:   blocks.forEach((block) => editor.append(createRichBlockElement(block)));
4590:   if (!editor.children.length) editor.append(createRichTextBlock(""));
4591: }
4592: }

```

## ***populateRichEditors - lines 4594-4596***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4594-4596 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function populateRichEditors(root = document) {                                                                                                                                                       |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelectorAll, forEach, populateRichEditor                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4594: function populateRichEditors(root = document) {
4595:     root.querySelectorAll("[data-rich-editor]").forEach((editor) => populateRichEditor(editor));
4596: }

```

## ***saveRichEditorToProject - lines 4598-4658***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4598-4658 (61 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | function saveRichEditorToProject(editor) {                                                                                                                                                            |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | syncFunFactsFromInput, setStatus, syncSiteRichField, syncProfileRichField, selectedProject, extractRichSummary, plainTextFromRich, wordCount, setByPath, scheduleAutosave, schedulePreviewRender      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 9 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4598: function saveRichEditorToProject(editor) {
4599:     if (!editor) return true;
4600:
4601:     if (editor.dataset.richEditor === "fun-facts") {
4602:         syncFunFactsFromInput();
4603:         setStatus("Unsaved local changes.");
4604:         return true;
4605:     }
4606:
4607:     if (editor.dataset.richEditor === "site-field") {
4608:         syncSiteRichField(editor);
4609:         setStatus("Front page text formatting updated locally. Click Save draft before applying to
site.");
4610:         return true;
4611:     }
4612:
4613:     if (editor.dataset.richEditor === "profile-field") {
4614:         syncProfileRichField(editor);
4615:         setStatus("Profile and contact formatting updated locally. Click Save draft before applying to
site.");
4616:         return true;
4617:     }
4618:
4619:     if (editor.dataset.richEditor === "standalone") {
4620:         setStatus("Unsaved dialog changes. Use the dialog save button to keep them.");
4621:         return true;
4622:     }
4623:
4624:     const project = selectedProject();
4625:
4626:     if (editor.dataset.richEditor === "pending-description" && pendingEditor) {
4627:         const rich = extractRichSummary(editor);

```

```

4628:     pendingEditor.richDescription = rich;
4629:     pendingEditor.description = plainTextFromRich(rich);
4630:     setStatus("Unsaved local changes.");
4631:     return true;
4632: }
4633:
4634: if (!project) return true;
4635:
4636: const rich = extractRichSummary(editor);
4637: const plain = plainTextFromRich(rich);
4638:
4639: if (wordCount(plain) > 1000) {
4640:     setStatus("Overview is limited to 1000 words. Shorten the text before saving.");
4641:     return false;
4642: }
4643:
4644: if (editor.dataset.richEditor === "summary" && !editor.dataset.richPath) {
4645:     project.summaryRich = rich;
4646:     project.summary = plain;
4647: } else {
4648:     const textPath = editor.dataset.richTextPath;
4649:     const richPath = editor.dataset.richPath;
4650:     if (textPath) setByPath(project, textPath, plain);
4651:     if (richPath) setByPath(project, richPath, rich);
4652: }
4653:
4654: setStatus("Unsaved local changes.");
4655: scheduleAutosave();
4656: schedulePreviewRender();
4657: return true;
4658: }

```

## ***saveVisibleRichEditors - lines 4660-4664***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4660-4664 (5 source lines).                                                                                                                                                                     |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | function saveVisibleRichEditors() {                                                                                                                                                                   |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelectorAll, forEach, saveRichEditorToProject                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4660: function saveVisibleRichEditors() {
4661:     document.querySelectorAll("[data-rich-editor]").forEach((editor) => {
4662:         saveRichEditorToProject(editor);
4663:     });
4664: }

```

## ***currentRichBlock - lines 4666-4672***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 4666-4672 (7 source lines).                            |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function currentRichBlock(editor) {                          |
| Touches            | browser/Electron DOM                                         |
| Calls detected     | getSelection, contains, querySelector, closest               |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 2 return statement(s).                |



| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4666: function currentRichBlock(editor) {
4667:   const selection = window.getSelection();
4668:   let node = selection?.anchorNode;
4669:   if (!node || !editor.contains(node)) return editor.querySelector(".rich-block:last-child");
4670:   if (node.nodeType === Node.TEXT_NODE) node = node.parentElement;
4671:   return node.closest(".rich-block") || editor.querySelector(".rich-block:last-child");
4672: }

```

## ***selectionRangeInsideEditor - lines 4674-4682***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4674-4682 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function selectionRangeInsideEditor(editor) {                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getSelection, getRangeAt, contains, cloneRange                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4674: function selectionRangeInsideEditor(editor) {
4675:   const selection = window.getSelection();
4676:   if (!selection?.rangeCount) return null;
4677:   const range = selection.getRangeAt(0);
4678:   const container = range.commonAncestorContainer.nodeType === Node.TEXT_NODE
4679:     ? range.commonAncestorContainer.parentElement
4680:     : range.commonAncestorContainer;
4681:   return container && editor.contains(container) ? range.cloneRange() : null;
4682: }

```

## ***captureRichSelection - lines 4684-4688***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4684-4688 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function captureRichSelection(editor) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectionRangeInsideEditor                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4684: function captureRichSelection(editor) {
4685:   const range = selectionRangeInsideEditor(editor);
4686:   if (range) activeRichSelectionRange = range;
4687:   return range;
4688: }

```

## ***rememberRichFormattingSelection - lines 4690-4700***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4690-4700 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function rememberRichFormattingSelection(editor = activeSummaryEditor) {                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | selectionRangeInsideEditor, rangeBelongsToEditor, cloneRange                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4690: function rememberRichFormattingSelection(editor = activeSummaryEditor) {
4691:   if (!editor) return null;
4692:   const range = selectionRangeInsideEditor(editor)
4693:     || (rangeBelongsToEditor(activeTextSelectionRange, editor) ? activeTextSelectionRange.cloneRange() :
null)
4694:     || (rangeBelongsToEditor(activeRichSelectionRange, editor) ? activeRichSelectionRange.cloneRange() :
null);
4695:   if (!range) return null;
4696:   activeSummaryEditor = editor;
4697:   activeRichSelectionRange = range.cloneRange();
4698:   if (!range.collapsed) activeTextSelectionRange = range.cloneRange();
4699:   return range;
4700: }
```

## ***restoreRichSelection - lines 4702-4713***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4702-4713 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function restoreRichSelection(editor) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | contains, getSelection, removeAllRanges, addRange, cloneRange, showPersistentTextSelection                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
4702: function restoreRichSelection(editor) {
4703:   if (!activeRichSelectionRange) return false;
4704:   const container = activeRichSelectionRange.commonAncestorContainer.nodeType === Node.TEXT_NODE
4705:     ? activeRichSelectionRange.commonAncestorContainer.parentElement
4706:     : activeRichSelectionRange.commonAncestorContainer;
4707:   if (!container || !editor.contains(container)) return false;
4708:   const selection = window.getSelection();
4709:   selection.removeAllRanges();
4710:   selection.addRange(activeRichSelectionRange.cloneRange());
4711:   showPersistentTextSelection(activeRichSelectionRange);
4712:   return true;
4713: }
```

## ***showPersistentTextSelection - lines 4715-4745***

| Item            | Explanation                        |
|-----------------|------------------------------------|
| Source location | Lines 4715-4745 (31 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function showPersistentTextSelection(range) {                                                                                                                                                         |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | set, Highlight, cloneRange, closest, createElement, setAttribute, append, replaceChildren, getClientRects, forEach                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4715: function showPersistentTextSelection(range) {
4716:   if (!range || range.collapsed) return;
4717:   if (window.CSS?.highlights && typeof window.Highlight === "function") {
4718:     window.CSS.highlights.set(persistentSelectionHighlightName, new
window.Highlight(range.cloneRange()));
4719:     if (persistentSelectionOverlay) persistentSelectionOverlay.hidden = true;
4720:     return;
4721:   }
4722:
4723:   const container = range.commonAncestorContainer.nodeType === Node.TEXT_NODE
4724:     ? range.commonAncestorContainer.parentElement
4725:     : range.commonAncestorContainer;
4726:   const editor = container?.closest?("[data-rich-editor]") || activeSummaryEditor;
4727:   const host = editor?.closest("dialog") || document.body;
4728:   if (!persistentSelectionOverlay) {
4729:     persistentSelectionOverlay = document.createElement("div");
4730:     persistentSelectionOverlay.className = "persistent-selection-overlay";
4731:     persistentSelectionOverlay.setAttribute("aria-hidden", "true");
4732:   }
4733:   if (persistentSelectionOverlay.parentElement !== host) host.append(persistentSelectionOverlay);
4734:   persistentSelectionOverlay.replaceChildren();
4735:   [...range.getClientRects()].forEach((rect) => {
4736:     if (rect.width < 1 || rect.height < 1) return;
4737:     const marker = document.createElement("span");
4738:     marker.style.left = `${rect.left}px`;
4739:     marker.style.top = `${rect.top}px`;
4740:     marker.style.width = `${rect.width}px`;
4741:     marker.style.height = `${rect.height}px`;
4742:     persistentSelectionOverlay.append(marker);
4743:   });
4744:   persistentSelectionOverlay.hidden = !persistentSelectionOverlay.childElementCount;
4745: }

```

## ***clearPersistentTextSelection - lines 4747-4756***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4747-4756 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function clearPersistentTextSelection(clearRanges = false) {                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | replaceChildren                                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4747: function clearPersistentTextSelection(clearRanges = false) {
4748:   window.CSS?.highlights?.delete?.(persistentSelectionHighlightName);
4749:   if (persistentSelectionOverlay) {
4750:     persistentSelectionOverlay.replaceChildren();
4751:     persistentSelectionOverlay.hidden = true;
4752:   }

```

```

4753:   if (!clearRanges) return;
4754:   activeRichSelectionRange = null;
4755:   activeTextSelectionRange = null;
4756: }

```

## ***refreshPersistentSelectionHighlight - lines 4758-4765***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4758-4765 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function refreshPersistentSelectionHighlight() {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | rangeBelongsToEditor, showPersistentTextSelection                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4758: function refreshPersistentSelectionHighlight() {
4759:   const range = rangeBelongsToEditor(activeTextSelectionRange, activeSummaryEditor)
4760:     ? activeTextSelectionRange
4761:     : rangeBelongsToEditor(activeRichSelectionRange, activeSummaryEditor)
4762:     ? activeRichSelectionRange
4763:     : null;
4764:   if (range) showPersistentTextSelection(range);
4765: }

```

## ***rangeBelongsToEditor - lines 4767-4773***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4767-4773 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function rangeBelongsToEditor(range, editor) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | contains                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4767: function rangeBelongsToEditor(range, editor) {
4768:   if (!range || range.collapsed || !editor) return false;
4769:   const container = range.commonAncestorContainer.nodeType === Node.TEXT_NODE
4770:     ? range.commonAncestorContainer.parentElement
4771:     : range.commonAncestorContainer;
4772:   return Boolean(container && editor.contains(container));
4773: }

```

## ***editorFromRange - lines 4775-4781***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 4775-4781 (7 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function editorFromRange(range) {                            |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4775: function editorFromRange(range) {
4776:   if (!range || range.collapsed) return null;
4777:   const container = range.commonAncestorContainer.nodeType === Node.TEXT_NODE
4778:     ? range.commonAncestorContainer.parentElement
4779:     : range.commonAncestorContainer;
4780:   return container?.closest?("[data-rich-editor]") || null;
4781: }

```

## ***pointInsideRange - lines 4783-4793***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4783-4793 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function pointInsideRange(range, x, y) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getClientRects, some                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4783: function pointInsideRange(range, x, y) {
4784:   if (!range || range.collapsed) return false;
4785:   return [...range.getClientRects()].some((rect) =>
4786:     rect.width > 0 &&
4787:     rect.height > 0 &&
4788:     x >= rect.left &&
4789:     x <= rect.right &&
4790:     y >= rect.top &&
4791:     y <= rect.bottom
4792:   );
4793: }

```

## ***richEditorFromContextEvent - lines 4795-4805***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 4795-4805 (11 source lines).                           |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function richEditorFromContextEvent(event) {                 |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | rangeBelongsToEditor, pointInsideRange, editorFromRange      |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 3 return statement(s).                |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4795: function richEditorFromContextEvent(event) {
4796:   const directEditor = event.target.closest?("[data-rich-editor]");
4797:   if (directEditor) return directEditor;
4798:   const savedRange = rangeBelongsToEditor(activeTextSelectionRange, activeSummaryEditor)
4799:     ? activeTextSelectionRange
4800:     : activeRichSelectionRange;
4801:   if (pointInsideRange(savedRange, event.clientX, event.clientY)) {
4802:     return editorFromRange(savedRange);
4803:   }
4804:   return null;
4805: }

```

## ***displayRichFontName - lines 4807-4812***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4807-4812 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function displayRichFontName(value = "") {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | split, replace, trim                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4807: function displayRichFontName(value = "") {
4808:   return String(value || "")
4809:     .split(",")[0]
4810:     .replace(/^[\'"]|[\']$/g, "")
4811:     .trim() || "Arial";
4812: }

```

## ***rgbColorToHex - lines 4814-4823***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4814-4823 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function rgbColorToHex(value = "") {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | trim, toLowerCase, slice, split, map, join, match, min, toString, padStart                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4814: function rgbColorToHex(value = "") {
4815:   const color = String(value || "").trim();
4816:   if (/^[0-9a-f]{6}$/i.test(color)) return color.toLowerCase();
4817:   if (/^[0-9a-f]{3}$/i.test(color)) {
4818:     return `#${color.slice(1).split("").map((part) => part + part).join("")}`.toLowerCase();
4819:   }

```

```

4820:   const match = color.match(/^rgba?\s*(\d+)\D+(\d+)\D+(\d+)\D+(\d+)/i);
4821:   if (!match) return color;
4822:   return `${match.slice(1, 4).map((part) => Math.min(255, Number(part))).toString(16).padStart(2,
"0")}.join("")}`;
4823: }

```

## textNodesInRichSelection - lines 4825-4844

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4825-4844 (20 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function textNodesInRichSelection(range, editor, includeWhitespace = false) {                                                                                                                         |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | createTreeWalker, acceptNode, trim, closest, intersectsNode, nextNode, push                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4825: function textNodesInRichSelection(range, editor, includeWhitespace = false) {
4826:   const walker = document.createTreeWalker(editor, NodeFilter.SHOW_TEXT, {
4827:     acceptNode(node) {
4828:       if (!node.textContent || (!includeWhitespace && !node.textContent.trim())) return
NodeFilter.FILTER_REJECT;
4829:       if (node.parentElement?.closest("[contenteditable='false']")) return NodeFilter.FILTER_REJECT;
4830:       try {
4831:         return range.intersectsNode(node) ? NodeFilter.FILTER_ACCEPT : NodeFilter.FILTER_REJECT;
4832:       } catch {
4833:         return NodeFilter.FILTER_REJECT;
4834:       }
4835:     }
4836:   });
4837:   const nodes = [];
4838:   let node = walker.nextNode();
4839:   while (node) {
4840:     nodes.push(node);
4841:     node = walker.nextNode();
4842:   }
4843:   return nodes;
4844: }

```

## richBlockFromRange - lines 4846-4853

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4846-4853 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richBlockFromRange(range, editor) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | contains                                                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4846: function richBlockFromRange(range, editor) {
4847:   if (!range || !editor) return null;
4848:   const container = range.startContainer?.nodeType === Node.TEXT_NODE
4849:     ? range.startContainer.parentElement
4850:     : range.startContainer;
4851:   const block = container?.closest?(".rich-block");

```

```

4852:   return block && editor.contains(block) ? block : null;
4853: }

```

## ***uniformSelectionValue - lines 4855-4861***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4855-4861 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function uniformSelectionValue(values) {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | map, trim, filter, every, toLowerCase                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4855: function uniformSelectionValue(values) {
4856:   const cleanValues = values.map((value) => String(value || "").trim()).filter(Boolean);
4857:   if (!cleanValues.length) return "";
4858:   return cleanValues.every((value) => value.toLowerCase() === cleanValues[0].toLowerCase())
4859:     ? cleanValues[0]
4860:     : "";
4861: }

```

## ***richSelectionStyleState - lines 4863-4871***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4863-4871 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richSelectionStyleState(range, editor) {                                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | textNodesInRichSelection, map, getComputedStyle, uniformSelectionValue, rgbColorToHex, displayRichFontName, round                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4863: function richSelectionStyleState(range, editor) {
4864:   const textNodes = textNodesInRichSelection(range, editor);
4865:   const styles = textNodes.map((node) => getComputedStyle(node.parentElement || editor));
4866:   return {
4867:     color: uniformSelectionValue(styles.map((style) => rgbColorToHex(style.color))),
4868:     font: uniformSelectionValue(styles.map((style) => displayRichFontName(style.fontFamily))),
4869:     size: uniformSelectionValue(styles.map((style) => String(Math.round(parseFloat(style.fontSize) ||
16))))
4870:   };
4871: }

```

## ***closeSelectionInspectorOptions - lines 4873-4877***

| Item            | Explanation                                               |
|-----------------|-----------------------------------------------------------|
| Source location | Lines 4873-4877 (5 source lines).                         |
| Purpose         | Controls a window, dialog, panel, menu, or visible state. |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function closeSelectionInspectorOptions(except = "") {                                                                                                                                                |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelectorAll, forEach                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4873: function closeSelectionInspectorOptions(except = "") {
4874:   textSelectionInspector?.querySelectorAll("[data-selection-options]").forEach((options) => {
4875:     if (options.dataset.selectionOptions !== except) options.hidden = true;
4876:   });
4877: }

```

## ***positionSelectionInspector - lines 4879-4900***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4879-4900 (22 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function positionSelectionInspector(range) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getBoundingClientRect, closest, min, max                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4879: function positionSelectionInspector(range) {
4880:   if (!textSelectionInspector || selectionInspectorWasMoved) return;
4881:   const rect = range.getBoundingClientRect();
4882:   const hostDialog = textSelectionInspector.closest("dialog");
4883:   const hostRect = hostDialog?.getBoundingClientRect();
4884:   const bounds = {
4885:     bottom: Math.min(window.innerHeight - 8, (hostRect?.bottom ?? window.innerHeight) - 8),
4886:     left: Math.max(8, (hostRect?.left ?? 0) + 8),
4887:     right: Math.min(window.innerWidth - 8, (hostRect?.right ?? window.innerWidth) - 8),
4888:     top: Math.max(8, (hostRect?.top ?? 0) + 8)
4889:   };
4890:   const inspectorRect = textSelectionInspector.getBoundingClientRect();
4891:   const inspectorWidth = Math.min(inspectorRect.width || 310, bounds.right - bounds.left);
4892:   const inspectorHeight = Math.min(inspectorRect.height || 260, bounds.bottom - bounds.top);
4893:   const left = Math.max(bounds.left, Math.min(rect.left, bounds.right - inspectorWidth));
4894:   const preferredTop = rect.bottom + 10;
4895:   const top = preferredTop + inspectorHeight <= bounds.bottom
4896:     ? preferredTop
4897:     : Math.max(bounds.top, rect.top - inspectorHeight - 10);
4898:   textSelectionInspector.style.left = `${left}px`;
4899:   textSelectionInspector.style.top = `${top}px`;
4900: }

```

## ***showTextSelectionInspector - lines 4902-4924***

| Item            | Explanation                                               |
|-----------------|-----------------------------------------------------------|
| Source location | Lines 4902-4924 (23 source lines).                        |
| Purpose         | Controls a window, dialog, panel, menu, or visible state. |
| Declaration     | function showTextSelectionInspector(editor, range) {      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | toString, trim, closest, append, cloneRange, showPersistentTextSelection, richSelectionStyleState, slice, positionSelectionInspector                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4902: function showTextSelectionInspector(editor, range) {
4903:   if (!textSelectionInspector || !editor || !range || range.collapsed) return;
4904:   const selectedText = range.toString().trim();
4905:   if (!selectedText) return;
4906:   const inspectorHost = editor.closest("dialog") || document.body;
4907:   if (textSelectionInspector.parentElement !== inspectorHost)
inspectorHost.append(textSelectionInspector);
4908:   activeSummaryEditor = editor;
4909:   activeRichSelectionRange = range.cloneRange();
4910:   activeTextSelectionRange = range.cloneRange();
4911:   showPersistentTextSelection(range);
4912:   const state = richSelectionStyleState(range, editor);
4913:   selectionTextPreview.textContent = selectedText.length > 120 ? `${selectedText.slice(0, 117)}...` :
selectedText;
4914:   selectionTextPreview.title = selectedText;
4915:   selectionFontInput.value = state.font;
4916:   selectionColorInput.value = state.color;
4917:   selectionSizeInput.value = state.size;
4918:   selectionFontInput.placeholder = state.font ? "" : "Mixed";
4919:   selectionColorInput.placeholder = state.color ? "" : "Mixed";
4920:   selectionSizeInput.placeholder = state.size ? "" : "Mixed";
4921:   if (state.color && /^[0-9a-f]{6}$/i.test(state.color)) selectionColorPicker.value = state.color;
4922:   textSelectionInspector.hidden = false;
4923:   positionSelectionInspector(range);
4924: }

```

## ***hideTextSelectionInspector - lines 4926-4931***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4926-4931 (6 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function hideTextSelectionInspector() {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closeSelectionInspectorOptions                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4926: function hideTextSelectionInspector() {
4927:   if (!textSelectionInspector) return;
4928:   textSelectionInspector.hidden = true;
4929:   closeSelectionInspectorOptions();
4930:   selectionInspectorWasMoved = false;
4931: }

```

## ***refreshTextSelectionInspector - lines 4933-4953***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 4933-4953 (21 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function refreshTextSelectionInspector() {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | matches, getSelection, showTextSelectionInspector, hideTextSelectionInspector, getRangeAt                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4933: function refreshTextSelectionInspector() {
4934:   if (!textSelectionInspector || selectionInspectorPointerInside ||
textSelectionInspector.matches(":focus-within")) return;
4935:   const selection = window.getSelection();
4936:   if (!selection?.rangeCount || selection.isCollapsed) {
4937:     if (activeTextSelectionRange && !activeTextSelectionRange.collapsed && activeSummaryEditor) {
4938:       showTextSelectionInspector(activeSummaryEditor, activeTextSelectionRange);
4939:       return;
4940:     }
4941:     hideTextSelectionInspector();
4942:     return;
4943:   }
4944:   const range = selection.getRangeAt(0);
4945:   let node = range.commonAncestorContainer;
4946:   if (node.nodeType === Node.TEXT_NODE) node = node.parentElement;
4947:   const editor = node?.closest?("[data-rich-editor]");
4948:   if (!editor) {
4949:     hideTextSelectionInspector();
4950:     return;
4951:   }
4952:   showTextSelectionInspector(editor, range);
4953: }

```

## ***scheduleTextSelectionInspectorRefresh - lines 4955-4958***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4955-4958 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function scheduleTextSelectionInspectorRefresh(delay = 0) {                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4955: function scheduleTextSelectionInspectorRefresh(delay = 0) {
4956:   clearTimeout(selectionInspectorRefreshTimer);
4957:   selectionInspectorRefreshTimer = setTimeout(refreshTextSelectionInspector, delay);
4958: }

```

## ***applySelectionInspectorFormat - lines 4960-4990***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 4960-4990 (31 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function applySelectionInspectorFormat(kind, rawValue) {     |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | cloneRange, cleanFontFamily, applyRichInlineCommand, normalizeTextColor, rgbColorToHex, normalizeFontPx, closeSelectionInspectorOptions, requestAnimationFrame, showTextSelectionInspector            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4960: function applySelectionInspectorFormat(kind, rawValue) {
4961:   if (!activeSummaryEditor || !activeTextSelectionRange) return;
4962:   activeRichSelectionRange = activeTextSelectionRange.cloneRange();
4963:   if (kind === "font") {
4964:     const font = cleanFontFamily(rawValue);
4965:     if (!font) return;
4966:     selectionFontInput.value = font;
4967:     applyRichInlineCommand(activeSummaryEditor, "fontName", font);
4968:   }
4969:   if (kind === "color") {
4970:     const color = normalizeTextColor(rawValue);
4971:     if (!color) return;
4972:     const normalized = rgbColorToHex(color);
4973:     selectionColorInput.value = normalized;
4974:     if (/^[0-9a-f]{6}$/i.test(normalized)) selectionColorPicker.value = normalized;
4975:     applyRichInlineCommand(activeSummaryEditor, "foreColor", normalized);
4976:   }
4977:   if (kind === "size") {
4978:     const size = normalizeFontPx(rawValue);
4979:     if (!size) return;
4980:     selectionSizeInput.value = size;
4981:     applyRichInlineCommand(activeSummaryEditor, "fontSize", size);
4982:   }
4983:   closeSelectionInspectorOptions();
4984:   requestAnimationFrame(() => {
4985:     if (activeRichSelectionRange && activeSummaryEditor) {
4986:       activeTextSelectionRange = activeRichSelectionRange.cloneRange();
4987:       showTextSelectionInspector(activeSummaryEditor, activeRichSelectionRange);
4988:     }
4989:   });
4990: }

```

## ***renderSelectionInspectorOptions - lines 4992-5007***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 4992-5007 (16 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderSelectionInspectorOptions() {                                                                                                                                                          |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, map, escapeHtml, join                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

4992: function renderSelectionInspectorOptions() {
4993:   const fontOptions = textSelectionInspector?.querySelector("[data-selection-options='font']");
4994:   const sizeOptions = textSelectionInspector?.querySelector("[data-selection-options='size']");
4995:   if (fontOptions) {
4996:     fontOptions.innerHTML = commonRichFonts.map((font) => `<button type="button" data-selection-
value="font" data-value="${escapeHtml(font)}" style="font-family:
${escapeHtml(font)}">${escapeHtml(font)}</button>`).join("");
4997:   }
4998:   if (sizeOptions) {
4999:     sizeOptions.innerHTML = commonRichSizes.map((size) => `<button type="button" data-selection-

```

```

value="size" data-value="${size}">${size} px</button>`).join("");
5000:   }
5001:   if (selectionColorSwatches) {
5002:     selectionColorSwatches.innerHTML = commonRichColors.map((color) => `<button type="button" data-
selection-value="color" data-value="${color}" style="--selection-swatch: ${color}" aria-label="Use
${color}"></button>`).join("");
5003:   }
5004:   if (richContextColorSwatches) {
5005:     richContextColorSwatches.innerHTML = commonRichColors.map((color) => `<button type="button" data-
rich-color-value="${color}" style="--selection-swatch: ${color}" aria-label="Use ${color}"></button>`).join("");
5006:   }
5007: }

```

## applyStyleToRichSelection - lines 5009-5055

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5009-5055 (47 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function applyStyleToRichSelection(editor, range, command, value = "") {                                                                                                                              |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | textNodesInRichSelection, every, getComputedStyle, includes, forEach, max, min, splitText, createElement, cleanFontFamily, normalizeTextColor, normalizeFontPx                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5009: function applyStyleToRichSelection(editor, range, command, value = "") {
5010:   const nodes = textNodesInRichSelection(range, editor, true);
5011:   if (!nodes.length) return null;
5012:   const makeBold = command === "bold"
5013:     ? !nodes.every((node) => Number.parseInt(getComputedStyle(node.parentElement || editor).fontWeight,
10) >= 600)
5014:     : false;
5015:   const makeItalic = command === "italic"
5016:     ? !nodes.every((node) => getComputedStyle(node.parentElement || editor).fontStyle === "italic")
5017:     : false;
5018:   const makeUnderline = command === "underline"
5019:     ? !nodes.every((node) => getComputedStyle(node.parentElement ||
editor).textDecorationLine.includes("underline"))
5020:     : false;
5021:   const formattedNodes = [];
5022:
5023:   nodes.forEach((node) => {
5024:     const length = node.textContent?.length || 0;
5025:     let start = node === range.startContainer ? range.startOffset : 0;
5026:     let end = node === range.endContainer ? range.endOffset : length;
5027:     start = Math.max(0, Math.min(start, length));
5028:     end = Math.max(start, Math.min(end, length));
5029:     if (start === end) return;
5030:
5031:     if (end < length) node.splitText(end);
5032:     const selectedNode = start > 0 ? node.splitText(start) : node;
5033:     const wrapper = document.createElement("span");
5034:     wrapper.className = "rich-inline-style";
5035:     wrapper.style.display = "inline";
5036:     if (command === "fontName") wrapper.style.fontFamily = cleanFontFamily(value);
5037:     if (command === "foreColor") wrapper.style.color = normalizeTextColor(value);
5038:     if (command === "fontSize") wrapper.style.fontSize = `${normalizeFontPx(value)}px`;
5039:     if (command === "bold") wrapper.style.fontWeight = makeBold ? "700" : "400";
5040:     if (command === "italic") wrapper.style.fontStyle = makeItalic ? "italic" : "normal";
5041:     if (command === "underline") wrapper.style.textDecoration = makeUnderline ? "underline" : "none";
5042:     selectedNode.replaceWith(wrapper);
5043:     wrapper.append(selectedNode);
5044:     formattedNodes.push(selectedNode);
5045:   });
5046:
5047:   if (!formattedNodes.length) return null;
5048:   normalizeInlineStyleSpans(editor);
5049:   const nextRange = document.createRange();
5050:   const firstNode = formattedNodes[0];
5051:   const lastNode = formattedNodes[formattedNodes.length - 1];
5052:   nextRange.setStart(firstNode, 0);
5053:   nextRange.setEnd(lastNode, lastNode.textContent?.length || 0);
5054:   return nextRange;
5055: }

```

## ***applyRichInlineCommand - lines 5057-5103***

| Item               | Explanation                                                                                                                                                                                                                                                |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5057-5103 (47 source lines).                                                                                                                                                                                                                         |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                                                                               |
| Declaration        | function applyRichInlineCommand(editor, command, value = "") {                                                                                                                                                                                             |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                                                                           |
| Calls detected     | selectionRangeInsideEditor, cloneRange, rangeBelongsToEditor, applyStyleToRichSelection, showPersistentTextSelection, richBlockFromRange, selectedRichBlock, currentRichBlock, setStatus, selectRichBlock, updateCurrentTextBlock, saveRichEditorToProject |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                                   |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                                    |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                                                                              |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                                      |

Source excerpt:

```
5057: function applyRichInlineCommand(editor, command, value = "") {
5058:   if (!editor) return;
5059:   const liveRange = selectionRangeInsideEditor(editor);
5060:   const savedRange = liveRange && !liveRange.collapsed
5061:     ? liveRange.cloneRange()
5062:     : rangeBelongsToEditor(activeTextSelectionRange, editor)
5063:     ? activeTextSelectionRange.cloneRange()
5064:     : rangeBelongsToEditor(activeRichSelectionRange, editor)
5065:     ? activeRichSelectionRange.cloneRange()
5066:     : null;
5067:
5068:   let appliedToSelection = false;
5069:   if (savedRange) {
5070:     const nextRange = applyStyleToRichSelection(editor, savedRange, command, value);
5071:     if (nextRange) {
5072:       activeRichSelectionRange = nextRange.cloneRange();
5073:       activeTextSelectionRange = nextRange.cloneRange();
5074:       showPersistentTextSelection(nextRange);
5075:       appliedToSelection = true;
5076:     }
5077:   }
5078:
5079:   if (!appliedToSelection) {
5080:     const block = richBlockFromRange(savedRange, editor) || selectedRichBlock(editor) ||
currentRichBlock(editor);
5081:     if (!block || block.dataset.type !== "paragraph") {
5082:       setStatus("Highlight text or click inside a text line before formatting.");
5083:       return;
5084:     }
5085:     selectRichBlock(block);
5086:     if (command === "fontName") updateCurrentTextBlock(editor, { fontFamily: value });
5087:     if (command === "foreColor") updateCurrentTextBlock(editor, { color: value });
5088:     if (command === "fontSize") updateCurrentTextBlock(editor, { fontPx: value });
5089:     if (command === "bold") updateCurrentTextBlock(editor, { bold: block.dataset.bold !== "true" });
5090:     if (command === "italic") {
5091:       updateCurrentTextBlock(editor, { italic: block.dataset.italic !== "true" });
5092:     }
5093:     if (command === "underline") {
5094:       updateCurrentTextBlock(editor, { underline: block.dataset.underline !== "true" });
5095:     }
5096:     saveRichEditorToProject(editor);
5097:     setStatus("Formatting applied and saved locally for the current text line.");
5098:     return;
5099:   }
5100:
5101:   saveRichEditorToProject(editor);
5102:   setStatus("Unsaved local changes. Click the section save button to keep this formatting.");
5103: }
```

## ***selectRichBlock - lines 5105-5113***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 5105-5113 (9 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function selectRichBlock(block) {                            |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelectorAll, forEach, remove, add, closest, indexOf                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5105: function selectRichBlock(block) {
5106:   document.querySelectorAll(".rich-block.selected").forEach((item) => item.classList.remove("selected"));
5107:   activeSummaryBlock = block || null;
5108:   if (block) {
5109:     block.classList.add("selected");
5110:     const editor = block.closest("[data-rich-editor]");
5111:     if (editor) editor.dataset.activeRichBlockIndex = String([...editor.children].indexOf(block));
5112:   }
5113: }

```

## ***selectedRichBlock - lines 5115-5123***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5115-5123 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function selectedRichBlock(editor) {                                                                                                                                                                  |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | isInteger, matches, contains, currentRichBlock                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5115: function selectedRichBlock(editor) {
5116:   if (!editor) return null;
5117:   const index = Number(editor.dataset.activeRichBlockIndex);
5118:   if (Number.isInteger(index) && index >= 0 && editor.children[index]?.matches(".rich-block")) {
5119:     return editor.children[index];
5120:   }
5121:   if (activeSummaryBlock && editor.contains(activeSummaryBlock)) return activeSummaryBlock;
5122:   return currentRichBlock(editor);
5123: }

```

## ***syncRichContextMenuControls - lines 5125-5144***

| Item               | Explanation                                                                                                                            |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5125-5144 (20 source lines).                                                                                                     |
| Purpose            | Keeps two places aligned, such as local data and target files.                                                                         |
| Declaration        | function syncRichContextMenuControls(block) {                                                                                          |
| Touches            | browser/Electron DOM, filesystem                                                                                                       |
| Calls detected     | getComputedStyle, displayRichFontName, round, normalizeTextColor, restoreRichSelection, closest, queryCommandState, some, setAttribute |
| API endpoints      | None detected inside this function body.                                                                                               |
| Storage keys       | No local/session storage keys detected.                                                                                                |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                          |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5125: function syncRichContextMenuControls(block) {
5126:   if (!block || block.dataset.type !== "paragraph") return;
5127:
5128:   const computed = getComputedStyle(block);
5129:   const family = displayRichFontName(computed.fontFamily) || "Arial";
5130:   const size = String(Math.round(parseFloat(computed.fontSize) || 16));
5131:   const color = normalizeTextColor(block.dataset.color || "") || "#17202a";
5132:   restoreRichSelection(block.closest("[data-rich-editor]"));
5133:   const bold = document.queryCommandState("bold") || block.dataset.bold === "true";
5134:
5135:   if (richFontSelect) {
5136:     richFontSelect.value = [...richFontSelect.options].some((option) => option.value === family ||
option.textContent === family)
5137:       ? family
5138:       : "Arial";
5139:   }
5140:
5141:   if (richFontSize) richFontSize.value = size;
5142:   if (richColorInput) richColorInput.value = color;
5143:   if (richBoldButton) richBoldButton.setAttribute("aria-pressed", bold ? "true" : "false");
5144: }

```

## ***focusTextBlock - lines 5146-5158***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5146-5158 (13 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function focusTextBlock(block) {                                                                                                                                                                      |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | selectRichBlock, closest, focus, createRange, selectNodeContents, collapse, getSelection, removeAllRanges, addRange, captureRichSelection                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5146: function focusTextBlock(block) {
5147:   if (!block) return;
5148:   selectRichBlock(block);
5149:   const editor = block.closest("[data-rich-editor]");
5150:   editor?.focus();
5151:   const range = document.createRange();
5152:   range.selectNodeContents(block);
5153:   range.collapse(false);
5154:   const selection = window.getSelection();
5155:   selection.removeAllRanges();
5156:   selection.addRange(range);
5157:   captureRichSelection(editor);
5158: }

```

## ***setTextBlockCaretOffset - lines 5160-5174***

| Item            | Explanation                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 5160-5174 (15 source lines).                                                                                                        |
| Purpose         | Part of the private builder frontend and editing experience.                                                                              |
| Declaration     | function setTextBlockCaretOffset(block, offset) {                                                                                         |
| Touches         | browser/Electron DOM                                                                                                                      |
| Calls detected  | selectRichBlock, closest, focus, max, min, createRange, setStart, collapse, getSelection, removeAllRanges, addRange, captureRichSelection |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5160: function setTextBlockCaretOffset(block, offset) {
5161:   if (!block) return;
5162:   selectRichBlock(block);
5163:   const editor = block.closest("[data-rich-editor]");
5164:   editor?.focus();
5165:   const textNode = block.firstChild || block;
5166:   const safeOffset = Math.max(0, Math.min(offset, textNode.textContent?.length || 0));
5167:   const range = document.createRange();
5168:   range.setStart(textNode, safeOffset);
5169:   range.collapse(true);
5170:   const selection = window.getSelection();
5171:   selection.removeAllRanges();
5172:   selection.addRange(range);
5173:   captureRichSelection(editor);
5174: }

```

### ***textBlockCaretOffset - lines 5176-5183***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5176-5183 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function textBlockCaretOffset(block) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getSelection, contains, getRangeAt, cloneRange, selectNodeContents, setEnd, toString                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5176: function textBlockCaretOffset(block) {
5177:   const selection = window.getSelection();
5178:   if (!selection?.rangeCount || !block.contains(selection.anchorNode)) return block.textContent.length;
5179:   const range = selection.getRangeAt(0).cloneRange();
5180:   range.selectNodeContents(block);
5181:   range.setEnd(selection.anchorNode, selection.anchorOffset);
5182:   return range.toString().length;
5183: }

```

### ***matchingTextBlock - lines 5185-5198***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 5185-5198 (14 source lines).                           |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | function matchingTextBlock(block, text = "") {               |
| Touches            | browser/Electron DOM, filesystem                             |
| Calls detected     | createRichTextBlock                                          |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 1 return statement(s).                |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5185: function matchingTextBlock(block, text = "") {
5186:     return createRichTextBlock(
5187:         text,
5188:         block?.dataset.fontSize || "normal",
5189:         block?.dataset.align || "left",
5190:         block?.dataset.fontFamily || "Arial",
5191:         block?.dataset.fontPx || "",
5192:         block?.dataset.color || "",
5193:         block?.dataset.bold === "true",
5194:         "",
5195:         block?.dataset.italic === "true",
5196:         block?.dataset.underline === "true"
5197:     );
5198: }

```

## ***richInsertionRange - lines 5200-5221***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5200-5221 (22 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richInsertionRange(editor) {                                                                                                                                                                 |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | selectionRangeInsideEditor, collapse, restoreRichSelection, querySelectorAll, at, createRange, selectNodeContents                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5200: function richInsertionRange(editor) {
5201:     const currentRange = selectionRangeInsideEditor(editor);
5202:     if (currentRange) {
5203:         currentRange.collapse(true);
5204:         return currentRange;
5205:     }
5206:     if (restoreRichSelection(editor)) {
5207:         const restored = selectionRangeInsideEditor(editor);
5208:         if (restored) {
5209:             restored.collapse(true);
5210:             return restored;
5211:         }
5212:     }
5213:     const lastText = [...editor.querySelectorAll(".rich-text-block")].at(-1);
5214:     if (lastText) {
5215:         const range = document.createRange();
5216:         range.selectNodeContents(lastText);
5217:         range.collapse(false);
5218:         return range;
5219:     }
5220:     return null;
5221: }

```

## ***insertRichBlockAtCursor - lines 5223-5257***

| Item            | Explanation                                                                     |
|-----------------|---------------------------------------------------------------------------------|
| Source location | Lines 5223-5257 (35 source lines).                                              |
| Purpose         | Part of the private builder frontend and editing experience.                    |
| Declaration     | function insertRichBlockAtCursor(editor, blockElement, insertionRange = null) { |
| Touches         | browser/Electron DOM                                                            |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | cloneRange, richInsertionRange, selectedRichBlock, contains, collapse, createRange, setStart, setEnd, extractContents, after, matchingTextBlock, append                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5223: function insertRichBlockAtCursor(editor, blockElement, insertionRange = null) {
5224:   const range = insertionRange?.cloneRange() || richInsertionRange(editor);
5225:   const rangeNode = range?.startContainer?.nodeType === Node.TEXT_NODE
5226:     ? range.startContainer.parentElement
5227:     : range?.startContainer;
5228:   const current = rangeNode?.closest?(".rich-block") || selectedRichBlock(editor);
5229:   let nextText = null;
5230:
5231:   if (range && current?.dataset.type === "paragraph" && current.contains(range.startContainer)) {
5232:     range.collapse(true);
5233:     const tail = document.createRange();
5234:     tail.setStart(range.startContainer, range.startOffset);
5235:     tail.setEnd(current, current.childNodes.length);
5236:     const trailingContent = tail.extractContents();
5237:     current.after(blockElement);
5238:     nextText = matchingTextBlock(current, "");
5239:     nextText.append(trailingContent);
5240:     blockElement.after(nextText);
5241:   } else if (range?.startContainer === editor) {
5242:     const reference = editor.children[range.startOffset] || null;
5243:     editor.insertBefore(blockElement, reference);
5244:     nextText = createRichTextBlock("");
5245:     blockElement.after(nextText);
5246:   } else if (current) {
5247:     current.after(blockElement);
5248:     nextText = matchingTextBlock(current, "");
5249:     blockElement.after(nextText);
5250:   } else {
5251:     editor.append(blockElement);
5252:     nextText = createRichTextBlock("");
5253:     editor.append(nextText);
5254:   }
5255:
5256:   focusTextBlock(nextText);
5257: }

```

### ***insertRichBlockAfterCursor - lines 5259-5261***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5259-5261 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function insertRichBlockAfterCursor(editor, blockElement) {                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | insertRichBlockAtCursor                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5259: function insertRichBlockAfterCursor(editor, blockElement) {
5260:   insertRichBlockAtCursor(editor, blockElement);
5261: }

```

### ***cleanupEmptyRichTextBlocks - lines 5263-5275***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5263-5275 (13 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function cleanupEmptyRichTextBlocks(editor) {                                                                                                                                                         |
| Touches            | browser/Electron DOM, authentication/security data                                                                                                                                                    |
| Calls detected     | querySelectorAll, forEach, richElementPlainText, remove, querySelector, append, createRichTextBlock                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5263: function cleanupEmptyRichTextBlocks(editor) {
5264:   if (!editor) return;
5265:   [...editor.querySelectorAll(":scope > .rich-text-block")].forEach((block) => {
5266:     const hasVisibleText = richElementPlainText(block);
5267:     if (hasVisibleText) return;
5268:     const previous = block.previousElementSibling;
5269:     const next = block.nextElementSibling;
5270:     const keepTypingBlockAfterImage = previous?.dataset.type === "image" && !next;
5271:     const isOnlyBlock = editor.children.length <= 1;
5272:     if (!isOnlyBlock && !keepTypingBlockAfterImage) block.remove();
5273:   });
5274:   if (!editor.querySelector(":scope > .rich-block")) editor.append(createRichTextBlock(""));
5275: }

```

## ***normalizeRichEditorStructure - lines 5277-5307***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5277-5307 (31 source lines).                                                                                                                                                                    |
| Purpose            | Validates or normalizes input so later code receives safe predictable values.                                                                                                                         |
| Declaration        | function normalizeRichEditorStructure(editor) {                                                                                                                                                       |
| Touches            | browser/Electron DOM, filesystem, authentication/security data                                                                                                                                        |
| Calls detected     | forEach, trim, remove, createRichTextBlock, replaceWith, matches, sanitizeRichInlineHtml, querySelectorAll, add, removeAttribute, applyRichTextBlockStyle, querySelector                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5277: function normalizeRichEditorStructure(editor) {
5278:   if (!editor) return;
5279:   [...editor.childNodes].forEach((node) => {
5280:     if (node.nodeType === Node.TEXT_NODE) {
5281:       if (!node.textContent.trim()) {
5282:         node.remove();
5283:         return;
5284:       }
5285:       const paragraph = createRichTextBlock(node.textContent);
5286:       node.replaceWith(paragraph);
5287:       return;
5288:     }
5289:   });
5290:   if (node.nodeType !== Node.ELEMENT_NODE || node.matches(".rich-block")) return;
5291:   const paragraph = createRichTextBlock("");
5292:   paragraph.innerHTML = sanitizeRichInlineHtml(node.innerHTML || node.textContent || "");
5293:   node.replaceWith(paragraph);
5294: });
5295:
5296: editor.querySelectorAll(".rich-text-block").forEach((block) => {
5297:   block.dataset.type = "paragraph";
5298:   block.dataset.fontSize ||= "normal";

```

```

5299:     block.dataset.align ||= "left";
5300:     block.dataset.fontFamily ||= "Arial";
5301:     block.classList.add("rich-block");
5302:     block.removeAttribute("contenteditable");
5303:     applyRichTextBlockStyle(block);
5304: });
5305:
5306: if (!editor.querySelector(":scope > .rich-block")) editor.append(createRichTextBlock(""));
5307: }

```

## ***richElementPlainText - lines 5309-5313***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5309-5313 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richElementPlainText(element) {                                                                                                                                                              |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | cloneNode, querySelectorAll, forEach, replaceWith, trim                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5309: function richElementPlainText(element) {
5310:   const cloneElement = element.cloneNode(true);
5311:   cloneElement.querySelectorAll("br").forEach((lineBreak) => lineBreak.replaceWith("\n"));
5312:   return String(cloneElement.textContent || "").trim();
5313: }

```

## ***updateCurrentTextBlock - lines 5315-5357***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5315-5357 (43 source lines).                                                                                                                                                                    |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateCurrentTextBlock(editor, updates) {                                                                                                                                                    |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | selectedRichBlock, remove, add, call, cleanFontFamily, normalizeFontPx, normalizeTextColor, applyRichTextBlockStyle, saveRichEditorToProject                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5315: function updateCurrentTextBlock(editor, updates) {
5316:   const block = selectedRichBlock(editor);
5317:   if (!block || block.dataset.type !== "paragraph") return;
5318:
5319:   if (updates.fontSize) {
5320:     block.dataset.fontSize = updates.fontSize;
5321:     block.classList.remove("rich-text-small", "rich-text-normal", "rich-text-large");
5322:     block.classList.add(`rich-text-${updates.fontSize}`);
5323:   }
5324:
5325:   if (updates.align) {
5326:     block.dataset.align = updates.align;
5327:     block.classList.remove("text-left", "text-center", "text-right");
5328:     block.classList.add(`text-${updates.align}`);
5329:   }
5330:
5331:   if (Object.prototype.hasOwnProperty.call(updates, "fontFamily")) {

```

```

5332:     block.dataset.fontFamily = cleanFontFamily(updates.fontFamily);
5333: }
5334:
5335: if (Object.prototype.hasOwnProperty.call(updates, "fontPx")) {
5336:     block.dataset.fontPx = normalizeFontPx(updates.fontPx);
5337: }
5338:
5339: if (Object.prototype.hasOwnProperty.call(updates, "color")) {
5340:     block.dataset.color = normalizeTextColor(updates.color);
5341: }
5342:
5343: if (Object.prototype.hasOwnProperty.call(updates, "bold")) {
5344:     block.dataset.bold = updates.bold ? "true" : "false";
5345: }
5346:
5347: if (Object.prototype.hasOwnProperty.call(updates, "italic")) {
5348:     block.dataset.italic = updates.italic ? "true" : "false";
5349: }
5350:
5351: if (Object.prototype.hasOwnProperty.call(updates, "underline")) {
5352:     block.dataset.underline = updates.underline ? "true" : "false";
5353: }
5354:
5355: applyRichTextBlockStyle(block);
5356: saveRichEditorToProject(editor);
5357: }

```

## extractRichSummary - lines 5359-5416

| Item               | Explanation                                                                                                                                                                                                                               |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5359-5416 (58 source lines).                                                                                                                                                                                                        |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                                                              |
| Declaration        | function extractRichSummary(editor) {                                                                                                                                                                                                     |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                                                          |
| Calls detected     | normalizeRichEditorStructure, normalizeInlineStyleSpans, querySelectorAll, map, normalizeCropAspect, normalizeCropPosition, normalizeCropZoom, unwrapFormula, normalizeCodeText, querySelector, normalizeCodeLanguage, detectCodeLanguage |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                  |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                   |
| Async/return clues | 0 await expression(s), 8 return statement(s).                                                                                                                                                                                             |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                     |

Source excerpt:

```

5359: function extractRichSummary(editor) {
5360:     normalizeRichEditorStructure(editor);
5361:     normalizeInlineStyleSpans(editor);
5362:     const blocks = [...editor.querySelectorAll(".rich-block")].map((element) => {
5363:         const align = element.dataset.align || "left";
5364:         if (element.dataset.type === "image") {
5365:             return {
5366:                 align,
5367:                 caption: element.dataset.caption || "",
5368:                 cropAspect: normalizeCropAspect(element.dataset.cropAspect),
5369:                 cropX: normalizeCropPosition(element.dataset.cropX),
5370:                 cropY: normalizeCropPosition(element.dataset.cropY),
5371:                 cropZoom: normalizeCropZoom(element.dataset.cropZoom),
5372:                 display: element.dataset.display === "download" ? "download" : "show",
5373:                 source: element.dataset.source || "local",
5374:                 title: element.dataset.title || "",
5375:                 type: "image",
5376:                 url: element.dataset.url || ""
5377:             };
5378:         }
5379:         if (element.dataset.type === "formula") {
5380:             return {
5381:                 align,
5382:                 formula: unwrapFormula(element.dataset.formula || element.textContent || ""),
5383:                 type: "formula"
5384:             };
5385:         }
5386:         if (element.dataset.type === "code") {
5387:             const code = normalizeCodeText(element.dataset.code || element.querySelector("code")?.textContent || "", element.dataset.pasteMode || "source");
5388:             if (!code) return null;
5389:             return {
5390:                 code,
5391:                 language: normalizeCodeLanguage(element.dataset.language || detectCodeLanguage(code)),

```

```

5392:         pasteMode: normalizeCodePasteMode(element.dataset.pasteMode),
5393:         type: "code"
5394:     };
5395: }
5396: const text = richElementPlainText(element);
5397: if (!text) return null;
5398: if (isFormulaOnly(text)) {
5399:     return { align: element.dataset.align || "center", formula: unwrapFormula(text), type: "formula" };
5400: }
5401: return {
5402:     align,
5403:     bold: element.dataset.bold === "true",
5404:     color: element.dataset.color || "",
5405:     fontFamily: element.dataset.fontFamily || "",
5406:     fontPx: element.dataset.fontPx || "",
5407:     fontSize: element.dataset.fontSize || "normal",
5408:     html: sanitizeRichInlineHtml(element.innerHTML),
5409:     italic: element.dataset.italic === "true",
5410:     text,
5411:     underline: element.dataset.underline === "true",
5412:     type: "paragraph"
5413: };
5414: }).filter(Boolean);
5415: return { blocks };
5416: }

```

## uploadSummaryImageFile - lines 5418-5458

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5418-5458 (41 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function uploadSummaryImageFile(file, options = {}) {                                                                                                                                           |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | selectedProject, isAllowedUpload, setStatus, call, withExtension, readFileAsDataURL, fetch, stringify, json, normalizeCropAspect, normalizeCropPosition, normalizeCropZoom                            |
| API endpoints      | /api/upload                                                                                                                                                                                           |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5418: async function uploadSummaryImageFile(file, options = {}) {
5419:     const project = selectedProject();
5420:     const projectId = options.projectId || project?.id;
5421:     if (!projectId || !file) return null;
5422:     const validation = isAllowedUpload("media", file.name, file.type);
5423:     if (!validation.ok) {
5424:         setStatus(validation.message);
5425:         return null;
5426:     }
5427:     const displayTitle = Object.prototype.hasOwnProperty.call(options, "title") ? options.title :
file.name;
5428:     const fileName = withExtension(displayTitle || file.name, file.name);
5429:     const data = await readFileAsDataURL(file);
5430:     const response = await fetch("/api/upload", {
5431:         method: "POST",
5432:         headers: { "Content-Type": "application/json" },
5433:         body: JSON.stringify({
5434:             projectId,
5435:             section: options.folder || "summary",
5436:             fileName,
5437:             data
5438:         })
5439:     });
5440:     const result = await response.json();
5441:     if (!response.ok) {
5442:         setStatus(result.error || "Image upload failed.");
5443:         return null;
5444:     }
5445:     return {
5446:         align: options.align || "center",
5447:         caption: options.caption || "",
5448:         cropAspect: normalizeCropAspect(options.cropAspect),
5449:         cropX: normalizeCropPosition(options.cropX),
5450:         cropY: normalizeCropPosition(options.cropY),
5451:         cropZoom: normalizeCropZoom(options.cropZoom),
5452:         display: options.display === "download" ? "download" : "show",

```

```

5453:     source: "local",
5454:     title: displayTitle || "",
5455:     type: "image",
5456:     url: result.url
5457:   };
5458: }

```

## updateSummaryImageDialogVisibility - lines 5460-5465

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5460-5465 (6 source lines).                                                                                                                                                                     |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateSummaryImageDialogVisibility() {                                                                                                                                                       |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector                                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5460: function updateSummaryImageDialogVisibility() {
5461:   const source = summaryImageSource?.value || "local";
5462:   const isLocal = source === "local";
5463:   document.querySelector(".summary-image-local-field").hidden = !isLocal;
5464:   document.querySelector(".summary-image-url-field").hidden = isLocal;
5465: }

```

## openSummaryImageDialog - lines 5467-5546

| Item               | Explanation                                                                                                                                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5467-5546 (80 source lines).                                                                                                                                                                                                       |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                                                                |
| Declaration        | async function openSummaryImageDialog(existingBlock = null, options = {}) {                                                                                                                                                              |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                                                         |
| Calls detected     | querySelector, normalizeCropAspect, normalizeCropZoom, updateSummaryImageDialogVisibility, dialogValue, normalizeLinkTarget, trim, setStatus, urlLooksLikeDirectImage, normalizeCropPosition, displayNameFromUrl, uploadSummaryImageFile |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                 |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                  |
| Async/return clues | 1 await expression(s), 7 return statement(s).                                                                                                                                                                                            |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                    |

Source excerpt:

```

5467: async function openSummaryImageDialog(existingBlock = null, options = {}) {
5468:   const heading = summaryImageDialog.querySelector("h2");
5469:   const submitButton = summaryImageDialog.querySelector("button[type='submit']");
5470:   if (heading) heading.textContent = existingBlock ? "Edit image" : "Add image to overview";
5471:   if (submitButton) submitButton.textContent = existingBlock ? "Save image" : "Add image";
5472:   summaryImageSource.value = existingBlock?.dataset.source || "local";
5473:   summaryImageFile.value = "";
5474:   summaryImageUrl.value = existingBlock?.dataset.source && existingBlock.dataset.source !== "local" ?
existingBlock.dataset.url || "" : "";
5475:   summaryImageTitle.value = existingBlock?.dataset.title || "";
5476:   summaryImageCaption.value = existingBlock?.dataset.caption || "";
5477:   summaryImageAlign.value = existingBlock?.dataset.align || "center";
5478:   summaryImageDisplay.value = existingBlock?.dataset.display || "show";
5479:   summaryImageCrop.value = normalizeCropAspect(existingBlock?.dataset.cropAspect);
5480:   summaryImageZoom.value = String(normalizeCropZoom(existingBlock?.dataset.cropZoom));
5481:   updateSummaryImageDialogVisibility();
5482:   const saved = await dialogValue(summaryImageDialog);
5483:   if (!saved) return null;

```



```

5484:   const file = summaryImageFile.files[0];
5485:   const source = summaryImageSource.value || "local";
5486:   const url = normalizeLinkTarget(summaryImageUrl.value.trim(), { assumeWeb: true });
5487:
5488:   if (source !== "local") {
5489:     const existingUrl = existingBlock?.dataset.url || "";
5490:     const nextUrl = url || existingUrl;
5491:     if (!nextUrl) {
5492:       setStatus("Paste an image URL or share link first.");
5493:       return null;
5494:     }
5495:     const display = summaryImageDisplay.value === "download" ? "download" : "show";
5496:     if (display === "show" && !urlLooksLikeDirectImage(nextUrl)) {
5497:       setStatus("Use a direct image URL for visible images, or choose Downloadable file only for share
links and non-image URLs.");
5498:       return null;
5499:     }
5500:     return {
5501:       align: summaryImageAlign.value,
5502:       caption: summaryImageCaption.value.trim(),
5503:       cropAspect: summaryImageCrop.value,
5504:       cropX: normalizeCropPosition(existingBlock?.dataset.cropX),
5505:       cropY: normalizeCropPosition(existingBlock?.dataset.cropY),
5506:       cropZoom: normalizeCropZoom(summaryImageZoom.value),
5507:       display,
5508:       source,
5509:       title: summaryImageTitle.value.trim() || displayNameFromUrl(nextUrl, source === "drive" ? "Google
Drive image" : "web image"),
5510:       type: "image",
5511:       url: nextUrl
5512:     };
5513:   }
5514:
5515:   if (!file && !existingBlock) {
5516:     setStatus("Choose an image first.");
5517:     return null;
5518:   }
5519:   if (!file && existingBlock) {
5520:     return {
5521:       align: summaryImageAlign.value,
5522:       caption: summaryImageCaption.value.trim(),
5523:       cropAspect: summaryImageCrop.value,
5524:       cropX: normalizeCropPosition(existingBlock.dataset.cropX),
5525:       cropY: normalizeCropPosition(existingBlock.dataset.cropY),
5526:       cropZoom: normalizeCropZoom(summaryImageZoom.value),
5527:       display: summaryImageDisplay.value === "download" ? "download" : "show",
5528:       source: existingBlock.dataset.source || "local",
5529:       title: summaryImageTitle.value.trim() || existingBlock.dataset.title || "",
5530:       type: "image",
5531:       url: existingBlock.dataset.url || ""
5532:     };
5533:   }
5534:   return uploadSummaryImageFile(file, {
5535:     align: summaryImageAlign.value,
5536:     caption: summaryImageCaption.value.trim(),
5537:     cropAspect: summaryImageCrop.value,
5538:     cropX: 50,
5539:     cropY: 50,
5540:     cropZoom: normalizeCropZoom(summaryImageZoom.value),
5541:     display: summaryImageDisplay.value === "download" ? "download" : "show",
5542:     folder: options.folder || "summary",
5543:     projectId: options.projectId || "",
5544:     title: summaryImageTitle.value.trim() || file.name
5545:   });
5546: }

```

## openSummaryFormulaDialog - lines 5548-5563

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5548-5563 (16 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | async function openSummaryFormulaDialog(existingBlock = null) {                                                                                                                                       |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | querySelector, dialogValue, unwrapFormula, trim, setStatus                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
5548: async function openSummaryFormulaDialog(existingBlock = null) {
5549:   const heading = summaryFormulaDialog.querySelector("h2");
5550:   const submitButton = summaryFormulaDialog.querySelector("button[type='submit']");
5551:   if (heading) heading.textContent = existingBlock ? "Edit formula" : "Add formula";
5552:   if (submitButton) submitButton.textContent = existingBlock ? "Save formula" : "Add formula";
5553:   summaryFormulaInput.value = existingBlock?.dataset.formula || "";
5554:   summaryFormulaAlign.value = existingBlock?.dataset.align || "center";
5555:   const saved = await dialogValue(summaryFormulaDialog);
5556:   if (!saved) return null;
5557:   const formula = unwrapFormula(summaryFormulaInput.value.trim());
5558:   if (!formula) {
5559:     setStatus("Paste or type a formula first.");
5560:     return null;
5561:   }
5562:   return { align: summaryFormulaAlign.value, formula, type: "formula" };
5563: }
```

## ***refreshSummaryCodeDialogPreview - lines 5565-5575***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5565-5575 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function refreshSummaryCodeDialogPreview() {                                                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | detectCodeLanguage, normalizeCodeLanguage, normalizeCodeText, renderRichCodeBlock                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
5565: function refreshSummaryCodeDialogPreview() {
5566:   if (!summaryCodePreview || !summaryCodeInput) return;
5567:   const rawCode = summaryCodeInput.value || "";
5568:   const language = summaryCodeLanguage?.value === "auto"
5569:     ? detectCodeLanguage(rawCode)
5570:     : normalizeCodeLanguage(summaryCodeLanguage?.value || "javascript");
5571:   const code = normalizeCodeText(rawCode, summaryCodePasteMode?.value || "source");
5572:   summaryCodePreview.innerHTML = code
5573:     ? renderRichCodeBlock({ code, language, pasteMode: summaryCodePasteMode?.value || "source" })
5574:     : `<p class="evidence-empty">Code preview appears here.</p>`;
5575: }
```

## ***beautifyCodeClient - lines 5577-5592***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5577-5592 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function beautifyCodeClient(code = "", language = "javascript") {                                                                                                                                     |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | replace, normalizeCodeLanguage, includes, split, map, trim, max, repeat, match, join, trimEnd                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
5577: function beautifyCodeClient(code = "", language = "javascript") {
5578:   const normalized = String(code || "").replace(/\r\n?/g, "\n").replace(/\t/g, " ");
5579:   const lang = normalizeCodeLanguage(language);
```

```

5580:   if (["c", "cpp", "java", "javascript", "verilog", "systemverilog"].includes(lang)) {
5581:     let depth = 0;
5582:     return normalized.split("\n").map((rawLine) => {
5583:       const line = rawLine.trim();
5584:       if (!line) return "";
5585:       if (/^[{}]\]/.test(line)) depth = Math.max(0, depth - 1);
5586:       const nextLine = `${"  ".repeat(depth)}${line}`;
5587:       const opens = (line.match(/\{([/g) || []).length;
5588:       const closes = (line.match(/[\}]/g) || []).length;
5589:       depth = Math.max(0, depth + opens - closes);
5590:       return nextLine;
5591:     }).join("\n").trimEnd();
5592:   }

```

## beautifySummaryCodeDialog - lines 5599-5618

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5599-5618 (20 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function beautifySummaryCodeDialog() {                                                                                                                                                          |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | detectCodeLanguage, normalizeCodeLanguage, fetch, now, stringify, json, Error, beautifyCodeClient, refreshSummaryCodeDialogPreview                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5599: async function beautifySummaryCodeDialog() {
5600:   if (!summaryCodeInput) return;
5601:   const language = summaryCodeLanguage?.value === "auto"
5602:     ? detectCodeLanguage(summaryCodeInput.value || "")
5603:     : normalizeCodeLanguage(summaryCodeLanguage?.value || "javascript");
5604:   try {
5605:     const response = await fetch(`/api/code/beautify?t=${Date.now()}`, {
5606:       method: "POST",
5607:       cache: "no-store",
5608:       headers: { "Content-Type": "application/json" },
5609:       body: JSON.stringify({ language, code: summaryCodeInput.value || "" })
5610:     });
5611:     const result = await response.json();
5612:     if (!response.ok || !result.ok) throw new Error(result.error || "Beautifier failed.");
5613:     summaryCodeInput.value = result.code || summaryCodeInput.value;
5614:   } catch {
5615:     summaryCodeInput.value = beautifyCodeClient(summaryCodeInput.value || "", language);
5616:   }
5617:   refreshSummaryCodeDialogPreview();
5618: }

```

## openSummaryCodeDialog - lines 5620-5640

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5620-5640 (21 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | async function openSummaryCodeDialog(existingBlock = null, options = {}) {                                                                                                                            |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | querySelector, detectCodeLanguage, refreshSummaryCodeDialogPreview, dialogValue, normalizeCodePasteMode, normalizeCodeText, setStatus, normalizeCodeLanguage                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
5620: async function openSummaryCodeDialog(existingBlock = null, options = {}) {
5621:   const heading = summaryCodeDialog.querySelector("h2");
5622:   const submitButton = summaryCodeDialog.querySelector("button[type='submit']");
5623:   if (heading) heading.textContent = existingBlock ? "Edit code block" : "Add code block";
5624:   if (submitButton) submitButton.textContent = existingBlock ? "Save code" : "Add code";
5625:   const defaultCode = options.code || "";
5626:   summaryCodeLanguage.value = existingBlock?.dataset.language || (defaultCode ?
detectCodeLanguage(defaultCode) : "auto");
5627:   summaryCodePasteMode.value = existingBlock?.dataset.pasteMode || options.pasteMode || "source";
5628:   summaryCodeInput.value = existingBlock?.dataset.code || defaultCode;
5629:   refreshSummaryCodeDialogPreview();
5630:   const saved = await dialogValue(summaryCodeDialog);
5631:   if (!saved) return null;
5632:   const pasteMode = normalizeCodePasteMode(summaryCodePasteMode.value);
5633:   const code = normalizeCodeText(summaryCodeInput.value, pasteMode);
5634:   if (!code) {
5635:     setStatus("Paste or type code before saving the block.");
5636:     return null;
5637:   }
5638:   const language = summaryCodeLanguage.value === "auto" ? detectCodeLanguage(code) :
normalizeCodeLanguage(summaryCodeLanguage.value);
5639:   return { code, language, pasteMode, type: "code" };
5640: }
```

## ***configureSummaryContextMenu - lines 5642-5665***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5642-5665 (24 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function configureSummaryContextMenu(mode = "rich", options = {}) {                                                                                                                                   |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | Set, querySelectorAll, forEach, has                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
5642: function configureSummaryContextMenu(mode = "rich", options = {}) {
5643:   const plainActions = new Set([
5644:     "copy-text",
5645:     "paste-text",
5646:     "cut-text",
5647:     "select-all-text",
5648:     "toggle-bold",
5649:     "toggle-italic",
5650:     "toggle-underline"
5651:   ]);
5652:   summaryContextMenu.querySelectorAll("[data-rich-action]").forEach((button) => {
5653:     const action = button.dataset.richAction;
5654:     button.hidden = mode === "plain" ? !plainActions.has(action) : false;
5655:   });
5656:   summaryContextMenu.querySelectorAll(".rich-menu-field").forEach((field) => {
5657:     field.hidden = false;
5658:   });
5659:   if (mode === "rich") {
5660:     const textOnly = Boolean(options.textOnly);
5661:     summaryContextMenu.querySelectorAll("[data-rich-action='add-image'], [data-rich-action='add-
formula'], [data-rich-action='add-code'], [data-rich-action='paste-code']").forEach((button) => {
5662:       button.hidden = textOnly;
5663:     });
5664:   }
5665: }
```

## ***applyPlainTextControlFormat - lines 5667-5701***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 5667-5701 (35 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function applyPlainTextControlFormat(control, updates = {}) {                                                                                                                                         |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | call, cleanFontFamily, normalizeFontPx, normalizeTextColor, rgbColorToHex, storePlainControlStyle, setStatus                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5667: function applyPlainTextControlFormat(control, updates = {}) {
5668:   if (!control) return;
5669:   if (Object.prototype.hasOwnProperty.call(updates, "fontFamily")) {
5670:     const font = cleanFontFamily(updates.fontFamily);
5671:     control.dataset.builderFontFamily = font;
5672:     control.style.fontFamily = font || "";
5673:   }
5674:   if (Object.prototype.hasOwnProperty.call(updates, "fontPx")) {
5675:     const size = normalizeFontPx(updates.fontPx);
5676:     control.dataset.builderFontPx = size;
5677:     control.style.fontSize = size ? `${size}px` : "";
5678:   }
5679:   if (Object.prototype.hasOwnProperty.call(updates, "color")) {
5680:     const color = normalizeTextColor(updates.color);
5681:     control.dataset.builderColor = color;
5682:     control.style.color = color || "";
5683:     if (richColorInput && color && /^[0-9a-f]{6}$/i.test(rgbColorToHex(color))) {
5684:       richColorInput.value = rgbColorToHex(color);
5685:     }
5686:   }
5687:   if (Object.prototype.hasOwnProperty.call(updates, "bold")) {
5688:     control.dataset.builderBold = updates.bold ? "true" : "false";
5689:     control.style.fontWeight = updates.bold ? "700" : "";
5690:   }
5691:   if (Object.prototype.hasOwnProperty.call(updates, "italic")) {
5692:     control.dataset.builderItalic = updates.italic ? "true" : "false";
5693:     control.style.fontStyle = updates.italic ? "italic" : "";
5694:   }
5695:   if (Object.prototype.hasOwnProperty.call(updates, "underline")) {
5696:     control.dataset.builderUnderline = updates.underline ? "true" : "false";
5697:     control.style.textDecoration = updates.underline ? "underline" : "";
5698:   }
5699:   storePlainControlStyle(control);
5700:   setStatus("Formatting applied and saved locally for this field.");
5701: }

```

## ***syncPlainContextMenuControls - lines 5703-5719***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5703-5719 (17 source lines).                                                                                                                                                                    |
| Purpose            | Keeps two places aligned, such as local data and target files.                                                                                                                                        |
| Declaration        | function syncPlainContextMenuControls(control) {                                                                                                                                                      |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | getComputedStyle, displayRichFontName, round, rgbColorToHex, some, setAttribute                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5703: function syncPlainContextMenuControls(control) {
5704:   if (!control) return;
5705:   const computed = getComputedStyle(control);
5706:   const family = displayRichFontName(control.dataset.builderFontFamily || computed.fontFamily) ||
"Arial";

```

```

5707:   const size = String(Math.round(parseFloat(control.dataset.builderFontPx || computed.fontSize) || 16));
5708:   const color = rgbColorToHex(control.dataset.builderColor || computed.color || "#17202a");
5709:   if (richFontSelect) {
5710:     richFontSelect.value = [...richFontSelect.options].some((option) => option.value === family ||
option.textContent === family)
5711:     ? family
5712:     : "Arial";
5713:   }
5714:   if (richFontSize) richFontSize.value = [...richFontSize.options].some((option) => option.value === size
|| option.textContent === size)
5715:   ? size
5716:   : "16";
5717:   if (richColorInput && /^[0-9a-f]{6}$/i.test(color)) richColorInput.value = color;
5718:   if (richBoldButton) richBoldButton.setAttribute("aria-pressed", control.dataset.builderBold === "true"
? "true" : "false");
5719: }

```

## plainTextControlFromContextEvent - lines 5721-5728

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5721-5728 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function plainTextControlFromContextEvent(event) {                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closest, toLowerCase, includes                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5721: function plainTextControlFromContextEvent(event) {
5722:   const control = event.target.closest?(".textarea, input");
5723:   if (!control || control.closest("#summary-context-menu")) return null;
5724:   const type = String(control.type || "text").toLowerCase();
5725:   if (control.disabled || control.readOnly) return null;
5726:   if (["button", "checkbox", "color", "file", "hidden", "image", "radio", "range", "reset",
"submit"].includes(type)) return null;
5727:   return control;
5728: }

```

## handlePlainTextAction - lines 5730-5778

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5730-5778 (49 source lines).                                                                                                                                                                    |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | async function handlePlainTextAction(action) {                                                                                                                                                        |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | focus, max, min, slice, writeText, setRangeText, dispatchEvent, Event, readText, select, applyPlainTextControlFormat                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 7 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5730: async function handlePlainTextAction(action) {
5731:   const control = activePlainTextControl;
5732:   if (!control) return;
5733:   control.focus({ preventScroll: true });
5734:   const selection = activePlainTextSelection || {
5735:     start: control.selectionStart ?? 0,
5736:     end: control.selectionEnd ?? control.value.length

```

```

5737:   };
5738:   const start = Math.max(0, Math.min(selection.start, control.value.length));
5739:   const end = Math.max(start, Math.min(selection.end, control.value.length));
5740:
5741:   if (action === "copy-text" || action === "cut-text") {
5742:     const text = control.value.slice(start, end) || control.value;
5743:     if (text) await navigator.clipboard.writeText(text);
5744:     if (action === "cut-text" && end > start) {
5745:       control.setRangeText("", start, end, "start");
5746:       control.dispatchEvent(new Event("input", { bubbles: true }));
5747:     }
5748:     return;
5749:   }
5750:
5751:   if (action === "paste-text") {
5752:     const text = await navigator.clipboard.readText();
5753:     if (text) {
5754:       control.setRangeText(text, start, end, "end");
5755:       control.dispatchEvent(new Event("input", { bubbles: true }));
5756:     }
5757:     return;
5758:   }
5759:
5760:   if (action === "select-all-text") {
5761:     control.select();
5762:     activePlainTextSelection = { start: 0, end: control.value.length };
5763:     return;
5764:   }
5765:
5766:   if (action === "toggle-bold") {
5767:     applyPlainTextControlFormat(control, { bold: control.dataset.builderBold !== "true" });
5768:     return;
5769:   }
5770:   if (action === "toggle-italic") {
5771:     applyPlainTextControlFormat(control, { italic: control.dataset.builderItalic !== "true" });
5772:     return;
5773:   }
5774:   if (action === "toggle-underline") {
5775:     applyPlainTextControlFormat(control, { underline: control.dataset.builderUnderline !== "true" });
5776:     return;
5777:   }
5778: }

```

## handleRichAction - lines 5780-5878

| Item               | Explanation                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5780-5878 (99 source lines).                                                                                                                                                                                       |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                                      |
| Declaration        | async function handleRichAction(action, editor) {                                                                                                                                                                        |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                                         |
| Calls detected     | applyRichInlineCommand, updateCurrentTextBlock, restoreRichSelection, getSelection, toString, selectedRichBlock, writeText, readText, insertPlainTextIntoRichEditor, saveRichEditorToProject, getRangeAt, deleteContents |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                 |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                  |
| Async/return clues | 9 await expression(s), 5 return statement(s).                                                                                                                                                                            |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                    |

Source excerpt:

```

5780: async function handleRichAction(action, editor) {
5781:   if (!editor) return;
5782:   activeSummaryEditor = editor;
5783:   if (action === "toggle-bold") {
5784:     applyRichInlineCommand(editor, "bold");
5785:   }
5786:   if (action === "toggle-italic") {
5787:     applyRichInlineCommand(editor, "italic");
5788:   }
5789:   if (action === "toggle-underline") {
5790:     applyRichInlineCommand(editor, "underline");
5791:   }
5792:   if (action === "align-left") updateCurrentTextBlock(editor, { align: "left" });
5793:   if (action === "align-center") updateCurrentTextBlock(editor, { align: "center" });
5794:   if (action === "align-right") updateCurrentTextBlock(editor, { align: "right" });
5795:   if (action === "copy-text") {
5796:     restoreRichSelection(editor);
5797:     const selectedText = window.getSelection()?.toString() || "";
5798:     const block = selectedRichBlock(editor);

```

```

5799:         const text = selectedText || block?.textContent || "";
5800:         if (text) await navigator.clipboard.writeText(text);
5801:         return;
5802:     }
5803:
5804:     if (action === "paste-text") {
5805:         const text = await navigator.clipboard.readText();
5806:         if (text) {
5807:             insertPlainTextIntoRichEditor(editor, text);
5808:             saveRichEditorToProject(editor);
5809:         }
5810:         return;
5811:     }
5812:     if (action === "cut-text") {
5813:         restoreRichSelection(editor);
5814:         const selection = window.getSelection();
5815:         const selectedText = selection?.toString() || "";
5816:         if (selectedText && selection.rangeCount) {
5817:             await navigator.clipboard.writeText(selectedText);
5818:             const range = selection.getRangeAt(0);
5819:             range.deleteContents();
5820:             captureRichSelection(editor);
5821:             cleanupEmptyRichTextBlocks(editor);
5822:             saveRichEditorToProject(editor);
5823:         }
5824:         return;
5825:     }
5826:     if (action === "select-all-text") {
5827:         const block = selectedRichBlock(editor);
5828:         const target = block?.dataset.type === "paragraph" ? block : editor;
5829:         const range = document.createRange();
5830:         range.selectNodeContents(target);
5831:         const selection = window.getSelection();
5832:         selection.removeAllRanges();
5833:         selection.addRange(range);
5834:         activeRichSelectionRange = range.cloneRange();
5835:         activeTextSelectionRange = range.cloneRange();
5836:         showPersistentTextSelection(range);
5837:         showTextSelectionInspector(editor, range);
5838:         return;
5839:     }
5840:     if (action === "add-image") {
5841:         const imageBlock = await openSummaryImageDialog(null, {
5842:             folder: editor.dataset.richFolder || "summary",
5843:             projectId: editor.dataset.richProjectId || ""
5844:         });
5845:         if (imageBlock) {
5846:             insertRichBlockAfterCursor(editor, createRichImageBlock(imageBlock));
5847:             saveRichEditorToProject(editor);
5848:         }
5849:     }
5850:     if (action === "add-formula") {
5851:         const formulaBlock = await openSummaryFormulaDialog();
5852:         if (formulaBlock) {
5853:             insertRichBlockAfterCursor(editor, createRichFormulaBlock(formulaBlock));
5854:             saveRichEditorToProject(editor);
5855:         }
5856:     }
5857:     if (action === "add-code") {
5858:         const selectedText = window.getSelection()?.toString() || "";
5859:         const codeBlock = await openSummaryCodeDialog(null, { code: selectedText, pasteMode: "source" });
5860:         if (codeBlock) {
5861:             insertRichBlockAfterCursor(editor, createRichCodeBlock(codeBlock));
5862:             saveRichEditorToProject(editor);
5863:         }
5864:     }
5865:     if (action === "paste-code") {
5866:         const text = await navigator.clipboard.readText();
5867:         const codeBlock = await openSummaryCodeDialog(null, { code: text, pasteMode: "source" });
5868:         if (codeBlock) {
5869:             insertRichBlockAfterCursor(editor, createRichCodeBlock(codeBlock));
5870:             saveRichEditorToProject(editor);
5871:         }
5872:     }
5873:     if (action === "edit-block") {
5874:         await editSelectedRichBlock(editor);
5875:         saveRichEditorToProject(editor);
5876:     }
5877:     if (action === "delete-block") deleteSelectedRichBlock(editor);
5878: }

```

## hideSummaryContextMenu - lines 5880-5884

| Item            | Explanation                                               |
|-----------------|-----------------------------------------------------------|
| Source location | Lines 5880-5884 (5 source lines).                         |
| Purpose         | Controls a window, dialog, panel, menu, or visible state. |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function hideSummaryContextMenu() {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
5880: function hideSummaryContextMenu() {
5881:   summaryContextMenu.hidden = true;
5882:   activePlainTextControl = null;
5883:   activePlainTextSelection = null;
5884: }
```

## ***editSelectedRichBlock - lines 5886-5909***

| Item               | Explanation                                                                                                                                                                                                  |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5886-5909 (24 source lines).                                                                                                                                                                           |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                                 |
| Declaration        | async function editSelectedRichBlock(editor) {                                                                                                                                                               |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                             |
| Calls detected     | selectedRichBlock, selectRichBlock, openSummaryImageDialog, refreshRichImageBlock, openSummaryFormulaDialog, refreshRichFormulaBlock, openSummaryCodeDialog, refreshRichCodeBlock, focusTextBlock, setStatus |
| API endpoints      | None detected inside this function body.                                                                                                                                                                     |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                      |
| Async/return clues | 3 await expression(s), 4 return statement(s).                                                                                                                                                                |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.        |

Source excerpt:

```
5886: async function editSelectedRichBlock(editor) {
5887:   const block = selectedRichBlock(editor);
5888:   if (!block) return;
5889:   selectRichBlock(block);
5890:   if (block.dataset.type === "image") {
5891:     const updated = await openSummaryImageDialog(block);
5892:     if (updated) refreshRichImageBlock(block, updated);
5893:     return;
5894:   }
5895:   if (block.dataset.type === "formula") {
5896:     const updated = await openSummaryFormulaDialog(block);
5897:     if (updated) refreshRichFormulaBlock(block, updated);
5898:     return;
5899:   }
5900:   if (block.dataset.type === "code") {
5901:     const updated = await openSummaryCodeDialog(block);
5902:     if (updated) refreshRichCodeBlock(block, updated);
5903:     return;
5904:   }
5905:   if (block.dataset.type === "paragraph") {
5906:     focusTextBlock(block);
5907:     setStatus("Text block selected. Type directly in the block to edit it.");
5908:   }
5909: }
```

## ***copyRichCodeBlock - lines 5911-5918***

| Item            | Explanation                       |
|-----------------|-----------------------------------|
| Source location | Lines 5911-5918 (8 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function copyRichCodeBlock(block) {                                                                                                                                                             |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, writeText, setStatus                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5911: async function copyRichCodeBlock(block) {
5912:   if (!block || block.dataset.type !== "code") return false;
5913:   const code = block.dataset.code || block.querySelector("code")?.textContent || "";
5914:   if (!code) return false;
5915:   await navigator.clipboard.writeText(code);
5916:   setStatus("Code copied to clipboard.");
5917:   return true;
5918: }

```

## ***removeRichBlock - lines 5920-5931***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5920-5931 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function removeRichBlock(block, editor) {                                                                                                                                                             |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | matches, remove, querySelector, append, createRichTextBlock, focusTextBlock                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5920: function removeRichBlock(block, editor) {
5921:   if (!block) return;
5922:   const nextFocus = block.previousElementSibling?.matches(".rich-text-block")
5923:     ? block.previousElementSibling
5924:     : block.nextElementSibling?.matches(".rich-text-block")
5925:     ? block.nextElementSibling
5926:     : null;
5927:   block.remove();
5928:   activeSummaryBlock = null;
5929:   if (!editor.querySelector(".rich-block")) editor.append(createRichTextBlock(""));
5930:   focusTextBlock(nextFocus || editor.querySelector(".rich-text-block"));
5931: }

```

## ***deleteSelectedRichBlock - lines 5933-5948***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 5933-5948 (16 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function deleteSelectedRichBlock(editor) {                   |
| Touches         | browser/Electron DOM, filesystem                             |
| Calls detected  | selectedRichBlock, openDeleteConfirm, removeRichBlock        |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5933: function deleteSelectedRichBlock(editor) {
5934:   const block = selectedRichBlock(editor);
5935:   if (!block) return;
5936:   const label = block.dataset.type === "image"
5937:     ? "this overview image"
5938:     : block.dataset.type === "formula"
5939:     ? "this formula"
5940:     : block.dataset.type === "code"
5941:     ? "this code block"
5942:     : "this text block";
5943:   openDeleteConfirm({
5944:     title: "Delete overview block",
5945:     message: `Are you sure you want to delete ${label}?`,
5946:     onConfirm: () => removeRichBlock(block, editor)
5947:   });
5948: }

```

## ***deleteProjectSummary - lines 5950-5966***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5950-5966 (17 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function deleteProjectSummary(project) {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | openDeleteConfirm, setStatus, scheduleAutosave, renderAll                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5950: function deleteProjectSummary(project) {
5951:   if (!project) return;
5952:   openDeleteConfirm({
5953:     title: "Delete project overview",
5954:     message: "Are you sure you want to delete the current project overview?",
5955:     onConfirm: () => {
5956:       project.summary = "";
5957:       project.summaryRich = null;
5958:       summaryEditorProjectId = "";
5959:       activeSummaryEditor = null;
5960:       activeSummaryBlock = null;
5961:       setStatus("Project overview deleted in the editor. Click Save project to update the portfolio
preview.");
5962:       scheduleAutosave();
5963:       renderAll();
5964:     }
5965:   });
5966: }

```

## ***saveRichSummary - lines 5968-5987***

| Item            | Explanation                                                          |
|-----------------|----------------------------------------------------------------------|
| Source location | Lines 5968-5987 (20 source lines).                                   |
| Purpose         | Persists data to local state, disk, credentials, or browser storage. |
| Declaration     | async function saveRichSummary() {                                   |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | selectedProject, querySelector, extractRichSummary, plainTextFromRich, wordCount, setStatus, scheduleAutosave, renderAll                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5968: async function saveRichSummary() {
5969:   const project = selectedProject();
5970:   const editor = projectFields.querySelector("[data-rich-editor='summary']");
5971:   if (!project || !editor) return true;
5972:   const rich = extractRichSummary(editor);
5973:   const plain = plainTextFromRich(rich);
5974:   if (wordCount(plain) > 1000) {
5975:     setStatus("Project overview is limited to 1000 words. Shorten the text before saving.");
5976:     return false;
5977:   }
5978:   project.summaryRich = rich;
5979:   project.summary = plain;
5980:   summaryEditorProjectId = "";
5981:   activeSummaryEditor = null;
5982:   activeSummaryBlock = null;
5983:   setStatus("Overview saved in the editor. Click Save project to include this version in the portfolio
preview.");
5984:   scheduleAutosave();
5985:   renderAll();
5986:   return true;
5987: }

```

## ***renderFields - lines 5989-6004***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 5989-6004 (16 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderFields(project) {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | renderSummaryBuilder, requestAnimationFrame, populateSummaryEditor                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

5989: function renderFields(project) {
5990:   if (!project) {
5991:     projectFields.hidden = false;
5992:     projectFields.innerHTML = `
```

## ***sectionOptions - lines 6006-6016***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6006-6016 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function sectionOptions(project) {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, filter, isAnalogProject                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6006: function sectionOptions(project) {
6007:   const custom = (project.sections || []).map((section) => ({
6008:     id: `custom:${section.id}`,
6009:     label: section.title,
6010:     folder: section.id
6011:   }));
6012:   return [
6013:     ...standardSections.filter((section) => !section.analogOnly || isAnalogProject(project)),
6014:     ...custom
6015:   ];
6016: }

```

### ***custom - lines 6007-6011***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6007-6011 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const custom = (project.sections    []).map((section) => ({                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map                                                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6007:   const custom = (project.sections || []).map((section) => ({
6008:     id: `custom:${section.id}`,
6009:     label: section.title,
6010:     folder: section.id
6011:   }));

```

### ***ensureSiteSections - lines 6018-6021***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 6018-6021 (4 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function ensureSiteSections() {                              |
| Touches         | Mostly local calculations or local UI state.                 |
| Calls detected  | clone                                                        |
| API endpoints   | None detected inside this function body.                     |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6018: function ensureSiteSections() {
6019:   catalog.siteSections = catalog.siteSections || [];
6020:   savedPortfolioCatalog.siteSections = savedPortfolioCatalog.siteSections || clone(catalog.siteSections
|| []);
6021: }

```

## ***siteSectionHasContent - lines 6023-6032***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6023-6032 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function siteSectionHasContent(section) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | richHasContent, some                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6023: function siteSectionHasContent(section) {
6024:   return Boolean(
6025:     section?.description ||
6026:     richHasContent(section?.richDescription) ||
6027:     section?.backgroundImage ||
6028:     (section?.links || []).some((link) => link.label || link.url) ||
6029:     (section?.assets || []).some((asset) => asset.title || asset.url) ||
6030:     (section?.subsections || []).some((item) => item.title || item.description ||
richHasContent(item.richDescription) || (item.links || []).length)
6031:   );
6032: }

```

## ***siteSectionRenderable - lines 6034-6036***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6034-6036 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function siteSectionRenderable(section) {                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | siteSectionHasContent                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6034: function siteSectionRenderable(section) {
6035:   return section?.visible !== false && siteSectionHasContent(section);
6036: }

```

## siteSectionLinkCount - lines 6038-6044

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6038-6044 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function siteSectionLinkCount(section) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | flatMap, filter                                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6038: function siteSectionLinkCount(section) {
6039:   return [
6040:     ...(section.links || []),
6041:     ...(section.assets || []),
6042:     ...(section.subsections || []).flatMap((item) => item.links || []),
6043:   ].filter((item) => item.label || item.url).length;
6044: }
```

## renderSiteSectionList - lines 6046-6094

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6046-6094 (49 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderSiteSectionList() {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureSiteSections, map, url, siteSectionLinkCount, richHasContent, renderRichContent, join                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6046: function renderSiteSectionList() {
6047:   ensureSiteSections();
6048:   const sections = catalog.siteSections || [];
6049:   siteSectionList.innerHTML = sections.length ? sections.map((section) => `
6050:     <article class="site-section-item ${section.visible === false ? "is-hidden-section" : ""}"
style="${section.backgroundImage ? '--site-section-bg: url('${section.backgroundImage}');` : ""}">
6051:       <div>
6052:         <strong>${section.title || "Untitled section"}</strong>
6053:         <span>${section.visible === false ? "Hidden from website" : "Visible on website"} .
${(section.subsections || []).length} subsection${(section.subsections || []).length === 1 ? "" : "s"} .
${siteSectionLinkCount(section)} link${siteSectionLinkCount(section) === 1 ? "" : "s"}</span>
6054:         ${section.description || richHasContent(section.richDescription) ?
renderRichContent(section.richDescription, section.description || "") : ""}
6055:       </div>
6056:       ${((section.subsections || []).length ? `
6057:         <div class="site-subsection-list">
6058:           ${((section.subsections || []).map((item, index) => `
6059:             <article class="site-subsection-item">
6060:               <div>
6061:                 <strong>${item.title || "Untitled subsection"}</strong>
6062:                 ${item.description || richHasContent(item.richDescription) ?
renderRichContent(item.richDescription, item.description || "") : ""}
6063:               </div>
6064:               <div class="site-section-actions">
6065:                 <button type="button" data-site-subsection-edit="${section.id}" data-
index="${index}>Edit</button>
```

```

6066:         <button class="danger-icon" type="button" data-site-subsection-delete="${section.id}"
data-index="${index}">Delete</button>
6067:     </div>
6068: </article>
6069: `).join("")}
6070: </div>
6071: ` : ""}
6072: ${section.links || []}.length ? `
6073: <div class="site-link-list">
6074:     ${section.links || []}.map((link, index) => `
6075:         <span>
6076:             ${link.label || "Untitled link"}
6077:             <button type="button" data-site-link-edit="${section.id}" data-
index="${index}">Edit</button>
6078:             <button type="button" data-site-link-delete="${section.id}" data-
index="${index}">Delete</button>
6079:         </span>
6080:     `).join("")}
6081: </div>
6082: ` : ""}
6083: <div class="site-section-actions">
6084:     <button type="button" data-site-section-toggle="${section.id}">${section.visible === false ?
"Show" : "Hide"}</button>
6085:     <button type="button" data-site-section-edit="${section.id}">Edit</button>
6086:     <button type="button" data-site-subsection-add="${section.id}">Add subsection</button>
6087:     <button type="button" data-site-link-add="${section.id}">Add link</button>
6088:     <button type="button" data-site-section-bg="${section.id}">Background</button>
6089:     ${section.backgroundImage ? `<button type="button" data-site-section-bg-
clear="${section.id}">Clear bg</button>` : ""}
6090:     <button class="danger-icon" type="button" data-site-section-
delete="${section.id}">Delete</button>
6091: </div>
6092: </article>
6093: `).join("") : `<p class="evidence-empty">No extra portfolio areas added yet.</p>`;
6094: }

```

## siteSectionDialogValue - lines 6096-6118

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6096-6118 (23 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function siteSectionDialogValue(section = {}) {                                                                                                                                                 |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | populateStandaloneRichEditor, dialogValue, trim, setStatus, extractRichSummary, plainTextFromRich, slugify                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6096: async function siteSectionDialogValue(section = {}) {
6097:   sectionDialogTitle.textContent = section.id ? "Edit section" : "Create section";
6098:   sectionTitle.placeholder = "Professional profile, Personal background, Sports...";
6099:   sectionDescription.dataset.placeholder = "Short section overview shown on the website";
6100:   sectionDescription.dataset.richFolder = section.id || "new-portfolio-section";
6101:   sectionDescription.dataset.richProjectId = "_site-sections";
6102:   sectionTitle.value = section.title || "";
6103:   populateStandaloneRichEditor(sectionDescription, section.richDescription, section.description || "");
6104:   const saved = await dialogValue(sectionDialog);
6105:   if (!saved) return null;
6106:   const title = sectionTitle.value.trim();
6107:   if (!title) {
6108:     setStatus("Type a section title before saving.");
6109:     return null;
6110:   }
6111:   const richDescription = extractRichSummary(sectionDescription);
6112:   return {
6113:     description: plainTextFromRich(richDescription, "\n").trim(),
6114:     id: section.id || slugify(title),
6115:     richDescription,
6116:     title
6117:   };
6118: }

```

## siteSubsectionDialogValue - lines 6120-6143



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6120-6143 (24 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function siteSubsectionDialogValue(subsection = {}) {                                                                                                                                           |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | populateStandaloneRichEditor, dialogValue, trim, setStatus, extractRichSummary, plainTextFromRich, slugify                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6120: async function siteSubsectionDialogValue(subsection = {}) {
6121:   sectionDialogTitle.textContent = subsection.id ? "Edit subsection" : "Create subsection";
6122:   sectionTitle.placeholder = "Sports, Hobbies, Resume links, Origin...";
6123:   sectionDescription.dataset.placeholder = "Short subsection text shown inside the portfolio area";
6124:   sectionDescription.dataset.richFolder = subsection.id || "new-portfolio-subsection";
6125:   sectionDescription.dataset.richProjectId = "_site-sections";
6126:   sectionTitle.value = subsection.title || "";
6127:   populateStandaloneRichEditor(sectionDescription, subsection.richDescription, subsection.description ||
  "");
6128:   const saved = await dialogValue(sectionDialog);
6129:   if (!saved) return null;
6130:   const title = sectionTitle.value.trim();
6131:   if (!title) {
6132:     setStatus("Type a subsection title before saving.");
6133:     return null;
6134:   }
6135:   const richDescription = extractRichSummary(sectionDescription);
6136:   return {
6137:     description: plainTextFromRich(richDescription, "\n").trim(),
6138:     id: subsection.id || slugify(title),
6139:     links: subsection.links || [],
6140:     richDescription,
6141:     title
6142:   };
6143: }

```

## ***dialogLinkValue - lines 6145-6169***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6145-6169 (25 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function dialogLinkValue(link = {}) {                                                                                                                                                                 |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | removeEventListener, resolve, trim, setStatus, normalizeLinkTarget, addEventListener, showModal                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6145: function dialogLinkValue(link = {}) {
6146:   return new Promise((resolve) => {
6147:     siteLinkEyebrow.textContent = link.url ? "Edit portfolio link" : "Add portfolio link";
6148:     siteLinkTitle.textContent = link.url ? "Edit link" : "Add link";
6149:     siteLinkLabel.value = link.label || "";
6150:     siteLinkUrl.value = link.url || "";
6151:     const onClose = () => {
6152:       siteLinkDialog.removeEventListener("close", onClose);
6153:       if (siteLinkDialog.returnValue !== "save") {
6154:         resolve(null);
6155:         return;

```

```

6156:     }
6157:     const label = siteLinkLabel.value.trim();
6158:     const url = siteLinkUrl.value.trim();
6159:     if (!label || !url) {
6160:         setStatus("Type both a link label and URL before saving.");
6161:         resolve(null);
6162:         return;
6163:     }
6164:     resolve({ ...link, label, url: normalizeLinkTarget(url, { assumeweb: true }) });
6165: };
6166: siteLinkDialog.addListener("close", onClose);
6167: siteLinkDialog.showModal();
6168: });
6169: }

```

## onClose - lines 6151-6165

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6151-6165 (15 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const onClose = () => {                                                                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | removeEventListener, resolve, trim, setStatus, normalizeLinkTarget                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6151:     const onClose = () => {
6152:         siteLinkDialog.removeEventListener("close", onClose);
6153:         if (siteLinkDialog.returnValue !== "save") {
6154:             resolve(null);
6155:             return;
6156:         }
6157:         const label = siteLinkLabel.value.trim();
6158:         const url = siteLinkUrl.value.trim();
6159:         if (!label || !url) {
6160:             setStatus("Type both a link label and URL before saving.");
6161:             resolve(null);
6162:             return;
6163:         }
6164:         resolve({ ...link, label, url: normalizeLinkTarget(url, { assumeweb: true }) });
6165:     };

```

## addSiteSection - lines 6171-6191

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6171-6191 (21 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function addSiteSection() {                                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureSiteSections, siteSectionDialogValue, slugify, push, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6171: async function addSiteSection() {
6172:     ensureSiteSections();

```

```

6173:   const input = await siteSectionDialogValue();
6174:   if (!input) return;
6175:   const id = input.id || slugify(input.title);
6176:   catalog.siteSections.push({
6177:     id,
6178:     title: input.title,
6179:     description: input.description,
6180:     richDescription: input.richDescription,
6181:     visible: false,
6182:     backgroundImage: "",
6183:     links: [],
6184:     assets: [],
6185:     subsections: []
6186:   });
6187:   markDraftNeedsSave();
6188:   setStatus("Portfolio area added locally.");
6189:   scheduleAutosave();
6190:   renderAll();
6191: }

```

## ***editSiteSection - lines 6193-6206***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6193-6206 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function editSiteSection(sectionId) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureSiteSections, find, siteSectionDialogValue, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6193: async function editSiteSection(sectionId) {
6194:   ensureSiteSections();
6195:   const section = catalog.siteSections.find((item) => item.id === sectionId);
6196:   if (!section) return;
6197:   const input = await siteSectionDialogValue(section);
6198:   if (!input) return;
6199:   section.title = input.title;
6200:   section.description = input.description;
6201:   section.richDescription = input.richDescription;
6202:   markDraftNeedsSave();
6203:   setStatus("Portfolio area edited locally.");
6204:   scheduleAutosave();
6205:   renderAll();
6206: }

```

## ***addSiteSubsection - lines 6208-6219***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6208-6219 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function addSiteSubsection(sectionId) {                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, siteSubsectionDialogValue, push, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6208: async function addSiteSubsection(sectionId) {
6209:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6210:   if (!section) return;
6211:   const input = await siteSubsectionDialogValue();
6212:   if (!input) return;
6213:   section.subsections = section.subsections || [];
6214:   section.subsections.push(input);
6215:   markDraftNeedsSave();
6216:   setStatus("Portfolio subsection added locally.");
6217:   scheduleAutosave();
6218:   renderAll();
6219: }

```

## section - lines 6209-6218

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6209-6218 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const section = (catalog.siteSections    []).find((item) => item.id === sectionId);                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, siteSubsectionDialogValue, push, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6209:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6210:   if (!section) return;
6211:   const input = await siteSubsectionDialogValue();
6212:   if (!input) return;
6213:   section.subsections = section.subsections || [];
6214:   section.subsections.push(input);
6215:   markDraftNeedsSave();
6216:   setStatus("Portfolio subsection added locally.");
6217:   scheduleAutosave();
6218:   renderAll();

```

## editSiteSubsection - lines 6221-6232

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6221-6232 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function editSiteSubsection(sectionId, index) {                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, siteSubsectionDialogValue, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6221: async function editSiteSubsection(sectionId, index) {
6222:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6223:   const subsection = section?.subsections?.[Number(index)];
6224:   if (!section || !subsection) return;
6225:   const input = await siteSubsectionDialogValue(subsection);
6226:   if (!input) return;
6227:   section.subsections[Number(index)] = input;
6228:   markDraftNeedsSave();
6229:   setStatus("Portfolio subsection edited locally.");
6230:   scheduleAutosave();

```

```

6231:   renderAll();
6232: }

```

## section - lines 6222-6231

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6222-6231 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>const section = (catalog.siteSections    []).find((item) =&gt; item.id === sectionId);</code>                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>find</code> , <code>siteSubsectionDialogValue</code> , <code>markDraftNeedsSave</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 <code>await</code> expression(s), 2 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6222:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6223:   const subsection = section?.subsections?.[Number(index)];
6224:   if (!section || !subsection) return;
6225:   const input = await siteSubsectionDialogValue(subsection);
6226:   if (!input) return;
6227:   section.subsections[Number(index)] = input;
6228:   markDraftNeedsSave();
6229:   setStatus("Portfolio subsection edited locally.");
6230:   scheduleAutosave();
6231:   renderAll();

```

## deleteSiteSubsection - lines 6234-6248

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6234-6248 (15 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>function deleteSiteSubsection(sectionId, index) {</code>                                                                                                                                        |
| Touches            | <code>filesystem</code>                                                                                                                                                                               |
| Calls detected     | <code>find</code> , <code>openDeleteConfirm</code> , <code>splice</code> , <code>markDraftNeedsSave</code> , <code>scheduleAutosave</code> , <code>renderAll</code>                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 <code>await</code> expression(s), 1 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6234: function deleteSiteSubsection(sectionId, index) {
6235:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6236:   const subsection = section?.subsections?.[Number(index)];
6237:   if (!section || !subsection) return;
6238:   openDeleteConfirm({
6239:     title: "Delete subsection",
6240:     message: `Are you sure you want to delete "${subsection.title}"?`,
6241:     onConfirm: () => {
6242:       section.subsections.splice(Number(index), 1);
6243:       markDraftNeedsSave();
6244:       scheduleAutosave();
6245:       renderAll();
6246:     }
6247:   });
6248: }

```

## section - lines 6235-6247

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6235-6247 (13 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const section = (catalog.siteSections    []).find((item) => item.id === sectionId);                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | find, openDeleteConfirm, splice, markDraftNeedsSave, scheduleAutosave, renderAll                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6235: const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6236: const subsection = section?.subsections?.[Number(index)];
6237: if (!section || !subsection) return;
6238: openDeleteConfirm({
6239:   title: "Delete subsection",
6240:   message: `Are you sure you want to delete "${subsection.title}"?`,
6241:   onConfirm: () => {
6242:     section.subsections.splice(Number(index), 1);
6243:     markDraftNeedsSave();
6244:     scheduleAutosave();
6245:     renderAll();
6246:   }
6247: });

```

## ***addSiteSectionLink - lines 6250-6261***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6250-6261 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function addSiteSectionLink(sectionId) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, dialogLinkValue, push, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6250: async function addSiteSectionLink(sectionId) {
6251:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6252:   if (!section) return;
6253:   const input = await dialogLinkValue();
6254:   if (!input) return;
6255:   section.links = section.links || [];
6256:   section.links.push(input);
6257:   markDraftNeedsSave();
6258:   setStatus("Portfolio link added locally.");
6259:   scheduleAutosave();
6260:   renderAll();
6261: }

```

## ***section - lines 6251-6260***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 6251-6260 (10 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | <code>const section = (catalog.siteSections    []).find((item) =&gt; item.id === sectionId);</code>                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>find</code> , <code>dialogLinkValue</code> , <code>push</code> , <code>markDraftNeedsSave</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 <code>await</code> expression(s), 2 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6251:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6252:   if (!section) return;
6253:   const input = await dialogLinkValue();
6254:   if (!input) return;
6255:   section.links = section.links || [];
6256:   section.links.push(input);
6257:   markDraftNeedsSave();
6258:   setStatus("Portfolio link added locally.");
6259:   scheduleAutosave();
6260:   renderAll();

```

## ***editSiteSectionLink - lines 6263-6274***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6263-6274 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>async function editSiteSectionLink(sectionId, index) {</code>                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>find</code> , <code>dialogLinkValue</code> , <code>markDraftNeedsSave</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 <code>await</code> expression(s), 2 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6263: async function editSiteSectionLink(sectionId, index) {
6264:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6265:   const link = section?.links?.[Number(index)];
6266:   if (!section || !link) return;
6267:   const input = await dialogLinkValue(link);
6268:   if (!input) return;
6269:   section.links[Number(index)] = input;
6270:   markDraftNeedsSave();
6271:   setStatus("Portfolio link edited locally.");
6272:   scheduleAutosave();
6273:   renderAll();
6274: }

```

## ***section - lines 6264-6273***

| Item            | Explanation                                                                                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 6264-6273 (10 source lines).                                                                                                                                   |
| Purpose         | Part of the private builder frontend and editing experience.                                                                                                         |
| Declaration     | <code>const section = (catalog.siteSections    []).find((item) =&gt; item.id === sectionId);</code>                                                                  |
| Touches         | Mostly local calculations or local UI state.                                                                                                                         |
| Calls detected  | <code>find</code> , <code>dialogLinkValue</code> , <code>markDraftNeedsSave</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code> |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6264:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6265:   const link = section?.links?.[Number(index)];
6266:   if (!section || !link) return;
6267:   const input = await dialogLinkValue(link);
6268:   if (!input) return;
6269:   section.links[Number(index)] = input;
6270:   markDraftNeedsSave();
6271:   setStatus("Portfolio link edited locally.");
6272:   scheduleAutosave();
6273:   renderAll();

```

## ***deleteSiteSectionLink - lines 6276-6284***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6276-6284 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function deleteSiteSectionLink(sectionId, index) {                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, splice, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6276: function deleteSiteSectionLink(sectionId, index) {
6277:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6278:   if (!section?.links?.[Number(index)]) return;
6279:   section.links.splice(Number(index), 1);
6280:   markDraftNeedsSave();
6281:   setStatus("Portfolio link deleted locally.");
6282:   scheduleAutosave();
6283:   renderAll();
6284: }

```

## ***section - lines 6277-6283***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6277-6283 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const section = (catalog.siteSections    []).find((item) => item.id === sectionId);                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, splice, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |



Source excerpt:

```
6277: const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6278: if (!section?.links?.[Number(index)]) return;
6279: section.links.splice(Number(index), 1);
6280: markDraftNeedsSave();
6281: setStatus("Portfolio link deleted locally.");
6282: scheduleAutosave();
6283: renderAll();
```

## ***toggleSiteSectionVisibility - lines 6286-6294***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6286-6294 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function toggleSiteSectionVisibility(sectionId) {                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6286: function toggleSiteSectionVisibility(sectionId) {
6287:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6288:   if (!section) return;
6289:   section.visible = section.visible === false;
6290:   markDraftNeedsSave();
6291:   setStatus(section.visible ? "Portfolio area will show on the website preview." : "Portfolio area hidden
from the website preview.");
6292:   scheduleAutosave();
6293:   renderAll();
6294: }
```

## ***section - lines 6287-6293***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6287-6293 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const section = (catalog.siteSections    []).find((item) => item.id === sectionId);                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6287: const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6288: if (!section) return;
6289: section.visible = section.visible === false;
6290: markDraftNeedsSave();
6291: setStatus(section.visible ? "Portfolio area will show on the website preview." : "Portfolio area hidden
from the website preview.");
6292: scheduleAutosave();
6293: renderAll();
```

## ***deleteSiteSection - lines 6296-6310***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6296-6310 (15 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function deleteSiteSection(sectionId) {                                                                                                                                                               |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | find, openDeleteConfirm, filter, markDraftNeedsSave, scheduleAutosave, renderAll                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6296: function deleteSiteSection(sectionId) {
6297:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6298:   if (!section) return;
6299:   openDeleteConfirm({
6300:     title: "Delete portfolio area",
6301:     message: `Are you sure you want to delete "${section.title}"?`,
6302:     onConfirm: () => {
6303:       catalog.siteSections = (catalog.siteSections || []).filter((item) => item.id !== sectionId);
6304:       savedPortfolioCatalog.siteSections = (savedPortfolioCatalog.siteSections || []).filter((item) =>
item.id !== sectionId);
6305:       markDraftNeedsSave();
6306:       scheduleAutosave();
6307:       renderAll();
6308:     }
6309:   });
6310: }

```

## ***section - lines 6297-6309***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6297-6309 (13 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const section = (catalog.siteSections    []).find((item) => item.id === sectionId);                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | find, openDeleteConfirm, filter, markDraftNeedsSave, scheduleAutosave, renderAll                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6297:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6298:   if (!section) return;
6299:   openDeleteConfirm({
6300:     title: "Delete portfolio area",
6301:     message: `Are you sure you want to delete "${section.title}"?`,
6302:     onConfirm: () => {
6303:       catalog.siteSections = (catalog.siteSections || []).filter((item) => item.id !== sectionId);
6304:       savedPortfolioCatalog.siteSections = (savedPortfolioCatalog.siteSections || []).filter((item) =>
item.id !== sectionId);
6305:       markDraftNeedsSave();
6306:       scheduleAutosave();
6307:       renderAll();
6308:     }
6309:   });

```

## ***setSiteSectionBackground - lines 6312-6343***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6312-6343 (32 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function setSiteSectionBackground(sectionId) {                                                                                                                                                  |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | find, chooseFile, startsWith, setStatus, readFileAsDataURL, fetch, now, stringify, json, markDraftNeedsSave, scheduleAutosave, renderAll                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 4 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6312: async function setSiteSectionBackground(sectionId) {
6313:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6314:   if (!section) return;
6315:   const file = await chooseFile();
6316:   if (!file) return;
6317:   if (!file.type.startsWith("image/")) {
6318:     setStatus("Choose an image file for a section background.");
6319:     return;
6320:   }
6321:   const data = await readFileAsDataURL(file);
6322:   const response = await fetch(`/api/upload?t=${Date.now()}`, {
6323:     method: "POST",
6324:     cache: "no-store",
6325:     headers: { "Content-Type": "application/json" },
6326:     body: JSON.stringify({
6327:       projectId: "_site-sections",
6328:       section: section.id,
6329:       fileName: file.name,
6330:       data
6331:     })
6332:   });
6333:   const result = await response.json();
6334:   if (!response.ok) {
6335:     setStatus(result.error || "Background upload failed.");
6336:     return;
6337:   }
6338:   section.backgroundImage = result.url;
6339:   markDraftNeedsSave();
6340:   setStatus("Portfolio area background saved locally.");
6341:   scheduleAutosave();
6342:   renderAll();
6343: }

```

## section - lines 6313-6320

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6313-6320 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const section = (catalog.siteSections    []).find((item) => item.id === sectionId);                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, chooseFile, startsWith, setStatus                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6313:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6314:   if (!section) return;
6315:   const file = await chooseFile();

```

```

6316:   if (!file) return;
6317:   if (!file.type.startsWith("image/")) {
6318:     setStatus("Choose an image file for a section background.");
6319:     return;
6320:   }

```

## ***clearSiteSectionBackground - lines 6345-6353***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6345-6353 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function clearSiteSectionBackground(sectionId) {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6345: function clearSiteSectionBackground(sectionId) {
6346:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6347:   if (!section) return;
6348:   section.backgroundImage = "";
6349:   markDraftNeedsSave();
6350:   setStatus("Portfolio area background cleared locally.");
6351:   scheduleAutosave();
6352:   renderAll();
6353: }

```

## ***section - lines 6346-6352***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6346-6352 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const section = (catalog.siteSections    []).find((item) => item.id === sectionId);                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6346:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6347:   if (!section) return;
6348:   section.backgroundImage = "";
6349:   markDraftNeedsSave();
6350:   setStatus("Portfolio area background cleared locally.");
6351:   scheduleAutosave();
6352:   renderAll();

```

## ***renderSectionTabs - lines 6355-6369***

| Item            | Explanation                            |
|-----------------|----------------------------------------|
| Source location | Lines 6355-6369 (15 source lines).     |
| Purpose         | Renders UI, markup, or display output. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function renderSectionTabs(project) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | sectionOptions, map, join                                                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6355: function renderSectionTabs(project) {
6356:   if (!project) {
6357:     sectionTabs.innerHTML = "";
6358:     return;
6359:   }
6360:
6361:   sectionTabs.innerHTML = `
6362:     ${sectionOptions(project).map((section) => `
6363:       <button class="${section.id === activeSectionId ? "active" : ""}" type="button" data-section-
id="${section.id}">
6364:         ${section.label}
6365:       </button>
6366:     `).join("")}
6367:     <button type="button" data-add-section="true">Add section</button>
6368:   `;
6369: }

```

## renderTextArray - lines 6371-6389

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6371-6389 (19 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderTextArray(project, key, label) {                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, join, toLowerCase                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6371: function renderTextArray(project, key, label) {
6372:   const items = project[key] || [];
6373:   return `
6374:     <div class="section-actions">
6375:       <button class="button primary" type="button" data-open-editor="text-array" data-key="${key}" data-
mode="add">Add ${label}</button>
6376:     </div>
6377:     <div class="builder-items">
6378:       ${items.map((item, index) => `
6379:         <article class="builder-item">
6380:           <div>
6381:             <strong>${typeof item === "string" ? item : item.title || item.name || item.label ||
""}</strong>
6382:           </div>
6383:           <button type="button" data-open-editor="text-array" data-key="${key}" data-mode="edit" data-
index="${index}">Edit</button>
6384:           <button class="danger-icon" type="button" data-delete-array="${key}" data-
index="${index}">Delete</button>
6385:         </article>
6386:       `).join("")} || `<p class="evidence-empty">No ${label.toLowerCase()} added yet.</p>`
6387:     </div>
6388:   `;
6389: }

```

## ***toolName - lines 6391-6393***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6391-6393 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function toolName(item) {                                                                                                                                                                             |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6391: function toolName(item) {  
6392:   return typeof item === "string" ? item : item.name || item.title || item.label || "";  
6393: }
```

## ***toolDescription - lines 6395-6397***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6395-6397 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function toolDescription(item) {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6395: function toolDescription(item) {  
6396:   return typeof item === "string" ? "" : item.description || item.usedFor || "";  
6397: }
```

## ***renderToolsSection - lines 6399-6420***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6399-6420 (22 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderToolsSection(project) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | map, toolName, richHasContent, renderRichContent, toolDescription, join                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6399: function renderToolsSection(project) {
6400:   const items = project.tools || [];
6401:   return `
6402:     <div class="section-actions">
6403:       <button class="button primary" type="button" data-open-editor="tool" data-mode="add">Add
tool</button>
6404:     </div>
6405:     <div class="builder-items tool-items">
6406:       ${items.map((item, index) => `
6407:         <article class="builder-item tool-item">
6408:           <div>
6409:             <strong>${toolName(item)} || "Untitled tool"</strong>
6410:           </div>
6411:           ${richHasContent(item?.richDescription)
6412:             ? renderRichContent(item.richDescription, toolDescription(item))
6413:             : `<p>${toolDescription(item)} || "No usage description added yet."</p>`}
6414:           <button type="button" data-open-editor="tool" data-mode="edit" data-
index="${index}">Edit</button>
6415:           <button class="danger-icon" type="button" data-delete-array="tools" data-
index="${index}">Delete</button>
6416:         </article>
6417:       `).join("")} || `<p class="evidence-empty">No tools added yet.</p>`}
6418:     </div>
6419:   `;
6420: }
```

## renderObjectSection - lines 6422-6450

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6422-6450 (29 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderObjectSection(project, key, folder, label) {                                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | map, richHasContent, renderRichContent, renderMultilineInlineText, join, toLowerCase                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6422: function renderObjectSection(project, key, folder, label) {
6423:   const items = project[key] || [];
6424:   return `
6425:     <div class="section-actions">
6426:       <button class="button primary" type="button" data-open-editor="object" data-key="${key}" data-
mode="add">Add text item</button>
6427:       ${folder ? `<button class="button secondary" type="button" data-upload="${key}" data-
folder="${folder}">Add file or image</button>` : ""}
6428:     </div>
6429:     <div class="builder-items">
6430:       ${items.map((item, index) => {
6431:         const title = item.title || item.name || item.label || "Untitled item";
6432:         const url = item.url || item.artifact || "";
6433:         const description = item.description || item.caption || item.result || item.method || "";
6434:         return `
6435:           <article class="builder-item">
6436:             <div>
6437:               <strong>${title}</strong>
6438:               <span>${url || "No file attached"}</span>
6439:               ${richHasContent(item.richDescription)
6440:                 ? renderRichContent(item.richDescription, description)
6441:                 : description ? `<p>${renderMultilineInlineText(description)}</p>` : ""}
6442:             </div>
6443:             <button type="button" data-open-editor="object" data-key="${key}" data-mode="edit" data-
index="${index}">Edit</button>
6444:             <button class="danger-icon" type="button" data-delete-item="${key}" data-
index="${index}">Delete</button>
6445:           </article>
6446:         `;
6447:       }).join("")} || `<p class="evidence-empty">No ${label.toLowerCase()} added yet.</p>`}
6448:     </div>
6449:   `;
6450: }
```

## ***designArray - lines 6452-6458***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6452-6458 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function designArray(project, pathValue) {                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureDesignModel, getByPath, isArray, setByPath                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6452: function designArray(project, pathValue) {
6453:   ensureDesignModel(project);
6454:   const items = getByPath(project, pathValue);
6455:   if (Array.isArray(items)) return items;
6456:   setByPath(project, pathValue, []);
6457:   return getByPath(project, pathValue);
6458: }
```

## ***makeDesignItem - lines 6460-6466***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6460-6466 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function makeDesignItem(kind, title, description = "", url = "") {                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6460: function makeDesignItem(kind, title, description = "", url = "") {
6461:   if (kind === "reference") return { title, description, url, type: "Reference", status: url ? "linked" :
"draft" };
6462:   if (kind === "analysis") return { title, description, type: "Math / Analysis", status: "draft" };
6463:   if (kind === "simulation-file") return { title, description, url, type: "Simulation file", status: url
? "uploaded" : "draft" };
6464:   if (kind === "simulation-result") return { title, description, url, type: "Simulation result", status:
url ? "uploaded" : "draft" };
6465:   return { title, description, url, type: "Document", status: url ? "uploaded" : "draft" };
6466: }
```

## ***renderDesignObjectSection - lines 6468-6496***

| Item            | Explanation                                                                                |
|-----------------|--------------------------------------------------------------------------------------------|
| Source location | Lines 6468-6496 (29 source lines).                                                         |
| Purpose         | Renders UI, markup, or display output.                                                     |
| Declaration     | function renderDesignObjectSection(project, pathValue, folder, label, kind = "document") { |
| Touches         | filesystem                                                                                 |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | designArray, map, richHasContent, renderRichContent, renderMultilineInlineText, join, toLowerCase                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6468: function renderDesignObjectSection(project, pathValue, folder, label, kind = "document") {
6469:   const items = designArray(project, pathValue);
6470:   return `
6471:     <div class="section-actions">
6472:       <button class="button primary" type="button" data-open-editor="design-object" data-design-
path="${pathValue}" data-design-kind="${kind}" data-mode="add">Add ${label}</button>
6473:       ${folder ? `<button class="button secondary" type="button" data-upload-design="${pathValue}" data-
folder="${folder}" data-design-kind="${kind}">Add file</button>` : ""}
6474:     </div>
6475:     <div class="builder-items">
6476:       ${items.map((item, index) => {
6477:         const title = item.title || item.name || item.label || "Untitled item";
6478:         const url = item.url || item.artifact || "";
6479:         const description = item.description || item.caption || item.result || item.method || "";
6480:         return `
6481:           <article class="builder-item">
6482:             <div>
6483:               <strong>${title}</strong>
6484:               <span>${url || "No file attached"}</span>
6485:               ${richHasContent(item.richDescription)
6486:                 ? renderRichContent(item.richDescription, description)
6487:                 : description ? `<p>${renderMultilineInlineText(description)}</p>` : ""}
6488:             </div>
6489:             <button type="button" data-open-editor="design-object" data-design-path="${pathValue}" data-
design-kind="${kind}" data-mode="edit" data-index="${index}">Edit</button>
6490:             <button class="danger-icon" type="button" data-delete-design-item="${pathValue}" data-
index="${index}">Delete</button>
6491:           </article>
6492:         `;
6493:       }).join("") || `<p class="evidence-empty">No ${label.toLowerCase()} added yet.</p>`}
6494:     </div>
6495:   `;
6496: }

```

## ***renderDesignSummaryField - lines 6498-6530***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6498-6530 (33 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderDesignSummaryField(project, pathValue, label, placeholder) {                                                                                                                           |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | getByPath, summaryRichPathFromTextPath, escapeHtml, overviewFolderFromPath                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6498: function renderDesignSummaryField(project, pathValue, label, placeholder) {
6499:   const textValue = getByPath(project, pathValue) || "";
6500:   const richPath = summaryRichPathFromTextPath(pathValue);
6501:   const richValue = getByPath(project, richPath);
6502:   const hasValue = Boolean(textValue || richValue?.blocks?.length);
6503:
6504:   return `
6505:     <div class="wide-field design-summary-field rich-design-summary-field">
6506:       <div class="summary-builder-heading">
6507:         <span>
6508:           ${label}
6509:           ${hasValue ? `<button class="inline-danger-button" type="button" data-clear-design-

```

```

summary="${pathValue}">Clear overview</button>` : ""}
6510:     </span>
6511:   </div>
6512:
6513:     <div class="rich-summary-panel">
6514:       <p class="rich-editor-hint">Right-click to add images, code blocks, apply bold, color, font,
numeric size, alignment, or formulas.</p>
6515:
6516:       <div
6517:         class="rich-summary-editor"
6518:         data-rich-editor="section-overview"
6519:         data-rich-text-path="${escapeHtml(pathValue)}"
6520:         data-rich-path="${escapeHtml(richPath)}"
6521:         data-rich-folder="${escapeHtml(overviewFolderFromPath(pathValue))}"
6522:         data-placeholder="${escapeHtml(placeholder || "")}"
6523:         contenteditable="true"
6524:         spellcheck="true"
6525:         aria-label="${escapeHtml(label)} rich editor"
6526:       ></div>
6527:     </div>
6528:   </div>
6529: `;
6530: }

```

## designPathHasContent - lines 6532-6536

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6532-6536 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function designPathHasContent(project, pathValue) {                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getByPath, isArray, some, itemTitle, richHasContent, summaryRichPathFromTextPath                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6532: function designPathHasContent(project, pathValue) {
6533:   const value = getByPath(project, pathValue);
6534:   if (Array.isArray(value)) return value.some((item) => itemTitle(item) || item?.description || item?.url
|| item?.artifact || richHasContent(item?.richDescription));
6535:   return Boolean(value || richHasContent(getByPath(project, summaryRichPathFromTextPath(pathValue))));
6536: }

```

## renderEmptyDynamicSectionPrompt - lines 6538-6548

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6538-6548 (11 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderEmptyDynamicSectionPrompt(label = "section") {                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | escapeHtml                                                                                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6538: function renderEmptyDynamicSectionPrompt(label = "section") {
6539:   return
6540:     <div class="compile-code-empty">

```

```

6541:         <h3>No ${escapeHtml(label)} content added yet</h3>
6542:         <p>Create your own sections and subsections with the Add section button. Empty predefined groups
stay hidden so the project structure is controlled by you.</p>
6543:         <div class="section-actions">
6544:             <button class="button primary" type="button" data-add-section="true">Add section</button>
6545:         </div>
6546:     </div>
6547: `;
6548: }

```

## renderAnalogDesignWorkspace - lines 6550-6593

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6550-6593 (44 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderAnalogDesignWorkspace(project) {                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureDesignModel, designPathHasContent, push, renderDesignSummaryField, renderDesignObjectSection, renderObjectSection, join, renderEmptyDynamicSectionPrompt, renderPendingEditor                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6550: function renderAnalogDesignWorkspace(project) {
6551:     ensureDesignModel(project);
6552:     const cards = [];
6553:     if (designPathHasContent(project, "design.brief.summary") || designPathHasContent(project,
"design.brief.files")) {
6554:         cards.push(`
6555:             <section class="design-workspace-card">
6556:                 <h3>Design Overview</h3>
6557:                 ${renderDesignSummaryField(project, "design.brief.summary", "Design overview", "State the analog
design objective, topology, constraints, and design decisions.")}
6558:                 ${renderDesignObjectSection(project, "design.brief.files", "design-brief", "brief file",
"document")}
6559:             </section>
6560:         `);
6561:     }
6562:     if (
6563:         designPathHasContent(project, "design.documentation.summary") ||
6564:         designPathHasContent(project, "design.documentation.files") ||
6565:         designPathHasContent(project, "design.documentation.references") ||
6566:         designPathHasContent(project, "design.documentation.mathAnalysis")
6567:     ) {
6568:         cards.push(`
6569:             <section class="design-workspace-card">
6570:                 <h3>Documentation</h3>
6571:                 ${renderDesignSummaryField(project, "design.documentation.summary", "Documentation overview",
"Summarize schematics, notes, datasheets, calculations, and supporting documents.")}
6572:                 ${renderDesignObjectSection(project, "design.documentation.files", "documentation", "document",
"document")}
6573:                 ${designPathHasContent(project, "design.documentation.references") ?
`<h4>References</h4>${renderDesignObjectSection(project, "design.documentation.references", "", "reference",
"reference")}` : ""}
6574:                 ${designPathHasContent(project, "design.documentation.mathAnalysis") ? `<h4>Math /
Analysis</h4>${renderDesignObjectSection(project, "design.documentation.mathAnalysis", "", "analysis item",
"analysis")}` : ""}
6575:             </section>
6576:         `);
6577:     }
6578:     if ((project.pcbcs || []).length || (project.media || []).length) {
6579:         cards.push(`
6580:             <section class="design-workspace-card">
6581:                 <h3>PCB and Visual Evidence</h3>
6582:                 ${((project.pcbcs || []).length ? renderObjectSection(project, "pcbcs", "pcbcs", "PCBs") : "")}
6583:                 ${((project.media || []).length ? renderObjectSection(project, "media", "images", "Images") : "")}
6584:             </section>
6585:         `);
6586:     }
6587:     return `
6588:         <div class="section-stack analog-design-workspace">
6589:             ${cards.join("")} || renderEmptyDynamicSectionPrompt("design")}
6590:             ${renderPendingEditor()}
6591:         </div>
6592:     `;

```

6593: }

## ***renderAnalogSimulationWorkspace - lines 6595-6628***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6595-6628 (34 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderAnalogSimulationWorkspace(project) {                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureDesignModel, designPathHasContent, push, renderDesignSummaryField, renderDesignObjectSection, join, renderEmptyDynamicSectionPrompt, renderPendingEditor                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6595: function renderAnalogSimulationWorkspace(project) {
6596:   ensureDesignModel(project);
6597:   const cards = [];
6598:   if (designPathHasContent(project, "design.simulation.summary")) {
6599:     cards.push(`
6600:       <section class="design-workspace-card">
6601:         <h3>Simulation Overview</h3>
6602:         ${renderDesignSummaryField(project, "design.simulation.summary", "Simulation overview", "Explain
what was simulated, why it mattered, and what the results proved.")}
6603:       </section>
6604:     `);
6605:   }
6606:   if (designPathHasContent(project, "design.simulation.files")) {
6607:     cards.push(`
6608:       <section class="design-workspace-card">
6609:         <h3>Simulation Files</h3>
6610:         ${renderDesignObjectSection(project, "design.simulation.files", "simulations", "simulation file",
"simulation-file")}
6611:       </section>
6612:     `);
6613:   }
6614:   if (designPathHasContent(project, "design.simulation.results")) {
6615:     cards.push(`
6616:       <section class="design-workspace-card">
6617:         <h3>Simulation Results</h3>
6618:         ${renderDesignObjectSection(project, "design.simulation.results", "simulation-results",
"simulation result", "simulation-result")}
6619:       </section>
6620:     `);
6621:   }
6622:   return `
6623:     <div class="section-stack analog-design-workspace">
6624:       ${cards.join("")} || renderEmptyDynamicSectionPrompt("simulation")
6625:       ${renderPendingEditor()}
6626:     </div>
6627:   `;
6628: }
```

## ***pendingEditorUsesRichDescription - lines 6630-6632***

| Item            | Explanation                                                         |
|-----------------|---------------------------------------------------------------------|
| Source location | Lines 6630-6632 (3 source lines).                                   |
| Purpose         | Part of the private builder frontend and editing experience.        |
| Declaration     | function pendingEditorUsesRichDescription(editor = pendingEditor) { |
| Touches         | Mostly local calculations or local UI state.                        |
| Calls detected  | No obvious named calls detected by the generator.                   |
| API endpoints   | None detected inside this function body.                            |
| Storage keys    | No local/session storage keys detected.                             |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6630: function pendingEditorUsesRichDescription(editor = pendingEditor) {
6631:   return Boolean(editor && editor.type !== "text-array");
6632: }
```

## ***pendingEditorFolder - lines 6634-6641***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6634-6641 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function pendingEditorFolder(editor = pendingEditor) {                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | slugify                                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6634: function pendingEditorFolder(editor = pendingEditor) {
6635:   if (!editor) return "content";
6636:   if (editor.type === "custom-section") return `custom-${slugify(editor.sectionId ||
"section")}-overview`;
6637:   if (editor.type === "custom") return `custom-${slugify(editor.sectionId || "section")}-subsection`;
6638:   if (editor.type === "design-object") return `design-${slugify(editor.kind || "content")}`;
6639:   if (editor.type === "object") return slugify(editor.key || "content");
6640:   return slugify(editor.type || "content");
6641: }
```

## ***renderPendingRichDescriptionEditor - lines 6643-6666***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6643-6666 (24 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPendingRichDescriptionEditor(editor = pendingEditor) {                                                                                                                                 |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | escapeHtml, pendingEditorFolder                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6643: function renderPendingRichDescriptionEditor(editor = pendingEditor) {
6644:   const label = editor?.type === "tool"
6645:     ? "What it was used for"
6646:     : editor?.type === "custom-section"
6647:       ? "Section content"
6648:       : "Content";
6649:   return `
6650:     <div class="pending-rich-description rich-design-summary-field">
6651:       <div class="summary-builder-heading"><span>${escapeHtml(label)}</span></div>
```

```

6652:         <div class="rich-summary-panel">
6653:             <p class="rich-editor-hint">Right-click to add images, code blocks, apply bold, color, font,
numeric size, alignment, or formulas.</p>
6654:             <div
6655:                 class="rich-summary-editor"
6656:                 data-rich-editor="pending-description"
6657:                 data-rich-folder="${escapeHtml(pendingEditorFolder(editor))}"
6658:                 data-placeholder="Type or paste the complete content here."
6659:                 contenteditable="true"
6660:                 spellcheck="true"
6661:                 aria-label="${escapeHtml(label)} rich editor"
6662:             ></div>
6663:         </div>
6664:     </div>
6665: `;
6666: }

```

## renderPendingEditor - lines 6668-6692

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6668-6692 (25 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPendingEditor() {                                                                                                                                                                      |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | renderPendingRichDescriptionEditor                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6668: function renderPendingEditor() {
6669:     if (!pendingEditor) return "";
6670:     const title = pendingEditor.title || "";
6671:     const isTool = pendingEditor.type === "tool";
6672:     const isSimpleText = pendingEditor.type === "text-array";
6673:     const isSectionDetails = pendingEditor.type === "custom-section";
6674:     return
6675:         <form class="pending-editor" id="pending-editor">
6676:             <div>
6677:                 <h3>${pendingEditor.mode === "edit" ? "Edit" : "Add"} ${isTool ? "tool" : isSimpleText ? "item" :
isSectionDetails ? "section" : "subsection"}</h3>
6678:             </div>
6679:             <div class="${isTool ? "tool-editor-grid" : "pending-editor-grid"}">
6680:                 <label>
6681:                     <span>${isTool ? "Tool name" : "Title"}</span>
6682:                     <input name="title" type="text" value="${title}" required>
6683:                 </label>
6684:                 ${isSimpleText ? "" : renderPendingRichDescriptionEditor(pendingEditor)}
6685:             </div>
6686:             <div class="pending-editor-actions">
6687:                 <button class="button primary" type="submit">Save</button>
6688:                 <button class="button secondary" type="button" data-cancel-pending="true">Cancel</button>
6689:             </div>
6690:         </form>
6691:     `;
6692: }

```

## openPendingEditor - lines 6694-6701

| Item            | Explanation                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 6694-6701 (8 source lines).                                                                                |
| Purpose         | Controls a window, dialog, panel, menu, or visible state.                                                        |
| Declaration     | function openPendingEditor(config) {                                                                             |
| Touches         | browser/Electron DOM                                                                                             |
| Calls detected  | renderSectionContent, selectedProject, requestAnimationFrame, populateRichEditors, querySelector, scrollIntoView |
| API endpoints   | None detected inside this function body.                                                                         |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6694: function openPendingEditor(config) {
6695:   pendingEditor = config;
6696:   renderSectionContent(selectedProject());
6697:   requestAnimationFrame(() => {
6698:     populateRichEditors(sectionContent);
6699:     document.querySelector("#pending-editor")?.scrollIntoView({ behavior: "smooth", block: "nearest" });
6700:   });
6701: }

```

## ***savePendingEditor - lines 6703-6785***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6703-6785 (83 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | function savePendingEditor(form) {                                                                                                                                                                    |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | selectedProject, trim, querySelector, extractRichSummary, plainTextFromRich, setStatus, push, makeItemForSection, designArray, makeDesignItem, find, scheduleAutosave                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6703: function savePendingEditor(form) {
6704:   const project = selectedProject();
6705:   if (!project || !pendingEditor) return;
6706:   const title = form.elements.title.value.trim();
6707:   const richEditor = form.querySelector("[data-rich-editor='pending-description']");
6708:   const richDescription = richEditor ? extractRichSummary(richEditor) : null;
6709:   const description = richEditor ? plainTextFromRich(richDescription) : "";
6710:   if (!title) {
6711:     setStatus("Type a title before saving.");
6712:     return;
6713:   }
6714:
6715:   if (pendingEditor.type === "tool") {
6716:     project.tools = project.tools || [];
6717:     const nextTool = { name: title, description, richDescription };
6718:     if (pendingEditor.mode === "edit") {
6719:       project.tools[Number(pendingEditor.index)] = nextTool;
6720:     } else {
6721:       project.tools.push(nextTool);
6722:     }
6723:   }
6724:
6725:   if (pendingEditor.type === "object") {
6726:     const key = pendingEditor.key;
6727:     project[key] = project[key] || [];
6728:     if (pendingEditor.mode === "edit") {
6729:       const existing = project[key][Number(pendingEditor.index)] || {};
6730:       const existingUrl = existing.url || existing.artifact || "";
6731:       const nextItem = makeItemForSection(key, title, description, existingUrl);
6732:       project[key][Number(pendingEditor.index)] = { ...existing, ...nextItem, richDescription };
6733:     } else {
6734:       const nextItem = makeItemForSection(key, title, description);
6735:       project[key].push({ ...nextItem, richDescription });
6736:     }
6737:   }
6738:
6739:   if (pendingEditor.type === "design-object") {
6740:     const items = designArray(project, pendingEditor.path);
6741:     const kind = pendingEditor.kind || "document";
6742:     if (pendingEditor.mode === "edit") {

```

```

6743:     const existing = items[Number(pendingEditor.index)] || {};
6744:     const existingUrl = existing.url || existing.artifact || "";
6745:     items[Number(pendingEditor.index)] = { ...existing, ...makeDesignItem(kind, title, description,
existingUrl), richDescription };
6746:   } else {
6747:     items.push({ ...makeDesignItem(kind, title, description), richDescription });
6748:   }
6749: }
6750:
6751: if (pendingEditor.type === "text-array") {
6752:   const key = pendingEditor.key;
6753:   project[key] = project[key] || [];
6754:   if (pendingEditor.mode === "edit") {
6755:     project[key][Number(pendingEditor.index)] = title;
6756:   } else {
6757:     project[key].push(title);
6758:   }
6759: }
6760:
6761: if (pendingEditor.type === "custom") {
6762:   const section = (project.sections || []).find((item) => item.id === pendingEditor.sectionId);
6763:   if (!section) return;
6764:   section.items = section.items || [];
6765:   const nextItem = { title, description, richDescription, type: "Text", status: "draft" };
6766:   if (pendingEditor.mode === "edit") {
6767:     section.items[Number(pendingEditor.index)] = { ...section.items[Number(pendingEditor.index)],
...nextItem };
6768:   } else {
6769:     section.items.push(nextItem);
6770:   }
6771: }
6772:
6773: if (pendingEditor.type === "custom-section") {
6774:   const section = (project.sections || []).find((item) => item.id === pendingEditor.sectionId);
6775:   if (!section) return;
6776:   section.title = title;
6777:   section.description = description;
6778:   section.richDescription = richDescription;
6779: }
6780:
6781: pendingEditor = null;
6782: setStatus("Saved in the project editor. Click Save project to include this version in the portfolio
preview.");
6783: scheduleAutosave();
6784: renderAll();
6785: }

```

## section - lines 6762-6765

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6762-6765 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>const section = (project.sections    []).find((item) =&gt; item.id === pendingEditor.sectionId);</code>                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>find</code>                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6762:   const section = (project.sections || []).find((item) => item.id === pendingEditor.sectionId);
6763:   if (!section) return;
6764:   section.items = section.items || [];
6765:   const nextItem = { title, description, richDescription, type: "Text", status: "draft" };

```

## section - lines 6774-6779

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 6774-6779 (6 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | const section = (project.sections    []).find((item) => item.id === pendingEditor.sectionId);                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | find                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6774:     const section = (project.sections || []).find((item) => item.id === pendingEditor.sectionId);
6775:     if (!section) return;
6776:     section.title = title;
6777:     section.description = description;
6778:     section.richDescription = richDescription;
6779: }

```

## ***openEditorFromButton - lines 6787-6879***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6787-6879 (93 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function openEditorFromButton(button) {                                                                                                                                                               |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | selectedProject, openPendingEditor, toolName, toolDescription, designArray, find                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6787: function openEditorFromButton(button) {
6788:   const project = selectedProject();
6789:   if (!project) return;
6790:   const type = button.dataset.openEditor;
6791:   const mode = button.dataset.mode || "add";
6792:   const index = button.dataset.index;
6793:
6794:   if (type === "tool") {
6795:     const item = mode === "edit" ? project.tools?.[Number(index)] : {};
6796:     openPendingEditor({
6797:       type,
6798:       mode,
6799:       index,
6800:       title: toolName(item),
6801:       description: toolDescription(item),
6802:       richDescription: item?.richDescription || null
6803:     });
6804:     return;
6805:   }
6806:
6807:   if (type === "object") {
6808:     const key = button.dataset.key;
6809:     const item = mode === "edit" ? project[key]?.[Number(index)] : {};
6810:     openPendingEditor({
6811:       type,
6812:       mode,
6813:       key,
6814:       index,
6815:       title: item?.title || item?.name || item?.label || "",
6816:       description: item?.description || item?.caption || item?.result || item?.method || "",
6817:       richDescription: item?.richDescription || null
6818:     });
6819:     return;
6820:   }

```

```

6821:
6822:   if (type === "design-object") {
6823:     const pathValue = button.dataset.designPath;
6824:     const items = designArray(project, pathValue);
6825:     const item = mode === "edit" ? items?.[Number(index)] : {};
6826:     openPendingEditor({
6827:       type,
6828:       mode,
6829:       path: pathValue,
6830:       kind: button.dataset.designKind || "document",
6831:       index,
6832:       title: item?.title || item?.name || item?.label || "",
6833:       description: item?.description || item?.caption || item?.result || item?.method || "",
6834:       richDescription: item?.richDescription || null
6835:     });
6836:     return;
6837:   }
6838:
6839:
6840:   if (type === "custom") {
6841:     const section = (project.sections || []).find((item) => item.id === button.dataset.sectionId);
6842:     const item = mode === "edit" ? section?.items?.[Number(index)] : {};
6843:     openPendingEditor({
6844:       type,
6845:       mode,
6846:       sectionId: button.dataset.sectionId,
6847:       index,
6848:       title: item?.title || "",
6849:       description: item?.description || "",
6850:       richDescription: item?.richDescription || null
6851:     });
6852:     return;
6853:   }
6854:
6855:   if (type === "text-array") {
6856:     const key = button.dataset.key;
6857:     const item = mode === "edit" ? project[key]?.[Number(index)] : "";
6858:     openPendingEditor({
6859:       type,
6860:       mode,
6861:       key,
6862:       index,
6863:       title: typeof item === "string" ? item : item?.title || item?.name || item?.label || ""
6864:     });
6865:     return;
6866:   }
6867:
6868:   if (type === "custom-section") {
6869:     const section = (project.sections || []).find((item) => item.id === button.dataset.sectionId);
6870:     openPendingEditor({
6871:       type,
6872:       mode: "edit",
6873:       sectionId: button.dataset.sectionId,
6874:       title: section?.title || "",
6875:       description: section?.description || "",
6876:       richDescription: section?.richDescription || null
6877:     });
6878:   }
6879: }

```

## section - lines 6841-6842

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6841-6842 (2 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const section = (project.sections    []).find((item) => item.id === button.dataset.sectionId);                                                                                                        |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | find                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6841:     const section = (project.sections || []).find((item) => item.id === button.dataset.sectionId);
6842:     const item = mode === "edit" ? section?.items?.[Number(index)] : {};

```

## section - lines 6869-6877

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6869-6877 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>const section = (project.sections    []).find((item) =&gt; item.id === button.dataset.sectionId);</code>                                                                                        |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | find, openPendingEditor                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6869:   const section = (project.sections || []).find((item) => item.id === button.dataset.sectionId);
6870:   openPendingEditor({
6871:     type,
6872:     mode: "edit",
6873:     sectionId: button.dataset.sectionId,
6874:     title: section?.title || "",
6875:     description: section?.description || "",
6876:     richDescription: section?.richDescription || null
6877:   });
```

## builderPreviewContent - lines 6881-6888

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6881-6888 (8 source lines).                                                                                                                                                                     |
| Purpose            | Creates an object, UI structure, process, payload, or generated artifact.                                                                                                                             |
| Declaration        | <code>function builderPreviewContent(rich, text = "", emptyText = "No content has been added yet.") {</code>                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | richHasContent, renderRichContent, renderMultilineInlineText, escapeHtml                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6881: function builderPreviewContent(rich, text = "", emptyText = "No content has been added yet.") {
6882:   const content = richHasContent(rich)
6883:     ? renderRichContent(rich, text || "")
6884:     : text
6885:     ? `

${renderMultilineInlineText(text)}</p>`
6886:     : `

${escapeHtml(emptyText)}</p>`;
6887:   return `

${content}</div>`;
6888: }


```

## renderCustomSection - lines 6890-6933

| Item            | Explanation                                                     |
|-----------------|-----------------------------------------------------------------|
| Source location | Lines 6890-6933 (44 source lines).                              |
| Purpose         | Renders UI, markup, or display output.                          |
| Declaration     | <code>function renderCustomSection(project, sectionId) {</code> |
| Touches         | Mostly local calculations or local UI state.                    |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | find, builderPreviewContent, map, escapeHtml, richHasContent, join, renderPendingEditor                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6890: function renderCustomSection(project, sectionId) {
6891:   const section = (project.sections || []).find((item) => item.id === sectionId);
6892:   if (!section) return `<p class="evidence-empty">Section not found.</p>`;
6893:
6894:   return `
6895:     <div class="section-window-heading">
6896:       <h3>${section.title || "Custom section"}</h3>
6897:       <div>
6898:         <button type="button" data-open-editor="custom-section" data-section-id="${section.id}">Edit
section</button>
6899:         <button class="danger-icon" type="button" data-delete-section="${section.id}">Delete
section</button>
6900:       </div>
6901:     </div>
6902:     ${builderPreviewContent(section.richDescription, section.description || "", "No section content has
been added yet.")}
6903:     <div class="section-actions">
6904:       <button class="button primary" type="button" data-add-custom-item="${section.id}">Add
subsection</button>
6905:       <button class="button secondary" type="button" data-upload-custom="${section.id}">Add file or
image</button>
6906:     </div>
6907:     <div class="builder-items">
6908:       ${(section.items || []).map((item, index) => `
6909:         <article class="builder-item">
6910:           <div>
6911:             <strong>${escapeHtml(item.title || (item.url ? "Untitled file" : "Untitled
subsection"))}</strong>
6912:             ${item.url ? `<span>${escapeHtml(item.url)}</span>` : ""}
6913:             ${!(richHasContent(item.richDescription) || item.description) ?
builderPreviewContent(item.richDescription, item.description || "", "") : ""}
6914:             ${!(item.children || []).length ? `
6915:               <div class="builder-child-files">
6916:                 ${(item.children || []).map((child, childIndex) => `
6917:                   <div>
6918:                     <span>${escapeHtml(child.title || "Untitled file")}</span>
6919:                     <button class="danger-icon" type="button" data-delete-custom-
child="${escapeHtml(section.id)}" data-index="${index}" data-child-index="${childIndex}">Delete</button>
6920:                   </div>
6921:                 `).join("")}
6922:               </div>
6923:             ` : ""}
6924:           </div>
6925:           ${item.url ? "" : `<button type="button" data-upload-custom-
subsection="${escapeHtml(section.id)}" data-index="${index}">Add file</button>`}
6926:           ${item.url ? "" : `<button type="button" data-open-editor="custom" data-mode="edit" data-
section-id="${escapeHtml(section.id)}" data-index="${index}">Edit</button>`}
6927:           <button class="danger-icon" type="button" data-delete-custom-item="${section.id}" data-
index="${index}">Delete</button>
6928:         </article>
6929:       `).join("")} || `<p class="evidence-empty">No subsections added yet.</p>`
6930:     </div>
6931:     ${renderPendingEditor()}
6932:   `;
6933: }

```

## section - lines 6891-6892

| Item            | Explanation                                                                     |
|-----------------|---------------------------------------------------------------------------------|
| Source location | Lines 6891-6892 (2 source lines).                                               |
| Purpose         | Part of the private builder frontend and editing experience.                    |
| Declaration     | const section = (project.sections    []).find((item) => item.id === sectionId); |
| Touches         | Mostly local calculations or local UI state.                                    |
| Calls detected  | find                                                                            |
| API endpoints   | None detected inside this function body.                                        |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6891:   const section = (project.sections || []).find((item) => item.id === sectionId);
6892:   if (!section) return `<p class="evidence-empty">Section not found.</p>`;
```

## ***compileLanguageOptions - lines 6935-6940***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6935-6940 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function compileLanguageOptions(selected = "javascript") {                                                                                                                                            |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeCodeLanguage, map, escapeHtml, join                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6935: function compileLanguageOptions(selected = "javascript") {
6936:   const normalized = normalizeCodeLanguage(selected);
6937:   return supportedCodeLanguages.map((language) => `
6938:     <option value="${escapeHtml(language.id)}"${language.id === normalized ? " selected" :
""}>${escapeHtml(language.label)}</option>
6939:   `).join("");
6940: }
```

## ***compileRoleOptions - lines 6942-6947***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6942-6947 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function compileRoleOptions(selected = "design") {                                                                                                                                                    |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeCompileFileRole, map, escapeHtml, compileFileRoleLabel, join                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
6942: function compileRoleOptions(selected = "design") {
6943:   const normalized = normalizeCompileFileRole(selected, "verilog");
6944:   return ["design", "testbench"].map((role) => `
6945:     <option value="${escapeHtml(role)}"${role === normalized ? " selected" :
""}>${escapeHtml(compileFileRoleLabel(role))}</option>
6946:   `).join("");
6947: }
```

## ***hdlTestbenchTemplate - lines 6949-6973***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6949-6973 (25 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function hdlTestbenchTemplate(language = "systemverilog", designName = "design") {                                                                                                                    |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | slugify, replace, dut, clk, reset_n, done, dumpfile, dumpvars, join                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6949: function hdlTestbenchTemplate(language = "systemverilog", designName = "design") {
6950:   const tbName = `tb_${slugify(designName || "design").replace(/-/g, "_")} || "design"`;
6951:   const dutName = slugify(designName || "design").replace(/-/g, "_") || "design";
6952:   const commentPrefix = language === "verilog" ? "/*" : "/*";
6953:   return [
6954:     `timescale 1ns/1ps`,
6955:     ``,
6956:     `module ${tbName};`,
6957:     ` ${commentPrefix} Declare testbench signals here.`,
6958:     ` ${commentPrefix} Example: reg clk = 0; reg reset_n = 0; wire done;`,
6959:     ``,
6960:     ` ${commentPrefix} Instantiate the design under test and connect the signals.`,
6961:     ` ${commentPrefix} Example: ${dutName} dut (.clk(clk), .reset_n(reset_n), .done(done));`,
6962:     ``,
6963:     ` initial begin`,
6964:     `   $dumpfile("waveform.vcd");`,
6965:     `   $dumpvars(0, ${tbName});`,
6966:     ``,
6967:     ` ${commentPrefix} Drive stimulus over time here.`,
6968:     `   #100;`,
6969:     `   $finish;`,
6970:     ` end`,
6971:     `endmodule`
6972:   ].join("\n");
6973: }

```

## ***firstHdlModuleName - lines 6975-6978***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6975-6978 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function firstHdlModuleName(code = "") {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | match                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6975: function firstHdlModuleName(code = "") {
6976:   const match = String(code || "").match(/\\bmodule\\s+([A-Za-z_$][\\w$]*)/);
6977:   return match?.[1] || "design";
6978: }

```

## ***compileAppendDestinations - lines 6980-7026***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 6980-7026 (47 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function compileAppendDestinations(project) {                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isAnalogProject, ensureDesignModel, forEach, push, summaryRichPathFromTextPath, walkItems                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

6980: function compileAppendDestinations(project) {
6981:   if (!project) return [];
6982:   const destinations = [{
6983:     id: "project-overview",
6984:     label: "Project overview",
6985:     richPath: "summaryRich",
6986:     textPath: "summary"
6987:   }];
6988:
6989:   if (isAnalogProject(project)) {
6990:     ensureDesignModel(project);
6991:     [
6992:       ["design-overview", "Design overview", "design.brief.summary"],
6993:       ["documentation-overview", "Documentation overview", "design.documentation.summary"],
6994:       ["simulation-overview", "Simulation overview", "design.simulation.summary"]
6995:     ].forEach(([id, label, textPath]) => {
6996:       destinations.push({ id, label, richPath: summaryRichPathFromTextPath(textPath), textPath });
6997:     });
6998:   }
6999:
7000:   (project.sections || []).forEach((section, index) => {
7001:     destinations.push({
7002:       id: `custom-${section.id || index}`,
7003:       label: `${section.title || `Custom section ${index + 1}`} overview`,
7004:       richPath: `sections.${index}.richDescription`,
7005:       textPath: `sections.${index}.description`
7006:     });
7007:     const walkItems = (items = [], prefix = `sections.${index}.items`, labelPrefix = section.title || `Custom section ${index + 1}`) => {
7008:       (items || []).forEach((item, itemIndex) => {
7009:         const itemPath = `${prefix}.${itemIndex}`;
7010:         const itemTitle = item.title || `Subsection ${itemIndex + 1}`;
7011:         if (!item.url) {
7012:           destinations.push({
7013:             id: `custom-${section.id || index}-${itemPath}`,
7014:             label: `${labelPrefix} / ${itemTitle} overview`,
7015:             richPath: `${itemPath}.richDescription`,
7016:             textPath: `${itemPath}.description`
7017:           });
7018:         }
7019:         walkItems(item.children || item.items || [], `${itemPath}.children`, `${labelPrefix} / ${itemTitle}`);
7020:       });
7021:     };
7022:     walkItems(section.items || []);
7023:   });
7024:
7025:   return destinations;
7026: }

```

## ***compileAppendDestinationOptions - lines 7028-7036***

| Item            | Explanation                                                          |
|-----------------|----------------------------------------------------------------------|
| Source location | Lines 7028-7036 (9 source lines).                                    |
| Purpose         | Part of the Compile Code language/tool/build/run flow.               |
| Declaration     | function compileAppendDestinationOptions(project, selectedId = "") { |
| Touches         | Mostly local calculations or local UI state.                         |
| Calls detected  | compileAppendDestinations, some, map, escapeHtml, join               |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7028: function compileAppendDestinationOptions(project, selectedId = "") {
7029:   const destinations = compileAppendDestinations(project);
7030:   const selected = destinations.some((destination) => destination.id === selectedId)
7031:     ? selectedId
7032:     : destinations[0]?.id || "";
7033:   return destinations.map((destination) => `
7034:     <option value="${escapeHtml(destination.id)}"${destination.id === selected ? " selected" :
""}>${escapeHtml(destination.label)}</option>
7035:   `).join("");
7036: }

```

## ***activeCompileFile - lines 7038-7041***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7038-7041 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function activeCompileFile(project) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureCompileCode, find                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7038: function activeCompileFile(project) {
7039:   const workspace = ensureCompileCode(project);
7040:   return workspace.files.find((file) => file.id === workspace.activeFileId) || workspace.files[0] ||
null;
7041: }

```

## ***compileLogTimestamp - lines 7043-7047***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7043-7047 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function compileLogTimestamp(value = new Date().toISOString()) {                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | toISOString, isNaN, getTime, toLocaleTimeString                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7043: function compileLogTimestamp(value = new Date().toISOString()) {
7044:   const date = new Date(value);

```



```

7045:   if (Number.isNaN(date.getTime())) return "";
7046:   return date.toLocaleTimeString([], { hour: "2-digit", minute: "2-digit", second: "2-digit" });
7047: }

```

## addCompileMessage - lines 7049-7062

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7049-7062 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function addCompileMessage(project, text, level = "info") {                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureCompileCode, slugify, now, random, toString, slice, toISOString, includes, updateCompileMessagesPanel                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7049: function addCompileMessage(project, text, level = "info") {
7050:   const workspace = ensureCompileCode(project);
7051:   if (!workspace || !text) return;
7052:   workspace.messages = [
7053:     {
7054:       id: slugify(`message-${Date.now()}-${Math.random().toString(16).slice(2)}`),
7055:       at: new Date().toISOString(),
7056:       level: ["info", "success", "warning", "error"].includes(level) ? level : "info",
7057:       text: String(text)
7058:     },
7059:     ...(workspace.messages || [])
7060:   ].slice(0, 80);
7061:   updateCompileMessagesPanel(project);
7062: }

```

## renderCompileMessages - lines 7064-7073

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7064-7073 (10 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderCompileMessages(workspace) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | isArray, map, escapeHtml, compileLogTimestamp, join                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7064: function renderCompileMessages(workspace) {
7065:   const messages = Array.isArray(workspace?.messages) ? workspace.messages : [];
7066:   if (!messages.length) return `<p class="compile-message-empty">No messages yet.</p>`;
7067:   return messages.map((message) => `
7068:     <div class="compile-message compile-message-${escapeHtml(message.level || "info")}">
7069:       <time>${escapeHtml(compileLogTimestamp(message.at))}</time>
7070:       <span>${escapeHtml(message.text)}</span>
7071:     </div>
7072:   `).join("");
7073: }

```

## updateCompileTerminalPanel - lines 7075-7084

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7075-7084 (10 source lines).                                                                                                                                                                    |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateCompileTerminalPanel(project, file = activeCompileFile(project)) {                                                                                                                     |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | activeCompileFile, ensureCompileCode, querySelector, requestAnimationFrame                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7075: function updateCompileTerminalPanel(project, file = activeCompileFile(project)) {
7076:   const workspace = ensureCompileCode(project);
7077:   const terminal = file?.lastResult?.terminal || workspace?.terminal || compileTerminalStatus ||
"Compiler output will appear here after you save or run a source file.";
7078:   const terminalElement = sectionContent.querySelector("[data-compile-terminal]");
7079:   if (!terminalElement) return;
7080:   terminalElement.textContent = terminal;
7081:   requestAnimationFrame(() => {
7082:     terminalElement.scrollTop = terminalElement.scrollHeight;
7083:   });
7084: }

```

## ***updateCompileMessagesPanel - lines 7086-7091***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7086-7091 (6 source lines).                                                                                                                                                                     |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateCompileMessagesPanel(project) {                                                                                                                                                        |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | ensureCompileCode, querySelector, renderCompileMessages                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7086: function updateCompileMessagesPanel(project) {
7087:   const workspace = ensureCompileCode(project);
7088:   const log = sectionContent.querySelector("[data-compile-messages]");
7089:   if (!log) return;
7090:   log.innerHTML = renderCompileMessages(workspace);
7091: }

```

## ***showCompileOutput - lines 7093-7105***

| Item            | Explanation                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------|
| Source location | Lines 7093-7105 (13 source lines).                                                                               |
| Purpose         | Part of the Compile Code language/tool/build/run flow.                                                           |
| Declaration     | function showCompileOutput(project, file = activeCompileFile(project)) {                                         |
| Touches         | browser/Electron DOM, filesystem                                                                                 |
| Calls detected  | activeCompileFile, updateCompileTerminalPanel, querySelector, add, scrollIntoView, requestAnimationFrame, remove |
| API endpoints   | None detected inside this function body.                                                                         |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7093: function showCompileOutput(project, file = activeCompileFile(project)) {
7094:   updateCompileTerminalPanel(project, file);
7095:   const panel = sectionContent.querySelector("[data-compile-terminal-panel]");
7096:   const terminal = sectionContent.querySelector("[data-compile-terminal]");
7097:   if (!panel || !terminal) return;
7098:   panel.classList.add("is-focused");
7099:   panel.scrollToView({ behavior: "smooth", block: "center" });
7100:   requestAnimationFrame(() => {
7101:     terminal.focus?({ preventScroll: true });
7102:     terminal.scrollTop = terminal.scrollHeight;
7103:   });
7104:   window.setTimeout(() => panel.classList.remove("is-focused"), 1400);
7105: }

```

## ***activeCompileWaveform - lines 7107-7110***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7107-7110 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function activeCompileWaveform(project, file = activeCompileFile(project)) {                                                                                                                          |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | activeCompileFile, ensureCompileCode                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7107: function activeCompileWaveform(project, file = activeCompileFile(project)) {
7108:   const workspace = ensureCompileCode(project);
7109:   return file?.lastResult?.waveform || file?.waveform || workspace?.waveform || null;
7110: }

```

## ***scopeSignalValueAt - lines 7112-7117***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7112-7117 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function scopeSignalValueAt(changes = [], index = 0) {                                                                                                                                                |
| Touches            | authentication/security data                                                                                                                                                                          |
| Calls detected     | toLowerCase                                                                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7112: function scopeSignalValueAt(changes = [], index = 0) {
7113:   const value = String(changes[index]?.value ?? "x").toLowerCase();

```

```

7114:   if (value === "1") return "1";
7115:   if (value === "0") return "0";
7116:   return "x";
7117: }

```

## renderScalarWavePath - lines 7119-7140

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7119-7140 (22 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderScalarWavePath(changes = [], maxTime = 1, rowY = 0, left = 150, width = 680) {                                                                                                         |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | isArray, map, sort, max, min, forEach, xFor, yFor, scopeSignalValueAt, toFixed                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7119: function renderScalarWavePath(changes = [], maxTime = 1, rowY = 0, left = 150, width = 680) {
7120:   const normalized = (Array.isArray(changes) && changes.length ? changes : [{ time: 0, value: "x" }])
7121:     .map((change) => ({ time: Number(change.time) || 0, value: String(change.value ?? "x") }));
7122:   .sort((a, b) => a.time - b.time);
7123:   const top = rowY + 5;
7124:   const bottom = rowY + 23;
7125:   const middle = rowY + 14;
7126:   const yFor = (value) => value === "1" ? top : value === "0" ? bottom : middle;
7127:   const xFor = (time) => left + Math.max(0, Math.min(1, (Number(time) || 0) / Math.max(1, maxTime))) *
width;
7128:   let path = "";
7129:   normalized.forEach((change, index) => {
7130:     const x = xFor(change.time);
7131:     const y = yFor(scopeSignalValueAt(normalized, index));
7132:     if (index === 0) {
7133:       path += `M ${x.toFixed(1)} ${y.toFixed(1)}`;
7134:     } else {
7135:       path += `H ${x.toFixed(1)} V ${y.toFixed(1)}`;
7136:     }
7137:   });
7138:   path += `H ${(left + width).toFixed(1)}`;
7139:   return path;
7140: }

```

## yFor - lines 7126-7137

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7126-7137 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const yFor = (value) => value === "1" ? top : value === "0" ? bottom : middle;                                                                                                                        |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | max, min, forEach, xFor, scopeSignalValueAt, toFixed                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7126:   const yFor = (value) => value === "1" ? top : value === "0" ? bottom : middle;
7127:   const xFor = (time) => left + Math.max(0, Math.min(1, (Number(time) || 0) / Math.max(1, maxTime))) *
width;
7128:   let path = "";
7129:   normalized.forEach((change, index) => {

```

```

7130:     const x = xFor(change.time);
7131:     const y = yFor(scopeSignalValueAt(normalized, index));
7132:     if (index === 0) {
7133:       path += `M ${x.toFixed(1)} ${y.toFixed(1)}`;
7134:     } else {
7135:       path += ` H ${x.toFixed(1)} V ${y.toFixed(1)}`;
7136:     }
7137:   });

```

## xFor - lines 7127-7137

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7127-7137 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const xFor = (time) => left + Math.max(0, Math.min(1, (Number(time)    0) / Math.max(1, maxTime))) * width;                                                                                           |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | max, min, forEach, yFor, scopeSignalValueAt, toFixed                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7127:   const xFor = (time) => left + Math.max(0, Math.min(1, (Number(time) || 0) / Math.max(1, maxTime))) *
width;
7128:   let path = "";
7129:   normalized.forEach((change, index) => {
7130:     const x = xFor(change.time);
7131:     const y = yFor(scopeSignalValueAt(normalized, index));
7132:     if (index === 0) {
7133:       path += `M ${x.toFixed(1)} ${y.toFixed(1)}`;
7134:     } else {
7135:       path += ` H ${x.toFixed(1)} V ${y.toFixed(1)}`;
7136:     }
7137:   });

```

## renderBusWaveLabels - lines 7142-7152

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7142-7152 (11 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderBusWaveLabels(changes = [], maxTime = 1, rowY = 0, left = 150, width = 680) {                                                                                                          |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | isArray, map, sort, slice, max, min, toFixed, escapeHtml, join                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7142: function renderBusWaveLabels(changes = [], maxTime = 1, rowY = 0, left = 150, width = 680) {
7143:   const normalized = (Array.isArray(changes) && changes.length ? changes : [{ time: 0, value: "x" }])
7144:     .map((change) => ({ time: Number(change.time) || 0, value: String(change.value ?? "x") }));
7145:   .sort((a, b) => a.time - b.time)
7146:   .slice(0, 12);
7147:   return normalized.map((change, index) => {
7148:     const x = left + Math.max(0, Math.min(1, change.time / Math.max(1, maxTime))) * width;
7149:     const value = change.value.length > 12 ? `${change.value.slice(0, 12)}...` : change.value;
7150:     return `<text x="${Math.min(left + width - 70, x + 4).toFixed(1)}" y="${rowY + 19}" class="scope-
value-label">${escapeHtml(value)}</text>${index ? `<line x1="${x.toFixed(1)}" x2="${x.toFixed(1)}" y1="${rowY +
5}" y2="${rowY + 25}" class="scope-transition" />` : ""}`;
7151:   }).join("");

```

7152: }

## renderHdlScope - lines 7154-7203

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7154-7203 (50 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderHdlScope(waveform) {                                                                                                                                                                   |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | isArray, slice, max, flatMap, map, round, toFixed, join, escapeHtml, renderScalarWavePath, renderBusWaveLabels                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7154: function renderHdlScope(waveform) {
7155:   const signals = Array.isArray(waveform?.signals) ? waveform.signals.slice(0, 16) : [];
7156:   if (!signals.length) {
7157:     return `
7158:       <div class="compile-scope-empty">
7159:         <strong>No waveform scope yet.</strong>
7160:         <span>Run a Verilog/SystemVerilog design with a testbench that calls <code>$dumpfile</code> and
<code>$dumpvars</code>.</span>
7161:       </div>
7162:     `;
7163:   }
7164:   const left = 165;
7165:   const width = 760;
7166:   const rowHeight = 34;
7167:   const top = 28;
7168:   const height = top + signals.length * rowHeight + 28;
7169:   const maxTime = Math.max(1, Number(waveform.maxTime) || Math.max(...signals.flatMap((signal) =>
(signal.changes || []).map((change) => Number(change.time) || 0)), 1));
7170:   const grid = [0, 0.25, 0.5, 0.75, 1].map((ratio) => {
7171:     const x = left + ratio * width;
7172:     const label = Math.round(maxTime * ratio);
7173:     return `<line x1="${x.toFixed(1)}" x2="${x.toFixed(1)}" y1="18" y2="${height - 18}" class="scope-
grid" /><text x="${x.toFixed(1)}" y="14" class="scope-time-label">${label}</text>`;
7174:   }).join("");
7175:   const rows = signals.map((signal, index) => {
7176:     const y = top + index * rowHeight;
7177:     const name = String(signal.name || signal.reference || `signal_${index + 1}`);
7178:     const changes = Array.isArray(signal.changes) ? signal.changes : [];
7179:     const scalar = Number(signal.width || 1) <= 1;
7180:     return `
7181:       <g class="scope-row">
7182:         <text x="12" y="${y + 20}" class="scope-signal-label">${escapeHtml(name.length > 24 ?
`${name.slice(0, 24)}...` : name)}</text>
7183:         <line x1="${left}" x2="${left + width}" y1="${y + 27}" y2="${y + 27}" class="scope-row-line" />
7184:         ${scalar
7185:           ? `<path d="${renderScalarWavePath(changes, maxTime, y, left, width)}" class="scope-wave" />`
7186:           : `<line x1="${left}" x2="${left + width}" y1="${y + 15}" y2="${y + 15}" class="scope-bus"
/>${renderBusWaveLabels(changes, maxTime, y, left, width)}`
7187:         }
7188:       </g>
7189:     `;
7190:   }).join("");
7191:   return `
7192:     <div class="compile-scope-meta">
7193:       <span>${escapeHtml(waveform.source || "waveform.vcd")}</span>
7194:       <span>${signals.length} signals${signals.length === 1 ? "" : "s"}</span>
7195:       <span>0 to ${escapeHtml(maxTime)} ${escapeHtml(waveform.timeScale || "ticks")}</span>
7196:     </div>
7197:     <div class="compile-scope-scroll">
7198:       <svg class="compile-scope-svg" viewBox="0 0 ${left + width + 20} ${height}" role="img" aria-
label="HDL waveform scope">
7199:         ${grid}
7200:         ${rows}
7201:       </svg>
7202:     </div>
7203:   `;
}
```

## renderCompileScopePanel - lines 7205-7217

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7205-7217 (13 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderCompileScopePanel(project, file = activeCompileFile(project)) {                                                                                                                        |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | activeCompileFile, activeCompileWaveform, isHdlLanguage, renderHdlScope                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7205: function renderCompileScopePanel(project, file = activeCompileFile(project)) {
7206:   const waveform = activeCompileWaveform(project, file);
7207:   if (!isHdlLanguage(file?.language) && !waveform) return "";
7208:   return `
7209:     <section class="compile-scope-panel" data-compile-scope-panel aria-label="HDL signal scope">
7210:       <div class="compile-terminal-heading">
7211:         <span class="compile-terminal-title"><span aria-hidden="true" class="scope-dot"></span> Signal
scope</span>
7212:         <button type="button" data-compile-show-scope>Show scope</button>
7213:       </div>
7214:       ${renderHdlScope(waveform)}
7215:     </section>
7216:   `;
7217: }
```

## ***showCompileScope - lines 7219-7228***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7219-7228 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function showCompileScope(project, file = activeCompileFile(project)) {                                                                                                                               |
| Touches            | browser/Electron DOM, authentication/security data                                                                                                                                                    |
| Calls detected     | activeCompileFile, querySelector, addCompileMessage, add, scrollIntoView, remove                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7219: function showCompileScope(project, file = activeCompileFile(project)) {
7220:   const panel = sectionContent.querySelector("[data-compile-scope-panel]");
7221:   if (!panel) {
7222:     addCompileMessage(project, "No HDL scope is available for the active source yet.", "warning");
7223:     return;
7224:   }
7225:   panel.classList.add("is-focused");
7226:   panel.scrollIntoView({ behavior: "smooth", block: "center" });
7227:   window.setTimeout(() => panel.classList.remove("is-focused"), 1400);
7228: }
```

## ***copyCompileOutput - lines 7230-7245***

| Item            | Explanation                                            |
|-----------------|--------------------------------------------------------|
| Source location | Lines 7230-7245 (16 source lines).                     |
| Purpose         | Part of the Compile Code language/tool/build/run flow. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | async function copyCompileOutput(project, file = activeCompileFile(project)) {                                                                                                                        |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | activeCompileFile, ensureCompileCode, trim, addCompileMessage, writeText, setStatus                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7230: async function copyCompileOutput(project, file = activeCompileFile(project)) {
7231:   const workspace = ensureCompileCode(project);
7232:   const text = file?.lastResult?.terminal || workspace?.terminal || compileTerminalStatus || "";
7233:   if (!text.trim()) {
7234:     addCompileMessage(project, "There is no terminal output to copy yet.", "warning");
7235:     return;
7236:   }
7237:   try {
7238:     await navigator.clipboard.writeText(text);
7239:     addCompileMessage(project, "Terminal output copied to the clipboard.", "success");
7240:     setStatus("Terminal output copied.");
7241:   } catch {
7242:     addCompileMessage(project, "Terminal output could not be copied by this browser.", "warning");
7243:     setStatus("Terminal output could not be copied.");
7244:   }
7245: }

```

## ***clearCompileOutput - lines 7247-7256***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7247-7256 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function clearCompileOutput(project, file = activeCompileFile(project)) {                                                                                                                             |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | activeCompileFile, ensureCompileCode, updateCompileTerminalPanel, addCompileMessage, setStatus, scheduleAutosave                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7247: function clearCompileOutput(project, file = activeCompileFile(project)) {
7248:   const workspace = ensureCompileCode(project);
7249:   if (file?.lastResult) file.lastResult.terminal = "";
7250:   if (workspace) workspace.terminal = "";
7251:   compileTerminalStatus = "";
7252:   updateCompileTerminalPanel(project, file);
7253:   addCompileMessage(project, "Terminal output cleared.", "info");
7254:   setStatus("Terminal output cleared.");
7255:   scheduleAutosave();
7256: }

```

## ***richBlocksForCompileAppend - lines 7258-7265***

| Item            | Explanation                                                 |
|-----------------|-------------------------------------------------------------|
| Source location | Lines 7258-7265 (8 source lines).                           |
| Purpose         | Part of the Compile Code language/tool/build/run flow.      |
| Declaration     | function richBlocksForCompileAppend(project, destination) { |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getByPath, richBlocksFromValue, filter, trim                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7258: function richBlocksForCompileAppend(project, destination) {
7259:   const existingRich = getByPath(project, destination.richPath);
7260:   const existingText = getByPath(project, destination.textPath) || "";
7261:   return richBlocksFromValue(existingRich, existingText).filter((block) => {
7262:     if (block.type !== "paragraph") return true;
7263:     return Boolean(String(block.text || "").trim() || block.html);
7264:   });
7265: }

```

### ***codeBlockForCompileFile - lines 7267-7274***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7267-7274 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function codeBlockForCompileFile(file) {                                                                                                                                                              |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeCodeText, normalizeCodeLanguage, detectCodeLanguage                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7267: function codeBlockForCompileFile(file) {
7268:   return {
7269:     code: normalizeCodeText(file?.code || "", "source"),
7270:     language: normalizeCodeLanguage(file?.language || detectCodeLanguage(file?.code || "", file?.fileName
|| "")),
7271:     pasteMode: "source",
7272:     type: "code"
7273:   };
7274: }

```

### ***compileFileDirtyLabel - lines 7276-7283***

| Item               | Explanation                                            |
|--------------------|--------------------------------------------------------|
| Source location    | Lines 7276-7283 (8 source lines).                      |
| Purpose            | Part of the Compile Code language/tool/build/run flow. |
| Declaration        | function compileFileDirtyLabel(file) {                 |
| Touches            | Mostly local calculations or local UI state.           |
| Calls detected     | No obvious named calls detected by the generator.      |
| API endpoints      | None detected inside this function body.               |
| Storage keys       | No local/session storage keys detected.                |
| Async/return clues | 0 await expression(s), 6 return statement(s).          |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7276: function compileFileDirtyLabel(file) {
7277:   if (!file) return "";
7278:   if (file.dirty) return "Unsaved";
7279:   if (file.lastResult?.ok === true) return "Compiled";
7280:   if (file.lastResult?.ok === false) return "Needs fix";
7281:   if (file.savedAt) return "Saved";
7282:   return "Draft";
7283: }

```

## ***renderCompileCodeSection - lines 7285-7408***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7285-7408 (124 source lines).                                                                                                                                                                   |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderCompileCodeSection(project) {                                                                                                                                                          |
| Touches            | filesystem, authentication/security data                                                                                                                                                              |
| Calls detected     | ensureCompileCode, activeCompileFile, escapeHtml, map, match, replace, compileFileDirtyLabel, join, defaultCodeFileName, compileLanguageOptions, isHdlLanguage, compileRoleOptions                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7285: function renderCompileCodeSection(project) {
7286:   const workspace = ensureCompileCode(project);
7287:   const activeFile = activeCompileFile(project);
7288:   const terminal = activeFile?.lastResult?.terminal || workspace.terminal || compileTerminalStatus ||
"Compiler output will appear here after you save or run a source file.";
7289:   return `
7290:     <div class="compile-code-workspace">
7291:       <aside class="compile-code-sidebar" aria-label="Compile code files">
7292:         <div class="compile-code-sidebar-heading">
7293:           <h3>Compile Code</h3>
7294:           <small>${workspace.files.length} source file${workspace.files.length === 1 ? "" : "s"}</small>
7295:         </div>
7296:         <div class="compile-code-workspace-tree" aria-label="Project workspace tree">
7297:           <details open>
7298:             <summary>${escapeHtml(project.title || "Project workspace")}</summary>
7299:             <div class="compile-code-tree-files">
7300:               ${workspace.files.map((file) => `
7301:                 <button class="compile-code-tree-file${file.id === workspace.activeFileId ? " is-active"
: ""}" type="button" data-compile-select="${escapeHtml(file.id)}">
7302:                   <span class="compile-file-extension">${escapeHtml((file.fileName.match(/\.[^.]?$/)?.[0]
|| "src").replace(".", ""))}</span>
7303:                   <span>${escapeHtml(file.fileName)}</span>
7304:                   <small>${escapeHtml(compileFileDirtyLabel(file))}</small>
7305:                 </button>
7306:               `).join("")} || `<p>No source files yet.</p>`}
7307:             </div>
7308:           </details>
7309:         </div>
7310:         <div class="compile-code-actions">
7311:           <button type="button" data-compile-add>Add code file</button>
7312:           <button type="button" data-compile-add-testbench>Add testbench</button>
7313:           <button type="button" data-compile-import>Import file</button>
7314:           <button type="button" data-compile-tools>Check compilers</button>
7315:         </div>
7316:       </aside>
7317:       <section class="compile-code-main" aria-label="Compile code editor">
7318:         ${activeFile ?
7319:           <div class="compile-code-meta-grid">
7320:             <label>
7321:               <span>Title</span>
7322:               <input type="text" data-compile-field="title" value="${escapeHtml(activeFile.title || "")}"
placeholder="Source title" />
7323:             </label>
7324:           </div>

```

```

7325:         <label>
7326:             <span>File name</span>
7327:             <input type="text" data-compile-field="fileName" value="{escapeHtml(activeFile.fileName ||
7328:             ""}" placeholder="{escapeHtml(defaultCodeFileName(activeFile.language))}" />
7329:         </label>
7330:         <label>
7331:             <span>Language</span>
7332:             <select data-compile-
7333:             field="language">${compileLanguageOptions(activeFile.language)}</select>
7334:         </label>
7335:         <label>
7336:             <span>${isHdlLanguage(activeFile.language) ? "HDL role" : "Source role"}</span>
7337:             ${isHdlLanguage(activeFile.language)
7338:             ? `<select data-compile-field="role">${compileRoleOptions(activeFile.role)}</select>`
7339:             : `<input type="text" value="Program source" disabled />`}
7340:         </label>
7341:     </div>
7342:     <div class="compile-code-editor-grid compile-code-editor-grid-single">
7343:         <section class="compile-code-preview-panel compile-code-active-panel" aria-label="Active
7344:         syntax highlighted code editor">
7345:             <div class="compile-code-preview-heading">
7346:                 <span>${escapeHtml(codeLanguageLabel(activeFile.language))} active editor</span>
7347:             </div>
7348:             <div class="compile-code-active-editor">
7349:                 <pre class="compile-code-preview" aria-hidden="true"><code data-compile-
7350:                 preview=${tokenizedCodeHtml(activeFile.code || "", activeFile.language)}</code></pre>
7351:                 <textarea data-compile-field="code" data-compile-active-editor spellcheck="false"
7352:                 wrap="off" rows="18" placeholder="Type or paste code here">${escapeHtml(activeFile.code || "")}</textarea>
7353:             </div>
7354:         </section>
7355:     </div>
7356:     <label class="compile-stdin-field">
7357:         <span>Program input, optional</span>
7358:         <textarea data-compile-field="stdin" rows="4" placeholder="Input passed to stdin when the
7359:         code runs">${escapeHtml(activeFile.stdin || "")}</textarea>
7360:     </label>
7361:     <div class="compile-code-command-bar">
7362:         <button type="button" data-compile-save>Save source</button>
7363:         <button type="button" data-compile-beautify>Beautify</button>
7364:         <button type="button" data-compile-compile>Compile</button>
7365:         <button type="button" data-compile-build>Build project</button>
7366:         ${isHdlLanguage(activeFile.language)
7367:         ? `<button type="button" data-compile-simulate>Simulate</button>`
7368:         : `<button type="button" data-compile-run>Run</button>`}
7369:         <button class="compile-output-button" type="button" data-compile-show-output title="Open the
7370:         terminal output for this source"><span class="compile-command-icon" aria-hidden="true">&gt;</span><span>Show
7371:         output</span></button>
7372:         ${isHdlLanguage(activeFile.language) ? `<button class="compile-output-button" type="button"
7373:         data-compile-show-scope title="Open the HDL waveform scope"><span class="compile-command-icon" aria-
7374:         hidden="true"></span><span>Show scope</span></button>` : ""}
7375:         <button type="button" data-compile-install>Install tools</button>
7376:         <button class="danger-icon" type="button" data-compile-delete>Delete source</button>
7377:     </div>
7378:     <section class="compile-append-panel" aria-label="Append code to project section">
7379:         <div>
7380:             <strong>Append code to project</strong>
7381:             <span>Insert the active source as a formatted code block into this project.</span>
7382:         </div>
7383:         <label>
7384:             <span>Destination</span>
7385:             <select data-compile-append-target>${compileAppendDestinationOptions(project,
7386:             workspace.appendTargetId)}</select>
7387:         </label>
7388:         <div class="compile-append-actions">
7389:             <button type="button" data-compile-append-code>Append code to project</button>
7390:         </div>
7391:     </section>
7392:     :
7393:     <div class="compile-code-empty">
7394:         <h3>No compile file selected</h3>
7395:         <p>Add a source file for C, C++, SystemVerilog, LTspice, Java, JavaScript, Python, or
7396:         HTML.</p>
7397:     </div>
7398: }
7399: <section class="compile-terminal-panel" data-compile-terminal-panel aria-label="Compiler terminal
7400: output">
7401:     <div class="compile-terminal-heading">
7402:         <span class="compile-terminal-title"><span aria-hidden="true" class="terminal-dot"></span></span>
7403:         OMB Local Terminal</span>
7404:         <span class="compile-terminal-controls">
7405:             <button type="button" data-compile-copy-output>Copy output</button>
7406:             <button type="button" data-compile-clear-output>Clear output</button>
7407:         </span>
7408:     </div>
7409:     <pre class="compile-terminal" data-compile-terminal tabindex="0" role="log" aria-
7410:     live="polite">${escapeHtml(terminal)}</pre>
7411: </section>
7412: ${renderCompileScopePanel(project, activeFile)}
7413: <section class="compile-messages-panel" aria-label="Compile messages log">
7414:     <div class="compile-terminal-heading">
7415:         <span>Messages log</span>
7416:         <button type="button" data-compile-clear-messages>Clear log</button>

```

```

7402:         </div>
7403:         <div class="compile-messages-log" data-compile-
messages>${renderCompileMessages(workspace)}</div>
7404:     </section>
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

## renderSectionContent - lines 7410-7472

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7410-7472 (63 source lines).                                                                                                                                                                    |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderSectionContent(project) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | sectionOptions, find, startsWith, renderCustomSection, replace, requestAnimationFrame, populateRichEditors, isAnalogProject, renderAnalogDesignWorkspace, renderObjectSection, filter, join           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7410: function renderSectionContent(project) {
7411:   if (!project) {
7412:     sectionContent.hidden = true;
7413:     sectionContent.innerHTML = "";
7414:     return;
7415:   }
7416:
7417:   const section = sectionOptions(project).find((item) => item.id === activeSectionId) ||
standardSections[0];
7418:   const isOverview = section.id === "brief";
7419:   projectFields.hidden = !isOverview;
7420:   sectionContent.hidden = isOverview;
7421:   if (isOverview) {
7422:     sectionContent.innerHTML = "";
7423:     return;
7424:   }
7425:
7426:   if (section.id.startsWith("custom:")) {
7427:     sectionContent.innerHTML = renderCustomSection(project, section.id.replace("custom:", ""));
7428:     requestAnimationFrame(() => populateRichEditors(sectionContent));
7429:     return;
7430:   }
7431:
7432:   const renderers = {
7433:     brief: () => "",
7434:     design: () => {
7435:       if (isAnalogProject(project)) return renderAnalogDesignWorkspace(project);
7436:       const cards = [
7437:         (project.documents || []).length ? `<section><h3>Documents</h3>${renderObjectSection(project,
"documents", "documents", "Documents")}</section>` : "",
7438:         (project.pcbS || []).length ? `<section><h3>PCBs Built</h3>${renderObjectSection(project, "pcbS",
"pcbS", "PCBs")}</section>` : "",
7439:         (project.media || []).length ? `<section><h3>Images</h3>${renderObjectSection(project, "media",
"images", "Images")}</section>` : ""
7440:       ].filter(Boolean);
7441:       return `
7442:         <div class="section-stack">
7443:           ${cards.join("")} || renderEmptyDynamicSectionPrompt("design")}
7444:           ${renderPendingEditor()}
7445:         </div>
7446:       `;
7447:     },
7448:     simulation: () => isAnalogProject(project) ? renderAnalogSimulationWorkspace(project) : "",
7449:     "compile-code": () => renderCompileCodeSection(project),
7450:     tests: () => `${renderObjectSection(project, "tests", "tests", "Tests")}${renderPendingEditor()}`,
7451:     tools: () => `
7452:       <div class="section-stack">
7453:         <section>
7454:           <h3>Tools Used</h3>
7455:           ${renderToolsSection(project)}
7456:         </section>
7457:         <section>
7458:           <h3>Languages</h3>
7459:           ${renderTextArray(project, "languages", "Language")}
7460:         </section>
7461:       </div>

```

```

7462:         <h3>Links</h3>
7463:         ${renderObjectSection(project, "links", "", "Links")}
7464:     </section>
7465:     ${renderPendingEditor()}
7466: </div>
7467:
7468: };
7469:
7470: sectionContent.innerHTML = renderers[section.id]();
7471: requestAnimationFrame(() => populateRichEditors(sectionContent));
7472: }

```

## ***fullPortfolioPreviewHtml - lines 7474-7476***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7474-7476 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function fullPortfolioPreviewHtml() {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | fullPortfolioPreviewHtmlExact                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7474: function fullPortfolioPreviewHtml() {
7475:     return fullPortfolioPreviewHtmlExact();
7476: }

```

## ***previewValue - lines 7478-7480***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7478-7480 (3 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function previewValue(source, key, fallback) {                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | call                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7478: function previewValue(source, key, fallback) {
7479:     return source && Object.prototype.hasOwnProperty.call(source, key) ? source[key] : fallback;
7480: }

```

## ***fullPortfolioPreviewHtmlExact - lines 7482-7702***

| Item            | Explanation                                                    |
|-----------------|----------------------------------------------------------------|
| Source location | Lines 7482-7702 (221 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience.   |
| Declaration     | function fullPortfolioPreviewHtmlExact(dataOverride = null) {  |
| Touches         | filesystem, network/API requests, authentication/security data |

| Item               | Explanation                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | <code>parsedPortfolioGlobals</code> , <code>normalizeSiteContent</code> , <code>previewValue</code> , <code>normalizeProfile</code> , <code>normalizeFieldStyles</code> , <code>stringify</code> , <code>replaceAll</code> , <code>escapeHtml</code> , <code>plainFieldStyleAttribute</code> , <code>renderRichFieldContent</code> , <code>getFullYear</code> |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                                                                                                                                      |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                                                                                                                                       |
| Async/return clues | 0 <code>await</code> expression(s), 1 <code>return</code> statement(s).                                                                                                                                                                                                                                                                                       |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                                                                                                                                         |

Source excerpt:

```

7482: function fullPortfolioPreviewHtmlExact(dataOverride = null) {
7483:   const baseHref = `${window.location.origin}/`;
7484:   const parsedGlobals = dataOverride ? null : parsedPortfolioGlobals();
7485:   const siteContent = normalizeSiteContent(previewValue(dataOverride, "siteContent",
parsedGlobals?.siteContent || {}));
7486:   const profile = normalizeProfile(previewValue(dataOverride, "profile", parsedGlobals?.profile || {}));
7487:   const fieldStyles = normalizeFieldStyles(previewValue(dataOverride, "fieldStyles",
parsedGlobals?.fieldStyles || {}));
7488:   const profileRich = previewValue(dataOverride, "profileRich", parsedGlobals?.profileRich || {});
7489:   const siteContentRich = previewValue(dataOverride, "siteContentRich", parsedGlobals?.siteContentRich ||
{});
7490:   const ownerName = profile.displayName || "Portfolio";
7491:   const portfolioLabel = profile.portfolioLabel || "Portfolio";
7492:   const previewDataObject = {
7493:     categories: previewValue(dataOverride, "categories", parsedGlobals?.categories || []),
7494:     fieldStyles,
7495:     funFacts: previewValue(dataOverride, "funFacts", parsedGlobals?.funFacts || []),
7496:     funFactsRich: previewValue(dataOverride, "funFactsRich", parsedGlobals?.funFactsRich || null),
7497:     profile,
7498:     profileRich,
7499:     projects: previewValue(dataOverride, "projects", savedPortfolioCatalog.projects || []),
7500:     siteContent,
7501:     siteContentRich,
7502:     siteSections: previewValue(dataOverride, "siteSections", parsedGlobals?.siteSections || [])
7503:   };
7504:   const previewData = JSON.stringify(previewDataObject).replaceAll("</", "<\\");
7505:
7506:   return `<!DOCTYPE html>
7507: <html lang="en">
7508:   <head>
7509:     <meta charset="UTF-8" />
7510:     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7511:     <base href="${baseHref}" />
7512:     <meta name="description" content="A portfolio website built with OMB Portfolio Builder for projects,
documents, diagrams, source code, tests, links, and profile information." />
7513:     <meta name="robots" content="index, follow" />
7514:     <meta name="author" content="${escapeHtml(ownerName)}" />
7515:     <meta name="portfolio-ai-endpoint" content="https://omb-portfolio-ai.maurice-baraza-
otieno.workers.dev/api/portfolio-ai" />
7516:     <title>${escapeHtml(ownerName)} | ${escapeHtml(portfolioLabel)}</title>
7517:     <link rel="stylesheet" href="styles.css" />
7518:   </head>
7519:   <body>
7520:     <header class="site-header" id="top">
7521:       <a class="brand" href="#top" aria-label="Go to top">
7522:         <span class="brand-lockup" aria-hidden="true">
7523:           <img class="brand-icon" alt="" hidden />
7524:           <span class="omb-engraving">${escapeHtml(profile.brandText || "Portfolio")}</span>
7525:         </span>
7526:         <span>
7527:           <strong>${plainFieldStyleAttribute(fieldStyles, "profile-display-
name")}>${renderRichFieldContent(profileRich.displayName, ownerName)}</strong>
7528:           <small>${plainFieldStyleAttribute(fieldStyles, "profile-portfolio-
label")}>${renderRichFieldContent(profileRich.portfolioLabel, portfolioLabel)}</small>
7529:         </span>
7530:       </a>
7531:
7532:       <div class="header-right">
7533:         <nav class="site-nav" aria-label="Primary navigation">
7534:           <a href="#ask-ai">Ask AI</a>
7535:           <a href="#projects">Projects</a>
7536:           <a href="#resume">Resume</a>
7537:           <a href="#process">Process</a>
7538:           <a href="#contact">Contact</a>
7539:         </nav>
7540:         <a class="header-avatar" href="#contact" aria-label="Go to contact information" hidden>
7541:           <img alt="" />
7542:         </a>
7543:       </div>
7544:     </header>
7545:
7546:     <main>
7547:       <section class="hero" aria-labelledby="hero-title">
7548:         <img class="hero-image" alt="" hidden />

```

```

7549:         <div class="hero-shade" aria-hidden="true"></div>
7550:         <div class="hero-content">
7551:             <p class="eyebrow"${plainFieldStyleAttribute(fieldStyles, "site-hero-eyebrow")}>${renderRichFieldContent(siteContentRich.heroEyebrow, siteContent.heroEyebrow)}</p>
7552:             <h1 id="hero-title"${plainFieldStyleAttribute(fieldStyles, "site-hero-title")}>${renderRichFieldContent(siteContentRich.heroTitle, siteContent.heroTitle)}</h1>
7553:             <p class="hero-copy"${plainFieldStyleAttribute(fieldStyles, "site-hero-copy")}>${renderRichFieldContent(siteContentRich.heroCopy, siteContent.heroCopy)}</p>
7554:             <div class="hero-actions">
7555:                 <a class="button primary" href="#projects">View projects</a>
7556:                 <a class="button secondary" href="#resume" hidden>View resume</a>
7557:                 <a class="button secondary" href="#" target="_blank" rel="noreferrer" hidden>GitHub
profile</a>
7558:                 <a class="ai-jump-link" href="#ask-ai" aria-label="Jump to the portfolio AI assistant">
7559:                     <svg viewBox="0 0 24 24" aria-hidden="true">
7560:                         <path d="M12 3v3m-5 7H4m16 0h-3M7.5 7.5 5.4 5.4m11.1 2.1 2.1-2.1M8 10h8v7a2 2 0 0 1-2
2h-4a2 2 0 0 1-2-2z" />
7561:                         <path d="M10 13h.01M14 13h.01M10 16h4" />
7562:                     </svg>
7563:                     <span>Ask AI</span>
7564:                 </a>
7565:             </div>
7566:         </div>
7567:     </section>
7568:
7569:     <section class="fun-facts-section" id="fun-facts-callout" aria-label="Fun facts" hidden></section>
7570:
7571:     <section class="builder-download-section" aria-label="Download portfolio builder">
7572:         <a class="builder-download-link" href="https://github.com/otienomaurice/omb-portfolio-builder/releases/latest/download/OMB-Portfolio-Builder-Setup-latest-x64.exe" download>
7573:             <svg viewBox="0 0 24 24" aria-hidden="true">
7574:                 <path d="M12 3v11m0 0 4-4m-4 4-4-4" />
7575:                 <path d="M5 17v2h14v-2" />
7576:             </svg>
7577:             <span>Download builder application</span>
7578:         </a>
7579:     </section>
7580:
7581:     <section class="summary-band" aria-label="Portfolio highlights">
7582:         <div class="metric">
7583:             <strong id="project-count">0</strong>
7584:             <span>presented projects</span>
7585:         </div>
7586:         <div class="metric">
7587:             <strong id="project-track-count">0 Tracks</strong>
7588:             <span id="project-track-labels">project categories</span>
7589:         </div>
7590:         <div class="metric">
7591:             <strong>Artifacts</strong>
7592:             <span>source, diagrams, and reports</span>
7593:         </div>
7594:     </section>
7595:
7596:     <section class="section" id="projects" aria-labelledby="projects-title">
7597:         <div class="section-heading">
7598:             <div>
7599:                 <p class="eyebrow">Selected work</p>
7600:                 <h2 id="projects-title">Engineering Projects</h2>
7601:             </div>
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

## renderPreview - lines 7704-7706

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7704-7706 (3 source lines).                                                                                                                                                                     |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                |
| Declaration        | function renderPreview() {                                                                                                                                                                            |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | fullPortfolioPreviewHtmlExact                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7704: function renderPreview() {
7705:     portfolioPreviewFrame.srdoc = fullPortfolioPreviewHtmlExact();

```

7706: }

## ***openPortfolioPreview - lines 7708-7713***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7708-7713 (6 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | async function openPortfolioPreview() {                                                                                                                                                               |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | saveAllSections, renderPreview, add, showModal                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7708: async function openPortfolioPreview() {
7709:   if (!(await saveAllSections({ refreshOpenPreviews: false }))) return;
7710:   renderPreview();
7711:   document.body.classList.add("full-window-open");
7712:   portfolioPreviewDialog.showModal();
7713: }
```

## ***renderAll - lines 7715-7732***

| Item               | Explanation                                                                                                                                                                                                             |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7715-7732 (18 source lines).                                                                                                                                                                                      |
| Purpose            | Renders UI, markup, or display output.                                                                                                                                                                                  |
| Declaration        | function renderAll() {                                                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                                            |
| Calls detected     | ensureSiteSections, selectedProject, sectionOptions, some, renderFunFactsEditor, renderSiteContentEditor, renderProfileEditor, renderSiteSectionList, renderTree, renderFields, renderSectionTabs, renderSectionContent |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                 |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                                           |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                   |

Source excerpt:

```
7715: function renderAll() {
7716:   ensureSiteSections();
7717:   if (!selectedProjectId && catalog.projects.length) selectedProjectId = catalog.projects[0].id;
7718:   const project = selectedProject();
7719:   if (project && !sectionOptions(project).some((section) => section.id === activeSectionId)) {
7720:     activeSectionId = "brief";
7721:   }
7722:   renderFunFactsEditor();
7723:   renderSiteContentEditor();
7724:   renderProfileEditor();
7725:   renderSiteSectionList();
7726:   renderTree();
7727:   renderFields(project);
7728:   renderSectionTabs(project);
7729:   renderSectionContent(project);
7730:   if (portfolioPreviewDialog?.open) renderPreview();
7731:   updateBuilderWorkflow();
7732: }
```

## ***updateProjectField - lines 7734-7768***



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7734-7768 (35 source lines).                                                                                                                                                                    |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateProjectField(field, value) {                                                                                                                                                           |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | selectedProject, split, map, trim, filter, limitWords, slugify, includes, setStatus, scheduleAutosave, scheduleChromeRender, schedulePreviewRender                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7734: function updateProjectField(field, value) {
7735:   const project = selectedProject();
7736:   if (!project) return;
7737:
7738:   let needsChromeRender = false;
7739:   let statusMessage = "Unsaved local changes.";
7740:   if (field === "focus") {
7741:     project.focus = value.split(",").map((item) => item.trim()).filter(Boolean);
7742:   } else if (field === "summary") {
7743:     const limited = limitWords(value, 1000);
7744:     project.summary = limited;
7745:     if (limited !== value && document.activeElement?.dataset?.field === "summary") {
7746:       document.activeElement.value = limited;
7747:       statusMessage = "Project overview is limited to 1000 words.";
7748:     }
7749:   } else if (field === "id") {
7750:     const nextId = slugify(value);
7751:     if (!nextId) return;
7752:     project.id = nextId;
7753:     selectedProjectId = nextId;
7754:     needsChromeRender = true;
7755:   } else {
7756:     project[field] = value;
7757:     needsChromeRender = ["title", "category", "status"].includes(field);
7758:   }
7759:
7760:   if (field === "title") projectWindowTitle.textContent = value || "Untitled project";
7761:   setStatus(statusMessage);
7762:   scheduleAutosave();
7763:   if (needsChromeRender) {
7764:     scheduleChromeRender();
7765:   } else {
7766:     schedulePreviewRender();
7767:   }
7768: }

```

## ***renameSelectedProject - lines 7770-7775***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7770-7775 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function renameSelectedProject() {                                                                                                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, beginTitleRename                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7770: function renameSelectedProject() {
7771:   const project = selectedProject();
7772:   if (!project) return;
7773:   projectTitleMenu.hidden = true;
7774:   beginTitleRename();
7775: }

```

## projectMetaRows - lines 7777-7813

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7777-7813 (37 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function projectMetaRows(project) {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | categoryById, projectTemplateFor, find, flatMap, filter, toLocaleString                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7777: function projectMetaRows(project) {
7778:   if (!project) return [];
7779:   const category = categoryById(project.category);
7780:   const template = projectTemplateFor(project);
7781:   const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project.id);
7782:   const design = project.design || {};
7783:   const designFileCount = [
7784:     ...(design.brief?.files || []),
7785:     ...(design.documentation?.files || []),
7786:     ...(design.documentation?.references || []),
7787:     ...(design.documentation?.mathAnalysis || []),
7788:     ...(design.simulation?.files || []),
7789:     ...(design.simulation?.results || [])
7790:   ].length;
7791:   const uploadedFileCount = [
7792:     ...(project.documents || []),
7793:     ...(project.tests || []),
7794:     ...(project.pcbns || []),
7795:     ...(project.media || []),
7796:     ...(project.links || []),
7797:     ...(project.sections || []).flatMap((section) => section.items || [])
7798:   ].filter((item) => item.url || item.artifact).length + designFileCount;
7799:   return [
7800:     ["Title", project.title || "Untitled project"],
7801:     ["Project ID / folder", project.id || "Not assigned"],
7802:     ["Category", category.label || project.category || "Not assigned"],
7803:     ["Status", project.status || "Draft"],
7804:     ["Appearance", template.label || project.templateLabel || "No appearance - white project layout"],
7805:     ["Appearance ID", project.templateId || "None"],
7806:     ["Portfolio preview", savedProject ? "Saved into parsed portfolio preview" : "Not yet saved into portfolio preview"],
7807:     ["Custom sections", String((project.sections || []).length)],
7808:     ["Project files / links", String(uploadedFileCount)],
7809:     ["Tools", String((project.tools || []).length)],
7810:     ["Languages", String((project.languages || []).length)],
7811:     ["Last parsed", savedProject?.portfolioView?.builtAt ? new
Date(savedProject.portfolioView.builtAt).toLocaleString() : "Not parsed yet"]
7812:   ];
7813: }

```

## savedProject - lines 7781-7782

| Item            | Explanation                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------------------|
| Source location | Lines 7781-7782 (2 source lines).                                                                   |
| Purpose         | Persists data to local state, disk, credentials, or browser storage.                                |
| Declaration     | const savedProject = (savedPortfolioCatalog.projects    []).find((item) => item.id === project.id); |
| Touches         | Mostly local calculations or local UI state.                                                        |
| Calls detected  | find                                                                                                |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7781: const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project.id);
7782: const design = project.design || {};
```

## ***openProjectMetaDetails - lines 7815-7827***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7815-7827 (13 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function openProjectMetaDetails() {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, projectMetaRows, map, join, showModal                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7815: function openProjectMetaDetails() {
7816:   const project = selectedProject();
7817:   if (!project) return;
7818:   projectTitleMenu.hidden = true;
7819:   projectMetaTitle.textContent = project.title || "Project details";
7820:   projectMetaGrid.innerHTML = projectMetaRows(project).map(([label, value]) => `
7821:     <article>
7822:       <span>${label}</span>
7823:       <strong>${value}</strong>
7824:     </article>
7825: `).join("");
7826:   if (!projectMetaDialog.open) projectMetaDialog.showModal();
7827: }
```

## ***selectTitleText - lines 7829-7835***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7829-7835 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function selectTitleText() {                                                                                                                                                                          |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | getSelection, createRange, selectNodeContents, removeAllRanges, addRange                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7829: function selectTitleText() {
7830:   const selection = window.getSelection();
```

```

7831:   const range = document.createRange();
7832:   range.selectNodeContents(projectWindowTitle);
7833:   selection.removeAllRanges();
7834:   selection.addRange(range);
7835: }

```

## ***beginTitleRename - lines 7837-7846***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7837-7846 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function beginTitleRename() {                                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, focus, selectTitleText, setStatus                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7837: function beginTitleRename() {
7838:   const project = selectedProject();
7839:   if (!project) return;
7840:   originalTitleDraft = project.title || "";
7841:   projectWindowTitle.contentEditable = "true";
7842:   projectWindowTitle.focus();
7843:   selectTitleText();
7844:   titleRenameActions.hidden = false;
7845:   setStatus("Edit the title, then click Save title.");
7846: }

```

## ***endTitleRename - lines 7848-7865***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7848-7865 (18 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function endTitleRename(saveChanges) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, trim, getSelection, removeAllRanges, setStatus, scheduleAutosave, scheduleChromeRender                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7848: function endTitleRename(saveChanges) {
7849:   const project = selectedProject();
7850:   if (!project) return;
7851:   const nextTitle = projectWindowTitle.textContent.trim();
7852:   projectWindowTitle.contentEditable = "false";
7853:   titleRenameActions.hidden = true;
7854:   window.getSelection()?.removeAllRanges();
7855:   if (saveChanges && nextTitle) {
7856:     project.title = nextTitle;
7857:     projectWindowTitle.textContent = nextTitle;
7858:     setStatus("Project title saved in the editor. Click Save project to include this version in the
portfolio preview.");
7859:     scheduleAutosave();
7860:     scheduleChromeRender();
7861:   } else {
7862:     projectWindowTitle.textContent = originalTitleDraft || project.title || "Untitled project";
7863:     setStatus("Title edit canceled.");

```

```

7864:   }
7865: }

```

## ***makeItemForSection - lines 7867-7874***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7867-7874 (8 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function makeItemForSection(key, title, description = "", url = "") {                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7867: function makeItemForSection(key, title, description = "", url = "") {
7868:   if (key === "documents") return { title, type: "Document", url, status: url ? "uploaded" : "draft" };
7869:   if (key === "tests") return { name: title, method: description || "Validation notes", status: url ?
"uploaded" : "draft", artifact: url, result: description };
7870:   if (key === "pcbs") return { name: title, revision: "Rev A", status: url ? "uploaded" : "draft",
artifact: url };
7871:   if (key === "media") return { title, url, status: url ? "uploaded" : "draft", caption: description };
7872:   if (key === "links") return { label: title, url };
7873:   return { title, description, url, status: url ? "uploaded" : "draft" };
7874: }

```

## ***addTextItem - lines 7876-7891***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7876-7891 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function addTextItem(key) {                                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, prompt, includes, push, makeItemForSection, setStatus, scheduleAutosave, renderAll                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7876: function addTextItem(key) {
7877:   const project = selectedProject();
7878:   const title = prompt("Type a title for this item.");
7879:   if (!project || !title) return;
7880:
7881:   project[key] = project[key] || [];
7882:   if (["highlights", "tools", "languages"].includes(key)) {
7883:     project[key].push(title);
7884:   } else {
7885:     const description = prompt("Add a short description if you want.") || "";
7886:     project[key].push(makeItemForSection(key, title, description));
7887:   }
7888:   setStatus("Item added in the editor. Click Save project to include this version in the portfolio
preview.");
7889:   scheduleAutosave();
7890:   renderAll();
7891: }

```

## ***editObjectItem - lines 7893-7914***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7893-7914 (22 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function editObjectItem(key, index) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, prompt, setStatus, scheduleAutosave, renderAll                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7893: function editObjectItem(key, index) {
7894:   const project = selectedProject();
7895:   const item = project?.[key]?.[index];
7896:   if (!item) return;
7897:
7898:   const currentTitle = item.title || item.name || item.label || "";
7899:   const nextTitle = prompt("Edit title.", currentTitle);
7900:   if (!nextTitle) return;
7901:   const nextDescription = prompt("Edit description.", item.description || item.caption || item.result ||
item.method || "" || "");
7902:
7903:   if ("title" in item) item.title = nextTitle;
7904:   if ("name" in item) item.name = nextTitle;
7905:   if ("label" in item) item.label = nextTitle;
7906:   if ("caption" in item) item.caption = nextDescription;
7907:   if ("result" in item) item.result = nextDescription;
7908:   if ("description" in item) item.description = nextDescription;
7909:   if (!("caption" in item) && !("result" in item) && !("description" in item)) item.description =
nextDescription;
7910:
7911:   setStatus("Item edited in the editor. Click Save project to include this version in the portfolio
preview.");
7912:   scheduleAutosave();
7913:   renderAll();
7914: }
```

## ***chooseFile - lines 7916-7922***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7916-7922 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function chooseFile() {                                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | resolve, click                                                                                                                                                                                        |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7916: function chooseFile() {
7917:   return new Promise((resolve) => {
7918:     filePicker.value = "";
7919:     filePicker.onchange = () => resolve(filePicker.files[0] || null);
7920:     filePicker.click();
7921:   });
7922: }
```

## ***newCompileFile - lines 7924-7942***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7924-7942 (19 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function newCompileFile(language = "javascript", seed = {}) {                                                                                                                                         |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | normalizeCodeLanguage, safeClientCodeFileName, defaultCodeFileName, normalizeCompileFileRole, inferCompileFileRole, slugify, now, random, toString, slice                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7924: function newCompileFile(language = "javascript", seed = {}) {
7925:   const normalized = normalizeCodeLanguage(language);
7926:   const fileName = safeClientCodeFileName(seed.fileName || defaultCodeFileName(normalized), normalized);
7927:   const role = normalizeCompileFileRole(seed.role || inferCompileFileRole(fileName, seed.code || "",
normalized), normalized);
7928:   return {
7929:     id: slugify(`${fileName}-${Date.now()}-${Math.random().toString(16).slice(2)}`),
7930:     title: seed.title || fileName,
7931:     fileName,
7932:     language: normalized,
7933:     role,
7934:     code: String(seed.code || ""),
7935:     stdin: "",
7936:     savedPath: "",
7937:     savedAt: "",
7938:     waveform: null,
7939:     lastResult: null,
7940:     dirty: true
7941:   };
7942: }
```

## ***activeCompileWorkspaceAndFile - lines 7944-7947***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7944-7947 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function activeCompileWorkspaceAndFile(project = selectedProject()) {                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, ensureCompileCode, activeCompileFile                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7944: function activeCompileWorkspaceAndFile(project = selectedProject()) {
7945:   const workspace = ensureCompileCode(project);
7946:   return { workspace, file: activeCompileFile(project) };
7947: }
```

## ***updateCompilePreview - lines 7949-7952***

| Item            | Explanation                       |
|-----------------|-----------------------------------|
| Source location | Lines 7949-7952 (4 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateCompilePreview(file) {                                                                                                                                                                 |
| Touches            | browser/Electron DOM, authentication/security data                                                                                                                                                    |
| Calls detected     | querySelector, tokenizedCodeHtml                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7949: function updateCompilePreview(file) {
7950:   const preview = sectionContent.querySelector("[data-compile-preview]");
7951:   if (preview && file) preview.innerHTML = tokenizedCodeHtml(file.code || "", file.language);
7952: }

```

## ***syncCompileEditorScroll - lines 7954-7960***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7954-7960 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function syncCompileEditorScroll(textarea) {                                                                                                                                                          |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | closest, querySelector                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

7954: function syncCompileEditorScroll(textarea) {
7955:   const editor = textarea?.closest(".compile-code-active-editor");
7956:   const preview = editor?.querySelector(".compile-code-preview");
7957:   if (!preview || !textarea) return;
7958:   preview.scrollTop = textarea.scrollTop;
7959:   preview.scrollLeft = textarea.scrollLeft;
7960: }

```

## ***compilePayload - lines 7962-7984***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7962-7984 (23 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function compilePayload(project, file, options = {}) {                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureCompileCode, inferCompileFileRole, map                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |



Source excerpt:

```
7962: function compilePayload(project, file, options = {}) {
7963:   const workspace = ensureCompileCode(project);
7964:   return {
7965:     projectId: project.id,
7966:     fileId: file.id,
7967:     title: file.title,
7968:     fileName: file.fileName,
7969:     language: file.language,
7970:     role: file.role || inferCompileFileRole(file.fileName, file.code, file.language),
7971:     code: file.code,
7972:     stdin: file.stdin || "",
7973:     workspaceFiles: (workspace.files || []).map((workspaceFile) => ({
7974:       id: workspaceFile.id,
7975:       title: workspaceFile.title,
7976:       fileName: workspaceFile.fileName,
7977:       language: workspaceFile.language,
7978:       role: workspaceFile.role || inferCompileFileRole(workspaceFile.fileName, workspaceFile.code,
workspaceFile.language),
7979:       code: workspaceFile.code || ""
7980:     })),
7981:     action: options.action || "",
7982:     forceRebuild: options.forceRebuild === true
7983:   };
7984: }
```

### ***selectedCompileAppendDestination - lines 7986-7990***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7986-7990 (5 source lines).                                                                                                                                                                     |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | function selectedCompileAppendDestination(project) {                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureCompileCode, compileAppendDestinations, find                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
7986: function selectedCompileAppendDestination(project) {
7987:   const workspace = ensureCompileCode(project);
7988:   const destinations = compileAppendDestinations(project);
7989:   return destinations.find((destination) => destination.id === workspace.appendTargetId) ||
destinations[0] || null;
7990: }
```

### ***appendCompileCodeToProject - lines 7992-8019***

| Item               | Explanation                                                                                                                                                                                                         |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 7992-8019 (28 source lines).                                                                                                                                                                                  |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                              |
| Declaration        | function appendCompileCodeToProject(project, file) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                                        |
| Calls detected     | codeBlockForCompileFile, trim, setStatus, addCompileMessage, selectedCompileAppendDestination, richBlocksForCompileAppend, push, setByPath, plainTextFromRich, toISOString, scheduleAutosave, schedulePreviewRender |
| API endpoints      | None detected inside this function body.                                                                                                                                                                            |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                             |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                                       |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.               |

Source excerpt:

```
7992: function appendCompileCodeToProject(project, file) {
7993:   const block = codeBlockForCompileFile(file);
7994:   if (!block.code.trim()) {
7995:     setStatus("Type or import code before appending it to the project.");
7996:     addCompileMessage(project, "Append stopped because the active source is empty.", "warning");
7997:     return false;
7998:   }
7999:
8000:   const destination = selectedCompileAppendDestination(project);
8001:   if (!destination) {
8002:     setStatus("No project destination is available for this code block.");
8003:     addCompileMessage(project, "Append stopped because no destination section is available.", "warning");
8004:     return false;
8005:   }
8006:
8007:   const blocks = richBlocksForCompileAppend(project, destination);
8008:   blocks.push(block);
8009:   setByPath(project, destination.richPath, { blocks });
8010:   setByPath(project, destination.textPath, plainTextFromRich({ blocks }));
8011:
8012:   file.lastAppendedAt = new Date().toISOString();
8013:   addCompileMessage(project, `${file.fileName || file.title || "Source code"} appended to
${destination.label}.`, "success");
8014:   setStatus(`Code appended to ${destination.label}. Save project to include it in the portfolio
preview.`);
8015:   scheduleAutosave();
8016:   schedulePreviewRender();
8017:   updateCompileMessagesPanel(project);
8018:   return true;
8019: }
```

## ***saveCompileFile - lines 8021-8038***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8021-8038 (18 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | async function saveCompileFile(project, file) {                                                                                                                                                       |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | fetch, now, stringify, compilePayload, json, Error, toISOString, addCompileMessage                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
8021: async function saveCompileFile(project, file) {
8022:   if (!project || !file) return null;
8023:   const response = await fetch(`/api/code/save?t=${Date.now()}`, {
8024:     method: "POST",
8025:     cache: "no-store",
8026:     headers: { "Content-Type": "application/json" },
8027:     body: JSON.stringify(compilePayload(project, file))
8028:   });
8029:   const result = await response.json();
8030:   if (!response.ok || !result.ok) throw new Error(result.error || "Code could not be saved.");
8031:   file.savedPath = result.saved?.sourcePath || "";
8032:   file.savedAt = result.saved?.savedAt || new Date().toISOString();
8033:   file.fileName = result.saved?.fileName || file.fileName;
8034:   file.language = result.saved?.language || file.language;
8035:   file.dirty = false;
8036:   addCompileMessage(project, `Saved ${file.fileName || "source file"}.`, "info");
8037:   return result.saved;
8038: }
```

## ***importCompileFile - lines 8040-8063***

| Item            | Explanation                                            |
|-----------------|--------------------------------------------------------|
| Source location | Lines 8040-8063 (24 source lines).                     |
| Purpose         | Part of the Compile Code language/tool/build/run flow. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | async function importCompileFile(project) {                                                                                                                                                           |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureCompileCode, chooseFile, flatMap, split, pop, toLowerCase, includes, setStatus, text, detectCodeLanguage, codeLanguageProfile, newCompileFile                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8040: async function importCompileFile(project) {
8041:   const workspace = ensureCompileCode(project);
8042:   const file = await chooseFile();
8043:   if (!file) return;
8044:   const allowedExtensions = supportedCodeLanguages.flatMap((item) => item.extensions || []);
8045:   const extension = `${String(file.name || "").split(".").pop().toLowerCase()}`;
8046:   if (!allowedExtensions.includes(extension)) {
8047:     setStatus("Choose a supported source file: C, C++, Verilog, SystemVerilog, LTspice, Java, JavaScript,
Python, or HTML.");
8048:     return;
8049:   }
8050:   const code = await file.text();
8051:   const language = detectCodeLanguage(code, file.name);
8052:   const profile = codeLanguageProfile(language);
8053:   const sourceFile = newCompileFile(profile.id, {
8054:     fileName: file.name,
8055:     title: file.name.replace(/\.([^.]+)$/, ""),
8056:     code
8057:   });
8058:   workspace.files.push(sourceFile);
8059:   workspace.activeFileId = sourceFile.id;
8060:   setStatus(`Imported ${file.name}. Click Save source before compiling.`);
8061:   scheduleAutosave();
8062:   renderSectionContent(project);
8063: }

```

## ***beautifyCompileFile - lines 8065-8085***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8065-8085 (21 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | async function beautifyCompileFile(project, file) {                                                                                                                                                   |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | fetch, now, stringify, compilePayload, json, Error, setStatus, scheduleAutosave, renderSectionContent                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8065: async function beautifyCompileFile(project, file) {
8066:   if (!file) return;
8067:   try {
8068:     const response = await fetch(`/api/code/beautify?t=${Date.now()}`, {
8069:       method: "POST",
8070:       cache: "no-store",
8071:       headers: { "Content-Type": "application/json" },
8072:       body: JSON.stringify(compilePayload(project, file))
8073:     });
8074:     const result = await response.json();
8075:     if (!response.ok || !result.ok) throw new Error(result.error || "Beautifier failed.");
8076:     file.code = result.code || file.code;
8077:     file.language = result.language || file.language;
8078:     file.dirty = true;

```

```

8079:     setStatus("Code beautified. Review it, then save source.");
8080:     scheduleAutosave();
8081:     renderSectionContent(project);
8082:   } catch (error) {
8083:     setStatus(error.message || "Code beautifier failed.");
8084:   }
8085: }

```

## compileActiveFile - lines 8087-8145

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8087-8145 (59 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | async function compileActiveFile(project, file, options = {}) {                                                                                                                                       |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | ensureCompileCode, isHdlLanguage, addCompileMessage, updateCompileTerminalPanel, updateCompileMessagesPanel, saveCompileFile, fetch, now, stringify, compilePayload, json, toISOString                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8087: async function compileActiveFile(project, file, options = {}) {
8088:   if (!file) return;
8089:   const workspace = ensureCompileCode(project);
8090:   const action = options.action || (isHdlLanguage(file.language) ? "simulate" : "run");
8091:   const actionLabel = action === "compile" ? "Compile" : action === "build" ? "Build project" : action
=== "simulate" ? "Simulate" : "Run";
8092:   file.lastResult = {
8093:     ok: null,
8094:     terminal: `${actionLabel} started. Please wait...`
8095:   };
8096:   workspace.terminal = file.lastResult.terminal;
8097:   addCompileMessage(project, `${actionLabel} started for ${file.fileName || file.title || "source
file"}`, "info");
8098:   updateCompileTerminalPanel(project, file);
8099:   updateCompileMessagesPanel(project);
8100:   try {
8101:     await saveCompileFile(project, file);
8102:     const response = await fetch(`/api/code/compile?t=${Date.now()}`, {
8103:       method: "POST",
8104:       cache: "no-store",
8105:       headers: { "Content-Type": "application/json" },
8106:       body: JSON.stringify(compilePayload(project, file, {
8107:         action,
8108:         forceRebuild: options.forceRebuild === true || action === "build"
8109:       })))
8110:   };
8111:   const result = await response.json();
8112:   const compileResult = result.result || {};
8113:   file.lastResult = {
8114:     ok: Boolean(result.ok),
8115:     language: compileResult.language || file.language,
8116:     terminal: compileResult.terminal || result.error || "No compiler output was returned.",
8117:     waveform: compileResult.waveform || null,
8118:     finishedAt: new Date().toISOString()
8119:   };
8120:   file.waveform = compileResult.waveform || null;
8121:   ensureCompileCode(project).waveform = compileResult.waveform || null;
8122:   if (compileResult.saved) {
8123:     file.savedPath = compileResult.saved.sourcePath || file.savedPath;
8124:     file.savedAt = compileResult.saved.savedAt || file.savedAt;
8125:   }
8126:   file.dirty = false;
8127:   ensureCompileCode(project).terminal = file.lastResult.terminal;
8128:   updateCompileTerminalPanel(project, file);
8129:   addCompileMessage(project, result.ok ? `${actionLabel} succeeded for ${file.fileName}` :
`${actionLabel} completed with errors for ${file.fileName}`, result.ok ? "success" : "error");
8130:   setStatus(result.ok ? `${actionLabel} succeeded.` : `${actionLabel} completed with errors.`);
8131:   scheduleAutosave();
8132:   renderSectionContent(project);
8133: } catch (error) {
8134:   file.lastResult = {
8135:     ok: false,
8136:     terminal: error.message || "Compile/run failed.",
8137:     finishedAt: new Date().toISOString()

```

```

8138:     };
8139:     ensureCompileCode(project).terminal = file.lastResult.terminal;
8140:     updateCompileTerminalPanel(project, file);
8141:     addCompileMessage(project, `${actionLabel} failed for ${file.fileName || "source file"}.`, "error");
8142:     setStatus(error.message || `${actionLabel} failed.`);
8143:     renderSectionContent(project);
8144:   }
8145: }

```

## checkCompileTools - lines 8147-8162

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8147-8162 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | async function checkCompileTools(project) {                                                                                                                                                           |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | fetch, now, json, Error, entries, map, join, ensureCompileCode, renderSectionContent, setStatus                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8147: async function checkCompileTools(project) {
8148:   try {
8149:     const response = await fetch(`/api/code/tools?t=${Date.now()}`, { cache: "no-store" });
8150:     const result = await response.json();
8151:     if (!response.ok || !result.ok) throw new Error(result.error || "Compiler status failed.");
8152:     const lines = Object.entries(result.tools?.languages || {}).map(([id, status]) => {
8153:       const toolText = Object.entries(status.tools || {}).map(([tool, value]) => `${tool}:
${value}`).join(", ") || "no compiler required";
8154:       return `${status.ready ? "READY" : "MISSING"} ${status.label} (${id}) - ${toolText}`;
8155:     });
8156:     compileTerminalStatus = [`Compile workspace: ${result.tools?.compileRoot || ""}`,
...lines].join("\n");
8157:     ensureCompileCode(project).terminal = compileTerminalStatus;
8158:     renderSectionContent(project);
8159:   } catch (error) {
8160:     setStatus(error.message || "Compiler status failed.");
8161:   }
8162: }

```

## installCompileTools - lines 8164-8187

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8164-8187 (24 source lines).                                                                                                                                                                    |
| Purpose            | Part of the Compile Code language/tool/build/run flow.                                                                                                                                                |
| Declaration        | async function installCompileTools(project, file) {                                                                                                                                                   |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | codeLanguageLabel, renderSectionContent, fetch, now, stringify, json, toISOString, setStatus                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8164: async function installCompileTools(project, file) {
8165:   if (!file) return;
8166:   file.lastResult = { ok: null, terminal: `Installing tools for ${codeLanguageLabel(file.language)}. This
can take several minutes...` };
8167:   renderSectionContent(project);
8168:   try {

```

```

8169:     const response = await fetch(`/api/code/install-tools?t=${Date.now()}`, {
8170:       method: "POST",
8171:       cache: "no-store",
8172:       headers: { "Content-Type": "application/json" },
8173:       body: JSON.stringify({ language: file.language })
8174:     });
8175:     const result = await response.json();
8176:     file.lastResult = {
8177:       ok: Boolean(result.ok),
8178:       terminal: result.terminal || result.error || "Tool installation finished.",
8179:       finishedAt: new Date().toISOString()
8180:     };
8181:     setStatus(result.ok ? "Compiler tool installation finished." : "Compiler tool installation could not
complete.");
8182:     renderSectionContent(project);
8183:   } catch (error) {
8184:     file.lastResult = { ok: false, terminal: error.message || "Tool installation failed." };
8185:     renderSectionContent(project);
8186:   }
8187: }

```

## ***readFileAsDataURL - lines 8189-8196***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8189-8196 (8 source lines).                                                                                                                                                                     |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | function readFileAsDataURL(file) {                                                                                                                                                                    |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | FileReader, resolve, readAsDataURL                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8189: function readFileAsDataURL(file) {
8190:   return new Promise((resolve, reject) => {
8191:     const reader = new FileReader();
8192:     reader.onload = () => resolve(reader.result);
8193:     reader.onerror = reject;
8194:     reader.readAsDataURL(file);
8195:   });
8196: }

```

## ***uploadProfileAsset - lines 8198-8242***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8198-8242 (45 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function uploadProfileAsset(kind) {                                                                                                                                                             |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | syncProfileFromInputs, chooseFile, Set, has, startsWith, setStatus, includes, extensionFor, toLowerCase, readFileAsDataURL, fetch, now                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 4 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8198: async function uploadProfileAsset(kind) {
8199:   const profile = syncProfileFromInputs();
8200:   const file = await chooseFile();

```

```

8201:   if (!file) return;
8202:
8203:   const imageKinds = new Set(["profileImage", "heroImage", "brandImage"]);
8204:   if (imageKinds.has(kind) && !file.type.startsWith("image/")) {
8205:     setStatus("Choose an image file for this profile asset.");
8206:     return;
8207:   }
8208:
8209:   if (kind === "resumeUrl") {
8210:     const allowedResume = [".pdf", ".doc", ".docx"].includes(extensionFor(file.name).toLowerCase());
8211:     if (!allowedResume) {
8212:       setStatus("Choose a PDF, DOC, or DOCX resume file.");
8213:       return;
8214:     }
8215:   }
8216:
8217:   const data = await readFileAsDataURL(file);
8218:   const response = await fetch(`/api/upload?t=${Date.now()}`, {
8219:     method: "POST",
8220:     cache: "no-store",
8221:     headers: { "Content-Type": "application/json" },
8222:     body: JSON.stringify({
8223:       projectId: "_site-profile",
8224:       section: kind,
8225:       fileName: file.name,
8226:       data
8227:     })
8228:   });
8229:   const result = await response.json();
8230:   if (!response.ok) {
8231:     setStatus(result.error || "Profile asset upload failed.");
8232:     return;
8233:   }
8234:
8235:   profile[kind] = result.url;
8236:   catalog.profile = normalizeProfile(profile);
8237:   markDraftNeedsSave();
8238:   setStatus("Profile asset saved locally. Click Save draft before applying to site.");
8239:   scheduleAutosave();
8240:   schedulePreviewRender();
8241:   renderProfileEditor();
8242: }

```

## ***extensionFor - lines 8244-8247***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8244-8247 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function extensionFor(fileName) {                                                                                                                                                                     |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | match                                                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8244: function extensionFor(fileName) {
8245:   const match = String(fileName || "").match(/(\.[a-z0-9]+)$/i);
8246:   return match ? match[1] : "";
8247: }

```

## ***withExtension - lines 8249-8253***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 8249-8253 (5 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function withExtension(name, fallbackFileName) {             |
| Touches         | Mostly local calculations or local UI state.                 |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | trim, extensionFor                                                                                                                                                                                    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8249: function withExtension(name, fallbackFileName) {
8250:   const trimmed = String(name || "").trim();
8251:   if (!trimmed) return fallbackFileName;
8252:   return /\.[a-z0-9]+$/.test(trimmed) ? trimmed : `${trimmed}${extensionFor(fallbackFileName)}`;
8253: }

```

## ***uploadProfile - lines 8255-8286***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8255-8286 (32 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function uploadProfile(key) {                                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8255: function uploadProfile(key) {
8256:   const anyProjectFile = {
8257:     label: "project file",
8258:     accept: "",
8259:     extensions: [],
8260:     mimeStarts: [],
8261:     allFiles: true
8262:   };
8263:   const profiles = {
8264:     media: {
8265:       ...anyProjectFile,
8266:       label: "image or file"
8267:     },
8268:     pcbs: {
8269:       ...anyProjectFile,
8270:       label: "PCB artifact"
8271:     },
8272:     tests: {
8273:       ...anyProjectFile,
8274:       label: "test artifact"
8275:     },
8276:     documents: {
8277:       ...anyProjectFile,
8278:       label: "document"
8279:     },
8280:     sections: {
8281:       ...anyProjectFile,
8282:       label: "section file"
8283:     }
8284:   };
8285:   return profiles[key] || profiles.sections;
8286: }

```

## ***allowedFileMessage - lines 8288-8291***



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8288-8291 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function allowedFileMessage(profile) {                                                                                                                                                                |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | join                                                                                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
8288: function allowedFileMessage(profile) {
8289:   if (profile.allFiles) return `Add any local file type for this ${profile.label}.`;
8290:   return `Add a ${profile.label}: ${profile.extensions.join(", ")}`;
8291: }
```

## ***isAllowedUpload - lines 8293-8306***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8293-8306 (14 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function isAllowedUpload(key, fileName, mimeType = "") {                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | uploadProfile, extensionFor, toLowerCase, includes, startsWith, some, allowedFileMessage                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
8293: function isAllowedUpload(key, fileName, mimeType = "") {
8294:   const profile = uploadProfile(key);
8295:   if (profile.allFiles) return { ok: true, message: "" };
8296:   const extension = extensionFor(fileName).toLowerCase();
8297:   if (key === "media" && !profile.extensions.includes(extension) && !mimeType.startsWith("image/")) {
8298:     return { ok: false, message: "Images can only accept image files. Add PDFs under Documents, Tests, or
a custom section instead." };
8299:   }
8300:   const allowedByExtension = profile.extensions.includes(extension);
8301:   const allowedByMime = profile.mimeStarts.some((prefix) => mimeType.startsWith(prefix));
8302:   return {
8303:     ok: allowedByExtension || allowedByMime,
8304:     message: `${fileName} || "This file" is not a supported ${profile.label}.
${allowedFileMessage(profile)}.`
8305:   };
8306: }
```

## ***validateAssetForSection - lines 8308-8317***

| Item            | Explanation                                                                   |
|-----------------|-------------------------------------------------------------------------------|
| Source location | Lines 8308-8317 (10 source lines).                                            |
| Purpose         | Validates or normalizes input so later code receives safe predictable values. |
| Declaration     | function validateAssetForSection(key, asset) {                                |
| Touches         | filesystem                                                                    |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calls detected     | isAllowedUpload, normalizeLinkTarget, extensionFromUrl, extensionFor                                                                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8308: function validateAssetForSection(key, asset) {
8309:   if (asset.source === "local") {
8310:     return isAllowedUpload(key, asset.fileName || asset.file?.name || "", asset.file?.type || "");
8311:   }
8312:   const url = normalizeLinkTarget(asset.url || "", { assumeWeb: true });
8313:   if (!extensionFromUrl(url) && !extensionFor(asset.title || "")) {
8314:     return { ok: true, message: "" };
8315:   }
8316:   return isAllowedUpload(key, url || asset.title || "", "");
8317: }

```

## ***dialogValue - lines 8319-8328***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8319-8328 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function dialogValue(dialog) {                                                                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | removeEventListener, resolve, addEventListener, showModal                                                                                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8319: function dialogValue(dialog) {
8320:   return new Promise((resolve) => {
8321:     const onClose = () => {
8322:       dialog.removeEventListener("close", onClose);
8323:       resolve(dialog.returnValue === "save");
8324:     };
8325:     dialog.addEventListener("close", onClose);
8326:     dialog.showModal();
8327:   });
8328: }

```

## ***onClose - lines 8321-8324***

| Item               | Explanation                                                  |
|--------------------|--------------------------------------------------------------|
| Source location    | Lines 8321-8324 (4 source lines).                            |
| Purpose            | Part of the private builder frontend and editing experience. |
| Declaration        | const onClose = () => {                                      |
| Touches            | Mostly local calculations or local UI state.                 |
| Calls detected     | removeEventListener, resolve                                 |
| API endpoints      | None detected inside this function body.                     |
| Storage keys       | No local/session storage keys detected.                      |
| Async/return clues | 0 await expression(s), 0 return statement(s).                |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8321:     const onClose = () => {
8322:         dialog.removeEventListener("close", onClose);
8323:         resolve(dialog.returnValue === "save");
8324:     };

```

## ***openAssetDialog - lines 8330-8392***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8330-8392 (63 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | async function openAssetDialog(key) {                                                                                                                                                                 |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | uploadProfile, updateAssetDialogVisibility, dialogValue, trim, setStatus, withExtension, isAllowedUpload, normalizeLinkTarget, displayNameFromUrl, validateAssetForSection, text, inferUrlAssetKind   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8330: async function openAssetDialog(key) {
8331:     const profile = uploadProfile(key);
8332:     assetDialogTitle.textContent = `Add ${profile.label}`;
8333:     assetSource.value = "local";
8334:     assetFile.value = "";
8335:     assetFile.accept = profile.accept;
8336:     assetUrl.value = "";
8337:     assetTitle.value = "";
8338:     assetCaption.value = "";
8339:     captionSource.value = "text";
8340:     captionFile.value = "";
8341:     updateAssetDialogVisibility();
8342:
8343:     const saved = await dialogValue(assetDialog);
8344:     if (!saved) return null;
8345:
8346:     let file = null;
8347:     let url = assetUrl.value.trim();
8348:     let fileName = "";
8349:     let title = assetTitle.value.trim();
8350:
8351:     if (assetSource.value === "local") {
8352:         file = assetFile.files[0];
8353:         if (!file) {
8354:             setStatus("Choose a local file first.");
8355:             return null;
8356:         }
8357:         fileName = withExtension(title, file.name);
8358:         title = title || file.name;
8359:         const validation = isAllowedUpload(key, fileName, file.type);
8360:         if (!validation.ok) {
8361:             setStatus(validation.message);
8362:             return null;
8363:         }
8364:     } else {
8365:         if (!url) {
8366:             setStatus("Paste a web or Google Drive link first.");
8367:             return null;
8368:         }
8369:         url = normalizeLinkTarget(url, { assumeWeb: true });
8370:         title = title || displayNameFromUrl(url, "Linked asset");
8371:         const validation = validateAssetForSection(key, { source: assetSource.value, url, title });
8372:         if (!validation.ok) {
8373:             setStatus(validation.message);
8374:             return null;
8375:         }
8376:     }
8377: }

```

```

8378:   let caption = assetCaption.value.trim();
8379:   if (captionSource.value === "file" && captionFile.files[0]) {
8380:     caption = await captionFile.files[0].text();
8381:   }
8382:
8383:   return {
8384:     caption,
8385:     file,
8386:     fileName,
8387:     kind: file ? "file" : inferUrlAssetKind(url, assetSource.value),
8388:     source: assetSource.value,
8389:     title,
8390:     url
8391:   };
8392: }

```

## uploadToSection - lines 8394-8454

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8394-8454 (61 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function uploadToSection(key, folder, customSectionId = "", customItemIndex = null) {                                                                                                           |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | selectedProject, openAssetDialog, validateAssetForSection, setStatus, labelForUrlAssetKind, readFileAsDataURL, fetch, stringify, json, find, push, makeItemForSection                                 |
| API endpoints      | /api/upload                                                                                                                                                                                           |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 4 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8394: async function uploadToSection(key, folder, customSectionId = "", customItemIndex = null) {
8395:   const project = selectedProject();
8396:   if (!project) return;
8397:
8398:   const asset = await openAssetDialog(key);
8399:   if (!asset) return;
8400:
8401:   const validation = validateAssetForSection(key, asset);
8402:   if (!validation.ok) {
8403:     setStatus(validation.message);
8404:     return;
8405:   }
8406:
8407:   const sectionFolder = customSectionId || folder || key;
8408:   let url = asset.url;
8409:   let type = asset.source === "local" ? "File" : labelForUrlAssetKind(asset.kind);
8410:
8411:   if (asset.file) {
8412:     const data = await readFileAsDataURL(asset.file);
8413:     const response = await fetch("/api/upload", {
8414:       method: "POST",
8415:       headers: { "Content-Type": "application/json" },
8416:       body: JSON.stringify({
8417:         projectId: project.id,
8418:         section: sectionFolder,
8419:         fileName: asset.fileName,
8420:         data
8421:       })
8422:     });
8423:     const result = await response.json();
8424:     if (!response.ok) {
8425:       setStatus(result.error || "Upload failed.");
8426:       return;
8427:     }
8428:     url = result.url;
8429:     type = asset.file.type || "File";
8430:   }
8431:
8432:   if (customSectionId) {
8433:     const section = (project.sections || []).find((item) => item.id === customSectionId);
8434:     const uploadedItem = { title: asset.title, description: asset.caption, type, url, status:
asset.source === "local" ? "uploaded" : "linked" };
8435:     if (customItemIndex !== null && customItemIndex !== "") {
8436:       const parent = section?.items?.[Number(customItemIndex)];
8437:       if (!parent || parent.url) return;
8438:       parent.children = parent.children || [];

```

```

8439:     parent.children.push(uploadedItem);
8440:   } else {
8441:     section.items = section.items || [];
8442:     section.items.push(uploadedItem);
8443:   }
8444: } else {
8445:   project[key] = project[key] || [];
8446:   project[key].push(makeItemForSection(key, asset.title, asset.caption, url));
8447: }
8448:
8449: setStatus(asset.source === "local"
8450:   ? `Saved ${asset.fileName} to ${url}. Click Save project to include it in the portfolio preview.`
8451:   : `Linked ${asset.title}. Click Save project to include it in the portfolio preview.`);
8452: scheduleAutosave();
8453: renderAll();
8454: }

```

## section - lines 8433-8434

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8433-8434 (2 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>const section = (project.sections    []).find((item) =&gt; item.id === customSectionId);</code>                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>find</code>                                                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8433:     const section = (project.sections || []).find((item) => item.id === customSectionId);
8434:     const uploadedItem = { title: asset.title, description: asset.caption, type, url, status:
asset.source === "local" ? "uploaded" : "linked" };

```

## uploadToDesignSection - lines 8456-8502

| Item               | Explanation                                                                                                                                                                                                                                                                                                                                     |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8456-8502 (47 source lines).                                                                                                                                                                                                                                                                                                              |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                                                                                                                                                                    |
| Declaration        | <code>async function uploadToDesignSection(pathValue, folder, kind = "document") {</code>                                                                                                                                                                                                                                                       |
| Touches            | filesystem, network/API requests                                                                                                                                                                                                                                                                                                                |
| Calls detected     | <code>selectedProject</code> , <code>openAssetDialog</code> , <code>validateAssetForSection</code> , <code>setStatus</code> , <code>labelForUrlAssetKind</code> , <code>readFileAsDataURL</code> , <code>fetch</code> , <code>stringify</code> , <code>json</code> , <code>designArray</code> , <code>push</code> , <code>makeDesignItem</code> |
| API endpoints      | <code>/api/upload</code>                                                                                                                                                                                                                                                                                                                        |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                                                                                                                         |
| Async/return clues | 4 await expression(s), 4 return statement(s).                                                                                                                                                                                                                                                                                                   |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                                                                                                                           |

Source excerpt:

```

8456: async function uploadToDesignSection(pathValue, folder, kind = "document") {
8457:   const project = selectedProject();
8458:   if (!project) return;
8459:
8460:   const asset = await openAssetDialog(kind === "simulation-result" ? "tests" : "sections");
8461:   if (!asset) return;
8462:
8463:   const validation = validateAssetForSection(kind === "simulation-result" ? "tests" : "sections", asset);
8464:   if (!validation.ok) {
8465:     setStatus(validation.message);
8466:     return;
8467:   }

```

```

8468:
8469:   let url = asset.url;
8470:   let type = asset.source === "local" ? "File" : labelForUrlAssetKind(asset.kind);
8471:   if (asset.file) {
8472:     const data = await readFileAsDataURL(asset.file);
8473:     const response = await fetch("/api/upload", {
8474:       method: "POST",
8475:       headers: { "Content-Type": "application/json" },
8476:       body: JSON.stringify({
8477:         projectId: project.id,
8478:         section: folder || "design",
8479:         fileName: asset.fileName,
8480:         data
8481:       })
8482:     });
8483:     const result = await response.json();
8484:     if (!response.ok) {
8485:       setStatus(result.error || "Upload failed.");
8486:       return;
8487:     }
8488:     url = result.url;
8489:     type = asset.file.type || "File";
8490:   }
8491:
8492:   const items = designArray(project, pathValue);
8493:   items.push({
8494:     ...makeDesignItem(kind, asset.title, asset.caption, url),
8495:     type
8496:   });
8497:   setStatus(asset.source === "local"
8498:     ? `Saved ${asset.fileName} to ${url}. Click Save project to include it in the portfolio preview.`
8499:     : `Linked ${asset.title}. Click Save project to include it in the portfolio preview.`);
8500:   scheduleAutosave();
8501:   renderAll();
8502: }

```

## openSectionDialog - lines 8504-8523

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8504-8523 (20 source lines).                                                                                                                                                                    |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | async function openSectionDialog() {                                                                                                                                                                  |
| Touces             | filesystem                                                                                                                                                                                            |
| Calls detected     | dialogValue, trim, setStatus                                                                                                                                                                          |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8504: async function openSectionDialog() {
8505:   sectionDialogTitle.textContent = "Create a section";
8506:   sectionTitle.placeholder = "Design, Measurements, Firmware...";
8507:   sectionDescription.placeholder = "Optional description shown to recruiters";
8508:   sectionTitle.value = "";
8509:   sectionDescription.value = "";
8510:   const saved = await dialogValue(sectionDialog);
8511:   if (!saved) return null;
8512:
8513:   const title = sectionTitle.value.trim();
8514:   if (!title) {
8515:     setStatus("Type a section title before saving.");
8516:     return null;
8517:   }
8518:
8519:   return {
8520:     description: sectionDescription.value.trim(),
8521:     title
8522:   };
8523: }

```

## openCategoryDialog - lines 8525-8531

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8525-8531 (7 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function openCategoryDialog() {                                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | showModal, focus                                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8525: function openCategoryDialog() {
8526:   categoryTitle.value = "";
8527:   categoryDescription.value = "";
8528:   categoryAccent.value = "#1677a8";
8529:   categoryDialog.showModal();
8530:   setTimeout(() => categoryTitle.focus(), 0);
8531: }

```

## ***saveCategoryFromDialog - lines 8533-8559***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8533-8559 (27 source lines).                                                                                                                                                                    |
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | function saveCategoryFromDialog() {                                                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | trim, setStatus, slugify, Set, map, has, normalizeCategory, add, refreshCategoryControls, renderAll, scheduleAutosave                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8533: function saveCategoryFromDialog() {
8534:   const label = categoryTitle.value.trim();
8535:   if (!label) {
8536:     setStatus("Type a category name before saving.");
8537:     return;
8538:   }
8539:   const baseId = slugify(label);
8540:   const ids = new Set((catalog.categories || []).map((category) => category.id));
8541:   let id = baseId;
8542:   let suffix = 2;
8543:   while (ids.has(id)) {
8544:     id = `${baseId}-${suffix}`;
8545:     suffix += 1;
8546:   }
8547:   const category = normalizeCategory({
8548:     accent: categoryAccent.value,
8549:     description: categoryDescription.value,
8550:     id,
8551:     label
8552:   });
8553:   catalog.categories = [...(catalog.categories || []), category];
8554:   expandedCategories.add(category.id);
8555:   refreshCategoryControls();
8556:   renderAll();
8557:   scheduleAutosave();
8558:   setStatus(`Category "${category.label}" added locally. It is now available under Add new project.`);
8559: }

```

## ***addCustomSection - lines 8561-8582***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8561-8582 (22 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | async function addCustomSection() {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, openSectionDialog, slugify, Set, map, has, push, setStatus, scheduleAutosave, renderAll                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
8561: async function addCustomSection() {
8562:   const project = selectedProject();
8563:   if (!project) return;
8564:
8565:   const sectionInput = await openSectionDialog();
8566:   if (!sectionInput) return;
8567:
8568:   let id = slugify(sectionInput.title);
8569:   const existingIds = new Set((project.sections || []).map((section) => section.id));
8570:   let suffix = 2;
8571:   while (existingIds.has(id)) {
8572:     id = `${slugify(sectionInput.title)}-${suffix}`;
8573:     suffix += 1;
8574:   }
8575:
8576:   project.sections = project.sections || [];
8577:   project.sections.push({ id, title: sectionInput.title, description: sectionInput.description, items: []
});
8578:   activeSectionId = `custom:${id}`;
8579:   setStatus("New section added in the editor. Click Save project to include this version in the portfolio
preview.");
8580:   scheduleAutosave();
8581:   renderAll();
8582: }
```

## ***addCustomItem - lines 8584-8596***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8584-8596 (13 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function addCustomItem(sectionId) {                                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | selectedProject, find, prompt, push, setStatus, scheduleAutosave, renderAll                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
8584: function addCustomItem(sectionId) {
8585:   const project = selectedProject();
8586:   const section = (project.sections || []).find((item) => item.id === sectionId);
8587:   const title = prompt("Type a subsection title.");
8588:   if (!section || !title) return;
8589:   const description = prompt("Paste or type a text description if you want.") || "";
8590:
8591:   section.items = section.items || [];
8592:   section.items.push({ title, description, type: "Text", status: "draft" });
```



```

8593:   setStatus("Subsection added locally.");
8594:   scheduleAutosave();
8595:   renderAll();
8596: }

```

## section - lines 8586-8589

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8586-8589 (4 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>const section = (project.sections    []).find((item) =&gt; item.id === sectionId);</code>                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>find</code> , <code>prompt</code>                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 <code>await</code> expression(s), 1 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8586:   const section = (project.sections || []).find((item) => item.id === sectionId);
8587:   const title = prompt("Type a subsection title.");
8588:   if (!section || !title) return;
8589:   const description = prompt("Paste or type a text description if you want.") || "";

```

## editCustomItem - lines 8598-8609

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8598-8609 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>function editCustomItem(sectionId, index) {</code>                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>selectedProject</code> , <code>find</code> , <code>prompt</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 <code>await</code> expression(s), 2 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8598: function editCustomItem(sectionId, index) {
8599:   const project = selectedProject();
8600:   const item = (project.sections || []).find((section) => section.id === sectionId)?.items?.[index];
8601:   if (!item) return;
8602:   const title = prompt("Edit subsection title.", item.title || "");
8603:   if (!title) return;
8604:   item.title = title;
8605:   item.description = prompt("Edit description.", item.description || "") || "";
8606:   setStatus("Subsection edited locally.");
8607:   scheduleAutosave();
8608:   renderAll();
8609: }

```

## item - lines 8600-8608

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 8600-8608 (9 source lines).                            |
| Purpose         | Part of the private builder frontend and editing experience. |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | <code>const item = (project.sections    []).find((section) =&gt; section.id === sectionId)?.items?.[index];</code>                                                                                    |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | <code>find</code> , <code>prompt</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 <code>await</code> expression(s), 2 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8600:   const item = (project.sections || []).find((section) => section.id === sectionId)?.items?.[index];
8601:   if (!item) return;
8602:   const title = prompt("Edit subsection title.", item.title || "");
8603:   if (!title) return;
8604:   item.title = title;
8605:   item.description = prompt("Edit description.", item.description || "") || "";
8606:   setStatus("Subsection edited locally.");
8607:   scheduleAutosave();
8608:   renderAll();

```

## ***catalogForSave - lines 8611-8633***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8611-8633 (23 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | <code>function catalogForSave(endpoint) {</code>                                                                                                                                                      |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | <code>parsedPortfolioGlobals</code> , <code>clone</code>                                                                                                                                              |
| API endpoints      | <code>/api/apply-catalog</code>                                                                                                                                                                       |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 <code>await</code> expression(s), 2 <code>return</code> statement(s).                                                                                                                               |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8611: function catalogForSave(endpoint) {
8612:   const parsedGlobals = parsedPortfolioGlobals();
8613:   if (endpoint === "/api/apply-catalog") {
8614:     return {
8615:       ...savedPortfolioCatalog,
8616:       ...parsedGlobals,
8617:       projects: clone(savedPortfolioCatalog.projects || [])
8618:     };
8619:   }
8620:   return {
8621:     ...catalog,
8622:     categories: parsedGlobals.categories,
8623:     fieldStyles: parsedGlobals.fieldStyles,
8624:     funFacts: parsedGlobals.funFacts,
8625:     funFactsRich: parsedGlobals.funFactsRich,
8626:     profile: parsedGlobals.profile,
8627:     profileRich: parsedGlobals.profileRich,
8628:     siteContent: parsedGlobals.siteContent,
8629:     siteContentRich: parsedGlobals.siteContentRich,
8630:     siteSections: clone(catalog.siteSections || []),
8631:     projects: clone(catalog.projects || [])
8632:   };
8633: }

```

## ***saveCatalog - lines 8635-8697***

| Item            | Explanation                        |
|-----------------|------------------------------------|
| Source location | Lines 8635-8697 (63 source lines). |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Purpose            | Persists data to local state, disk, credentials, or browser storage.                                                                                                                                  |
| Declaration        | async function saveCatalog(endpoint, message) {                                                                                                                                                       |
| Touches            | network/API requests                                                                                                                                                                                  |
| Calls detected     | catalogForSave, showBuilderError, setStatus, setSaveState, fetch, now, stringify, json, showPublishResult                                                                                             |
| API endpoints      | /api/apply-catalog, /api/save-draft                                                                                                                                                                   |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8635: async function saveCatalog(endpoint, message) {
8636:   const targetCatalog = catalogForSave(endpoint);
8637:   if (endpoint === "/api/apply-catalog" && !draftSavedSinceChanges) {
8638:     showBuilderError(
8639:       "Save draft required",
8640:       "Please click Save draft before applying this portfolio to the live site.",
8641:       "Apply to site was stopped before commit and push. Save the draft, then click Apply to site again."
8642:     );
8643:     setStatus("Apply to site stopped. Save draft first.");
8644:     setSaveState("dirty", "Save draft needed");
8645:     return;
8646:   }
8647:   if (endpoint === "/api/apply-catalog" && !(targetCatalog.projects || []).length) {
8648:     showBuilderError(
8649:       "No saved project preview",
8650:       "Save a project or use Save all sections before applying to the live site.",
8651:       "Apply to site was stopped before commit and push."
8652:     );
8653:     setStatus("Apply to site stopped. Save a project or use Save all sections first.");
8654:     setSaveState("dirty", "Preview needs projects");
8655:     return;
8656:   }
8657:   try {
8658:     clearTimeout(autosaveTimer);
8659:     saveDraftButton.disabled = true;
8660:     applyCatalogButton.disabled = true;
8661:     const applying = endpoint === "/api/apply-catalog";
8662:     setStatus(applying ? "Applying site changes..." : "Saving draft...");
8663:     setSaveState(applying ? "publishing" : "saving", applying ? "Publishing..." : "Saving...");
8664:     const response = await fetch(`${endpoint}?t=${Date.now()}`, {
8665:       method: "POST",
8666:       cache: "no-store",
8667:       headers: { "Content-Type": "application/json" },
8668:       body: JSON.stringify({ catalog: targetCatalog })
8669:     });
8670:     const result = await response.json();
8671:     if (!response.ok) {
8672:       setStatus(result.error || "Save failed.");
8673:       setSaveState("error", endpoint === "/api/apply-catalog" ? "Publish failed" : "Save failed");
8674:       return;
8675:     }
8676:     if (result.publish) {
8677:       setStatus(result.publish.pushed
8678:         ? `${message}: ${result.file}. Pushed to ${result.publish.branch}.`
8679:         : `${message}: ${result.file}. Git push failed: ${result.publish.error}`);
8680:       setSaveState(result.publish.pushed ? "published" : "error", result.publish.pushed ? "Site applied"
8681:         : "Push failed");
8682:       showPublishResult(result);
8683:       return;
8684:     }
8685:     if (endpoint === "/api/save-draft") {
8686:       draftSavedSinceChanges = true;
8687:       setSaveState("saved", "Draft saved");
8688:     }
8689:     setStatus(`${message}: ${result.file}`);
8690:   } catch {
8691:     setStatus("Save failed. Make sure the local server window is still running.");
8692:     setSaveState("error", "Save failed");
8693:   } finally {
8694:     saveDraftButton.disabled = false;
8695:     applyCatalogButton.disabled = false;
8696:   }
8697: }

```

**loadData - lines 8699-8745**

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8699-8745 (47 source lines).                                                                                                                                                                    |
| Purpose            | Loads data from disk, network, browser storage, or a service.                                                                                                                                         |
| Declaration        | async function loadData() {                                                                                                                                                                           |
| Touches            | filesystem, network/API requests                                                                                                                                                                      |
| Calls detected     | all, fetch, now, json, call, normalizeFunFacts, clone, normalizeFieldStyles, normalizeProfile, normalizeSiteContent, filter, has                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 3 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

#### Source excerpt:

```

8699: async function loadData() {
8700:   const [catalogResponse, templateResponse] = await Promise.all([
8701:     fetch(`/api/catalog?t=${Date.now()}`, { cache: "no-store" }),
8702:     fetch(`/api/templates?t=${Date.now()}`, { cache: "no-store" })
8703:   ]);
8704:   const catalogData = await catalogResponse.json();
8705:   const templateData = await templateResponse.json();
8706:
8707:   catalog = catalogData.catalog;
8708:   catalog.funFacts = Object.prototype.hasOwnProperty.call(catalog, "funFacts")
8709:     ? normalizeFunFacts(catalog.funFacts)
8710:     : clone(defaultFunFacts);
8711:   catalog.funFactsRich = catalog.funFactsRich || null;
8712:   catalog.fieldStyles = normalizeFieldStyles(catalog.fieldStyles || {});
8713:   catalog.siteContentRich = catalog.siteContentRich || {};
8714:   catalog.profileRich = catalog.profileRich || {};
8715:   catalog.profile = normalizeProfile(catalog.profile || {});
8716:   catalog.siteContent = normalizeSiteContent(catalog.siteContent || {});
8717:   catalog.siteSections = catalog.siteSections || [];
8718:   catalog.projects = (catalog.projects || []).filter((project) => !starterProjectIds.has(project.id));
8719:   catalog.projects.forEach((project) => {
8720:     if (isAnalogProject(project)) ensureDesignModel(project);
8721:     ensureCompileCode(project);
8722:   });
8723:   savedPortfolioCatalog = {
8724:     categories: clone(catalog.categories || []),
8725:     fieldStyles: clone(catalog.fieldStyles || {}),
8726:     funFacts: clone(catalog.funFacts || []),
8727:     funFactsRich: clone(catalog.funFactsRich || null),
8728:     profile: clone(normalizeProfile(catalog.profile || {})),
8729:     profileRich: clone(catalog.profileRich || {}),
8730:     siteContent: clone(catalog.siteContent || defaultSiteContent),
8731:     siteContentRich: clone(catalog.siteContentRich || {}),
8732:     projects: [],
8733:     siteSections: clone(catalog.siteSections || [])
8734:   };
8735:   templates = templateData.templates || [];
8736:   setStatus(`Loaded ${catalogData.source} catalog. Builder controls are local-only.`);
8737:   draftSavedSinceChanges = true;
8738:   setSaveState("loaded", "Loaded");
8739:
8740:   templateSelect.innerHTML = groupedTemplateOptions();
8741:   refreshCategoryControls();
8742:   selectedProjectId = catalog.projects[0]?.id || "";
8743:   expandedCategories = new Set(catalog.categories.map((category) => category.id));
8744:   renderAll();
8745: }

```

### ***prepareBuilderGuideTopicLinks - lines 8758-8768***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 8758-8768 (11 source lines).                           |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function prepareBuilderGuideTopicLinks() {                   |
| Touches         | browser/Electron DOM                                         |
| Calls detected  | querySelectorAll, forEach, querySelector, setAttribute, trim |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8758: function prepareBuilderGuideTopicLinks() {
8759:   builderGuideSections?.querySelectorAll("[data-guide-section]").forEach((section) => {
8760:     const summary = section.querySelector("summary");
8761:     if (!summary) return;
8762:     summary.setAttribute("role", "link");
8763:     summary.setAttribute("tabindex", "0");
8764:     summary.setAttribute("aria-label", `Open ${summary.textContent.trim()}`);
8765:     summary.title = "Open this guide topic";
8766:     section.open = false;
8767:   });
8768: }

```

## ***updateBuilderGuideStats - lines 8770-8849***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8770-8849 (80 source lines).                                                                                                                                                                    |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateBuilderGuideStats() {                                                                                                                                                                  |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | replace, entries, forEach, querySelector, guideSectionText, join, toLowerCase, filterBuilderGuide, trim, querySelectorAll, split, includes                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8770: function updateBuilderGuideStats() {
8771:   if (!builderGuideDialog) return;
8772:   const projectCount = catalog.projects?.length || 0;
8773:   const categoryCount = catalog.categories?.length || 0;
8774:   const parsedCount = savedPortfolioCatalog.projects?.length || 0;
8775:   const targetLabel = currentPublishTarget?.remote
8776:     ? `Target: ${currentPublishTarget.remote.replace(/^https:\/\/github\.com\/\//i, "")}`
8777:     : "Target: not connected";
8778:   const values = {
8779:     projects: `Projects: ${projectCount} (${parsedCount} parsed)`,
8780:     categories: `Categories: ${categoryCount}`,
8781:     draft: draftSavedSinceChanges ? "Draft: saved" : "Draft: unsaved changes",
8782:     target: targetLabel
8783:   };
8784:   Object.entries(values).forEach(([key, value]) => {
8785:     const node = builderGuideDialog.querySelector(`[data-guide-stat='${key}']`);
8786:     if (node) node.textContent = value;
8787:   });
8788: }
8789:
8790: function guideSectionText(section) {
8791:   return [
8792:     section.querySelector("summary")?.textContent || "",
8793:     section.textContent || "",
8794:     section.dataset.guideSection || ""
8795:   ].join(" ").toLowerCase();
8796: }
8797:
8798: function filterBuilderGuide() {
8799:   if (!builderGuideSections) return;
8800:   const query = String(builderGuideSearch?.value || "").trim().toLowerCase();
8801:   const sections = [...builderGuideSections.querySelectorAll("[data-guide-section]")];
8802:   let visibleCount = 0;
8803:   sections.forEach((section) => {
8804:     const topicMatch = activeBuilderGuideTopic === "all" || (section.dataset.guideSection ||

```

```

"").split(/\s+/).includes(activeBuilderGuideTopic);
8805:   const queryMatch = !query || guideSectionText(section).includes(query);
8806:   const visible = topicMatch && queryMatch;
8807:   section.hidden = !visible;
8808:   if (visible) {
8809:     visibleCount += 1;
8810:     section.open = false;
8811:   }
8812: });
8813: if (builderGuideResults) {
8814:   const topicText = activeBuilderGuideTopic === "all" ? "all topics" : activeBuilderGuideTopic;
8815:   builderGuideResults.textContent = visibleCount
8816:   ? `Showing ${visibleCount} guide topic${visibleCount === 1 ? "" : "s"} for ${topicText}${query ? `
matching "${query}"` : ""}` : `No guide topics match "${query}"`;
8817:   : `No guide topics match "${query}"`;
8818: }
8819: }
8820:
8821: function setBuilderGuideTopic(topic = "all") {
8822:   activeBuilderGuideTopic = topic || "all";
8823:   builderGuideDialog?.querySelectorAll("[data-guide-topic]").forEach((button) => {
8824:     button.setAttribute("aria-pressed", button.dataset.guideTopic === activeBuilderGuideTopic ? "true" :
"false");
8825:   });
8826:   filterBuilderGuide();
8827: }
8828:
8829: function closeGuideAndFocus(target) {
8830:   closeDialogElement(builderGuideDialog);
8831:   requestAnimationFrame(() => {
8832:     target?.scrollIntoView?({ behavior: "smooth", block: "center" });
8833:     if (target?.focus) target.focus({ preventScroll: true });
8834:   });
8835: }
8836:
8837: function handleBuilderGuideAction(action = "") {
8838:   if (action === "profile") closeGuideAndFocus(profileDisplayNameInput);
8839:   if (action === "site-content") closeGuideAndFocus(siteHeroTitleInput);
8840:   if (action === "fun-facts") closeGuideAndFocus(funFactsInput);
8841:   if (action === "projects") closeGuideAndFocus(projectTree);
8842:   if (action === "preview") {
8843:     closeDialogElement(builderGuideDialog);
8844:     openPortfolioPreviewButton?.click();
8845:   }
8846:   if (action === "target") {
8847:     closeDialogElement(builderGuideDialog);
8848:     publishTargetOpen?.click();
8849:   }

```

## guideSectionText - lines 8790-8796

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8790-8796 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function guideSectionText(section) {                                                                                                                                                                  |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelector, join, toLowerCase                                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8790: function guideSectionText(section) {
8791:   return [
8792:     section.querySelector("summary")?.textContent || "",
8793:     section.textContent || "",
8794:     section.dataset.guideSection || ""
8795:   ].join(" ").toLowerCase();
8796: }

```

## filterBuilderGuide - lines 8798-8819

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8798-8819 (22 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function filterBuilderGuide() {                                                                                                                                                                       |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | trim, toLowerCase, querySelectorAll, forEach, split, includes, guideSectionText                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8798: function filterBuilderGuide() {
8799:   if (!builderGuideSections) return;
8800:   const query = String(builderGuideSearch?.value || "").trim().toLowerCase();
8801:   const sections = [...builderGuideSections.querySelectorAll("[data-guide-section]")];
8802:   let visibleCount = 0;
8803:   sections.forEach((section) => {
8804:     const topicMatch = activeBuilderGuideTopic === "all" || (section.dataset.guideSection ||
8805:       "").split(/\s+/).includes(activeBuilderGuideTopic);
8806:     const queryMatch = !query || guideSectionText(section).includes(query);
8807:     const visible = topicMatch && queryMatch;
8808:     section.hidden = !visible;
8809:     if (visible) {
8810:       visibleCount += 1;
8811:       section.open = false;
8812:     }
8813:   });
8814:   if (builderGuideResults) {
8815:     const topicText = activeBuilderGuideTopic === "all" ? "all topics" : activeBuilderGuideTopic;
8816:     builderGuideResults.textContent = visibleCount
8817:       ? `Showing ${visibleCount} guide topic${visibleCount === 1 ? "" : "s"} for ${topicText}${query ? `
8818:         matching "${query}"` : ""}.`
8819:       : `No guide topics match "${query}"`;
8820:   }
8821: }

```

## setBuilderGuideTopic - lines 8821-8827

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8821-8827 (7 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function setBuilderGuideTopic(topic = "all") {                                                                                                                                                        |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | querySelectorAll, forEach, setAttribute, filterBuilderGuide                                                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8821: function setBuilderGuideTopic(topic = "all") {
8822:   activeBuilderGuideTopic = topic || "all";
8823:   builderGuideDialog?.querySelectorAll("[data-guide-topic]").forEach((button) => {
8824:     button.setAttribute("aria-pressed", button.dataset.guideTopic === activeBuilderGuideTopic ? "true" :
8825:       "false");
8826:   });
8827:   filterBuilderGuide();
8828: }

```

## closeGuideAndFocus - lines 8829-8835

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8829-8835 (7 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function closeGuideAndFocus(target) {                                                                                                                                                                 |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closeDialogElement, requestAnimationFrame, focus                                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8829: function closeGuideAndFocus(target) {
8830:   closeDialogElement(builderGuideDialog);
8831:   requestAnimationFrame(() => {
8832:     target?.scrollIntoView?({ behavior: "smooth", block: "center" });
8833:     if (target?.focus) target.focus({ preventScroll: true });
8834:   });
8835: }

```

## ***handleBuilderGuideAction - lines 8837-8854***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 8837-8854 (18 source lines).                                                                                                                                                                    |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handleBuilderGuideAction(action = "") {                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closeGuideAndFocus, closeDialogElement, click                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8837: function handleBuilderGuideAction(action = "") {
8838:   if (action === "profile") closeGuideAndFocus(profileDisplayNameInput);
8839:   if (action === "site-content") closeGuideAndFocus(siteHeroTitleInput);
8840:   if (action === "fun-facts") closeGuideAndFocus(funFactsInput);
8841:   if (action === "projects") closeGuideAndFocus(projectTree);
8842:   if (action === "preview") {
8843:     closeDialogElement(builderGuideDialog);
8844:     openPortfolioPreviewButton?.click();
8845:   }
8846:   if (action === "target") {
8847:     closeDialogElement(builderGuideDialog);
8848:     publishTargetOpen?.click();
8849:   }
8850:   if (action === "updates") {
8851:     closeDialogElement(builderGuideDialog);
8852:     appUpdateCheckButton?.click();
8853:   }
8854: }

```

## ***openBuilderGuideTopic - lines 8856-8878***

| Item            | Explanation                                               |
|-----------------|-----------------------------------------------------------|
| Source location | Lines 8856-8878 (23 source lines).                        |
| Purpose         | Controls a window, dialog, panel, menu, or visible state. |



| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Declaration        | function openBuilderGuideTopic(section) {                                                                                                                                                             |
| Touches            | browser/Electron DOM, authentication/security data                                                                                                                                                    |
| Calls detected     | querySelector, trim, replaceChildren, cloneNode, add, append, createElement, showModal, requestAnimationFrame, focus, updateDialogWindowButtons                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

8856: function openBuilderGuideTopic(section) {
8857:   if (!section || !builderGuideTopicDialog || !builderGuideTopicTitle || !builderGuideTopicBody) return;
8858:   const summary = section.querySelector("summary");
8859:   const source = section.querySelector(":scope > div");
8860:   const title = summary?.textContent?.trim() || "Builder guide topic";
8861:   builderGuideTopicTitle.textContent = title;
8862:   builderGuideTopicBody.replaceChildren();
8863:   if (source) {
8864:     const copy = source.cloneNode(true);
8865:     copy.classList.add("builder-guide-topic-copy");
8866:     builderGuideTopicBody.append(copy);
8867:   } else {
8868:     const empty = document.createElement("p");
8869:     empty.textContent = "This guide topic does not have written instructions yet.";
8870:     builderGuideTopicBody.append(empty);
8871:   }
8872:   if (!builderGuideTopicDialog.open) builderGuideTopicDialog.showModal();
8873:   requestAnimationFrame(() => {
8874:     builderGuideTopicBody.scrollTop = 0;
8875:     builderGuideTopicBody.focus({ preventScroll: true });
8876:     updateDialogWindowButtons(builderGuideTopicDialog);
8877:   });
8878: }

```

## ***handleRichEditorContextMenu - lines 9312-9372***

| Item               | Explanation                                                                                                                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9312-9372 (61 source lines).                                                                                                                                                                                                                        |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                                                                       |
| Declaration        | function handleRichEditorContextMenu(event) {                                                                                                                                                                                                             |
| Touches            | browser/Electron DOM                                                                                                                                                                                                                                      |
| Calls detected     | richEditorFromContextEvent, preventDefault, stopPropagation, closest, clearPersistentTextSelection, hideTextSelectionInspector, configureSummaryContextMenu, selectionRangeInsideEditor, cloneRange, contains, restoreRichSelection, captureRichSelection |
| API endpoints      | None detected inside this function body.                                                                                                                                                                                                                  |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                                                                   |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                                                                             |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.                                                     |

Source excerpt:

```

9312: function handleRichEditorContextMenu(event) {
9313:   const editor = richEditorFromContextEvent(event);
9314:   const textBlock = event.target.closest?(".rich-text-block");
9315:
9316:   if (!editor && !textBlock) return;
9317:
9318:   event.preventDefault();
9319:   event.stopPropagation();
9320:   const nextEditor = editor || textBlock.closest("[data-rich-editor]");
9321:   if (activeSummaryEditor && nextEditor && activeSummaryEditor !== nextEditor) {
9322:     activeRichSelectionRange = null;
9323:     activeTextSelectionRange = null;
9324:     clearPersistentTextSelection(true);
9325:     summaryContextMenu.hidden = true;

```

```

9326:     }
9327:     hideTextSelectionInspector();
9328:
9329:     activePlainTextControl = null;
9330:     activePlainTextSelection = null;
9331:     activeSummaryEditor = nextEditor;
9332:     configureSummaryContextMenu("rich", { textOnly: activeSummaryEditor.dataset.richTextOnly === "true"
});
9333:     const liveRange = selectionRangeInsideEditor(activeSummaryEditor);
9334:     const savedRange = activeTextSelectionRange && !activeTextSelectionRange.collapsed
9335:       ? activeTextSelectionRange.cloneRange()
9336:       : null;
9337:     const savedContainer = savedRange?.commonAncestorContainer?.nodeType === Node.TEXT_NODE
9338:       ? savedRange.commonAncestorContainer.parentElement
9339:       : savedRange?.commonAncestorContainer;
9340:     if (liveRange && !liveRange.collapsed) {
9341:       activeRichSelectionRange = liveRange.cloneRange();
9342:       activeTextSelectionRange = liveRange.cloneRange();
9343:     } else if (savedRange && savedContainer && activeSummaryEditor.contains(savedContainer)) {
9344:       activeRichSelectionRange = savedRange.cloneRange();
9345:       restoreRichSelection(activeSummaryEditor);
9346:     } else {
9347:       captureRichSelection(activeSummaryEditor);
9348:     }
9349:     selectRichBlock(event.target.closest?(".rich-block") || currentRichBlock(activeSummaryEditor));
9350:     syncRichContextMenuControls(activeSummaryBlock);
9351:
9352:     const menuHost = activeSummaryEditor.closest("dialog") || document.body;
9353:     if (summaryContextMenu.parentElement !== menuHost) menuHost.append(summaryContextMenu);
9354:
9355:     summaryContextMenu.style.left = `${event.clientX}px`;
9356:     summaryContextMenu.style.top = `${event.clientY}px`;
9357:     summaryContextMenu.style.maxHeight = "";
9358:     summaryContextMenu.hidden = false;
9359:     const hostRect = menuHost.getBoundingClientRect();
9360:     const bounds = {
9361:       bottom: Math.min(window.innerHeight - 8, hostRect.bottom - 8),
9362:       left: Math.max(8, hostRect.left + 8),
9363:       right: Math.min(window.innerWidth - 8, hostRect.right - 8),
9364:       top: Math.max(8, hostRect.top + 8)
9365:     };
9366:     summaryContextMenu.style.maxHeight = `${Math.max(180, bounds.bottom - bounds.top)}px`;
9367:     const menuRect = summaryContextMenu.getBoundingClientRect();
9368:     const left = Math.max(bounds.left, Math.min(event.clientX, bounds.right - menuRect.width));
9369:     const top = Math.max(bounds.top, Math.min(event.clientY, bounds.bottom - menuRect.height));
9370:     summaryContextMenu.style.left = `${left}px`;
9371:     summaryContextMenu.style.top = `${top}px`;
9372: }

```

## handlePlainTextContextMenu - lines 9398-9433

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9398-9433 (36 source lines).                                                                                                                                                                    |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handlePlainTextContextMenu(event) {                                                                                                                                                          |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | plainTextControlFromContextEvent, preventDefault, hideTextSelectionInspector, configureSummaryContextMenu, syncPlainTextMenuControls, closest, append, getBoundingClientRect, min, max                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9398: function handlePlainTextContextMenu(event) {
9399:   const control = plainTextControlFromContextEvent(event);
9400:   if (!control) return;
9401:
9402:   event.preventDefault();
9403:   hideTextSelectionInspector();
9404:   activeSummaryEditor = null;
9405:   activeSummaryBlock = null;
9406:   activePlainTextControl = control;
9407:   activePlainTextSelection = {
9408:     start: control.selectionStart ?? 0,
9409:     end: control.selectionEnd ?? control.value.length
9410:   };

```

```

9411: configureSummaryContextMenu("plain");
9412: syncPlainContextMenuControls(control);
9413:
9414: const menuHost = control.closest("dialog") || document.body;
9415: if (summaryContextMenu.parentElement !== menuHost) menuHost.append(summaryContextMenu);
9416: summaryContextMenu.style.left = `${event.clientX}px`;
9417: summaryContextMenu.style.top = `${event.clientY}px`;
9418: summaryContextMenu.style.maxHeight = "";
9419: summaryContextMenu.hidden = false;
9420: const hostRect = menuHost.getBoundingClientRect();
9421: const bounds = {
9422:   bottom: Math.min(window.innerHeight - 8, hostRect.bottom - 8),
9423:   left: Math.max(8, hostRect.left + 8),
9424:   right: Math.min(window.innerWidth - 8, hostRect.right - 8),
9425:   top: Math.max(8, hostRect.top + 8)
9426: };
9427: summaryContextMenu.style.maxHeight = `${Math.max(140, bounds.bottom - bounds.top)}px`;
9428: const menuRect = summaryContextMenu.getBoundingClientRect();
9429: const left = Math.max(bounds.left, Math.min(event.clientX, bounds.right - menuRect.width));
9430: const top = Math.max(bounds.top, Math.min(event.clientY, bounds.bottom - menuRect.height));
9431: summaryContextMenu.style.left = `${left}px`;
9432: summaryContextMenu.style.top = `${top}px`;
9433: }

```

## handleRichEditorPaste - lines 9437-9480

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9437-9480 (44 source lines).                                                                                                                                                                    |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | async function handleRichEditorPaste(event) {                                                                                                                                                         |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | closest, find, startsWith, preventDefault, setStatus, uploadSummaryImageFile, getAsFile, insertRichBlockAtCursor, createRichImageBlock, saveRichEditorToProject, getData, insertHtmlIntoRichEditor    |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 5 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9437: async function handleRichEditorPaste(event) {
9438:   const editor = event.target.closest("[data-rich-editor]");
9439:   if (!editor) return;
9440:
9441:   const imageItem = [...(event.clipboardData?.items || [])].find((item) =>
item.type.startsWith("image/"));
9442:
9443:   if (imageItem) {
9444:     event.preventDefault();
9445:     if (editor.dataset.richTextOnly === "true") {
9446:       setStatus("Images can only be pasted into overview editors, not title, profile, or contact
fields.");
9447:       return;
9448:     }
9449:
9450:     const imageBlock = await uploadSummaryImageFile(imageItem.getAsFile(), {
9451:       align: "center",
9452:       display: "show",
9453:       folder: editor.dataset.richFolder || "summary",
9454:       projectId: editor.dataset.richProjectId || "",
9455:       title: ""
9456:     });
9457:
9458:     if (imageBlock) {
9459:       insertRichBlockAtCursor(editor, createRichImageBlock(imageBlock));
9460:       saveRichEditorToProject(editor);
9461:     }
9462:
9463:     return;
9464:   }
9465:
9466:   const pastedHtml = event.clipboardData?.getData("text/html") || "";
9467:   if (pastedHtml) {
9468:     event.preventDefault();
9469:     insertHtmlIntoRichEditor(editor, pastedHtml);
9470:     saveRichEditorToProject(editor);
9471:     return;
9472:   }

```

```

9473:
9474:     const pastedText = event.clipboardData?.getData("text/plain") || "";
9475:     if (!pastedText) return;
9476:
9477:     event.preventDefault();
9478:     insertPlainTextIntoRichEditor(editor, pastedText);
9479:     saveRichEditorToProject(editor);
9480: }

```

## ***restoreRichInsertionPoint - lines 9494-9503***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9494-9503 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function restoreRichInsertionPoint(editor) {                                                                                                                                                          |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | restoreRichSelection, focusTextBlock, selectedRichBlock, querySelector, focus, selectionRangeInsideEditor, deleteContents                                                                             |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9494: function restoreRichInsertionPoint(editor) {
9495:     if (!restoreRichSelection(editor)) focusTextBlock(selectedRichBlock(editor) ||
editor.querySelector(".rich-text-block"));
9496:     editor.focus({ preventScroll: true });
9497:
9498:     const range = selectionRangeInsideEditor(editor);
9499:     if (!range) return null;
9500:
9501:     range.deleteContents();
9502:     return range;
9503: }

```

## ***insertFragmentIntoRichEditor - lines 9505-9521***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9505-9521 (17 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function insertFragmentIntoRichEditor(editor, fragment) {                                                                                                                                             |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | restoreRichInsertionPoint, insertNode, createRange, setStartAfter, setStart, collapse, getSelection, removeAllRanges, addRange, captureRichSelection                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9505: function insertFragmentIntoRichEditor(editor, fragment) {
9506:     const range = restoreRichInsertionPoint(editor);
9507:     if (!range || !fragment) return;
9508:
9509:     const lastInsertedNode = fragment.lastChild;
9510:     range.insertNode(fragment);
9511:
9512:     const nextRange = document.createRange();
9513:     if (lastInsertedNode) nextRange.setStartAfter(lastInsertedNode);
9514:     else nextRange.setStart(range.startContainer, range.startOffset);
9515:     nextRange.collapse(true);

```

```

9516:
9517:     const selection = window.getSelection();
9518:     selection.removeAllRanges();
9519:     selection.addRange(nextRange);
9520:     captureRichSelection(editor);
9521: }

```

## ***insertPlainTextIntoRichEditor - lines 9523-9533***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9523-9533 (11 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function insertPlainTextIntoRichEditor(editor, pastedText) {                                                                                                                                          |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | replace, createDocumentFragment, split, forEach, append, createElement, createTextNode, insertFragmentIntoRichEditor                                                                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9523: function insertPlainTextIntoRichEditor(editor, pastedText) {
9524:     const textToInsert = String(pastedText || "").replace(/\r\n?/g, "\n");
9525:     if (!textToInsert) return;
9526:
9527:     const fragment = document.createDocumentFragment();
9528:     textToInsert.split("\n").forEach((line, index) => {
9529:         if (index > 0) fragment.append(document.createElement("br"));
9530:         if (line) fragment.append(document.createTextNode(line));
9531:     });
9532:     insertFragmentIntoRichEditor(editor, fragment);
9533: }

```

## ***insertHtmlIntoRichEditor - lines 9535-9543***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9535-9543 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function insertHtmlIntoRichEditor(editor, pastedHtml) {                                                                                                                                               |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | replace, createElement, sanitizeRichInlineHtml, insertFragmentIntoRichEditor, cloneNode                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9535: function insertHtmlIntoRichEditor(editor, pastedHtml) {
9536:     const normalizedHtml = String(pastedHtml || "")
9537:     .replace(/<\/(?<:p|div|li|h[1-6]|tr)>/gi, "<br>")
9538:     .replace(/<(?<:p|div|li|h[1-6]|tr|td|th)[^>]*>/gi, "")
9539:     .replace(/<br\s*\/?>\s*(<br\s*\/?>\s*){2,}/gi, "<br><br>");
9540:     const template = document.createElement("template");
9541:     template.innerHTML = sanitizeRichInlineHtml(normalizedHtml);
9542:     insertFragmentIntoRichEditor(editor, template.content.cloneNode(true));
9543: }

```

## ***handleRichEditorKeydown - lines 9565-9610***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9565-9610 (46 source lines).                                                                                                                                                                    |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handleRichEditorKeydown(event) {                                                                                                                                                             |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | closest, includes, toLowerCase, preventDefault, applyRichInlineCommand, execCommand, captureRichSelection, selectionRangeInsideEditor, contains, deleteContents, createRange, setStart                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 6 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

#### Source excerpt:

```

9565: function handleRichEditorKeydown(event) {
9566:   const editor = event.target.closest("[data-rich-editor]");
9567:   if (!editor) return;
9568:
9569:   if ((event.ctrlKey || event.metaKey) && ["b", "i", "u"].includes(event.key.toLowerCase())) {
9570:     event.preventDefault();
9571:     const command = event.key.toLowerCase() === "b"
9572:       ? "bold"
9573:       : event.key.toLowerCase() === "i"
9574:       ? "italic"
9575:       : "underline";
9576:     applyRichInlineCommand(editor, command);
9577:     return;
9578:   }
9579:
9580:   if (event.key !== "Enter") return;
9581:
9582:   const block = event.target.closest(".rich-text-block");
9583:   if (!block) return;
9584:
9585:   event.preventDefault();
9586:   if (event.shiftKey) {
9587:     document.execCommand("insertLineBreak", false);
9588:     captureRichSelection(editor);
9589:     return;
9590:   }
9591:
9592:   const range = selectionRangeInsideEditor(editor);
9593:   if (!range || !block.contains(range.startContainer)) return;
9594:   range.deleteContents();
9595:   const tail = document.createRange();
9596:   tail.setStart(range.startContainer, range.startOffset);
9597:   tail.setEnd(block, block.childNodes.length);
9598:   const trailingContent = tail.extractContents();
9599:   const nextBlock = matchingTextBlock(block, "");
9600:   nextBlock.append(trailingContent);
9601:   block.after(nextBlock);
9602:   const nextRange = document.createRange();
9603:   nextRange.selectNodeContents(nextBlock);
9604:   nextRange.collapse(true);
9605:   const selection = window.getSelection();
9606:   selection.removeAllRanges();
9607:   selection.addRange(nextRange);
9608:   captureRichSelection(editor);
9609:   setStatus("Unsaved local changes.");
9610: }

```

### ***handleRichEditorFocus - lines 9632-9637***

| Item            | Explanation                                         |
|-----------------|-----------------------------------------------------|
| Source location | Lines 9632-9637 (6 source lines).                   |
| Purpose         | Handles an event, request, command, or user action. |
| Declaration     | function handleRichEditorFocus(event) {             |
| Touches         | browser/Electron DOM                                |
| Calls detected  | closest, execCommand, captureRichSelection          |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9632: function handleRichEditorFocus(event) {
9633:   const editor = event.target.closest("[data-rich-editor]");
9634:   if (!editor) return;
9635:   document.execCommand("defaultParagraphSeparator", false, "p");
9636:   captureRichSelection(editor);
9637: }

```

### ***handleRichEditorInput - lines 9639-9644***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9639-9644 (6 source lines).                                                                                                                                                                     |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handleRichEditorInput(event) {                                                                                                                                                               |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closest, captureRichSelection, setStatus                                                                                                                                                              |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9639: function handleRichEditorInput(event) {
9640:   const editor = event.target.closest("[data-rich-editor]");
9641:   if (!editor) return;
9642:   captureRichSelection(editor);
9643:   setStatus("Unsaved local changes.");
9644: }

```

### ***handleRichEditorBeforeInput - lines 9646-9653***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9646-9653 (8 source lines).                                                                                                                                                                     |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handleRichEditorBeforeInput(event) {                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closest, clearPersistentTextSelection                                                                                                                                                                 |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9646: function handleRichEditorBeforeInput(event) {
9647:   const editor = event.target.closest("[data-rich-editor]");
9648:   if (!editor) return;
9649:   const inputType = event.inputType || "";
9650:   if (!/insertFromPaste|insertFromDrop/i.test(inputType) && /^(insert|delete|history)/i.test(inputType))

```

```

{
  9651:    clearPersistentTextSelection(true);
  9652:  }
  9653: }

```

## ***finishTextSelectionGesture - lines 9728-9753***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9728-9753 (26 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function finishTextSelectionGesture(event) {                                                                                                                                                          |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | getSelection, selectionRangeInsideEditor, cloneRange, showPersistentTextSelection, rangeBelongsToEditor, clearPersistentTextSelection, refreshTextSelectionInspector                                  |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9728: function finishTextSelectionGesture(event) {
9729:   if (event?.button === 2 || event?.target?.closest?("#text-selection-inspector, #summary-context-
menu")) return;
9730:   selectionGestureActive = false;
9731:   clearTimeout(selectionGestureFinishTimer);
9732:   selectionGestureFinishTimer = setTimeout(() => {
9733:     const selection = window.getSelection();
9734:     let node = selection?.anchorNode;
9735:     if (node?.nodeType === Node.TEXT_NODE) node = node.parentElement;
9736:     const editor = node?.closest?("[data-rich-editor]");
9737:     const range = editor ? selectionRangeInsideEditor(editor) : null;
9738:     if (editor && range && !range.collapsed) {
9739:       activeSummaryEditor = editor;
9740:       activeRichSelectionRange = range.cloneRange();
9741:       activeTextSelectionRange = range.cloneRange();
9742:       selectionGestureProducedRange = true;
9743:       showPersistentTextSelection(range);
9744:     } else if (!selectionGestureProducedRange) {
9745:       if (editor && rangeBelongsToEditor(activeTextSelectionRange, editor)) {
9746:         showPersistentTextSelection(activeTextSelectionRange);
9747:       } else {
9748:         clearPersistentTextSelection(true);
9749:       }
9750:     }
9751:     refreshTextSelectionInspector();
9752:   }, 40);
9753: }

```

## ***endSelectionInspectorDrag - lines 9814-9819***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9814-9819 (6 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function endSelectionInspectorDrag() {                                                                                                                                                                |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | remove                                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:



```

9814: function endSelectionInspectorDrag() {
9815:   if (!activeSelectionInspectorDrag) return;
9816:   activeSelectionInspectorDrag.target?.releasePointerCapture?.(activeSelectionInspectorDrag.pointerId);
9817:   textSelectionInspector?.classList.remove("is-dragging");
9818:   activeSelectionInspectorDrag = null;
9819: }

```

## ***handleSelectionInspectorTextField - lines 9843-9849***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9843-9849 (7 source lines).                                                                                                                                                                     |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handleSelectionInspectorTextField(event) {                                                                                                                                                   |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | preventDefault, applySelectionInspectorFormat                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9843: function handleSelectionInspectorTextField(event) {
9844:   if (event.type === "keydown" && event.key !== "Enter") return;
9845:   if (event.type === "keydown") event.preventDefault();
9846:   if (event.target === selectionFontInput) applySelectionInspectorFormat("font", event.target.value);
9847:   if (event.target === selectionColorInput) applySelectionInspectorFormat("color", event.target.value);
9848:   if (event.target === selectionSizeInput) applySelectionInspectorFormat("size", event.target.value);
9849: }

```

## ***scheduleSelectionInspectorTextField - lines 9851-9859***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9851-9859 (9 source lines).                                                                                                                                                                     |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function scheduleSelectionInspectorTextField(event) {                                                                                                                                                 |
| Touches            | filesystem                                                                                                                                                                                            |
| Calls detected     | cleanFontFamily, applySelectionInspectorFormat, normalizeTextColor, normalizeFontPx                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9851: function scheduleSelectionInspectorTextField(event) {
9852:   clearTimeout(selectionInspectorInputTimer);
9853:   const target = event.target;
9854:   selectionInspectorInputTimer = setTimeout(() => {
9855:     if (target === selectionFontInput && cleanFontFamily(target.value))
applySelectionInspectorFormat("font", target.value);
9856:     if (target === selectionColorInput && normalizeTextColor(target.value))
applySelectionInspectorFormat("color", target.value);
9857:     if (target === selectionSizeInput && normalizeFontPx(target.value))
applySelectionInspectorFormat("size", target.value);
9858:   }, 350);
9859: }

```

## ***focusAdjacentImageText - lines 9871-9882***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9871-9882 (12 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function focusAdjacentImageText(block, direction) {                                                                                                                                                   |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closest, matches, matchingTextBlock, before, after, focusTextBlock, saveRichEditorToProject                                                                                                           |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9871: function focusAdjacentImageText(block, direction) {
9872:   const editor = block?.closest("[data-rich-editor]");
9873:   if (!editor) return;
9874:   const sibling = direction === "before" ? block.previousElementSibling : block.nextElementSibling;
9875:   const textBlock = sibling?.matches(".rich-text-block") ? sibling : matchingTextBlock(block, "");
9876:   if (textBlock !== sibling) {
9877:     if (direction === "before") block.before(textBlock);
9878:     else block.after(textBlock);
9879:   }
9880:   focusTextBlock(textBlock);
9881:   saveRichEditorToProject(editor);
9882: }

```

## ***handleRichCaretClick - lines 9884-9889***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9884-9889 (6 source lines).                                                                                                                                                                     |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handleRichCaretClick(event) {                                                                                                                                                                |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | closest, focusAdjacentImageText                                                                                                                                                                       |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9884: function handleRichCaretClick(event) {
9885:   const caretButton = event.target.closest("[data-rich-caret]");
9886:   if (!caretButton) return false;
9887:   focusAdjacentImageText(caretButton.closest(".rich-image-block"), caretButton.dataset.richCaret);
9888:   return true;
9889: }

```

## ***handleRichDragStart - lines 9891-9910***

| Item            | Explanation                                                       |
|-----------------|-------------------------------------------------------------------|
| Source location | Lines 9891-9910 (20 source lines).                                |
| Purpose         | Handles an event, request, command, or user action.               |
| Declaration     | function handleRichDragStart(event) {                             |
| Touches         | browser/Electron DOM                                              |
| Calls detected  | closest, preventDefault, selectRichBlock, add, clearData, setData |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9891: function handleRichDragStart(event) {
9892:   const handle = event.target.closest("[data-rich-drag-handle]");
9893:   const imageBlock = event.target.closest(".rich-image-block");
9894:   const isControl = event.target.closest("button, input, select, textarea, a, .rich-block-actions, [data-rich-caret]");
9895:   const block = handle?.closest(".rich-block") || (!isControl ? imageBlock : null);
9896:   if (!block) {
9897:     if (imageBlock) event.preventDefault();
9898:     return;
9899:   }
9900:   activeRichDragBlock = block;
9901:   selectRichBlock(activeRichDragBlock);
9902:   activeRichDragBlock.classList.add("is-dragging");
9903:   if (event.dataTransfer) {
9904:     event.dataTransfer.clearData();
9905:     event.dataTransfer.effectAllowed = "move";
9906:     event.dataTransfer.dropEffect = "move";
9907:     event.dataTransfer.setData("text/x-rich-block", "move");
9908:     event.dataTransfer.setData("text/plain", activeRichDragBlock.dataset.title || activeRichDragBlock.dataset.type || "image");
9909:   }
9910: }

```

## ***handleRichDragOver - lines 9912-9919***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9912-9919 (8 source lines).                                                                                                                                                                     |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handleRichDragOver(event) {                                                                                                                                                                  |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closest, preventDefault, updateRichDropMarker                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9912: function handleRichDragOver(event) {
9913:   if (!activeRichDragBlock) return;
9914:   const editor = event.target.closest("[data-rich-editor]");
9915:   if (!editor || editor !== activeRichDragBlock.closest("[data-rich-editor]")) return;
9916:   event.preventDefault();
9917:   if (event.dataTransfer) event.dataTransfer.dropEffect = "move";
9918:   updateRichDropMarker(editor, activeRichDragBlock, event.clientX, event.clientY, event.target);
9919: }

```

## ***richRangeFromPoint - lines 9921-9933***

| Item            | Explanation                                                                  |
|-----------------|------------------------------------------------------------------------------|
| Source location | Lines 9921-9933 (13 source lines).                                           |
| Purpose         | Part of the private builder frontend and editing experience.                 |
| Declaration     | function richRangeFromPoint(x, y) {                                          |
| Touches         | browser/Electron DOM                                                         |
| Calls detected  | caretRangeFromPoint, caretPositionFromPoint, createRange, setStart, collapse |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9921: function richRangeFromPoint(x, y) {
9922:   if (document.caretRangeFromPoint) return document.caretRangeFromPoint(x, y);
9923:   if (document.caretPositionFromPoint) {
9924:     const position = document.caretPositionFromPoint(x, y);
9925:     if (position) {
9926:       const range = document.createRange();
9927:       range.setStart(position.offsetNode, position.offset);
9928:       range.collapse(true);
9929:       return range;
9930:     }
9931:   }
9932:   return null;
9933: }

```

## ***richDropLocation - lines 9935-9950***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9935-9950 (16 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function richDropLocation(editor, movingBlock, x, y, eventTarget = null) {                                                                                                                            |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | validDropRangeForEditor, richRangeFromPoint, contains, elementFromPoint                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9935: function richDropLocation(editor, movingBlock, x, y, eventTarget = null) {
9936:   const previousPointerEvents = movingBlock?.style.pointerEvents || "";
9937:   if (movingBlock) movingBlock.style.pointerEvents = "none";
9938:   try {
9939:     const dropRange = validDropRangeForEditor(editor, movingBlock, richRangeFromPoint(x, y));
9940:     const pointTarget = eventTarget && !movingBlock?.contains(eventTarget)
9941:       ? eventTarget
9942:       : document.elementFromPoint(x, y);
9943:     return {
9944:       dropRange,
9945:       target: pointTarget?.closest?(".rich-block") || null
9946:     };
9947:   } finally {
9948:     if (movingBlock) movingBlock.style.pointerEvents = previousPointerEvents;
9949:   }
9950: }

```

## ***validDropRangeForEditor - lines 9952-9960***

| Item            | Explanation                                                    |
|-----------------|----------------------------------------------------------------|
| Source location | Lines 9952-9960 (9 source lines).                              |
| Purpose         | Part of the private builder frontend and editing experience.   |
| Declaration     | function validDropRangeForEditor(editor, movingBlock, range) { |
| Touches         | Mostly local calculations or local UI state.                   |
| Calls detected  | contains, closest                                              |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9952: function validDropRangeForEditor(editor, movingBlock, range) {
9953:   const rangeNode = range?.startContainer?.nodeType === Node.TEXT_NODE
9954:     ? range.startContainer.parentElement
9955:     : range?.startContainer;
9956:   if (!range || !rangeNode || !editor.contains(rangeNode)) return null;
9957:   if (movingBlock?.contains(rangeNode)) return null;
9958:   if (rangeNode.closest("[contenteditable='false']")) return null;
9959:   return range;
9960: }

```

## ***ensureRichDropMarker - lines 9962-9971***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9962-9971 (10 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function ensureRichDropMarker() {                                                                                                                                                                     |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | createElement, setAttribute, append                                                                                                                                                                   |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

9962: function ensureRichDropMarker() {
9963:   if (!activeRichDropMarker) {
9964:     activeRichDropMarker = document.createElement("div");
9965:     activeRichDropMarker.className = "rich-drop-marker";
9966:     activeRichDropMarker.setAttribute("aria-hidden", "true");
9967:     document.body.append(activeRichDropMarker);
9968:   }
9969:   activeRichDropMarker.hidden = false;
9970:   return activeRichDropMarker;
9971: }

```

## ***hideRichDropMarker - lines 9973-9975***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9973-9975 (3 source lines).                                                                                                                                                                     |
| Purpose            | Controls a window, dialog, panel, menu, or visible state.                                                                                                                                             |
| Declaration        | function hideRichDropMarker() {                                                                                                                                                                       |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | No obvious named calls detected by the generator.                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 0 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
9973: function hideRichDropMarker() {
9974:   if (activeRichDropMarker) activeRichDropMarker.hidden = true;
9975: }
```

## ***updateRichDropMarker - lines 9977-10006***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 9977-10006 (30 source lines).                                                                                                                                                                   |
| Purpose            | Updates UI, state, cache, or stored data.                                                                                                                                                             |
| Declaration        | function updateRichDropMarker(editor, movingBlock, x, y, eventTarget = null) {                                                                                                                        |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | ensureRichDropMarker, richDropLocation, getClientRects, getBoundingClientRect, max, min, contains                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```
9977: function updateRichDropMarker(editor, movingBlock, x, y, eventTarget = null) {
9978:   if (!editor || !movingBlock) return;
9979:   const marker = ensureRichDropMarker();
9980:   const { dropRange, target } = richDropLocation(editor, movingBlock, x, y, eventTarget);
9981:   if (dropRange) {
9982:     const rangeRect = dropRange.getClientRects()[0];
9983:     const anchor = dropRange.startContainer.nodeType === Node.TEXT_NODE
9984:       ? dropRange.startContainer.parentElement
9985:       : dropRange.startContainer;
9986:     const anchorRect = anchor?.getBoundingClientRect?();
9987:     const rect = rangeRect || anchorRect || editor.getBoundingClientRect();
9988:     marker.className = "rich-drop-marker caret";
9989:     marker.style.left = `${Math.max(rect.left, Math.min(x, rect.right || rect.left))}px`;
9990:     marker.style.top = `${rect.top}px`;
9991:     marker.style.width = "2px";
9992:     marker.style.height = `${Math.max(22, rect.height || 22)}px`;
9993:     return;
9994:   }
9995:
9996:   const targetRect = (target && editor.contains(target) && target !== movingBlock)
9997:     ? target.getBoundingClientRect()
9998:     : editor.getBoundingClientRect();
9999:   const imageRect = movingBlock.getBoundingClientRect();
10000:   const beforeTarget = !target || y < targetRect.top + targetRect.height / 2;
10001:   marker.className = "rich-drop-marker block";
10002:   marker.style.left = `${targetRect.left}px`;
10003:   marker.style.top = `${beforeTarget ? targetRect.top : targetRect.bottom}px`;
10004:   marker.style.width = `${Math.min(targetRect.width, Math.max(160, imageRect.width ||
targetRect.width))}px`;
10005:   marker.style.height = "3px";
10006: }
```

## ***moveRichBlockToPoint - lines 10008-10029***

| Item               | Explanation                                                                                                                                                                                             |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10008-10029 (22 source lines).                                                                                                                                                                    |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                            |
| Declaration        | function moveRichBlockToPoint(editor, movingBlock, x, y, eventTarget = null, options = {}) {                                                                                                            |
| Touches            | browser/Electron DOM                                                                                                                                                                                    |
| Calls detected     | richDropLocation, remove, insertRichBlockAtCursor, contains, append, getBoundingClientRect, before, after, cleanupEmptyRichTextBlocks, focusAdjacentImageText, saveRichEditorToProject, selectRichBlock |
| API endpoints      | None detected inside this function body.                                                                                                                                                                |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                                 |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                           |

| Item           | Explanation                                                                                                                                                                                           |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Debug approach | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10008: function moveRichBlockToPoint(editor, movingBlock, x, y, eventTarget = null, options = {}) {
10009:   if (!editor || !movingBlock) return;
10010:   const { dropRange, target } = richDropLocation(editor, movingBlock, x, y, eventTarget);
10011:
10012:   movingBlock.classList.remove("is-dragging");
10013:   if (target === movingBlock) return;
10014:
10015:   if (dropRange) {
10016:     insertRichBlockAtCursor(editor, movingBlock, dropRange);
10017:   } else if (!target || !editor.contains(target)) {
10018:     editor.append(movingBlock);
10019:   } else if (y < target.getBoundingClientRect().top + target.getBoundingClientRect().height / 2) {
10020:     target.before(movingBlock);
10021:   } else {
10022:     target.after(movingBlock);
10023:   }
10024:
10025:   cleanupEmptyRichTextBlocks(editor);
10026:   if (options.focusAfter !== false) focusAdjacentImageText(movingBlock, "after");
10027:   if (options.commit !== false) saveRichEditorToProject(editor);
10028:   if (options.focusAfter === false) selectRichBlock(movingBlock);
10029: }

```

## ***handleRichDrop - lines 10031-10039***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10031-10039 (9 source lines).                                                                                                                                                                   |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | function handleRichDrop(event) {                                                                                                                                                                      |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closest, preventDefault, moveRichBlockToPoint, hideRichDropMarker                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10031: function handleRichDrop(event) {
10032:   if (!activeRichDragBlock) return;
10033:   const editor = event.target.closest("[data-rich-editor]");
10034:   if (!editor || editor !== activeRichDragBlock.closest("[data-rich-editor]")) return;
10035:   event.preventDefault();
10036:   moveRichBlockToPoint(editor, activeRichDragBlock, event.clientX, event.clientY, event.target);
10037:   hideRichDropMarker();
10038:   activeRichDragBlock = null;
10039: }

```

## ***imageMoveDragControl - lines 10041-10043***

| Item            | Explanation                                                  |
|-----------------|--------------------------------------------------------------|
| Source location | Lines 10041-10043 (3 source lines).                          |
| Purpose         | Part of the private builder frontend and editing experience. |
| Declaration     | function imageMoveDragControl(target) {                      |
| Touches         | Mostly local calculations or local UI state.                 |
| Calls detected  | closest                                                      |
| API endpoints   | None detected inside this function body.                     |
| Storage keys    | No local/session storage keys detected.                      |

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10041: function imageMoveDragControl(target) {
10042:   return target.closest("button, input, select, textarea, a, .rich-block-actions, [data-rich-caret]");
10043: }

```

## ***beginRichImageMoveDrag - lines 10045-10061***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10045-10061 (17 source lines).                                                                                                                                                                  |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function beginRichImageMoveDrag(event) {                                                                                                                                                              |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | closest, imageMoveDragControl                                                                                                                                                                         |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 4 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10045: function beginRichImageMoveDrag(event) {
10046:   if (event.button !== 0) return;
10047:   const block = event.target.closest(".rich-image-block");
10048:   if (!block || imageMoveDragControl(event.target)) return;
10049:   if (event.target.closest(".rich-image-viewport.crop-active")) return;
10050:   const editor = block.closest("[data-rich-editor]");
10051:   if (!editor) return;
10052:   activeRichImageMoveDrag = {
10053:     block,
10054:     editor,
10055:     moved: false,
10056:     pointerId: event.pointerId,
10057:     startX: event.clientX,
10058:     startY: event.clientY
10059:   };
10060:   block.setPointerCapture?.(event.pointerId);
10061: }

```

## ***moveRichImageMoveDrag - lines 10063-10077***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10063-10077 (15 source lines).                                                                                                                                                                  |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function moveRichImageMoveDrag(event) {                                                                                                                                                               |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | hypot, selectRichBlock, add, updateRichDropMarker, preventDefault                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 2 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10063: function moveRichImageMoveDrag(event) {

```



```

10064:   if (!activeRichImageMoveDrag) return;
10065:   const drag = activeRichImageMoveDrag;
10066:   const distance = Math.hypot(event.clientX - drag.startX, event.clientY - drag.startY);
10067:   if (!drag.moved && distance < 6) return;
10068:   if (!drag.moved) {
10069:     drag.moved = true;
10070:     activeRichDragBlock = drag.block;
10071:     selectRichBlock(drag.block);
10072:     drag.block.classList.add("is-dragging");
10073:     document.body.classList.add("rich-image-dragging");
10074:   }
10075:   updateRichDropMarker(drag.editor, drag.block, event.clientX, event.clientY, event.target);
10076:   event.preventDefault();
10077: }

```

## ***endRichImageMoveDrag - lines 10079-10092***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10079-10092 (14 source lines).                                                                                                                                                                  |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function endRichImageMoveDrag(event) {                                                                                                                                                                |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | moveRichBlockToPoint, hideRichDropMarker, preventDefault, remove                                                                                                                                      |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10079: function endRichImageMoveDrag(event) {
10080:   if (!activeRichImageMoveDrag) return;
10081:   const drag = activeRichImageMoveDrag;
10082:   drag.block.releasePointerCapture?.(drag.pointerId);
10083:   if (drag.moved) {
10084:     moveRichBlockToPoint(drag.editor, drag.block, event.clientX, event.clientY);
10085:     hideRichDropMarker();
10086:     event.preventDefault();
10087:   }
10088:   drag.block.classList.remove("is-dragging");
10089:   document.body.classList.remove("rich-image-dragging");
10090:   activeRichImageMoveDrag = null;
10091:   activeRichDragBlock = null;
10092: }

```

## ***beginImageCropDrag - lines 10094-10109***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10094-10109 (16 source lines).                                                                                                                                                                  |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function beginImageCropDrag(event) {                                                                                                                                                                  |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | closest, normalizeCropPosition, max, getBoundingClientRect, preventDefault                                                                                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10094: function beginImageCropDrag(event) {
10095:   const viewport = event.target.closest(".rich-image-viewport.crop-active");
10096:   if (!viewport || event.button !== 0) return;
10097:   const block = viewport.closest(".rich-image-block");

```

```

10098:   activeImageCropDrag = {
10099:     block,
10100:     editor: block.closest("[data-rich-editor]"),
10101:     startX: event.clientX,
10102:     startY: event.clientY,
10103:     cropX: normalizeCropPosition(block.dataset.cropX),
10104:     cropY: normalizeCropPosition(block.dataset.cropY),
10105:     width: Math.max(1, viewport.getBoundingClientRect().width),
10106:     height: Math.max(1, viewport.getBoundingClientRect().height)
10107:   };
10108:   event.preventDefault();
10109: }

```

## ***moveImageCropDrag - lines 10111-10121***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10111-10121 (11 source lines).                                                                                                                                                                  |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function moveImageCropDrag(event) {                                                                                                                                                                   |
| Touches            | browser/Electron DOM, filesystem                                                                                                                                                                      |
| Calls detected     | normalizeCropPosition, querySelector, setProperty                                                                                                                                                     |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10111: function moveImageCropDrag(event) {
10112:   if (!activeImageCropDrag) return;
10113:   const { block, startX, startY, cropX, cropY, width, height } = activeImageCropDrag;
10114:   const nextX = normalizeCropPosition(cropX - ((event.clientX - startX) / width) * 100);
10115:   const nextY = normalizeCropPosition(cropY - ((event.clientY - startY) / height) * 100);
10116:   block.dataset.cropX = String(nextX);
10117:   block.dataset.cropY = String(nextY);
10118:   const viewport = block.querySelector(".rich-image-viewport");
10119:   viewport?.style.setProperty("--crop-x", `${nextX}%`);
10120:   viewport?.style.setProperty("--crop-y", `${nextY}%`);
10121: }

```

## ***endImageCropDrag - lines 10123-10127***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10123-10127 (5 source lines).                                                                                                                                                                   |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | function endImageCropDrag() {                                                                                                                                                                         |
| Touches            | Mostly local calculations or local UI state.                                                                                                                                                          |
| Calls detected     | saveRichEditorToProject                                                                                                                                                                               |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 0 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10123: function endImageCropDrag() {
10124:   if (!activeImageCropDrag) return;
10125:   saveRichEditorToProject(activeImageCropDrag.editor);
10126:   activeImageCropDrag = null;
10127: }

```

## ***handleStandaloneRichClick - lines 10211-10226***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10211-10226 (16 source lines).                                                                                                                                                                  |
| Purpose            | Handles an event, request, command, or user action.                                                                                                                                                   |
| Declaration        | async function handleStandaloneRichClick(event) {                                                                                                                                                     |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | handleRichCaretClick, closest, selectRichBlock, copyRichCodeBlock, editSelectedRichBlock, deleteSelectedRichBlock, saveRichEditorToProject                                                            |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 2 await expression(s), 3 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10211: async function handleStandaloneRichClick(event) {
10212:   if (handleRichCaretClick(event)) return;
10213:   const blockAction = event.target.closest("[data-rich-block-action]");
10214:   const richBlock = event.target.closest(".rich-block");
10215:   if (richBlock) selectRichBlock(richBlock);
10216:   if (!blockAction) return;
10217:   const editor = blockAction.closest("[data-rich-editor]");
10218:   selectRichBlock(blockAction.closest(".rich-block"));
10219:   if (blockAction.dataset.richBlockAction === "copy-code") {
10220:     await copyRichCodeBlock(blockAction.closest(".rich-block"));
10221:     return;
10222:   }
10223:   if (blockAction.dataset.richBlockAction === "edit") await editSelectedRichBlock(editor);
10224:   if (blockAction.dataset.richBlockAction === "delete") deleteSelectedRichBlock(editor);
10225:   saveRichEditorToProject(editor);
10226: }

```

## ***hasDataset - lines 10323-10327***

| Item               | Explanation                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source location    | Lines 10323-10327 (5 source lines).                                                                                                                                                                   |
| Purpose            | Part of the private builder frontend and editing experience.                                                                                                                                          |
| Declaration        | const hasDataset = (key) => Object.prototype.hasOwnProperty.call(button.dataset, key);                                                                                                                |
| Touches            | browser/Electron DOM                                                                                                                                                                                  |
| Calls detected     | call, addCustomSection                                                                                                                                                                                |
| API endpoints      | None detected inside this function body.                                                                                                                                                              |
| Storage keys       | No local/session storage keys detected.                                                                                                                                                               |
| Async/return clues | 1 await expression(s), 1 return statement(s).                                                                                                                                                         |
| Debug approach     | Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs. |

Source excerpt:

```

10323: const hasDataset = (key) => Object.prototype.hasOwnProperty.call(button.dataset, key);
10324: if (button.dataset.addSection) {
10325:   await addCustomSection();
10326:   return;
10327: }

```

## **Representative opening snippet**

```

1: const builderGuideOpen = document.querySelector("#builder-guide-open");
2: const builderGuideDialog = document.querySelector("#builder-guide-dialog");
3: const builderGuideClose = document.querySelector("#builder-guide-close");
4: const builderGuideSearch = document.querySelector("#builder-guide-search");
5: const builderGuideResults = document.querySelector("#builder-guide-results");
6: const builderGuideSections = document.querySelector("#builder-guide-sections");
7: const builderGuideTopicDialog = document.querySelector("#builder-guide-topic-dialog");
8: const builderGuideTopicTitle = document.querySelector("#builder-guide-topic-title");
9: const builderGuideTopicBody = document.querySelector("#builder-guide-topic-body");

```

```
10: const builderGuideTopicClose = document.querySelector("#builder-guide-topic-close");
11: const templateSelect = document.querySelector("#template-select");
12: const newCategorySelect = document.querySelector("#new-category-select");
13: const placementSelect = document.querySelector("#placement-select");
14: const newProjectTitle = document.querySelector("#new-project-title");
15: const addProjectButton = document.querySelector("#add-project");
16: const addCategoryButton = document.querySelector("#add-category");
17: const addSiteSectionButton = document.querySelector("#add-site-section");
18: const funFactsInput = document.querySelector("#fun-facts-input");
19: const funFactsCount = document.querySelector("#fun-facts-count");
20: const saveFunFactsButton = document.querySelector("#save-fun-facts");
21: const clearFunFactsButton = document.querySelector("#clear-fun-facts");
22: const siteHeroEyebrowInput = document.querySelector("#site-hero-eyebrow");
23: const siteHeroTitleInput = document.querySelector("#site-hero-title");
24: const siteHeroCopyInput = document.querySelector("#site-hero-copy");
25: const saveSiteContentButton = document.querySelector("#save-site-content");
26: const profileDisplayNameInput = document.querySelector("#profile-display-name");
27: const profilePortfolioLabelInput = document.querySelector("#profile-portfolio-label");
28: const profileContactIntroInput = document.querySelector("#profile-contact-intro");
29: const profileEmailInput = document.querySelector("#profile-email");
30: const profilePhoneInput = document.querySelector("#profile-phone");
31: const profileGithubInput = document.querySelector("#profile-github");
32: const profileLinkedinInput = document.querySelector("#profile-linkedin");
33: const profileWebsiteInput = document.querySelector("#profile-website");
34: const saveProfileContentButton = document.querySelector("#save-profile-content");
35: const profileImageUploadButton = document.querySelector("#profile-image-upload");
36: const profileHeroUploadButton = document.querySelector("#profile-hero-upload");
37: const profileResumeUploadButton = document.querySelector("#profile-resume-upload");
38: const profileBrandUploadButton = document.querySelector("#profile-brand-upload");
39: const profileAssetStatus = document.querySelector("#profile-asset-status");
40: const categoryDropdown = document.querySelector("#category-dropdown");
41: const siteSectionList = document.querySelector("#site-section-list");
42: const projectCreateDialog = document.querySelector("#project-create-dialog");
43: const projectCreateForm = document.querySelector("#project-create-form");
44: const projectCreateCategory = document.querySelector("#project-create-category");
45: const projectCreateCancel = document.querySelector("#project-create-cancel");
46: const categoryDialog = document.querySelector("#category-dialog");
47: const categoryForm = document.querySelector("#category-form");
48: const categoryTitle = document.querySelector("#category-title");
49: const categoryDescription = document.querySelector("#category-description");
50: const categoryAccent = document.querySelector("#category-accent");
51: const categoryCancel = document.querySelector("#category-cancel");
52: const projectTree = document.querySelector("#project-tree");
53: const projectSearchInput = document.querySelector("#project-search");
54: const projectSearchClear = document.querySelector("#project-search-clear");
55: const projectSearchStatus = document.querySelector("#project-search-status");
56: const projectDialog = document.querySelector("#project-dialog");
57: const projectWindowTitle = document.querySelector("#project-window-title");
58: const projectWindowClose = document.querySelector("#project-window-close");
59: const projectWindowDelete = document.querySelector("#project-window-delete");
60: const viewProjectPreviewButton = document.querySelector("#view-project-preview");
61: const saveProjectButton = document.querySelector("#save-project");
62: const saveProjectCloseButton = document.querySelector("#save-project-close");
63: const projectTitleMenu = document.querySelector("#project-title-menu");
64: const titleRenameActions = document.querySelector("#title-rename-actions");
65: const titleRenameSave = document.querySelector("#title-rename-save");
66: const titleRenameCancel = document.querySelector("#title-rename-cancel");
67: const projectMetaDialog = document.querySelector("#project-meta-dialog");
68: const projectMetaTitle = document.querySelector("#project-meta-title");
69: const projectMetaGrid = document.querySelector("#project-meta-grid");
70: const projectMetaClose = document.querySelector("#project-meta-close");
71: const projectFields = document.querySelector("#project-fields");
72: const sectionTabs = document.querySelector("#section-tabs");
```